GOOGLE WORKSPACE TRAINING

Google Sheets Advanced

Two-day course review draft

Analyse a fictional procurement register. Labs1-9 include306 captured stages covering source audit, calculations, rules, text/lookup repair, formula/pivot tests, database criteria and tested dashboards. Native chart, sparkline and live checks reconcile; +45/+90 refresh tests were restored. Labs10-12, independent/reviewer work and learner timing remain pending.

Delivered by Asraf · Microsoft Certified Trainer

Google Sheets Advanced

Two-day course review draft · Review draft, version 0.4

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Resources: Use the practice files supplied with this review package.

Google, Google Workspace, Gmail, Google Drive, Google Docs, Google Sheets, Google Slides, Google Sites, Gemini and NotebookLM are names associated with Google. This independent training guide is not published or endorsed by Google.

Screenshots show the product at the time of writing. Google updates its apps often, so labels, availability and layouts in your account may differ. This review draft still requires full lab validation.

About the author

Asraf is a Microsoft Certified Trainer and HRD Corp-certified IT trainer based in Malaysia. He has been teaching working professionals since 2008, helping teams at banks, pharmaceutical companies, construction firms, and retailers get real work done with Microsoft 365, Power Platform, Excel, Google Workspace, and, more recently, AI tools like Copilot.

He holds a Bachelor of Information Technology (Multimedia) from the University of Malaya, along with Microsoft certifications in Power BI (PL-300) and AI for business (AB-730). Outside the classroom he writes his own automation tools in Python and Node.js, mostly to save himself from doing the same task twice.

He runs his training practice solo, which means he has personally felt every inefficiency this book tries to fix.

Contents

Before you begin 6

1 Audit the Source Register 7
1.1 Prepare the Workbook 7
1.2 Audit Key Fields 10
1.3 Protect the Analysis Trail 12
1.4 Quarantine Bad Rows and Write a Data Contract 14
Prompts and values 24

2 Build Conditional Calculations 30
2.1 Calculate Line Amounts 30
2.2 Create a Summary Grid 34
2.3 Test Logic Boundaries 36
2.4 Reconcile Three Populations and Test Conditional Boundaries 38
Prompts and values 51

3 Validate Entries and Flag Exceptions 54
3.1 Validate New Entries 54
3.2 Flag Pending High Amounts 56
3.3 Test the Rules 57
3.4 Create a Rule Test Harness and Reconcile Exceptions 60
Prompts and values 76

4 Combine and Clean Text 78
4.1 Extract Identifier Parts 78
4.2 Combine and Clean Text 81
4.3 Check Categorisation 82
4.4 Build a Reversible Text Normalisation Pipeline 85
Prompts and values 96

5 Match Records and Handle Errors 101
5.1 Look Up a Rate 101
5.2 Practise a Horizontal Lookup 104
5.3 Handle an Expected Missing Key 106
5.4 Challenge Exact Lookups With Missing and Duplicate Keys 108
Prompts and values 121

6 Complete the Formula Clinic 125
6.1 Create Three Test Defects 125
6.2 Repair and Test 129
6.3 Reconcile and Explain 132
6.4 Run the Formula Fault Clinic 135
Prompts and values 143

7 Summarise with Pivot Tables 147
7.1 Build the Pivot 147
7.2 Filter and Compare 150
7.3 Inspect a Calculated Field 151
7.4 Reconcile Pivot Semantics and New-Row Coverage 154
Prompts and values 166

8 Query Structured Criteria 170
8.1 Build the Criteria 170
8.2 Count and Average 172
8.3 Check Criteria Semantics 174
8.4 Prove AND, OR and Overlapping Criteria 176
Prompts and values 191

9 Build a Dashboard 197
9.1 Build a Compact View 197
9.2 Show the Weekly Sequence 200
9.3 Test a Refresh 202
9.4 Design a Dashboard With a Visible Refresh Contract 206
Prompts and values 215

10 Use Gemini as an Analysis Partner 219
10.1 Request a Formula Proposal 219
10.2 Check an Interpretation 220
10.3 Inspect Generated Visuals 220
10.4 Evaluate AI Proposals Against Known Test Cases 220
Prompts and values 222

11 Complete the Analysis Capstone 223
11.1 Define and Calculate 223
11.2 Challenge the Result 224
11.3 Present the Analysis 224
11.4 Complete a Decision Analysis With Independent Evidence 224
Prompts and values 226

12 Review and Handover 227
12.1 Review the Baseline 227
12.2 Write the Handover 227
12.3 Demonstrate One Update 228
12.4 Make the Analysis Operable by Another Analyst 228
Prompts and values 230

INTRODUCTION

Before you begin

This review book accompanies Google Sheets Advanced. Work through its chapters in order and keep the original practice inputs unchanged. The same notes and prompts also appear in the local course website.

Current Status

All chapters in this book are written drafts. Most have not yet been executed end to end in the application. The masterclass's reference brief chapter has a tested core path with a recorded recovery prompt; that does not validate the other exercises. Treat VERIFY notes and draft notices as production work still to complete.

What You Need

Use a desktop browser and the Google account approved for the class. Confirm app and AI access beforehand. Edition, administrator controls and language can affect availability.

Download the supplied fictional practice pack. No real client or personnel record is needed. Dataset week fields are relative practice weeks, not an agreed pilot start date.

How to Use the Exercises

Follow the numbered steps, attempt the output and then open the expected-result box. AI wording can vary; source facts and approval conditions must remain correct.

Values and prompts follow the notes. Replace source placeholders before submitting. Inspect proposed edits before applying them. Drafting does not authorise sending, inviting people or publishing.

Course Timing

This course is planned for 2 days, with 390 teaching minutes per day, excluding breaks and lunch. Estimates include practice, checks and discussion and need a classroom pilot.

Each course supplies independent starting inputs. Use a stated recovery file if an earlier output is missing.

Expanded Casework

Revision 0.4 adds casework, independent challenges and evidence gates to every chapter. The original short guided path alone is insufficient. Timings are targets, not measured learner results. Read each case input and keep baseline and scenario files separate.

01

CHAPTER 1

Audit the Source Register

In this chapter you will check types, blanks, duplicates and keys before analysis.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 45 minutes: 10 explanation/demonstration, 15 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 35 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: An audited source, preserved snapshot and empty Analysis workspace.

> **Status:** author reference with 30 staged application captures through 1.4. Native conversion, baseline diagnostics, preserved snapshot, empty Analysis, six raw intake cases, training contract and evidence-backed proposed dispositions checked. Upload required instructor handoff. Independent disposition/release-condition work, reviewer challenge and learner timing remain pending.

What You Will Learn

- Inspect dataset grain and keys.
- Check types and baseline controls.
- Preserve the source before analysis.

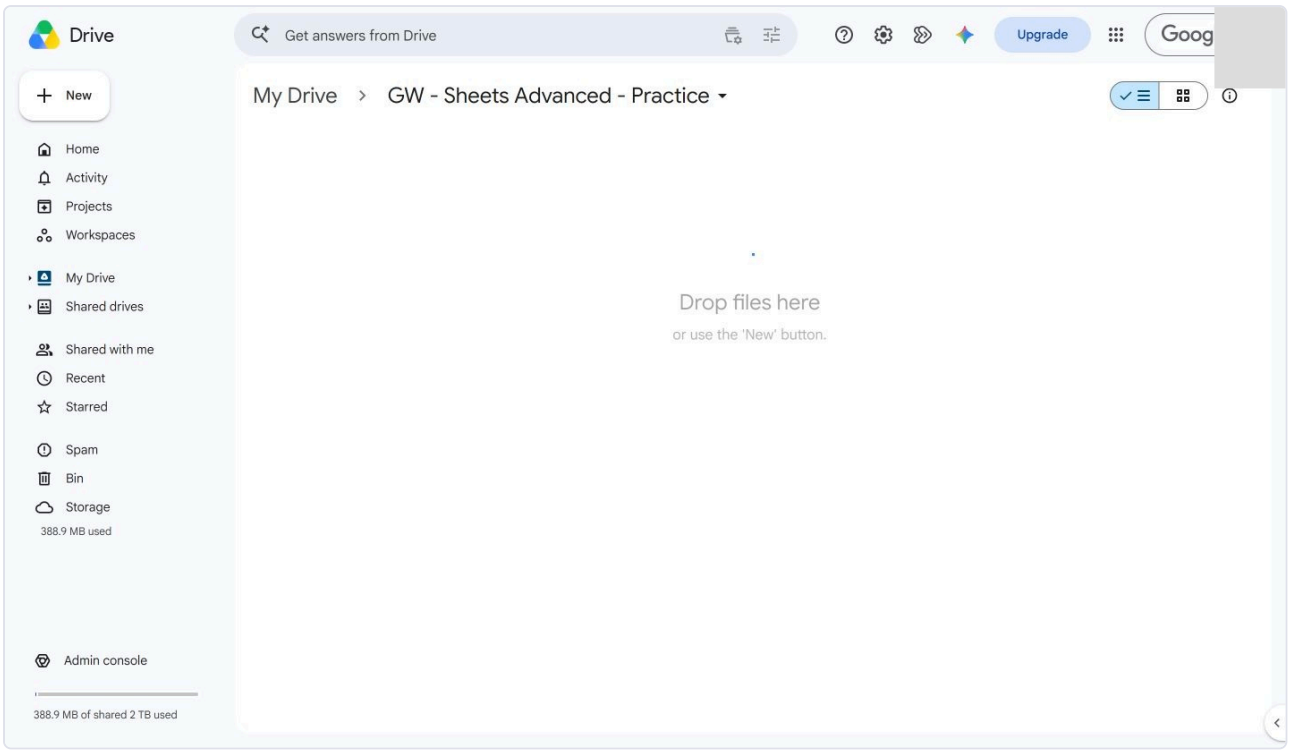
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

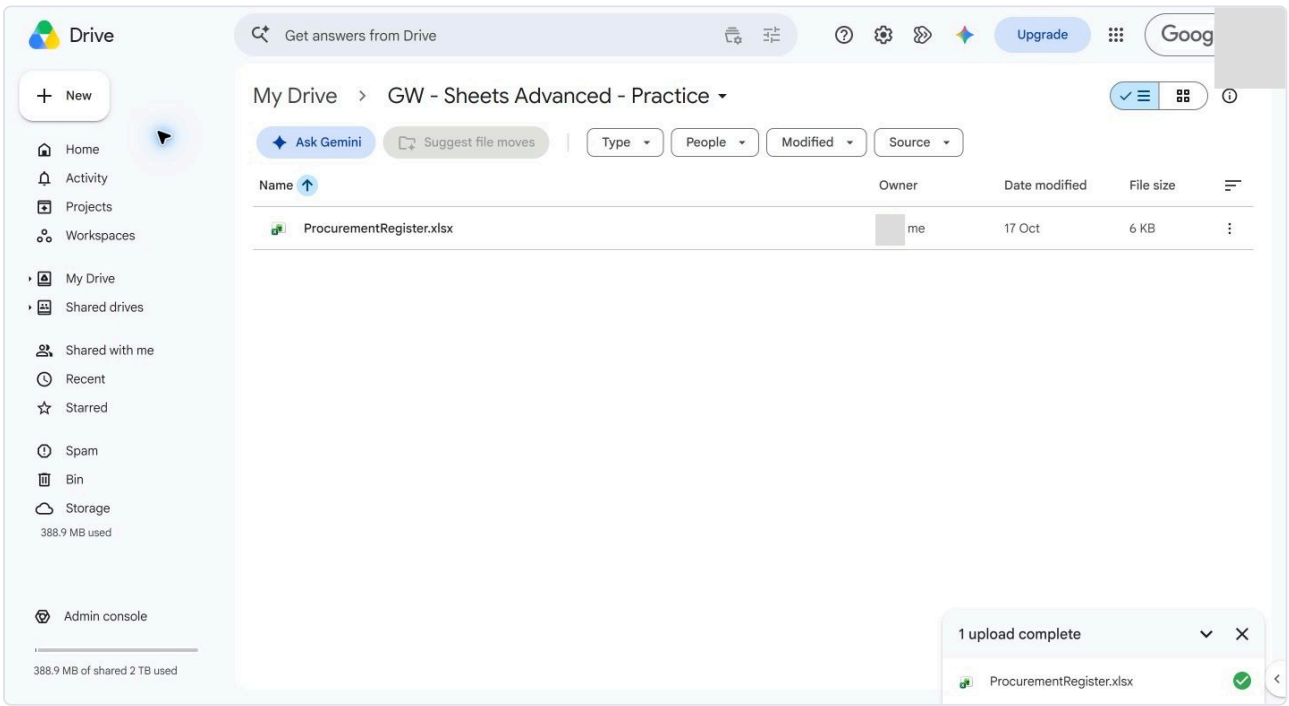
Download ProcurementRegister.xlsx. It contains 24 fictional order lines and a six-item rate table, separate from the pilot budget.

1.1 Prepare the Workbook

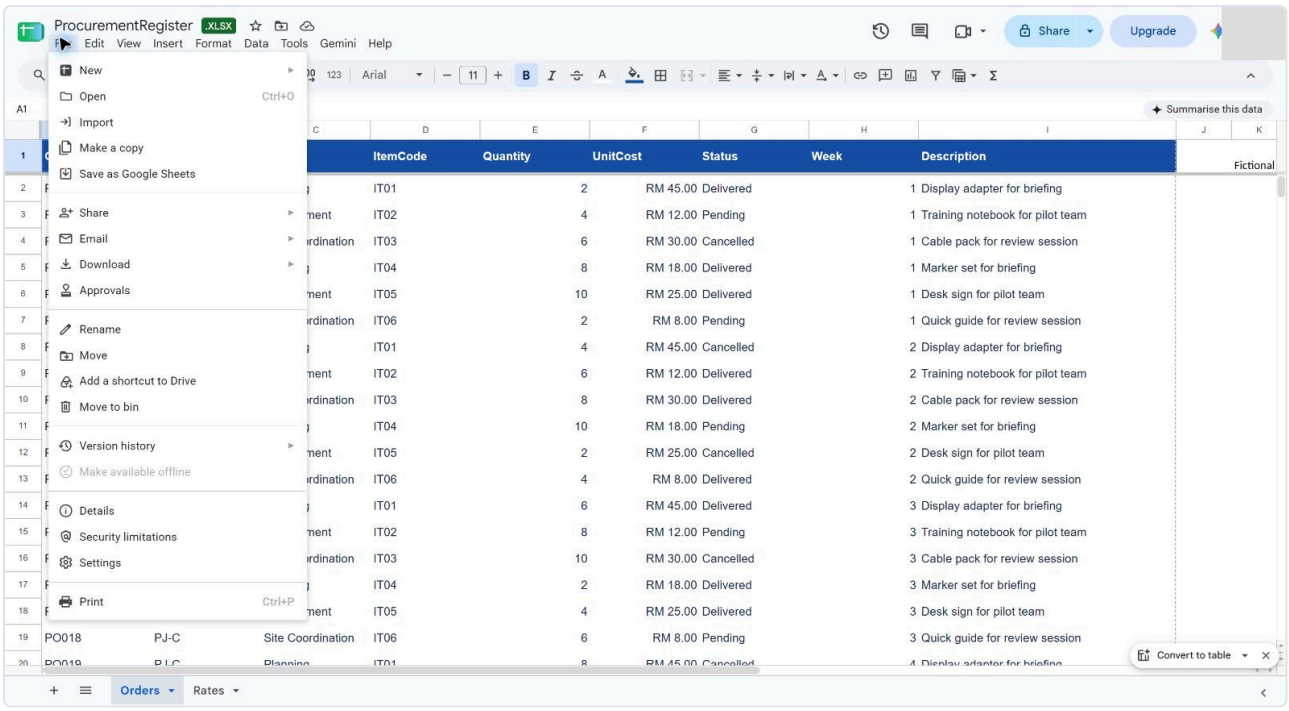
1. In the training account, create a separate **GW - Sheets Advanced - Practice** folder in My Drive. Keep it separate from the Foundation register.



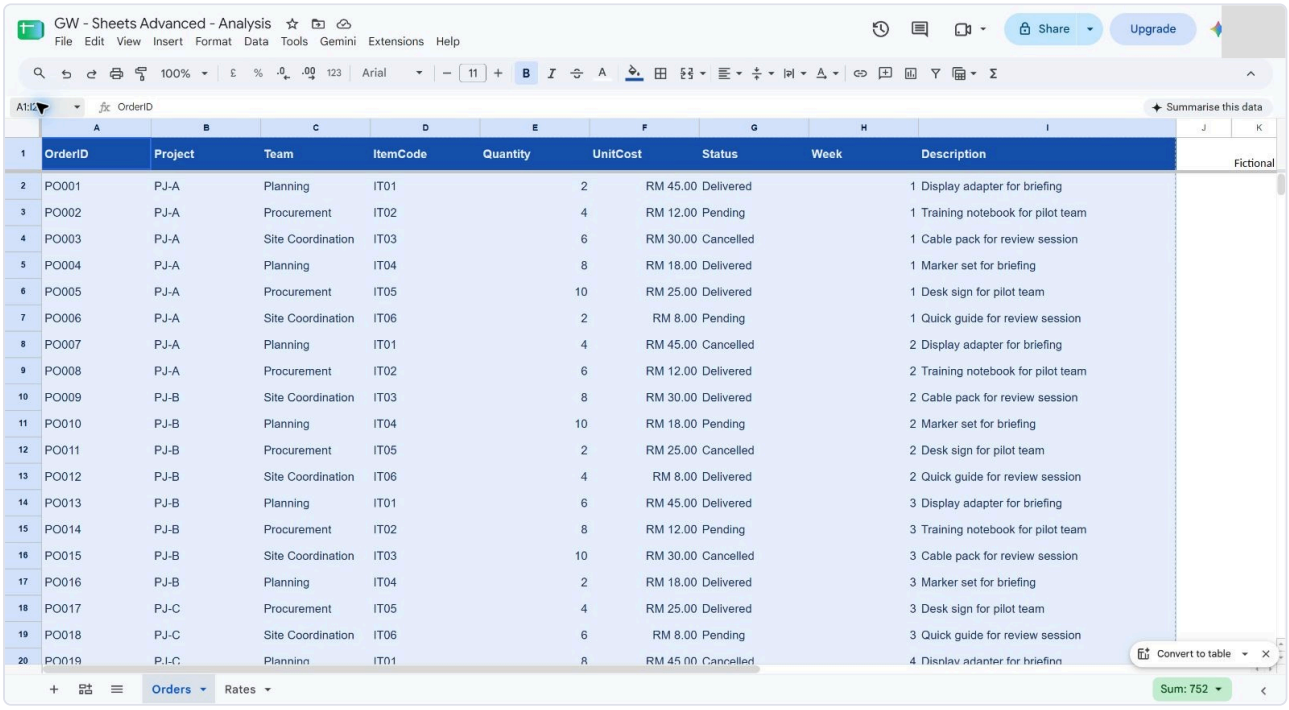
2. Upload **ProcurementRegister.xlsx** from the supplied practice files. In this author run the extension blocked local-file uploads, so the instructor uploaded the file manually. This is a handoff, not a verified automated upload.



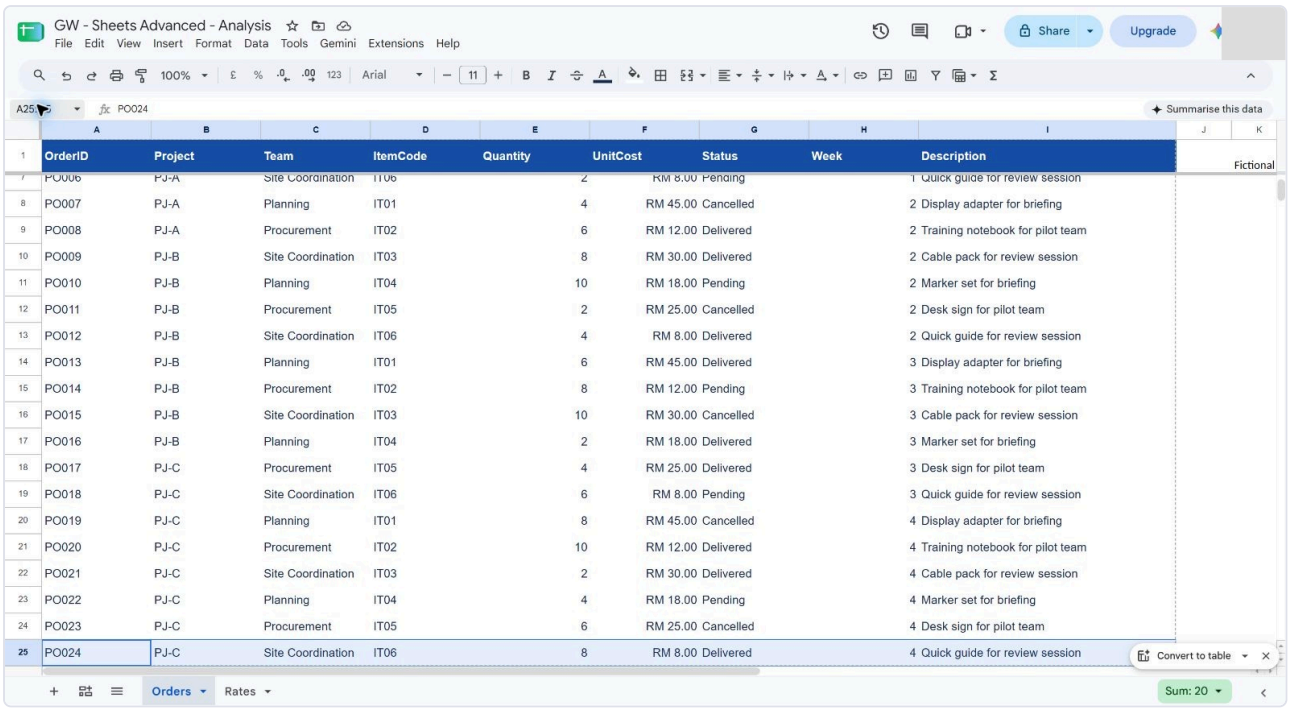
3. Open the Excel workbook and use **File > Save as Google Sheets**. Preserve the original XLSX. This creates a distinct native file; do not simply rename the Excel source.



4. Rename the native file **GW - Sheets Advanced - Analysis**. On **Orders**, confirm the nine working headers in **A1:I1** and inspect the records. The supplied **K1** fictional-data label is outside that working table.

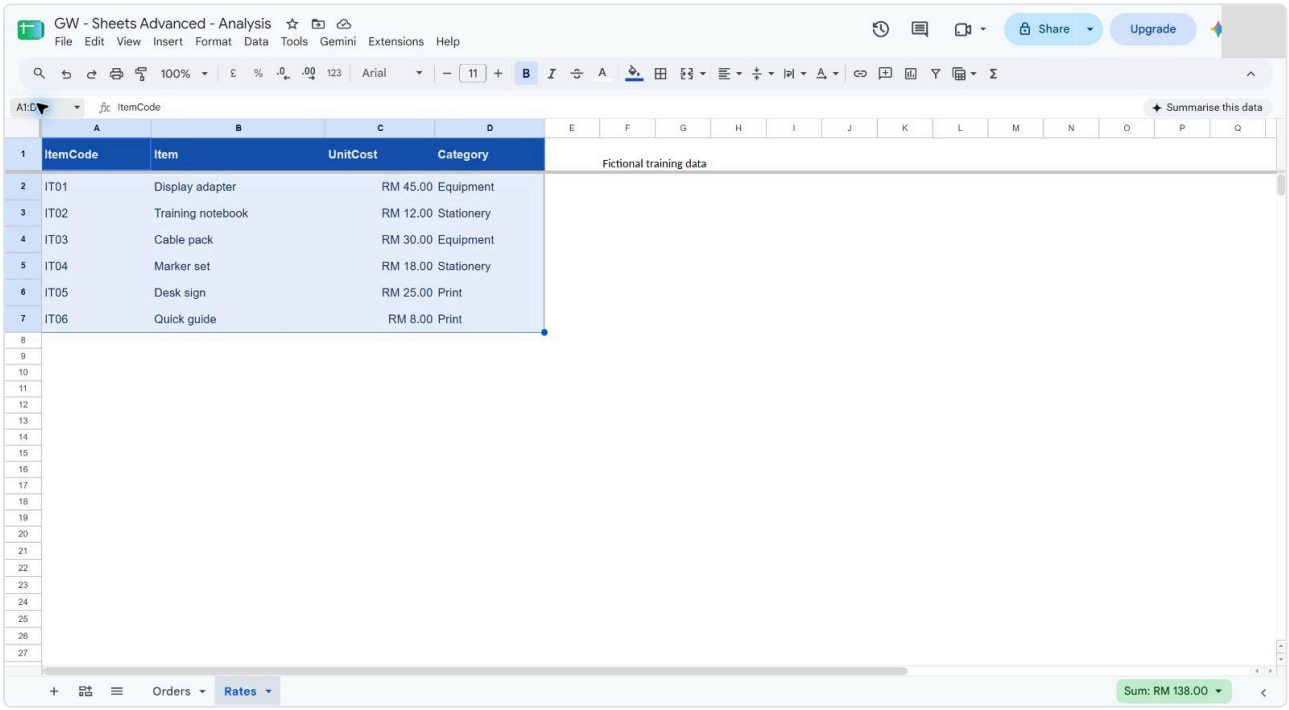


5. Navigate to **A25:I25**. Confirm **PO024** is the final supplied order, with 24 data rows in **2:25**. One row represents a fictional order line, not a project subtotal or purchase commitment.



OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description
PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session
PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing
PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team
PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session
PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing
PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team
PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session
PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing
PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team
PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session
PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing
PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team
PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session
PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing
PO020	PJ-C	Procurement	IT02	10	RM 12.00	Delivered	4	Training notebook for pilot team
PO021	PJ-C	Site Coordination	IT03	2	RM 30.00	Delivered	4	Cable pack for review session
PO022	PJ-C	Planning	IT04	4	RM 18.00	Pending	4	Marker set for briefing
PO023	PJ-C	Procurement	IT05	6	RM 25.00	Cancelled	4	Desk sign for pilot team
PO024	PJ-C	Site Coordination	IT06	8	RM 8.00	Delivered	4	Quick guide for review session

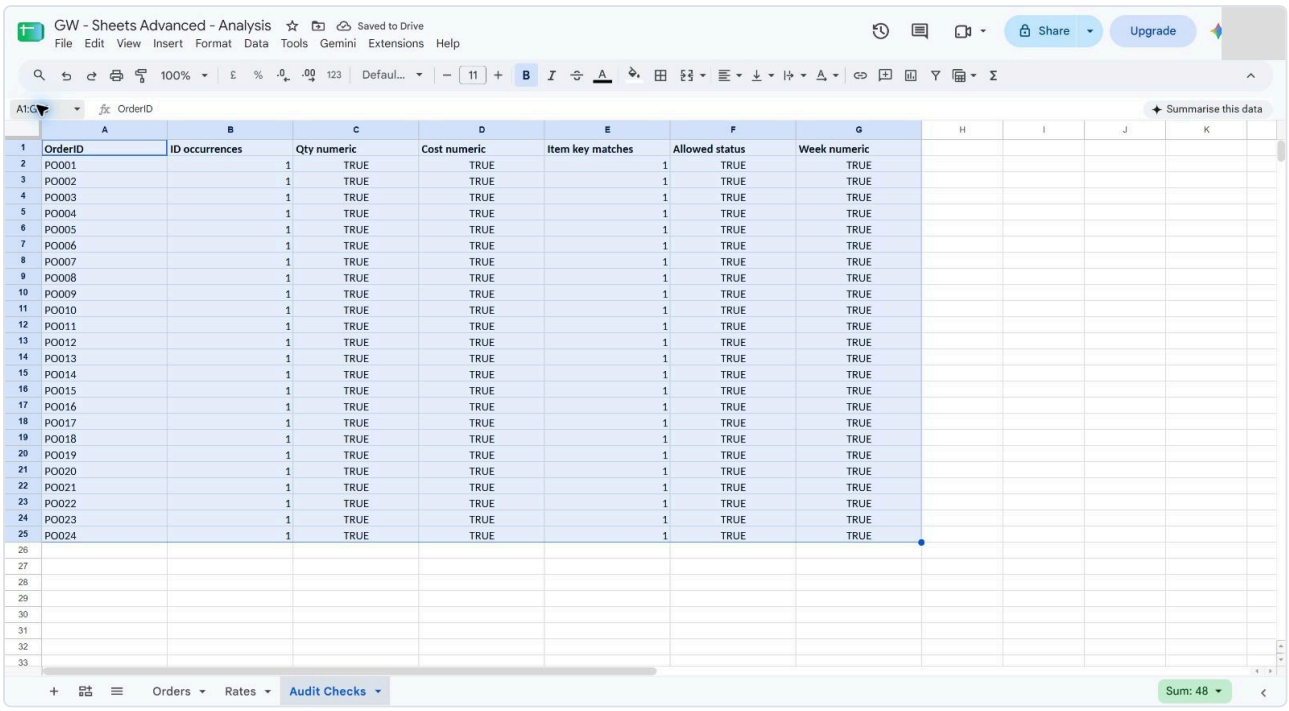
6. Open **Rates** and inspect **A1:D7**: six item codes **IT01–IT06** in A2:A7, with item, unit cost and category fields. Keep the supplied **F1** label outside the table.



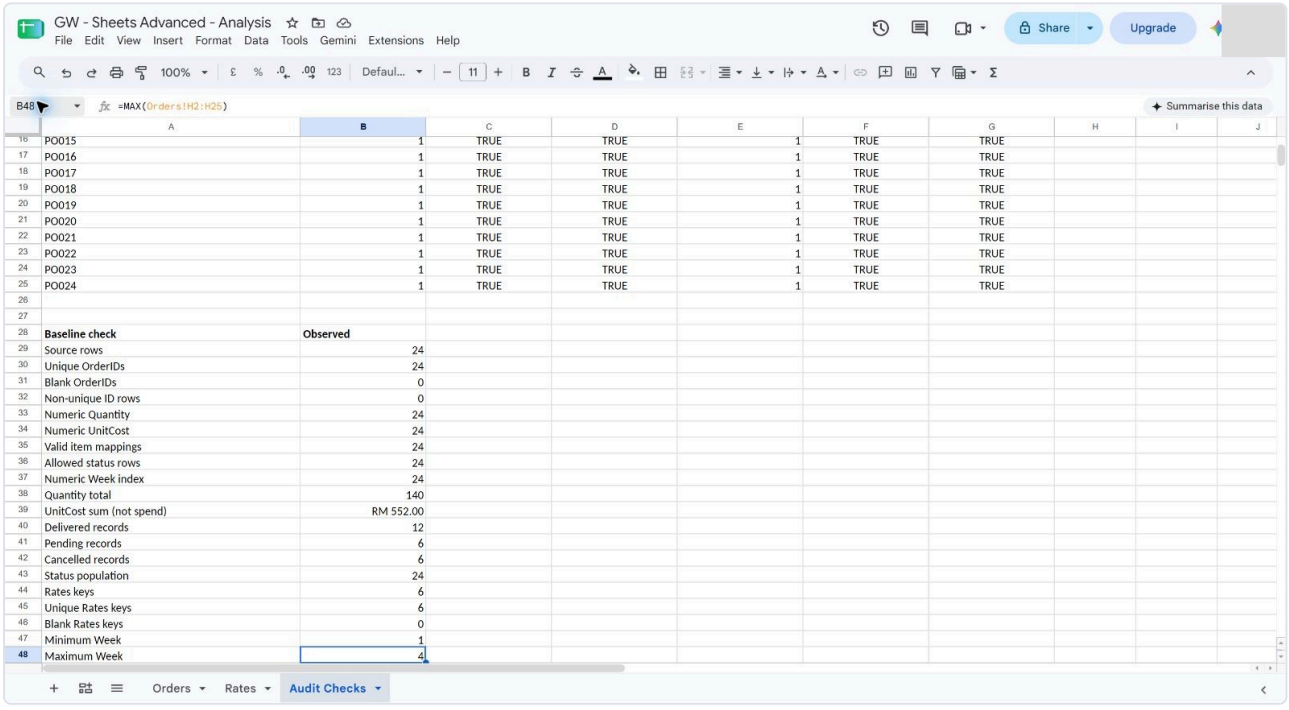
ItemCode	Item	UnitCost	Category
IT01	Display adapter	RM 45.00	Equipment
IT02	Training notebook	RM 12.00	Stationery
IT03	Cable pack	RM 30.00	Equipment
IT04	Marker set	RM 18.00	Stationery
IT05	Desk sign	RM 25.00	Print
IT06	Quick guide	RM 8.00	Print

1.2 Audit Key Fields

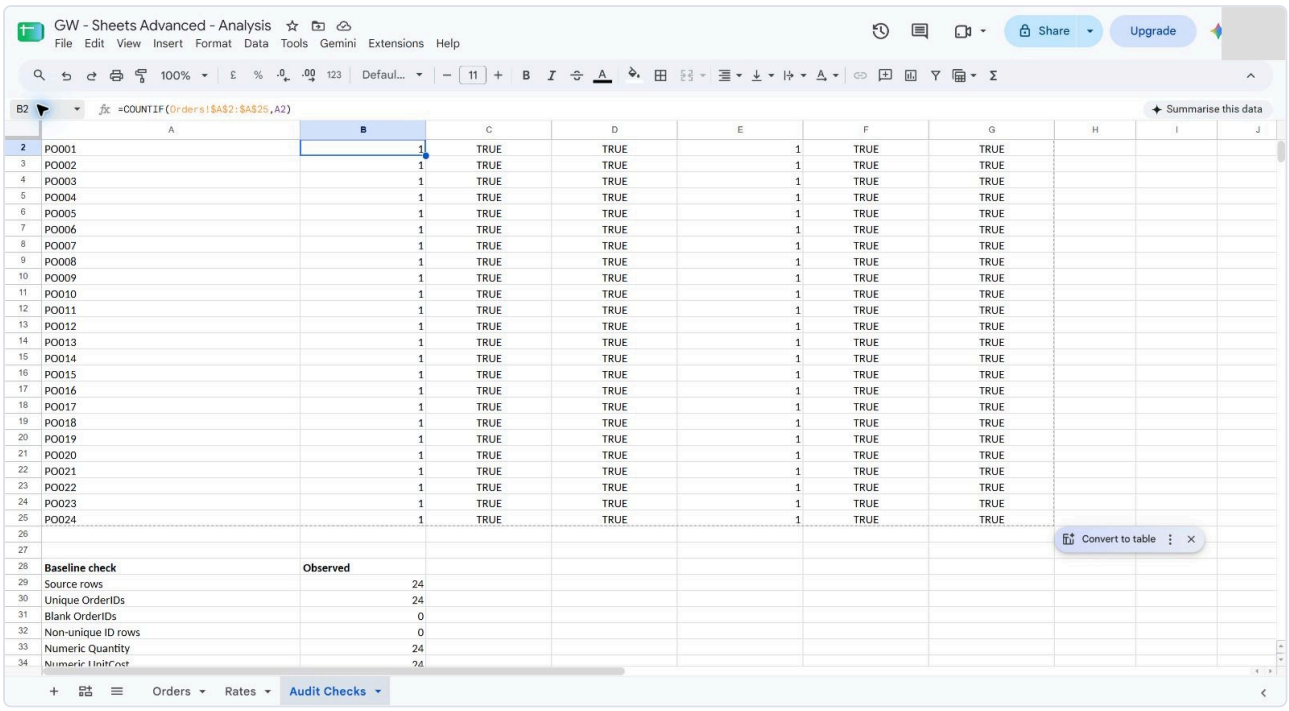
1. Create **Audit Checks** so the diagnostics do not alter Orders. Paste the baseline diagnostic card from the copy-paste page into **A1:G25**. It references each source row and checks ID occurrences, numeric Quantity/UnitCost/Week, exact item-key counts and allowed statuses. Make A:G wide enough to read the headings; the first captured stage uses 155-pixel columns.



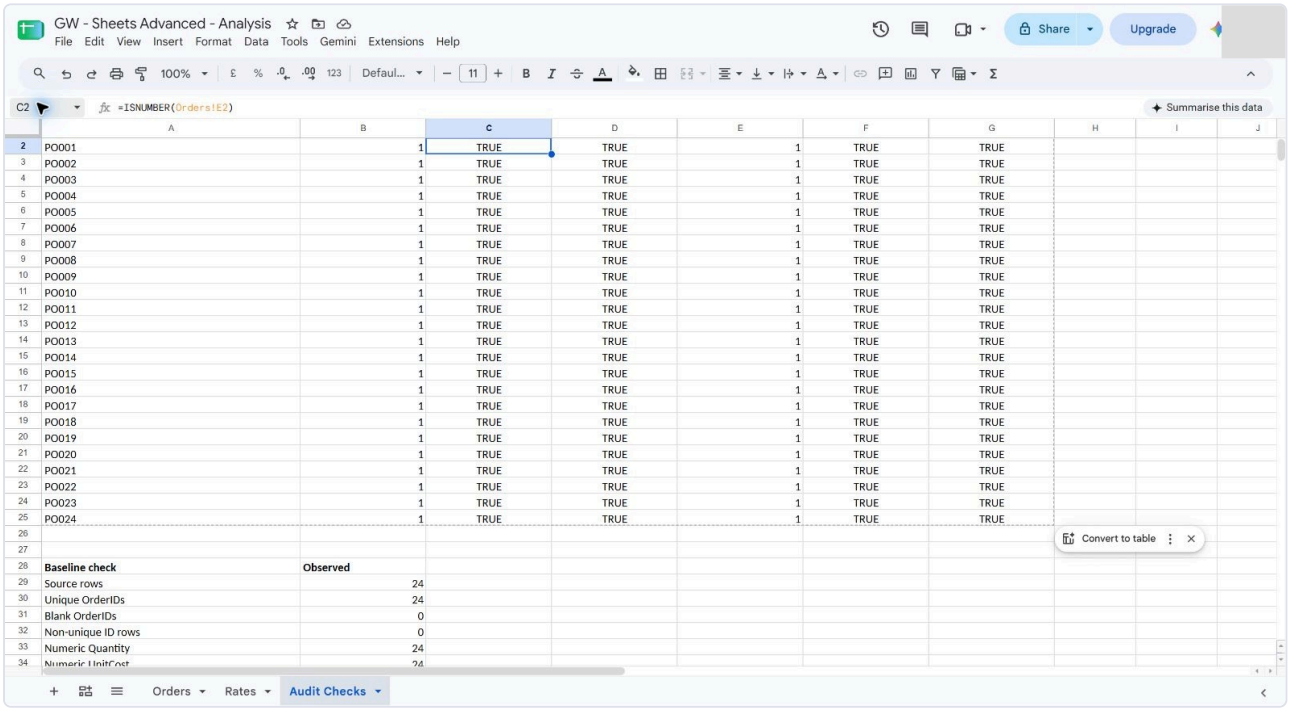
2. Paste the population-control card into **A28:B48**. Confirm **24 rows / 24 unique IDs / 0 blanks / 0 non-unique ID rows**, 24 numeric values in each required field, 24 matched item keys, and six unique nonblank Rates keys. The status population is **12 Delivered / 6 Pending / 6 Cancelled**, total 24. Quantity sums to **140**; the unit-price sum is **RM552**, which is a diagnostic checksum, not spend. Week is the numeric practice index **1–4**, not a date. Use a wider label column and navigate to B48 to see the complete control block.



3. Inspect **B2 =COUNTIF(Orders!$A$2:$A$25,A2)**. Each baseline ID occurs once. The anchored source population stays fixed while each row tests its own relative ID. A count of one inside intake alone would not prove that an ID is new to Orders; the later casework checks both populations.

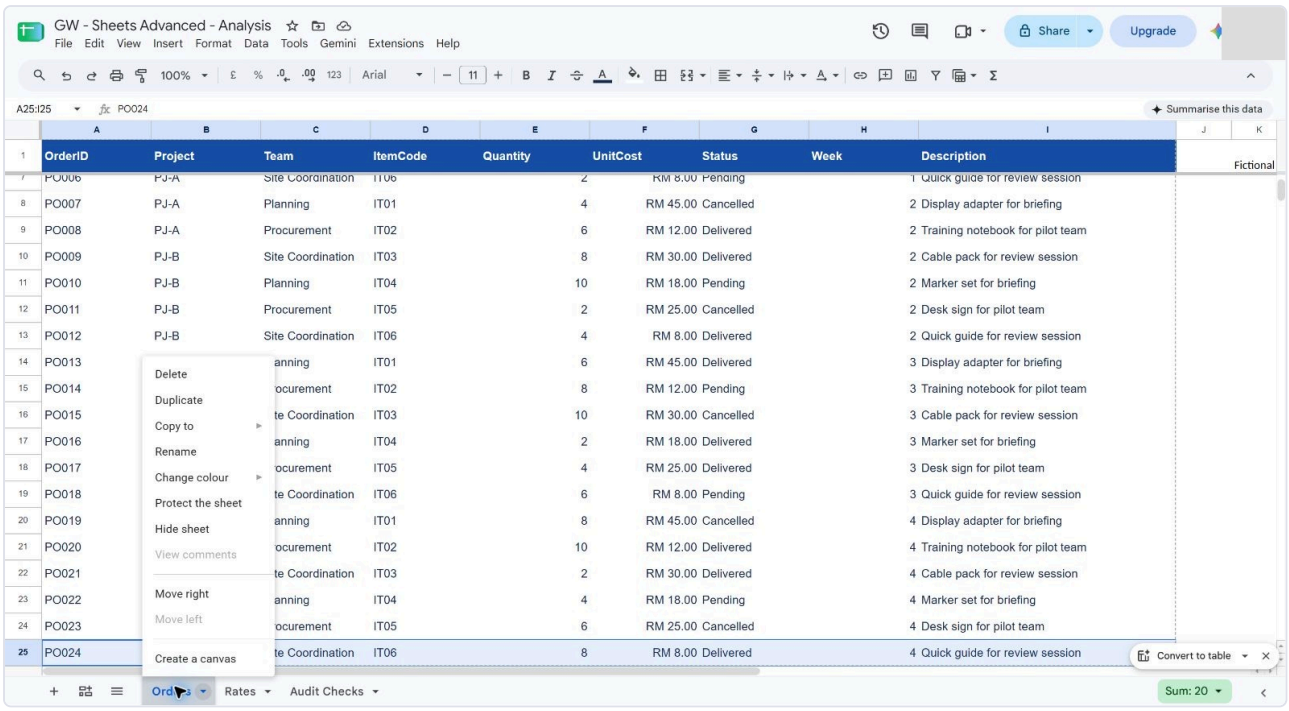


4. Inspect **C2 =ISNUMBER(Orders!E2)** and its results down to row 25. Verify Quantity and UnitCost are actual numbers. Keep Cancelled records visible; validity and the population chosen for a later question are separate decisions.

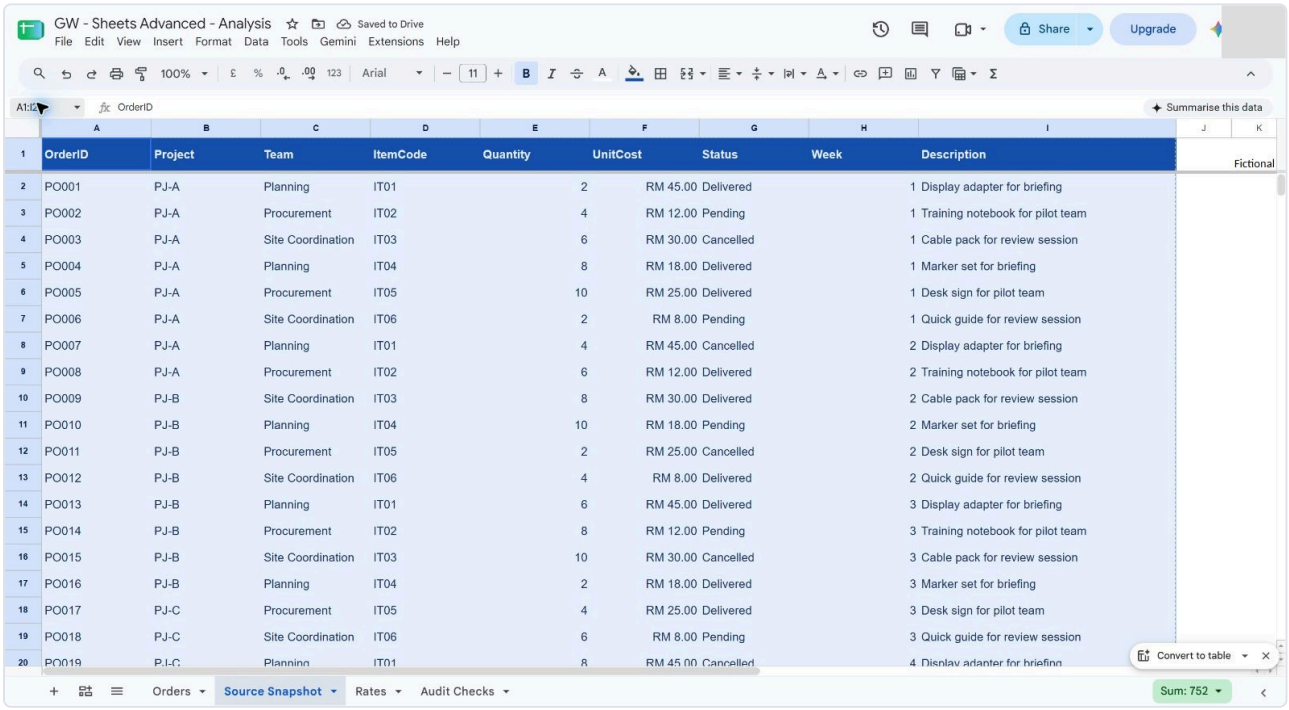


1.3 Protect the Analysis Trail

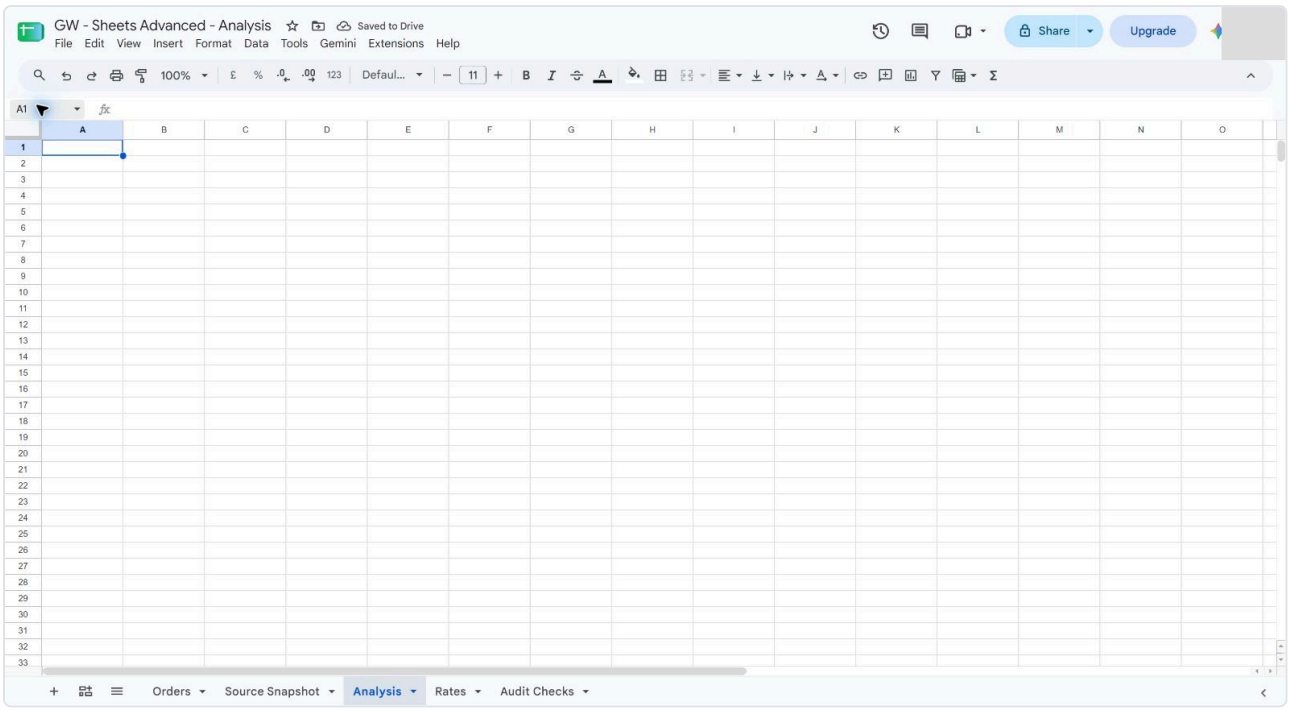
1. Return to **Orders**. Open its tab menu and choose **Duplicate** before adding any analysis calculation to that source.



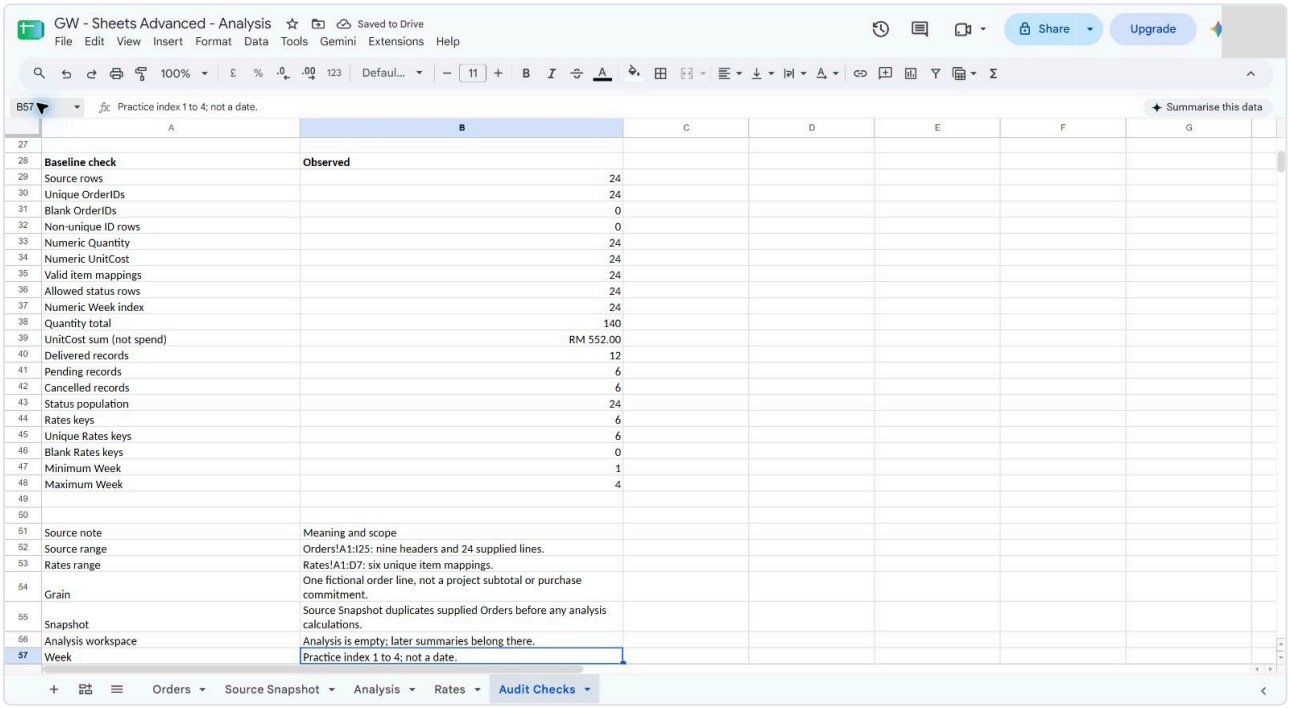
2. Rename the duplicate **Source Snapshot**. It retains the supplied Orders values and formatting. This is a preserved comparison copy; naming it does not create a protected-range permission or automatic version archive.



3. Create **Analysis** and leave it empty for later summaries. Keep diagnostics in Audit Checks rather than prebuilding later course outputs.



4. In **Audit Checks A51:B57**, record the source **Orders!A1:I25**, rate table **Rates!A1:D7**, order-line grain, snapshot purpose and empty Analysis workspace. Wrap the note text; in the captured workbook A is 320 pixels and B is 400 pixels. Source facts, training rules and unresolved questions must remain distinguishable.

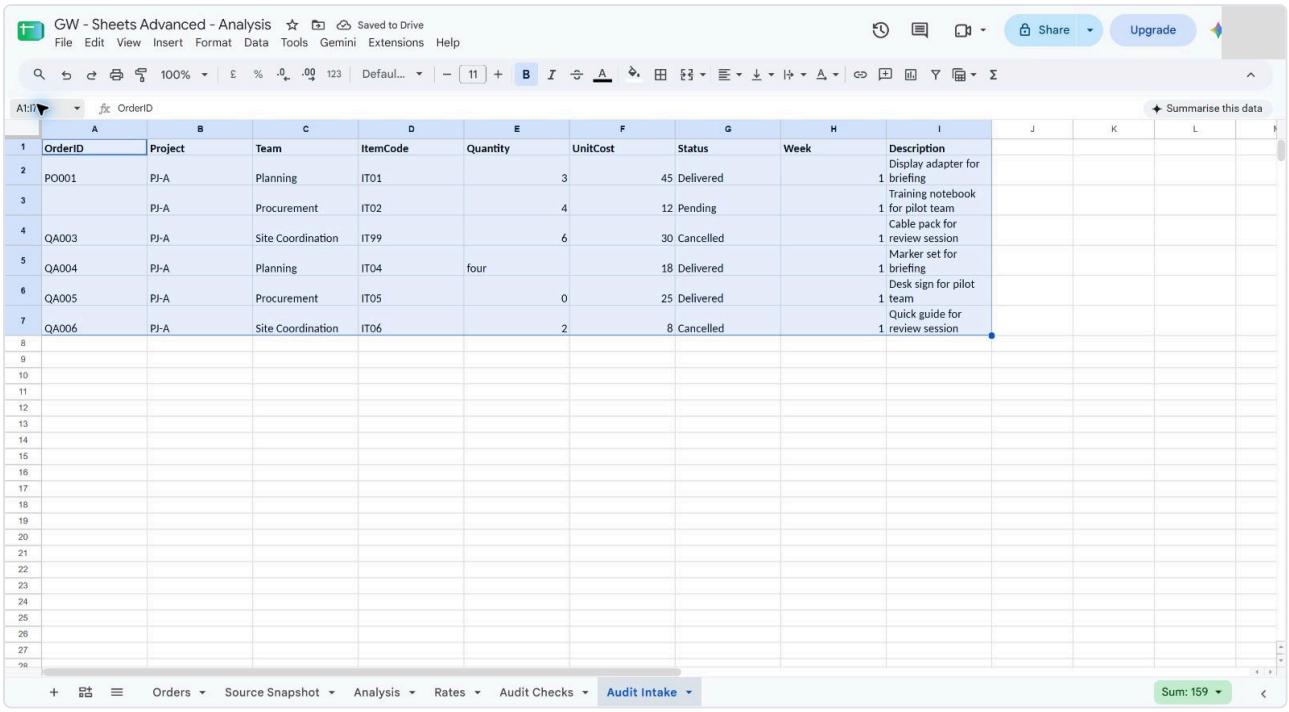


	A	B
27		
28	**Baseline check**	**Observed**
29	Source rows	24
30	Unique OrderIDs	24
31	Blank OrderIDs	0
32	Non-unique ID rows	0
33	Numeric Quantity	24
34	Numeric UnitCost	24
35	Valid item mappings	24
36	Allowed status rows	24
37	Numeric Week index	24
38	Quantity total	140
39	UnitCost sum (not spend)	RM 552.00
40	Delivered records	12
41	Pending records	6
42	Cancelled records	6
43	Status population	24
44	Rates keys	6
45	Unique Rates keys	6
46	Blank Rates keys	0
47	Minimum Week	1
48	Maximum Week	4
49		
50		
51	Source note	Meaning and scope
52	Source range	Orders!A1:I25: nine headers and 24 supplied lines.
53	Rates range	Rates!A1:D7: six unique item mappings.
54	Grain	One fictional order line, not a project subtotal or purchase commitment.
55	Snapshot	Source Snapshot duplicates supplied Orders before any analysis calculations.
56	Analysis workspace	Analysis is empty; later summaries belong there.
57	Week	Practice index 1 to 4; not a date.

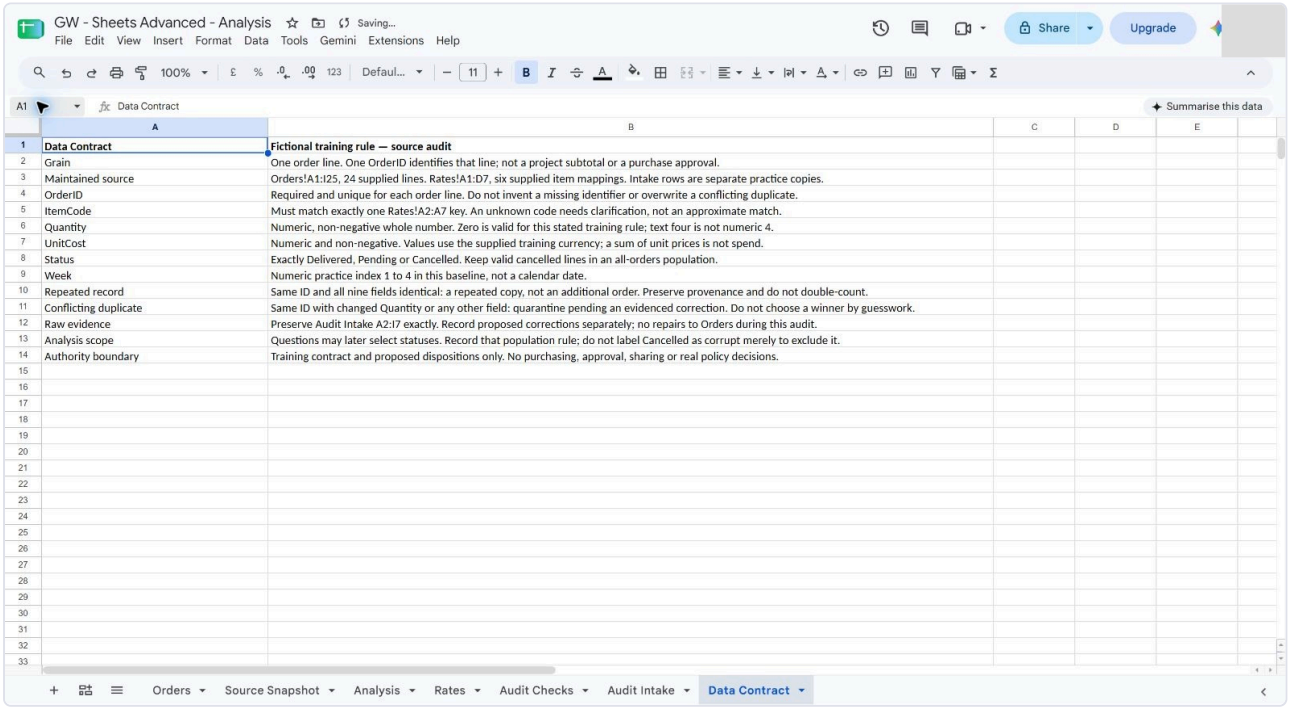
1.4 Quarantine Bad Rows and Write a Data Contract

1. Create **Audit Intake** and paste the six-row raw card into **A1:I7**. These are labelled practice copies of PO001–PO006, altered only as described in the card. Preserve the blank ID, **IT99**, text **four**, numeric **0** and **Cancelled** exactly. Use the practice IDs QA003–QA006 only in intake. Add **Issue / Evidence /**

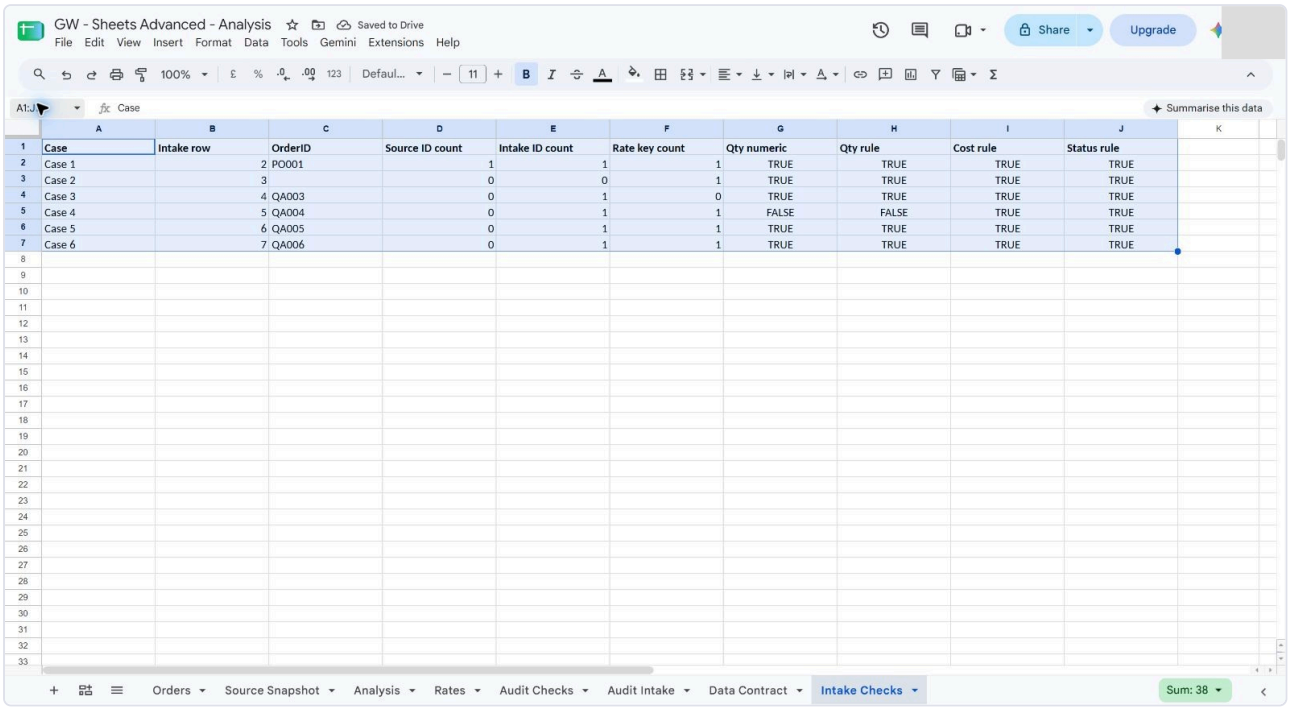
Disposition in J1:L1. Do not append any of these rows to Orders. The captured raw view uses 130-pixel A:I columns with wrapped Description cells.



2. Create **Data Contract** and record the stated training rules: one order line per required unique OrderID; exactly one Rates mapping; numeric non-negative whole-number Quantity; numeric non-negative UnitCost; statuses Delivered, Pending or Cancelled. **Zero Quantity is explicitly valid for this fictional training rule.** Distinguish a repeated identical row from the same key with a conflicting field. Preserve raw evidence and record proposed corrections separately. This is a training contract, not a real purchasing policy. Use A about 280 pixels and B about 900 pixels, with wrapped text.

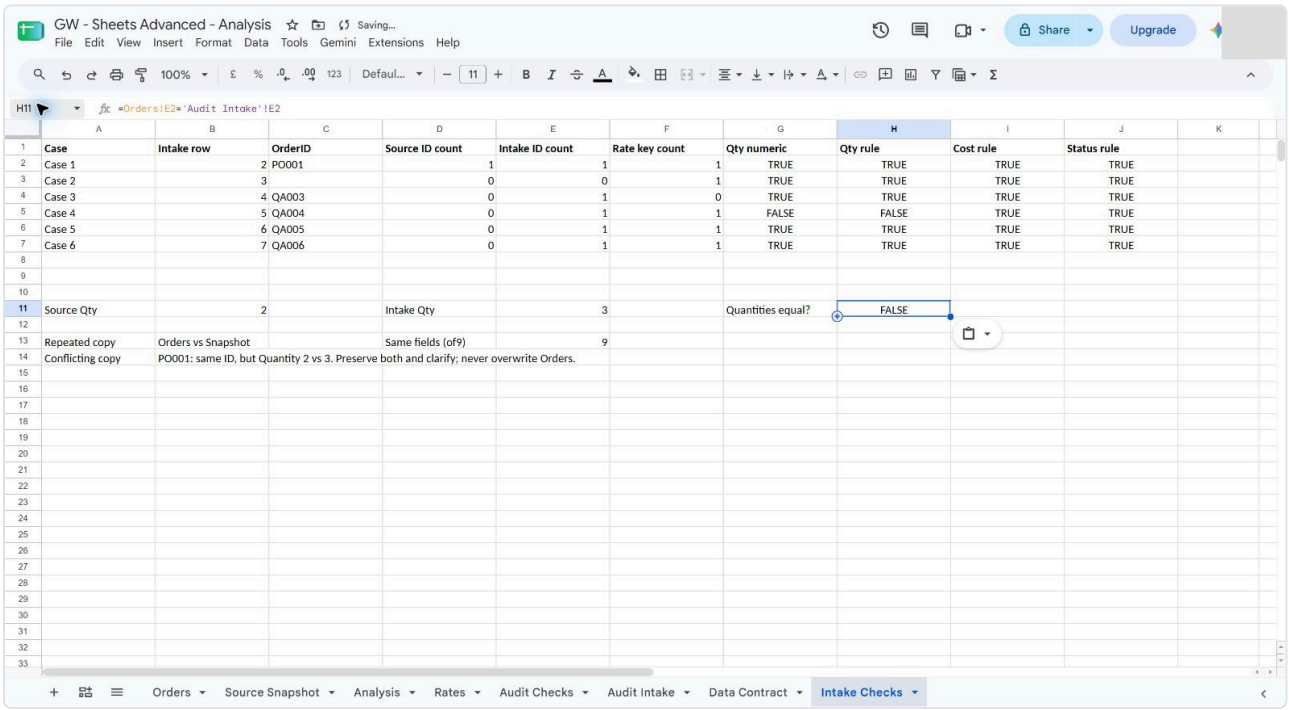


3. Create **Intake Checks** and paste its diagnostic card into **A1:J7**. The source-ID count tests Orders, the intake-ID count tests the separate six-row intake, and the rate-key count tests Rates. Inspect both numeric type and the non-negative whole-number rule. A zero key count or FALSE result is evidence to investigate; a set of TRUE checks does not by itself approve a record for release.



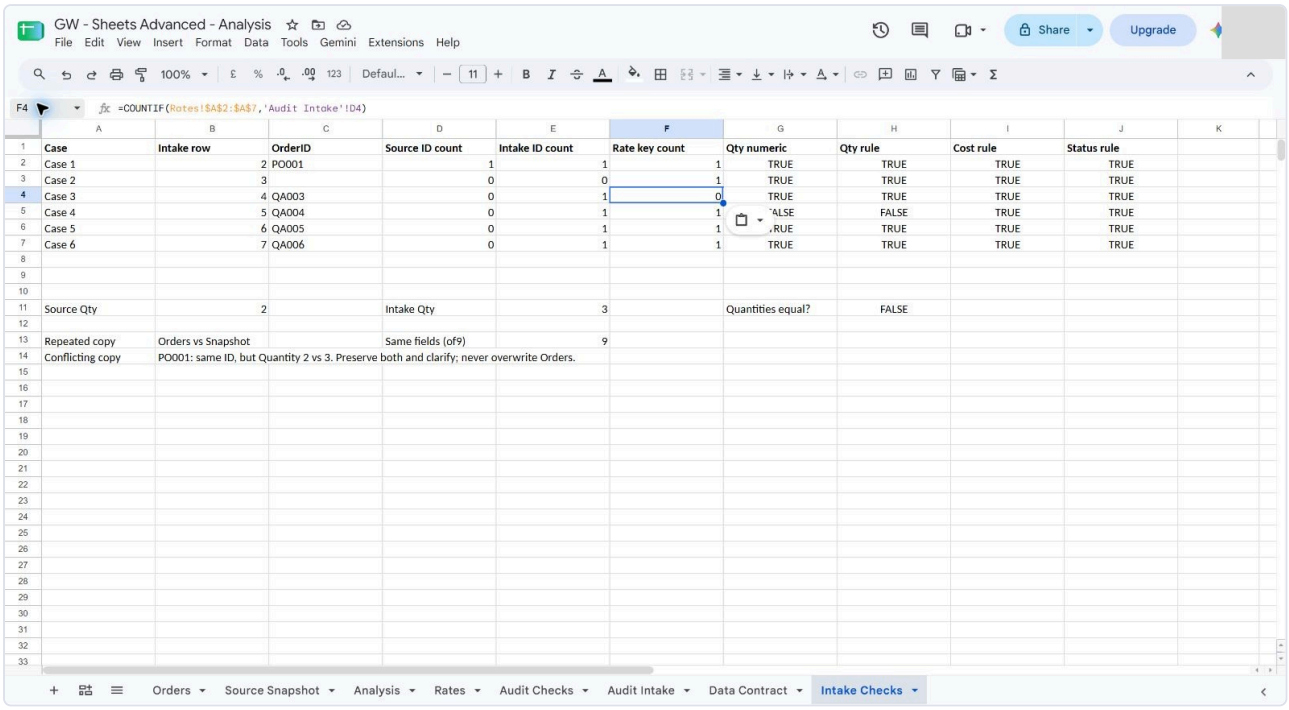
	A	B	C	D	E	F	G	H	I	J
1	Case	Intake row	OrderID	Source ID count	Intake ID count	Rate key count	Qty numeric	Qty rule	Cost rule	Status rule
2	Case 1	2	PO001	1	1	1	TRUE	TRUE	TRUE	TRUE
3	Case 2	3		0	0	1	TRUE	TRUE	TRUE	TRUE
4	Case 3	4	QA003	0	1	0	TRUE	TRUE	TRUE	TRUE
5	Case 4	5	QA004	0	1	1	FALSE	FALSE	TRUE	TRUE
6	Case 5	6	QA005	0	1	1	TRUE	TRUE	TRUE	TRUE
7	Case 6	7	QA006	0	1	1	TRUE	TRUE	TRUE	TRUE

4. Compare the conflicting **PO001**: source Quantity **2**, intake Quantity **3**, equality **FALSE**. The supplied Orders row and its unaltered Source Snapshot have **9 of 9 identical fields**, illustrating a repeated copy. Preserve the conflicting intake row and request evidence; do not overwrite the baseline or count both as separate orders. The supporting comparison cells are on the copy-paste page.

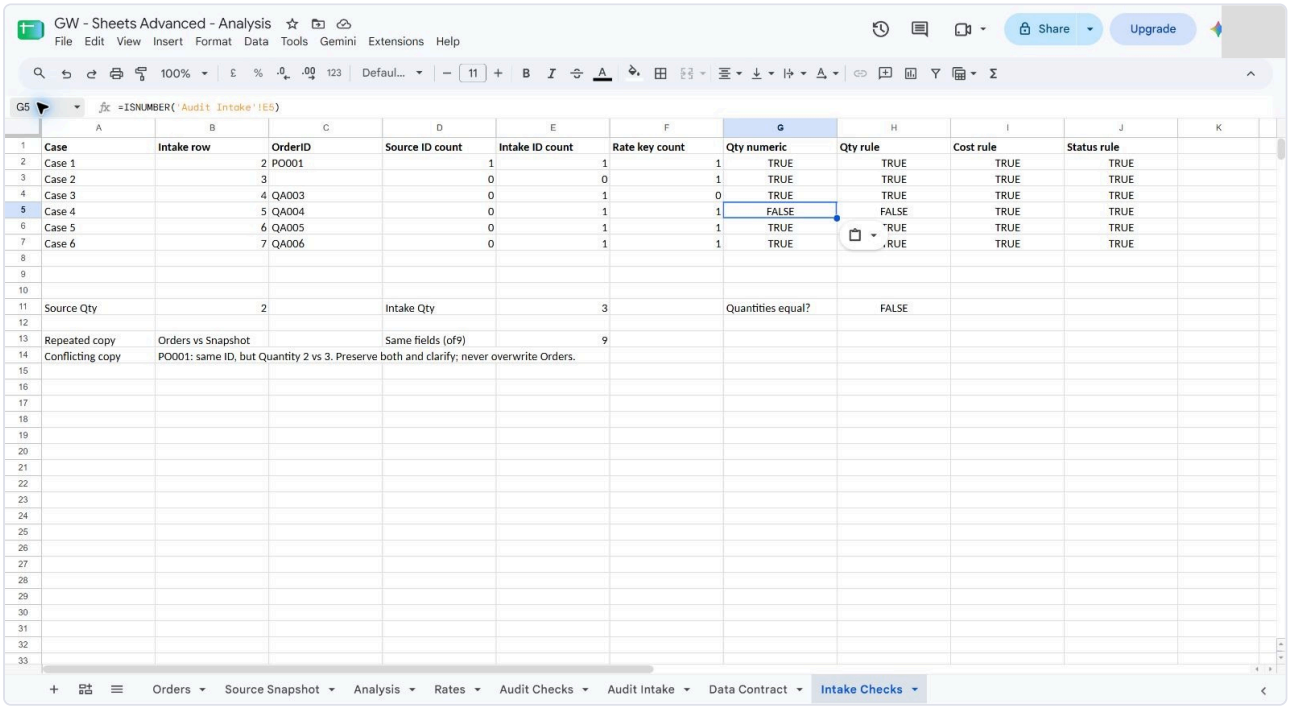


	A	B	C	D	E	F	G	H	I	J
1	Case	Intake row	OrderID	Source ID count	Intake ID count	Rate key count	Qty numeric	Qty rule	Cost rule	Status rule
2	Case 1	2	PO001	1	1	1	TRUE	TRUE	TRUE	TRUE
3	Case 2	3		0	0	1	TRUE	TRUE	TRUE	TRUE
4	Case 3	4	QA003	0	1	0	TRUE	TRUE	TRUE	TRUE
5	Case 4	5	QA004	0	1	1	FALSE	FALSE	TRUE	TRUE
6	Case 5	6	QA005	0	1	1	TRUE	TRUE	TRUE	TRUE
7	Case 6	7	QA006	0	1	1	TRUE	TRUE	TRUE	TRUE
11	Source Qty	2		Intake Qty	3		Quantities equal?	FALSE		
13	Repeated copy	Orders vs Snapshot		Same fields (of9)	9					
14	Conflicting copy	PO001: same ID, but Quantity 2 vs 3. Preserve both and clarify; never overwrite Orders.								

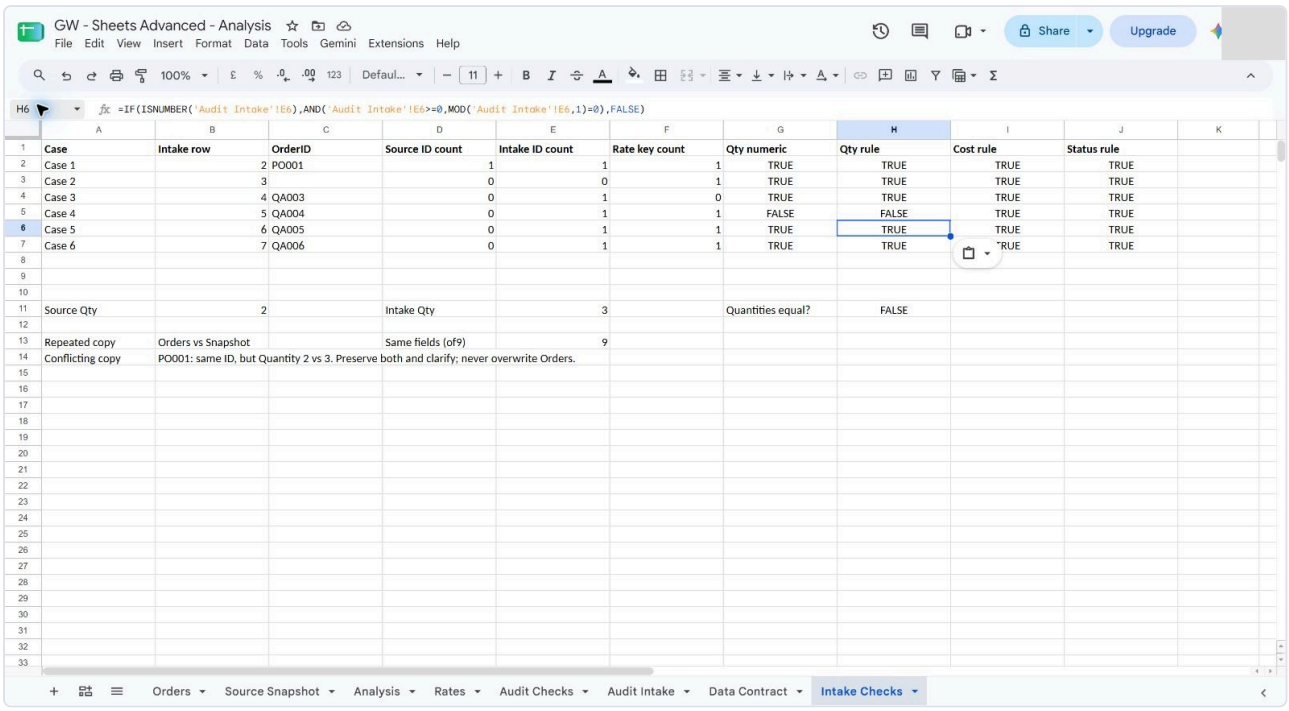
5. Inspect **F4 =COUNTIF(Rates!$A$2:$A$7,'Audit Intake'!D4)**. **IT99** has **0** exact matches. Its Cancelled status is valid, but that does not repair the unknown item key. Keep it quarantined pending clarification.



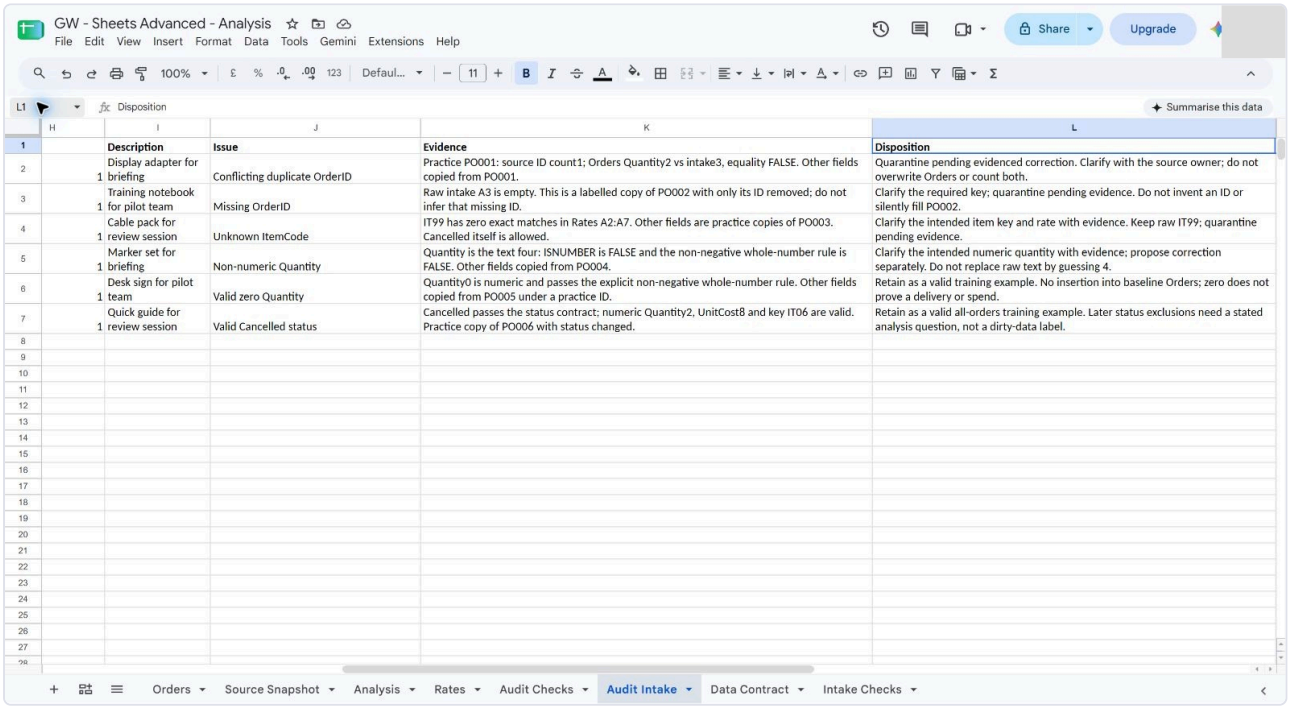
6. Inspect **G5 =ISNUMBER('Audit Intake'!E5)**: text **four** returns **FALSE**, as does its Quantity rule. Do not replace it with 4 without source evidence. Record the proposed next action separately.



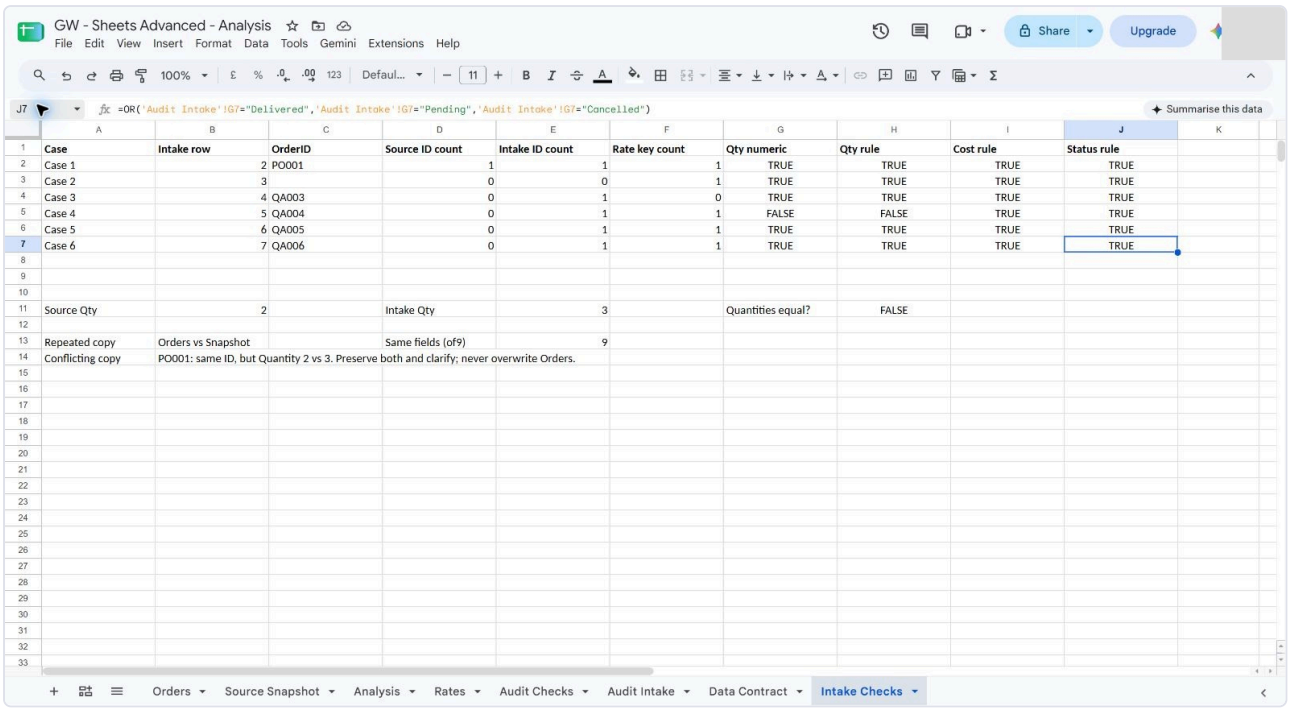
7. Inspect the zero-quantity row: ISNUMBER and the non-negative whole-number rule are both **TRUE**. Retain it as a valid training example under the stated contract; do not infer a delivery or purchasing decision from a zero.



8. Complete **Audit Intake J2:L7** with each Issue, supporting cell evidence and proposed Disposition. The first four cases remain quarantined pending evidence; the last two are valid training cases retained separately. Clarification requests are recorded as next actions, not sent. Wrap the text, widen J/K/L to about **260/560/500 pixels**, and navigate to **L1** to see the complete disposition column.

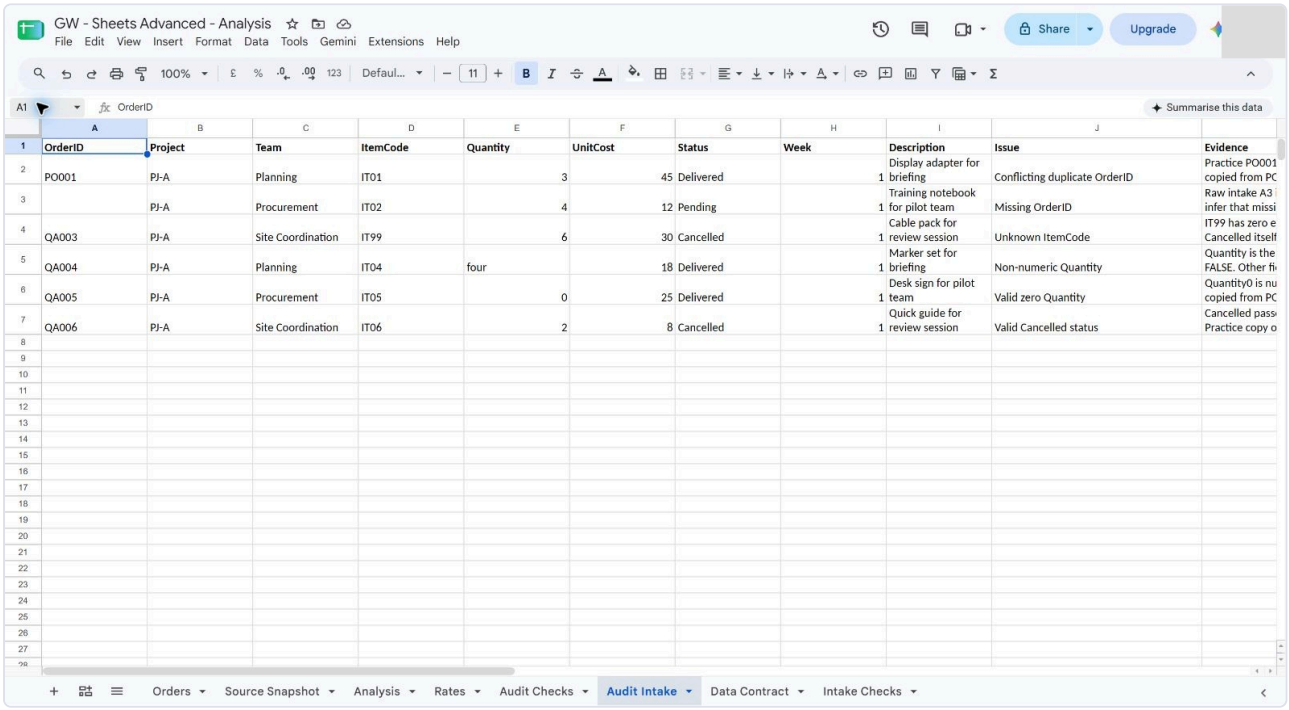


9. Inspect **J7** on Intake Checks. The QA006 Cancelled example passes the allowed-status test, as do its numeric and key checks. A later analysis may choose a status population, but a valid Cancelled record is not automatically corrupt.



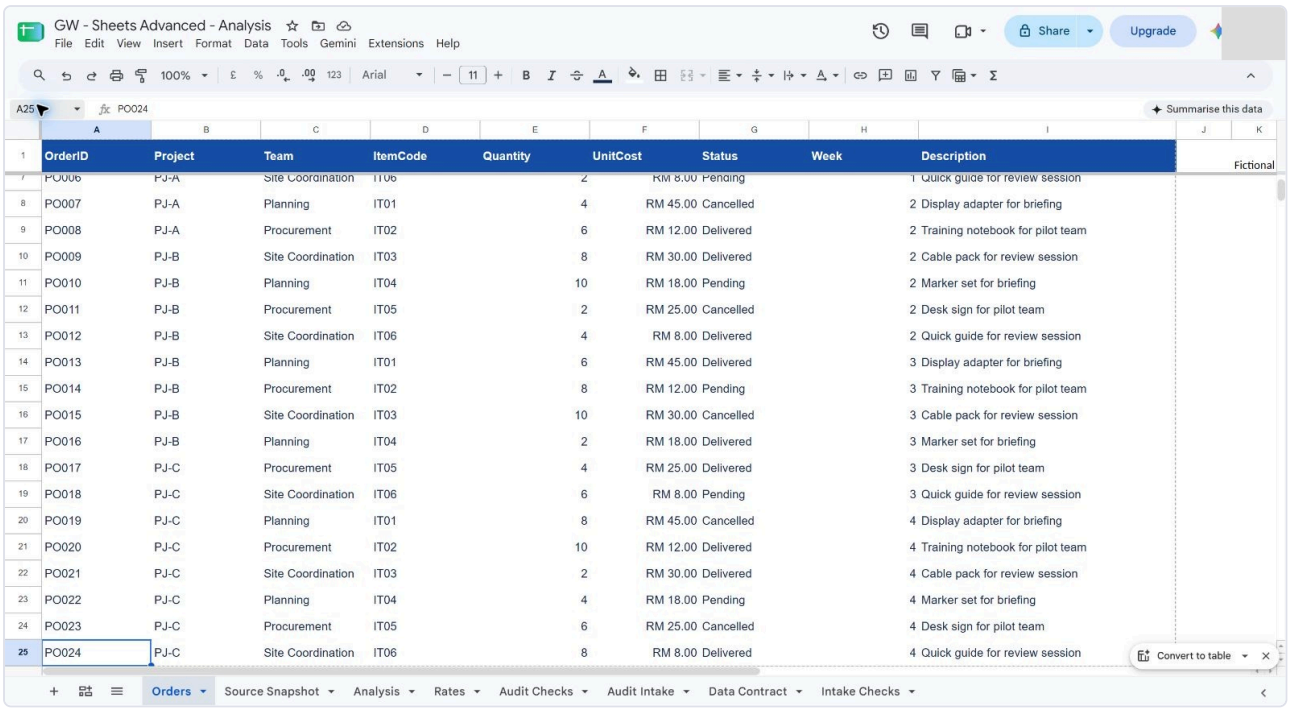
Case	Intake row	OrderID	Source ID count	Intake ID count	Rate key count	Qty numeric	Qty rule	Cost rule	Status rule
Case 1	2	PO001	1	1	1	TRUE	TRUE	TRUE	TRUE
Case 2	3		0	0	1	TRUE	TRUE	TRUE	TRUE
Case 3	4	QA003	0	1	0	TRUE	TRUE	TRUE	TRUE
Case 4	5	QA004	0	1	1	FALSE	FALSE	TRUE	TRUE
Case 5	6	QA005	0	1	1	TRUE	TRUE	TRUE	TRUE
Case 6	7	QA006	0	1	1	TRUE	TRUE	TRUE	TRUE
Source Qty	2		Intake Qty	3		Quantities equal?	FALSE		
Repeated copy	Orders vs Snapshot		Same fields (of9)	9					
Conflicting copy	PO001: same ID, but Quantity 2 vs 3. Preserve both and clarify; never overwrite Orders.								

10. Reopen **Audit Intake A1:I7** and verify every raw value remains intact beside the recorded issues. No repair has been applied to the raw examples or maintained source.



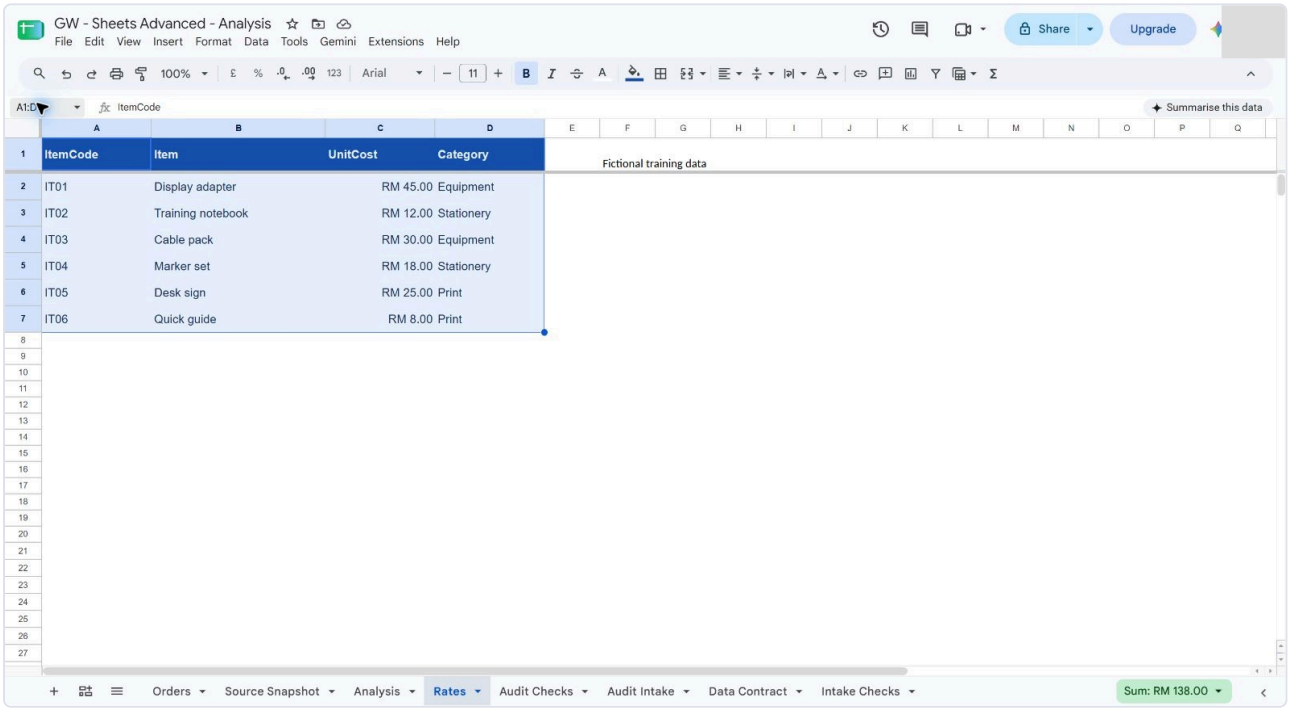
OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Issue	Evidence
PO001	PJ-A	Planning	IT01	3	45	Delivered	1	Display adapter for briefing	Conflicting duplicate OrderID	Practice PO001 copied from PC
	PJ-A	Procurement	IT02	4	12	Pending	1	Training notebook for pilot team	Missing OrderID	Raw intake A3 infer that missi
QA003	PJ-A	Site Coordination	IT99	6	30	Cancelled	1	Cable pack for review session	Unknown ItemCode	IT99 has zero e Cancelled itself
QA004	PJ-A	Planning	IT04	four	18	Delivered	1	Marker set for briefing	Non-numeric Quantity	Quantity is the FALSE. Other fi
QA005	PJ-A	Procurement	IT05	0	25	Delivered	1	Desk sign for pilot team	Valid zero Quantity	Quantity0 is nu copied from PC
QA006	PJ-A	Site Coordination	IT06	2	8	Cancelled	1	Quick guide for review session	Valid Cancelled status	Cancelled pass Practice copy o

11. Return to **Orders** and recheck its original 24 records through PO024. Native readback confirms its supplied values and formatting are unchanged.



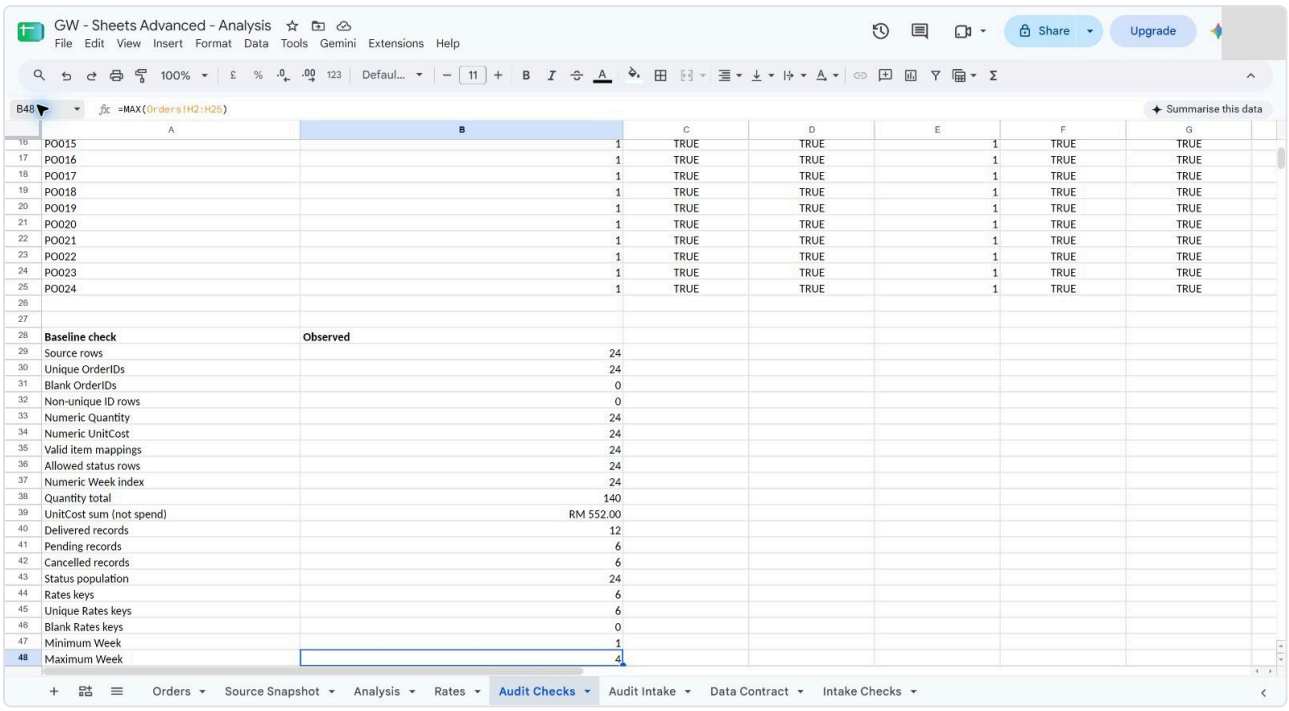
OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description
PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session
PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing
PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team
PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session
PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing
PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team
PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session
PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing
PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team
PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session
PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing
PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team
PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session
PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing
PO020	PJ-C	Procurement	IT02	10	RM 12.00	Delivered	4	Training notebook for pilot team
PO021	PJ-C	Site Coordination	IT03	2	RM 30.00	Delivered	4	Cable pack for review session
PO022	PJ-C	Planning	IT04	4	RM 18.00	Pending	4	Marker set for briefing
PO023	PJ-C	Procurement	IT05	6	RM 25.00	Cancelled	4	Desk sign for pilot team
PO024	PJ-C	Site Coordination	IT06	8	RM 8.00	Delivered	4	Quick guide for review session

12. Recheck the original six **Rates** keys and supplied values. The unknown intake code has not been added to the reference table to make a test pass.

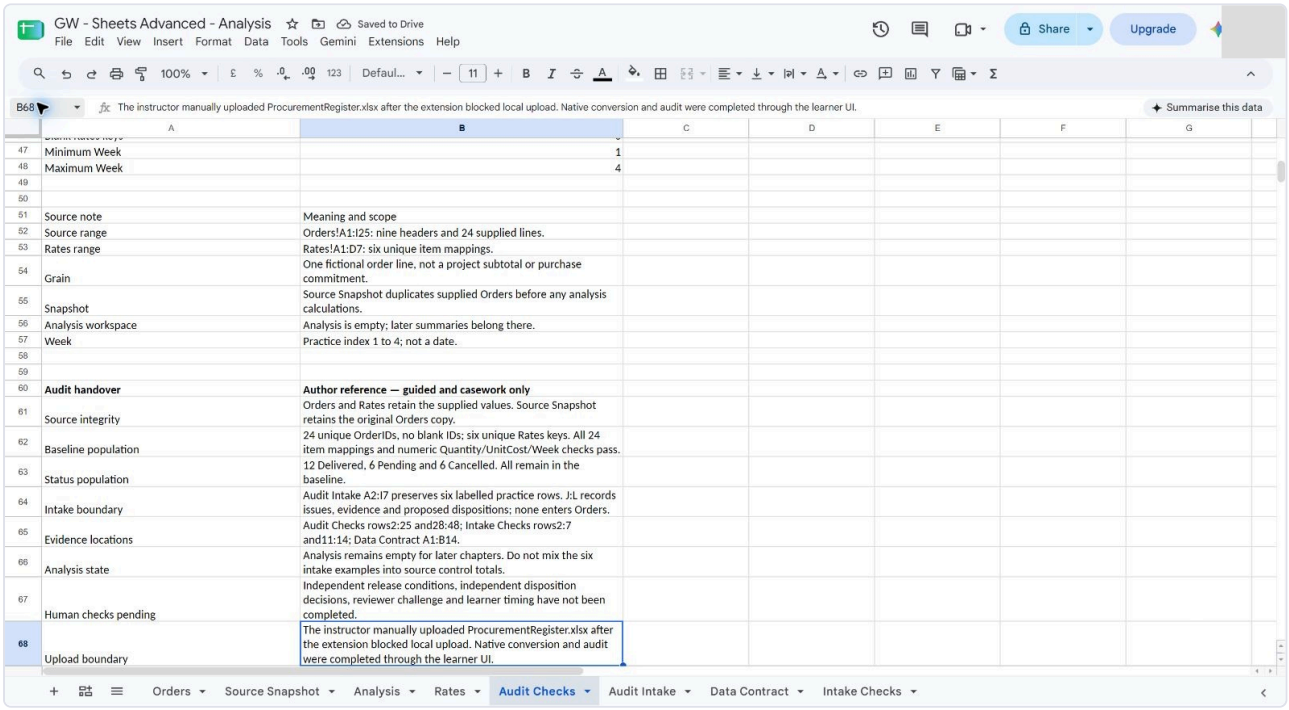


ItemCode	Item	UnitCost	Category
IT01	Display adapter	RM 45.00	Equipment
IT02	Training notebook	RM 12.00	Stationery
IT03	Cable pack	RM 30.00	Equipment
IT04	Marker set	RM 18.00	Stationery
IT05	Desk sign	RM 25.00	Print
IT06	Quick guide	RM 8.00	Print

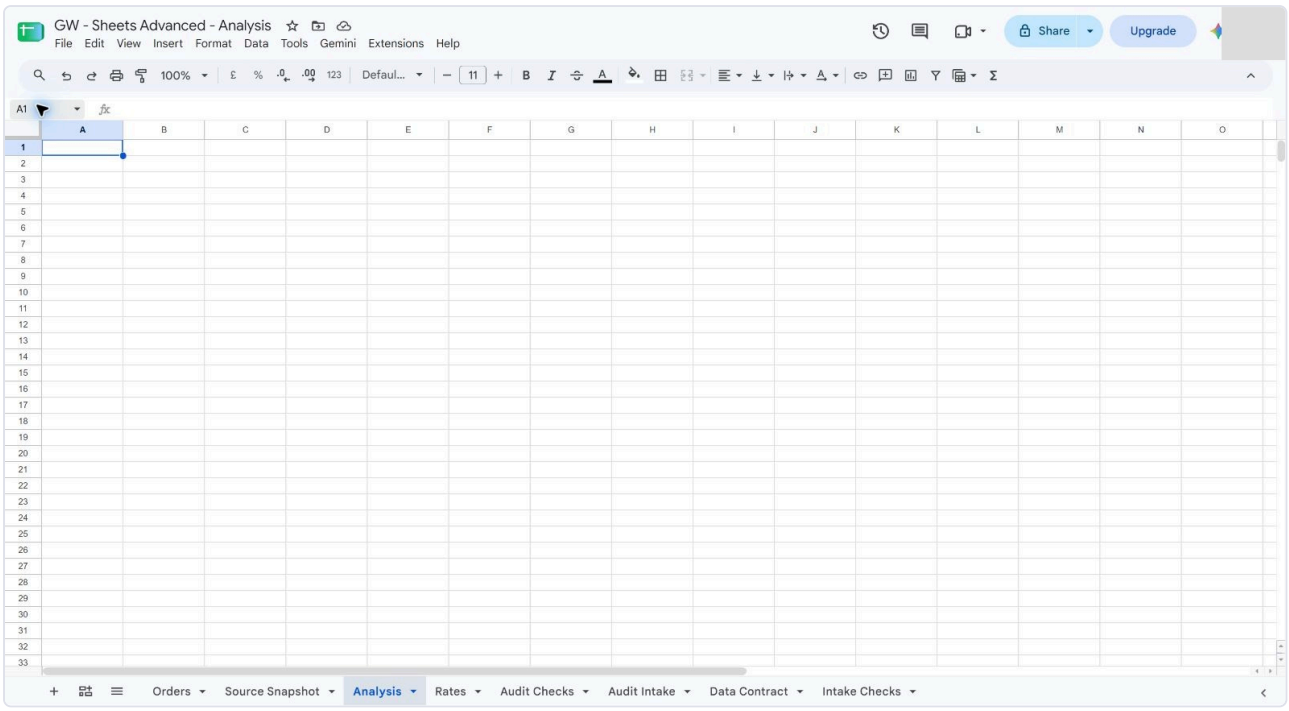
13. Reopen the baseline population controls. Counts, types, key coverage, status population, Quantity 140 and the unit-price checksum 552 remain unchanged. Intake anomalies are excluded from these bounded controls.



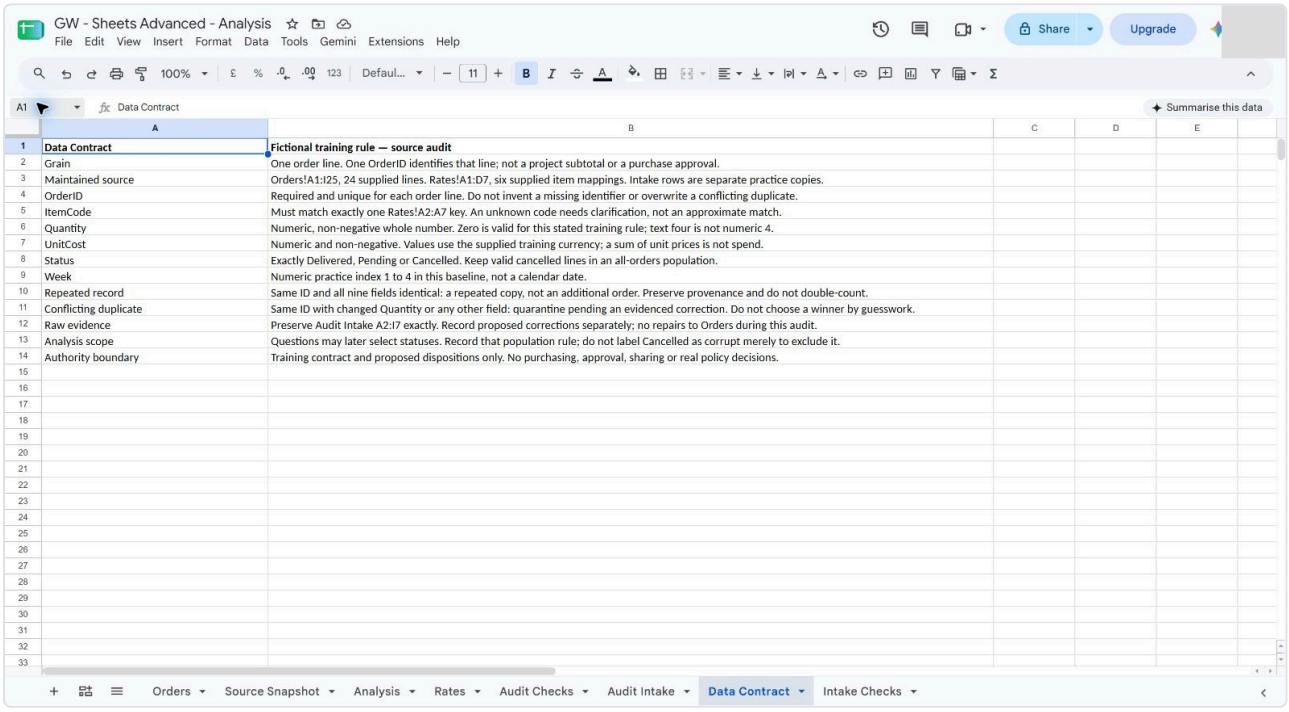
14. Record the current source integrity, evidence locations and intake boundary in **Audit Checks A60:B68**. Label independent disposition decisions, release conditions, reviewer challenge and learner timing as pending. The author walkthrough does not substitute for those checks.



15. Confirm **Analysis** remains empty. Leave the current workbook ready for conditional calculations in the next lab, with no later summary or dashboard prebuilt.



16. Leave the saved workbook with its **Data Contract**, isolated intake and diagnostics available. The source snapshot and original XLSX remain available for comparison.



> **Validation boundary:** guided 1.1–1.3 and author casework 1.4 are checked in the learner UI, supplemented by bounded read-only native-cell verification. Upload used an instructor handoff. Independent release conditions and disposition decisions, reviewer challenge and learner timing are not completed.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently decide whether each anomaly should enter the analysis, remain pending or be retained as a valid case. Justify every decision against the contract.
2. Write a release condition for the source: what must be true before conditional calculations and lookups can be trusted?
3. Have a reviewer challenge one disposition and resolve it using source evidence or an explicit unresolved label.

Evidence to Hand Over

- Data contract and six-row anomaly audit with preserved raw values.
- Baseline uniqueness/type/key checks and a justified quarantine policy.
- Independent release conditions and a recorded challenged decision.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A cancelled row is removed as dirty data	Separate data validity from the inclusion rule of the analysis question.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

An analysis begins with data meaning, types and source coverage.

Check yourself: What does one row represent, and which field identifies it?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
GW - Sheets Advanced - Analysis
```

```
Source Snapshot
```

```
Analysis
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Separate intake anomalies: duplicate PO001 with changed quantity; missing OrderID; ItemCode
IT99; Quantity “four”; valid zero quantity; valid Cancelled status. Do not inject these into
Orders. Other fields can be copied from a source row only in Audit Intake and labelled as
practice.
```

Captured Source-Audit Reference

The supplied source has 24 unique OrderIDs PO001–PO024, six unique Rates keys IT01–IT06, Quantity total140 and unit-price checksum552 (not spend). All24 Quantity/UnitCost/Week entries are numeric, all24 item keys map, statuses are12 Delivered/6 Pending/6 Cancelled. Source tables retain their supplied values and formats; Analysis is empty. Manual upload handoff was required; native conversion and audit were executed in the application.

Baseline row diagnostics — Audit Checks A1:G25

Paste this bounded card into the named diagnostic tab, not Orders. It supplies the exact formulas observed in the author run; inspect representative formulas and explain the anchored population.

```
OrderID ID occurrences  Qty numeric     Cost numeric    Item key matches        Allowed status
Week numeric
=Orders!A2      =COUNTIF(Orders!$A$2:$A$25,A2)  =ISNUMBER(Orders!E2)    =ISNUMBER(Orders!F2)
=COUNTIF(Rates!$A$2:$A$7,Orders!D2)
=OR(Orders!G2="Delivered",Orders!G2="Pending",Orders!G2="Cancelled")    =ISNUMBER(Orders!H2)
=Orders!A3      =COUNTIF(Orders!$A$2:$A$25,A3)  =ISNUMBER(Orders!E3)    =ISNUMBER(Orders!F3)
=COUNTIF(Rates!$A$2:$A$7,Orders!D3)
=OR(Orders!G3="Delivered",Orders!G3="Pending",Orders!G3="Cancelled")    =ISNUMBER(Orders!H3)
=Orders!A4      =COUNTIF(Orders!$A$2:$A$25,A4)  =ISNUMBER(Orders!E4)    =ISNUMBER(Orders!F4)
=COUNTIF(Rates!$A$2:$A$7,Orders!D4)
=OR(Orders!G4="Delivered",Orders!G4="Pending",Orders!G4="Cancelled")    =ISNUMBER(Orders!H4)
=Orders!A5      =COUNTIF(Orders!$A$2:$A$25,A5)  =ISNUMBER(Orders!E5)    =ISNUMBER(Orders!F5)
=COUNTIF(Rates!$A$2:$A$7,Orders!D5)
=OR(Orders!G5="Delivered",Orders!G5="Pending",Orders!G5="Cancelled")    =ISNUMBER(Orders!H5)
```

```
=Orders!A6      =COUNTIF(Orders!$A$2:$A$25,A6)  =ISNUMBER(Orders!E6)    =ISNUMBER(Orders!F6)
=COUNTIF(Rates!$A$2:$A$7,Orders!D6)
=OR(Orders!G6="Delivered",Orders!G6="Pending",Orders!G6="Cancelled")    =ISNUMBER(Orders!H6)
=Orders!A7      =COUNTIF(Orders!$A$2:$A$25,A7)  =ISNUMBER(Orders!E7)    =ISNUMBER(Orders!F7)
=COUNTIF(Rates!$A$2:$A$7,Orders!D7)
=OR(Orders!G7="Delivered",Orders!G7="Pending",Orders!G7="Cancelled")    =ISNUMBER(Orders!H7)
=Orders!A8      =COUNTIF(Orders!$A$2:$A$25,A8)  =ISNUMBER(Orders!E8)    =ISNUMBER(Orders!F8)
=COUNTIF(Rates!$A$2:$A$7,Orders!D8)
=OR(Orders!G8="Delivered",Orders!G8="Pending",Orders!G8="Cancelled")    =ISNUMBER(Orders!H8)
=Orders!A9      =COUNTIF(Orders!$A$2:$A$25,A9)  =ISNUMBER(Orders!E9)    =ISNUMBER(Orders!F9)
=COUNTIF(Rates!$A$2:$A$7,Orders!D9)
=OR(Orders!G9="Delivered",Orders!G9="Pending",Orders!G9="Cancelled")    =ISNUMBER(Orders!H9)
=Orders!A10     =COUNTIF(Orders!$A$2:$A$25,A10) =ISNUMBER(Orders!E10)   =ISNUMBER(Orders!F10)
=COUNTIF(Rates!$A$2:$A$7,Orders!D10)
=OR(Orders!G10="Delivered",Orders!G10="Pending",Orders!G10="Cancelled") =ISNUMBER(Orders!H10)
=Orders!A11     =COUNTIF(Orders!$A$2:$A$25,A11) =ISNUMBER(Orders!E11)   =ISNUMBER(Orders!F11)
=COUNTIF(Rates!$A$2:$A$7,Orders!D11)
=OR(Orders!G11="Delivered",Orders!G11="Pending",Orders!G11="Cancelled") =ISNUMBER(Orders!H11)
=Orders!A12     =COUNTIF(Orders!$A$2:$A$25,A12) =ISNUMBER(Orders!E12)   =ISNUMBER(Orders!F12)
=COUNTIF(Rates!$A$2:$A$7,Orders!D12)
=OR(Orders!G12="Delivered",Orders!G12="Pending",Orders!G12="Cancelled") =ISNUMBER(Orders!H12)
=Orders!A13     =COUNTIF(Orders!$A$2:$A$25,A13) =ISNUMBER(Orders!E13)   =ISNUMBER(Orders!F13)
=COUNTIF(Rates!$A$2:$A$7,Orders!D13)
=OR(Orders!G13="Delivered",Orders!G13="Pending",Orders!G13="Cancelled") =ISNUMBER(Orders!H13)
=Orders!A14     =COUNTIF(Orders!$A$2:$A$25,A14) =ISNUMBER(Orders!E14)   =ISNUMBER(Orders!F14)
=COUNTIF(Rates!$A$2:$A$7,Orders!D14)
=OR(Orders!G14="Delivered",Orders!G14="Pending",Orders!G14="Cancelled") =ISNUMBER(Orders!H14)
=Orders!A15     =COUNTIF(Orders!$A$2:$A$25,A15) =ISNUMBER(Orders!E15)   =ISNUMBER(Orders!F15)
=COUNTIF(Rates!$A$2:$A$7,Orders!D15)
=OR(Orders!G15="Delivered",Orders!G15="Pending",Orders!G15="Cancelled") =ISNUMBER(Orders!H15)
=Orders!A16     =COUNTIF(Orders!$A$2:$A$25,A16) =ISNUMBER(Orders!E16)   =ISNUMBER(Orders!F16)
=COUNTIF(Rates!$A$2:$A$7,Orders!D16)
=OR(Orders!G16="Delivered",Orders!G16="Pending",Orders!G16="Cancelled") =ISNUMBER(Orders!H16)
=Orders!A17     =COUNTIF(Orders!$A$2:$A$25,A17) =ISNUMBER(Orders!E17)   =ISNUMBER(Orders!F17)
=COUNTIF(Rates!$A$2:$A$7,Orders!D17)
=OR(Orders!G17="Delivered",Orders!G17="Pending",Orders!G17="Cancelled") =ISNUMBER(Orders!H17)
=Orders!A18     =COUNTIF(Orders!$A$2:$A$25,A18) =ISNUMBER(Orders!E18)   =ISNUMBER(Orders!F18)
=COUNTIF(Rates!$A$2:$A$7,Orders!D18)
=OR(Orders!G18="Delivered",Orders!G18="Pending",Orders!G18="Cancelled") =ISNUMBER(Orders!H18)
=Orders!A19     =COUNTIF(Orders!$A$2:$A$25,A19) =ISNUMBER(Orders!E19)   =ISNUMBER(Orders!F19)
=COUNTIF(Rates!$A$2:$A$7,Orders!D19)
=OR(Orders!G19="Delivered",Orders!G19="Pending",Orders!G19="Cancelled") =ISNUMBER(Orders!H19)
=Orders!A20     =COUNTIF(Orders!$A$2:$A$25,A20) =ISNUMBER(Orders!E20)   =ISNUMBER(Orders!F20)
=COUNTIF(Rates!$A$2:$A$7,Orders!D20)
=OR(Orders!G20="Delivered",Orders!G20="Pending",Orders!G20="Cancelled") =ISNUMBER(Orders!H20)
=Orders!A21     =COUNTIF(Orders!$A$2:$A$25,A21) =ISNUMBER(Orders!E21)   =ISNUMBER(Orders!F21)
=COUNTIF(Rates!$A$2:$A$7,Orders!D21)
=OR(Orders!G21="Delivered",Orders!G21="Pending",Orders!G21="Cancelled") =ISNUMBER(Orders!H21)
=Orders!A22     =COUNTIF(Orders!$A$2:$A$25,A22) =ISNUMBER(Orders!E22)   =ISNUMBER(Orders!F22)
=COUNTIF(Rates!$A$2:$A$7,Orders!D22)
=OR(Orders!G22="Delivered",Orders!G22="Pending",Orders!G22="Cancelled") =ISNUMBER(Orders!H22)
=Orders!A23     =COUNTIF(Orders!$A$2:$A$25,A23) =ISNUMBER(Orders!E23)   =ISNUMBER(Orders!F23)
=COUNTIF(Rates!$A$2:$A$7,Orders!D23)
=OR(Orders!G23="Delivered",Orders!G23="Pending",Orders!G23="Cancelled") =ISNUMBER(Orders!H23)
=Orders!A24     =COUNTIF(Orders!$A$2:$A$25,A24) =ISNUMBER(Orders!E24)   =ISNUMBER(Orders!F24)
=COUNTIF(Rates!$A$2:$A$7,Orders!D24)
=OR(Orders!G24="Delivered",Orders!G24="Pending",Orders!G24="Cancelled") =ISNUMBER(Orders!H24)
=Orders!A25     =COUNTIF(Orders!$A$2:$A$25,A25) =ISNUMBER(Orders!E25)   =ISNUMBER(Orders!F25)
```

```
=COUNTIF(Rates!$A$2:$A$7,Orders!D25)
=OR(Orders!G25="Delivered",Orders!G25="Pending",Orders!G25="Cancelled") =ISNUMBER(Orders!H25)
```

Baseline controls — Audit Checks A28:B48

```
Baseline check  Observed
Source rows     =COUNTA(Orders!A2:A25)
Unique OrderIDs =COUNTUNIQUE(Orders!A2:A25)
Blank OrderIDs  =COUNTBLANK(Orders!A2:A25)
Non-unique ID rows      =COUNTIF(B2:B25,">1")
Numeric Quantity        =COUNT(Orders!E2:E25)
Numeric UnitCost        =COUNT(Orders!F2:F25)
Valid item mappings     =COUNTIF(E2:E25,1)
Allowed status rows     =COUNTIF(F2:F25,TRUE)
Numeric Week index      =COUNT(Orders!H2:H25)
Quantity total  =SUM(Orders!E2:E25)
UnitCost sum (not spend)        =SUM(Orders!F2:F25)
Delivered records       =COUNTIF(Orders!G2:G25,"Delivered")
Pending records =COUNTIF(Orders!G2:G25,"Pending")
Cancelled records       =COUNTIF(Orders!G2:G25,"Cancelled")
Status population       =SUM(B40:B42)
Rates keys      =COUNTA(Rates!A2:A7)
Unique Rates keys       =COUNTUNIQUE(Rates!A2:A7)
Blank Rates keys        =COUNTBLANK(Rates!A2:A7)
Minimum Week    =MIN(Orders!H2:H25)
Maximum Week    =MAX(Orders!H2:H25)
```

Raw practice intake — Audit Intake A1:I7

Rows2–7 are practice copies of source PO001–PO006. Changes: PO001 Quantity2→3; PO002 ID removed; PO003 practice ID QA003 and ItemCode IT99; PO004 practice ID QA004 and Quantity text four; PO005 practice ID QA005 and numeric Quantity0; PO006 practice ID QA006 and status Cancelled. Other fields are copied only as fictional practice context. Do not insert these into Orders or replace the blank with PO002.

```
OrderID Project Team    ItemCode        Quantity        UnitCost        Status  Week
Description
PO001   PJ-A    Planning        IT01    3       45      Delivered       1       Display
adapter for briefing
        PJ-A    Procurement     IT02    4       12      Pending 1       Training notebook for
pilot team
QA003   PJ-A    Site Coordination       IT99    6       30      Cancelled       1       Cable
pack for review session
QA004   PJ-A    Planning        IT04    four    18      Delivered       1       Marker set for
briefing
QA005   PJ-A    Procurement     IT05    0       25      Delivered       1       Desk sign for
pilot team
QA006   PJ-A    Site Coordination       IT06    2       8       Cancelled       1       Quick
guide for review session
```

Training contract — Data Contract A1:B14

```
Data Contract   Fictional training rule — source audit
Grain   One order line. One OrderID identifies that line; not a project subtotal or a purchase approval.
Maintained source       Orders!A1:I25, 24 supplied lines. Rates!A1:D7, six supplied item mappings. Intake rows are separate practice copies.
OrderID Required and unique for each order line. Do not invent a missing identifier or overwrite a conflicting duplicate.
ItemCode        Must match exactly one Rates!A2:A7 key. An unknown code needs clarification, not an approximate match.
Quantity        Numeric, non-negative whole number. Zero is valid for this stated training rule; text four is not numeric 4.
UnitCost        Numeric and non-negative. Values use the supplied training currency; a sum of unit prices is not spend.
Status  Exactly Delivered, Pending or Cancelled. Keep valid cancelled lines in an all-orders population.
Week    Numeric practice index 1 to 4 in this baseline, not a calendar date.
Repeated record Same ID and all nine fields identical: a repeated copy, not an additional order. Preserve provenance and do not double-count.
Conflicting duplicate   Same ID with changed Quantity or any other field: quarantine pending an evidenced correction. Do not choose a winner by guesswork.
Raw evidence    Preserve Audit Intake A2:I7 exactly. Record proposed corrections separately; no repairs to Orders during this audit.
Analysis scope  Questions may later select statuses. Record that population rule; do not label Cancelled as corrupt merely to exclude it.
Authority boundary      Training contract and proposed dispositions only. No purchasing, approval, sharing or real policy decisions.
```

Intake diagnostics — Intake Checks A1:J7

```
Case    Intake row      OrderID Source ID count Intake ID count Rate key count  Qty numeric
Qty rule        Cost rule       Status rule
Case 1  2       ='Audit Intake'!A2      =IF(C2="",0,COUNTIF(Orders!$A$2:$A$25,C2))
=IF(C2="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C2))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D2)     =ISNUMBER('Audit Intake'!E2)    =IF(ISNUMBER('Audit Intake'!E2),AND('Audit
Intake'!E2≥0,MOD('Audit Intake'!E2,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F2),'Audit Intake'!F2≥0)       =OR('Audit Intake'!G2="Delivered",'Audit
Intake'!G2="Pending",'Audit Intake'!G2="Cancelled")
Case 2  3       ='Audit Intake'!A3      =IF(C3="",0,COUNTIF(Orders!$A$2:$A$25,C3))
=IF(C3="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C3))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D3)     =ISNUMBER('Audit Intake'!E3)    =IF(ISNUMBER('Audit Intake'!E3),AND('Audit
Intake'!E3≥0,MOD('Audit Intake'!E3,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F3),'Audit Intake'!F3≥0)       =OR('Audit Intake'!G3="Delivered",'Audit
Intake'!G3="Pending",'Audit Intake'!G3="Cancelled")
Case 3  4       ='Audit Intake'!A4      =IF(C4="",0,COUNTIF(Orders!$A$2:$A$25,C4))
=IF(C4="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C4))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D4)     =ISNUMBER('Audit Intake'!E4)    =IF(ISNUMBER('Audit Intake'!E4),AND('Audit
Intake'!E4≥0,MOD('Audit Intake'!E4,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F4),'Audit Intake'!F4≥0)       =OR('Audit Intake'!G4="Delivered",'Audit
Intake'!G4="Pending",'Audit Intake'!G4="Cancelled")
Case 4  5       ='Audit Intake'!A5      =IF(C5="",0,COUNTIF(Orders!$A$2:$A$25,C5))
=IF(C5="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C5))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D5)     =ISNUMBER('Audit Intake'!E5)    =IF(ISNUMBER('Audit Intake'!E5),AND('Audit
Intake'!E5≥0,MOD('Audit Intake'!E5,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F5),'Audit Intake'!F5≥0)       =OR('Audit Intake'!G5="Delivered",'Audit
Intake'!G5="Pending",'Audit Intake'!G5="Cancelled")
Case 5  6       ='Audit Intake'!A6      =IF(C6="",0,COUNTIF(Orders!$A$2:$A$25,C6))
=IF(C6="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C6))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D6)     =ISNUMBER('Audit Intake'!E6)    =IF(ISNUMBER('Audit Intake'!E6),AND('Audit
Intake'!E6≥0,MOD('Audit Intake'!E6,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F6),'Audit Intake'!F6≥0)       =OR('Audit Intake'!G6="Delivered",'Audit
Intake'!G6="Pending",'Audit Intake'!G6="Cancelled")
Case 6  7       ='Audit Intake'!A7      =IF(C7="",0,COUNTIF(Orders!$A$2:$A$25,C7))
=IF(C7="",0,COUNTIF('Audit Intake'!$A$2:$A$7,C7))       =COUNTIF(Rates!$A$2:$A$7,'Audit
Intake'!D7)     =ISNUMBER('Audit Intake'!E7)    =IF(ISNUMBER('Audit Intake'!E7),AND('Audit
Intake'!E7≥0,MOD('Audit Intake'!E7,1)=0),FALSE)        =AND(ISNUMBER('Audit
Intake'!F7),'Audit Intake'!F7≥0)       =OR('Audit Intake'!G7="Delivered",'Audit
Intake'!G7="Pending",'Audit Intake'!G7="Cancelled")
```

Supporting comparison cells: B11 **=Orders!E2** gives2; E11 **='Audit Intake'!E2** gives3; H11 **=Orders!E2='Audit Intake'!E2** givesFALSE. E13 **=SUMPRODUCT(--(Orders!A2:I2='Source Snapshot'!A2:I2))** gives9 identical fields in the preserved copy. A repeated copy is not an additional order; the conflicting PO001 intake remains quarantined.

Proposed exception register — Audit Intake J1:L7

These are guided author-reference dispositions, not completed independent decisions or messages to a source owner.

```
Issue   Evidence        Disposition
Conflicting duplicate OrderID   Practice PO001: source ID count1; Orders Quantity2 vs intake3,
equality FALSE. Other fields copied from PO001. Quarantine pending evidenced correction.
Clarify with the source owner; do not overwrite Orders or count both.
Missing OrderID Raw intake A3 is empty. This is a labelled copy of PO002 with only its ID
removed; do not infer that missing ID.  Clarify the required key; quarantine pending evidence.
Do not invent an ID or silently fill PO002.
Unknown ItemCode        IT99 has zero exact matches in Rates A2:A7. Other fields are practice
copies of PO003. Cancelled itself is allowed.   Clarify the intended item key and rate with
evidence. Keep raw IT99; quarantine pending evidence.
Non-numeric Quantity    Quantity is the text four: ISNUMBER is FALSE and the non-negative
whole-number rule is FALSE. Other fields copied from PO004.     Clarify the intended numeric
quantity with evidence; propose correction separately. Do not replace raw text by guessing 4.
Valid zero Quantity     Quantity0 is numeric and passes the explicit non-negative whole-number
rule. Other fields copied from PO005 under a practice ID.       Retain as a valid training
example. No insertion into baseline Orders; zero does not prove a delivery or spend.
Valid Cancelled status  Cancelled passes the status contract; numeric Quantity2, UnitCost8 and
key IT06 are valid. Practice copy of PO006 with status changed. Retain as a valid all-orders
training example. Later status exclusions need a stated analysis question, not a dirty-data
label.
```

Independent release conditions, independent disposition reasoning, reviewer challenge and learner timing remain pending. The six intake examples never enter baseline controls. A preserved Source Snapshot is a copy, not a protected-range permission or automatic archive.

02

CHAPTER 2

Build Conditional Calculations

In this chapter you will revise mixed references and apply IF, SUMIF, COUNTIF, SUMIFS and COUNTIFS with test cases.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 75 minutes: 15 explanation/demonstration, 32 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 60 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Checked line amounts, a project/status summary grid and tested logic bands.

> **Status:** author reference with37 staged application captures through2.4. Row formulas, mixed-reference grids, distinct populations, manual six-ID checks,21 expected-versus-actual logic cases and isolated anchor-fault repair verified. Original input fields and Lab1 evidence preserved. Independent threshold summary,250/restoration test, reviewer and learner timing remain pending.

What You Will Learn

- Build row calculations.
- Apply conditional totals with mixed references.
- Test a nested rule at its boundaries.

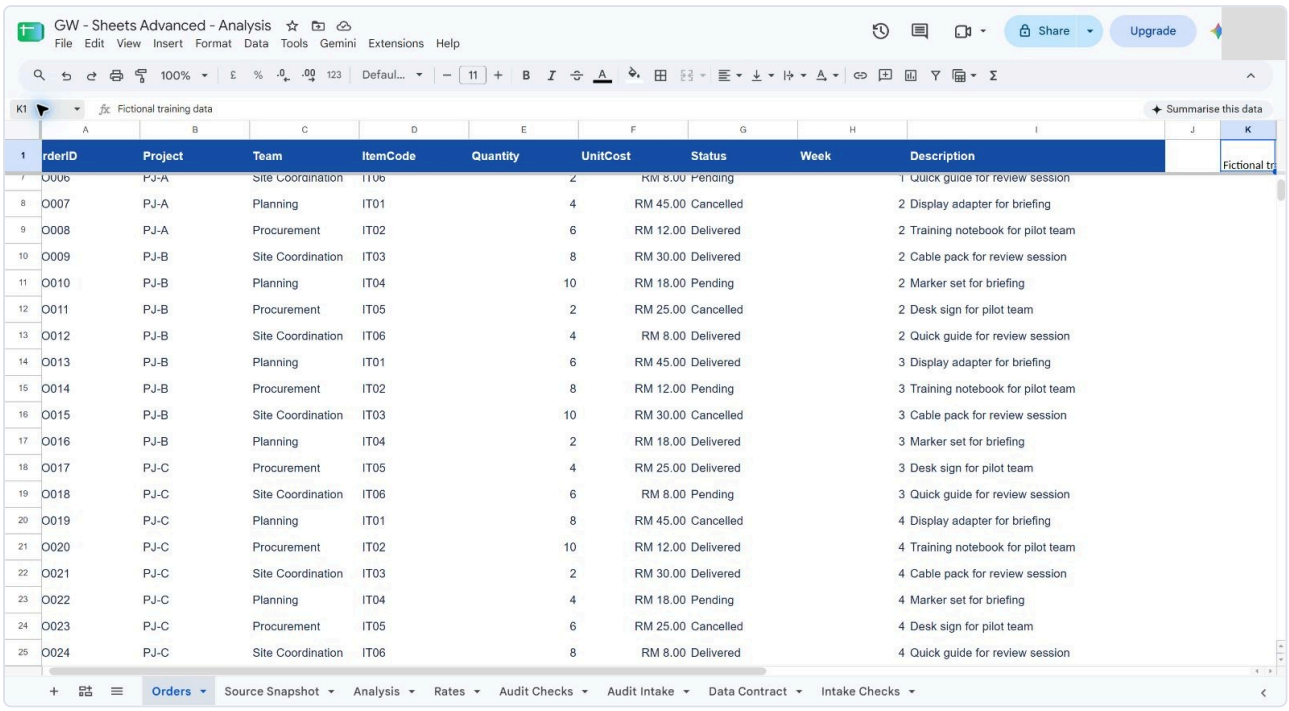
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders A1:I25 and the Analysis tab from Chapter 1.

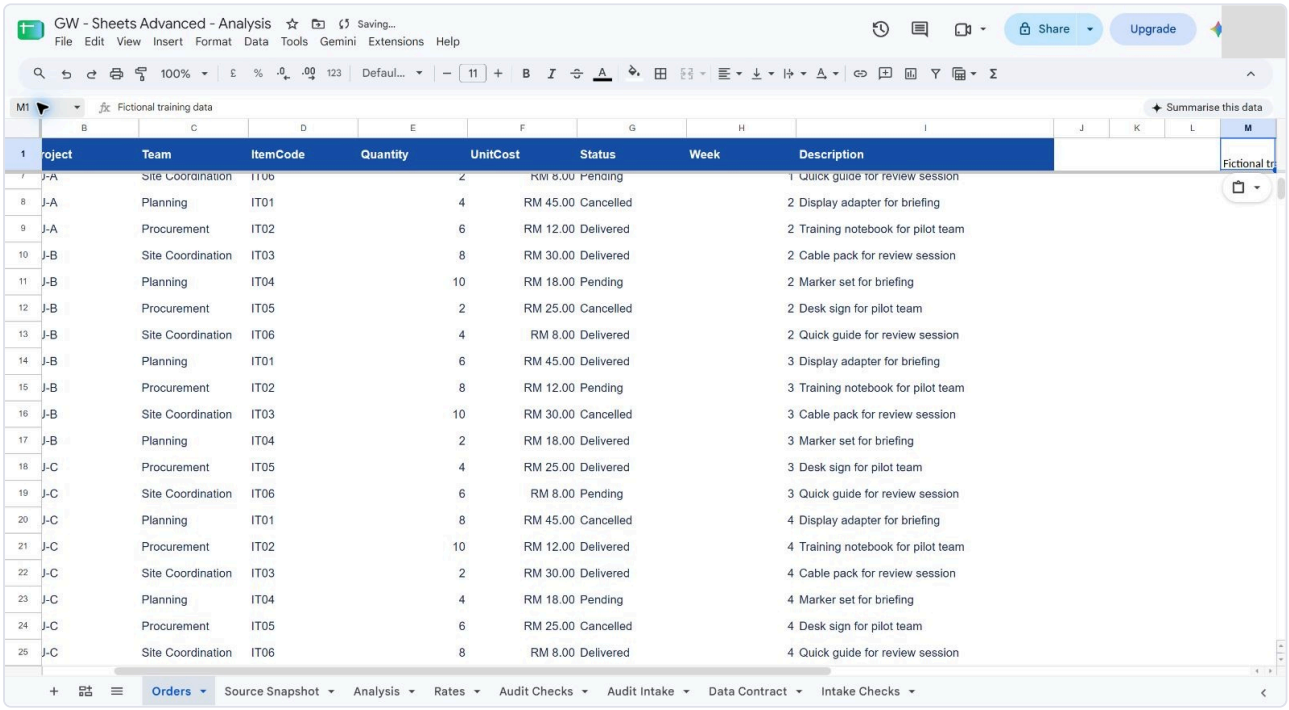
2.1 Calculate Line Amounts

1. Reopen **GW - Sheets Advanced - Analysis** from Chapter1. Inspect Orders. The supplied **K1** contains a fictional-data label, so preserve it before using K for a calculation. Keep A:I, Rates and Source Snapshot unchanged.



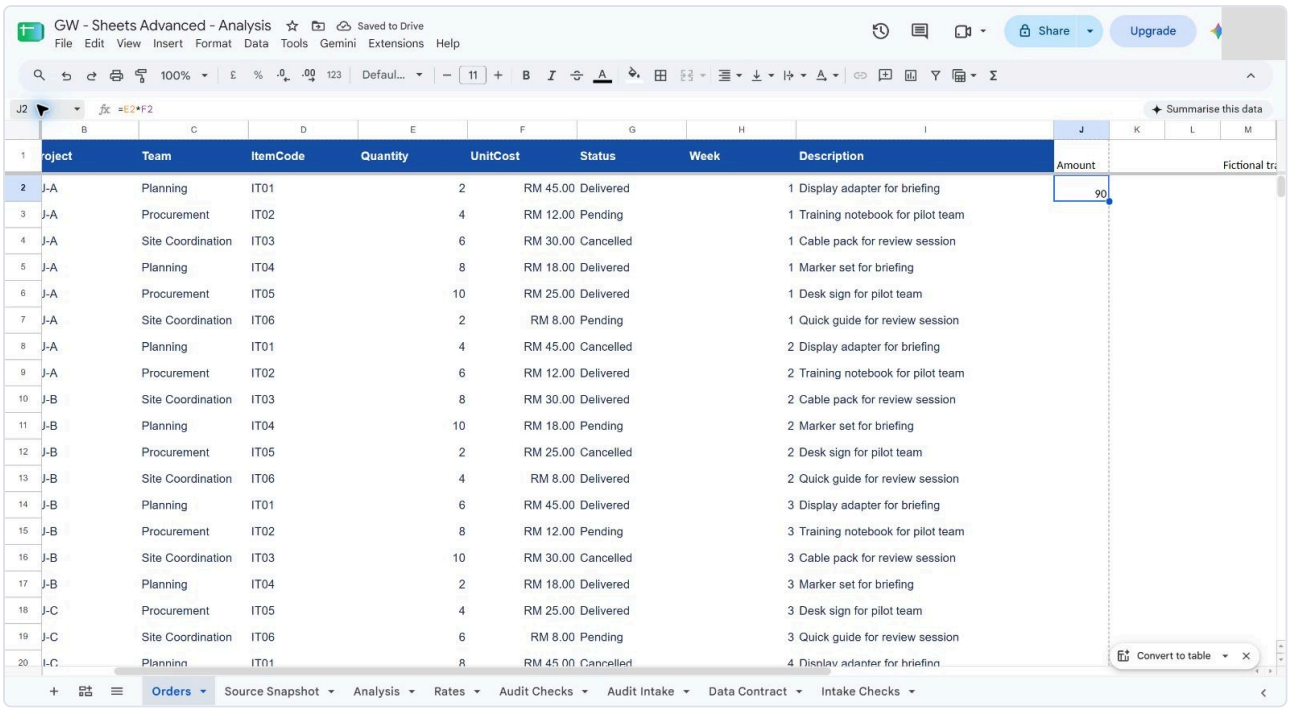
	rderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description
7	O006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session
8	O007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing
9	O008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team
10	O009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session
11	O010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing
12	O011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team
13	O012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session
14	O013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing
15	O014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team
16	O015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session
17	O016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing
18	O017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team
19	O018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session
20	O019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing
21	O020	PJ-C	Procurement	IT02	10	RM 12.00	Delivered	4	Training notebook for pilot team
22	O021	PJ-C	Site Coordination	IT03	2	RM 30.00	Delivered	4	Cable pack for review session
23	O022	PJ-C	Planning	IT04	4	RM 18.00	Pending	4	Marker set for briefing
24	O023	PJ-C	Procurement	IT05	6	RM 25.00	Cancelled	4	Desk sign for pilot team
25	O024	PJ-C	Site Coordination	IT06	8	RM 8.00	Delivered	4	Quick guide for review session

2. Cut **K1** and paste it into **M1**, outside the calculation columns. This preserves the label rather than silently overwriting it with ActiveAmount. Source Snapshot retains the original label at K1.

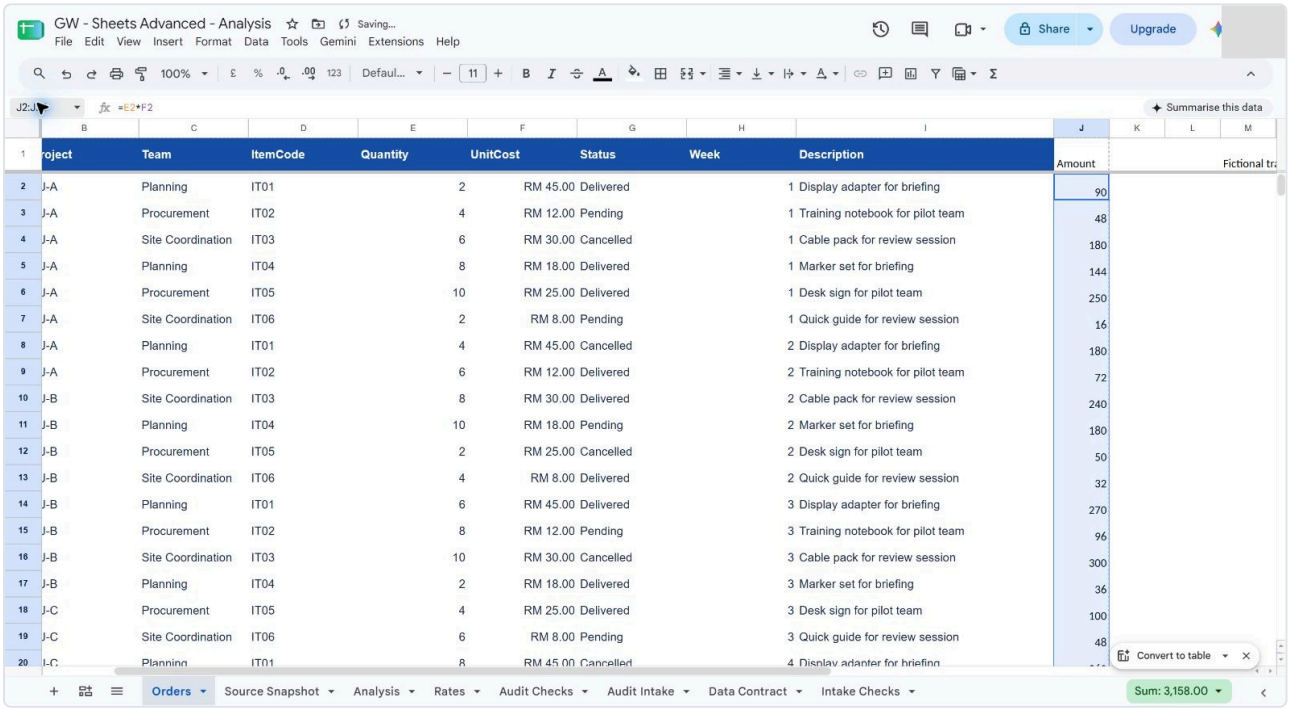


	oject	Team	ItemCode	Quantity	UnitCost	Status	Week	Description
7	J-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session
8	J-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing
9	J-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team
10	J-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session
11	J-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing
12	J-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team
13	J-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session
14	J-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing
15	J-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team
16	J-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session
17	J-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing
18	J-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team
19	J-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session
20	J-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing
21	J-C	Procurement	IT02	10	RM 12.00	Delivered	4	Training notebook for pilot team
22	J-C	Site Coordination	IT03	2	RM 30.00	Delivered	4	Cable pack for review session
23	J-C	Planning	IT04	4	RM 18.00	Pending	4	Marker set for briefing
24	J-C	Procurement	IT05	6	RM 25.00	Cancelled	4	Desk sign for pilot team
25	J-C	Site Coordination	IT06	8	RM 8.00	Delivered	4	Quick guide for review session

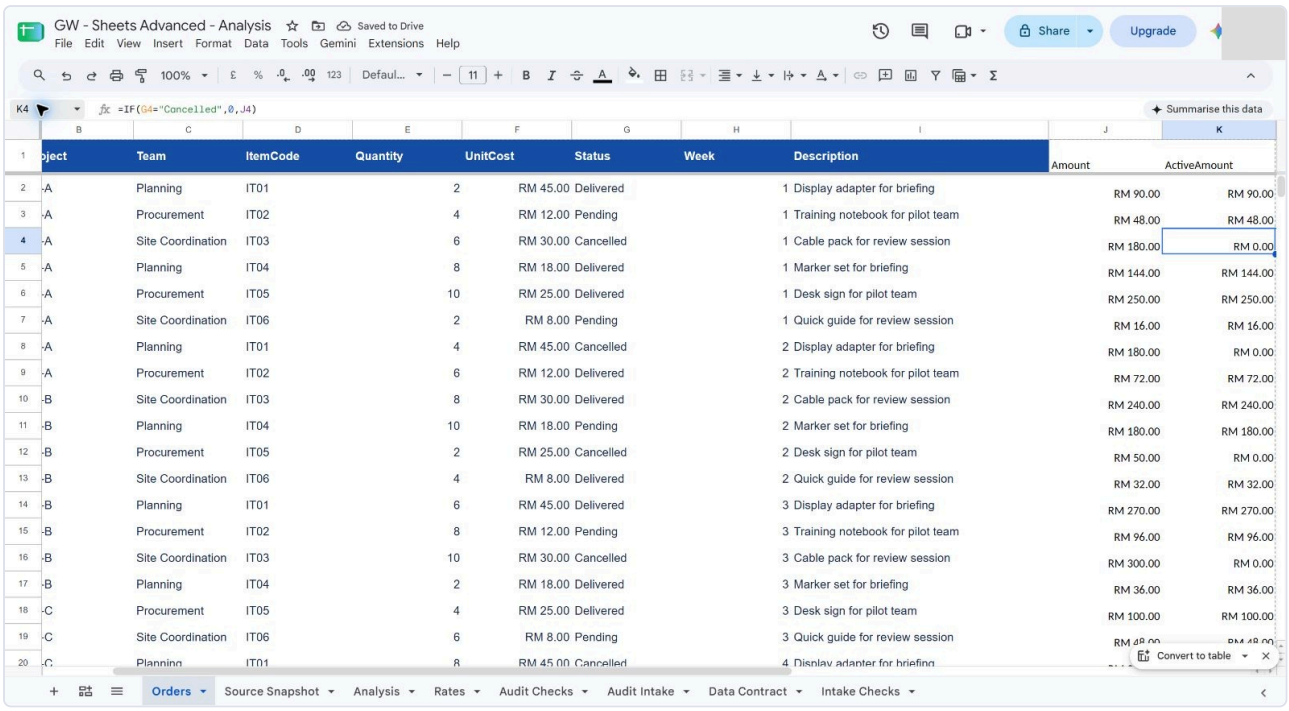
3. Enter **Amount** in J1 and **=E2*F2** in J2. For PO001, Quantity2 times UnitCost45 gives **90**.



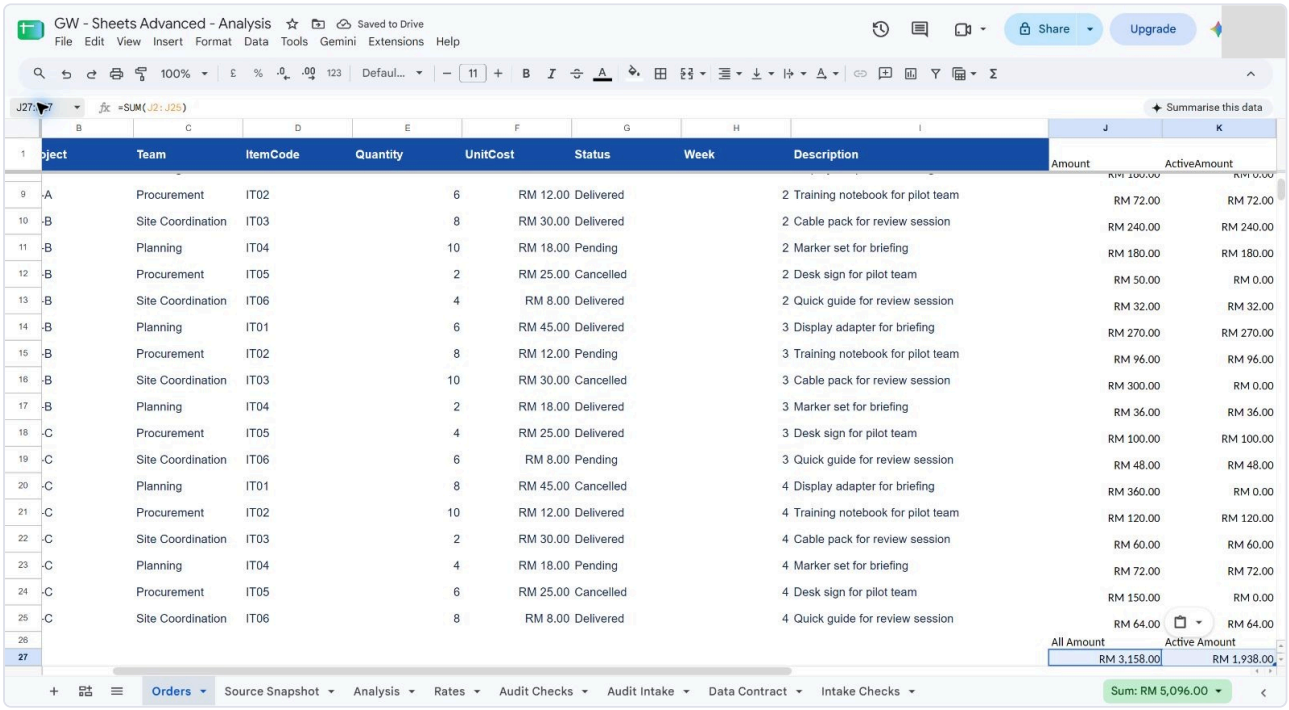
4. Select **J2:J25** and use **Fill down** (Ctrl+D). Check that every formula multiplies its own row's Quantity and UnitCost.



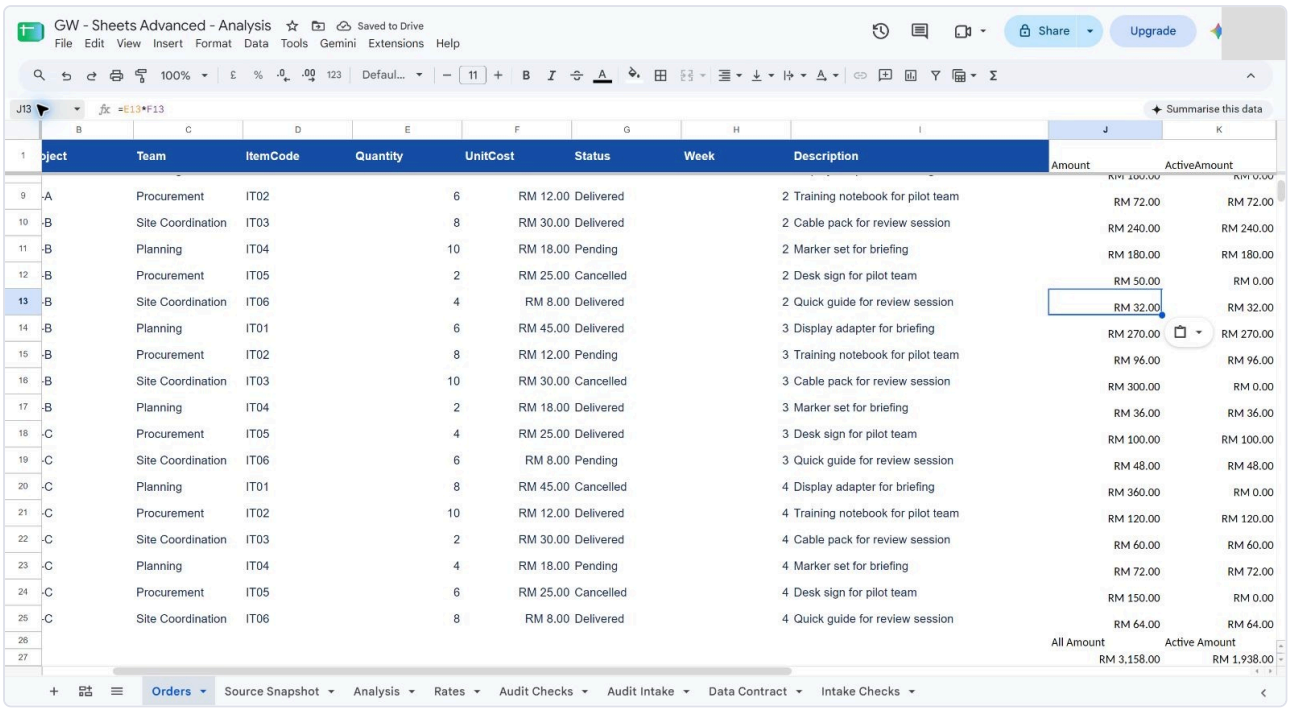
5. Enter **ActiveAmount** in K1 and **=IF(G2="Cancelled",0,J2)** in K2, then fill K2:K25 down. Inspect **K4**: PO003 remains in Orders with Amount180, while ActiveAmount is **0**. This applies a population rule; it does not delete or declare the source row corrupt. Widen new J/K columns as needed. For explicit RM display, apply custom number format **"RM" #,##0.00** to J2:K25; do not change the supplied numeric values.



6. Outside the 24-row table, put **All Amount / Active Amount** in J26:K26 and **=SUM(J2:J25) / =SUM(K2:K25)** in J27:K27. Expect **RM3158 / RM1938**. Keep these helper totals outside later bounded source calculations.

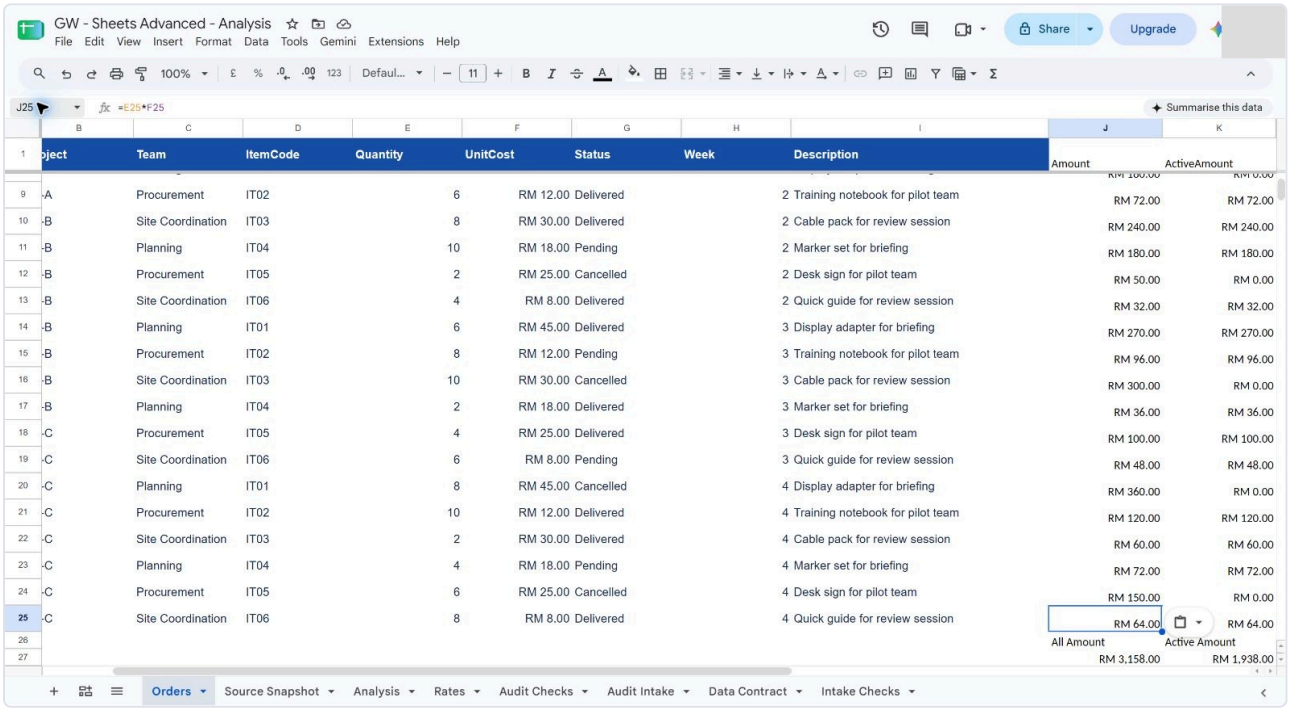


7. Inspect **J13 =E13*F13** for PO012: **4 × 8 =32**. Compare the referenced source fields, not only the displayed amount.



	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount
9	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00	RM 72.00
10	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00	RM 240.00
11	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00	RM 180.00
12	Procurement	IT05	2	RM 25.00	Cancelled		2 Desk sign for pilot team	RM 50.00	RM 0.00
13	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00	RM 32.00
14	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00	RM 270.00
15	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00	RM 96.00
16	Site Coordination	IT03	10	RM 30.00	Cancelled		3 Cable pack for review session	RM 300.00	RM 0.00
17	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00	RM 36.00
18	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00	RM 100.00
19	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	RM 48.00	RM 48.00
20	Planning	IT01	8	RM 45.00	Cancelled		4 Display adapter for briefing	RM 360.00	RM 0.00
21	Procurement	IT02	10	RM 12.00	Delivered		4 Training notebook for pilot team	RM 120.00	RM 120.00
22	Site Coordination	IT03	2	RM 30.00	Delivered		4 Cable pack for review session	RM 60.00	RM 60.00
23	Planning	IT04	4	RM 18.00	Pending		4 Marker set for briefing	RM 72.00	RM 72.00
24	Procurement	IT05	6	RM 25.00	Cancelled		4 Desk sign for pilot team	RM 150.00	RM 0.00
25	Site Coordination	IT06	8	RM 8.00	Delivered		4 Quick guide for review session	RM 64.00	RM 64.00
26								All Amount	Active Amount
27								RM 3,158.00	RM 1,938.00

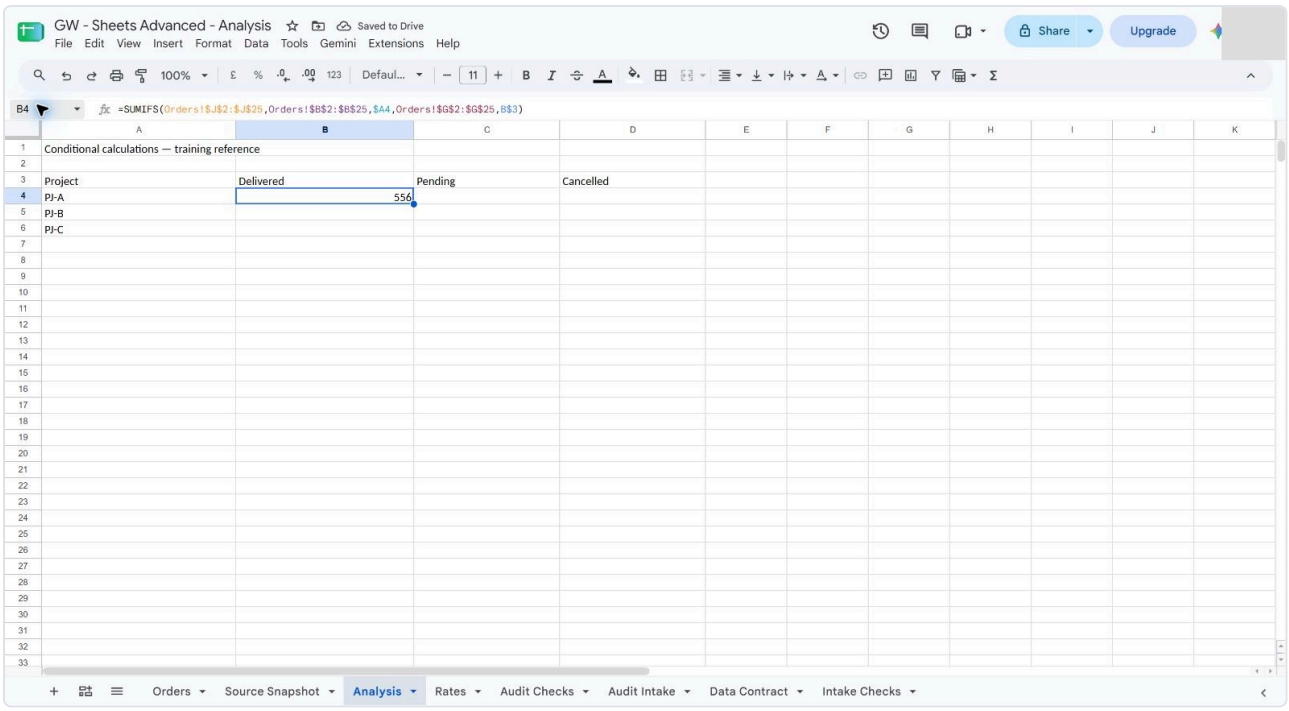
8. Inspect **J25 =E25*F25** for PO024: **8 × 8 =64**. The last row must be included in both filled columns and totals.



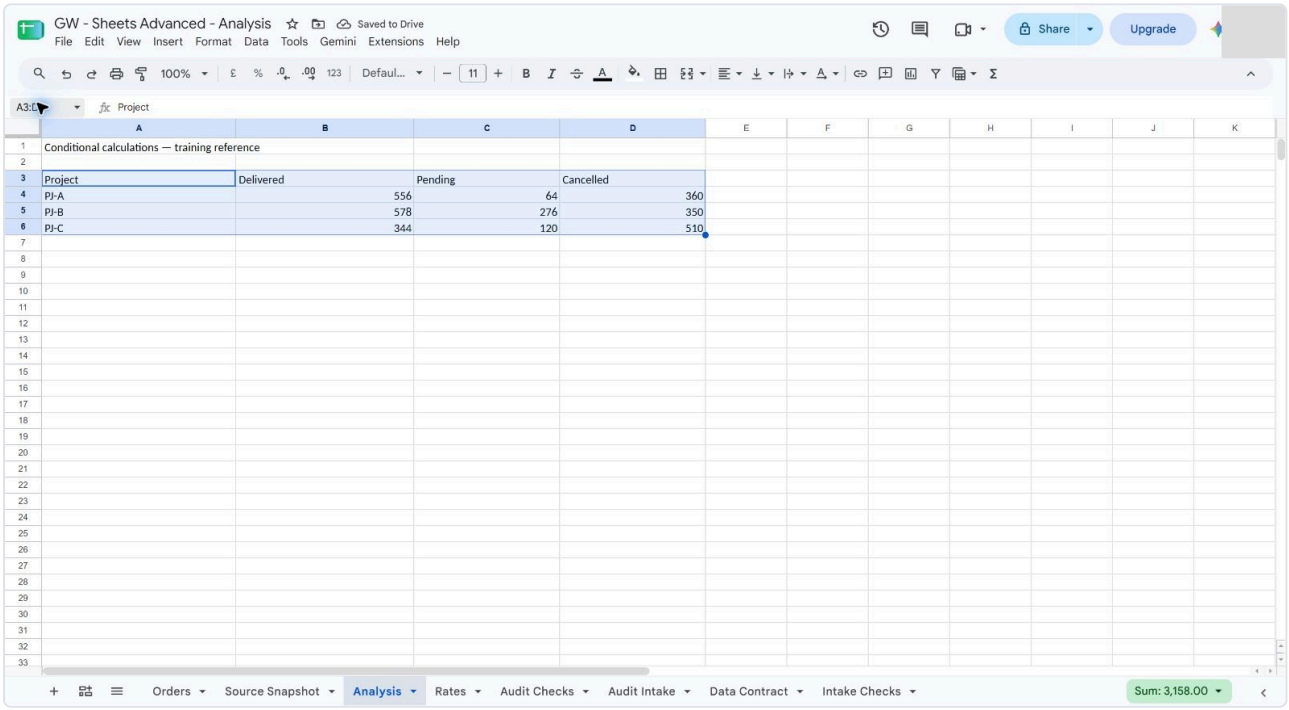
	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount
9	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00	RM 72.00
10	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00	RM 240.00
11	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00	RM 180.00
12	Procurement	IT05	2	RM 25.00	Cancelled		2 Desk sign for pilot team	RM 50.00	RM 0.00
13	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00	RM 32.00
14	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00	RM 270.00
15	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00	RM 96.00
16	Site Coordination	IT03	10	RM 30.00	Cancelled		3 Cable pack for review session	RM 300.00	RM 0.00
17	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00	RM 36.00
18	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00	RM 100.00
19	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	RM 48.00	RM 48.00
20	Planning	IT01	8	RM 45.00	Cancelled		4 Display adapter for briefing	RM 360.00	RM 0.00
21	Procurement	IT02	10	RM 12.00	Delivered		4 Training notebook for pilot team	RM 120.00	RM 120.00
22	Site Coordination	IT03	2	RM 30.00	Delivered		4 Cable pack for review session	RM 60.00	RM 60.00
23	Planning	IT04	4	RM 18.00	Pending		4 Marker set for briefing	RM 72.00	RM 72.00
24	Procurement	IT05	6	RM 25.00	Cancelled		4 Desk sign for pilot team	RM 150.00	RM 0.00
25	Site Coordination	IT06	8	RM 8.00	Delivered		4 Quick guide for review session	RM 64.00	RM 64.00
26								All Amount	Active Amount
27								RM 3,158.00	RM 1,938.00

2.2 Create a Summary Grid

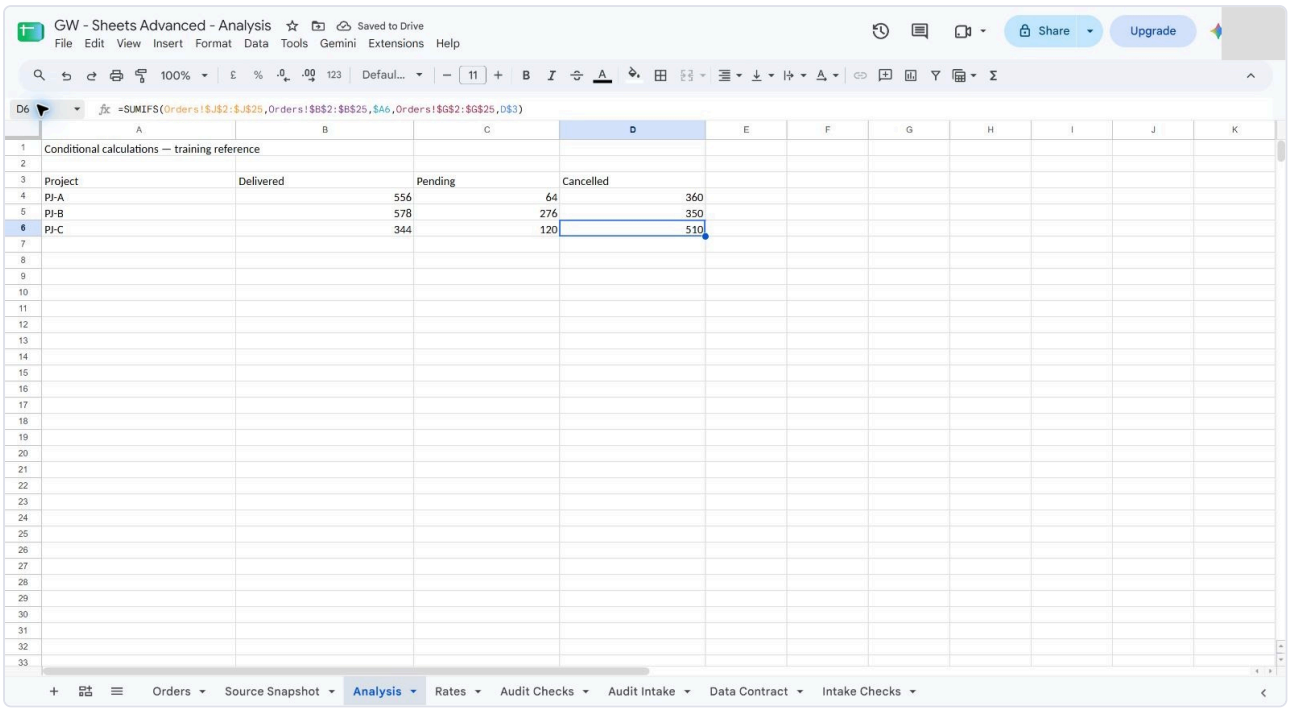
1. On **Analysis**, enter Project in A3, Delivered/Pending/Cancelled in B3:D3, and PJ-A/PJ-B/PJ-C in A4:A6. In B4 enter **=SUMIFS(Orders!$J$2:$J$25,Orders!$B$2:$B$25,$A4,Orders!$G$2:$G$25,B$3)**. Its first result is **556**. The source ranges are fully anchored; the project column is fixed but its row can move; the status row is fixed but its column can move.



2. Select **B4:D4** and **Fill right** (Ctrl+R), then select **B4:D6** and **Fill down** (Ctrl+D). Expected rows are **556/64/360**, **578/276/350** and **344/120/510**. Label the amount grid as RM. These nine intersections sum to **3158** across every status.

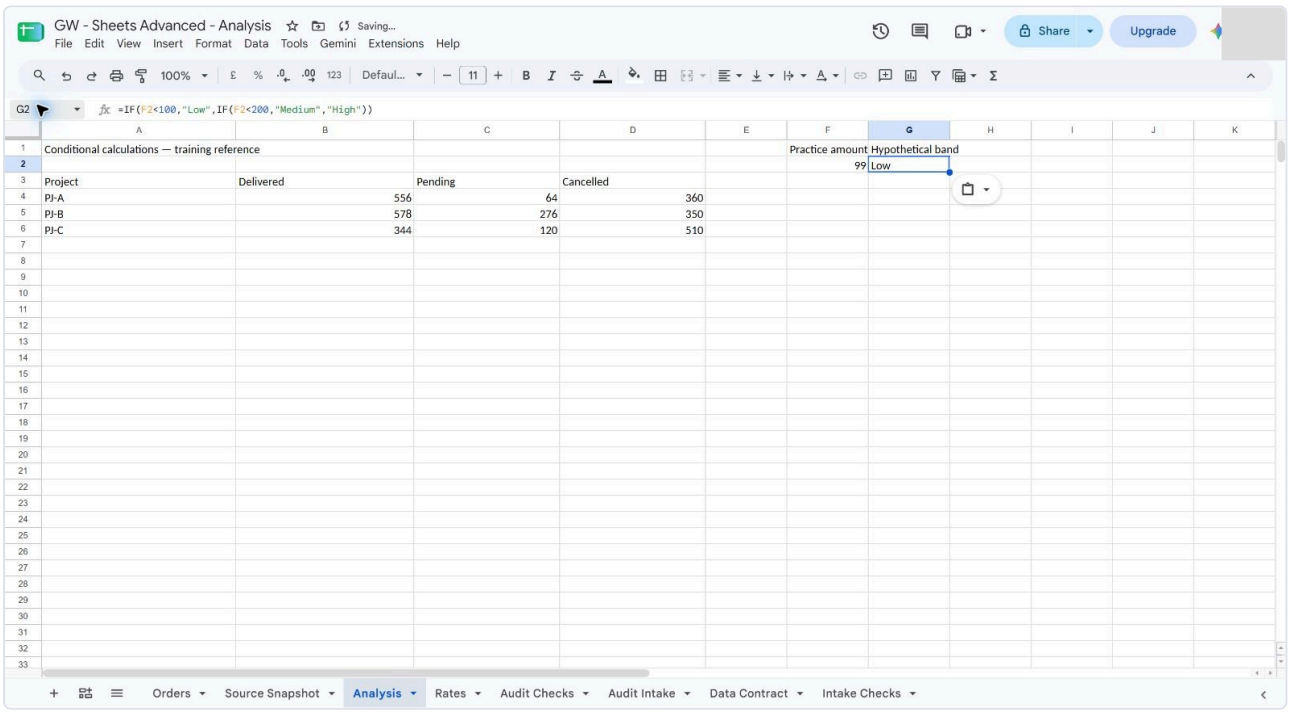


3. Inspect **D6**. It must use **$A6** for PJ-C and **D$3** for Cancelled, while both source criteria ranges and the sum range remain fixed. Its value is **510**. After populating Analysis, update the Chapter1 Audit Checks notes that previously described Analysis as empty; preserve the Chapter1 audit evidence as a historical checkpoint.

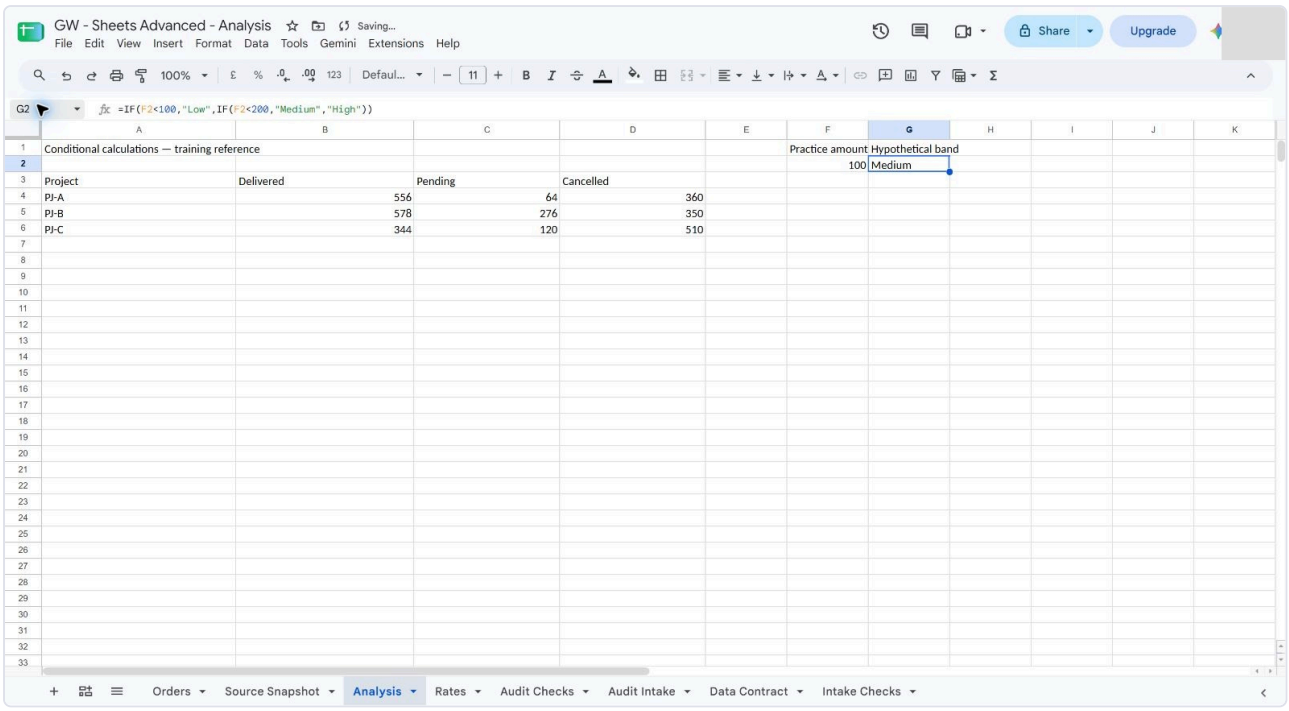


2.3 Test Logic Boundaries

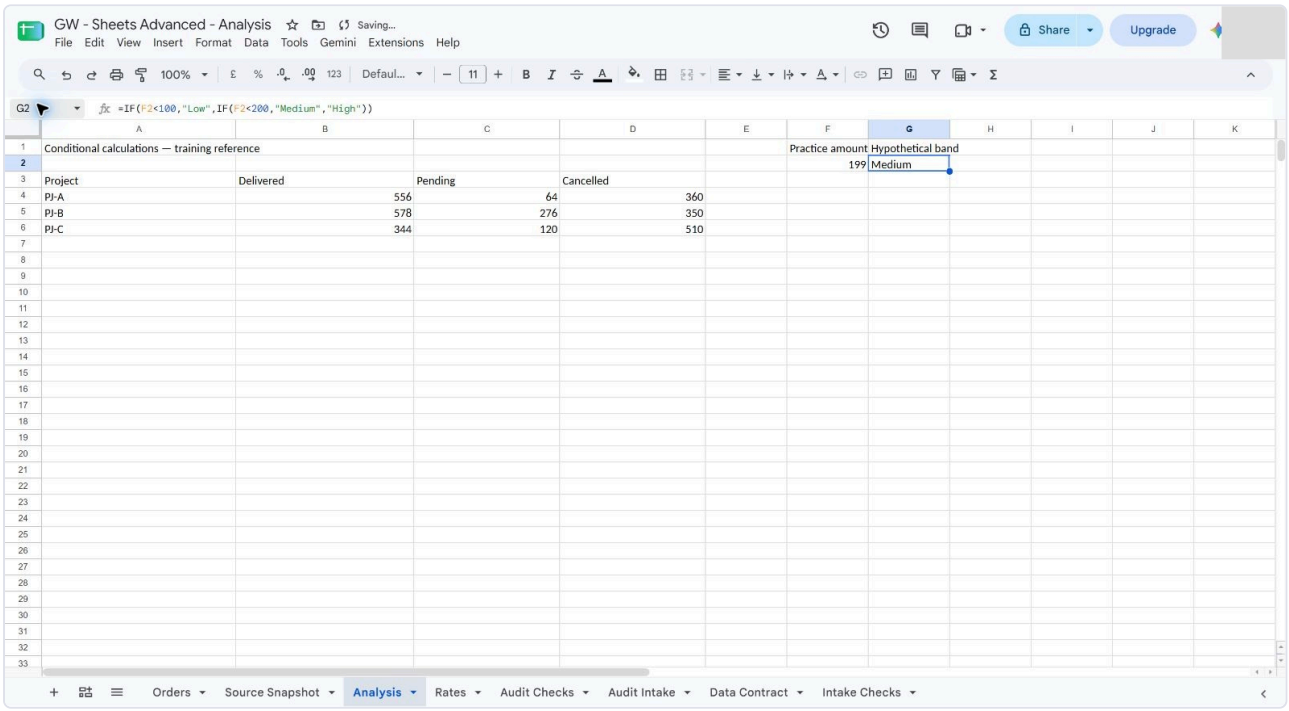
1. In the separate practice area on Analysis, label **F1/G1** Practice amount / Hypothetical band. Put **99** in F2 and **=IF(F2<100,"Low",IF(F2<200,"Medium","High"))** in G2. Expect **Low**. These are fictional training bands, not procurement policy.



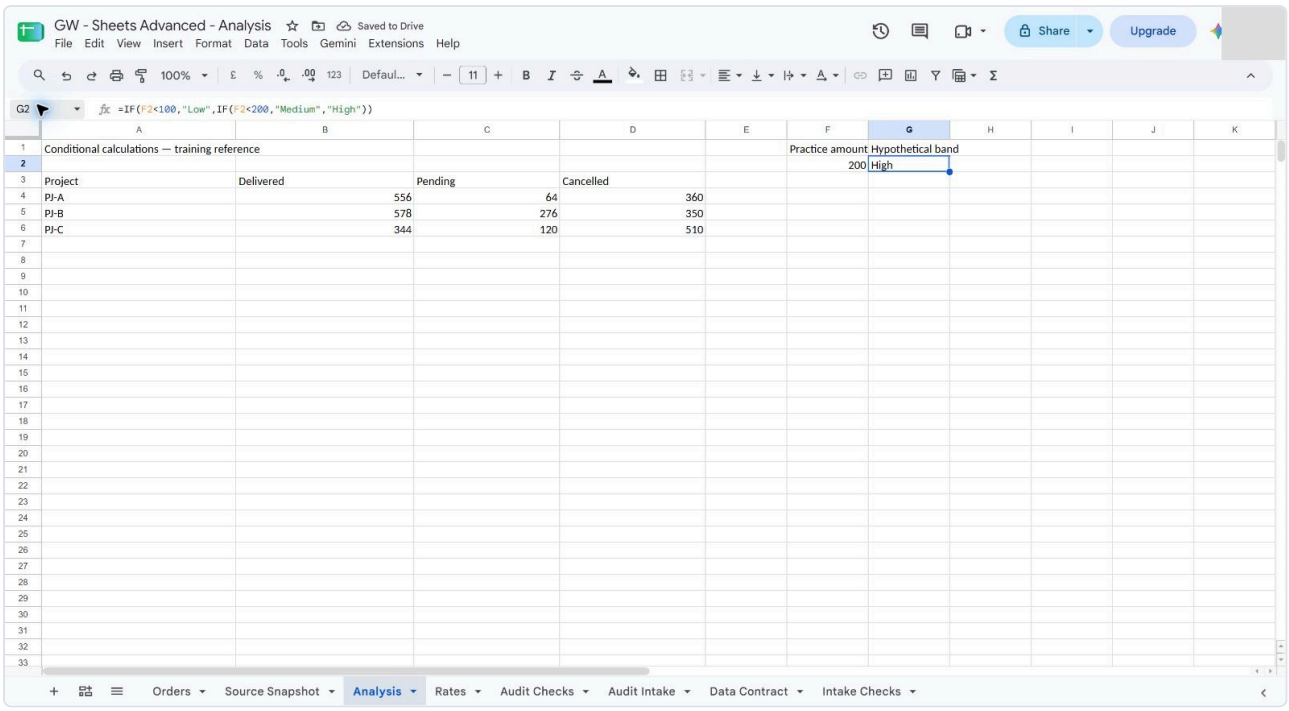
2. Change only F2 to **100**. The lower Medium boundary is inclusive: expect **Medium**.



3. Change F2 to **199**. It remains below200: expect **Medium**.

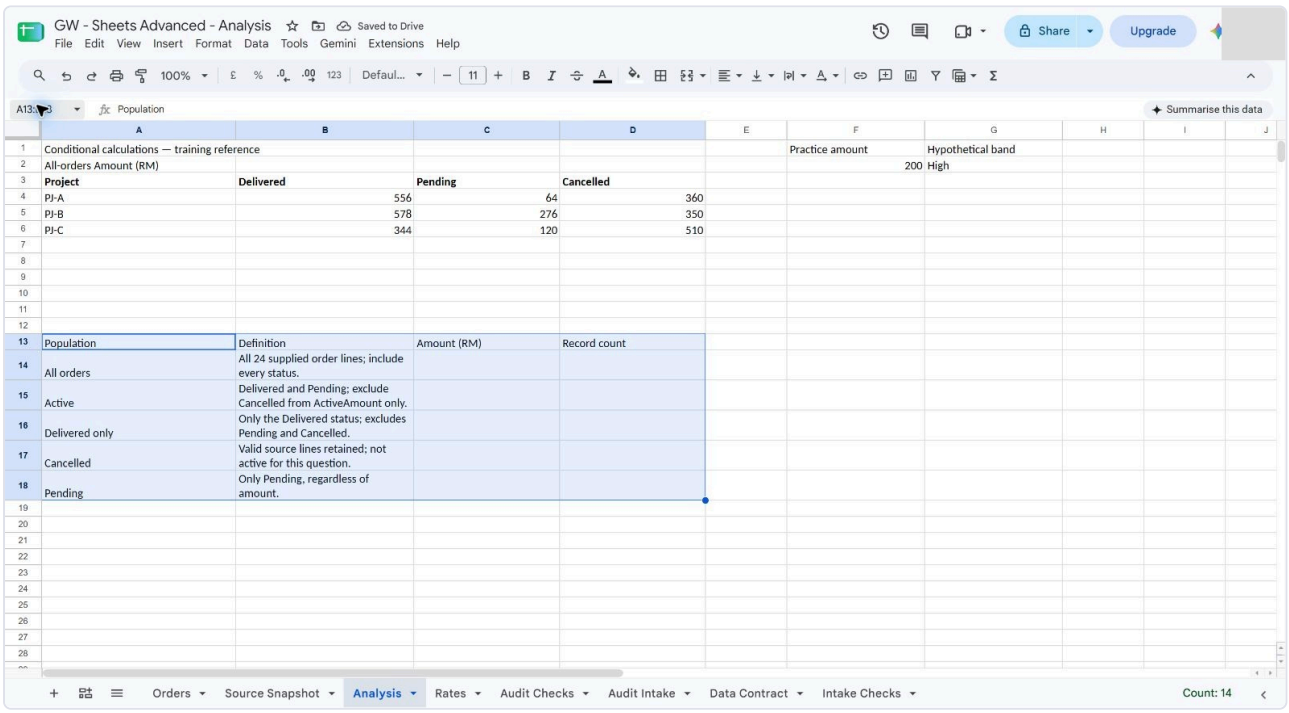


4. Change F2 to **200**. The High boundary is inclusive: expect **High**. Retain the tested formula; the cross-stage captures record the four separate results.

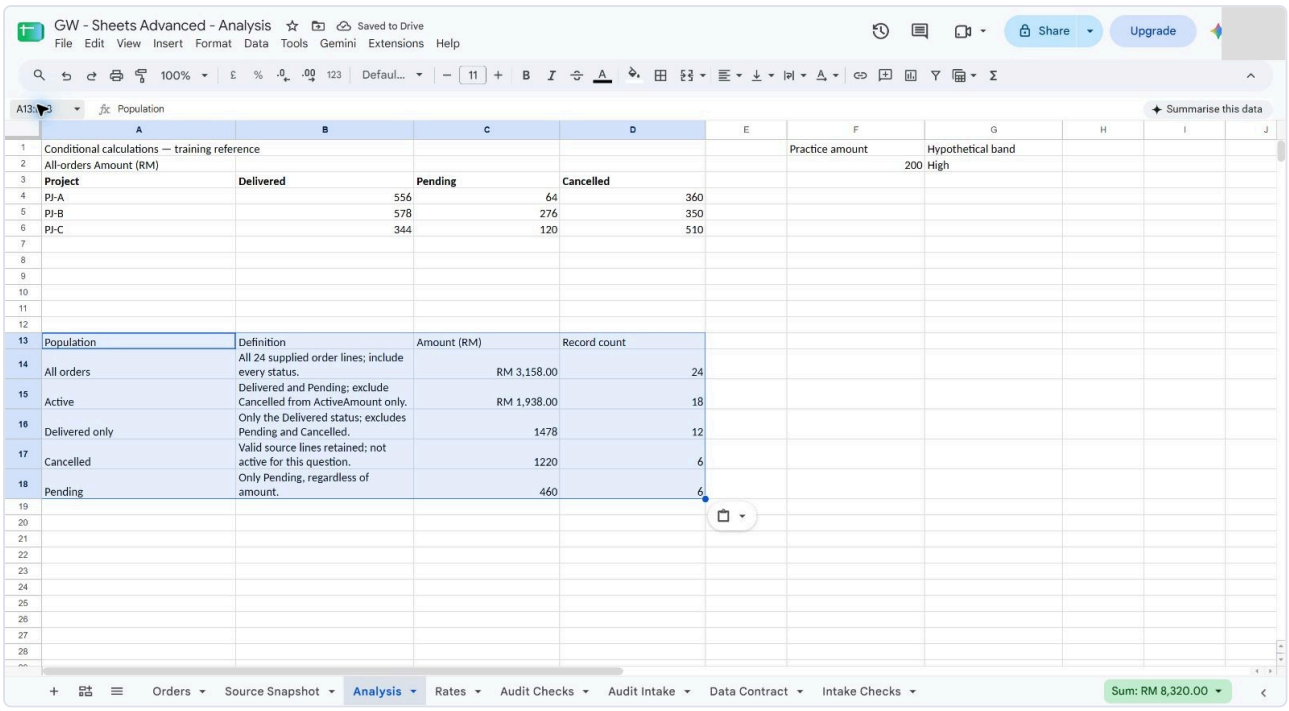


2.4 Reconcile Three Populations and Test Conditional Boundaries

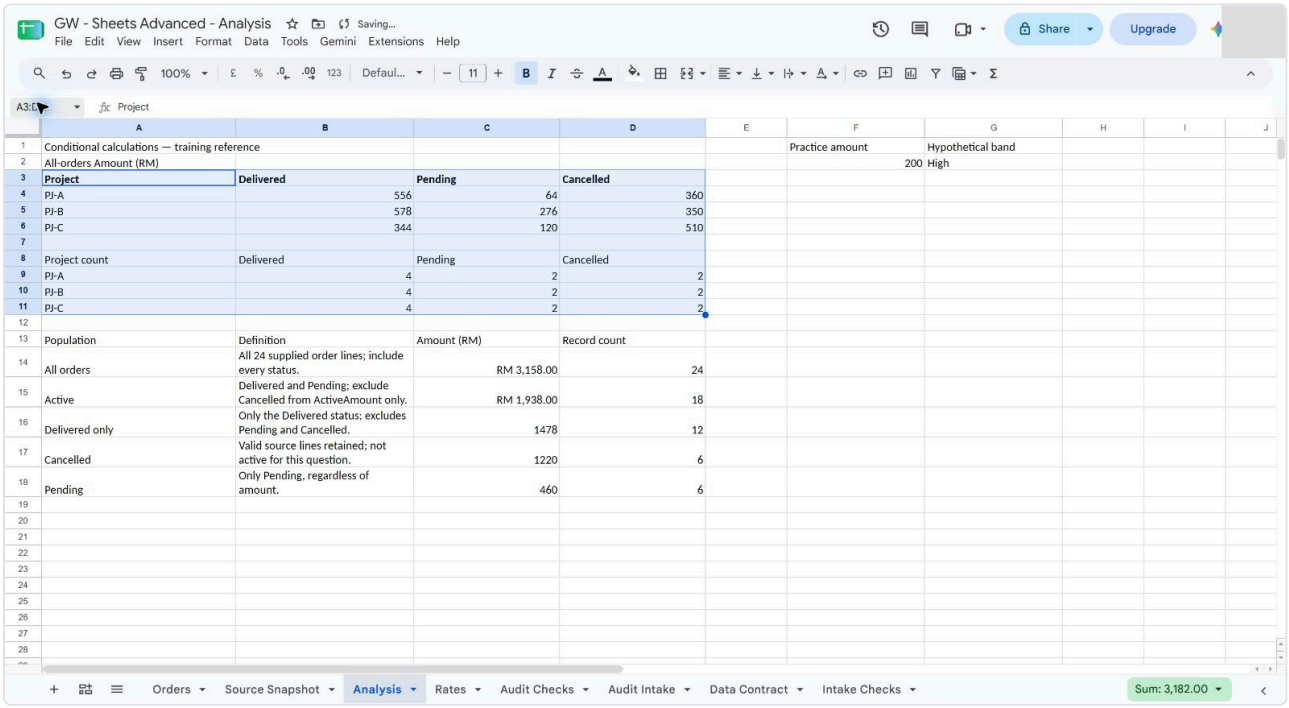
1. Write the population definitions on **Analysis A13:D18 before calculating**: all24 order lines; Active excludes Cancelled; Delivered-only excludes both Pending and Cancelled. Keep Cancelled and Pending controls explicit too. Do not treat totals for different populations as interchangeable.



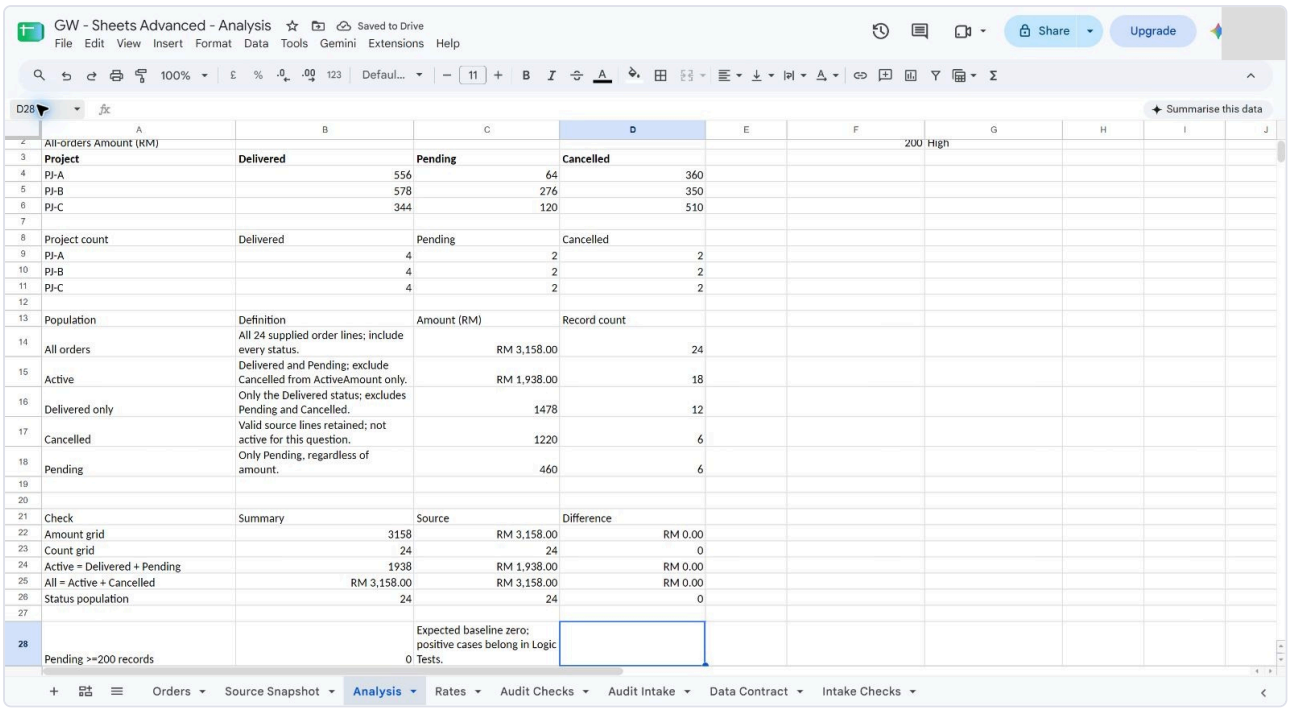
2. Fill the amount and count controls using the card on the copy-paste page. Expected amount/count pairs are **All3158/24**, **Active1938/18**, **Delivered1478/12**, **Cancelled1220/6**, **Pending460/6**. The SUMIF/COUNTIF checks use the exact status vocabulary; ActiveAmount is summed separately from all-orders Amount.



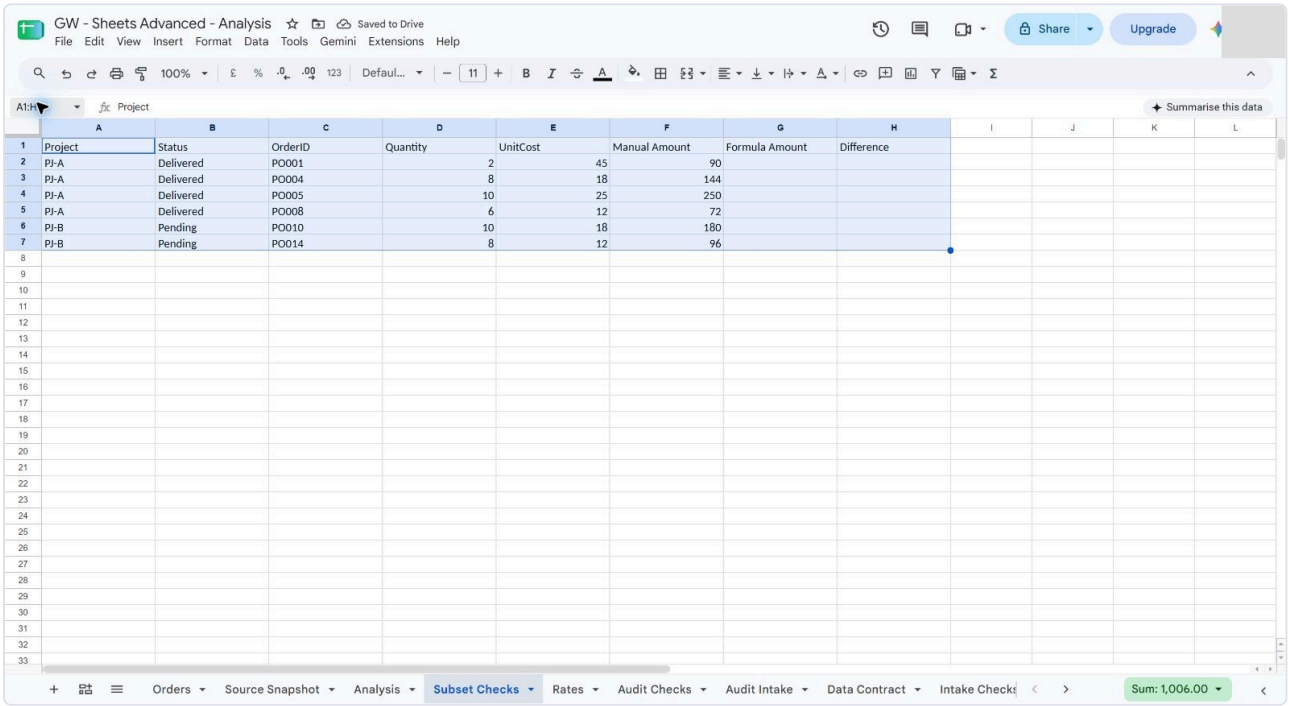
3. Build the matching COUNTIFS grid in **A8:D11**. Use Project count / Delivered / Pending / Cancelled in row8 and the three projects in A9:A11. Enter **=COUNTIFS(Orders!$B$2:$B$25,$A9,Orders!$G$2:$G$25,B$8)** in B9, fill right through D9 and down through row11. Each project has **4/2/2** records. Keep the amount and count grids separate, with clear units.



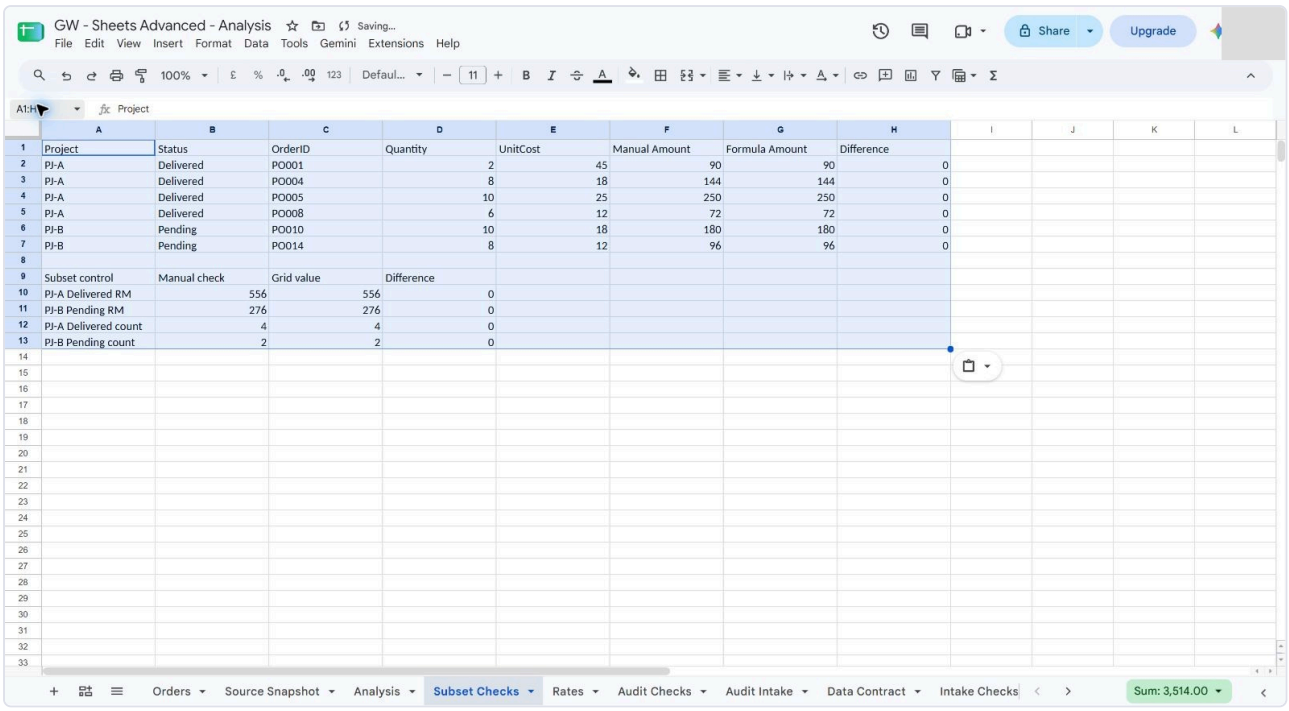
4. In **A21:D26**, reconcile the amount grid to3158 and count grid to24. Verify **Active = Delivered + Pending** and **All = Active + Cancelled**, with zero differences. In B28, **=COUNTIFS(Orders!G2:G25,"Pending",Orders!J2:J25,">=200")** returns **0**. This is a valid baseline result; positive-branch tests belong in the separate synthetic matrix.



5. Create **Subset Checks** and list the qualifying OrderIDs for two intersections. PJ-A Delivered is **PO001/PO004/PO005/PO008**, with manual products **90/144/250/72**. PJ-B Pending is **PO010/PO014**, with **180/96**. Inspect their actual source Quantity and UnitCost values. Record the manually checked products before entering the formula comparisons.



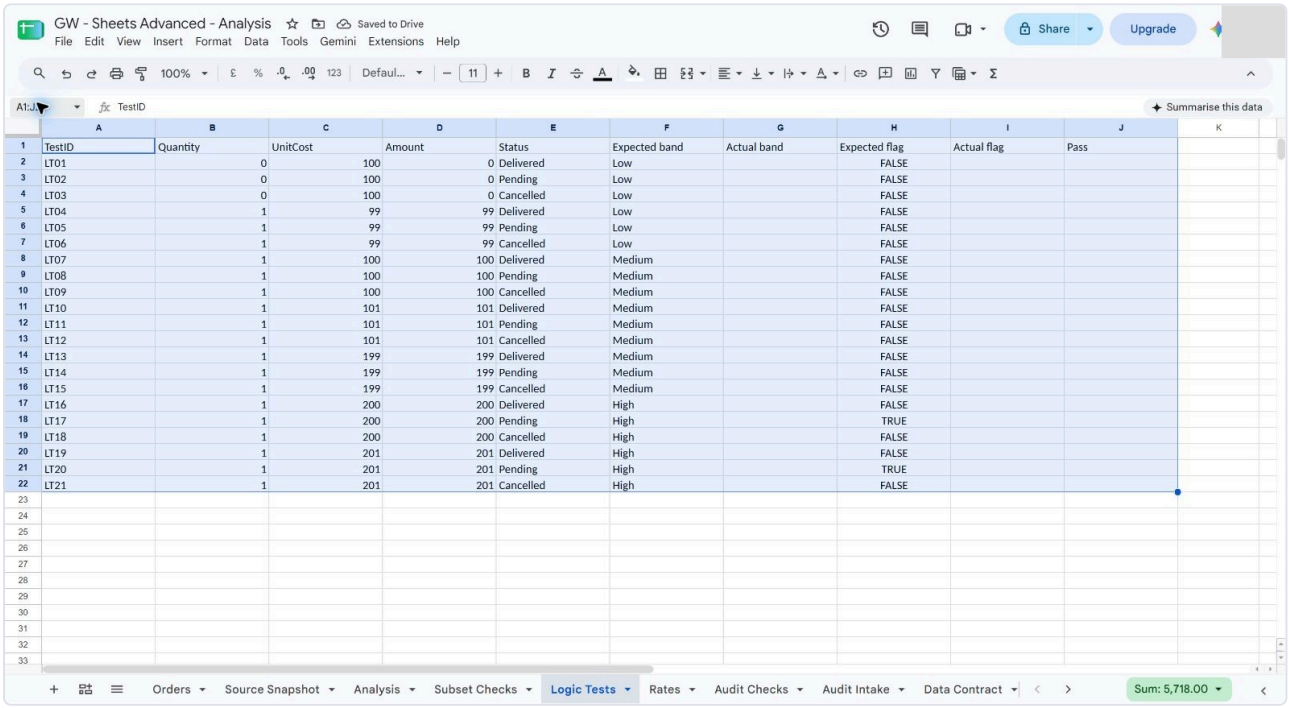
6. In Subset Checks G2 enter **=D2*E2**, H2 **=G2-F2**, and fill G2:H7 down. Compare the two manual totals to the maintained grid: **556** to Analysis B4, **276** to Analysis C5; compare qualifying-ID counts **4** and **2** to Analysis B9/C10. Every line and subset difference is zero. This is author-supported manual casework, not an independent learner validation.



Project	Status	OrderID	Quantity	UnitCost	Manual Amount	Formula Amount	Difference
PJ-A	Delivered	PO001	2	45	90	90	0
PJ-A	Delivered	PO004	8	18	144	144	0
PJ-A	Delivered	PO005	10	25	250	250	0
PJ-A	Delivered	PO008	6	12	72	72	0
PJ-B	Pending	PO010	10	18	180	180	0
PJ-B	Pending	PO014	8	12	96	96	0

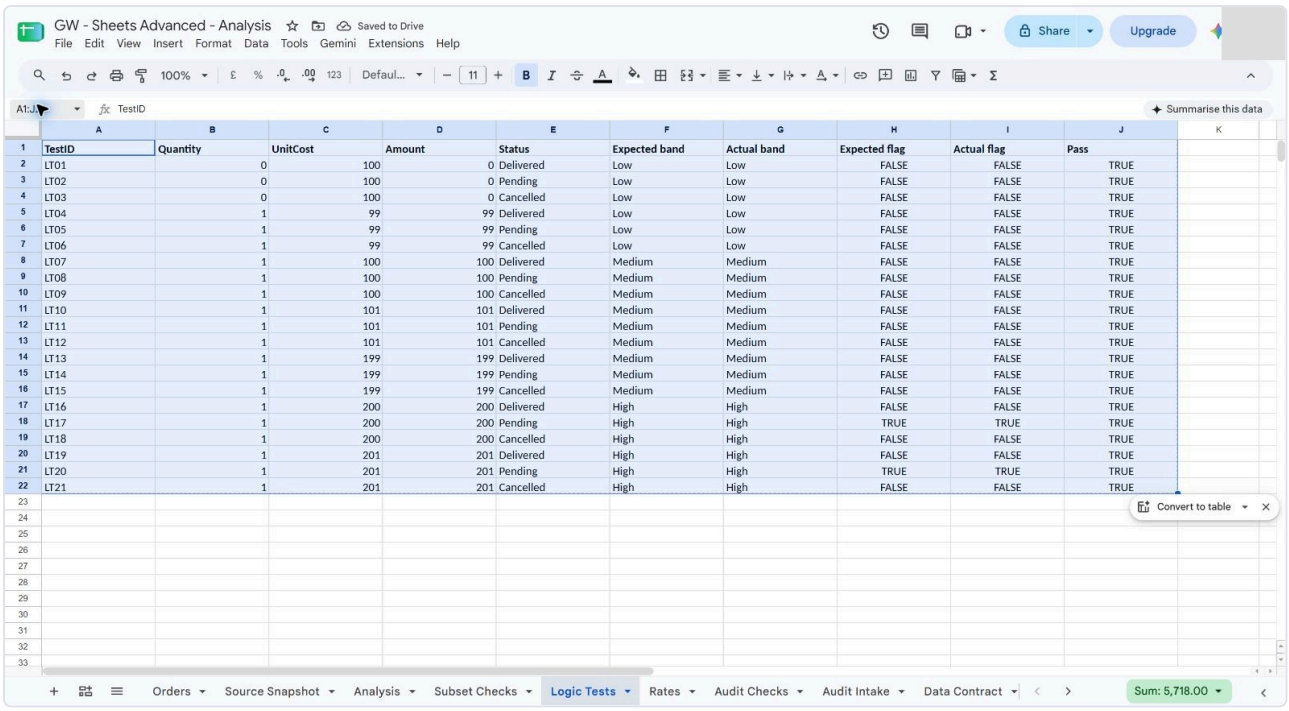
Subset control	Manual check	Grid value	Difference
PJ-A Delivered RM	556	556	0
PJ-B Pending RM	276	276	0
PJ-A Delivered count	4	4	0
PJ-B Pending count	2	2	0

7. Create **Logic Tests**. Paste the21-case expected-input card, leaving actual band/flag/pass columns empty initially. It crosses amounts **0,99,100,101,199,200,201** with all three statuses. The zero cases use **Quantity0 × known UnitCost100**, rather than an unknown rate disguised as0. Type the expected classifications and flags before running the actual formulas.



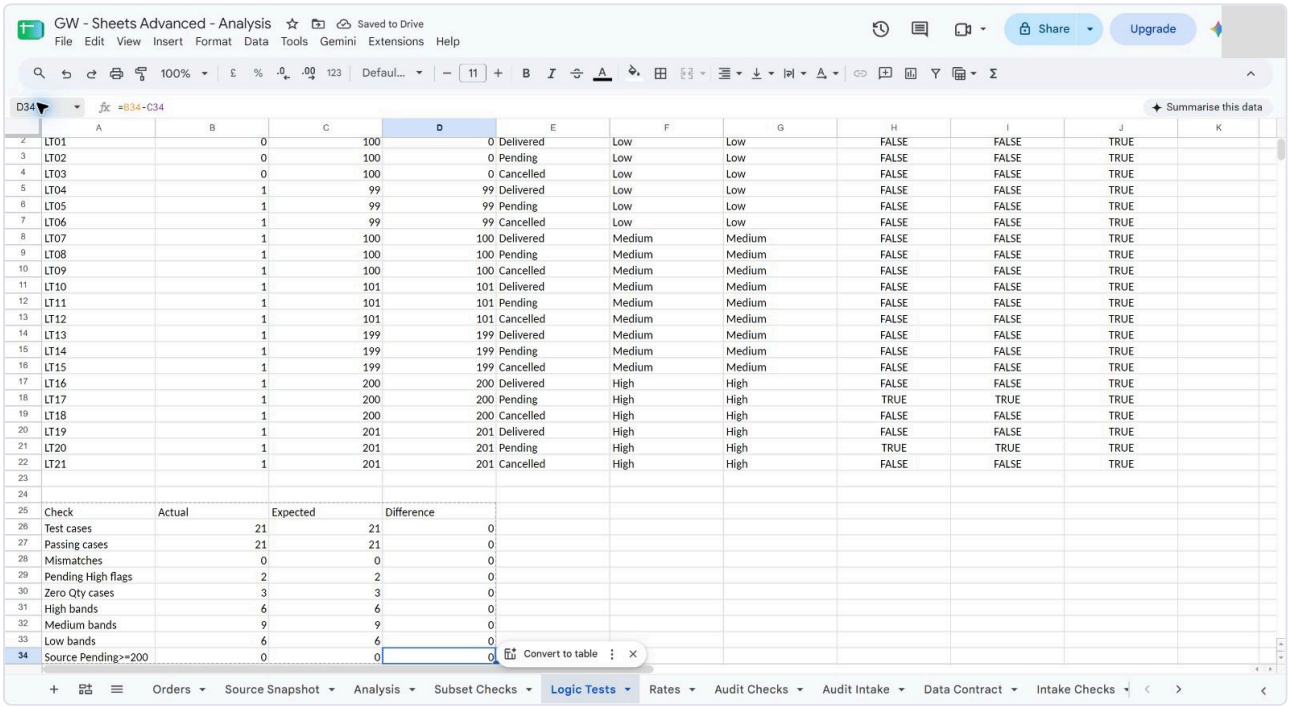
TestID	Quantity	UnitCost	Amount	Status	Expected band	Actual band	Expected flag	Actual flag	Pass
LT01	0	100	0	Delivered	Low		FALSE		
LT02	0	100	0	Pending	Low		FALSE		
LT03	0	100	0	Cancelled	Low		FALSE		
LT04	1	99	99	Delivered	Low		FALSE		
LT05	1	99	99	Pending	Low		FALSE		
LT06	1	99	99	Cancelled	Low		FALSE		
LT07	1	100	100	Delivered	Medium		FALSE		
LT08	1	100	100	Pending	Medium		FALSE		
LT09	1	100	100	Cancelled	Medium		FALSE		
LT10	1	101	101	Delivered	Medium		FALSE		
LT11	1	101	101	Pending	Medium		FALSE		
LT12	1	101	101	Cancelled	Medium		FALSE		
LT13	1	199	199	Delivered	Medium		FALSE		
LT14	1	199	199	Pending	Medium		FALSE		
LT15	1	199	199	Cancelled	Medium		FALSE		
LT16	1	200	200	Delivered	High		FALSE		
LT17	1	200	200	Pending	High		TRUE		
LT18	1	200	200	Cancelled	High		FALSE		
LT19	1	201	201	Delivered	High		FALSE		
LT20	1	201	201	Pending	High		TRUE		
LT21	1	201	201	Cancelled	High		FALSE		

8. Enter **G2 =IF(D2<100,"Low",IF(D2<200,"Medium","High"))**, **I2 =AND(E2="Pending",D2>=200)** and **J2 =AND(F2=G2,H2=I2)**. Fill each actual-formula column through row22. Compare expected and actual results; inspect the boundary cases rather than trusting only a total pass count.



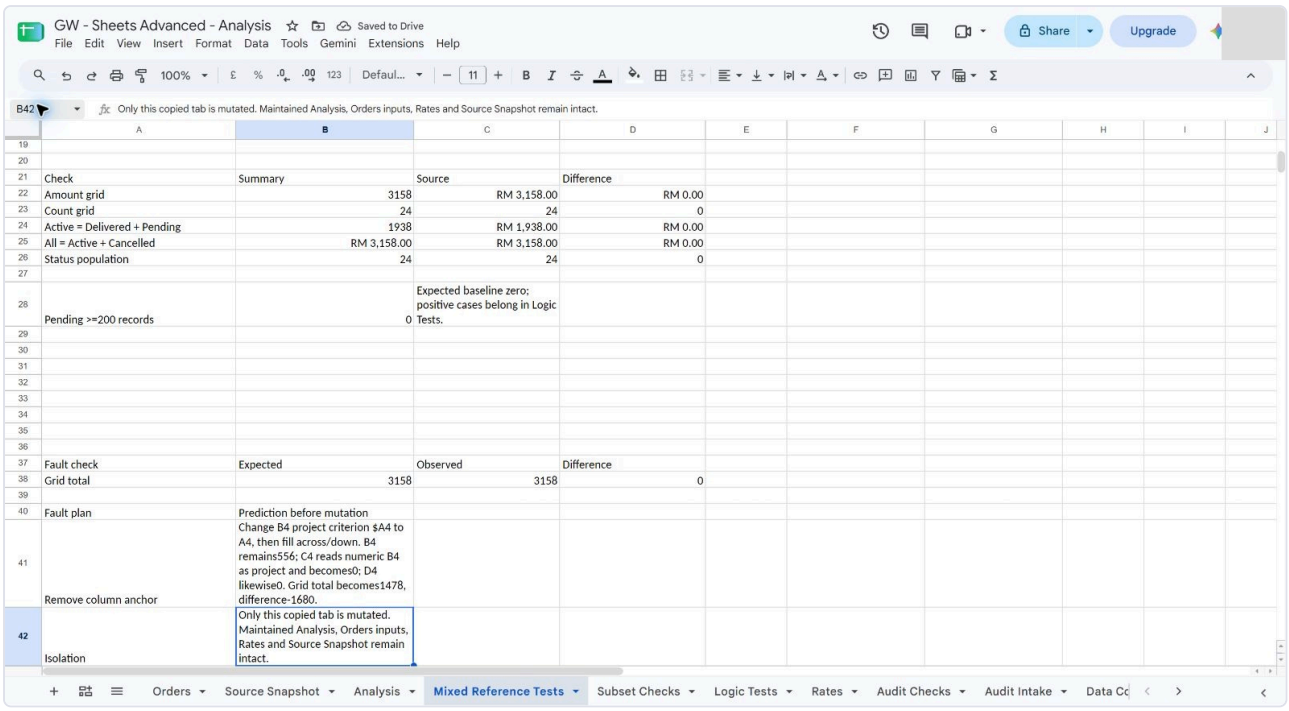
TestID	Quantity	UnitCost	Amount	Status	Expected band	Actual band	Expected flag	Actual flag	Pass
LT01	0	100	0	Delivered	Low	Low	FALSE	FALSE	TRUE
LT02	0	100	0	Pending	Low	Low	FALSE	FALSE	TRUE
LT03	0	100	0	Cancelled	Low	Low	FALSE	FALSE	TRUE
LT04	1	99	99	Delivered	Low	Low	FALSE	FALSE	TRUE
LT05	1	99	99	Pending	Low	Low	FALSE	FALSE	TRUE
LT06	1	99	99	Cancelled	Low	Low	FALSE	FALSE	TRUE
LT07	1	100	100	Delivered	Medium	Medium	FALSE	FALSE	TRUE
LT08	1	100	100	Pending	Medium	Medium	FALSE	FALSE	TRUE
LT09	1	100	100	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
LT10	1	101	101	Delivered	Medium	Medium	FALSE	FALSE	TRUE
LT11	1	101	101	Pending	Medium	Medium	FALSE	FALSE	TRUE
LT12	1	101	101	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
LT13	1	199	199	Delivered	Medium	Medium	FALSE	FALSE	TRUE
LT14	1	199	199	Pending	Medium	Medium	FALSE	FALSE	TRUE
LT15	1	199	199	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
LT16	1	200	200	Delivered	High	High	FALSE	FALSE	TRUE
LT17	1	200	200	Pending	High	High	TRUE	TRUE	TRUE
LT18	1	200	200	Cancelled	High	High	FALSE	FALSE	TRUE
LT19	1	201	201	Delivered	High	High	FALSE	FALSE	TRUE
LT20	1	201	201	Pending	High	High	TRUE	TRUE	TRUE
LT21	1	201	201	Cancelled	High	High	FALSE	FALSE	TRUE

9. Reconcile the matrix controls in **A25:D34**: **21 cases /21 passes /0 mismatches**, two Pending High flags, three zero-quantity cases, **6 High /9 Medium /6 Low**. The two positive flag cases are synthetic Pending amounts200 and201; the maintained source still has0 qualifying records. No baseline row is changed to force a positive result.



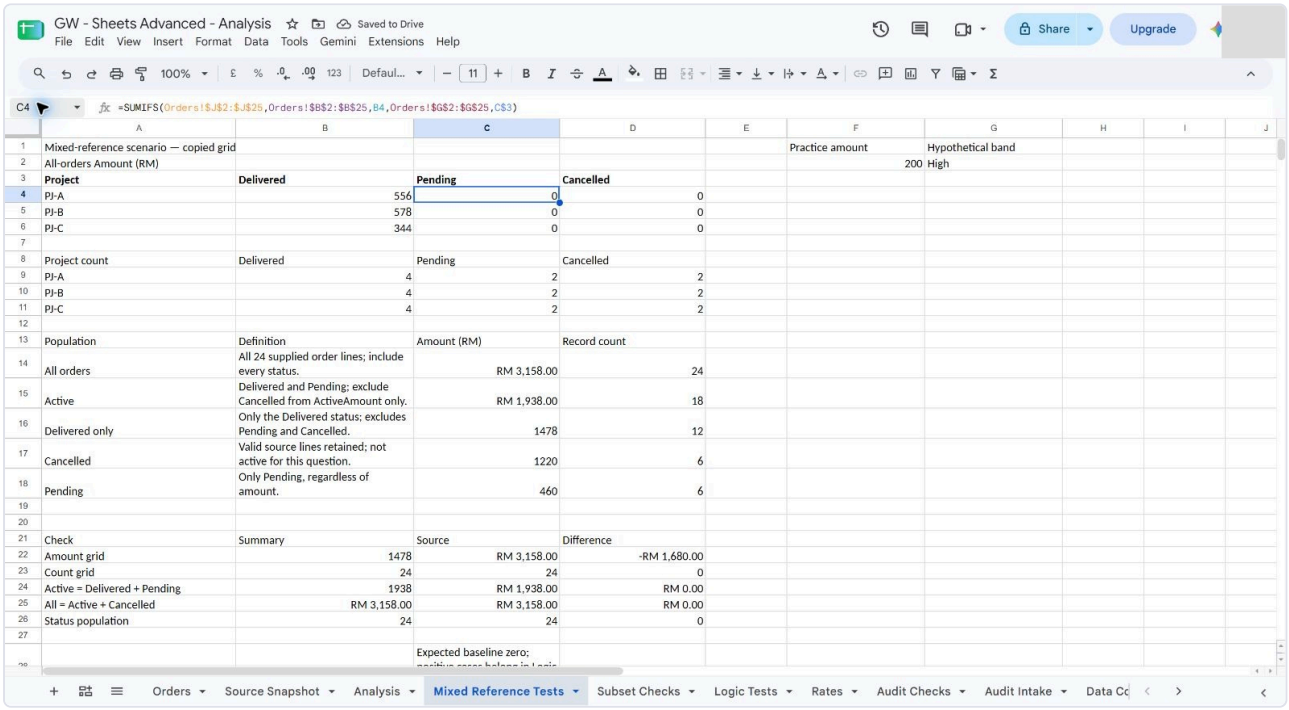
Check	Actual	Expected	Difference
Test cases	21	21	0
Passing cases	21	21	0
Mismatches	0	0	0
Pending High flags	2	2	0
Zero Qty cases	3	3	0
High bands	6	6	0
Medium bands	9	9	0
Low bands	6	6	0
Source Pending>=200	0	0	0

10. Duplicate the **Analysis** tab as **Mixed Reference Tests**, a named scenario copy. Record the fault prediction before editing: removing the column anchor from **$A4** will leave B4 at556, make C4 use numeric B4 as its project criterion and return0, and reduce the copied grid total to **1478**, a **-1680** discrepancy from3158. Add the copied-grid control **A37:D38** and prediction notes outside the working grid.



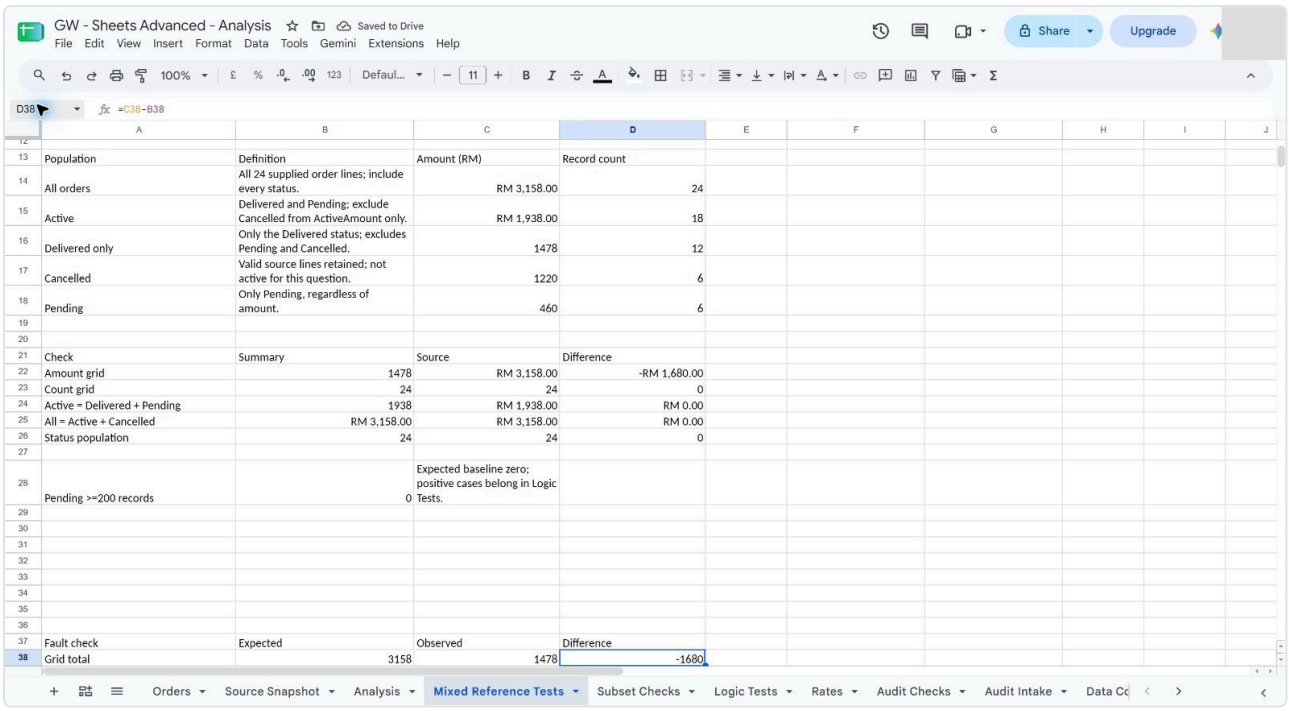
	A	B	C	D
21	Check	Summary	Source	Difference
22	Amount grid	3158	RM 3,158.00	RM 0.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				
30				
31				
32				
33				
34				
35				
36				
37	Fault check	Expected	Observed	Difference
38	Grid total	3158	3158	0
39				
40	Fault plan	Prediction before mutation		
41	Remove column anchor	Change B4 project criterion $A4 to A4, then fill across/down. B4 remains556; C4 reads numeric B4 as project and becomes0; D4 likewise0. Grid total becomes1478, difference-1680.		
42	Isolation	Only this copied tab is mutated. Maintained Analysis, Orders inputs, Rates and Source Snapshot remain intact.		

11. On the copied tab only, change B4's project criterion from **$A4** to **A4**. Fill right and down again. Inspect **C4**: the project criterion has shifted to **B4**, a numeric amount rather than PJ-A. The first incorrect intersection is **0**. Do not damage the maintained Analysis grid.



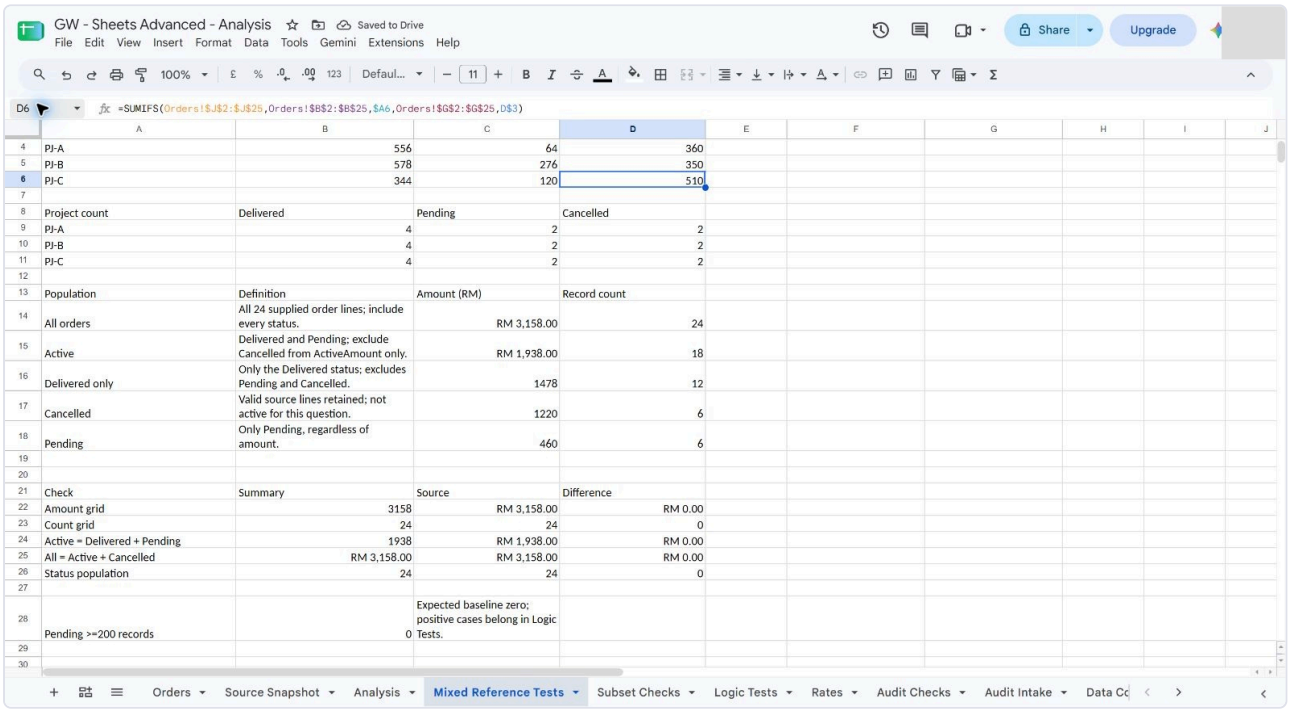
	A	B	C	D	E	F	G
1	Mixed-reference scenario — copied grid					Practice amount	Hypothetical band
2	All-orders Amount (RM)					200	High
3	**Project**	**Delivered**	**Pending**	**Cancelled**			
4	PJ-A	556	0	0			
5	PJ-B	578	0	0			
6	PJ-C	344	0	0			
7							
8	Project count	Delivered	Pending	Cancelled			
9	PJ-A	4	2	2			
10	PJ-B	4	2	2			
11	PJ-C	4	2	2			
12							
13	Population	Definition	Amount (RM)	Record count			
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24			
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18			
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12			
17	Cancelled	Valid source lines retained; not active for this question.	1220	6			
18	Pending	Only Pending, regardless of amount.	460	6			
19							
20							
21	Check	Summary	Source	Difference			
22	Amount grid	1478	RM 3,158.00	-RM 1,680.00			
23	Count grid	24	24	0			
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00			
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00			
26	Status population	24	24	0			
27							
28			Expected baseline zero;				

12. Inspect the copied-grid total control: expected3158, observed **1478**, difference **-1680**. The actual fault state was captured and read back before repair; a correct first cell alone did not prove that copying preserved the logic.



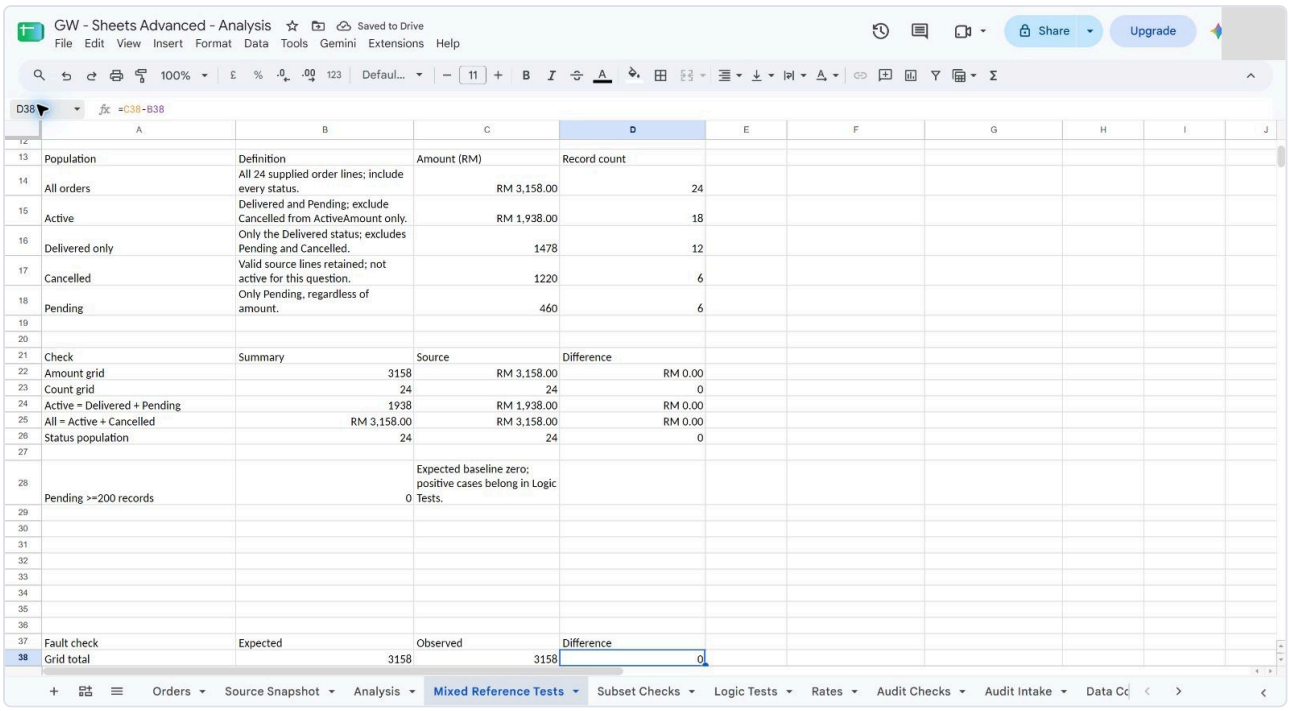
	A	B	C	D
13	Population	Definition	Amount (RM)	Record count
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12
17	Cancelled	Valid source lines retained; not active for this question.	1220	6
18	Pending	Only Pending, regardless of amount.	460	6
19				
20				
21	Check	Summary	Source	Difference
22	Amount grid	1478	RM 3,158.00	-RM 1,680.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				
30				
31				
32				
33				
34				
35				
36				
37	Fault check	Expected	Observed	Difference
38	Grid total	3158	1478	-1680

13. Restore **$A4** in copied B4 and refill right/down. Inspect the repaired D6 using **$A6 / D$3**, value510.



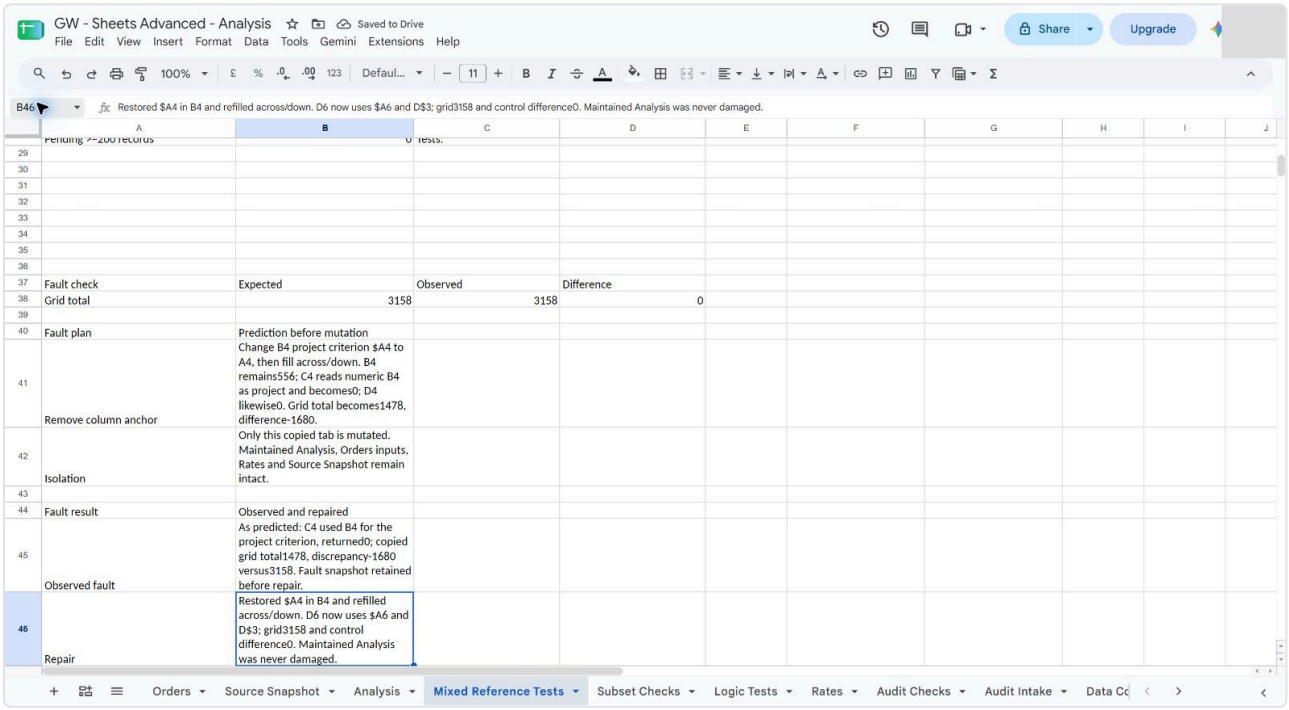
	A	B	C	D
4	PJ-A	556	64	360
5	PJ-B	578	276	350
6	PJ-C	344	120	510
7				
8	Project count	Delivered	Pending	Cancelled
9	PJ-A	4	2	2
10	PJ-B	4	2	2
11	PJ-C	4	2	2
12				
13	Population	Definition	Amount (RM)	Record count
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12
17	Cancelled	Valid source lines retained; not active for this question.	1220	6
18	Pending	Only Pending, regardless of amount.	460	6
19				
20				
21	Check	Summary	Source	Difference
22	Amount grid	3158	RM 3,158.00	RM 0.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				
30				

14. Recheck the copied total: **3158**, difference **0**. Keep the scenario clearly named as a repaired copy.



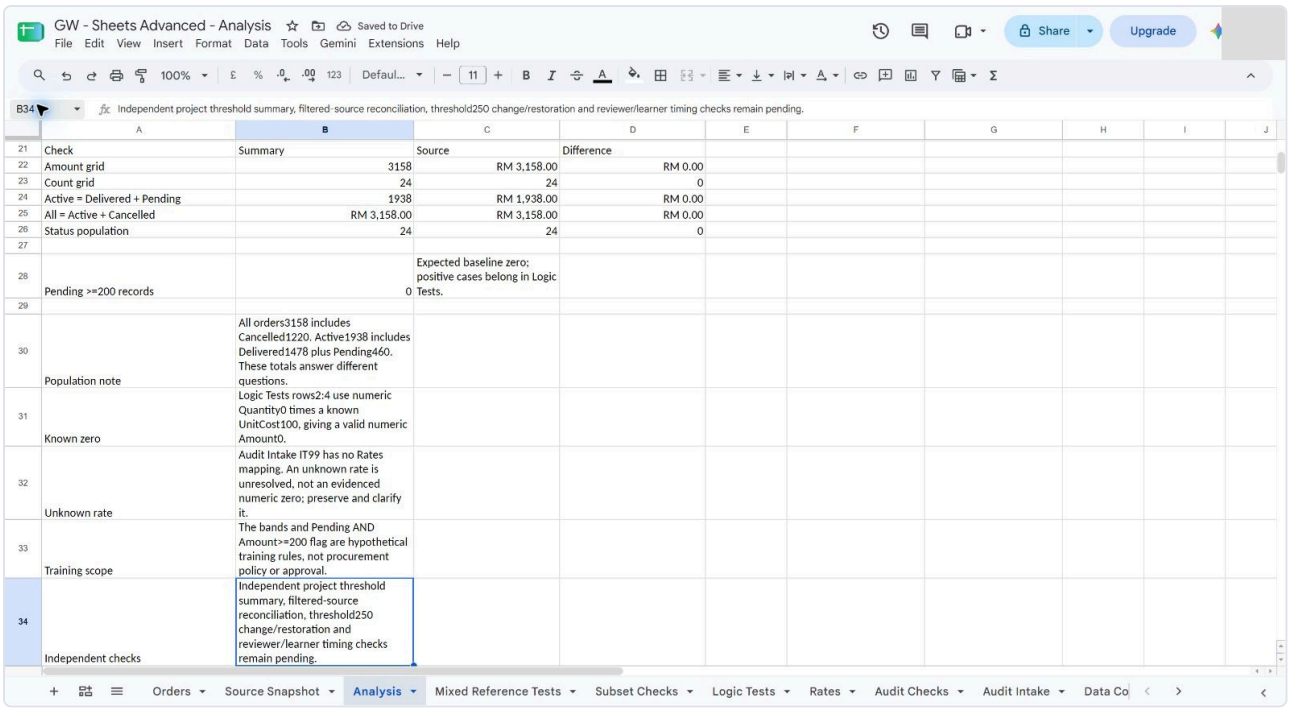
	A	B	C	D
13	Population	Definition	Amount (RM)	Record count
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12
17	Cancelled	Valid source lines retained; not active for this question.	1220	6
18	Pending	Only Pending, regardless of amount.	460	6
19				
20				
21	Check	Summary	Source	Difference
22	Amount grid	3158	RM 3,158.00	RM 0.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				
30				
31				
32				
33				
34				
35				
36				
37	Fault check	Expected	Observed	Difference
38	Grid total	3158	3158	0

15. Record the predicted error, observed criterion/value and repair in **A40:B46**. Label the tab's title cell **Mixed-reference scenario — repaired copy**. The maintained Analysis never required repair.



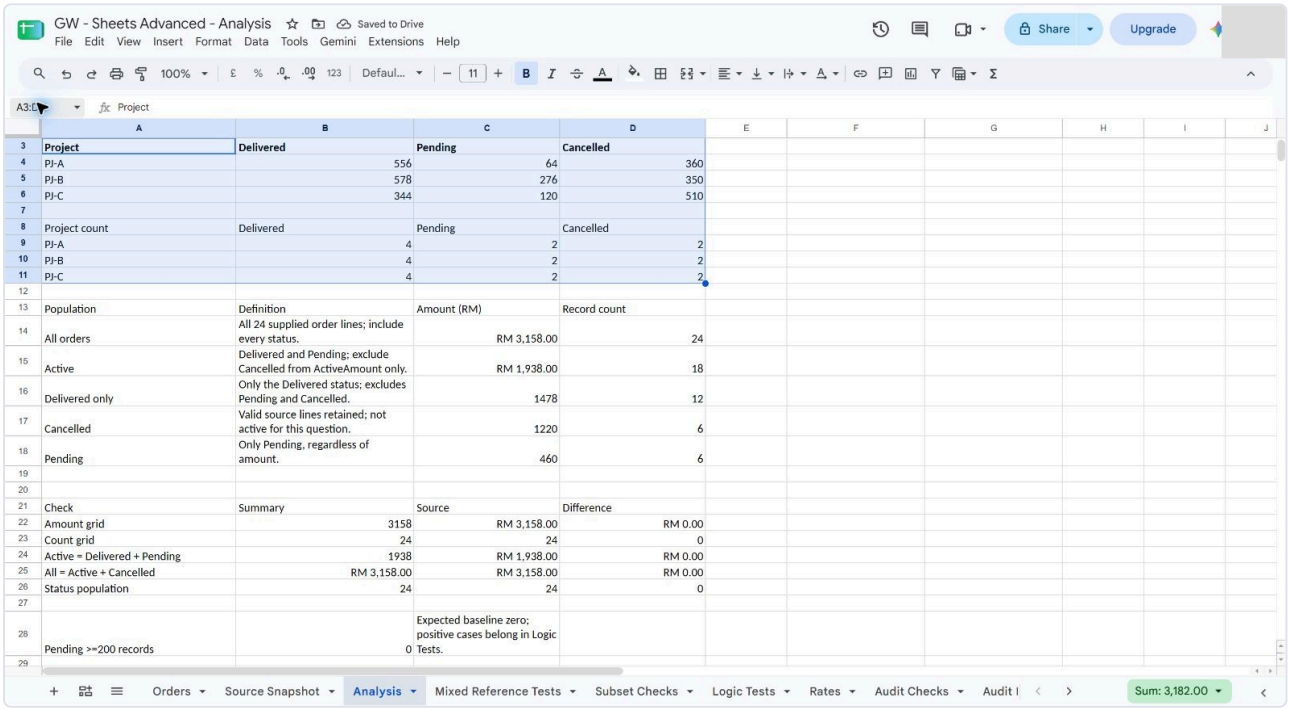
	A	B	C	D
37	Fault check	Expected	Observed	Difference
38	Grid total	3158	3158	0
39				
40	Fault plan	Prediction before mutation		
41	Remove column anchor	Change B4 project criterion $A4 to A4, then fill across/down. B4 remains556; C4 reads numeric B4 as project and becomes0; D4 likewise0. Grid total becomes1478, difference-1680.		
42	Isolation	Only this copied tab is mutated. Maintained Analysis, Orders inputs, Rates and Source Snapshot remain intact.		
43				
44	Fault result	Observed and repaired		
45	Observed fault	As predicted: C4 used B4 for the project criterion, returned0; copied grid total1478, discrepancy-1680 versus3158. Fault snapshot retained before repair.		
46	Repair	Restored $A4 in B4 and refilled across/down. D6 now uses $A6 and D$3; grid3158 and control difference0. Maintained Analysis was never damaged.		

16. On maintained Analysis, record why a **known numeric zero** differs from an unresolved **IT99** rate mapping, and why the all-orders, active and delivered-only totals answer different questions. Keep the training bands separate from source policy. Label the independent project threshold summary, filtered-source reconciliation and250/restoration challenge as pending.



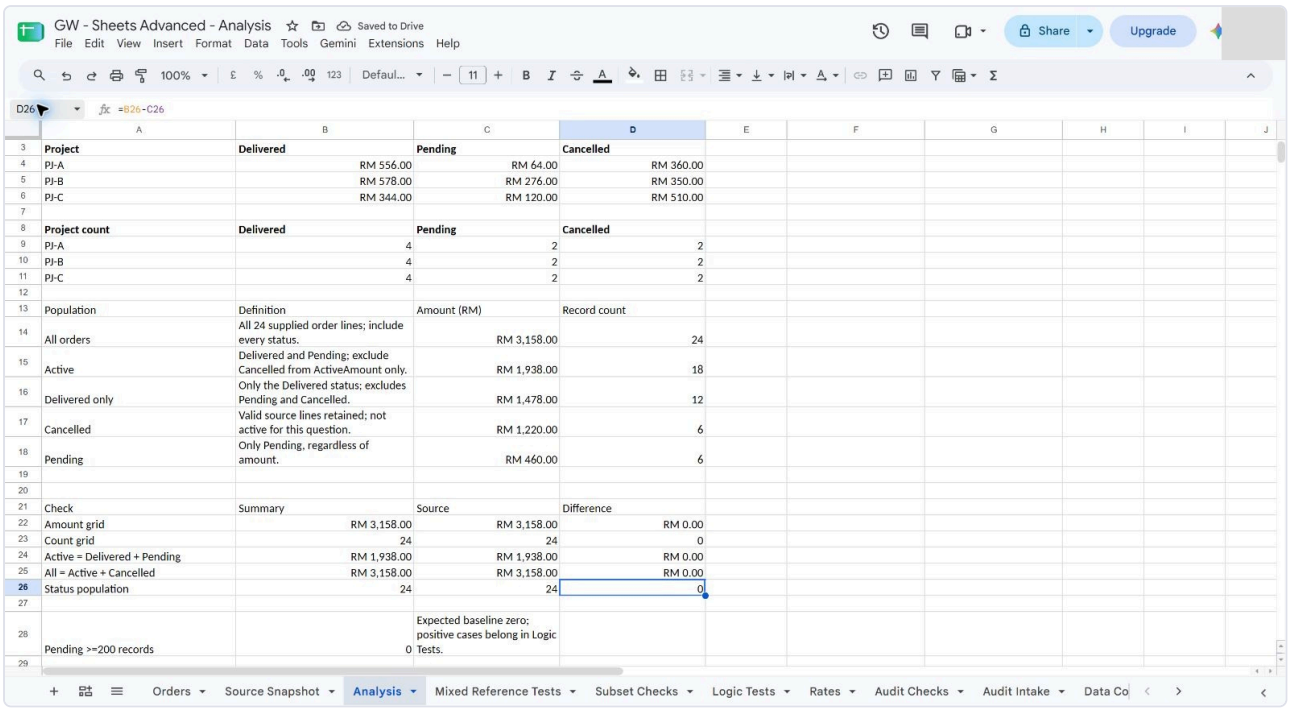
	A	B	C	D
21	Check	Summary	Source	Difference
22	Amount grid	3158	RM 3,158.00	RM 0.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				
30	Population note	All orders3158 includes Cancelled1220. Active1938 includes Delivered1478 plus Pending460. These totals answer different questions.		
31	Known zero	Logic Tests rows2:4 use numeric Quantity0 times a known UnitCost100, giving a valid numeric Amount0.		
32	Unknown rate	Audit Intake IT99 has no Rates mapping. An unknown rate is unresolved, not an evidenced numeric zero; preserve and clarify it.		
33	Training scope	The bands and Pending AND Amount>=200 flag are hypothetical training rules, not procurement policy or approval.		
34	Independent checks	Independent project threshold summary, filtered-source reconciliation, threshold250 change/restoration and reviewer/learner timing checks remain pending.		

17. Reinspect the maintained amount and count grids. The nine intersections still match the source, with amount total3158 and count24.

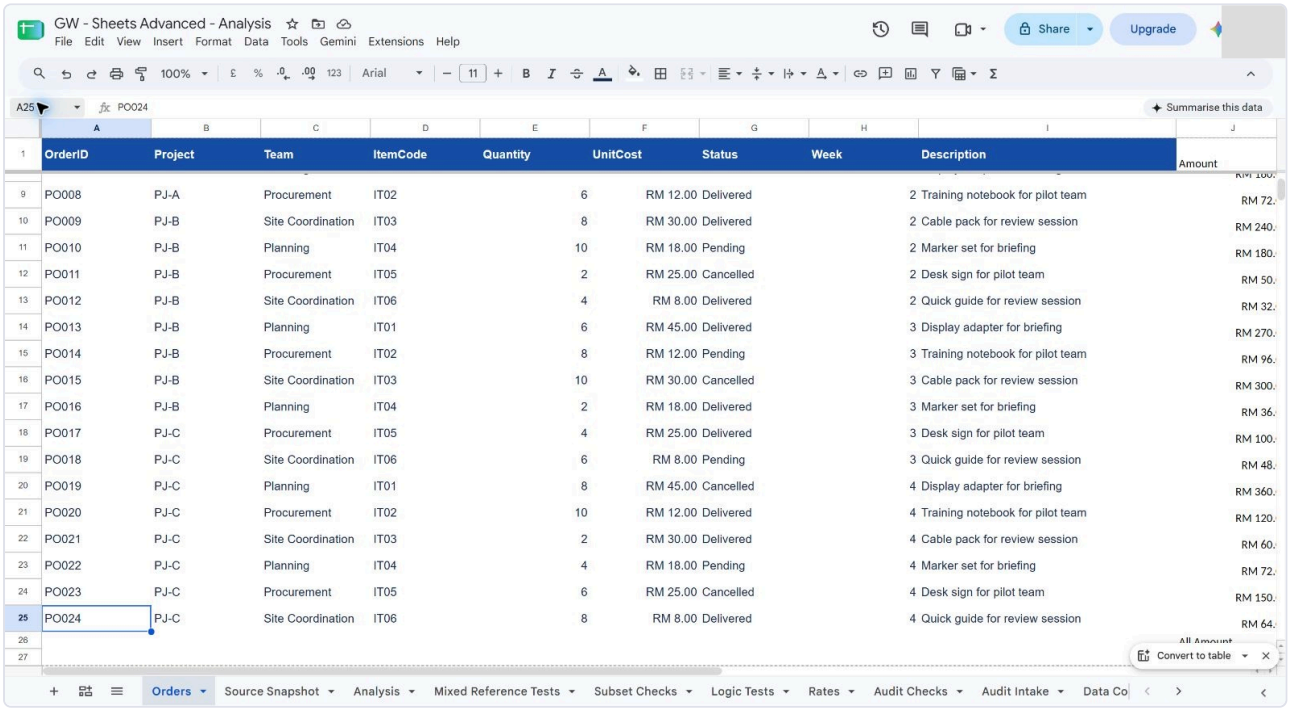


	A	B	C	D
3	**Project**	**Delivered**	**Pending**	**Cancelled**
4	PJ-A	556	64	360
5	PJ-B	578	276	350
6	PJ-C	344	120	510
7				
8	Project count	Delivered	Pending	Cancelled
9	PJ-A	4	2	2
10	PJ-B	4	2	2
11	PJ-C	4	2	2
12				
13	Population	Definition	Amount (RM)	Record count
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	1478	12
17	Cancelled	Valid source lines retained; not active for this question.	1220	6
18	Pending	Only Pending, regardless of amount.	460	6
19				
20				
21	Check	Summary	Source	Difference
22	Amount grid	3158	RM 3,158.00	RM 0.00
23	Count grid	24	24	0
24	Active = Delivered + Pending	1938	RM 1,938.00	RM 0.00
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00
26	Status population	24	24	0
27				
28	Pending >=200 records	0	Expected baseline zero; positive cases belong in Logic Tests.	
29				

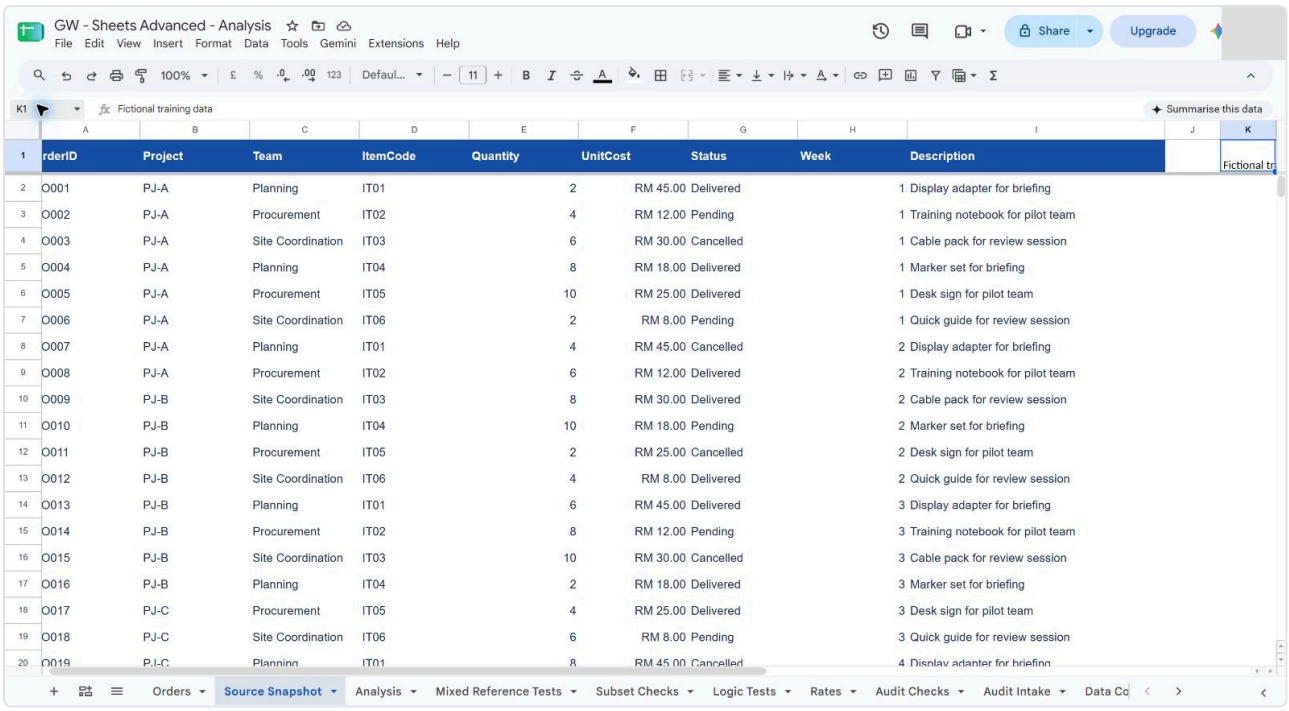
18. Finish readable amount formats and inspect all five zero-difference control rows. In the captured Analysis layout A/B/C/D are approximately240/220/180/180pixels, with wrapped definition and note cells; money formats are applied only to amount ranges, not the count grid.



19. Recheck Orders A:I through PO024. Original input values and formats, Rates, intake evidence and the training contract remain unchanged; only the new calculation columns, outside-table controls and preserved label location were added to Orders.

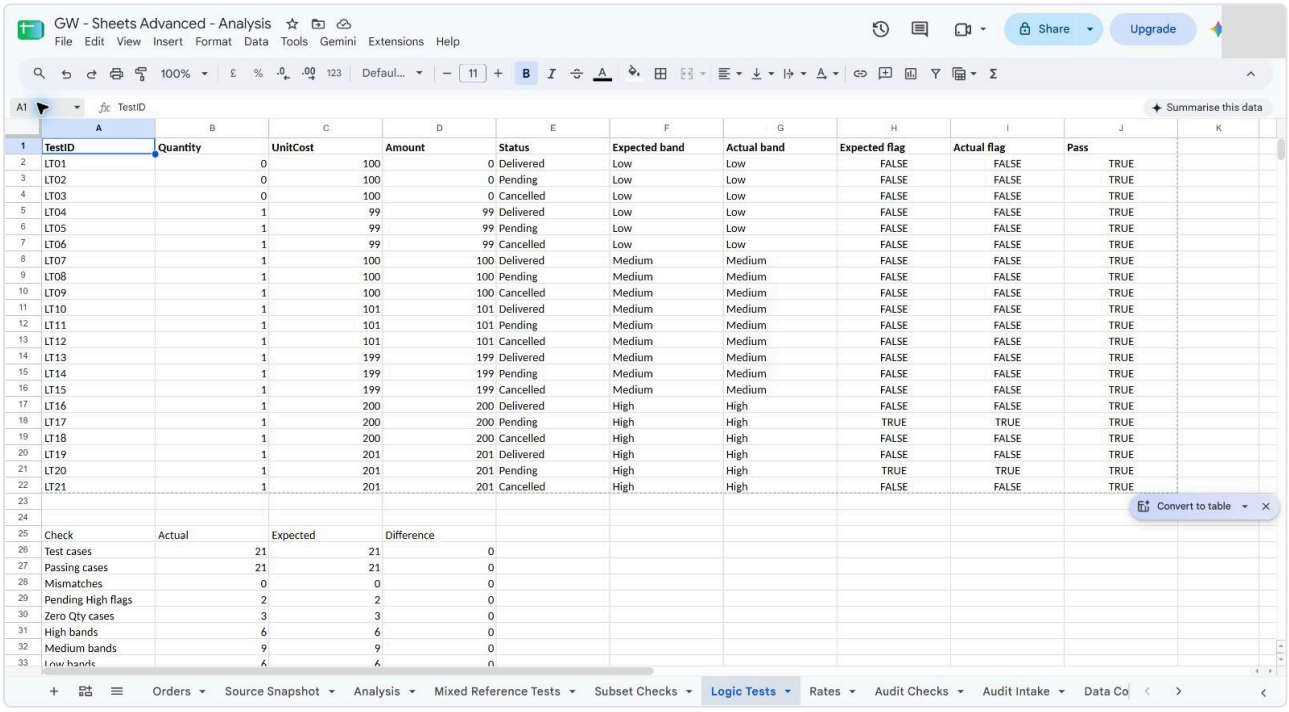


20. Inspect **Source Snapshot**. It retains the original source values and original K1 label, without the new calculation columns. It remains a comparison copy, not the maintained calculation source.



	rderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description
2	O001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter for briefing
3	O002	PJ-A	Procurement	IT02	4	RM 12.00	Pending	1	Training notebook for pilot team
4	O003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled	1	Cable pack for review session
5	O004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for briefing
6	O005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team
7	O006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session
8	O007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing
9	O008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team
10	O009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session
11	O010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing
12	O011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team
13	O012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session
14	O013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing
15	O014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team
16	O015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session
17	O016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing
18	O017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team
19	O018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session
20	O019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing

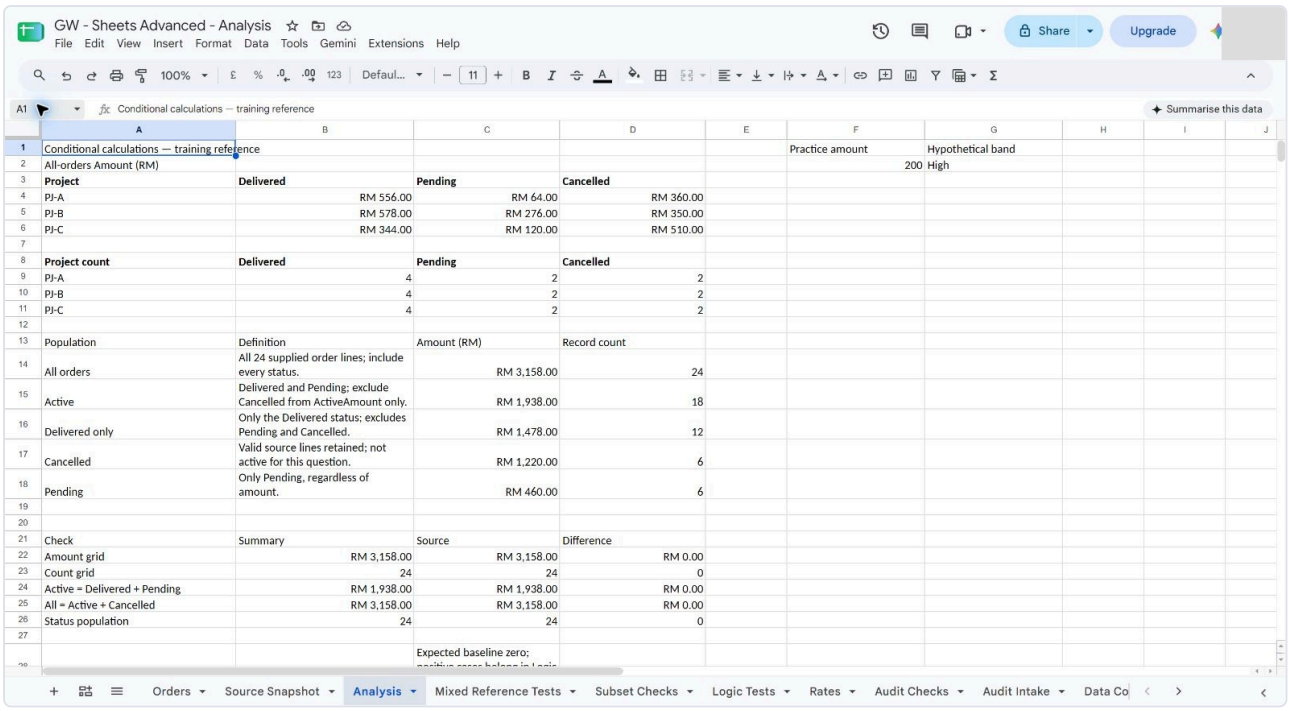
21. Leave all21 Logic Tests passing. Preserve the synthetic cases separately from Orders and the repaired reference scenario separately from Analysis.



	TestID	Quantity	UnitCost	Amount	Status	Expected band	Actual band	Expected flag	Actual flag	Pass
2	LT01	0	100	0	Delivered	Low	Low	FALSE	FALSE	TRUE
3	LT02	0	100	0	Pending	Low	Low	FALSE	FALSE	TRUE
4	LT03	0	100	0	Cancelled	Low	Low	FALSE	FALSE	TRUE
5	LT04	1	99	99	Delivered	Low	Low	FALSE	FALSE	TRUE
6	LT05	1	99	99	Pending	Low	Low	FALSE	FALSE	TRUE
7	LT06	1	99	99	Cancelled	Low	Low	FALSE	FALSE	TRUE
8	LT07	1	100	100	Delivered	Medium	Medium	FALSE	FALSE	TRUE
9	LT08	1	100	100	Pending	Medium	Medium	FALSE	FALSE	TRUE
10	LT09	1	100	100	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
11	LT10	1	101	101	Delivered	Medium	Medium	FALSE	FALSE	TRUE
12	LT11	1	101	101	Pending	Medium	Medium	FALSE	FALSE	TRUE
13	LT12	1	101	101	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
14	LT13	1	199	199	Delivered	Medium	Medium	FALSE	FALSE	TRUE
15	LT14	1	199	199	Pending	Medium	Medium	FALSE	FALSE	TRUE
16	LT15	1	199	199	Cancelled	Medium	Medium	FALSE	FALSE	TRUE
17	LT16	1	200	200	Delivered	High	High	FALSE	FALSE	TRUE
18	LT17	1	200	200	Pending	High	High	TRUE	TRUE	TRUE
19	LT18	1	200	200	Cancelled	High	High	FALSE	FALSE	TRUE
20	LT19	1	201	201	Delivered	High	High	FALSE	FALSE	TRUE
21	LT20	1	201	201	Pending	High	High	TRUE	TRUE	TRUE
22	LT21	1	201	201	Cancelled	High	High	FALSE	FALSE	TRUE

	Check	Actual	Expected	Difference
26	Test cases	21	21	0
27	Passing cases	21	21	0
28	Mismatches	0	0	0
29	Pending High flags	2	2	0
30	Zero Qty cases	3	3	0
31	High bands	6	6	0
32	Medium bands	9	9	0
33	Low bands	6	6	0

22. Leave the saved maintained **Analysis** ready for the next lab. Source inputs are preserved; hypothetical tests and historical audit notes are clearly scoped. Independent threshold work, reviewer checks and learner timing remain unfinished.



	A	B	C	D	E	F	G	H	I	J
1	Conditional calculations — training reference					Practice amount	Hypothetical band			
2	All-orders Amount (RM)					200	High			
3	**Project**	**Delivered**	**Pending**	**Cancelled**						
4	PJ-A	RM 556.00	RM 64.00	RM 360.00						
5	PJ-B	RM 578.00	RM 276.00	RM 350.00						
6	PJ-C	RM 344.00	RM 120.00	RM 510.00						
7										
8	**Project count**	**Delivered**	**Pending**	**Cancelled**						
9	PJ-A	4	2	2						
10	PJ-B	4	2	2						
11	PJ-C	4	2	2						
12										
13	Population	Definition	Amount (RM)	Record count						
14	All orders	All 24 supplied order lines; include every status.	RM 3,158.00	24						
15	Active	Delivered and Pending; exclude Cancelled from ActiveAmount only.	RM 1,938.00	18						
16	Delivered only	Only the Delivered status; excludes Pending and Cancelled.	RM 1,478.00	12						
17	Cancelled	Valid source lines retained; not active for this question.	RM 1,220.00	6						
18	Pending	Only Pending, regardless of amount.	RM 460.00	6						
19										
20										
21	Check	Summary	Source	Difference						
22	Amount grid	RM 3,158.00	RM 3,158.00	RM 0.00						
23	Count grid	24	24	0						
24	Active = Delivered + Pending	RM 1,938.00	RM 1,938.00	RM 0.00						
25	All = Active + Cancelled	RM 3,158.00	RM 3,158.00	RM 0.00						
26	Status population	24	24	0						
27										
28			Expected baseline zero;							

> **Validation boundary:** guided2.1–2.3 and author casework2.4 are verified through actual learner UI and bounded read-only native-cell evidence. The independent project/threshold summary, filtered-source reconciliation,250/restoration test and human checks were not performed. Planned75-minute classroom timing remains unmeasured.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Build a second summary answering count and amount for Pending orders of at least 200 by project. Keep threshold logic explicit.
2. Use a filtered source list as an independent reconciliation. A zero-match baseline is a valid result; create a separate synthetic qualifying case and a non-qualifying boundary case to test the rule.
3. Change the threshold to 250 in a labelled assumption cell, repair references so the grid follows it, then restore 200 and verify restoration.

Evidence to Hand Over

- Reconciled count/amount grids for clearly defined populations.
- Expected-versus-actual boundary matrix and repaired mixed-reference defect.
- Independent threshold summary with supporting OrderIDs and a restored assumption.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check

Correct totals conceal the wrong population	Reconcile counts and contributing IDs as well as the monetary total.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Visible row calculations and explicit criteria make advanced logic checkable.

Check yourself: Which references in the summary grid must stay fixed when copied across and down?

Zero-Match Check

The supplied baseline has no Pending record with Amount at least 200. Record that result explicitly. Use a separately labelled synthetic case to prove the positive branch; do not edit baseline records just to obtain a highlight.

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Amount
```

```
=E2*F2
```

```
ActiveAmount
```

```
=IF(G2="Cancelled",0,J2)
```

```
=SUMIFS(Orders!$J$2:$J$25,Orders!$B$2:$B$25,$A4,Orders!$G$2:$G$25,B$3)
```

```
=IF(F2<100,"Low",IF(F2<200,"Medium","High"))
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Logic matrix must cover below/on/above thresholds, zero quantity, Delivered, Pending and
Cancelled. Practice bands: Low <100; Medium 100 to <200; High ⩾200. Exception: Pending AND
Amount ⩾200. These are training rules, not procurement policy.
```

Captured Conditional-Calculation Reference

Preserve the supplied K1 fictional-data label in M1 before using K for ActiveAmount. Original A:I inputs, Rates, Source Snapshot and intake audit remain intact. J2=E2*F2 and K2=IF(G2="Cancelled",0,J2), filled to25; source totals3158 and1938. Spot checks J2/J13/J25 are90/32/64.

Amount grid B4:D6:556/64/360;578/276/350;344/120/510. D6 uses $A6 and D$3. Count grid B9:D11:4/2/2 for each project. B9 seed:

```
=COUNTIFS(Orders!$B$2:$B$25,$A9,Orders!$G$2:$G$25,B$8)
```

Population control card — Analysis A13:D18

Write each population definition before calculating the final two columns.

```
Population      Definition      Amount (RM)     Record count
All orders      All 24 supplied order lines; include every status.      =SUM(Orders!J2:J25)
=COUNTA(Orders!A2:A25)
Active  Delivered and Pending; exclude Cancelled from ActiveAmount only.
=SUM(Orders!K2:K25)     =COUNTIF(Orders!G2:G25,"<>Cancelled")
Delivered only  Only the Delivered status; excludes Pending and Cancelled.
=SUMIF(Orders!G2:G25,"Delivered",Orders!J2:J25) =COUNTIF(Orders!G2:G25,"Delivered")
Cancelled       Valid source lines retained; not active for this question.
=SUMIF(Orders!G2:G25,"Cancelled",Orders!J2:J25) =COUNTIF(Orders!G2:G25,"Cancelled")
Pending Only Pending, regardless of amount.     =SUMIF(Orders!G2:G25,"Pending",Orders!J2:J25)
=COUNTIF(Orders!G2:G25,"Pending")
```

Reconciliation card — Analysis A21:D26

```
Check   Summary Source  Difference
Amount grid     =SUM(B4:D6)     =C14    =B22-C22
Count grid      =SUM(B9:D11)    =D14    =B23-C23
Active = Delivered + Pending    =C16+C18        =C15    =B24-C24
All = Active + Cancelled        =C15+C17        =C14    =B25-C25
Status population       =SUM(D16:D18)   =D14    =B26-C26
```

B28 baseline check **=COUNTIFS(Orders!G2:G25,"Pending",Orders!J2:J25,">=200")** gives0. All3158/24, Active1938/18, Delivered1478/12, Cancelled1220/6, Pending460/6. Every reconciliation difference0.

Manual subset inputs — Subset Checks A1:H7

Record the six manually checked products before G/H formula comparisons. Inputs match the named source OrderIDs; all amounts use training RM.

```
Project Status  OrderID Quantity        UnitCost        Manual Amount   Formula Amount
Difference
PJ-A    Delivered       PO001   2       45      90
PJ-A    Delivered       PO004   8       18      144
PJ-A    Delivered       PO005   10      25      250
PJ-A    Delivered       PO008   6       12      72
PJ-B    Pending PO010   10      18      180
PJ-B    Pending PO014   8       12      96
```

G2 **=D2*E2**, H2 **=G2-F2**, fill to7. The two subsets total556 and276; their counts4 and2 also match the maintained grids.

Logic matrix input/expectations — Logic Tests A1:J22

Actual columns G/I/J are deliberately empty on this input card. D calculates Quantity times a known UnitCost. Zero Quantity is numeric0; it is not an unresolved rate lookup. Expected flags are Boolean TRUE/FALSE.

```
TestID  Quantity        UnitCost        Amount  Status  Expected band   Actual band
Expected flag   Actual flag     Pass
LT01    0       100     =B2*C2  Delivered       Low             FALSE
LT02    0       100     =B3*C3  Pending Low             FALSE
LT03    0       100     =B4*C4  Cancelled       Low             FALSE
LT04    1       99      =B5*C5  Delivered       Low             FALSE
LT05    1       99      =B6*C6  Pending Low             FALSE
```

```
LT06    1       99      =B7*C7  Cancelled       Low             FALSE
LT07    1       100     =B8*C8  Delivered       Medium          FALSE
LT08    1       100     =B9*C9  Pending Medium          FALSE
LT09    1       100     =B10*C10        Cancelled       Medium          FALSE
LT10    1       101     =B11*C11        Delivered       Medium          FALSE
LT11    1       101     =B12*C12        Pending Medium          FALSE
LT12    1       101     =B13*C13        Cancelled       Medium          FALSE
LT13    1       199     =B14*C14        Delivered       Medium          FALSE
LT14    1       199     =B15*C15        Pending Medium          FALSE
LT15    1       199     =B16*C16        Cancelled       Medium          FALSE
LT16    1       200     =B17*C17        Delivered       High            FALSE
LT17    1       200     =B18*C18        Pending High            TRUE
LT18    1       200     =B19*C19        Cancelled       High            FALSE
LT19    1       201     =B20*C20        Delivered       High            FALSE
LT20    1       201     =B21*C21        Pending High            TRUE
LT21    1       201     =B22*C22        Cancelled       High            FALSE
```

G2 **=IF(D2<100,"Low",IF(D2<200,"Medium","High"))**; I2 **=AND(E2="Pending",D2>=200)**; J2 **=AND(F2=G2,H2=I2)**. Fill each to22. Expected21passes/0mismatches,2positiveflags,3zero-quantitycases,6High/9Medium/6Low. Synthetic Pending200/201 exercise the positive flag branch; baseline remains0.

Repaired mixed-reference scenario

Duplicate Analysis as Mixed Reference Tests. On that copy only, remove the column anchor from B4's $A4 criterion, then refill across/down. C4 incorrectly uses B4 as project criterion and gives0; total1478, discrepancy-1680 from3158. Restore $A4 and refill: grid3158, difference0; D6 uses $A6 and D$3. The actual fault snapshot was retained before repair. The maintained Analysis grid was never damaged.

Independent project threshold summary, filtered-source reconciliation, threshold250 change/restoration and reviewer/learner-timing checks remain pending. Practice bands and flags are fictional training rules, not procurement policy.

03

CHAPTER 3

Validate Entries and Flag Exceptions

In this chapter you will use dropdowns, custom validation and formula-based conditional formatting.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 65 minutes: 15 explanation/demonstration, 22 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 50 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Validation rules and a tested pending-order exception highlight.

> **Status:** author reference with39 staged application captures through3.4. Separate practice copy, strict categorical/numeric rules, actual rejection behavior, ten input observations including blank clearing, five exception cases, first/final-row coverage and flag/highlight/filtered-ID agreement checked. Temporary values restored; maintained source preserved. Independent appended-row repair, bulk-paste/CSV numeric-text behavior, reviewer and learner timing remain pending.

What You Will Learn

- Validate categorical and numeric inputs.
- Highlight exceptions using a rule.
- Test an invalid and boundary value.

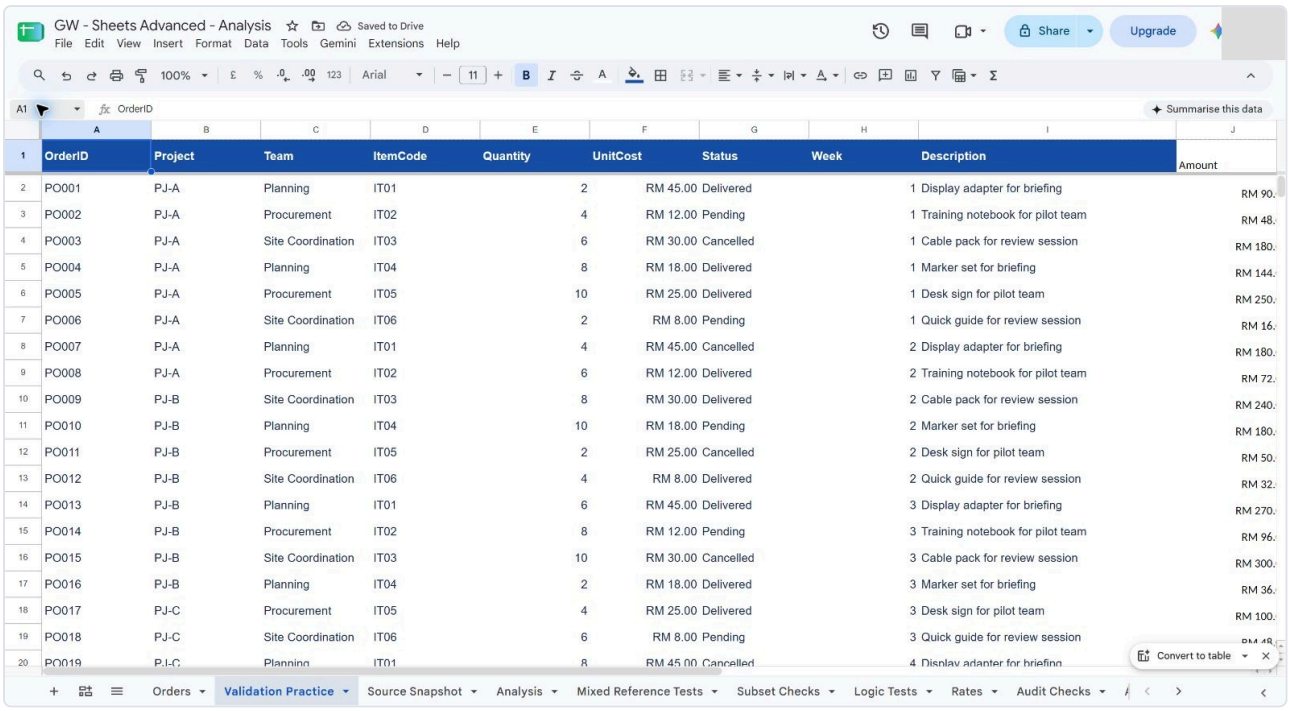
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use a practice copy of Orders. Restore all temporary tests before completing the chapter.

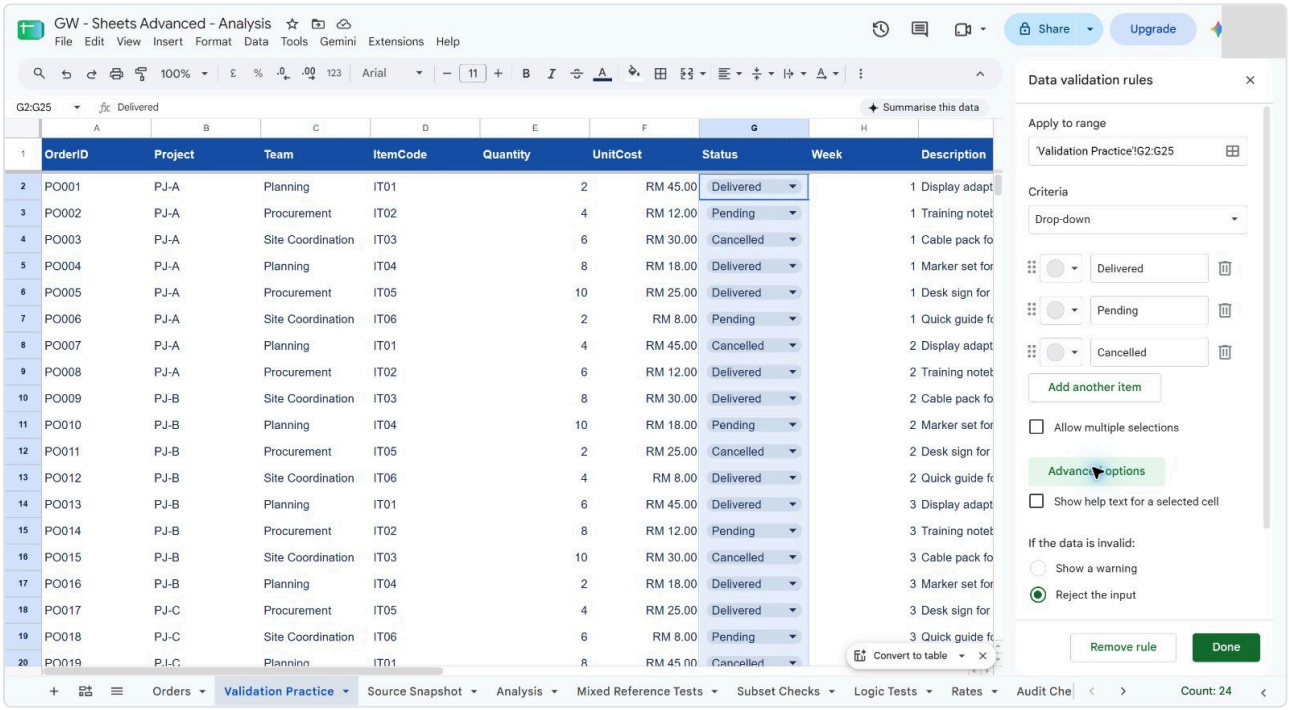
3.1 Validate New Entries

1. Duplicate **Orders** as **Validation Practice** in the existing Advanced workbook. Verify the separate tab name before editing. It retains the24 original records, Amount/ActiveAmount formulas and outside-table totals. Maintain Orders, Analysis, Rates and Source Snapshot unchanged.

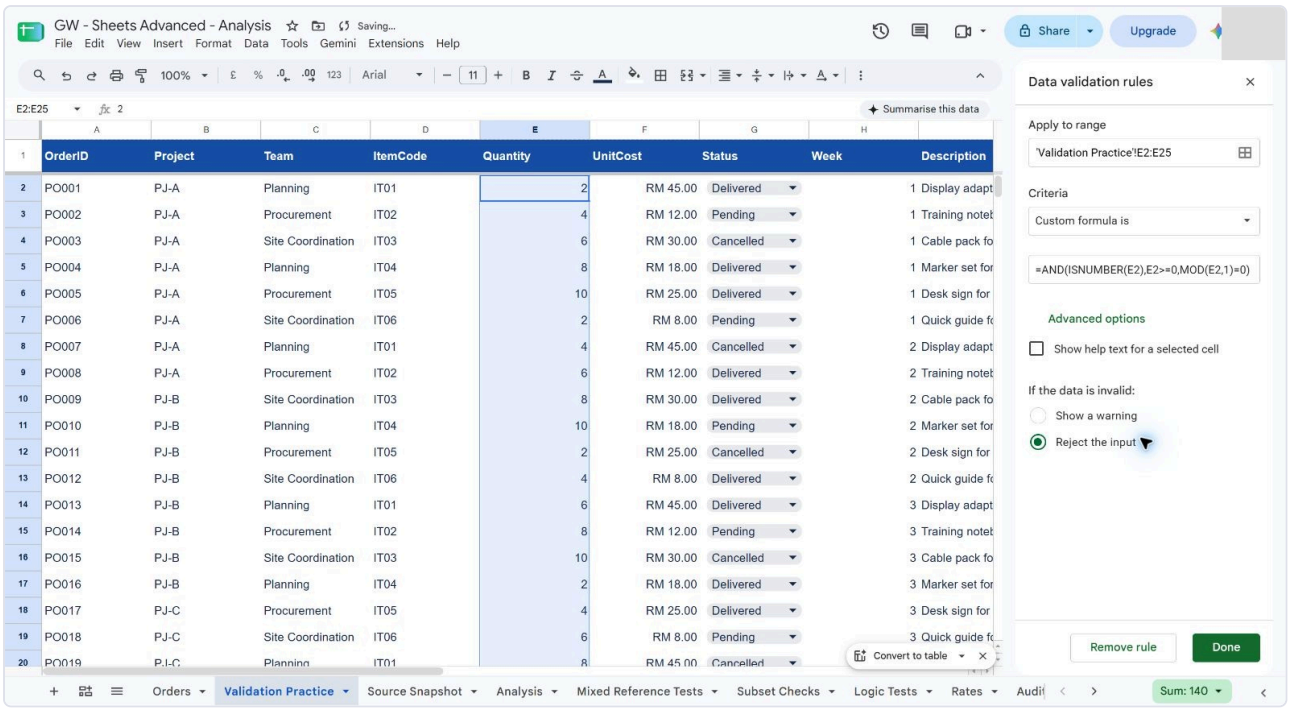


OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount
PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter for briefing	RM 90.
PO002	PJ-A	Procurement	IT02	4	RM 12.00	Pending	1	Training notebook for pilot team	RM 48.
PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled	1	Cable pack for review session	RM 180.
PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for briefing	RM 144.
PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team	RM 250.
PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session	RM 16.
PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing	RM 180.
PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team	RM 72.
PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session	RM 240.
PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing	RM 180.
PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team	RM 50.
PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session	RM 32.
PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing	RM 270.
PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team	RM 96.
PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session	RM 300.
PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing	RM 36.
PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team	RM 100.
PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session	
PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing	

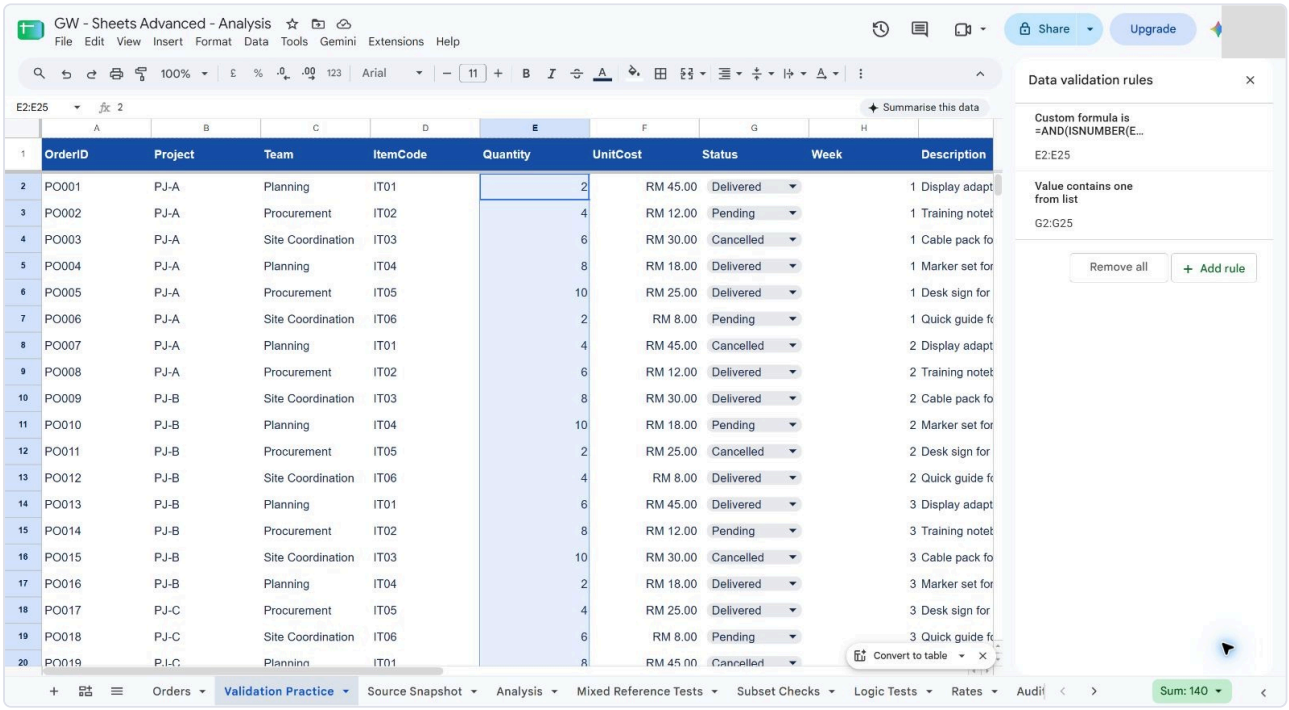
2. Select **G2:G25**, open **Data > Data validation** and add a rule. Use a single-choice **Drop-down** with **Delivered / Pending / Cancelled**. Inspect **Advanced options > Reject the input**, then save with **Done**. The canonical set matters; auto-populated order may differ.



3. Select **E2:E25** and add a separate rule. Choose **Custom formula is** and enter **=AND(ISNUMBER(E2),E2>=0,MOD(E2,1)=0)**. Set **Reject the input** explicitly: this custom-rule editor defaulted to **Show a warning** in the author account. Save with Done. The row-relative E2 corresponds to the first cell of the apply-to range. Numeric zero is permitted; do not substitute a positive-only restriction.

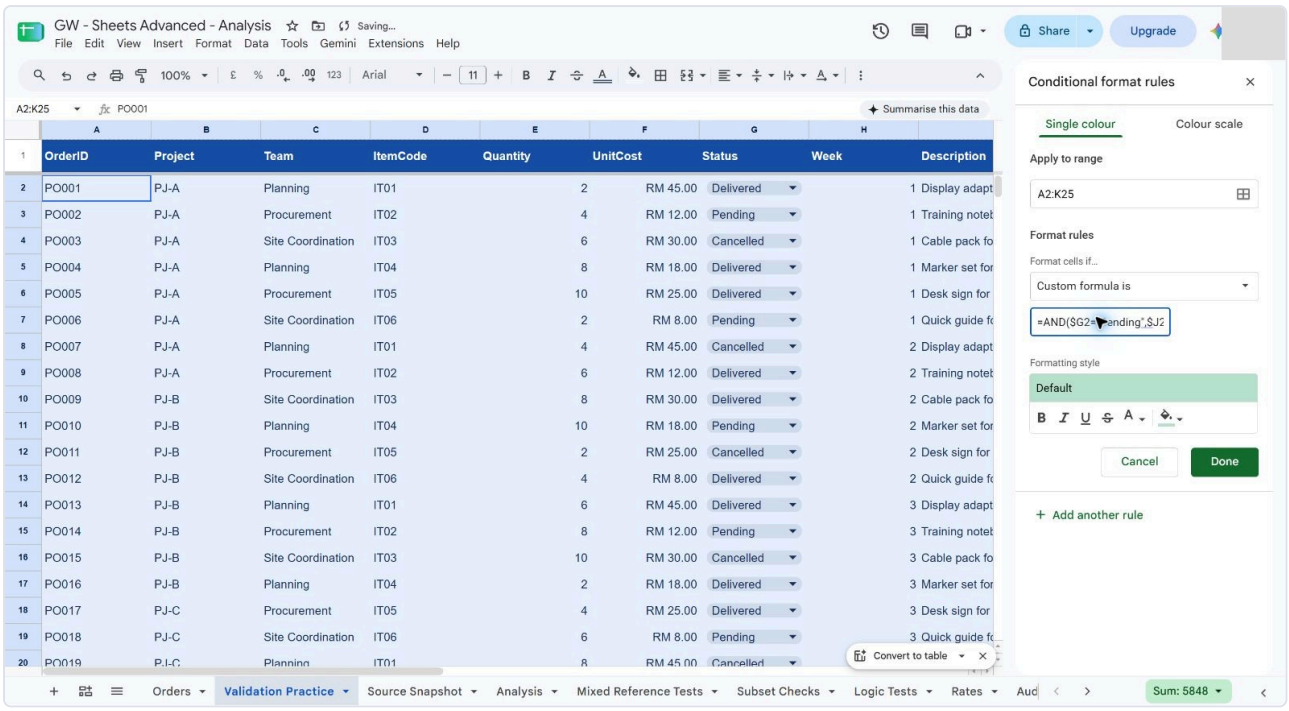


4. Inspect the saved rules for **E2:E25 / G2:G25**. Native readback later confirms strict rules on every cell through row25, with the quantity reference adjusted per row. A visible dropdown or coloured cell alone does not prove rejection behavior.

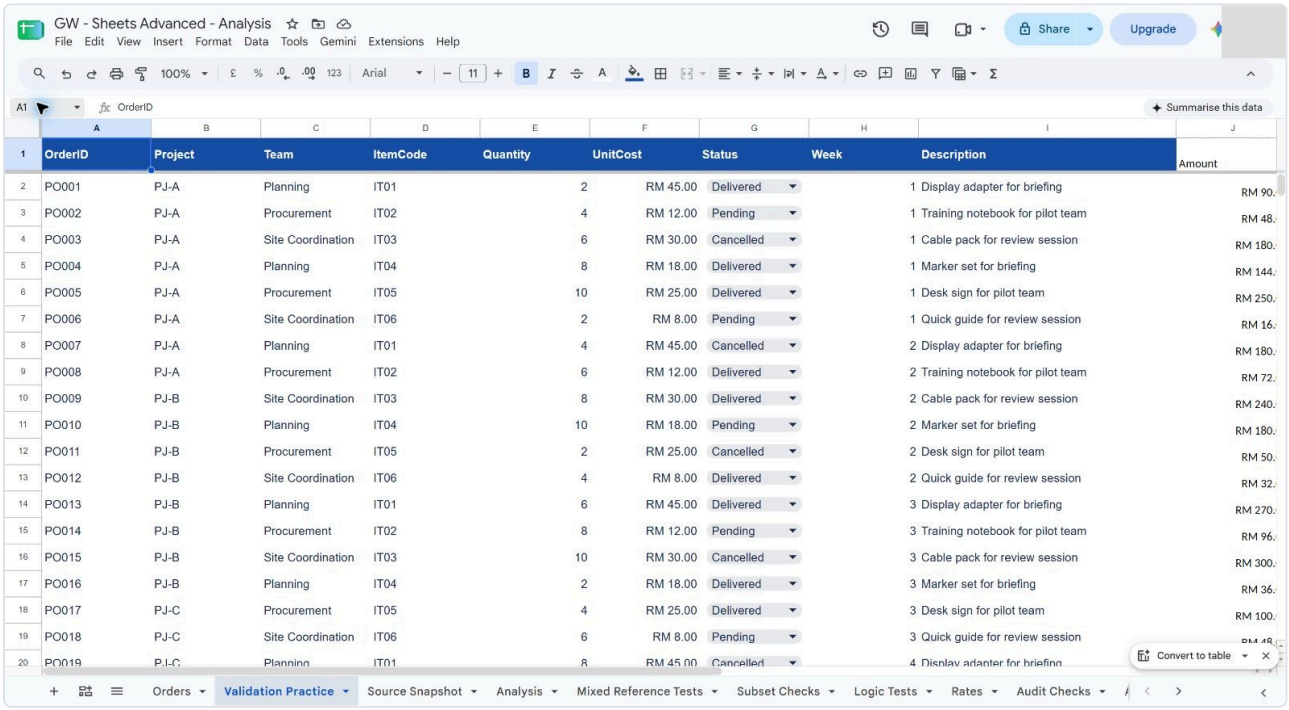


3.2 Flag Pending High Amounts

1. Select **A2:K25**, open **Format > Conditional formatting**, choose **Custom formula is**, and enter **=AND($G2="Pending",$J2>=200)**. Use a readable light fill; the captured rule uses **#b7e1cd**. Save with Done. Columns G/J are fixed while the row follows each record, starting at source row2.

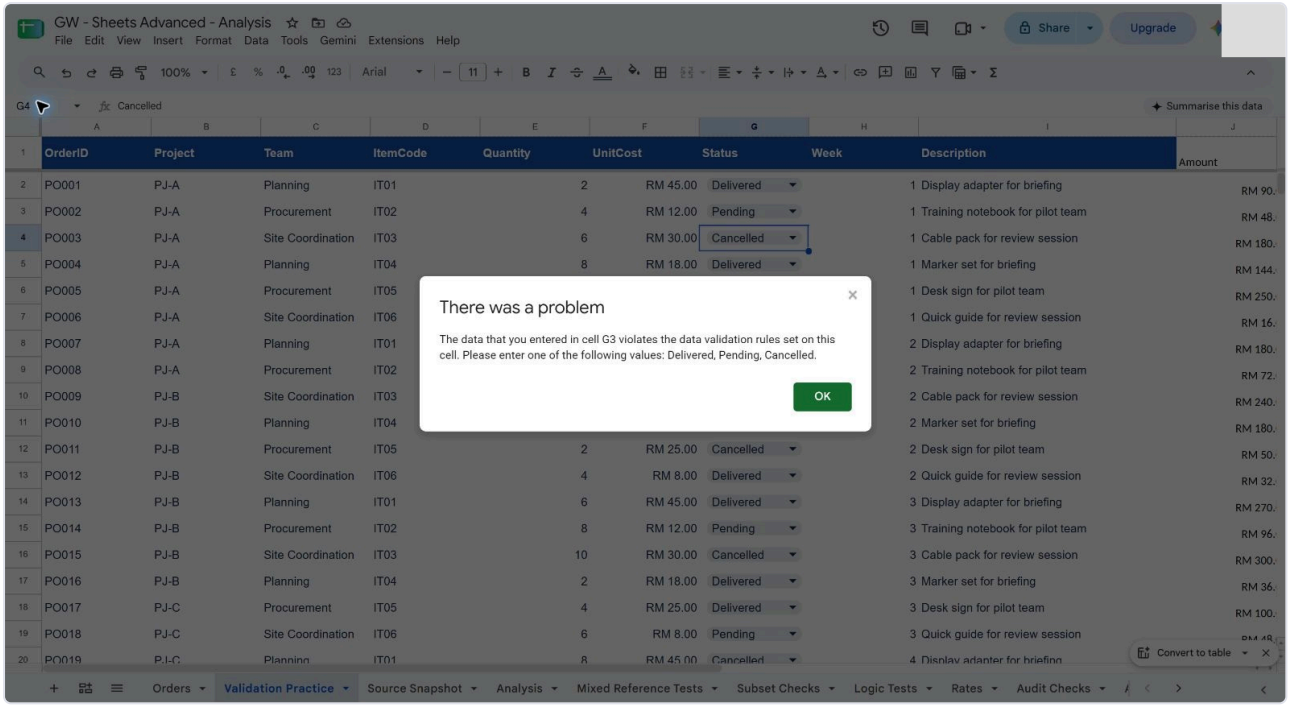


2. Close the rule sidebar and inspect the baseline. It contains **no Pending record with Amount at least200**, so no row qualifies. Keep this valid zero-match result; do not change the maintained source merely to obtain a highlight.

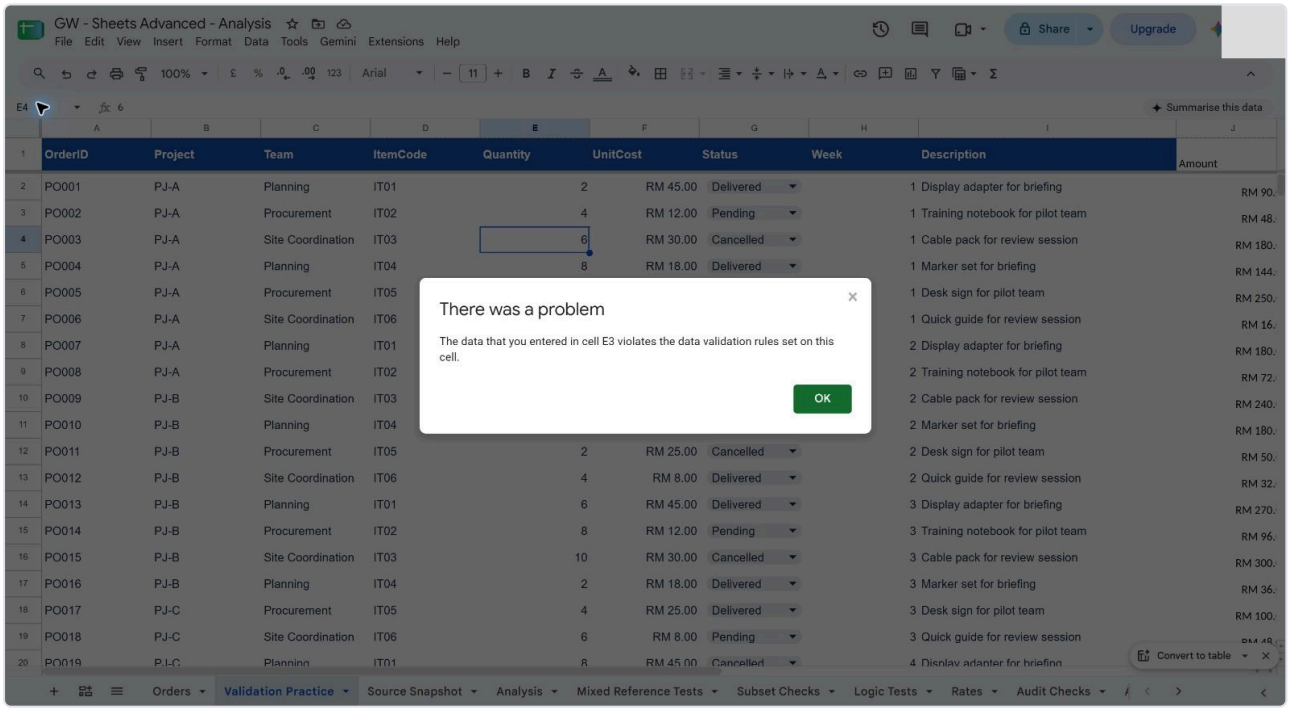


3.3 Test the Rules

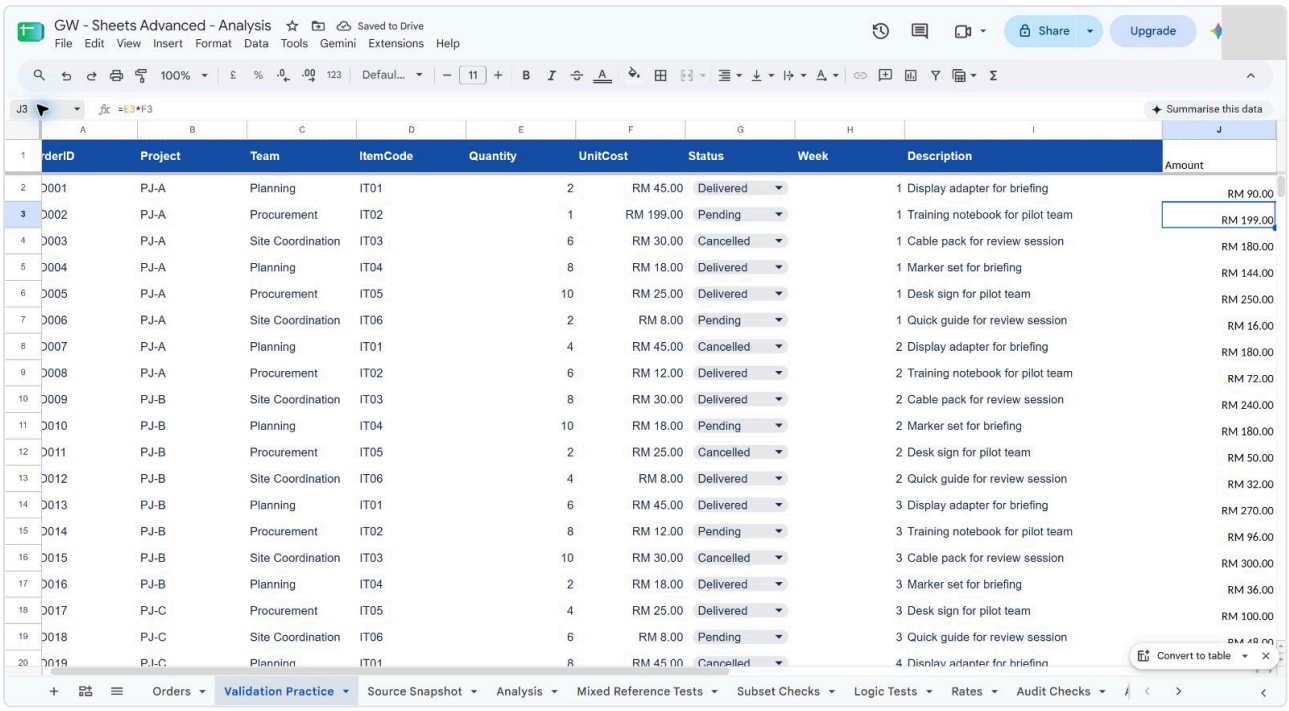
1. On the practice copy, edit **G3** to **Ready**. The actual native dialog rejects it and lists the three allowed statuses. Dismiss **OK**, then **Escape** from editing so Pending is retained. The input method in this walkthrough is cell-edit mode (F2), pasted text inside the editor, then Enter; this is not a bulk-range paste test.



2. Try **1.5** in E3 using the same method. The native validation dialog rejects it. Dismiss and exit editing; original Quantity4 remains. A rejected value was not saved, so do not blindly undo the newly configured rule.

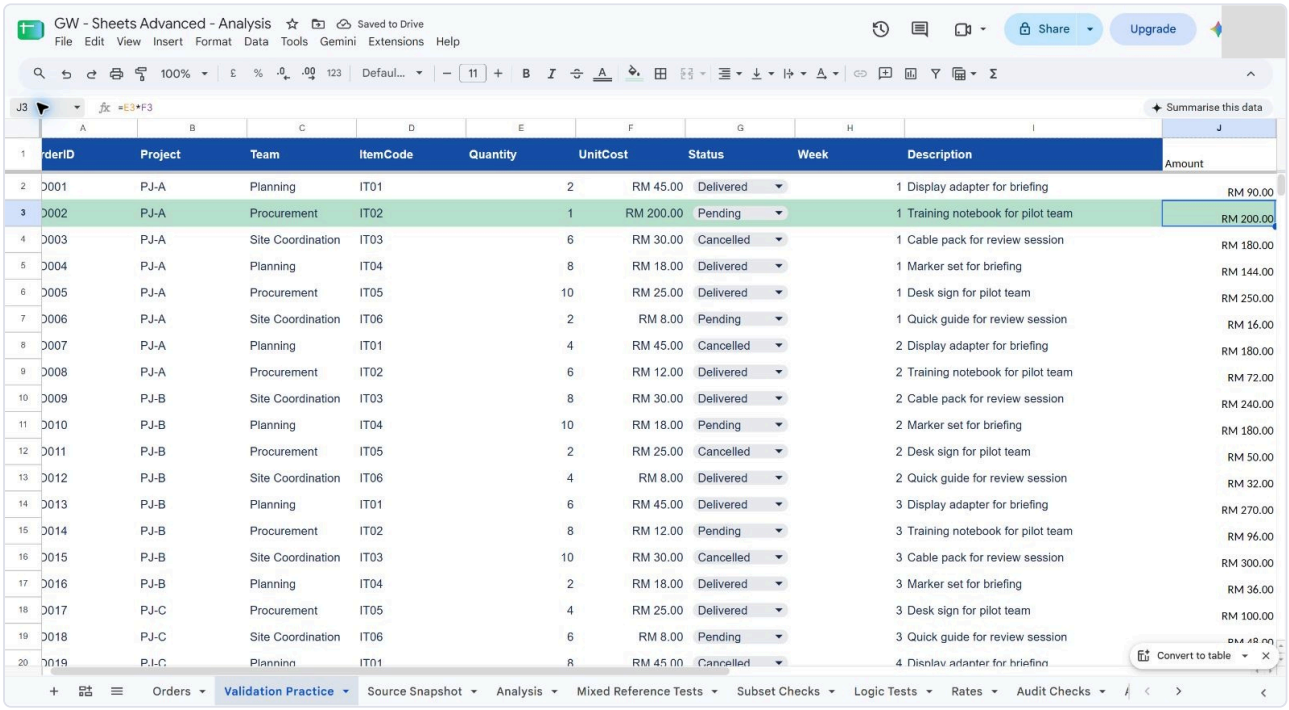


3. In the practice row only, set **E3=1 / F3=199**, keeping Pending. J3 is199 and the row is not highlighted.



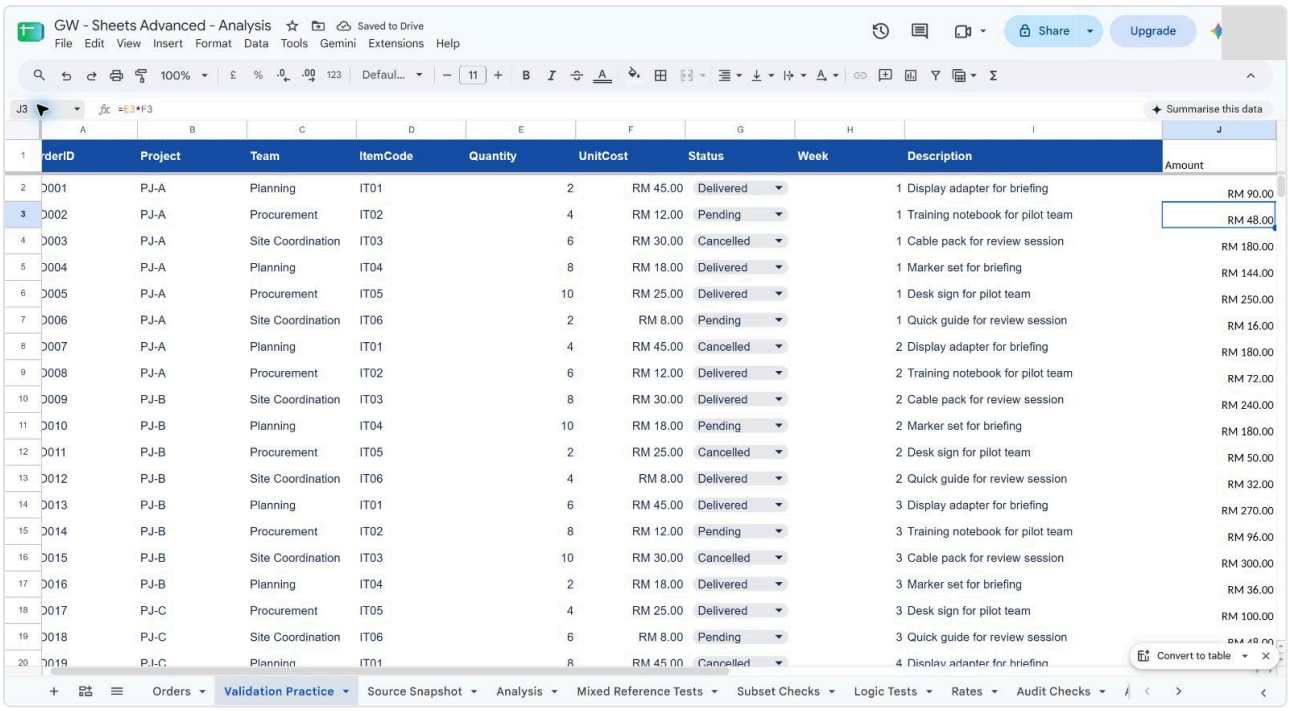
	rderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount
2	0001	PJ-A	Planning	IT01	2	RM 45.00	Delivered		1 Display adapter for briefing	RM 90.00
3	0002	PJ-A	Procurement	IT02	1	RM 199.00	Pending		1 Training notebook for pilot team	RM 199.00
4	0003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled		1 Cable pack for review session	RM 180.00
5	0004	PJ-A	Planning	IT04	8	RM 18.00	Delivered		1 Marker set for briefing	RM 144.00
6	0005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered		1 Desk sign for pilot team	RM 250.00
7	0006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00
8	0007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled		2 Display adapter for briefing	RM 180.00
9	0008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00
10	0009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00
11	0010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00
12	0011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled		2 Desk sign for pilot team	RM 50.00
13	0012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00
14	0013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00
15	0014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00
16	0015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled		3 Cable pack for review session	RM 300.00
17	0016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00
18	0017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00
19	0018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	
20	0019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled		4 Display adapter for briefing	

4. Change F3 to **200**. J3 becomes200 and the intended row is highlighted. Delivered and Cancelled records elsewhere do not qualify merely because their amount is high.



	rderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount
2	0001	PJ-A	Planning	IT01	2	RM 45.00	Delivered		1 Display adapter for briefing	RM 90.00
3	0002	PJ-A	Procurement	IT02	1	RM 200.00	Pending		1 Training notebook for pilot team	RM 200.00
4	0003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled		1 Cable pack for review session	RM 180.00
5	0004	PJ-A	Planning	IT04	8	RM 18.00	Delivered		1 Marker set for briefing	RM 144.00
6	0005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered		1 Desk sign for pilot team	RM 250.00
7	0006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00
8	0007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled		2 Display adapter for briefing	RM 180.00
9	0008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00
10	0009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00
11	0010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00
12	0011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled		2 Desk sign for pilot team	RM 50.00
13	0012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00
14	0013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00
15	0014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00
16	0015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled		3 Cable pack for review session	RM 300.00
17	0016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00
18	0017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00
19	0018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	
20	0019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled		4 Display adapter for briefing	

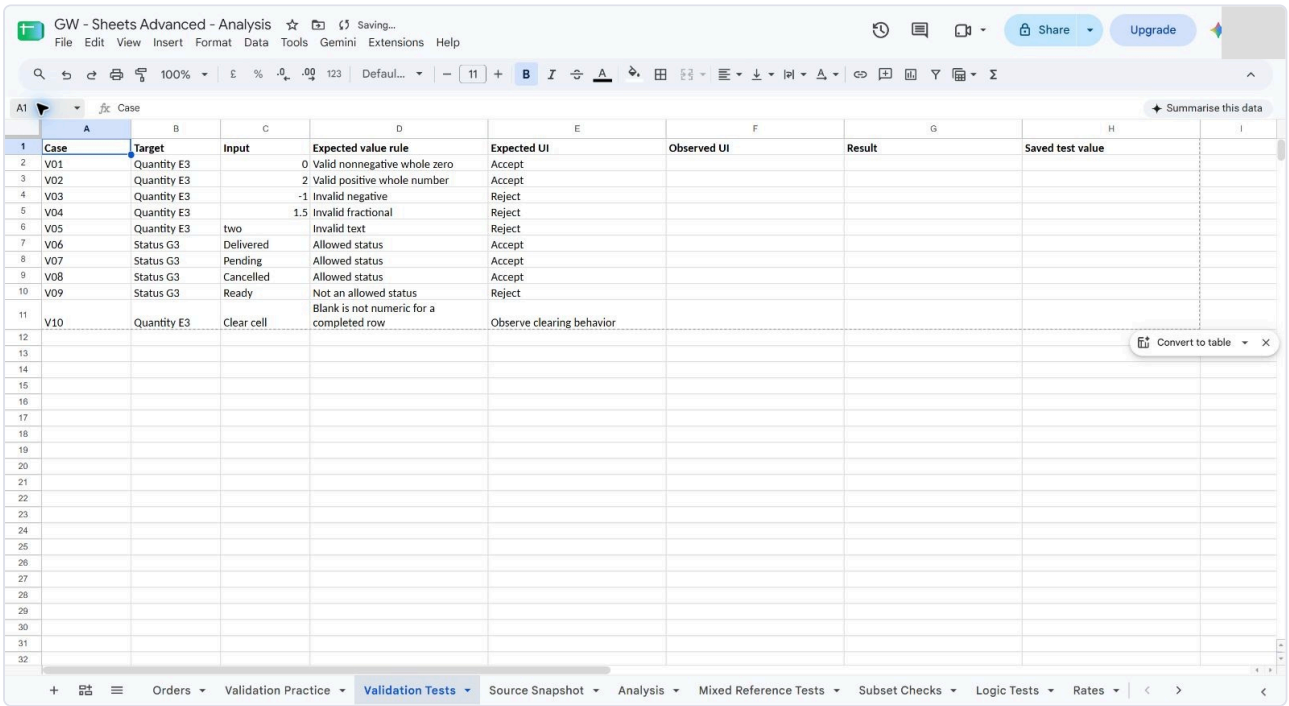
5. Restore **E3=4 / F3=12 / G3=Pending**. Amount returns to48 and the highlight disappears. The maintained source was never changed.



	OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount
2	0001	PJ-A	Planning	IT01	2	RM 45.00	Delivered		1 Display adapter for briefing	RM 90.00
3	0002	PJ-A	Procurement	IT02	4	RM 12.00	Pending		1 Training notebook for pilot team	RM 48.00
4	0003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled		1 Cable pack for review session	RM 180.00
5	0004	PJ-A	Planning	IT04	8	RM 18.00	Delivered		1 Marker set for briefing	RM 144.00
6	0005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered		1 Desk sign for pilot team	RM 250.00
7	0006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00
8	0007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled		2 Display adapter for briefing	RM 180.00
9	0008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00
10	0009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00
11	0010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00
12	0011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled		2 Desk sign for pilot team	RM 50.00
13	0012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00
14	0013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00
15	0014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00
16	0015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled		3 Cable pack for review session	RM 300.00
17	0016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00
18	0017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00
19	0018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	
20	0019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled		4 Display adapter for briefing	

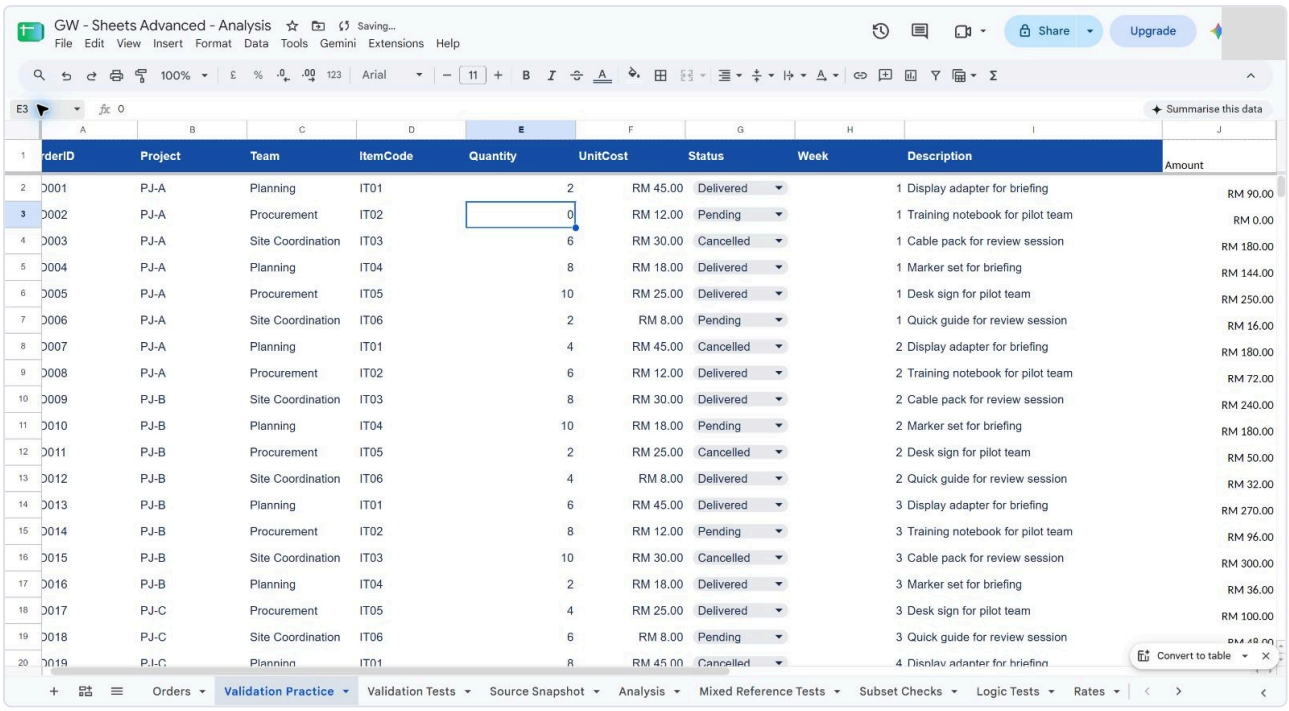
3.4 Create a Rule Test Harness and Reconcile Exceptions

1. Create **Validation Tests** and paste the input-expectations card into **A1:H11**. It includes nine ordinary entry cases and one separate blank-clearing observation. Record expected results before trying each input; leave observed-result cells empty until after testing. A blank is not a numeric Quantity for a completed row, but clearing behavior must be observed separately from entered-value rejection.



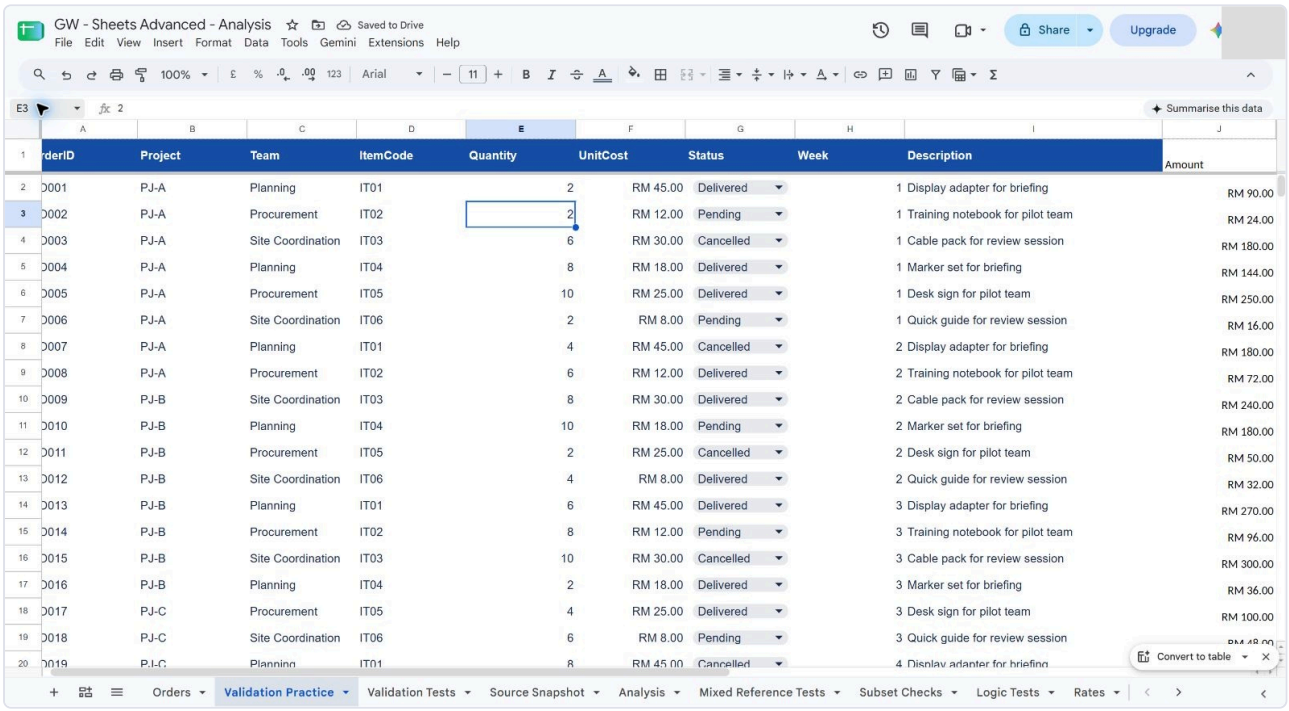
Case	Target	Input	Expected value rule	Expected UI	Observed UI	Result	Saved test value
V01	Quantity E3	0	Valid nonnegative whole zero	Accept			
V02	Quantity E3	2	Valid positive whole number	Accept			
V03	Quantity E3	-1	Invalid negative	Reject			
V04	Quantity E3	1.5	Invalid fractional	Reject			
V05	Quantity E3	two	Invalid text	Reject			
V06	Status G3	Delivered	Allowed status	Accept			
V07	Status G3	Pending	Allowed status	Accept			
V08	Status G3	Cancelled	Allowed status	Accept			
V09	Status G3	Ready	Not an allowed status	Reject			
V10	Quantity E3	Clear cell	Blank is not numeric for a completed row	Observe clearing behavior			

2. On practice E3, enter numeric **0**. It is accepted by the nonnegative whole-number rule. Record the result, then restore4.



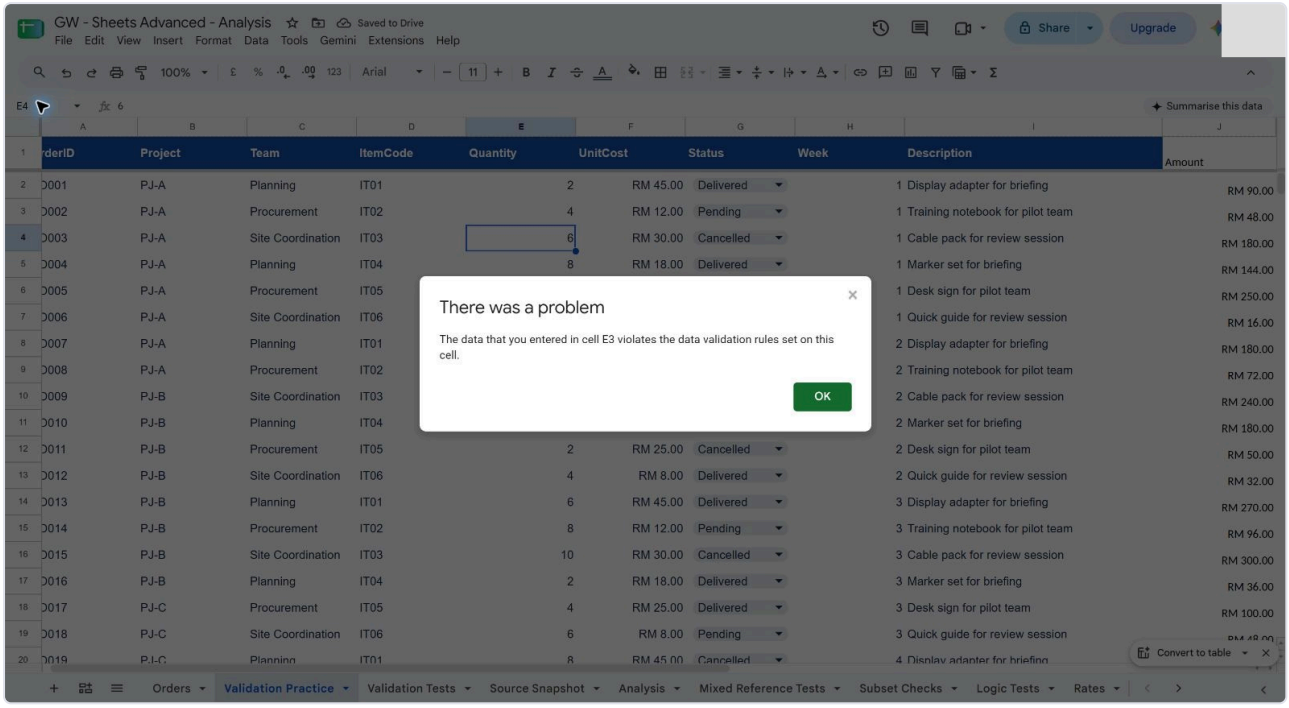
	derID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount
2	0001	PJ-A	Planning	IT01	2	RM 45.00	Delivered		1 Display adapter for briefing	RM 90.00
3	0002	PJ-A	Procurement	IT02	0	RM 12.00	Pending		1 Training notebook for pilot team	RM 0.00
4	0003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled		1 Cable pack for review session	RM 180.00
5	0004	PJ-A	Planning	IT04	8	RM 18.00	Delivered		1 Marker set for briefing	RM 144.00
6	0005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered		1 Desk sign for pilot team	RM 250.00
7	0006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00
8	0007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled		2 Display adapter for briefing	RM 180.00
9	0008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00
10	0009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00
11	0010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00
12	0011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled		2 Desk sign for pilot team	RM 50.00
13	0012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00
14	0013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00
15	0014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00
16	0015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled		3 Cable pack for review session	RM 300.00
17	0016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00
18	0017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00
19	0018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	
20	0019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled		4 Display adapter for briefing	

3. Enter **2**. It is accepted. Record the result, then restore4.

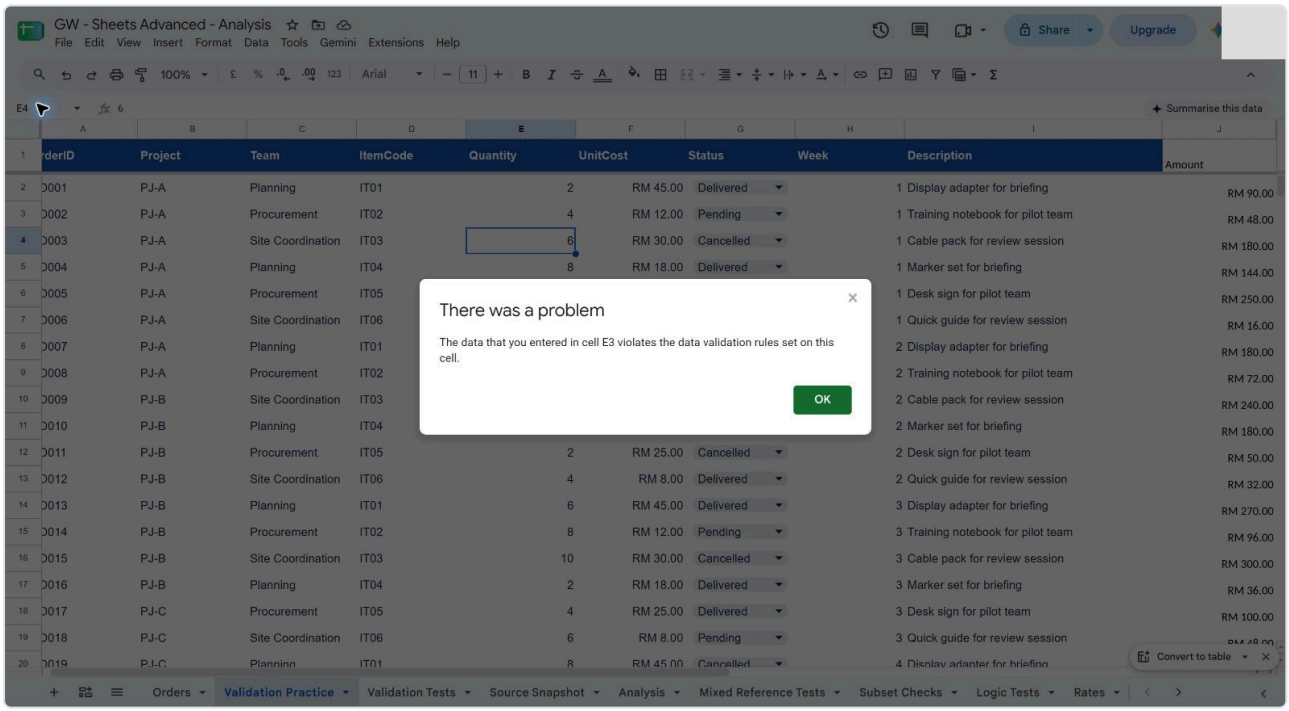


	derID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount
2	0001	PJ-A	Planning	IT01	2	RM 45.00	Delivered		1 Display adapter for briefing	RM 90.00
3	0002	PJ-A	Procurement	IT02	2	RM 12.00	Pending		1 Training notebook for pilot team	RM 24.00
4	0003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled		1 Cable pack for review session	RM 180.00
5	0004	PJ-A	Planning	IT04	8	RM 18.00	Delivered		1 Marker set for briefing	RM 144.00
6	0005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered		1 Desk sign for pilot team	RM 250.00
7	0006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00
8	0007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled		2 Display adapter for briefing	RM 180.00
9	0008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00
10	0009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00
11	0010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00
12	0011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled		2 Desk sign for pilot team	RM 50.00
13	0012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00
14	0013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00
15	0014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00
16	0015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled		3 Cable pack for review session	RM 300.00
17	0016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00
18	0017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00
19	0018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	
20	0019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled		4 Display adapter for briefing	

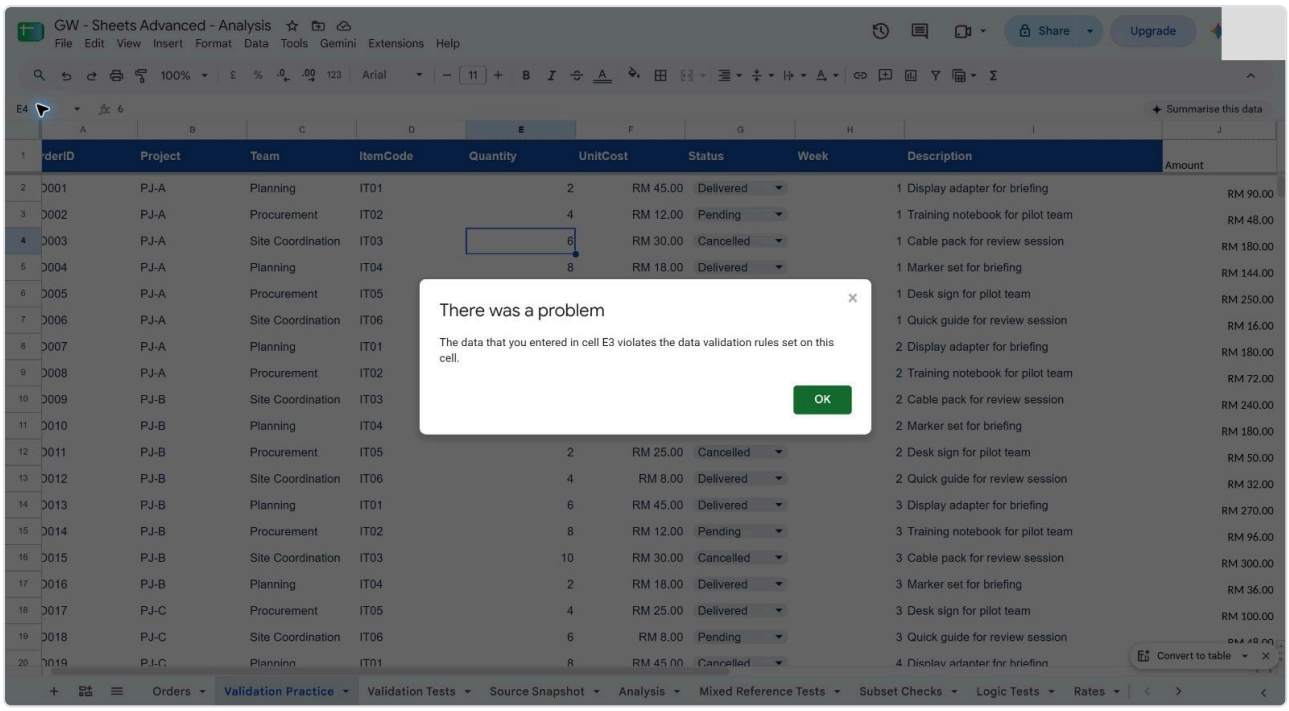
4. Enter **-1**. The actual dialog rejects it; dismiss and retain4.



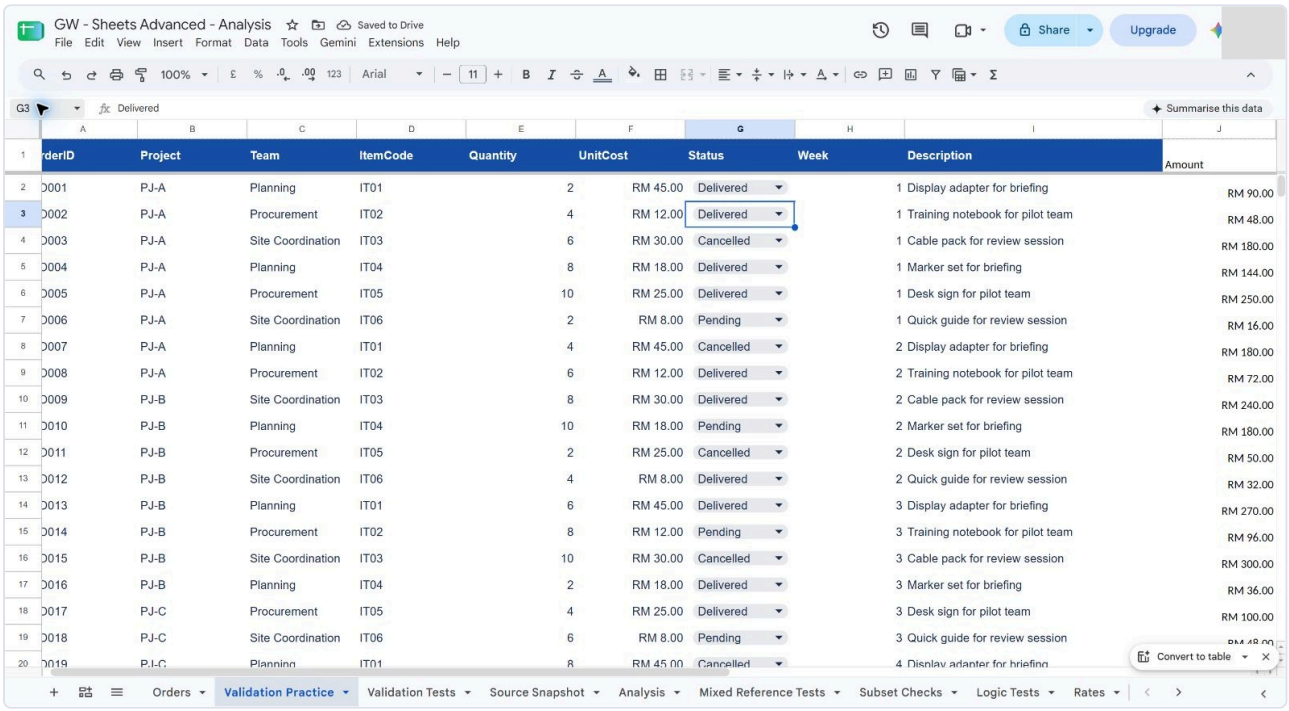
5. Enter **1.5**. It is rejected in the harness test too; retain4.



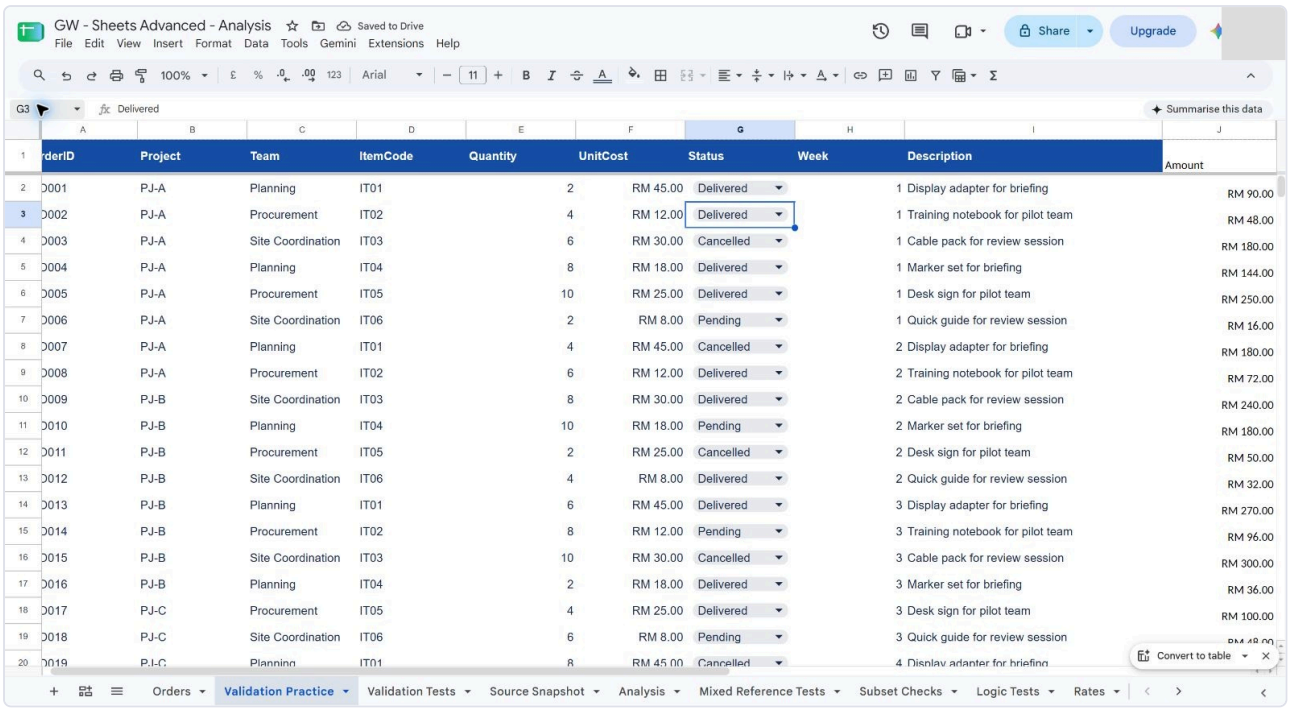
6. Enter the text **two**. It is rejected. Preserve the observed input and result in the report rather than silently translating it into numeric2.



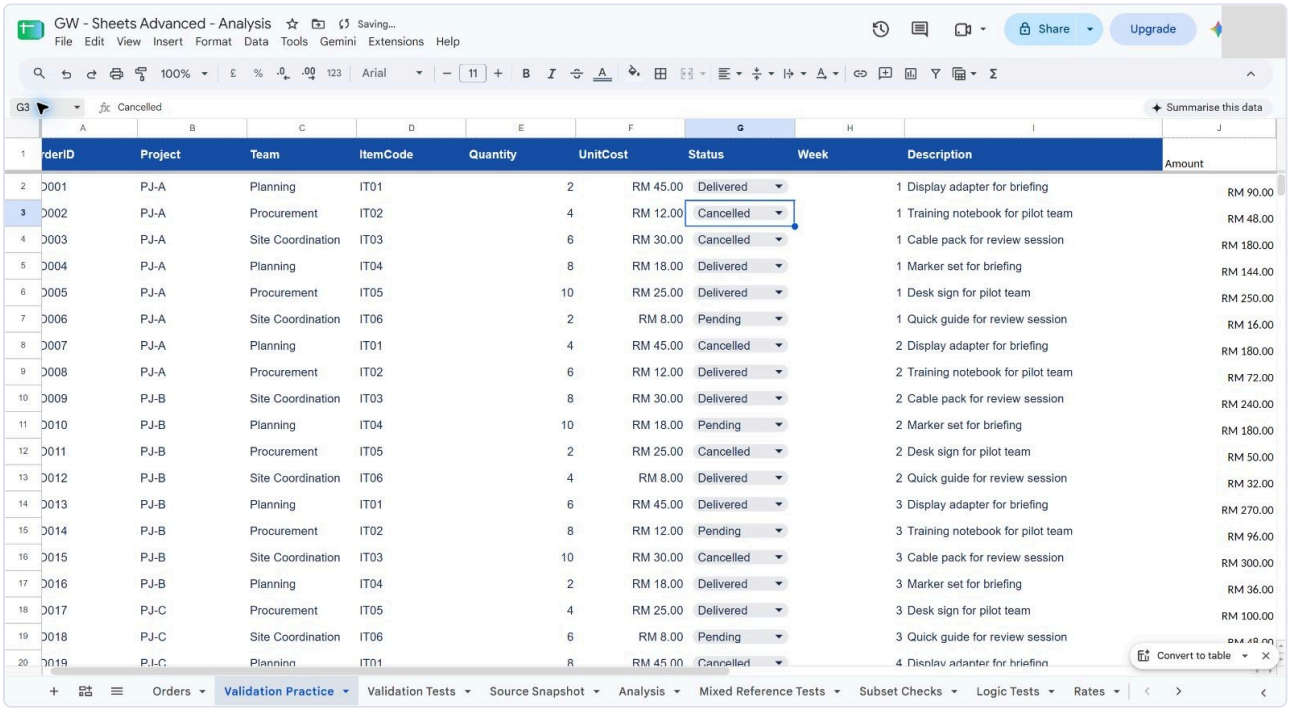
7. Test **Delivered** in G3. It is accepted; restore Pending after recording the result.



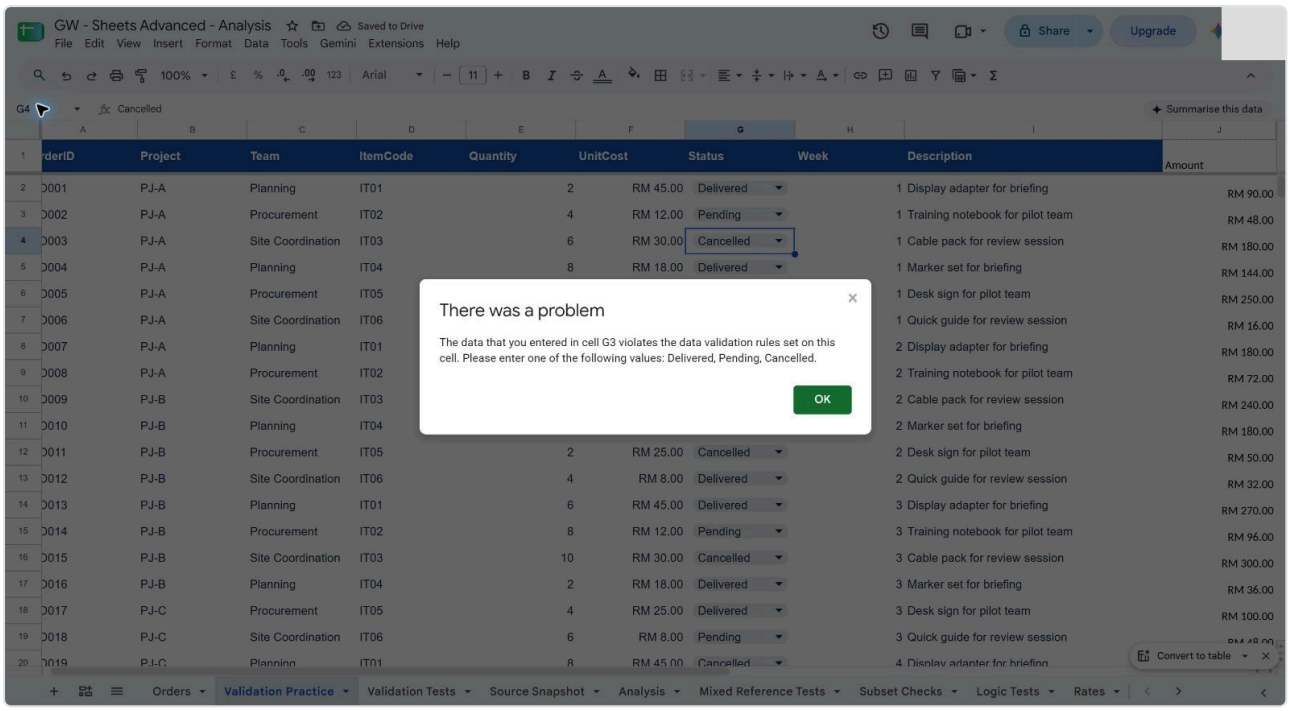
8. Test **Pending**. It is an allowed value and remains selected.



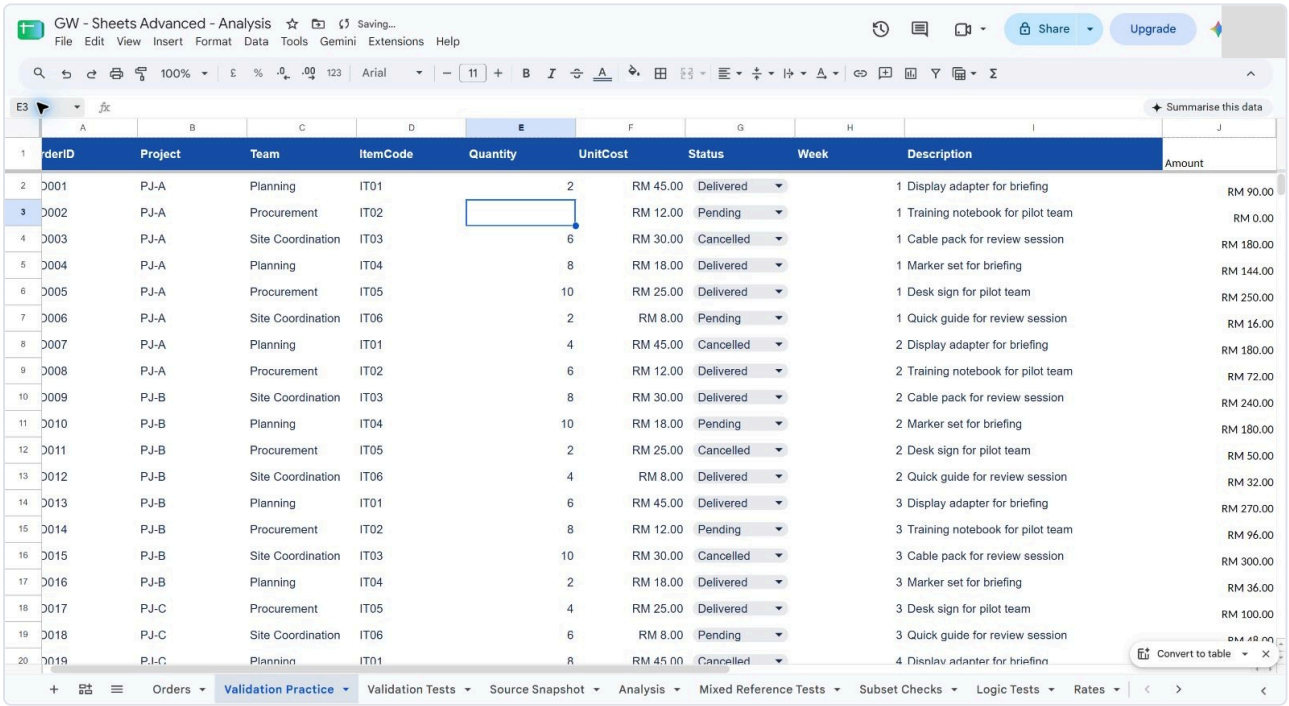
9. Test **Cancelled**. It is accepted; restore Pending. Cancelled remains a valid source status, even though the current exception condition excludes it.



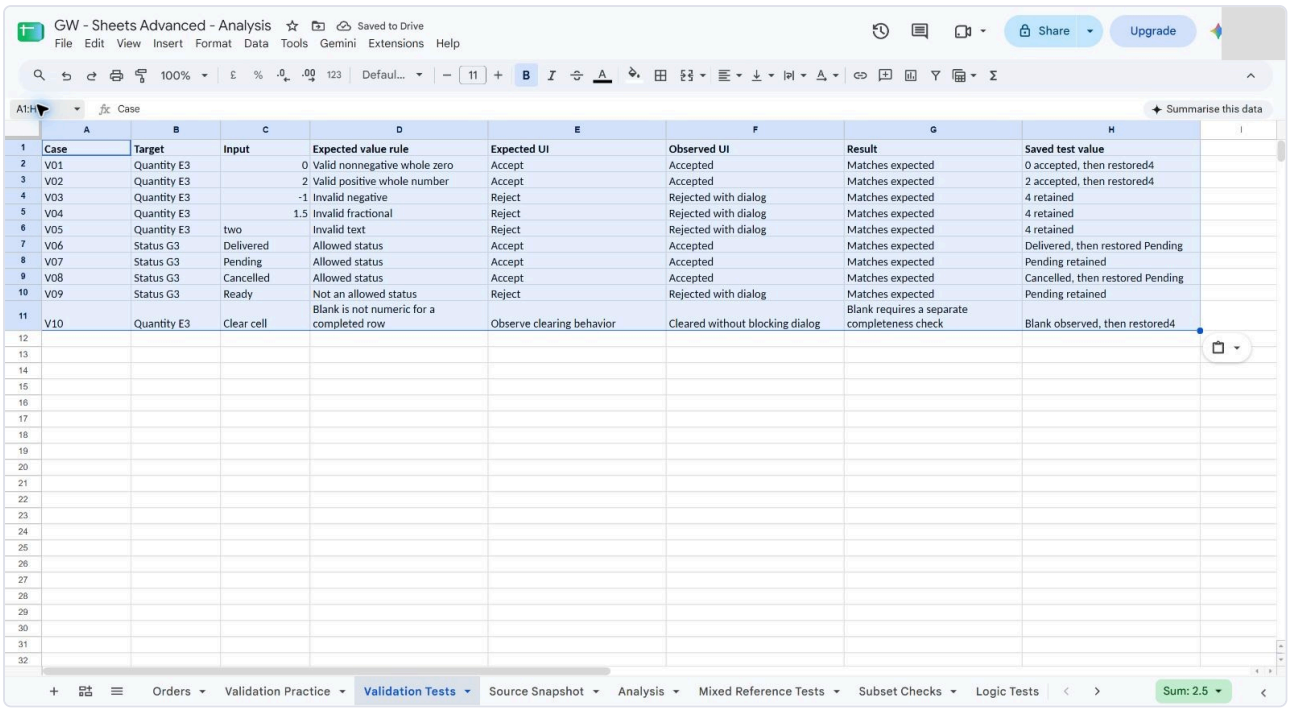
10. Test **Ready**. It is rejected and Pending is retained.



11. Select E3 and use **Delete** to clear it. In this account the cell became blank **without a blocking dialog**, while the custom validation rule remained. Record this limit: the input rule does not enforce requiredness against clearing, and a complete row still needs a numeric Quantity check. Save the blank observation before restoring4. Do not report this as a successful blank-rejection test.

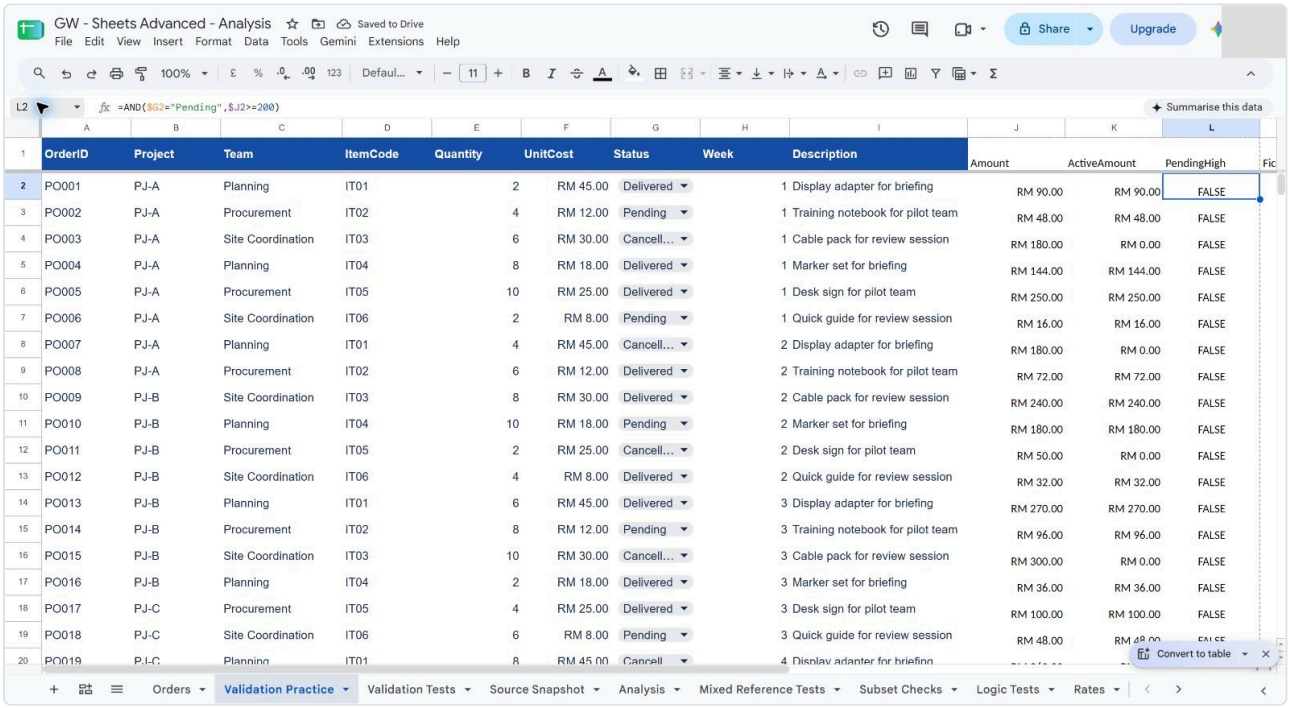


12. Complete the observed-result columns. Nine ordinary cases match the expected accept/reject behavior; the tenth records the separate completeness gap. Restore E3/F3/G3 to **4/12/Pending**. Bulk-range paste and numeric text imported from CSV were not tested.



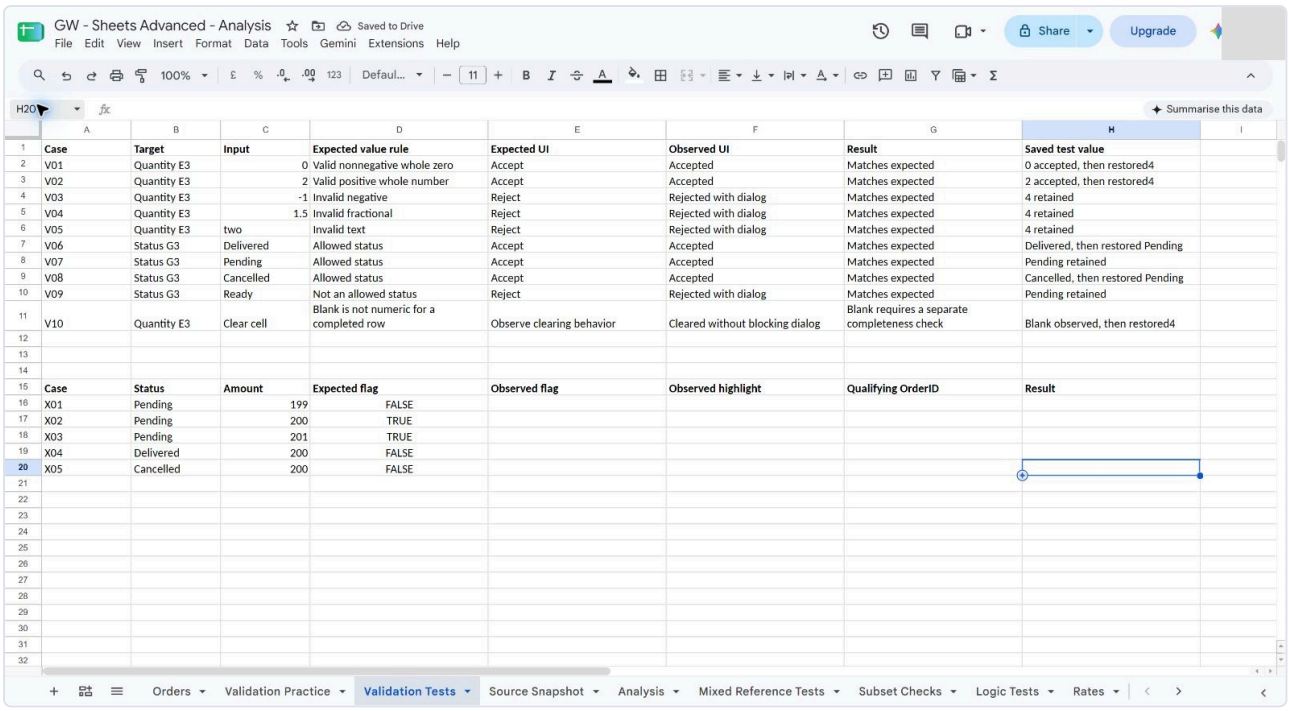
	A	B	C	D	E	F	G	H
1	Case	Target	Input	Expected value rule	Expected UI	Observed UI	Result	Saved test value
2	V01	Quantity E3	0	Valid nonnegative whole zero	Accept	Accepted	Matches expected	0 accepted, then restored4
3	V02	Quantity E3	2	Valid positive whole number	Accept	Accepted	Matches expected	2 accepted, then restored4
4	V03	Quantity E3	-1	Invalid negative	Reject	Rejected with dialog	Matches expected	4 retained
5	V04	Quantity E3	1.5	Invalid fractional	Reject	Rejected with dialog	Matches expected	4 retained
6	V05	Quantity E3	two	Invalid text	Reject	Rejected with dialog	Matches expected	4 retained
7	V06	Status G3	Delivered	Allowed status	Accept	Accepted	Matches expected	Delivered, then restored Pending
8	V07	Status G3	Pending	Allowed status	Accept	Accepted	Matches expected	Pending retained
9	V08	Status G3	Cancelled	Allowed status	Accept	Accepted	Matches expected	Cancelled, then restored Pending
10	V09	Status G3	Ready	Not an allowed status	Reject	Rejected with dialog	Matches expected	Pending retained
11	V10	Quantity E3	Clear cell	Blank is not numeric for a completed row	Observe clearing behavior	Cleared without blocking dialog	Blank requires a separate completeness check	Blank observed, then restored4

13. In Validation Practice, enter **PendingHigh** in L1 and **=AND($G2="Pending",$J2>=200)** in L2, then fill through L25. The restored baseline has24 FALSE flags, agreeing with no highlighted rows. For a full-width view, adjust only practice columns: A:H about110pixels, C150, I220, J:L120; later G140/H80 makes the status labels readable. Keep maintained-source widths unchanged.



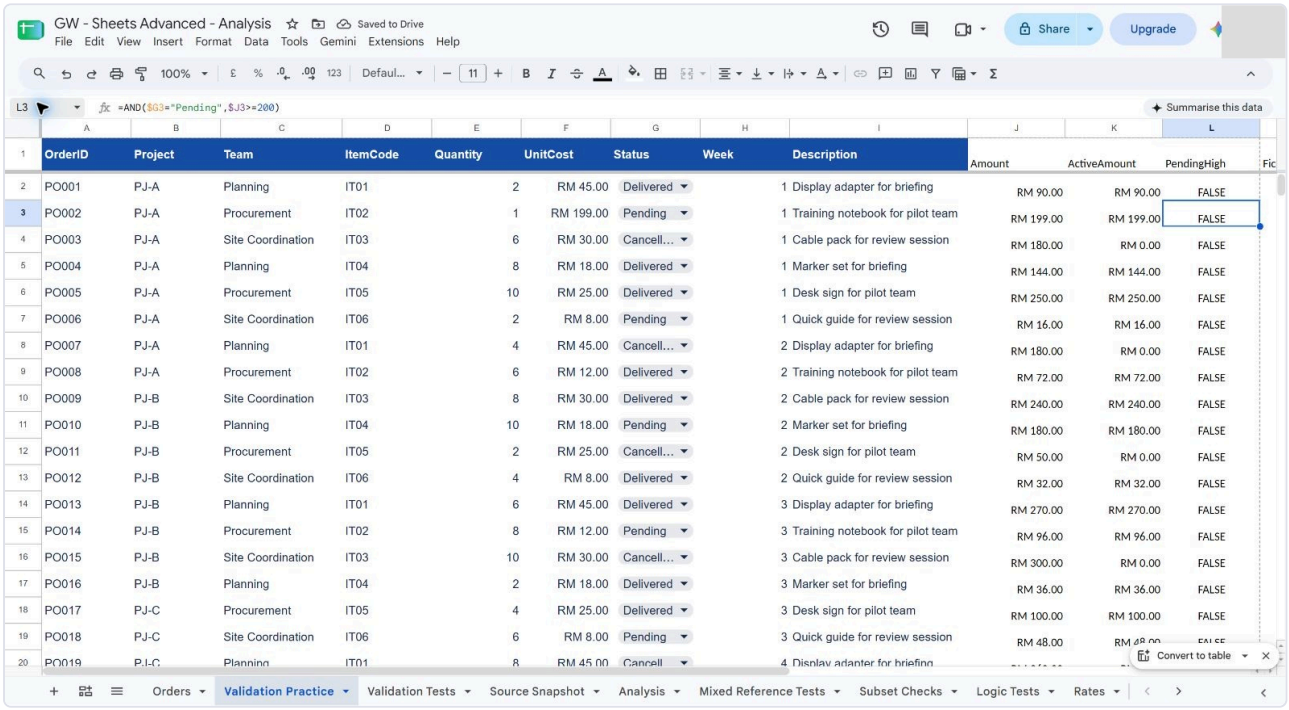
	A	B	C	D	E	F	G	H	I	J	K	L
1	OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	PendingHigh
2	PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter for briefing	RM 90.00	RM 90.00	FALSE
3	PO002	PJ-A	Procurement	IT02	4	RM 12.00	Pending	1	Training notebook for pilot team	RM 48.00	RM 48.00	FALSE
4	PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancell...	1	Cable pack for review session	RM 180.00	RM 0.00	FALSE
5	PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for briefing	RM 144.00	RM 144.00	FALSE
6	PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team	RM 250.00	RM 250.00	FALSE
7	PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session	RM 16.00	RM 16.00	FALSE
8	PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancell...	2	Display adapter for briefing	RM 180.00	RM 0.00	FALSE
9	PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team	RM 72.00	RM 72.00	FALSE
10	PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session	RM 240.00	RM 240.00	FALSE
11	PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing	RM 180.00	RM 180.00	FALSE
12	PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancell...	2	Desk sign for pilot team	RM 50.00	RM 0.00	FALSE
13	PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session	RM 32.00	RM 32.00	FALSE
14	PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing	RM 270.00	RM 270.00	FALSE
15	PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team	RM 96.00	RM 96.00	FALSE
16	PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancell...	3	Cable pack for review session	RM 300.00	RM 0.00	FALSE
17	PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing	RM 36.00	RM 36.00	FALSE
18	PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team	RM 100.00	RM 100.00	FALSE
19	PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session	RM 48.00	RM 48.00	FALSE
20	PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancell...	4	Display adapter for briefing			

14. On Validation Tests, enter the five exception expectations in **A15:H20** before trying them: Pending199 FALSE; Pending200 TRUE; Pending201 TRUE; Delivered200 FALSE; Cancelled200 FALSE. Actual flag/highlight/ID/result columns start empty. These are sequential temporary versions of practice PO002, not additional source records.



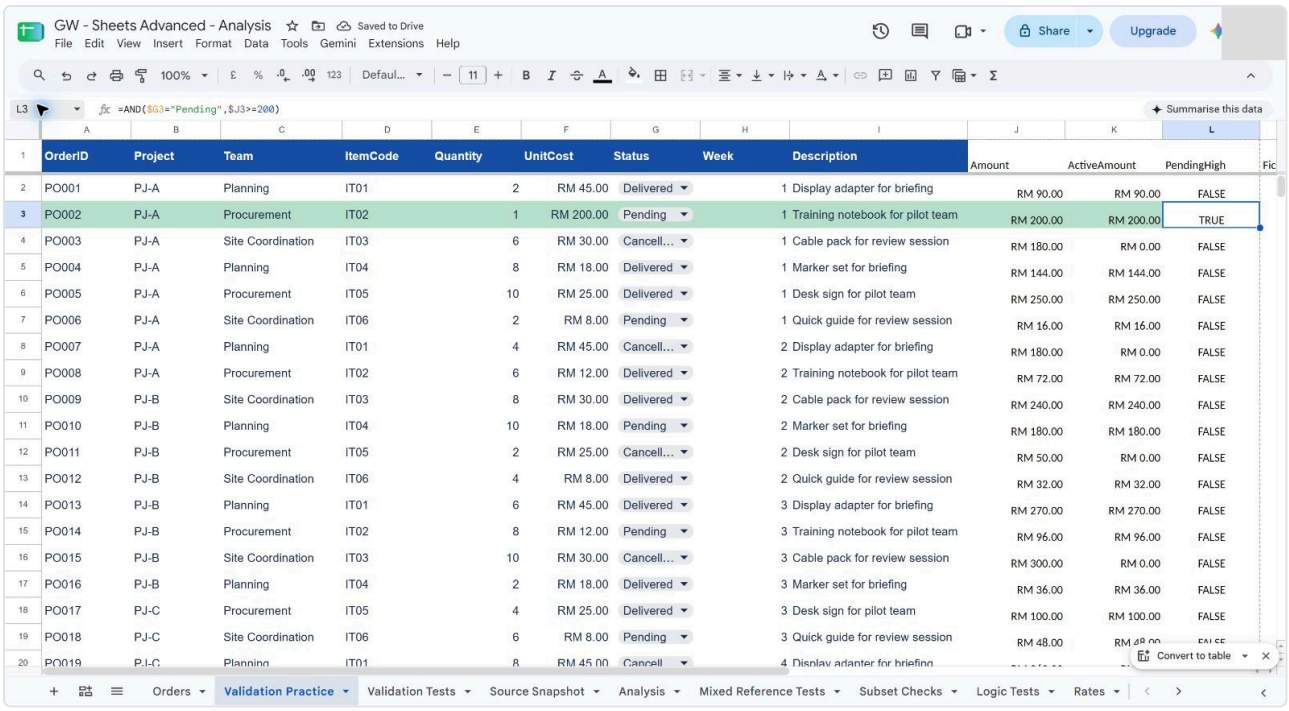
	A	B	C	D	E	F	G	H	I
1	Case	Target	Input	Expected value rule	Expected UI	Observed UI	Result	Saved test value	
2	V01	Quantity E3	0	Valid nonnegative whole zero	Accept	Accepted	Matches expected	0 accepted, then restored4	
3	V02	Quantity E3	2	Valid positive whole number	Accept	Accepted	Matches expected	2 accepted, then restored4	
4	V03	Quantity E3	-1	Invalid negative	Reject	Rejected with dialog	Matches expected	4 retained	
5	V04	Quantity E3	1.5	Invalid fractional	Reject	Rejected with dialog	Matches expected	4 retained	
6	V05	Quantity E3	two	Invalid text	Reject	Rejected with dialog	Matches expected	4 retained	
7	V06	Status G3	Delivered	Allowed status	Accept	Accepted	Matches expected	Delivered, then restored Pending	
8	V07	Status G3	Pending	Allowed status	Accept	Accepted	Matches expected	Pending retained	
9	V08	Status G3	Cancelled	Allowed status	Accept	Accepted	Matches expected	Cancelled, then restored Pending	
10	V09	Status G3	Ready	Not an allowed status	Reject	Rejected with dialog	Matches expected	Pending retained	
11	V10	Quantity E3	Clear cell	Blank is not numeric for a completed row	Observe clearing behavior	Cleared without blocking dialog	Blank requires a separate completeness check	Blank observed, then restored4	
12									
13									
14									
15	Case	Status	Amount	Expected flag	Observed flag	Observed highlight	Qualifying OrderID	Result	
16	X01	Pending	199	FALSE					
17	X02	Pending	200	TRUE					
18	X03	Pending	201	TRUE					
19	X04	Delivered	200	FALSE					
20	X05	Cancelled	200	FALSE					

15. With E3=1, test **Pending / F3=199**. L3 is FALSE and the row is not highlighted. The amount condition excludes it.



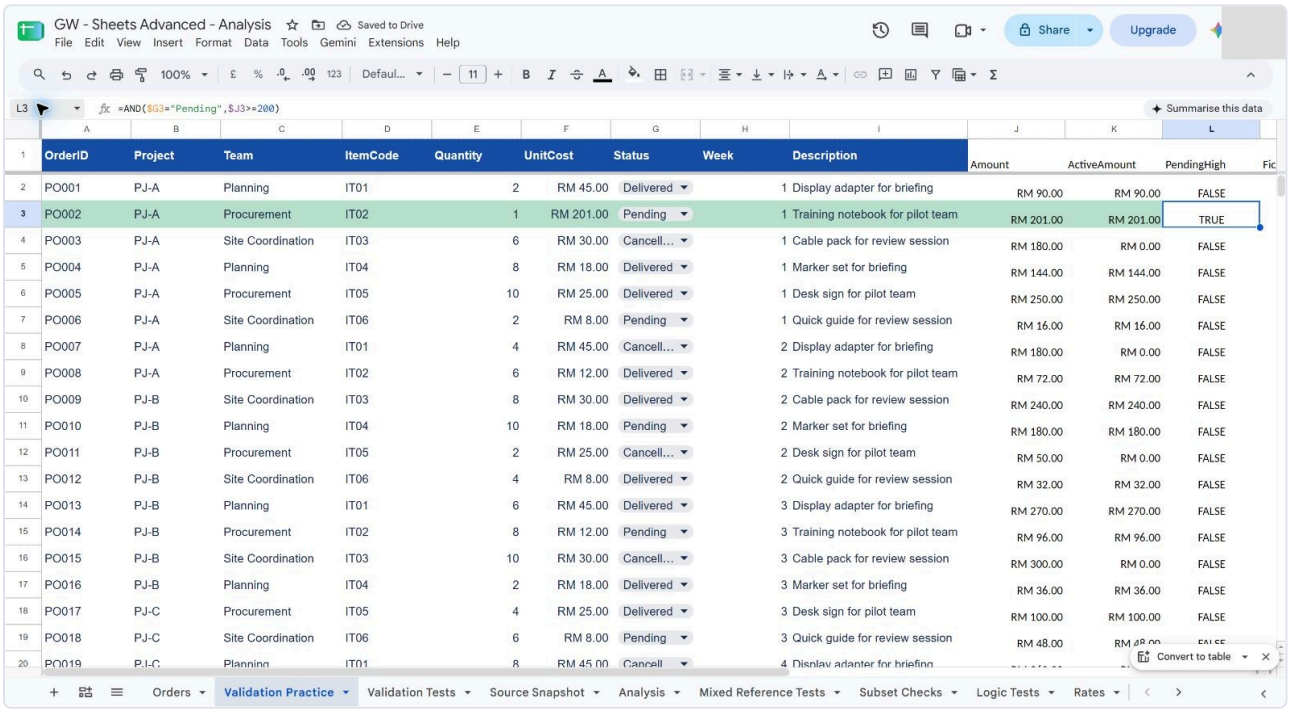
	A	B	C	D	E	F	G	H	I	J	K	L
1	OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	PendingHigh
2	PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter for briefing	RM 90.00	RM 90.00	FALSE
3	PO002	PJ-A	Procurement	IT02	1	RM 199.00	Pending	1	Training notebook for pilot team	RM 199.00	RM 199.00	FALSE
4	PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancell...	1	Cable pack for review session	RM 180.00	RM 0.00	FALSE
5	PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for briefing	RM 144.00	RM 144.00	FALSE
6	PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team	RM 250.00	RM 250.00	FALSE
7	PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session	RM 16.00	RM 16.00	FALSE
8	PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancell...	2	Display adapter for briefing	RM 180.00	RM 0.00	FALSE
9	PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team	RM 72.00	RM 72.00	FALSE
10	PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session	RM 240.00	RM 240.00	FALSE
11	PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing	RM 180.00	RM 180.00	FALSE
12	PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancell...	2	Desk sign for pilot team	RM 50.00	RM 0.00	FALSE
13	PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session	RM 32.00	RM 32.00	FALSE
14	PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing	RM 270.00	RM 270.00	FALSE
15	PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team	RM 96.00	RM 96.00	FALSE
16	PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancell...	3	Cable pack for review session	RM 300.00	RM 0.00	FALSE
17	PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing	RM 36.00	RM 36.00	FALSE
18	PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team	RM 100.00	RM 100.00	FALSE
19	PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session	RM 48.00	RM 48.00	FALSE
20	PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancell...	4	Display adapter for briefing			

16. Test **Pending /200**. L3 is TRUE and PO002 is highlighted; the threshold is inclusive.



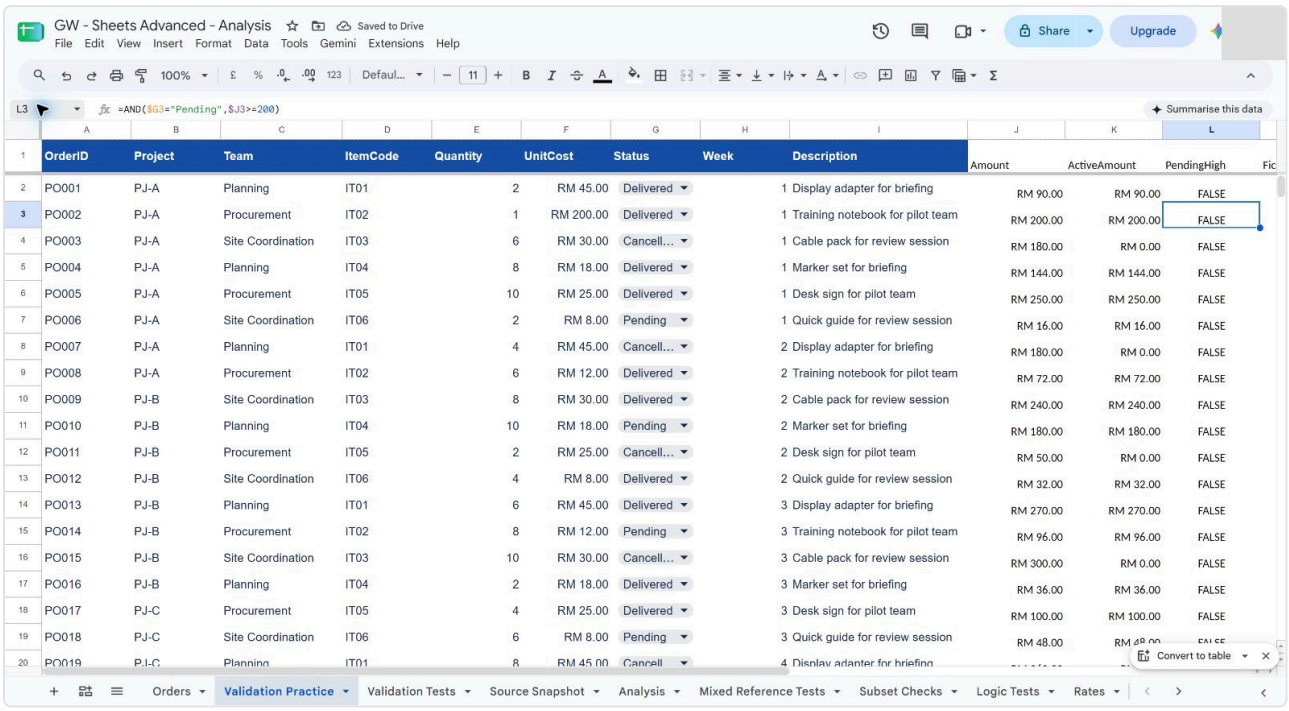
	A	B	C	D	E	F	G	H	I	J	K	L
1	OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	PendingHigh
2	PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered		1 Display adapter for briefing	RM 90.00	RM 90.00	FALSE
3	PO002	PJ-A	Procurement	IT02	1	RM 200.00	Pending		1 Training notebook for pilot team	RM 200.00	RM 200.00	TRUE
4	PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancell...		1 Cable pack for review session	RM 180.00	RM 0.00	FALSE
5	PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered		1 Marker set for briefing	RM 144.00	RM 144.00	FALSE
6	PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered		1 Desk sign for pilot team	RM 250.00	RM 250.00	FALSE
7	PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00	RM 16.00	FALSE
8	PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancell...		2 Display adapter for briefing	RM 180.00	RM 0.00	FALSE
9	PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00	RM 72.00	FALSE
10	PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00	RM 240.00	FALSE
11	PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00	RM 180.00	FALSE
12	PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancell...		2 Desk sign for pilot team	RM 50.00	RM 0.00	FALSE
13	PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00	RM 32.00	FALSE
14	PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00	RM 270.00	FALSE
15	PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00	RM 96.00	FALSE
16	PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancell...		3 Cable pack for review session	RM 300.00	RM 0.00	FALSE
17	PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00	RM 36.00	FALSE
18	PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00	RM 100.00	FALSE
19	PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	RM 48.00	RM 48.00	FALSE
20	PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancell		4 Display adapter for briefing			

17. Test **Pending /201**. L3 remains TRUE and the row remains highlighted.



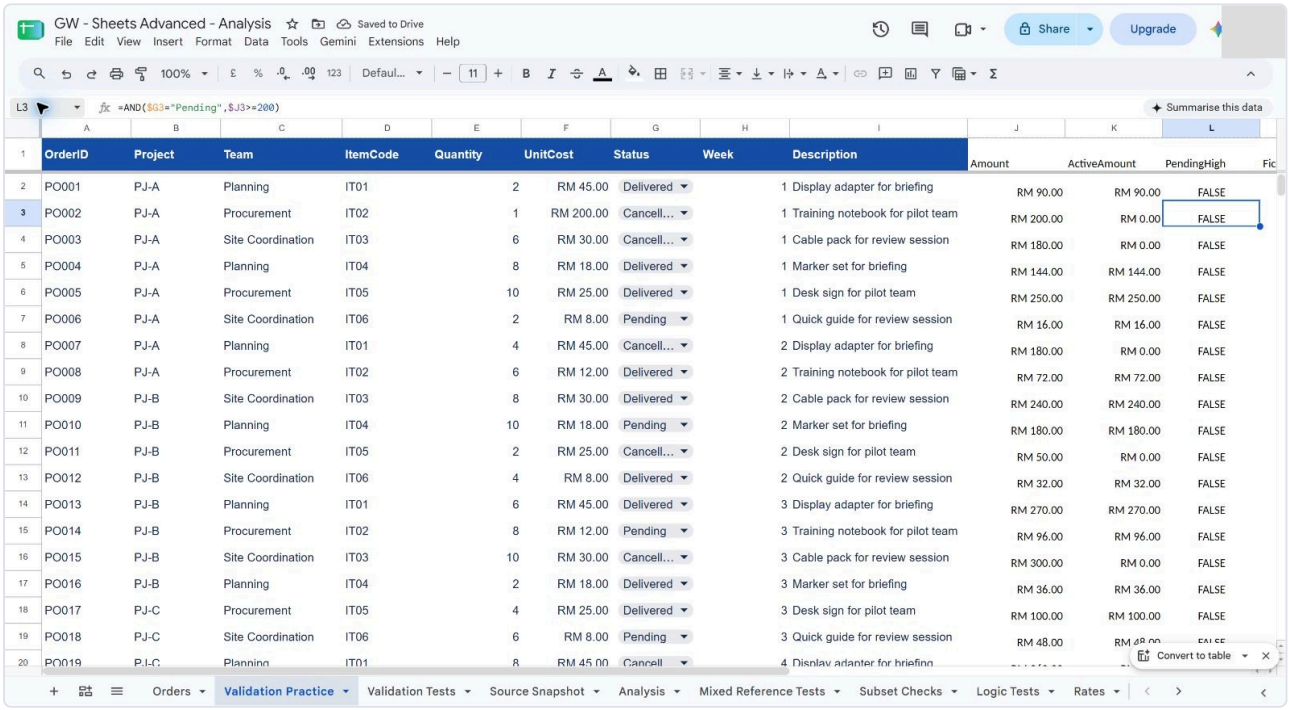
	A	B	C	D	E	F	G	H	I	J	K	L
1	OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	PendingHigh
2	PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered		1 Display adapter for briefing	RM 90.00	RM 90.00	FALSE
3	PO002	PJ-A	Procurement	IT02	1	RM 201.00	Pending		1 Training notebook for pilot team	RM 201.00	RM 201.00	TRUE
4	PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancell...		1 Cable pack for review session	RM 180.00	RM 0.00	FALSE
5	PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered		1 Marker set for briefing	RM 144.00	RM 144.00	FALSE
6	PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered		1 Desk sign for pilot team	RM 250.00	RM 250.00	FALSE
7	PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00	RM 16.00	FALSE
8	PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancell...		2 Display adapter for briefing	RM 180.00	RM 0.00	FALSE
9	PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00	RM 72.00	FALSE
10	PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00	RM 240.00	FALSE
11	PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00	RM 180.00	FALSE
12	PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancell...		2 Desk sign for pilot team	RM 50.00	RM 0.00	FALSE
13	PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00	RM 32.00	FALSE
14	PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00	RM 270.00	FALSE
15	PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00	RM 96.00	FALSE
16	PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancell...		3 Cable pack for review session	RM 300.00	RM 0.00	FALSE
17	PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00	RM 36.00	FALSE
18	PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00	RM 100.00	FALSE
19	PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	RM 48.00	RM 48.00	FALSE
20	PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancell		4 Display adapter for briefing			

18. Test **Delivered /200**. L3 is FALSE and the highlight is absent because the status condition fails.



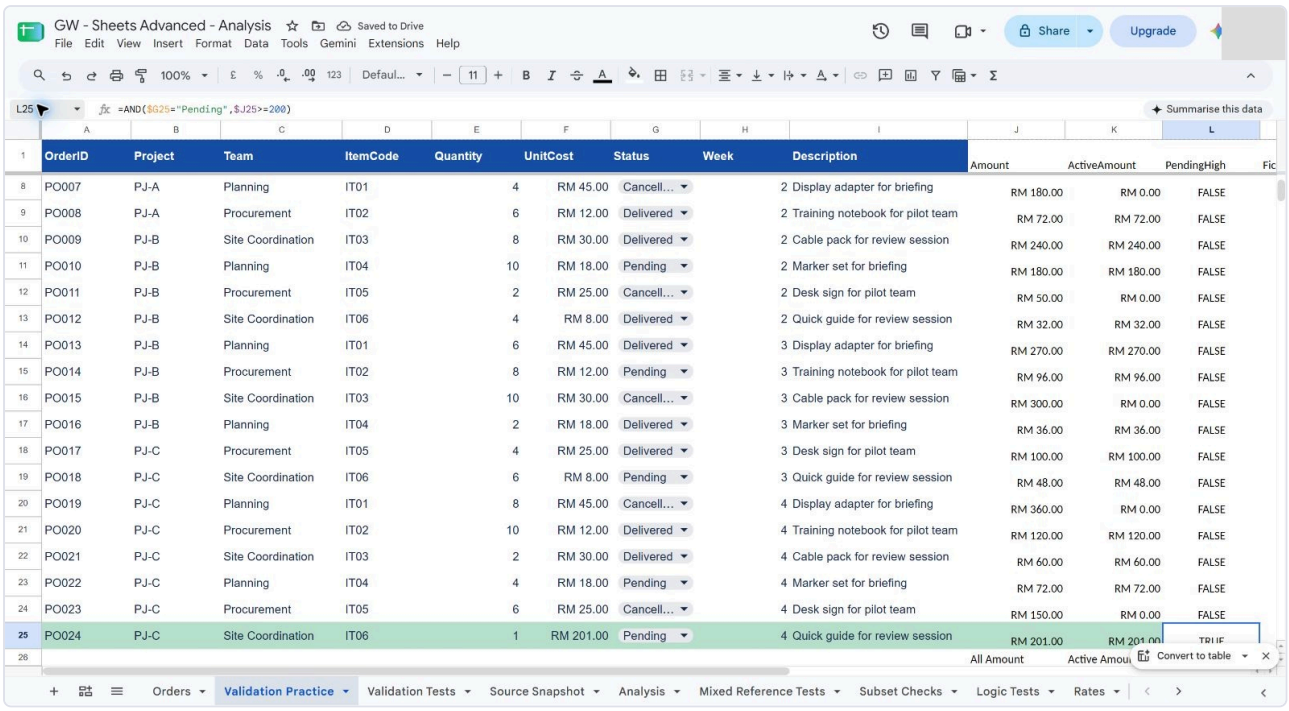
	A	B	C	D	E	F	G	H	I	J	K	L
1	OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	PendingHigh
2	PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered		1 Display adapter for briefing	RM 90.00	RM 90.00	FALSE
3	PO002	PJ-A	Procurement	IT02	1	RM 200.00	Delivered		1 Training notebook for pilot team	RM 200.00	RM 200.00	FALSE
4	PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancell...		1 Cable pack for review session	RM 180.00	RM 0.00	FALSE
5	PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered		1 Marker set for briefing	RM 144.00	RM 144.00	FALSE
6	PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered		1 Desk sign for pilot team	RM 250.00	RM 250.00	FALSE
7	PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00	RM 16.00	FALSE
8	PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancell...		2 Display adapter for briefing	RM 180.00	RM 0.00	FALSE
9	PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00	RM 72.00	FALSE
10	PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00	RM 240.00	FALSE
11	PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00	RM 180.00	FALSE
12	PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancell...		2 Desk sign for pilot team	RM 50.00	RM 0.00	FALSE
13	PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00	RM 32.00	FALSE
14	PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00	RM 270.00	FALSE
15	PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00	RM 96.00	FALSE
16	PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancell...		3 Cable pack for review session	RM 300.00	RM 0.00	FALSE
17	PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00	RM 36.00	FALSE
18	PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00	RM 100.00	FALSE
19	PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	RM 48.00		
20	PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancell		4 Display adapter for briefing			

19. Test **Cancelled /200**. L3 is FALSE, the highlight is absent and ActiveAmount is0. Record the observed cases, then restore E3/F3/G3 to **4/12/Pending**.

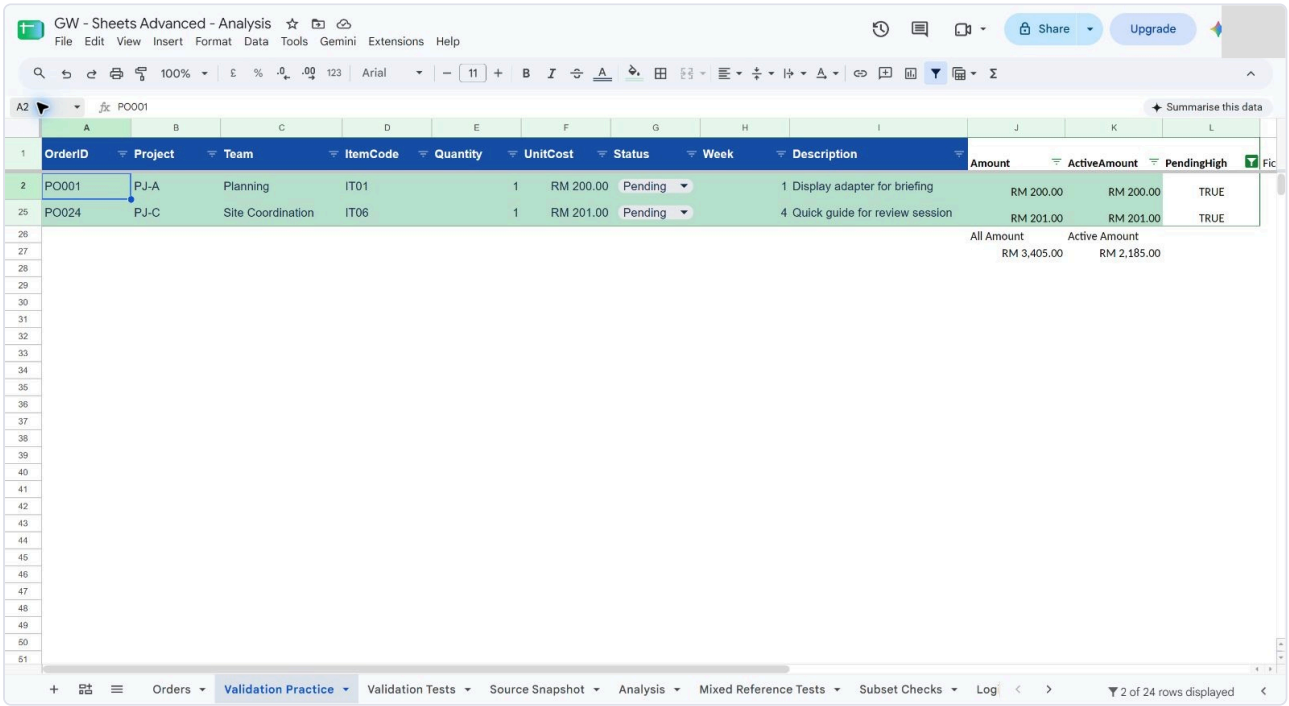


	A	B	C	D	E	F	G	H	I	J	K	L
1	OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	PendingHigh
2	PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered		1 Display adapter for briefing	RM 90.00	RM 90.00	FALSE
3	PO002	PJ-A	Procurement	IT02	1	RM 200.00	Cancell...		1 Training notebook for pilot team	RM 200.00	RM 0.00	FALSE
4	PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancell...		1 Cable pack for review session	RM 180.00	RM 0.00	FALSE
5	PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered		1 Marker set for briefing	RM 144.00	RM 144.00	FALSE
6	PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered		1 Desk sign for pilot team	RM 250.00	RM 250.00	FALSE
7	PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00	RM 16.00	FALSE
8	PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancell...		2 Display adapter for briefing	RM 180.00	RM 0.00	FALSE
9	PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00	RM 72.00	FALSE
10	PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00	RM 240.00	FALSE
11	PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00	RM 180.00	FALSE
12	PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancell...		2 Desk sign for pilot team	RM 50.00	RM 0.00	FALSE
13	PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00	RM 32.00	FALSE
14	PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00	RM 270.00	FALSE
15	PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00	RM 96.00	FALSE
16	PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancell...		3 Cable pack for review session	RM 300.00	RM 0.00	FALSE
17	PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00	RM 36.00	FALSE
18	PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00	RM 100.00	FALSE
19	PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	RM 48.00		
20	PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancell		4 Display adapter for briefing			

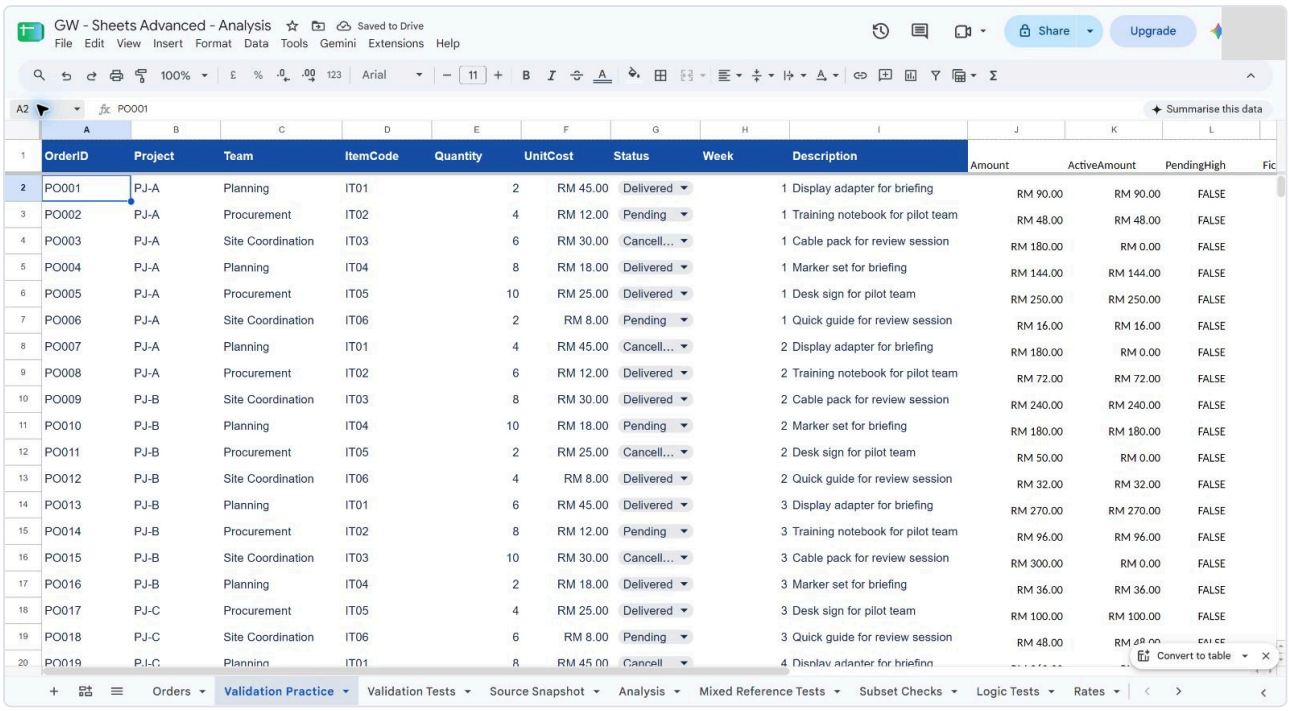
20. Check both range boundaries without appending a record. Temporarily set **PO001 row2** to Quantity1, UnitCost200, Pending, and **PO024 row25** to Quantity1, UnitCost201, Pending. Both flags and row highlights are TRUE. Inspect the final row explicitly; the first cell alone does not prove full coverage. Native readback retained both positive boundary states before restoration.



21. Create a native filter over **A1:L25** and filter PendingHigh to **TRUE**. Navigate to A2 so both records are visible. Expect **2 of24 rows displayed: PO001 and PO024**, with matching TRUE flags and highlights. The ordinary SUM controls still include the full practice population while rows are hidden; they are not filtered subtotals.

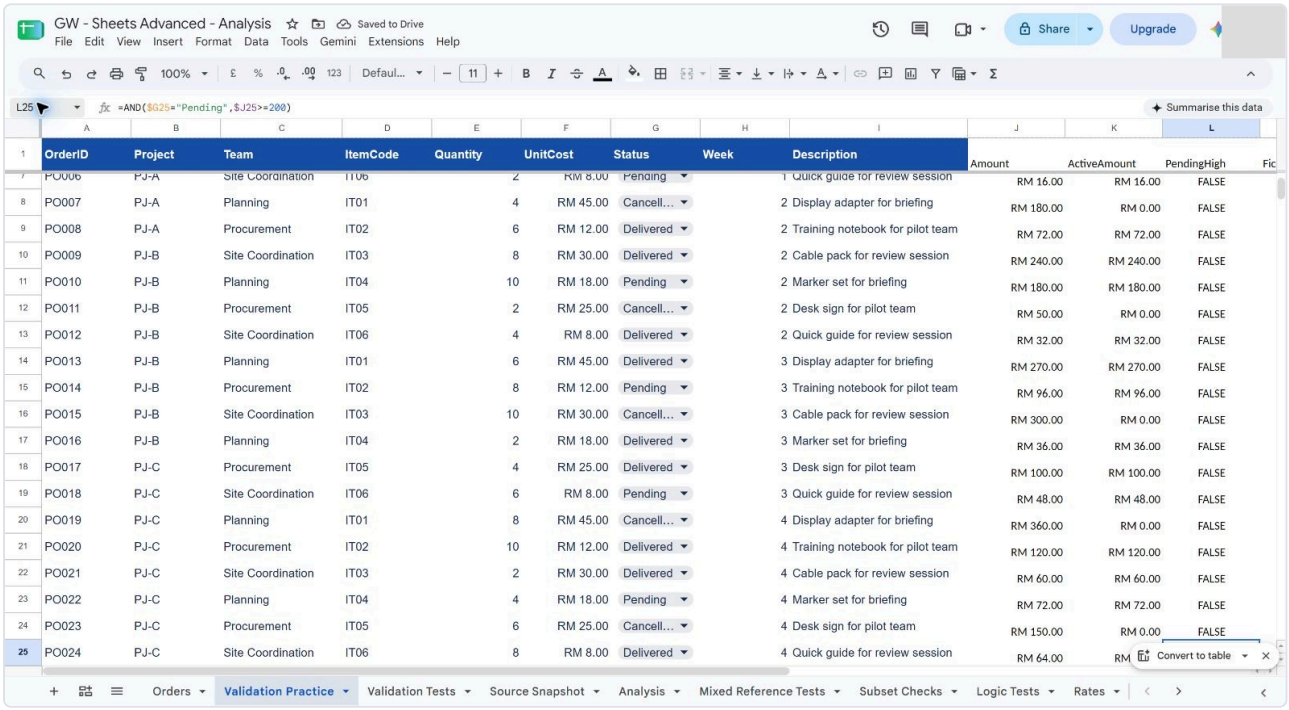


22. Remove the filter. Restore row2 to **2/45/Delivered** and row25 to **8/8/Delivered**; row3 remains **4/12/Pending**. Leave every original OrderID intact.



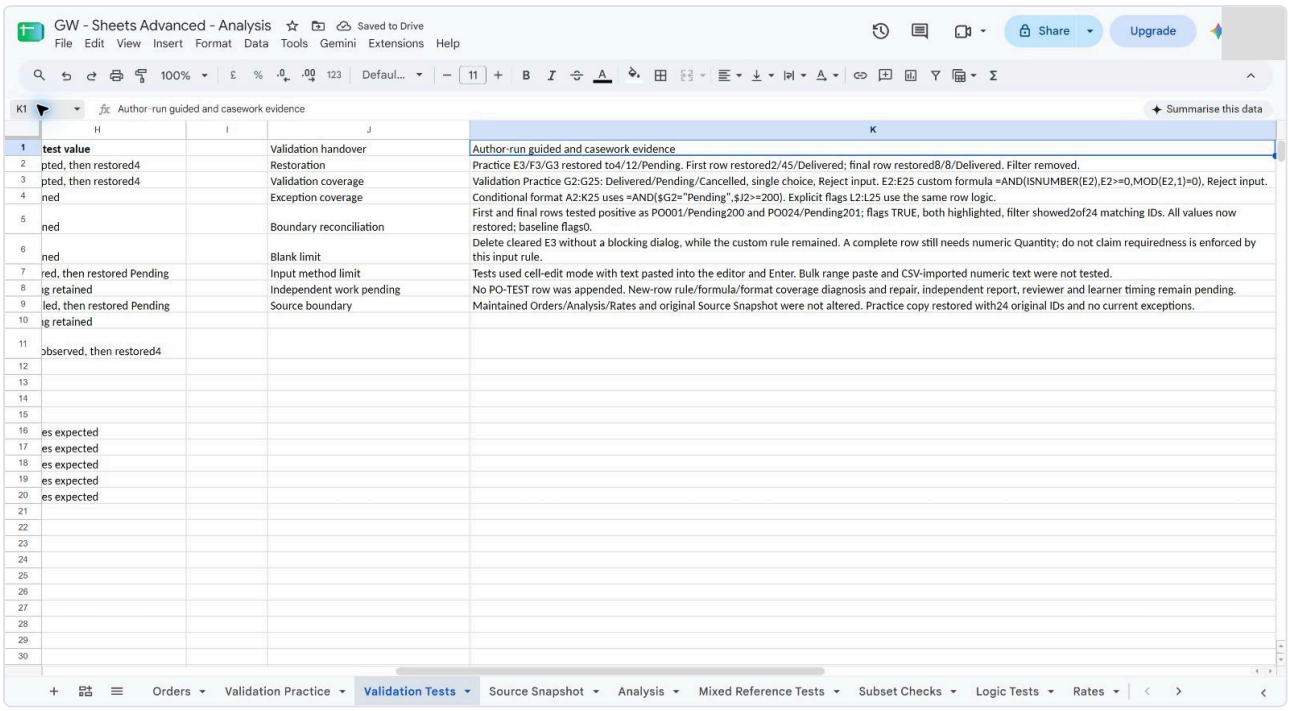
	A	B	C	D	E	F	G	H	I	J	K	L
1	OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	PendingHigh
2	PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered		1 Display adapter for briefing	RM 90.00	RM 90.00	FALSE
3	PO002	PJ-A	Procurement	IT02	4	RM 12.00	Pending		1 Training notebook for pilot team	RM 48.00	RM 48.00	FALSE
4	PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancell...		1 Cable pack for review session	RM 180.00	RM 0.00	FALSE
5	PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered		1 Marker set for briefing	RM 144.00	RM 144.00	FALSE
6	PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered		1 Desk sign for pilot team	RM 250.00	RM 250.00	FALSE
7	PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00	RM 16.00	FALSE
8	PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancell...		2 Display adapter for briefing	RM 180.00	RM 0.00	FALSE
9	PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00	RM 72.00	FALSE
10	PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00	RM 240.00	FALSE
11	PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00	RM 180.00	FALSE
12	PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancell...		2 Desk sign for pilot team	RM 50.00	RM 0.00	FALSE
13	PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00	RM 32.00	FALSE
14	PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00	RM 270.00	FALSE
15	PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00	RM 96.00	FALSE
16	PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancell...		3 Cable pack for review session	RM 300.00	RM 0.00	FALSE
17	PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00	RM 36.00	FALSE
18	PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00	RM 100.00	FALSE
19	PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	RM 48.00	RM 48.00	FALSE
20	PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancell		4 Display adapter for briefing			

23. Reinspect the final row's FALSE flag after restoration. The practice totals return to **3158 /1938**, with no current exceptions. No temporary test value is retained in the source table.

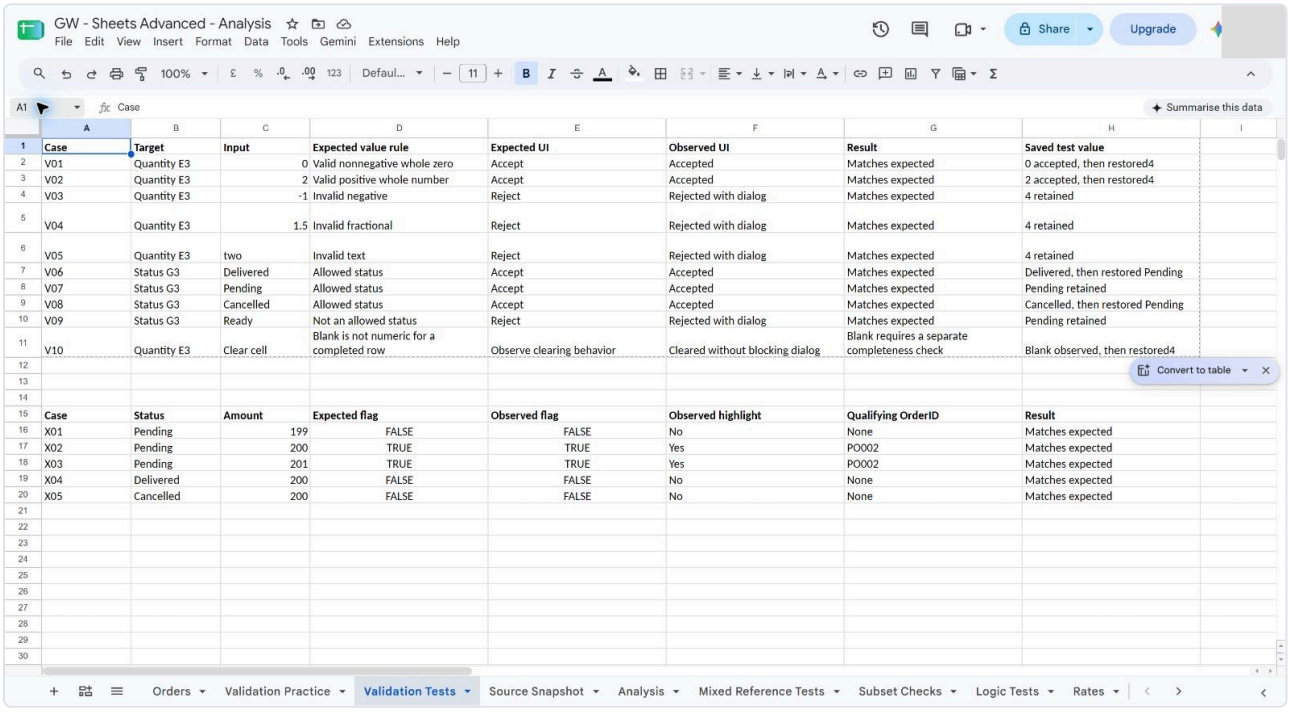


	A	B	C	D	E	F	G	H	I	J	K	L
1	OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	PendingHigh
7	PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending		1 Quick guide for review session	RM 16.00	RM 16.00	FALSE
8	PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancell...		2 Display adapter for briefing	RM 180.00	RM 0.00	FALSE
9	PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered		2 Training notebook for pilot team	RM 72.00	RM 72.00	FALSE
10	PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered		2 Cable pack for review session	RM 240.00	RM 240.00	FALSE
11	PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending		2 Marker set for briefing	RM 180.00	RM 180.00	FALSE
12	PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancell...		2 Desk sign for pilot team	RM 50.00	RM 0.00	FALSE
13	PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered		2 Quick guide for review session	RM 32.00	RM 32.00	FALSE
14	PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered		3 Display adapter for briefing	RM 270.00	RM 270.00	FALSE
15	PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending		3 Training notebook for pilot team	RM 96.00	RM 96.00	FALSE
16	PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancell...		3 Cable pack for review session	RM 300.00	RM 0.00	FALSE
17	PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered		3 Marker set for briefing	RM 36.00	RM 36.00	FALSE
18	PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered		3 Desk sign for pilot team	RM 100.00	RM 100.00	FALSE
19	PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending		3 Quick guide for review session	RM 48.00	RM 48.00	FALSE
20	PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancell...		4 Display adapter for briefing	RM 360.00	RM 0.00	FALSE
21	PO020	PJ-C	Procurement	IT02	10	RM 12.00	Delivered		4 Training notebook for pilot team	RM 120.00	RM 120.00	FALSE
22	PO021	PJ-C	Site Coordination	IT03	2	RM 30.00	Delivered		4 Cable pack for review session	RM 60.00	RM 60.00	FALSE
23	PO022	PJ-C	Planning	IT04	4	RM 18.00	Pending		4 Marker set for briefing	RM 72.00	RM 72.00	FALSE
24	PO023	PJ-C	Procurement	IT05	6	RM 25.00	Cancell...		4 Desk sign for pilot team	RM 150.00	RM 0.00	FALSE
25	PO024	PJ-C	Site Coordination	IT06	8	RM 8.00	Delivered		4 Quick guide for review session	RM 64.00		

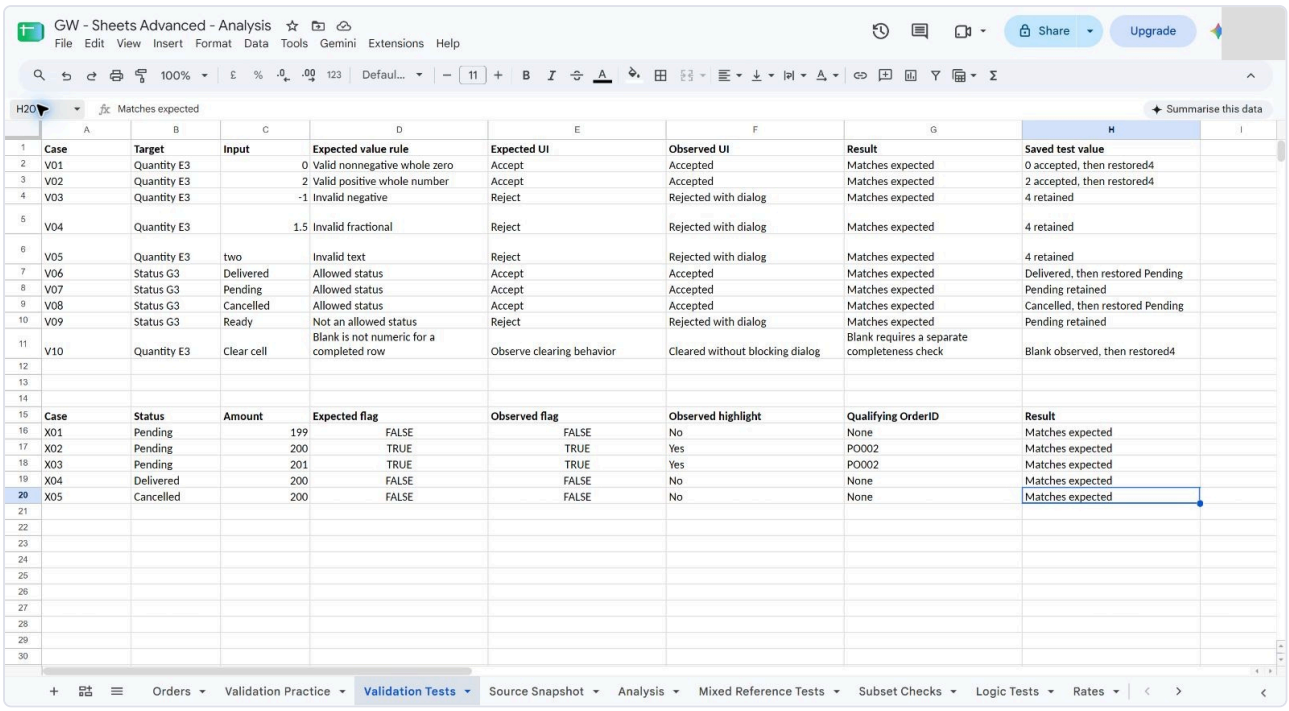
24. Record exact ranges, restoration and limitations in **Validation Tests J1:K9**, outside the matrices. Use wide wrapped note columns (about250/1000pixels) so they remain readable without widening the test tables. State that the independent appended PO-TEST row, new-row coverage repair, bulk paste, CSV numeric text and human/timing checks remain pending.



25. Inspect the completed ten input observations. Separate the nine expected entry results from the blank-clearing limit; a no-highlight row is not proof that every field is valid.



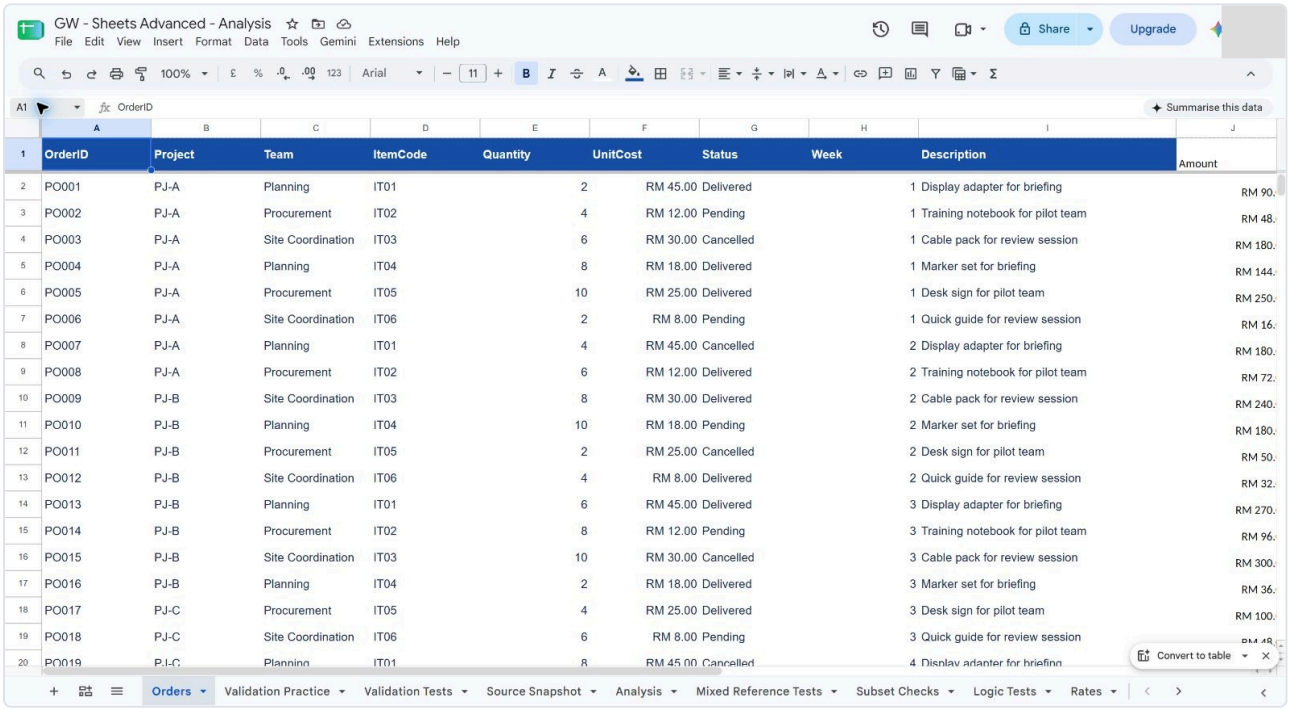
26. Inspect all five exception cases. Expected flags, actual flags and actual highlights agree. Two qualifying entries in the sequential matrix refer to two different temporary amounts on the same PO002, not two simultaneous source records.



Case	Target	Input	Expected value rule	Expected UI	Observed UI	Result	Saved test value
V01	Quantity E3	0	Valid nonnegative whole zero	Accept	Accepted	Matches expected	0 accepted, then restored4
V02	Quantity E3	2	Valid positive whole number	Accept	Accepted	Matches expected	2 accepted, then restored4
V03	Quantity E3	-1	Invalid negative	Reject	Rejected with dialog	Matches expected	4 retained
V04	Quantity E3	1.5	Invalid fractional	Reject	Rejected with dialog	Matches expected	4 retained
V05	Quantity E3	two	Invalid text	Reject	Rejected with dialog	Matches expected	4 retained
V06	Status G3	Delivered	Allowed status	Accept	Accepted	Matches expected	Delivered, then restored Pending
V07	Status G3	Pending	Allowed status	Accept	Accepted	Matches expected	Pending retained
V08	Status G3	Cancelled	Allowed status	Accept	Accepted	Matches expected	Cancelled, then restored Pending
V09	Status G3	Ready	Not an allowed status	Reject	Rejected with dialog	Matches expected	Pending retained
V10	Quantity E3	Clear cell	Blank is not numeric for a completed row	Observe clearing behavior	Cleared without blocking dialog	Blank requires a separate completeness check	Blank observed, then restored4

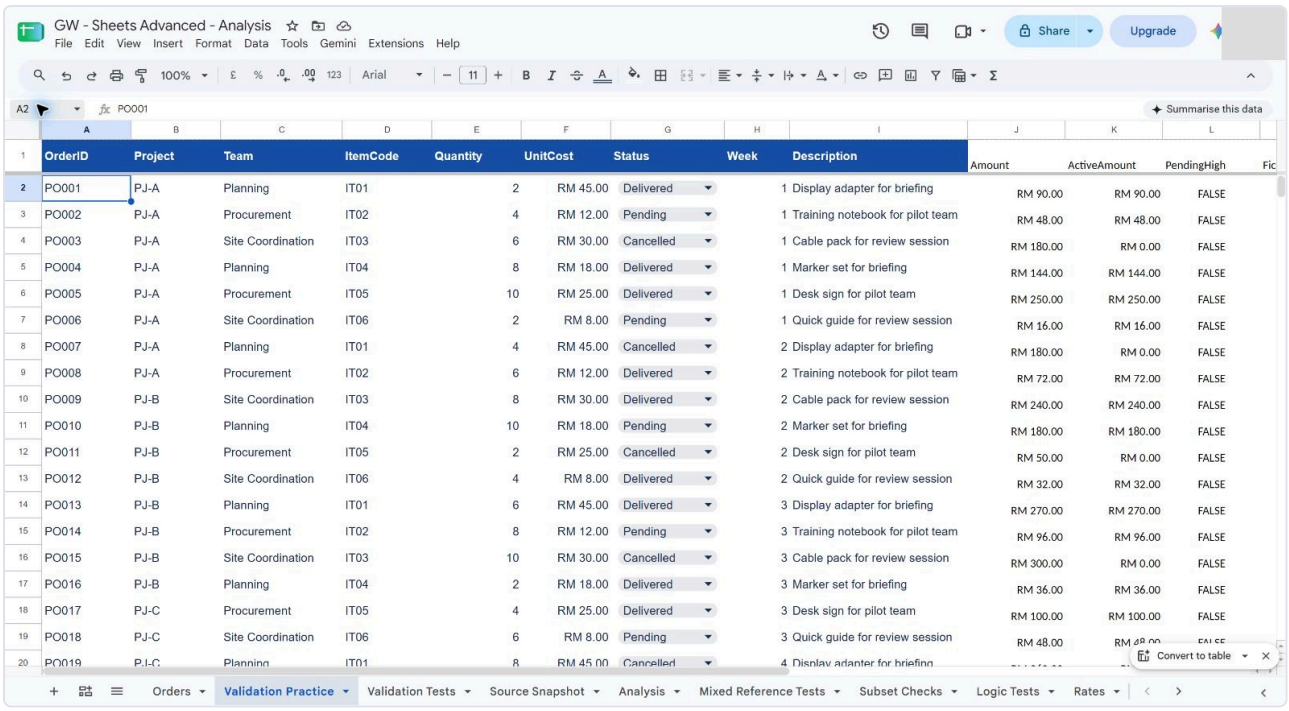
Case	Status	Amount	Expected flag	Observed flag	Observed highlight	Qualifying OrderID	Result
X01	Pending	199	FALSE	FALSE	No	None	Matches expected
X02	Pending	200	TRUE	TRUE	Yes	PO002	Matches expected
X03	Pending	201	TRUE	TRUE	Yes	PO002	Matches expected
X04	Delivered	200	FALSE	FALSE	No	None	Matches expected
X05	Cancelled	200	FALSE	FALSE	No	None	Matches expected

27. Recheck maintained **Orders**. Native verification confirms its values, formulas, formats and original controls remain unchanged, as do maintained Analysis, Rates and Source Snapshot.



OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount
PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter for briefing	RM 90.
PO002	PJ-A	Procurement	IT02	4	RM 12.00	Pending	1	Training notebook for pilot team	RM 48.
PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled	1	Cable pack for review session	RM 180.
PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for briefing	RM 144.
PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team	RM 250.
PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session	RM 16.
PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing	RM 180.
PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team	RM 72.
PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session	RM 240.
PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing	RM 180.
PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team	RM 50.
PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session	RM 32.
PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing	RM 270.
PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team	RM 96.
PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session	RM 300.
PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing	RM 36.
PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team	RM 100.
PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session	
PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing	

28. Leave **Validation Practice** saved with24 original records, original Amount/ActiveAmount formulas, the two strict validation rules, the conditional format and24 FALSE explicit flags. All temporary values and the filter are removed.



OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	PendingHigh	Fic
PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter for briefing	RM 90.00	RM 90.00	FALSE	
PO002	PJ-A	Procurement	IT02	4	RM 12.00	Pending	1	Training notebook for pilot team	RM 48.00	RM 48.00	FALSE	
PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled	1	Cable pack for review session	RM 180.00	RM 0.00	FALSE	
PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for briefing	RM 144.00	RM 144.00	FALSE	
PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team	RM 250.00	RM 250.00	FALSE	
PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session	RM 16.00	RM 16.00	FALSE	
PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing	RM 180.00	RM 0.00	FALSE	
PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team	RM 72.00	RM 72.00	FALSE	
PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session	RM 240.00	RM 240.00	FALSE	
PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing	RM 180.00	RM 180.00	FALSE	
PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team	RM 50.00	RM 0.00	FALSE	
PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session	RM 32.00	RM 32.00	FALSE	
PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing	RM 270.00	RM 270.00	FALSE	
PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team	RM 96.00	RM 96.00	FALSE	
PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session	RM 300.00	RM 0.00	FALSE	
PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing	RM 36.00	RM 36.00	FALSE	
PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team	RM 100.00	RM 100.00	FALSE	
PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session	RM 48.00	RM 48.00	FALSE	
PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing				

> **Validation boundary:** guided3.1–3.3 and author casework3.4 are checked in the learner UI. Nine ordinary entry cases match expectations; blank clearing is allowed and explicitly documented as a completeness limit. Five exception cases and first/final-row flag/highlight/filtered-ID agreement are checked. Independent appended-row coverage diagnosis/repair, bulk-paste/CSV numeric-text behavior, reviewer and learner timing remain pending. The65-minute classroom allocation is unmeasured.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Append a synthetic pending record only to the test copy and diagnose whether input validation, flag formulas and conditional formatting extend to it.
2. Repair coverage and produce a test report containing at least eight input cases and five exception cases.
3. State whether blanks and pasted values were actually tested and list any behaviour still requiring confirmation in the customer account.

Evidence to Hand Over

- Eight-case input validation matrix and five-case exception matrix.
- Agreement between explicit flags, highlighted rows and filtered IDs.
- New-row coverage repair and restored source values.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The colour rule flags the wrong row	Check the apply-to start row and the relative row in the custom formula.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Data rules constrain entry; conditional formatting draws attention to defined exceptions.

Check yourself: How does a validation rule differ from a coloured warning?

Zero-Match Check

The supplied baseline has no Pending record with Amount at least 200. Record that result explicitly. Use a separately labelled synthetic case to prove the positive branch; do not edit baseline records just to obtain a highlight.

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=AND($G2="Pending",$J2⩾200)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Synthetic coverage record, test copy only: PO-TEST | PJ-A | Planning | IT01 | Quantity 5 |
UnitCost 45 | Pending | week 4 | Coverage test. Quantity candidates: 0, 2, -1, 1.5, text
“two”, blank and numeric text imported from a CSV.
```

Captured Validation Reference

Use the separate Validation Practice tab. G2:G25: single-choice Delivered/Pending/Cancelled, strict Reject input. E2:E25: Custom formula is **=AND(ISNUMBER(E2),E2>=0,MOD(E2,1)=0)**, explicitly select Reject input (the custom editor defaulted to warning). Conditional format A2:K25: **=AND($G2="Pending",$J2>=200)**, light fill#b7e1cd. L2 uses that same formula, filled to25.

Input expectations — Validation Tests A1:H11

Observed columns F:H start empty. Each entry test uses cell-edit mode, not a bulk-range paste. Restore E3=4 or G3=Pending after each accepted test; rejected input retains the original value. Test clearing separately with Delete and restore4 afterward.

```
Case    Target  Input   Expected value rule     Expected UI     Observed UI     Result  Saved
test value
V01     Quantity E3     0       Valid nonnegative whole zero    Accept
V02     Quantity E3     2       Valid positive whole number     Accept
V03     Quantity E3     -1      Invalid negative        Reject
V04     Quantity E3     1.5     Invalid fractional      Reject
V05     Quantity E3     two     Invalid text    Reject
V06     Status G3       Delivered       Allowed status  Accept
V07     Status G3       Pending Allowed status  Accept
V08     Status G3       Cancelled       Allowed status  Accept
V09     Status G3       Ready   Not an allowed status   Reject
V10     Quantity E3     Clear cell      Blank is not numeric for a completed row
Observe clearing behavior
```

Nine ordinary cases matched accept/reject expectations. Delete cleared E3 without a blocking dialog while the custom rule remained. A blank is not a completed numeric Quantity: record the completeness limit rather than claiming requiredness is enforced. Bulk-range paste and CSV-imported numeric text remain untested.

Exception expectations — Validation Tests A15:H20

Use temporary practice PO002 (row3), Quantity1 and the listed UnitCost/Amount; restore4/12/Pending after all five cases. Observed columns E:H start empty.

```
Case    Status  Amount  Expected flag   Observed flag   Observed highlight      Qualifying
OrderID Result
X01     Pending 199     FALSE
X02     Pending 200     TRUE
X03     Pending 201     TRUE
X04     Delivered       200     FALSE
X05     Cancelled       200     FALSE
```

Actual flags/highlights were FALSE/none, TRUE/PO002, TRUE/PO002, FALSE/none, FALSE/none. The two positive rows in the report are sequential tests of one record. Delivered/Cancelled fail the status condition even at200.

First/final-row coverage check

On the practice copy only, set row2 PO001 to Quantity1/UnitCost200/Pending and row25 PO024 to Quantity1/UnitCost201/Pending. Flags L2/L25 and row highlights are TRUE. Filter A1:L25 on PendingHigh TRUE:2of24 records, PO001 and PO024. Remove the filter; restore row2 to2/45/Delivered and row25 to8/8/Delivered. Leave row3 at4/12/Pending. Original24IDs and J/K formulas remain, totals3158/1938 and24FALSE flags.

Handover notes — Validation Tests J1:K9

```
Validation handover     Author-run guided and casework evidence
Restoration     Practice E3/F3/G3 restored to4/12/Pending. First row restored2/45/Delivered;
final row restored8/8/Delivered. Filter removed.
Validation coverage     Validation Practice G2:G25: Delivered/Pending/Cancelled, single
choice, Reject input. E2:E25 custom formula =AND(ISNUMBER(E2),E2⩾0,MOD(E2,1)=0), Reject
input.
Exception coverage      Conditional format A2:K25 uses =AND($G2="Pending",$J2⩾200). Explicit
flags L2:L25 use the same row logic.
Boundary reconciliation First and final rows tested positive as PO001/Pending200 and
PO024/Pending201; flags TRUE, both highlighted, filter showed2of24 matching IDs. All values
now restored; baseline flags0.
Blank limit     Delete cleared E3 without a blocking dialog, while the custom rule remained. A
complete row still needs numeric Quantity; do not claim requiredness is enforced by this input
rule.
Input method limit      Tests used cell-edit mode with text pasted into the editor and Enter.
Bulk range paste and CSV-imported numeric text were not tested.
Independent work pending        No PO-TEST row was appended. New-row rule/formula/format
coverage diagnosis and repair, independent report, reviewer and learner timing remain pending.
Source boundary Maintained Orders/Analysis/Rates and original Source Snapshot were not
altered. Practice copy restored with24 original IDs and no current exceptions.
```

No PO-TEST row was appended. Independent new-row validation/formula/format coverage repair, bulk paste, CSV numeric text, reviewer and learner timing remain pending. Maintained Orders/Analysis/Rates and Source Snapshot were unchanged.

04

CHAPTER 4

Combine and Clean Text

In this chapter you will use LEFT, RIGHT, MID and concatenation; check AI categorisation against explicit rules.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 60 minutes: 15 explanation/demonstration, 17 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Text extraction and labels with checked category mappings.

> **Status:** author reference with32 staged application captures through4.4. Leading-zero text, raw/clean provenance, actual Gemini proposals, explicit category comparisons, NBSP exception and separate identifier syntax/reference checks verified. Maintained source preserved. Independent mixed intake, reviewer reconstruction, native file import and learner timing remain pending.

What You Will Learn

- Extract parts of identifiers.
- Combine readable labels.
- Check AI categories against a visible mapping.

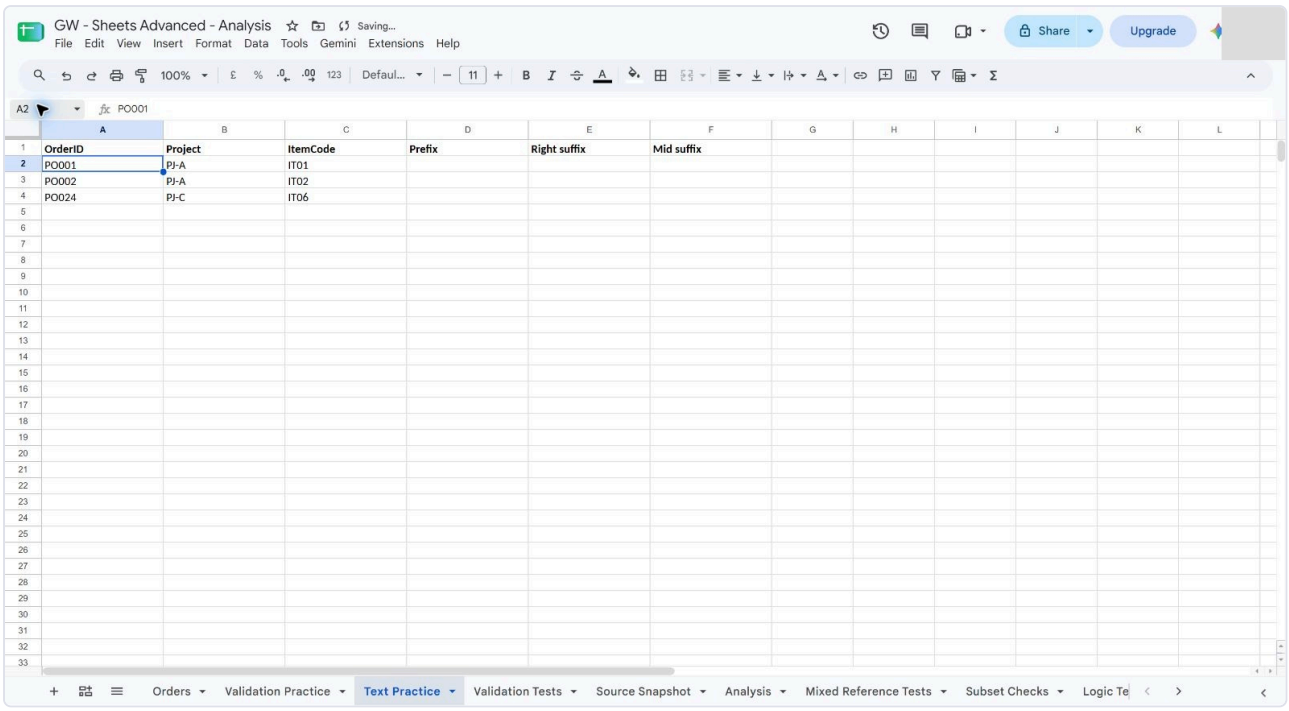
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

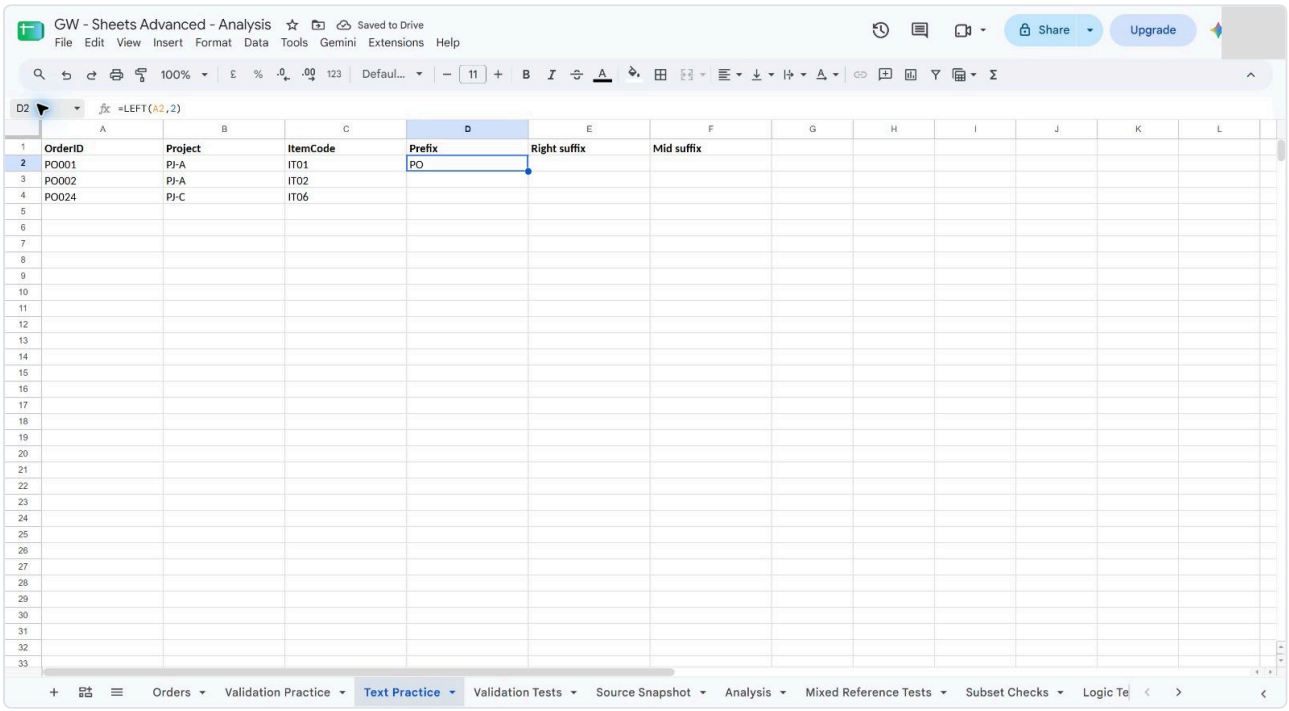
Use Orders and Rates. The supplied IDs follow PO plus three digits; this is a training convention.

4.1 Extract Identifier Parts

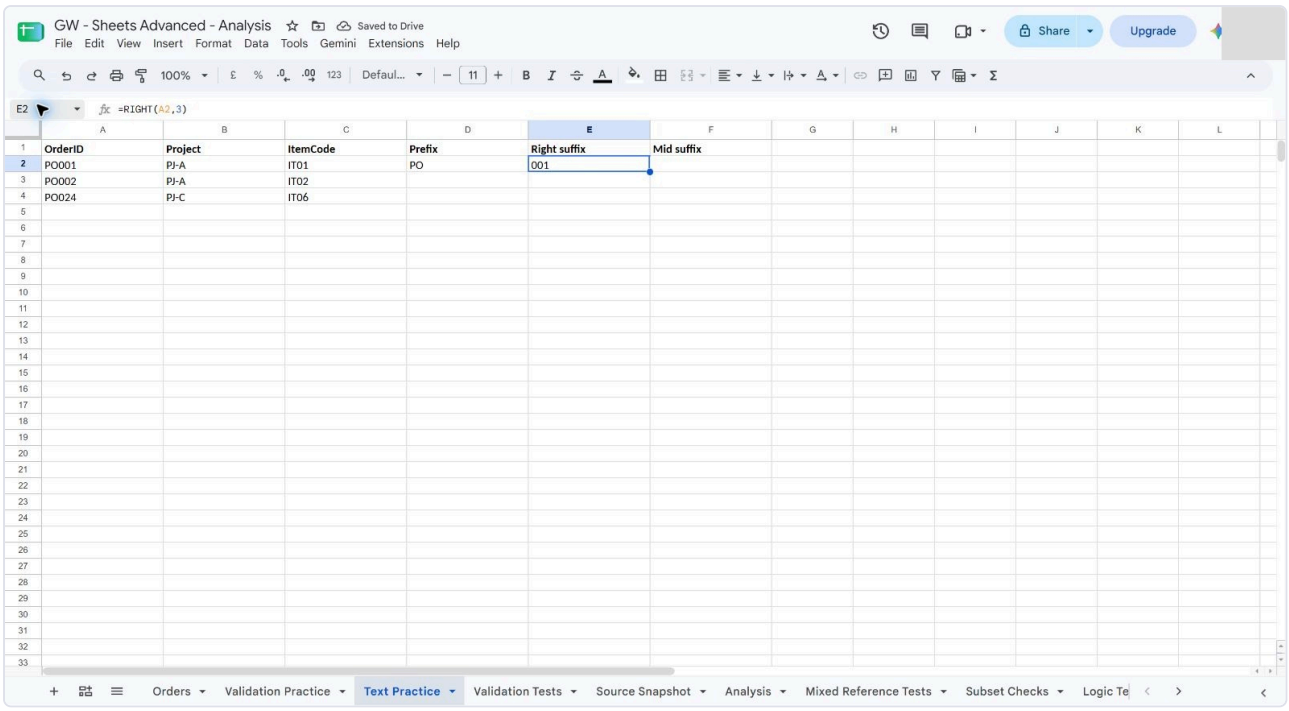
1. Create a separate **Text Practice** tab. Copy OrderID, Project and ItemCode values from Orders rows2,3,25 using the card. Put headers OrderID / Project / ItemCode / Prefix / Right suffix / Mid suffix in A1:F1. Leave the original Orders fields unchanged.



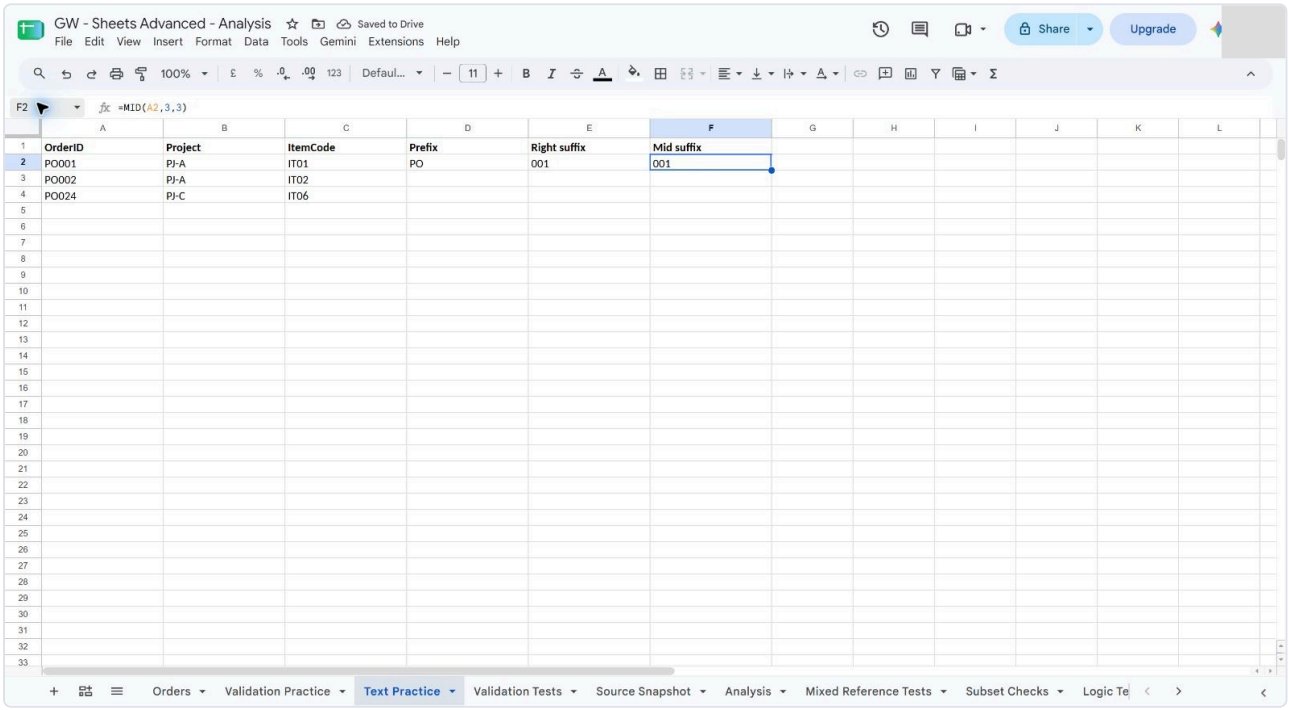
2. In D2 enter **=LEFT(A2,2)**. It returns **PO**.



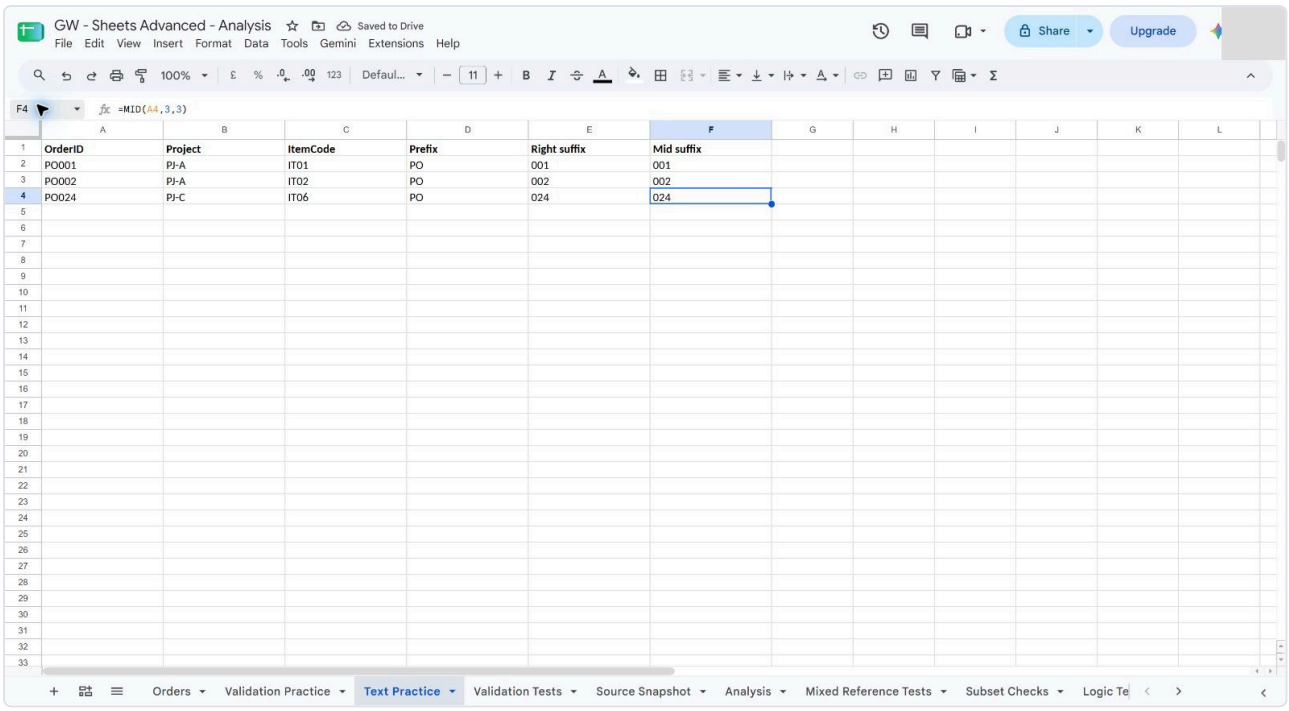
3. In E2 enter **=RIGHT(A2,3)**. The result is the text **001**, retaining its leading zeros.



4. In F2 enter **=MID(A2,3,3)**. It also returns **001**. Extracted text is not a newly assigned identifier and does not prove that the entire key is valid.

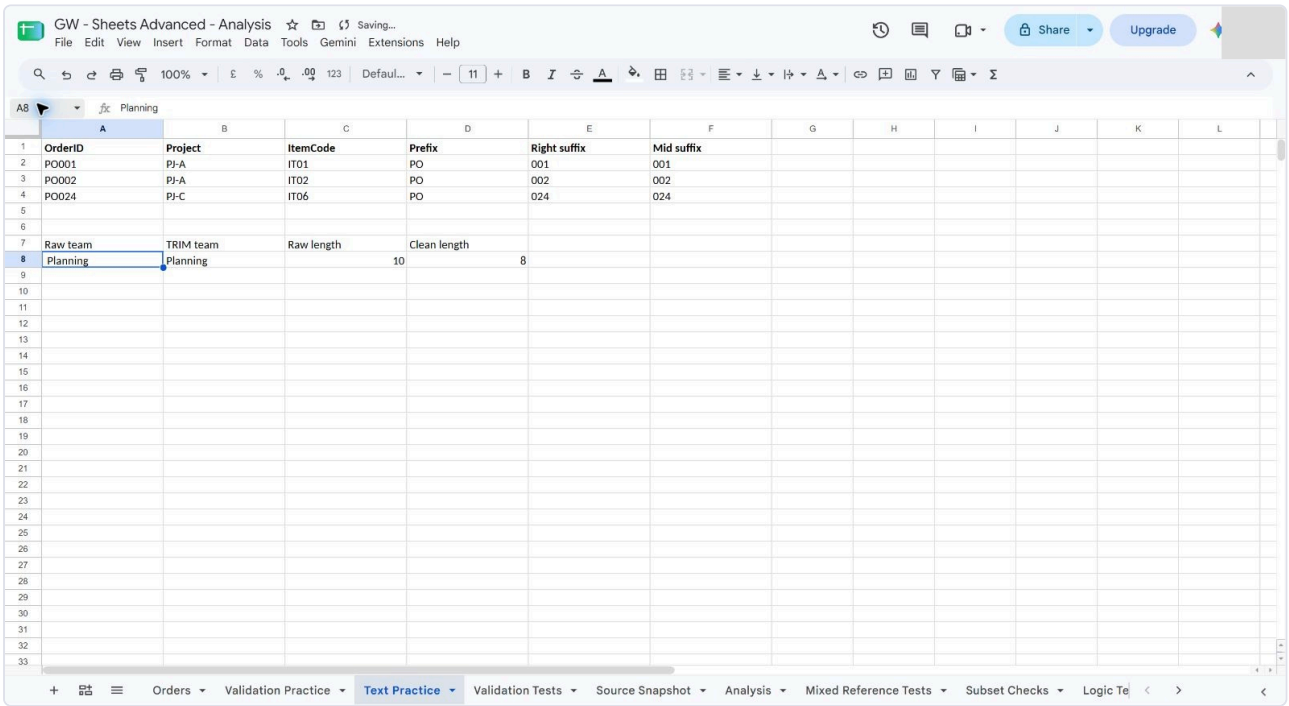


5. Select D2:F4 and **Fill down** (Ctrl+D). The two suffix columns show **001 /002 /024**. Inspect the final sample's own-row formula.

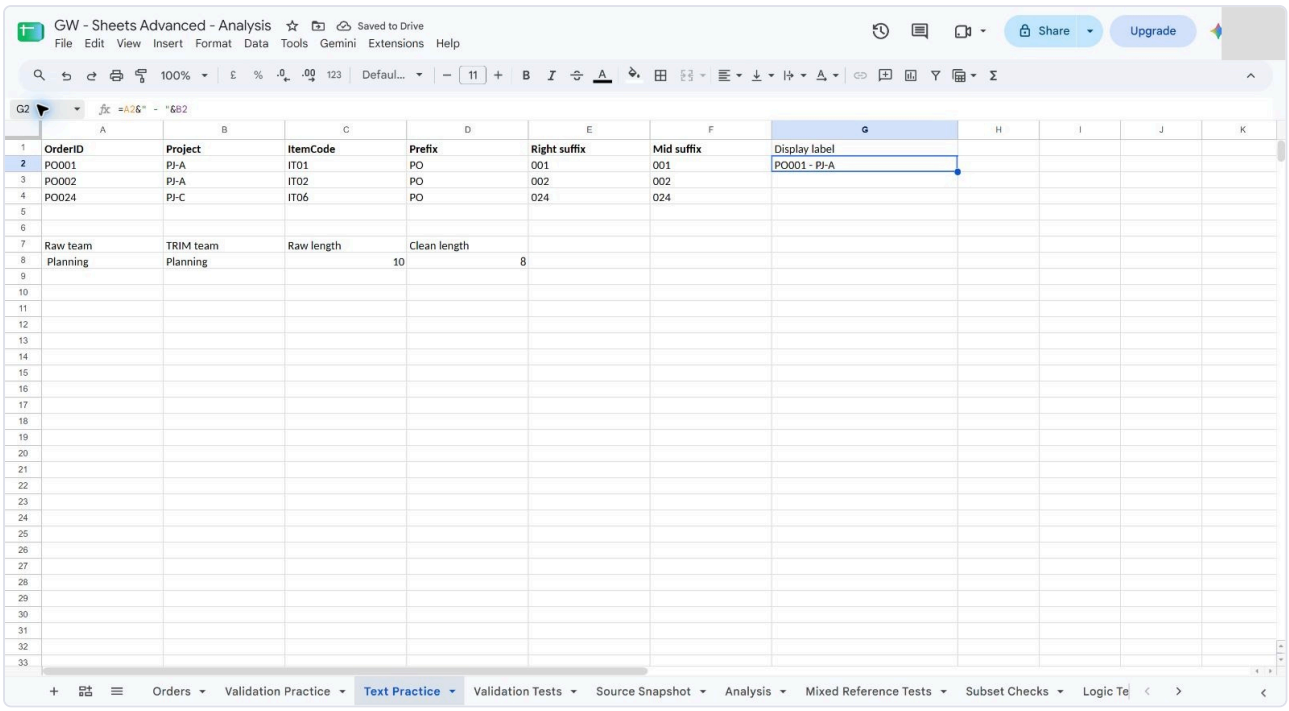


4.2 Combine and Clean Text

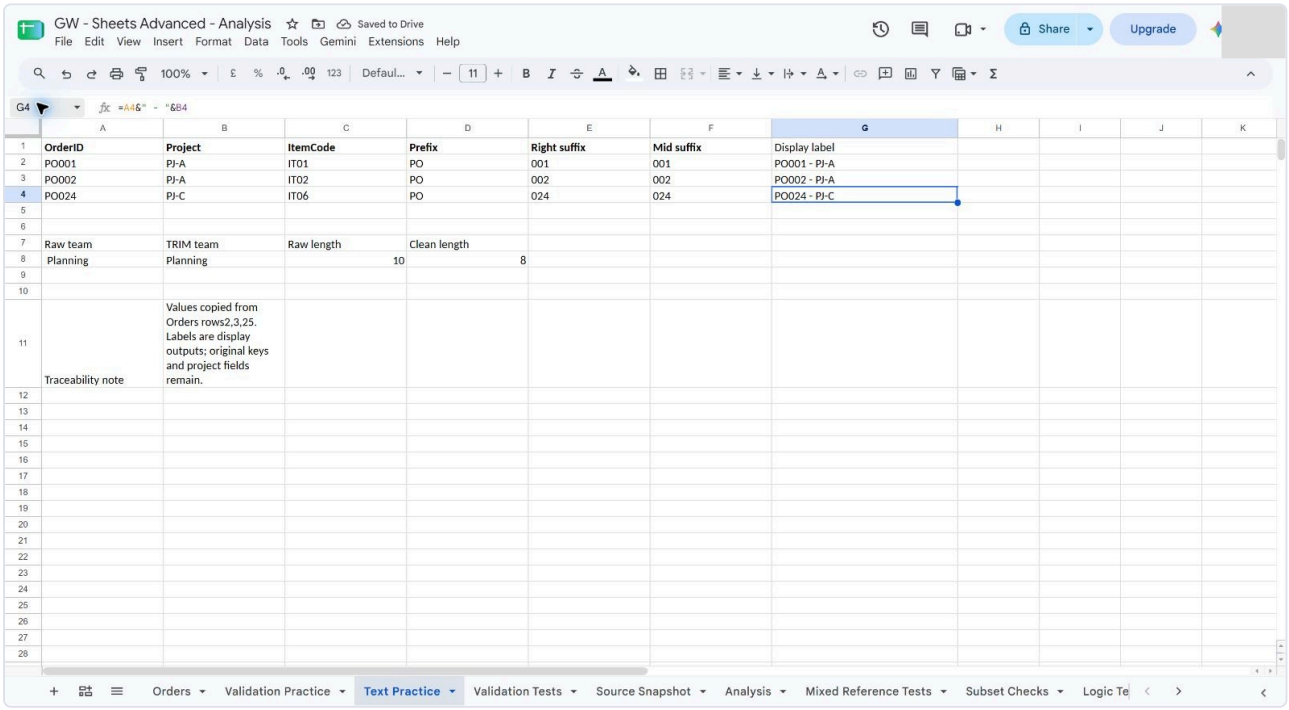
1. On Text Practice put Raw team / TRIM team / Raw length / Clean length in A7:D7. Enter Planning from DirtyText.csv into A8 with one leading and one trailing ordinary space, as a single-cell value. Use **=TRIM(A8)** in B8, **=LEN(A8)** in C8 and **=LEN(B8)** in D8. Expect Planning and lengths **10 /8**. Do not lose the raw spaces during entry.



2. Add Display label in G1 and **=A2&" - "&B2** in G2. This combines OrderID and Project on the practice tab; do not add display-only fields to the maintained source for this test.

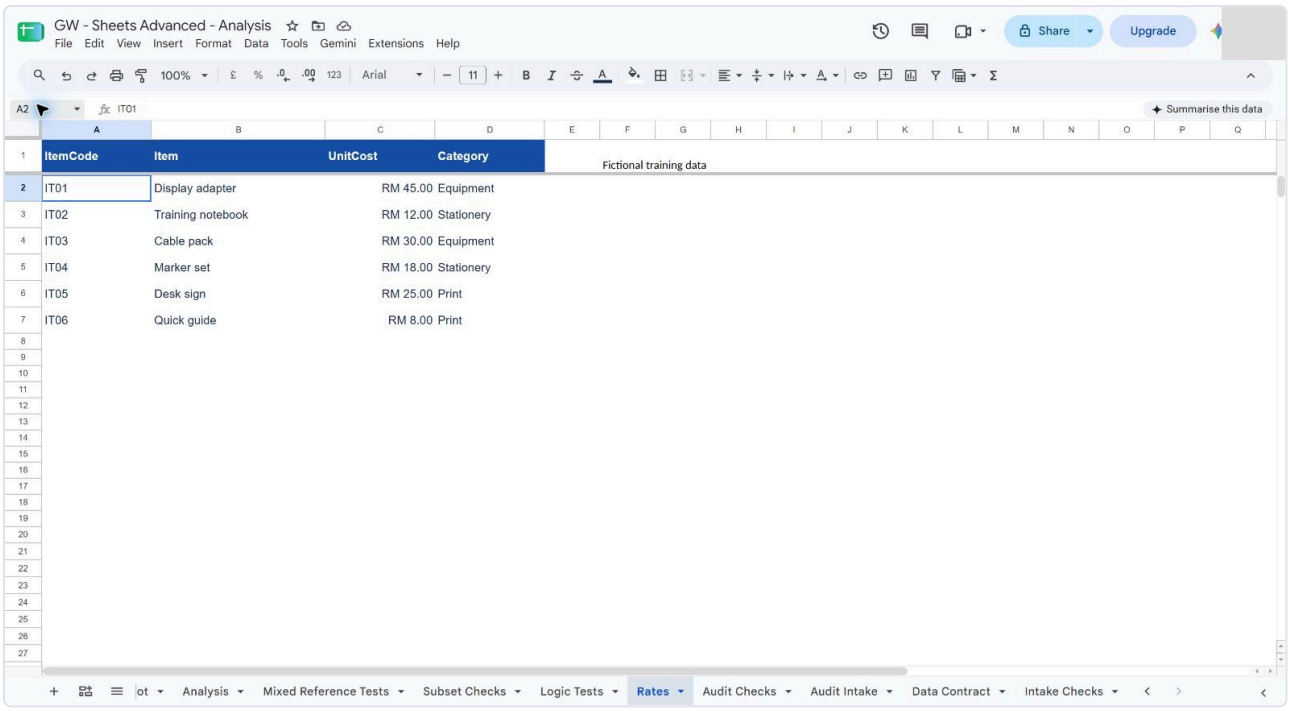


3. Fill G2:G4 down. Verify labels **PO001 - PJ-A**, **PO002 - PJ-A**, **PO024 - PJ-C** while separate OrderID/Project/ItemCode fields remain available. Add the source-row traceability note outside the table.

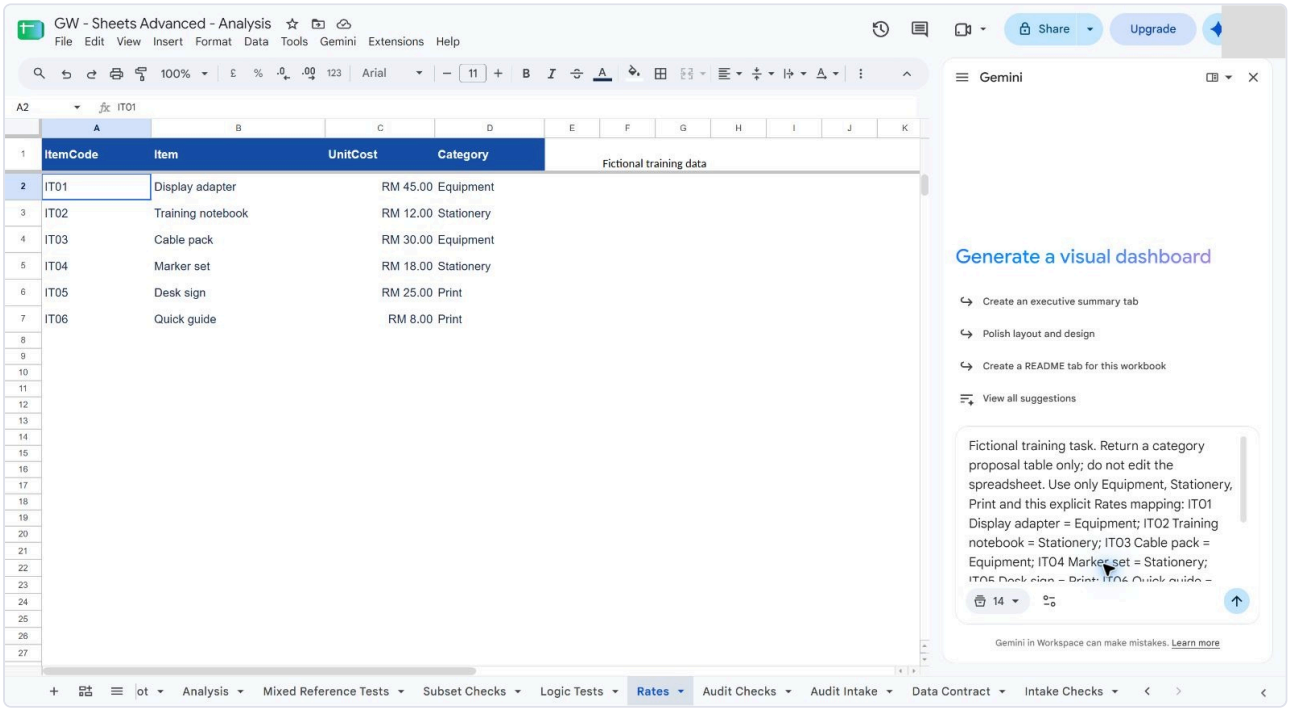


4.3 Check Categorisation

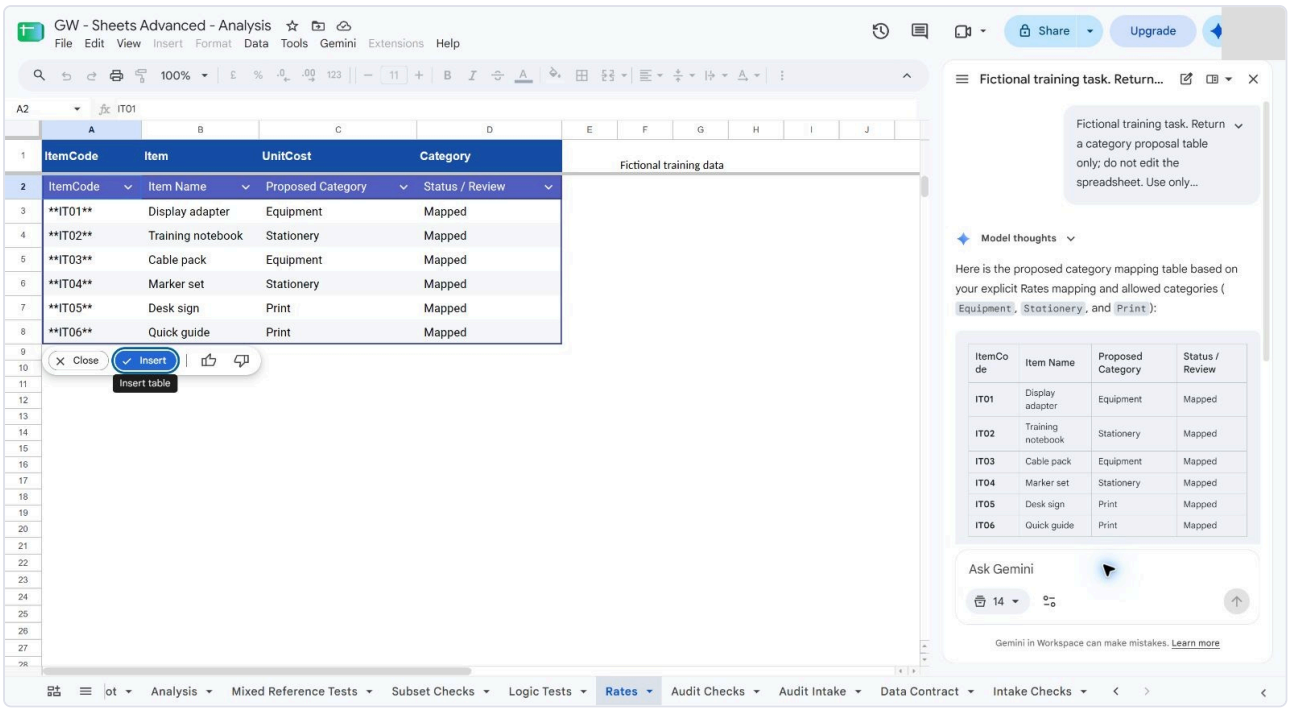
1. Inspect **Rates A1:D7**. Its six ItemCodes map to Equipment / Stationery / Equipment / Stationery / Print / Print in code order IT01–IT06. Rates is the reference, not an AI answer.



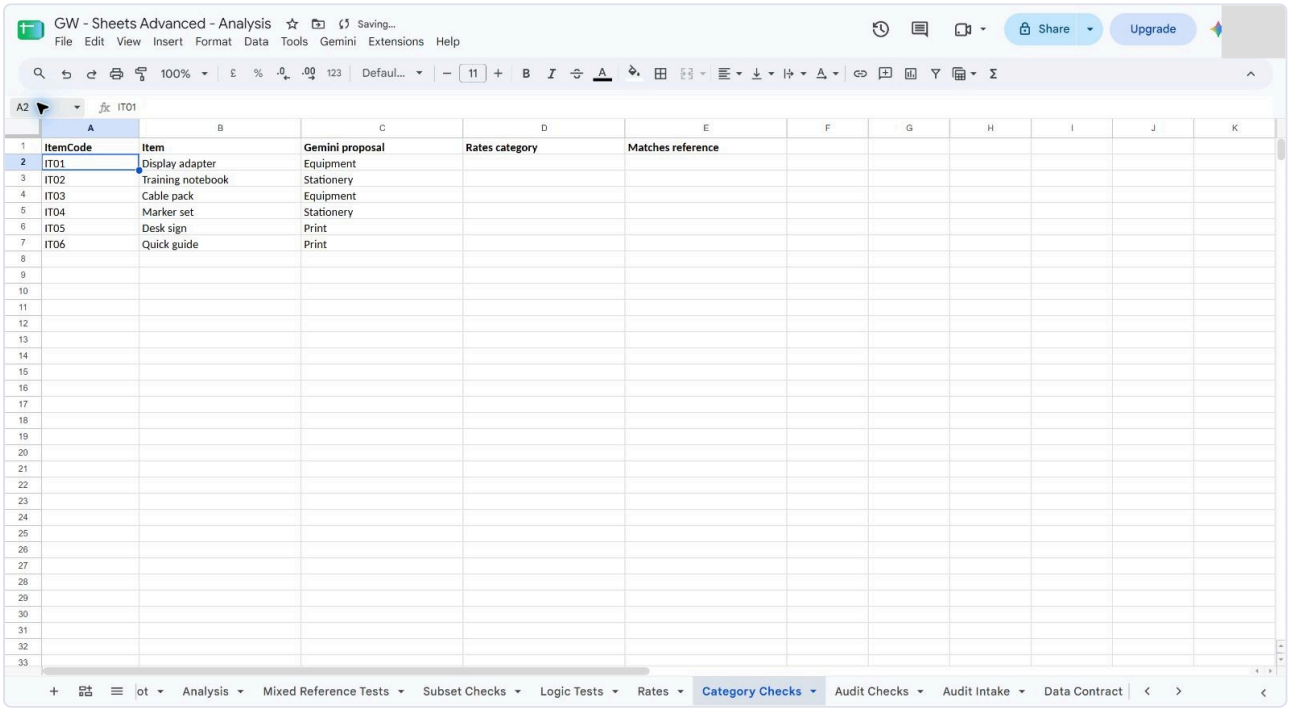
2. If Gemini is available, open **Ask Gemini** and submit the supplied proposal-only prompt. It includes the explicit six-item mapping and requests no spreadsheet edits. Keep the request visible before submitting; do not apply an Insert suggestion to the maintained Rates table.



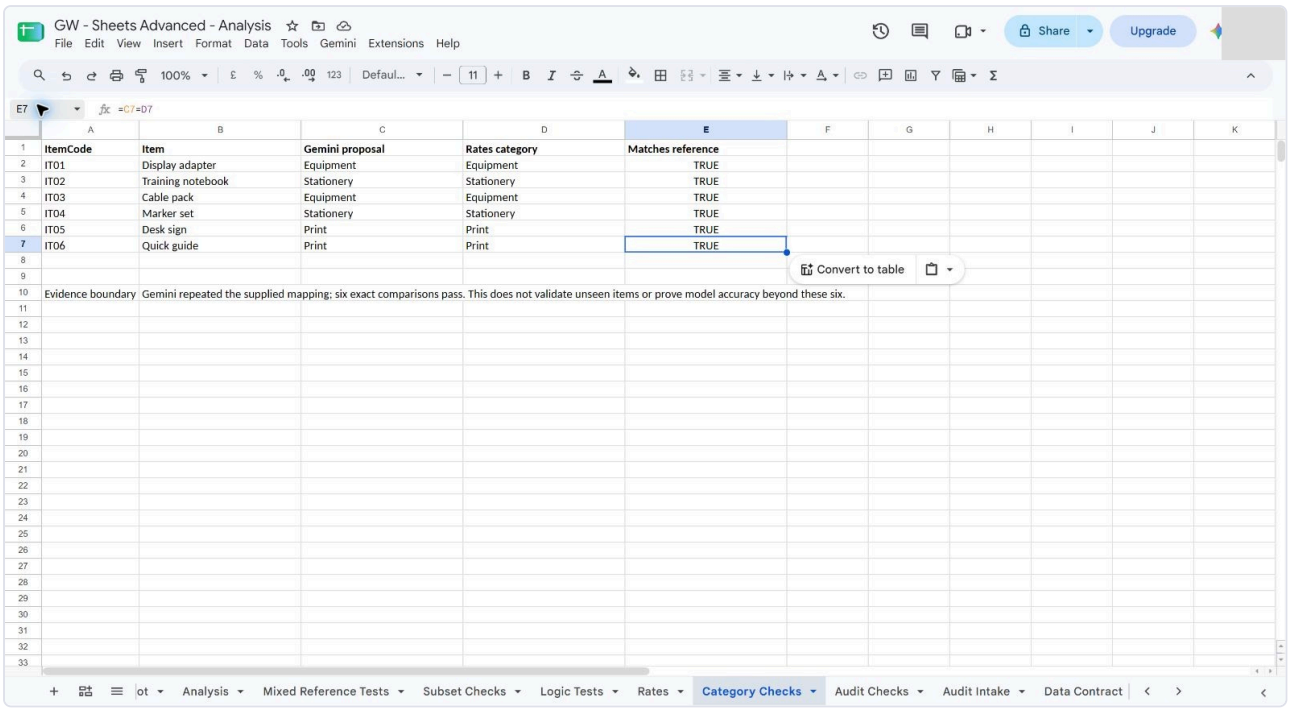
3. Read the actual six proposals. In the captured author run Gemini repeated all six supplied categories and advised review for unmapped or conflicting inputs. Model output can vary. This prompt tests adherence to a provided mapping, not independent classification of unseen items.



4. Create **Category Checks** with ItemCode / Item / Gemini proposal / Rates category / Matches reference in A1:E1. Record the six actual proposals in A2:C7 before calculating comparisons. If Gemini is unavailable, label the fallback as a manual reference exercise; do not invent an AI response.

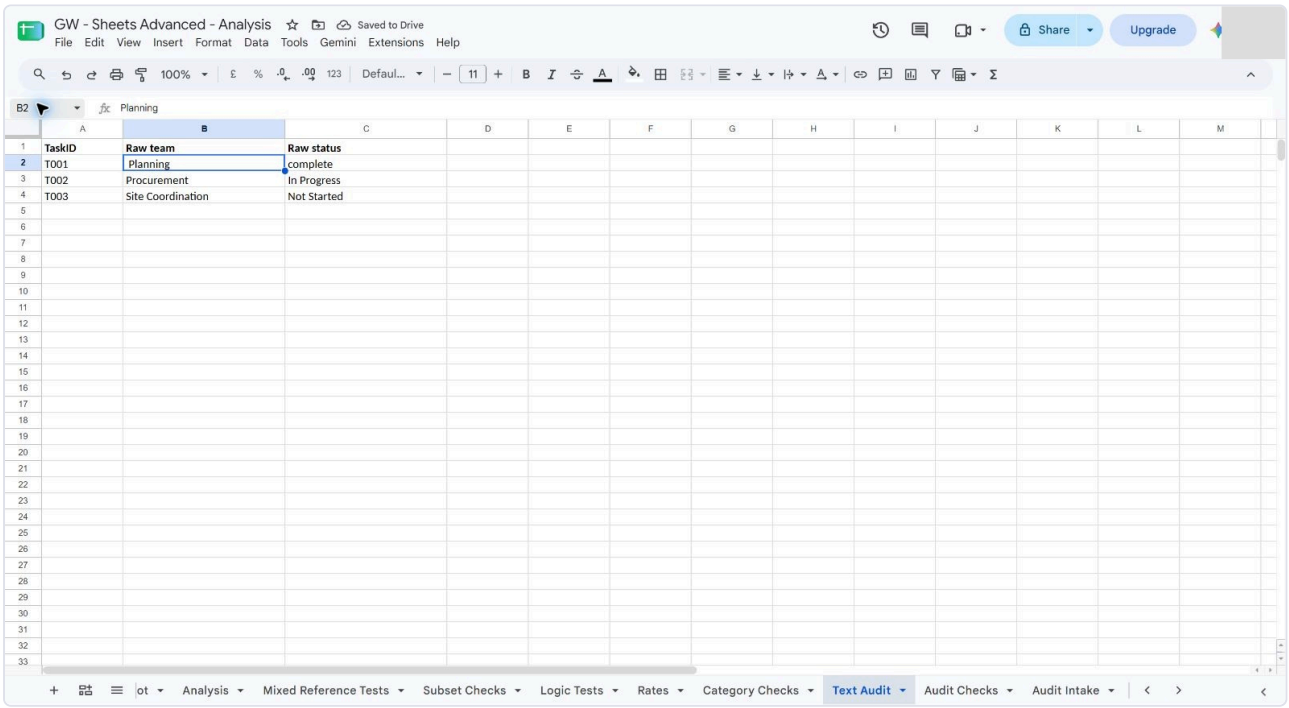


5. In D2 enter **=VLOOKUP(A2,Rates!$A$2:$D$7,4,FALSE)** and E2 **=C2=D2**. Fill D2:E7 down and inspect every row. All six comparisons are TRUE in this author run. Record the limited scope of that evidence.



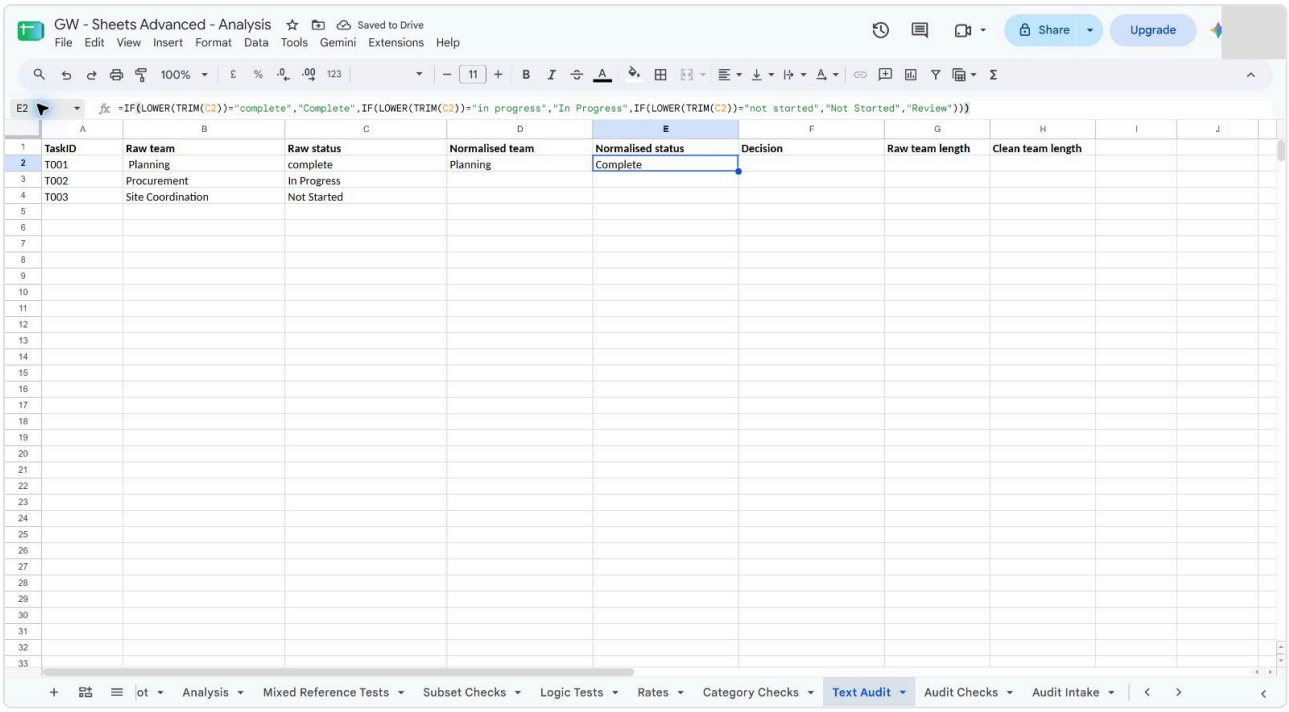
4.4 Build a Reversible Text Normalisation Pipeline

1. Create **Text Audit**. Retain exact DirtyText.csv raw columns TaskID / Raw team / Raw status in A1:C4. In this author run, TSV block paste removed surrounding spaces; single-cell entry restored B2 to Planning with one leading and one trailing ordinary space, and B4 to Site Coordination with one trailing ordinary space, before adding formulas. The verified route is clipboard entry, not a completed native CSV file-import test. Use the raw-length checks to detect entry loss.

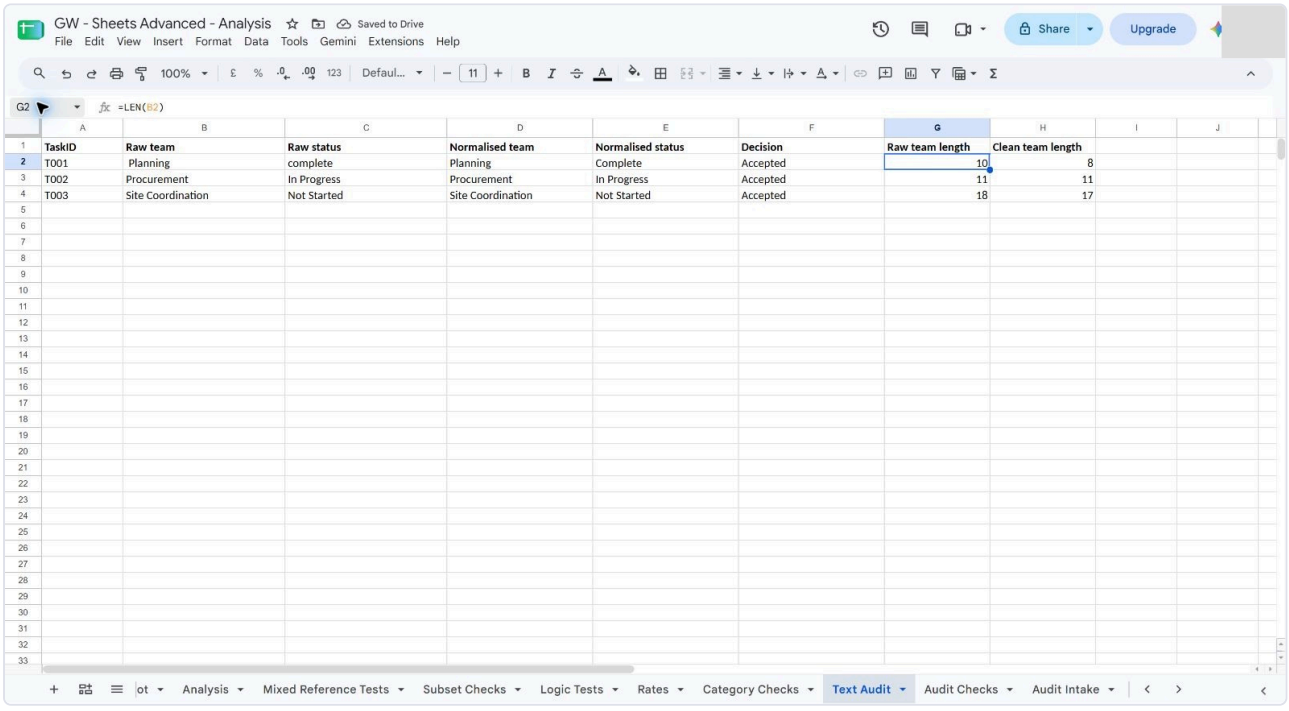


2. Add D:H headers Normalised team / Normalised status / Decision / Raw team length / Clean team length. In D2 use **=TRIM(B2)**. E2 uses the exact explicit-case formula on the card for **Complete /In**

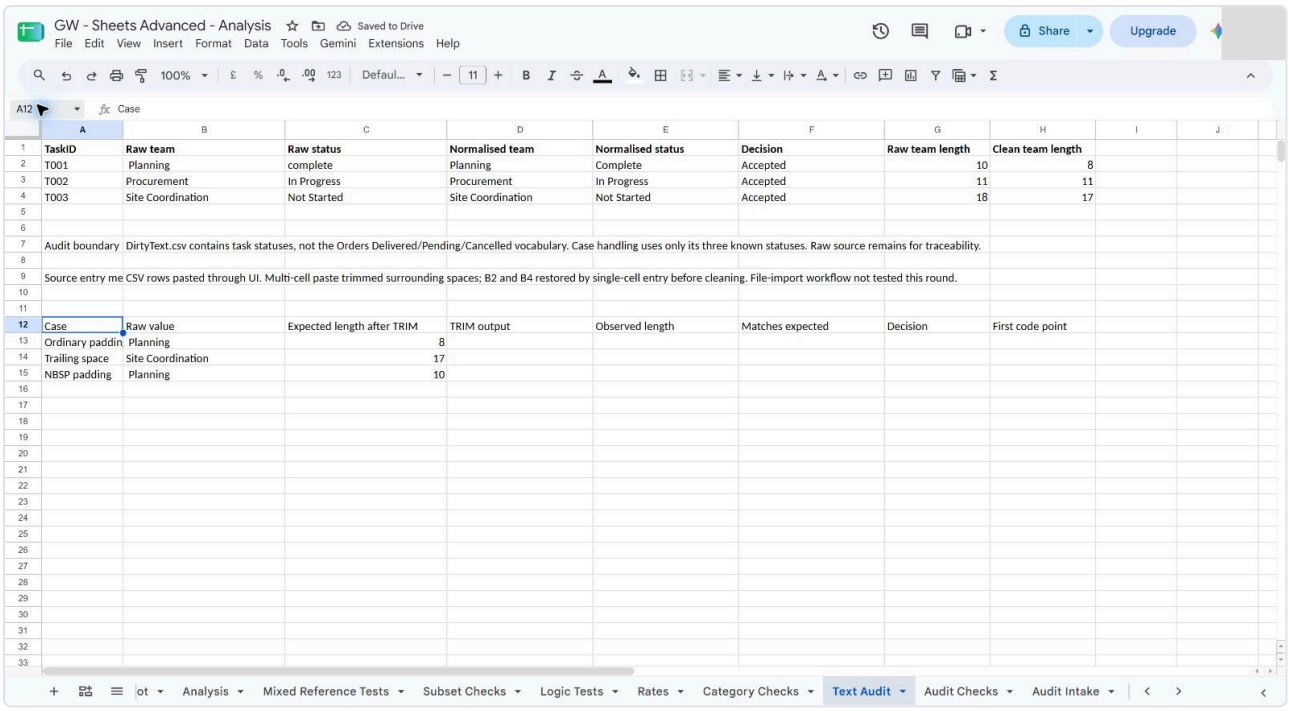
Progress /Not Started. These are task statuses from DirtyText.csv, separate from the Orders status vocabulary; do not infer a mapping to Delivered/Pending/Cancelled.



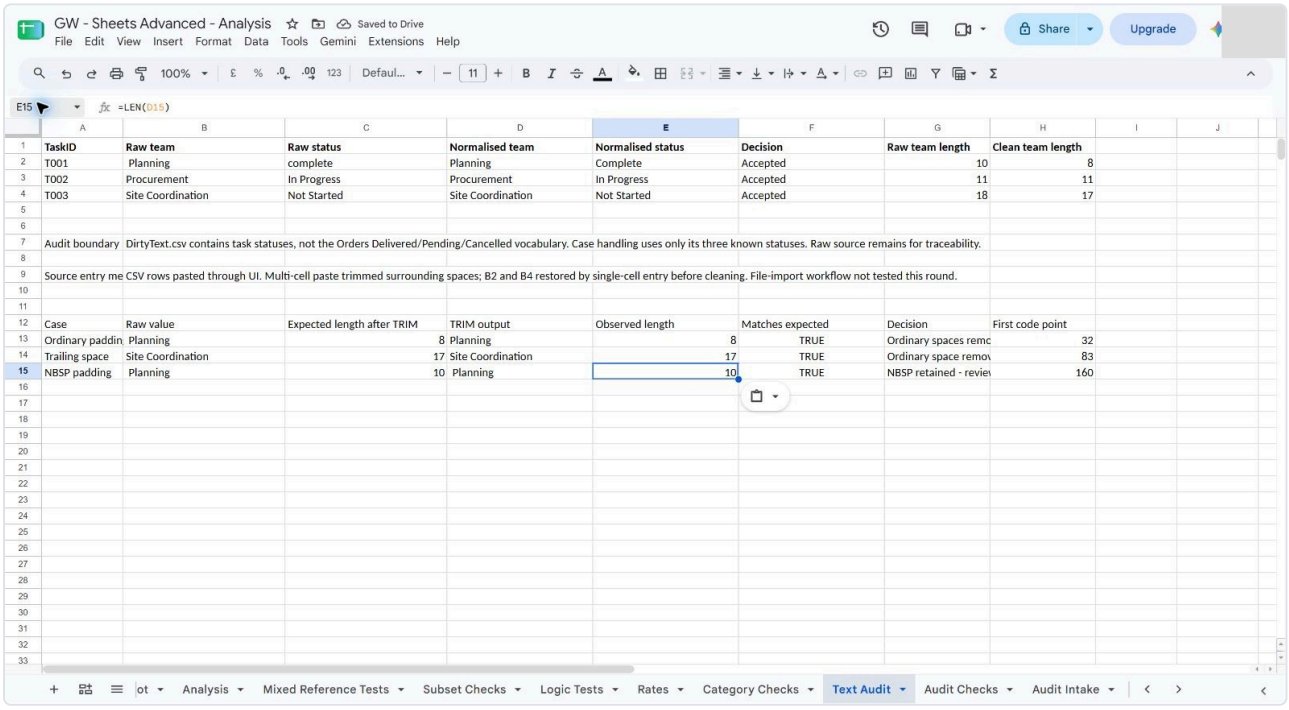
3. F2 accepts only the three known team names and a recognised task status. G2/H2 compare LEN(B2)/LEN(D2). Fill D2:H4 down. All three sample rows are Accepted; raw/clean team lengths are **10/8**, **11/11**, **18/17**. Add the source-entry and vocabulary notes outside the data.



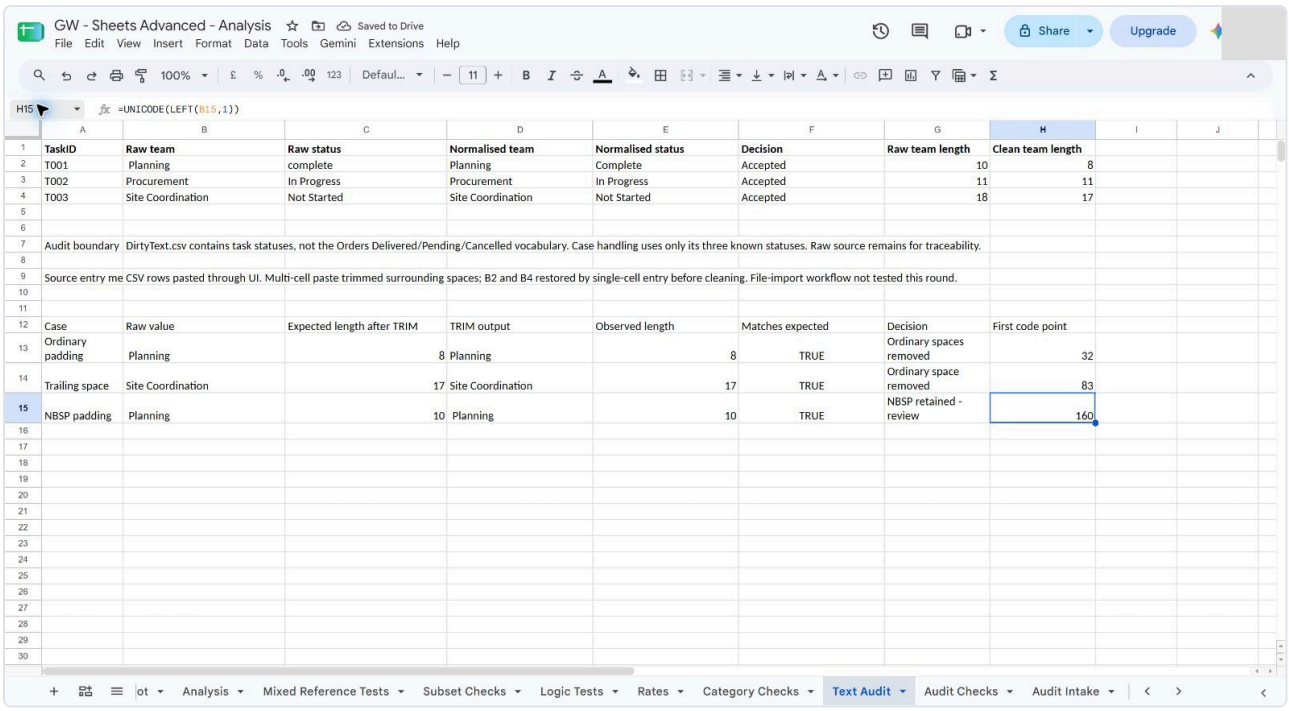
4. In A12:H15 create the whitespace test card. Record expected post-TRIM lengths **8 /17 /10** before actual formulas. Enter raw values individually to retain their characters. The third case is a literal non-breaking-space character before and after Planning; it is labelled as synthetic author casework, not an additional CSV row.



5. Use D13 **=TRIM(B13)**, E13 **=LEN(D13)**, F13 **=E13=C13**, H13 **=UNICODE(LEFT(B13,1))**; fill the formula columns through row15. Ordinary spaces are removed. The NBSP result remains length **10**, with first code point **160**. Keep that exception for review rather than silently treating TRIM as universal whitespace cleaning.



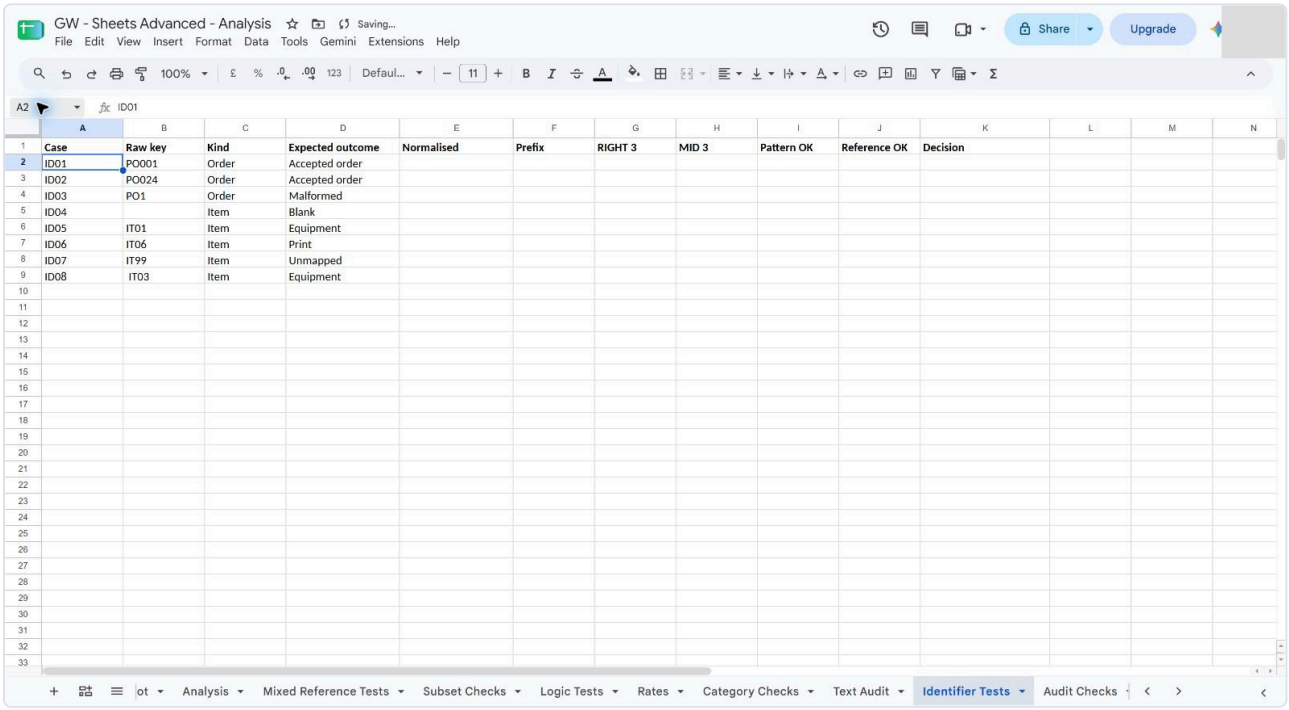
6. Wrap A12:H15 to make every case and decision readable. All three expected-length comparisons pass, including the expected retained-NBSP behavior; this does not make the unusual space an accepted team value.



TaskID	Raw team	Raw status	Normalised team	Normalised status	Decision	Raw team length	Clean team length
T001	Planning	complete	Planning	Complete	Accepted	10	8
T002	Procurement	In Progress	Procurement	In Progress	Accepted	11	11
T003	Site Coordination	Not Started	Site Coordination	Not Started	Accepted	18	17

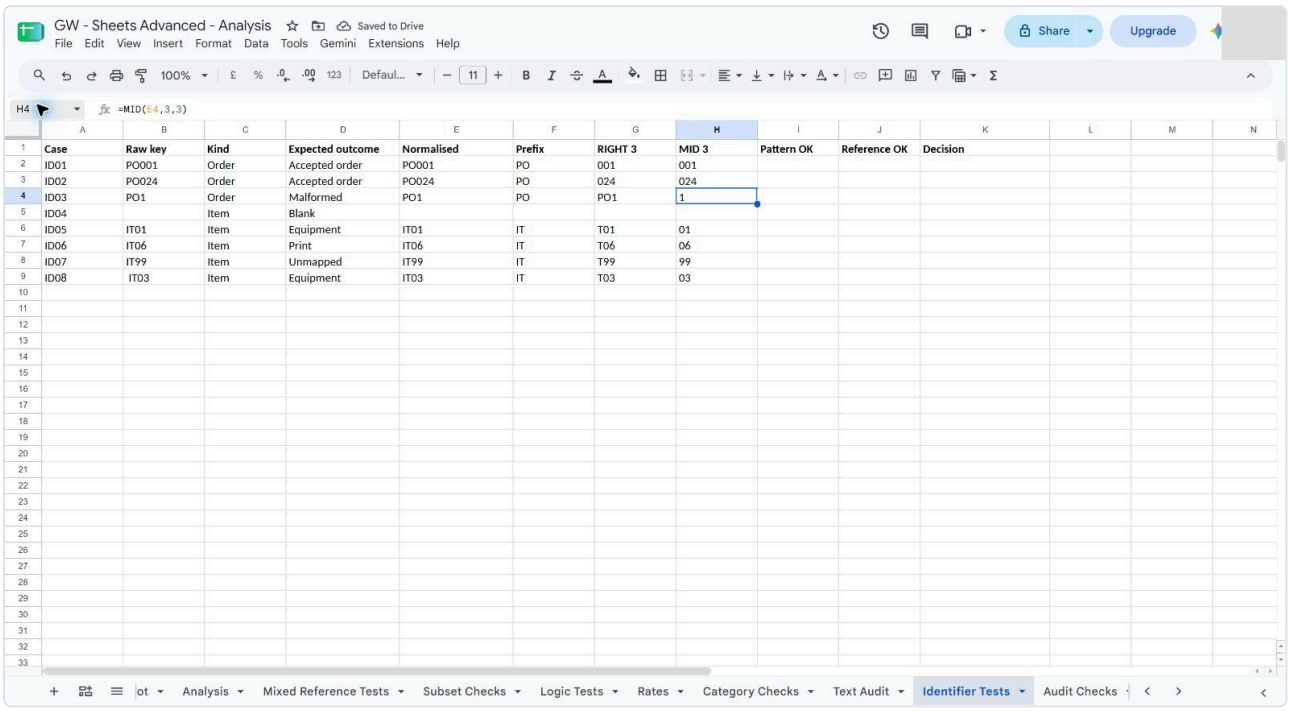
Case	Raw value	Expected length after TRIM	TRIM output	Observed length	Matches expected	Decision	First code point
Ordinary padding	Planning	8	Planning	8	TRUE	Ordinary spaces removed	32
Trailing space	Site Coordination	17	Site Coordination	17	TRUE	Ordinary space removed	83
NBSP padding	Planning	10	Planning	10	TRUE	NBSP retained - review	160

7. Create **Identifier Tests** using the eight-row card, with expected outcomes recorded before formulas. Keep B5 truly blank and enter B9 as IT03 with one leading and one trailing ordinary space separately. Order keys require PO plus three digits; item keys require IT plus two digits and a mapping entry.

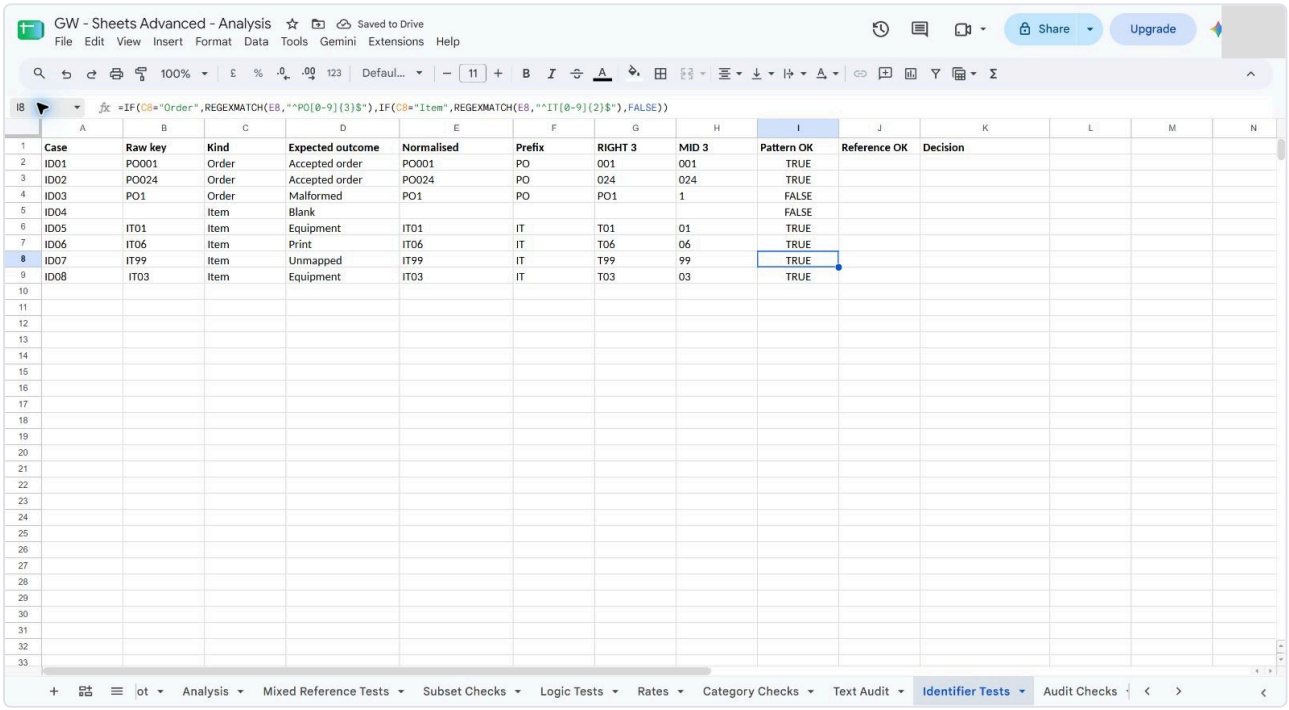


Case	Raw key	Kind	Expected outcome	Normalised	Prefix	RIGHT 3	MID 3	Pattern OK	Reference OK	Decision
ID01	PO001	Order	Accepted order							
ID02	PO024	Order	Accepted order							
ID03	PO1	Order	Malformed							
ID04		Item	Blank							
ID05	IT01	Item	Equipment							
ID06	IT06	Item	Print							
ID07	IT99	Item	Unmapped							
ID08	IT03	Item	Equipment							

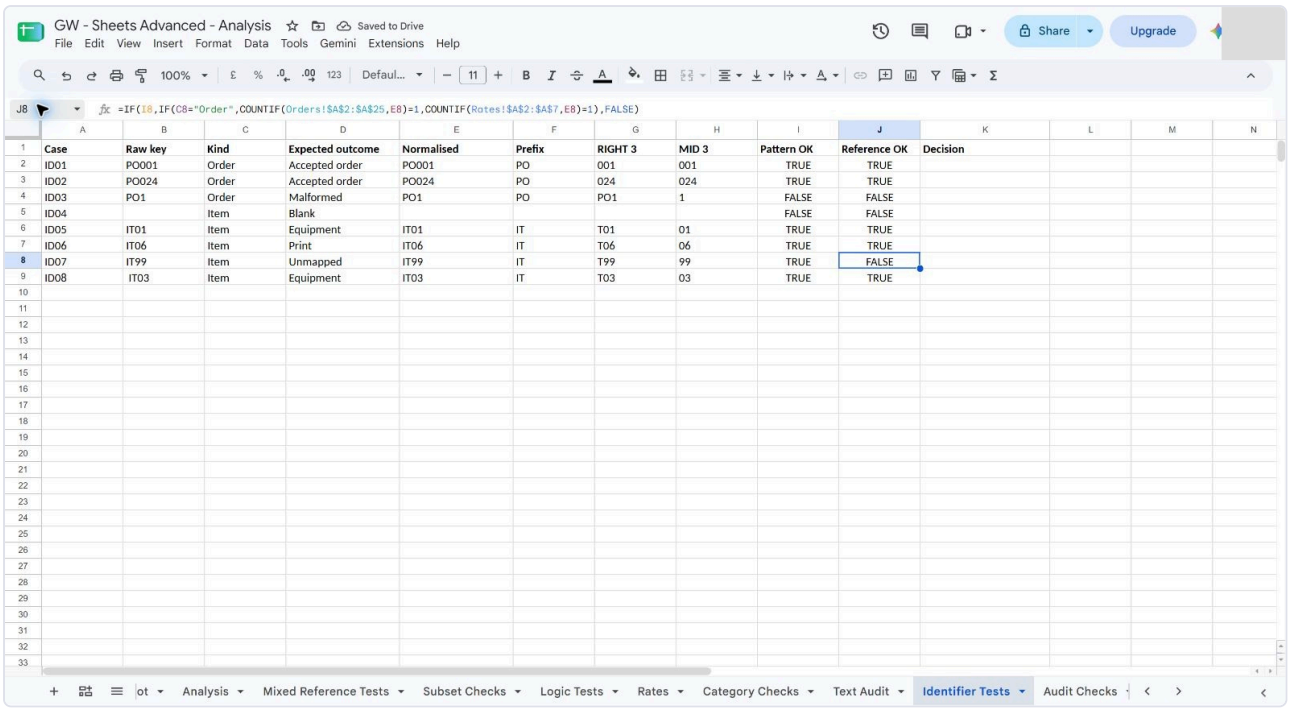
8. Normalise B with TRIM in E; apply LEFT(E2,2), RIGHT(E2,3) and MID(E2,3,3) in F:H and fill through row9. **PO1** still yields substrings despite being malformed. For IT01, RIGHT3 returns **T01**, while MID returns **01**; the three-digit order-suffix convention must not be applied blindly to item keys.



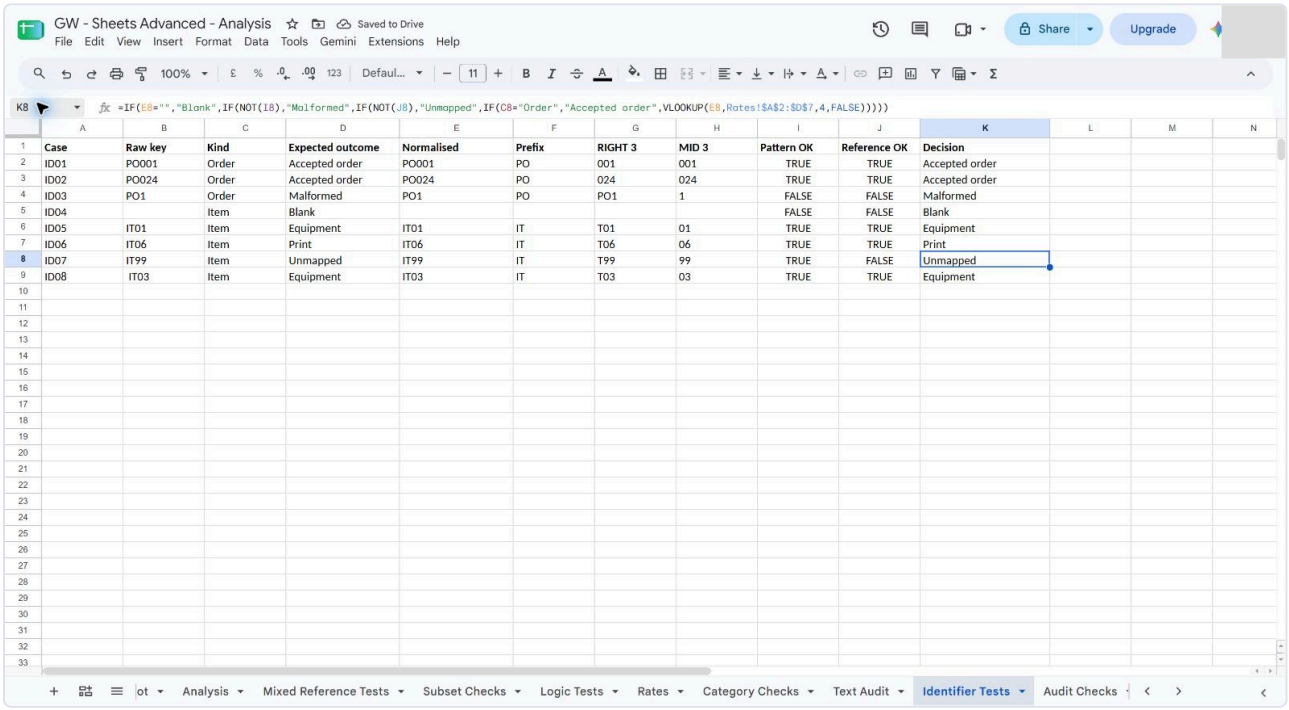
9. Use the card's complete-pattern REGEXMATCH formula in I2 and fill down. PO1 and the blank fail. IT99 passes its pattern, which proves syntax only.



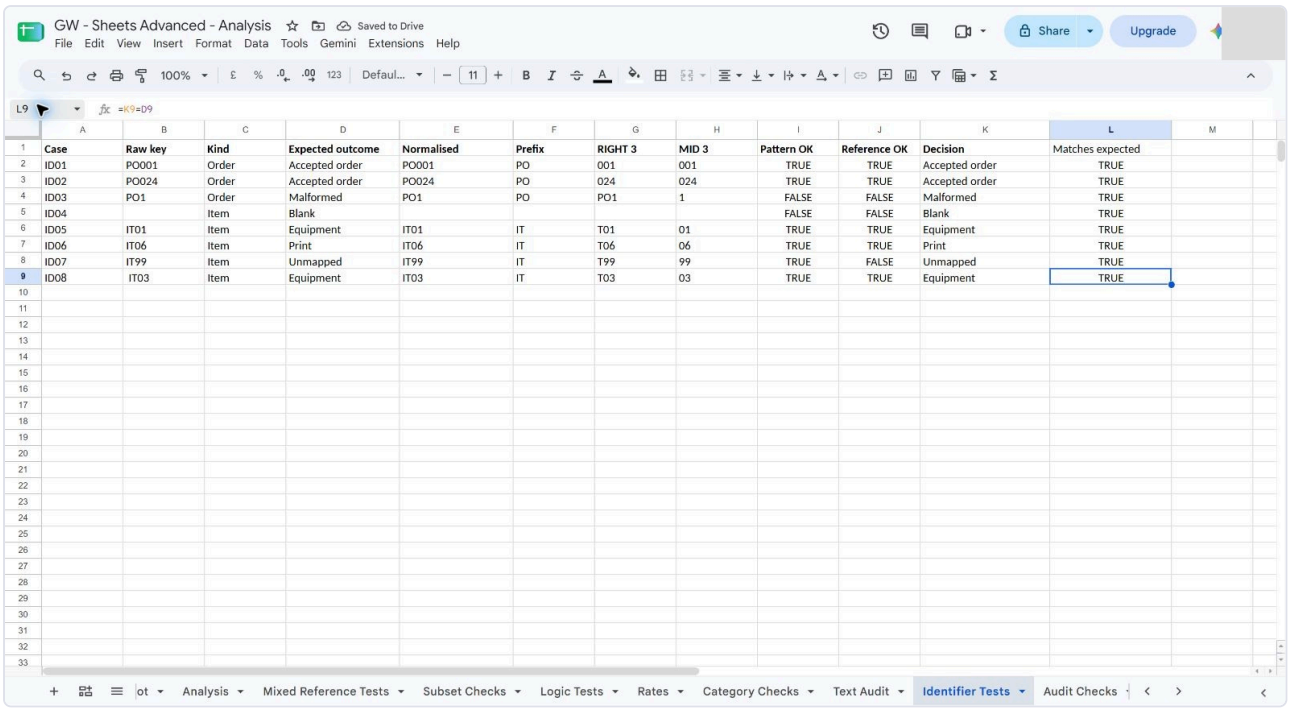
10. In J2 check membership against bounded Orders A2:A25 or Rates A2:A7 only after the pattern passes. Fill down. **IT99 is TRUE for pattern and FALSE for reference membership**. A plausible-looking code is not a valid category mapping.



11. Use K2's decision formula and fill down. It keeps **Blank /Malformed /Unmapped** separate, accepts the two existing order keys and returns exact Rates categories for the three mapped item cases. Raw values and extraction columns remain available for audit.

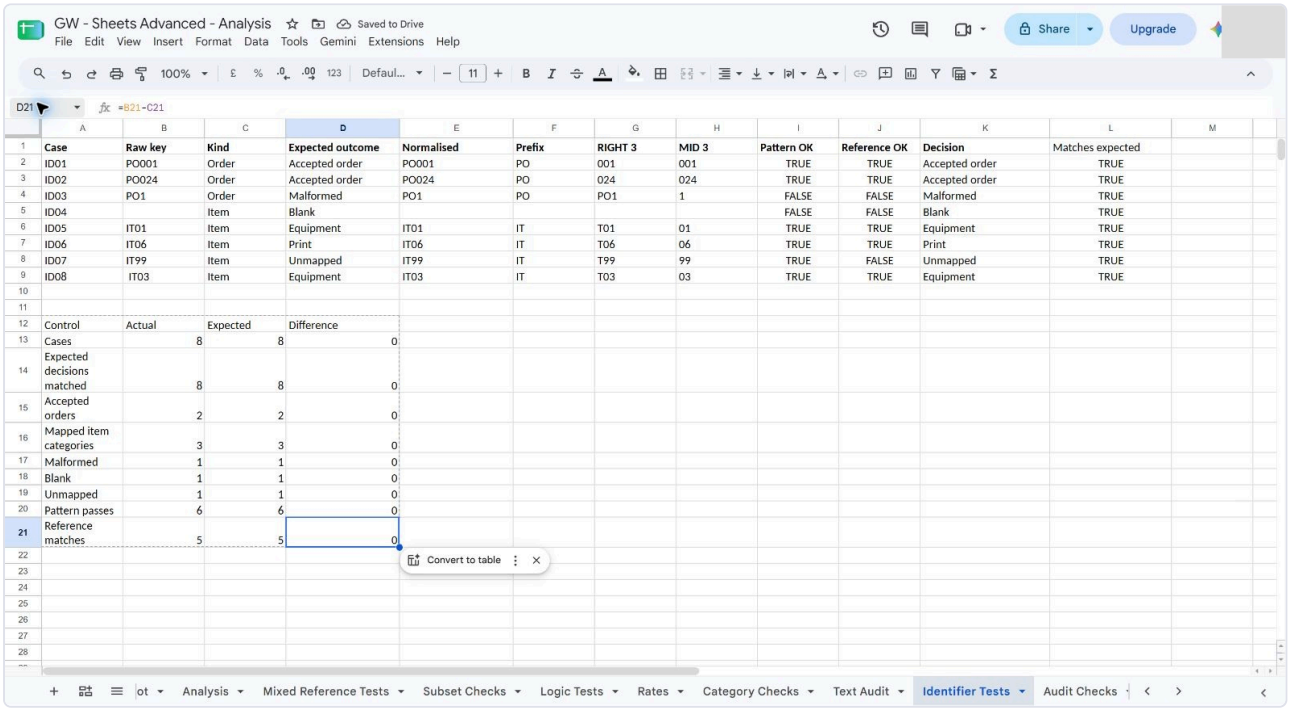


12. In L2 compare **=K2=D2** and fill through row9. All eight decisions match their recorded expected outcomes. These are guided author cases, not the independent mixed-intake challenge.



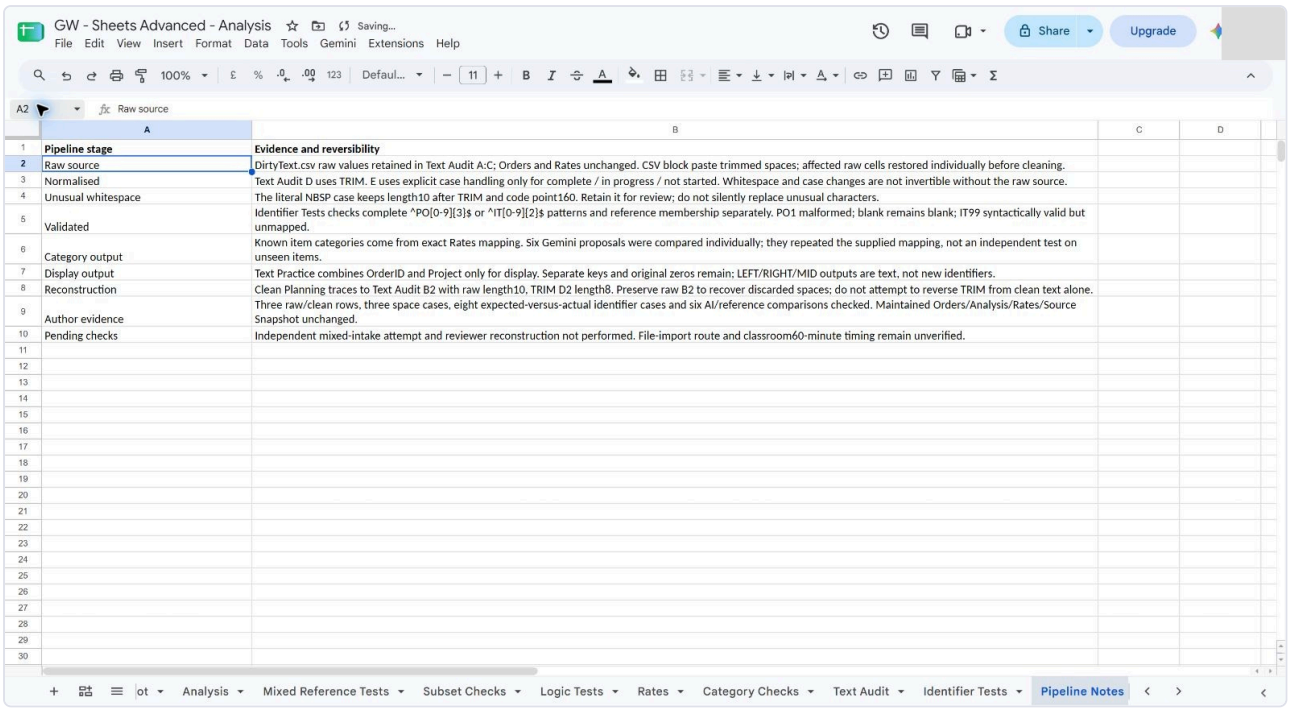
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Case	Raw key	Kind	Expected outcome	Normalised	Prefix	RIGHT 3	MID 3	Pattern OK	Reference OK	Decision	Matches expected	
2	ID01	PO001	Order	Accepted order	PO001	PO	001	001	TRUE	TRUE	Accepted order	TRUE	
3	ID02	PO024	Order	Accepted order	PO024	PO	024	024	TRUE	TRUE	Accepted order	TRUE	
4	ID03	PO1	Order	Malformed	PO1	PO	PO1	1	FALSE	FALSE	Malformed	TRUE	
5	ID04		Item	Blank					FALSE	FALSE	Blank	TRUE	
6	ID05	IT01	Item	Equipment	IT01	IT	T01	01	TRUE	TRUE	Equipment	TRUE	
7	ID06	IT06	Item	Print	IT06	IT	T06	06	TRUE	TRUE	Print	TRUE	
8	ID07	IT99	Item	Unmapped	IT99	IT	T99	99	TRUE	FALSE	Unmapped	TRUE	
9	ID08	IT03	Item	Equipment	IT03	IT	T03	03	TRUE	TRUE	Equipment	TRUE	

13. Add A12:D21 controls: **8 cases /8 matches /2 accepted orders /3 mapped item cases /1 malformed /1 blank /1 unmapped /6 pattern passes /5 reference matches**. Compare each actual to its expected count. All nine differences are zero; inspect contributing rows as well as totals.

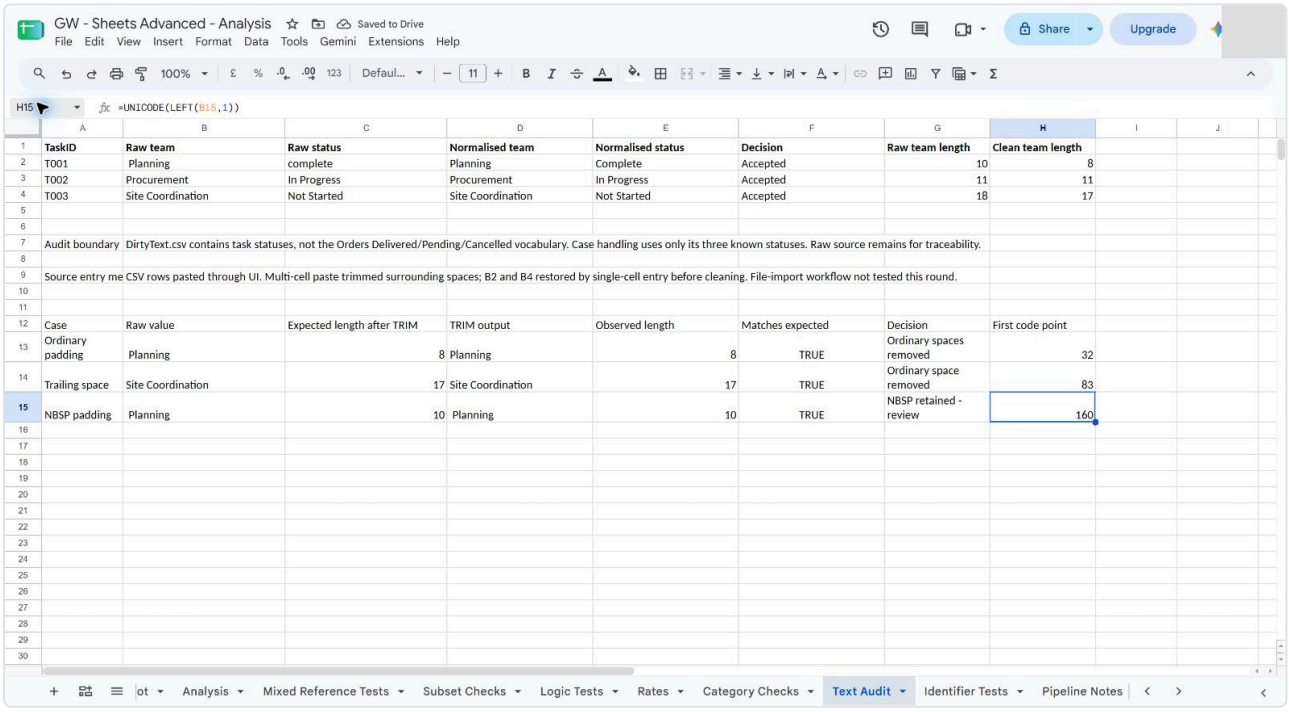


	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Case	Raw key	Kind	Expected outcome	Normalised	Prefix	RIGHT 3	MID 3	Pattern OK	Reference OK	Decision	Matches expected	
2	ID01	PO001	Order	Accepted order	PO001	PO	001	001	TRUE	TRUE	Accepted order	TRUE	
3	ID02	PO024	Order	Accepted order	PO024	PO	024	024	TRUE	TRUE	Accepted order	TRUE	
4	ID03	PO1	Order	Malformed	PO1	PO	PO1	1	FALSE	FALSE	Malformed	TRUE	
5	ID04		Item	Blank					FALSE	FALSE	Blank	TRUE	
6	ID05	IT01	Item	Equipment	IT01	IT	T01	01	TRUE	TRUE	Equipment	TRUE	
7	ID06	IT06	Item	Print	IT06	IT	T06	06	TRUE	TRUE	Print	TRUE	
8	ID07	IT99	Item	Unmapped	IT99	IT	T99	99	TRUE	FALSE	Unmapped	TRUE	
9	ID08	IT03	Item	Equipment	IT03	IT	T03	03	TRUE	TRUE	Equipment	TRUE	
10													
11													
12	Control	Actual	Expected	Difference									
13	Cases	8	8	0									
14	Expected decisions matched	8	8	0									
15	Accepted orders	2	2	0									
16	Mapped item categories	3	3	0									
17	Malformed	1	1	0									
18	Blank	1	1	0									
19	Unmapped	1	1	0									
20	Pattern passes	6	6	0									
21	Reference matches	5	5	0									

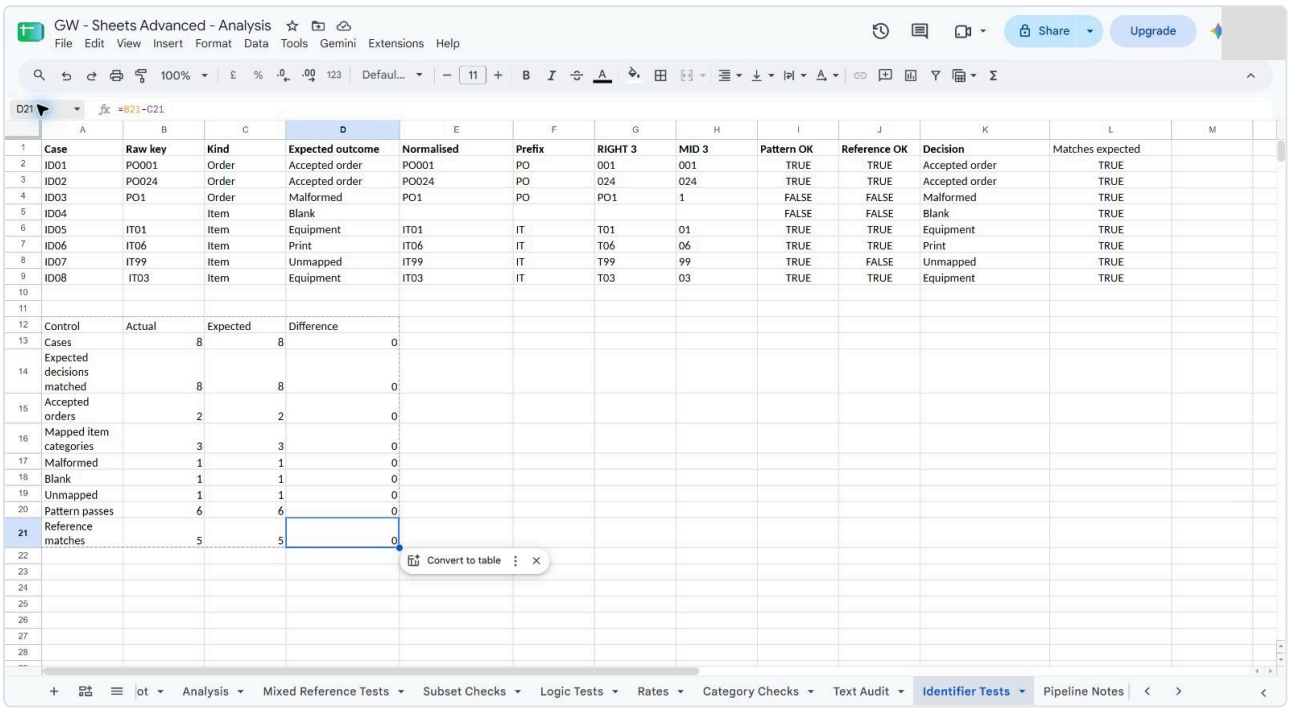
14. Create **Pipeline Notes** with the traceability card. Explain raw → normalised → validated → category/display output. TRIM and case changes require raw input for reconstruction; keeping raw alongside output preserves traceability but does not make TRIM itself invertible. The display label never replaces the separate keys.



15. Inspect the final Text Audit with wrapped headings and decisions. Check raw spaces remain and the NBSP example still records review rather than a silently repaired source value.



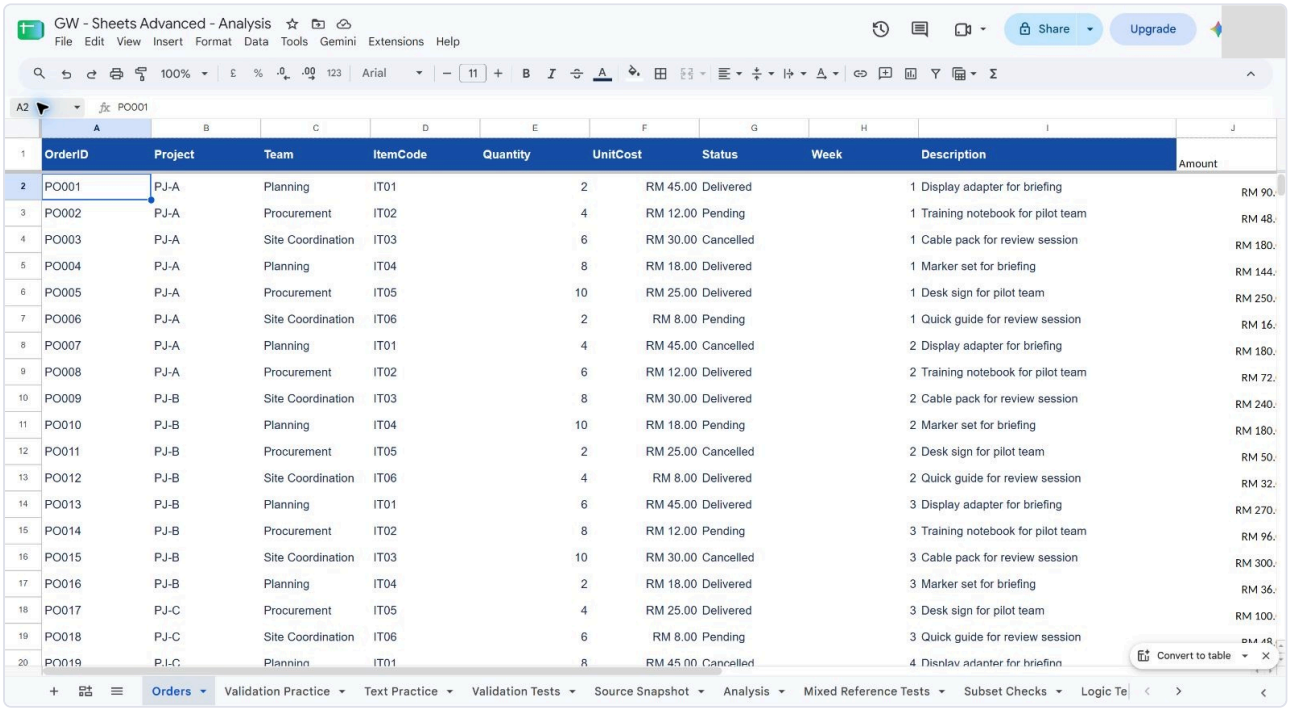
16. Inspect the final Identifier Tests table and zero-difference controls. Blank is not an item category; unmapped is not zero; extracted text is not membership proof.



Case	Raw key	Kind	Expected outcome	Normalised	Prefix	RIGHT 3	MID 3	Pattern OK	Reference OK	Decision	Matches expected
ID01	PO001	Order	Accepted order	PO001	PO	001	001	TRUE	TRUE	Accepted order	TRUE
ID02	PO024	Order	Accepted order	PO024	PO	024	024	TRUE	TRUE	Accepted order	TRUE
ID03	PO1	Order	Malformed	PO1	PO	PO1	1	FALSE	FALSE	Malformed	TRUE
ID04		Item	Blank					FALSE	FALSE	Blank	TRUE
ID05	IT01	Item	Equipment	IT01	IT	T01	01	TRUE	TRUE	Equipment	TRUE
ID06	IT06	Item	Print	IT06	IT	T06	06	TRUE	TRUE	Print	TRUE
ID07	IT99	Item	Unmapped	IT99	IT	T99	99	TRUE	FALSE	Unmapped	TRUE
ID08	IT03	Item	Equipment	IT03	IT	T03	03	TRUE	TRUE	Equipment	TRUE

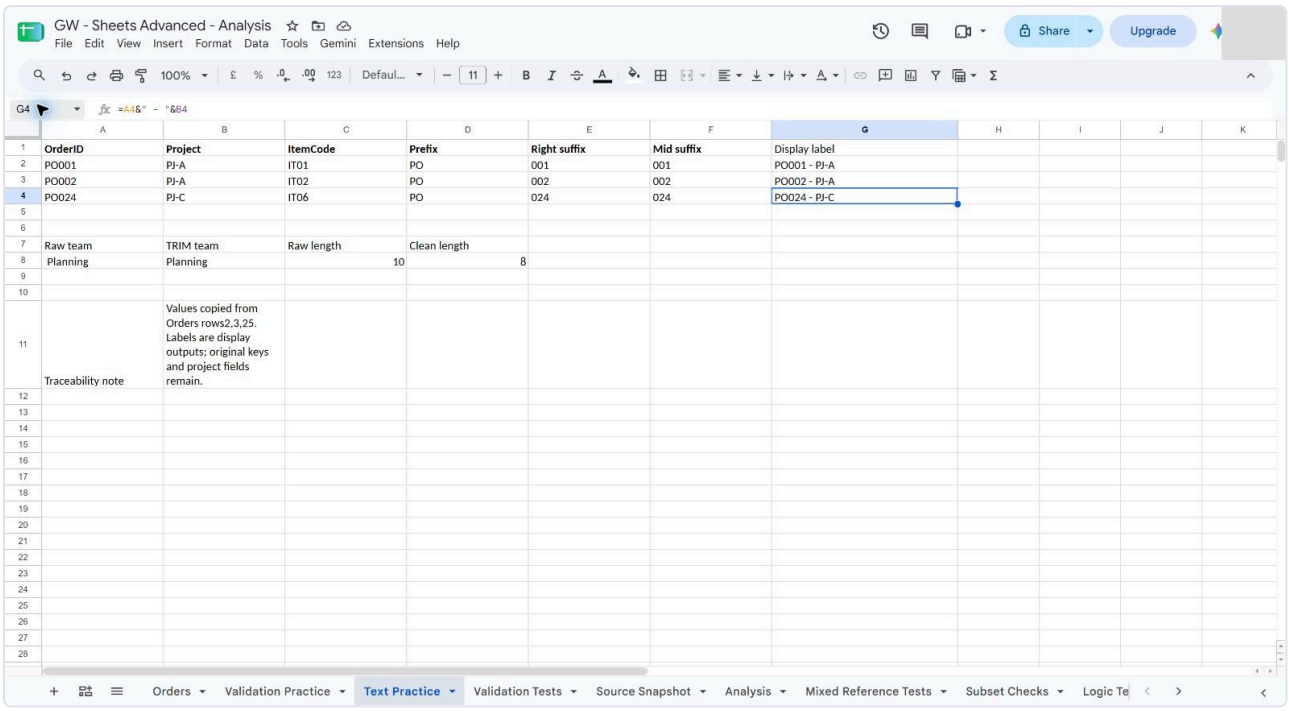
Control	Actual	Expected	Difference
Cases	8	8	0
Expected decisions matched	8	8	0
Accepted orders	2	2	0
Mapped item categories	3	3	0
Malformed	1	1	0
Blank	1	1	0
Unmapped	1	1	0
Pattern passes	6	6	0
Reference matches	5	5	0

17. Reinspect maintained Orders. This lab adds practice/audit tabs only; Orders, Analysis, Rates and Source Snapshot retain their original values and formats.

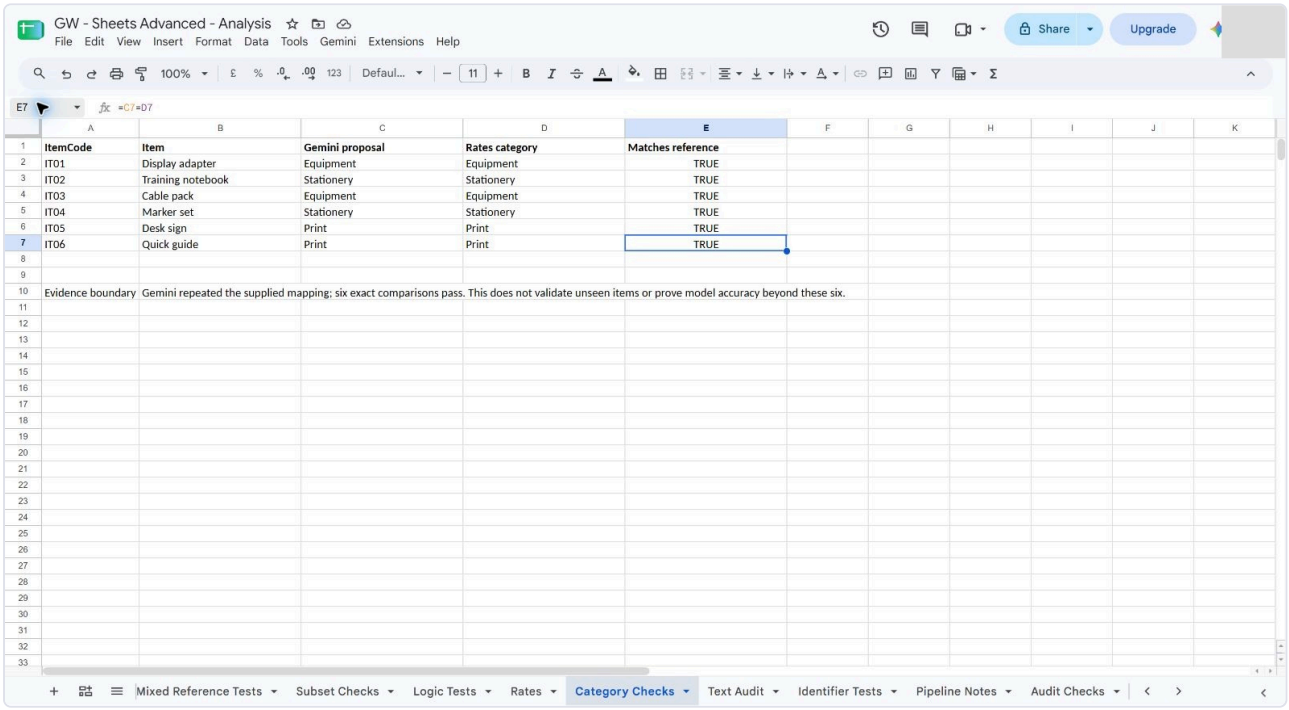


OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount
PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter for briefing	RM 90.
PO002	PJ-A	Procurement	IT02	4	RM 12.00	Pending	1	Training notebook for pilot team	RM 48.
PO003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled	1	Cable pack for review session	RM 180.
PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for briefing	RM 144.
PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team	RM 250.
PO006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session	RM 16.
PO007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing	RM 180.
PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team	RM 72.
PO009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session	RM 240.
PO010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing	RM 180.
PO011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team	RM 50.
PO012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session	RM 32.
PO013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing	RM 270.
PO014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team	RM 96.
PO015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session	RM 300.
PO016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing	RM 36.
PO017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team	RM 100.
PO018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session	
PO019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing	

18. Reinspect Text Practice's leading-zero suffixes and separate label inputs.



19. Leave the six Category Checks comparisons and saved Pipeline Notes ready for handover. Retain the actual Gemini response as author evidence without claiming reliability on untested items.



> **Validation boundary:** guided4.1–4.3 and author casework4.4 were executed through learner UI with bounded read-only native-cell proof. Independent mixed-intake work, human reviewer reconstruction, native CSV file import and planned60-minute learner timing remain unverified. Clipboard whitespace loss was detected and repaired before cleaning; the raw three-row CSV now matches exactly.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently process the mixed identifier intake into accepted keys and an exception list. Do not rely on string length alone for validity.
2. Demonstrate that IT99 is syntactically plausible but absent from the supplied mapping, and that a blank key is not a category.
3. Have a reviewer reconstruct one clean value's origin from your raw and transformation columns. Repair any missing trace.

Evidence to Hand Over

- Raw/clean text audit and identifier extraction tests.
- Separate syntax and mapping validation with explicit exceptions.
- Traceable category/display outputs and an independent mixed-intake result.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
Text extraction succeeds on a bad identifier	Validate the complete pattern and reference mapping separately from substring extraction.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Text functions reshape fields while source keys and mappings preserve meaning.

Check yourself: Why should an ID suffix such as 001 remain text?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=LEFT(A2,2)
```

```
=RIGHT(A2,3)
```

```
=MID(A2,3,3)
```

```
=A2&" - "&B2
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Identifier intake: PO001; PO024; PO1; blank; IT01; IT06; IT99; “ IT03 ”. Order IDs require PO plus three digits; mapped item keys are the six supplied Rates codes. Do not treat every extracted suffix as a valid key.
```

Captured Author-Run Reference

Fictional training material. Expectations precede actual formulas; actual results are recorded separately. Preserve spaces using single-cell entry where noted. This is not independent learner validation.

Text Practice A1:C4

```
OrderID Project ItemCode
PO001   PJ-A    IT01
PO002   PJ-A    IT02
PO024   PJ-C    IT06
```

Text Practice source-padded team A8

Enter one ordinary space, then `Planning`, then one ordinary space. The raw value must have length10; enter this cell separately from a multi-cell paste.

Gemini proposal-only prompt

```
Fictional training task. Return a category proposal table only; do not edit the spreadsheet. Use only Equipment, Stationery, Print and this explicit Rates mapping: IT01 Display adapter = Equipment; IT02 Training notebook = Stationery; IT03 Cable pack = Equipment; IT04 Marker set = Stationery; IT05 Desk sign = Print; IT06 Quick guide = Print. Propose a category for each of these six ItemCodes and names. Mark an unmapped or ambiguous input for review rather than inventing a category.
```

Category Checks A1:E7 — actual proposal observations, blank comparison columns

```
ItemCode        Item    Gemini proposal Rates category  Matches reference
IT01    Display adapter Equipment
IT02    Training notebook       Stationery
IT03    Cable pack      Equipment
IT04    Marker set      Stationery
IT05    Desk sign       Print
IT06    Quick guide     Print
```

Text Audit A1:C4 — raw CSV, restore B2/B4 single-cell spaces

```
TaskID  Raw team        Raw status
T001     Planning       complete
T002    Procurement     In Progress
T003    Site Coordination       Not Started
```

Whitespace expectations A12:H15 — blank actual columns

```
Case    Raw value       Expected length after TRIM      TRIM output     Observed length
Matches expected        Decision        First code point
Ordinary padding         Planning       8
Trailing space  Site Coordination       17
NBSP padding     Planning       10
```

Identifier expectations A1:L9 — blank actual columns

```
Case    Raw key Kind    Expected outcome        Normalised      Prefix  RIGHT 3 MID 3
Pattern OK      Reference OK    Decision        Matches expected
ID01    PO001   Order   Accepted order
ID02    PO024   Order   Accepted order
ID03    PO1     Order   Malformed
ID04            Item    Blank
ID05    IT01    Item    Equipment
ID06    IT06    Item    Print
ID07    IT99    Item    Unmapped
ID08     IT03   Item    Equipment
```

Identifier controls A12:D21

```
Control Actual  Expected        Difference
Cases   =COUNTA(A2:A9)  8       =B13-C13
Expected decisions matched      =COUNTIF(L2:L9,TRUE)    8       =B14-C14
Accepted orders =COUNTIF(K2:K9,"Accepted order")        2       =B15-C15
Mapped item categories
=COUNTIF(K2:K9,"Equipment")+COUNTIF(K2:K9,"Stationery")+COUNTIF(K2:K9,"Print")  3       =B16-
C16
Malformed       =COUNTIF(K2:K9,"Malformed")     1       =B17-C17
Blank   =COUNTIF(K2:K9,"Blank") 1       =B18-C18
Unmapped        =COUNTIF(K2:K9,"Unmapped")      1       =B19-C19
Pattern passes  =COUNTIF(I2:I9,TRUE)    6       =B20-C20
Reference matches       =COUNTIF(J2:J9,TRUE)    5       =B21-C21
```

Pipeline Notes A1:B10

```
Pipeline stage  Evidence and reversibility
Raw source      DirtyText.csv raw values retained in Text Audit A:C; Orders and Rates
unchanged. CSV block paste trimmed spaces; affected raw cells restored individually before
cleaning.
Normalised      Text Audit D uses TRIM. E uses explicit case handling only for complete / in
progress / not started. Whitespace and case changes are not invertible without the raw source.
Unusual whitespace      The literal NBSP case keeps length10 after TRIM and code point160.
Retain it for review; do not silently replace unusual characters.
Validated       Identifier Tests checks complete ^PO[0-9]{3}$ or ^IT[0-9]{2}$ patterns and
reference membership separately. PO1 malformed; blank remains blank; IT99 syntactically valid
but unmapped.
Category output Known item categories come from exact Rates mapping. Six Gemini proposals were
compared individually; they repeated the supplied mapping, not an independent test on unseen
items.
Display output  Text Practice combines OrderID and Project only for display. Separate keys and
original zeros remain; LEFT/RIGHT/MID outputs are text, not new identifiers.
Reconstruction  Clean Planning traces to Text Audit B2 with raw length10, TRIM D2 length8.
Preserve raw B2 to recover discarded spaces; do not attempt to reverse TRIM from clean text
alone.
Author evidence Three raw/clean rows, three space cases, eight expected-versus-actual
identifier cases and six AI/reference comparisons checked. Maintained
Orders/Analysis/Rates/Source Snapshot unchanged.
Pending checks  Independent mixed-intake attempt and reviewer reconstruction not performed.
File-import route and classroom60-minute timing remain unverified.
```

Seed formulas — fill each stated column down

Text Practice D2:D4

```
=LEFT(A2,2)
```

Text Practice E2:E4

```
=RIGHT(A2,3)
```

Text Practice F2:F4

```
=MID(A2,3,3)
```

Text Practice G2:G4

```
=A2&" - "&B2
```

Text Audit D2:D4

```
=TRIM(B2)
```

Text Audit E2:E4

```
=IF(LOWER(TRIM(C2))="complete","Complete",IF(LOWER(TRIM(C2))="in progress","In
Progress",IF(LOWER(TRIM(C2))="not started","Not Started","Review")))
```

Text Audit F2:F4

```
=IF(AND(OR(D2="Planning",D2="Procurement",D2="Site
Coordination"),E2<>"Review"),"Accepted","Review")
```

Text Audit G2:G4

```
=LEN(B2)
```

Text Audit H2:H4

```
=LEN(D2)
```

Text Audit D13:D15

```
=TRIM(B13)
```

Text Audit E13:E15

```
=LEN(D13)
```

Text Audit F13:F15

```
=E13=C13
```

Text Audit H13:H15

```
=UNICODE(LEFT(B13,1))
```

Identifier Tests E2:E9

```
=TRIM(B2)
```

Identifier Tests F2:F9

```
=LEFT(E2,2)
```

Identifier Tests G2:G9

```
=RIGHT(E2,3)
```

Identifier Tests H2:H9

```
=MID(E2,3,3)
```

Identifier Tests I2:I9

```
=IF(C2="Order",REGEXMATCH(E2,"^PO[0-9]{3}$"),IF(C2="Item",REGEXMATCH(E2,"^IT[0-9]
{2}$"),FALSE))
```

Identifier Tests J2:J9

```
=IF(I2,IF(C2="Order",COUNTIF(Orders!$A$2:$A$25,E2)=1,COUNTIF(Rates!$A$2:$A$7,E2)=1),FALSE)
```

Identifier Tests K2:K9

```
=IF(E2="","Blank",IF(NOT(I2),"Malformed",IF(NOT(J2),"Unmapped",IF(C2="Order","Accepted
order",VLOOKUP(E2,Rates!$A$2:$D$7,4,FALSE)))))
```

Identifier Tests L2:L9

```
=K2=D2
```

Category Checks D2:D7

```
=VLOOKUP(A2,Rates!$A$2:$D$7,4,FALSE)
```

Category Checks E2:E7

```
=C2=D2
```

Raw whitespace cases include literal NBSP (U+00A0) padding; if copying changes it, verify the first code point160 and raw length10. Do not substitute ordinary spaces and claim the same test. Independent mixed intake, reviewer reconstruction, native file import and learner timing remain pending.

05

CHAPTER 5

Match Records and Handle Errors

In this chapter you will compare exact-match VLOOKUP, HLOOKUP and INDEX/MATCH; distinguish a missing key from bad input.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 80 minutes: 15 explanation/demonstration, 37 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 65 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Exact-match lookup examples and an explicit missing-item result.

> **Status:** author reference with36 staged application captures through5.4.24 order rates reconcile, nine key cases match expectations, duplicate first-match and omitted-mode failures are observed, and distinct formula defects are captured and repaired. Maintained source preserved. Independent zero-rate/output/column-move work, reviewer and learner timing remain pending.

What You Will Learn

- Use exact-match lookups.
- Compare vertical and horizontal lookup layouts.
- Distinguish missing keys from broken formulas.

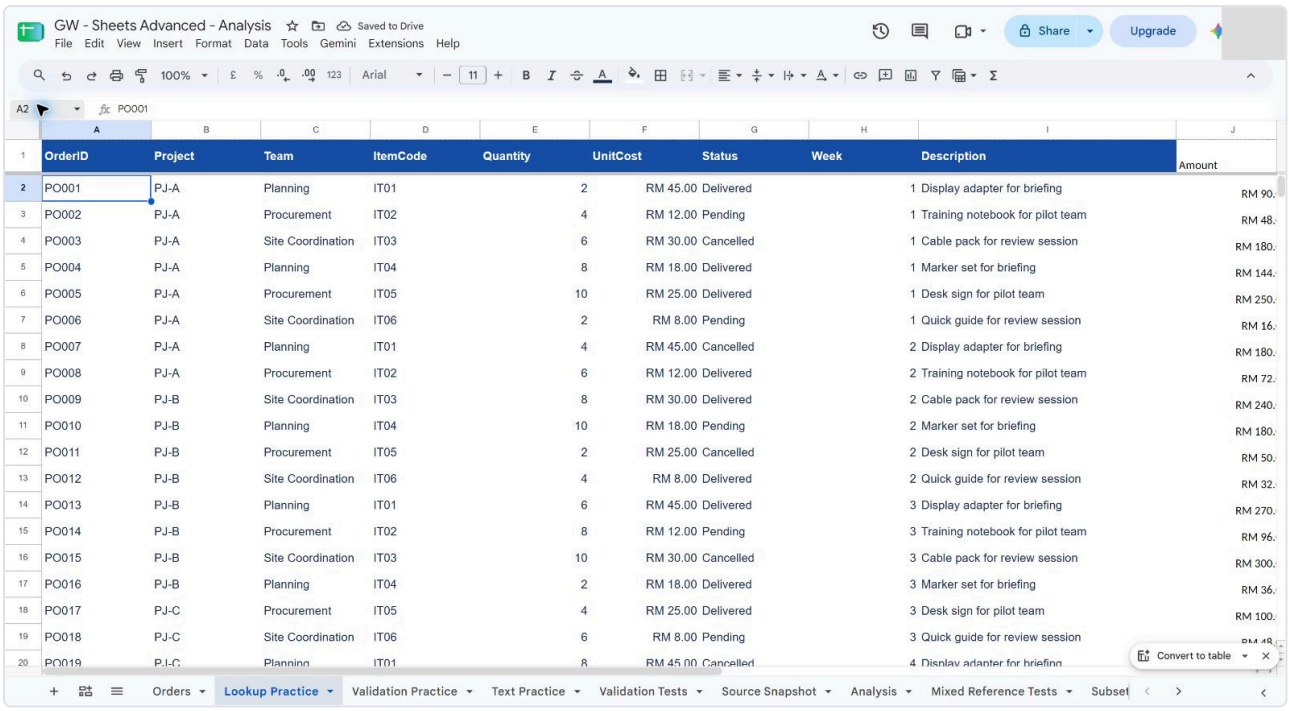
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

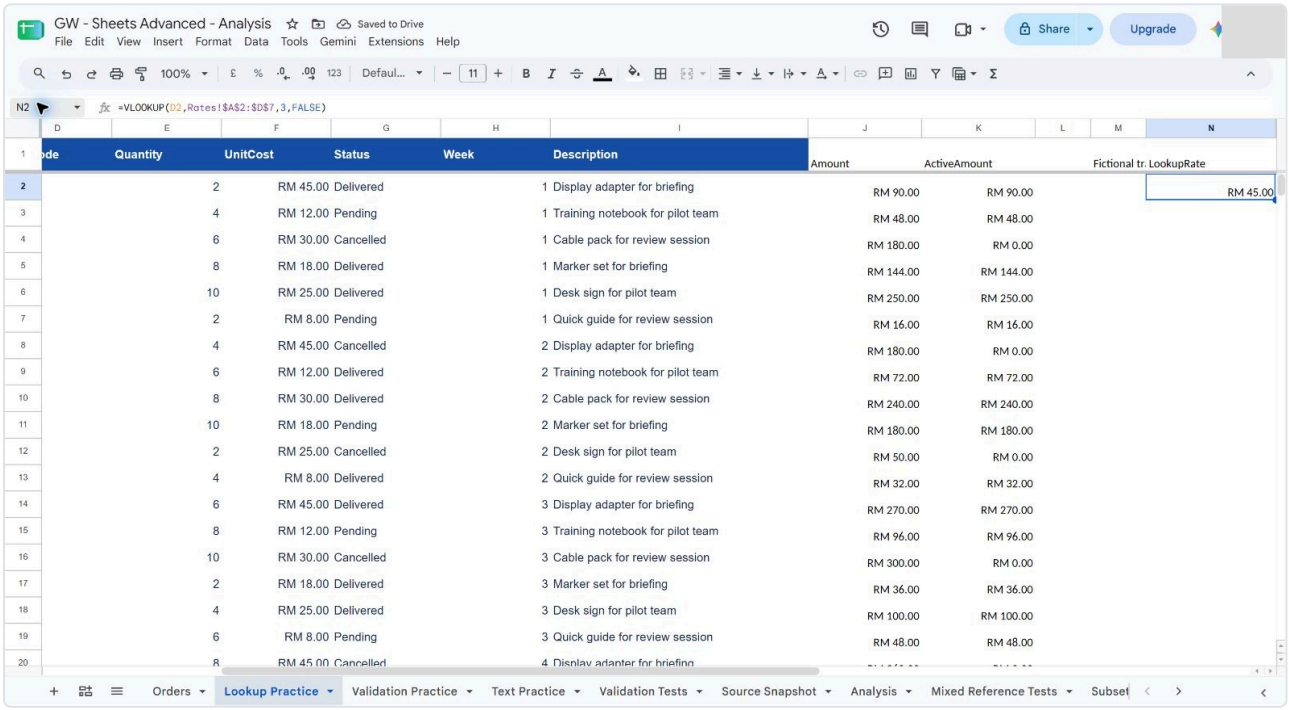
Use Orders and Rates. Rates A2:D7 has unique item codes and numeric unit costs.

5.1 Look Up a Rate

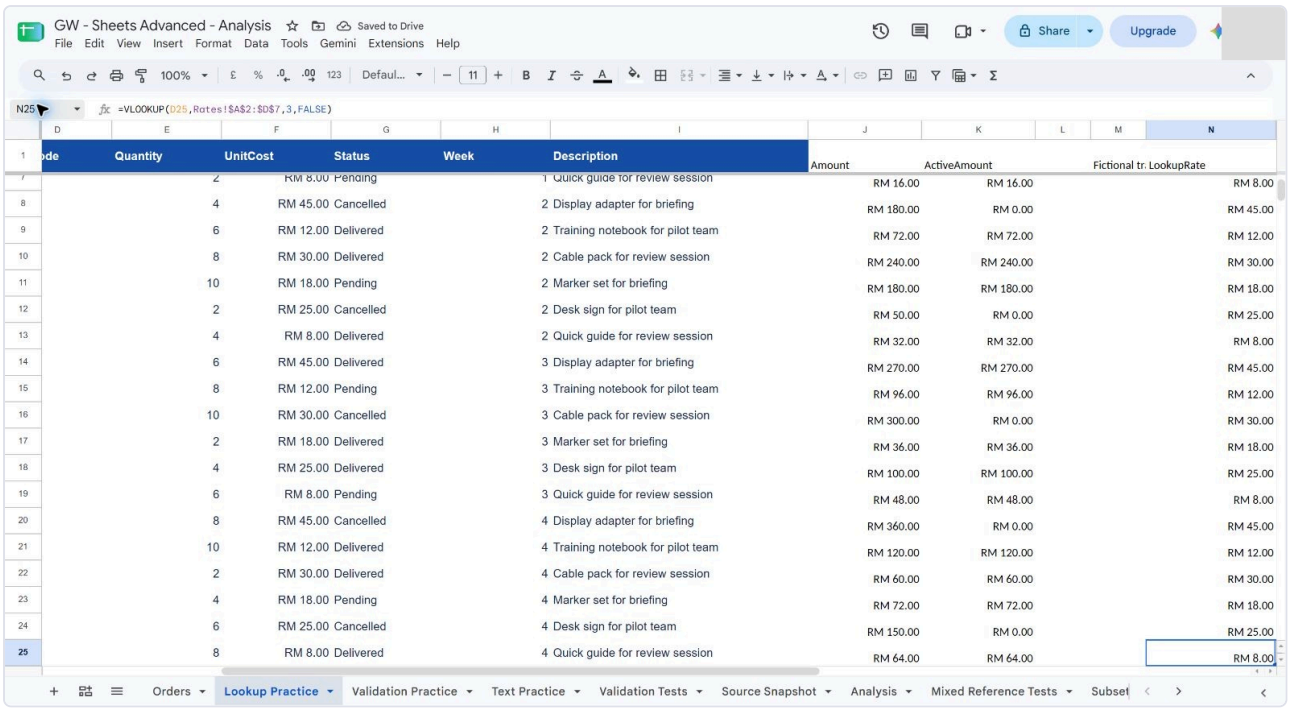
1. Duplicate **Orders** as **Lookup Practice**. Verify the separate tab before editing. Keep maintained Orders, Rates, Analysis and Source Snapshot unchanged. The training-data label remains in M1, so new practice calculations start at **N1**.



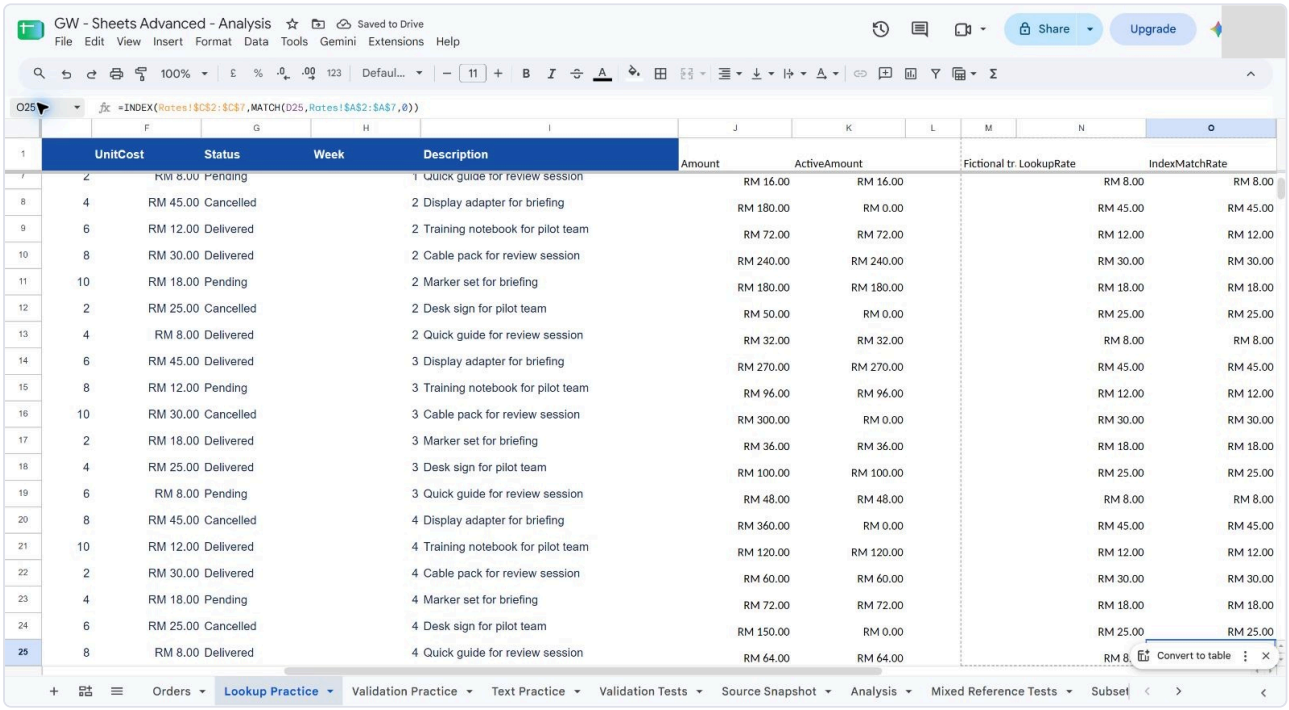
2. Label N1 **LookupRate** and enter **=VLOOKUP(D2,Rates!$A$2:$D$7,3,FALSE)** in N2. IT01 returns45. FALSE requests an exact key match; the key must be in the first column of A:D and the requested return column is3.



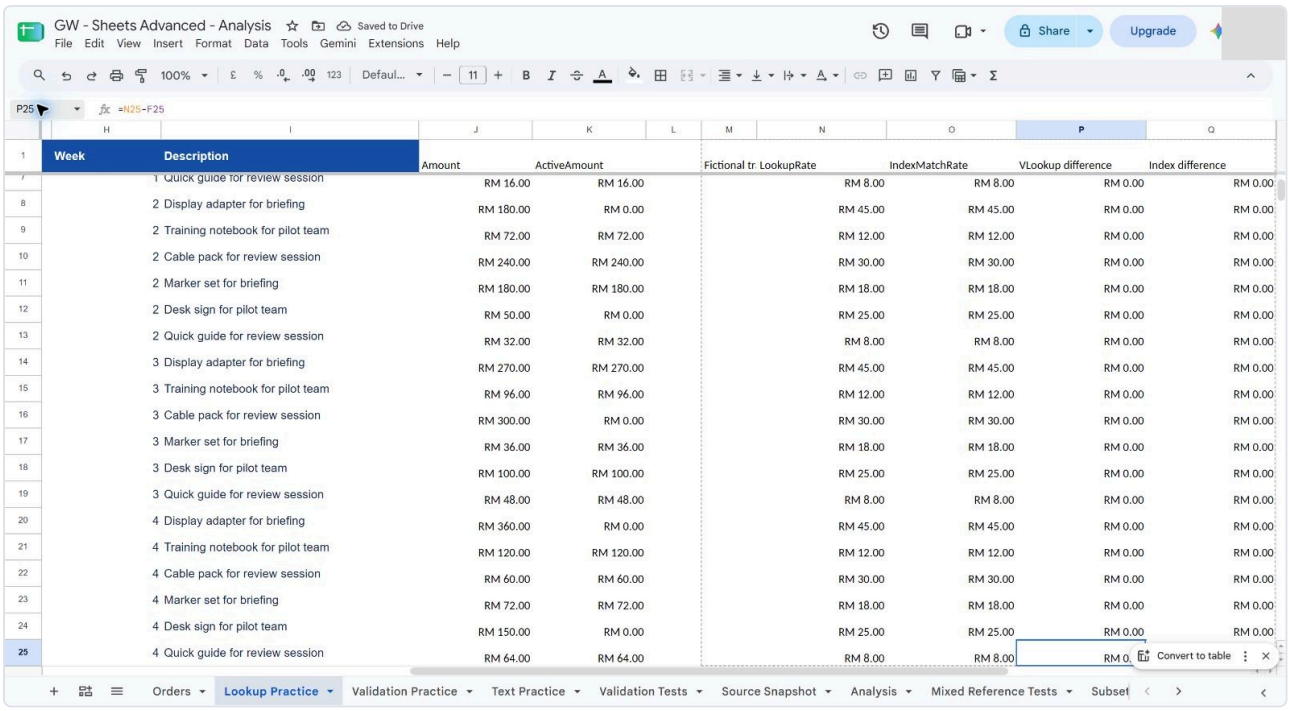
3. Fill **N2:N25** down. Inspect the final IT06 rate8 as well as the first result; a correct first row alone is not the complete check.



4. Label O1 **IndexMatchRate**. In O2 enter **=INDEX(Rates!$C$2:$C$7,MATCH(D2,Rates!$A$2:$A$7,0))** and fill O2:O25 down. MATCH0 is exact; the key and return ranges must be aligned.

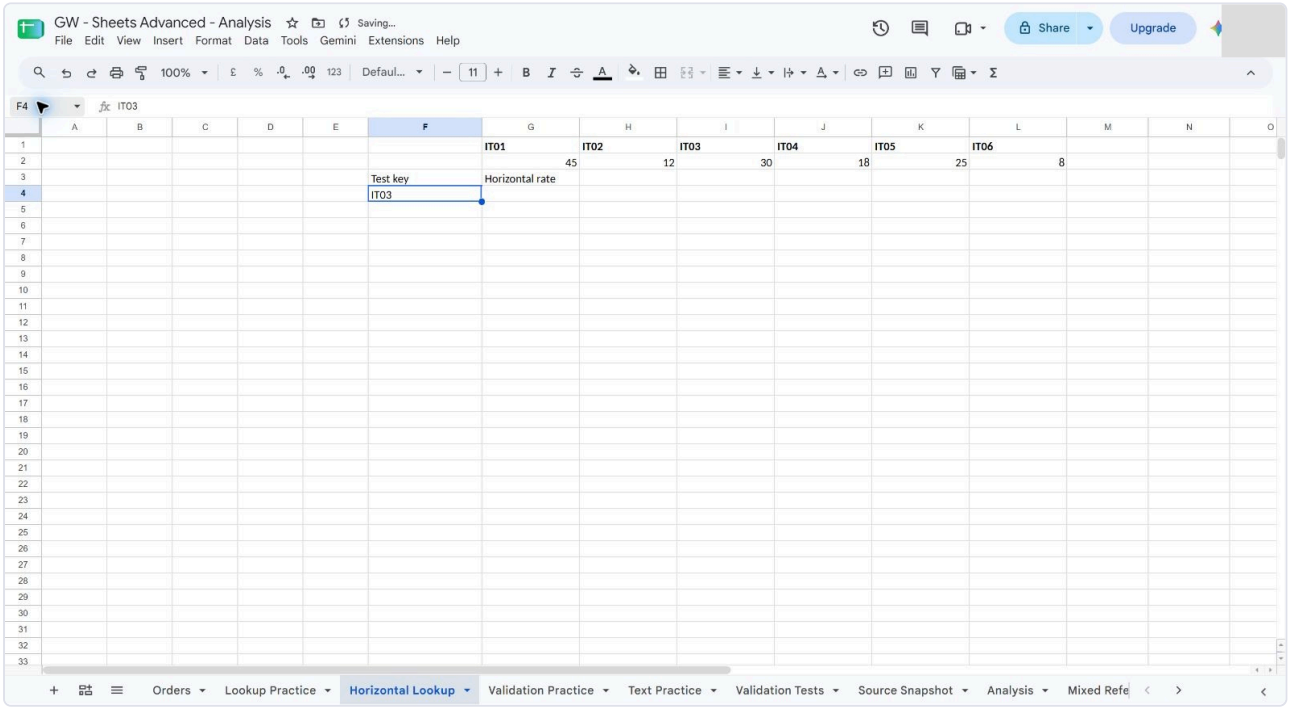


5. Label P1/Q1 **VLookup difference /Index difference**. Enter **=N2-F2 / =O2-F2** in P2/Q2 and fill through row25. All24 pairs are zero against the original UnitCost values. Widen only the new practice columns as needed.

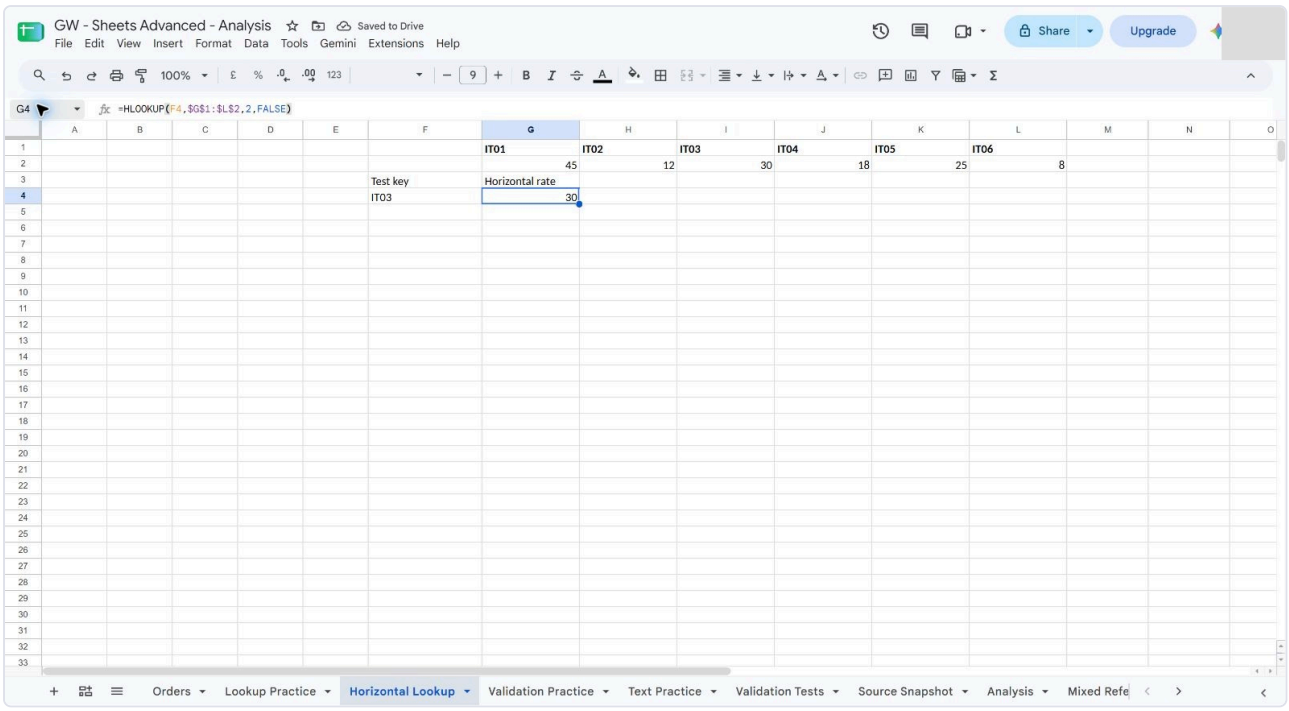


5.2 Practise a Horizontal Lookup

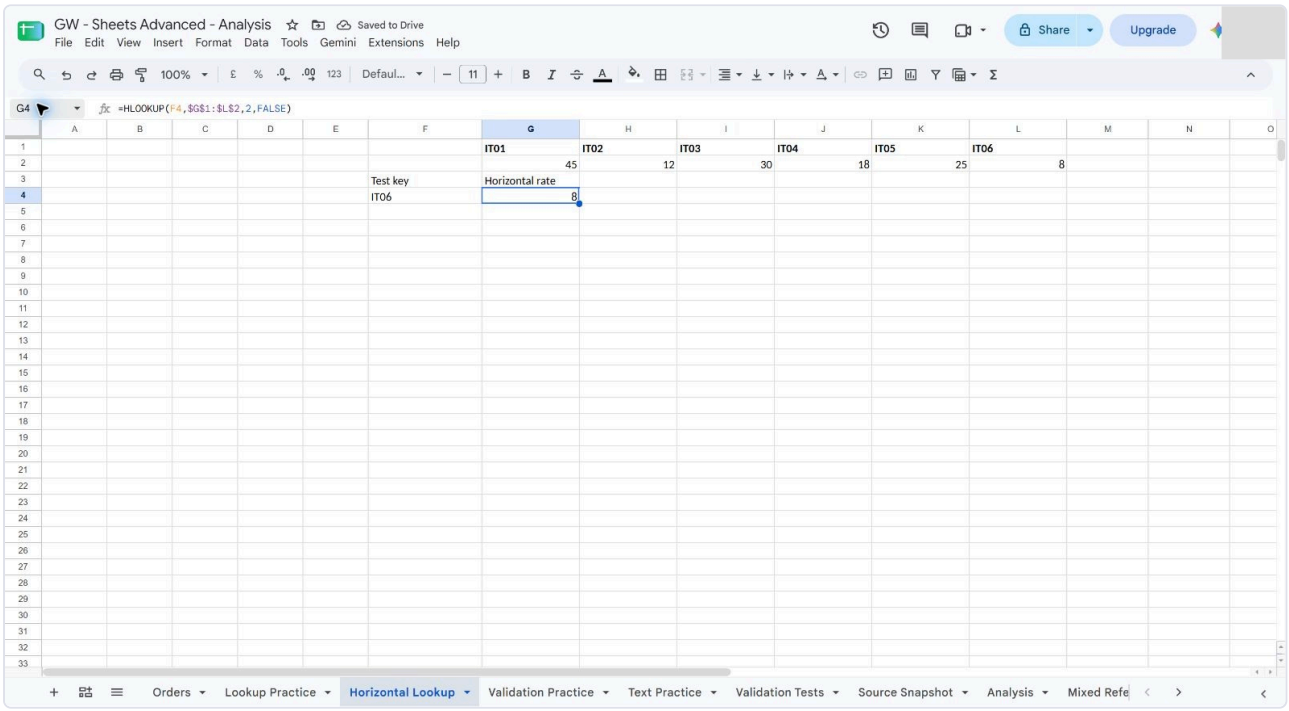
1. Create a blank **Horizontal Lookup** tab. The original guide's Analysis F/G cells now contain Lab2 work, so use the same address pattern on this isolated tab. Enter IT01–IT06 in **G1:L1** and rates **45 /12 /30 /18 /25 /8** in G2:L2. Put Test key / Horizontal rate in F3:G3 and **IT03** in F4.



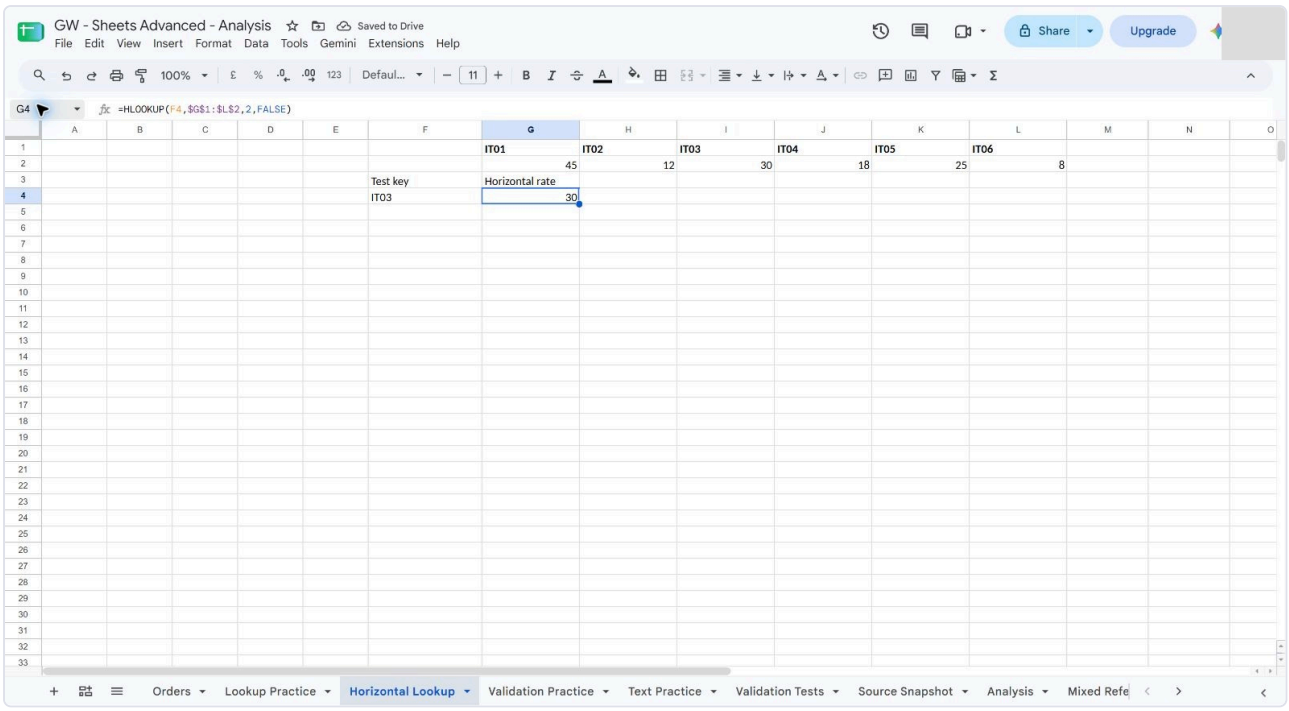
2. In G4 enter **=HLOOKUP(F4,$G$1:$L$2,2,FALSE)**. The exact lookup searches the first row and returns from the second row: **30**.



3. Change only F4 to **IT06**. Expect **8**.

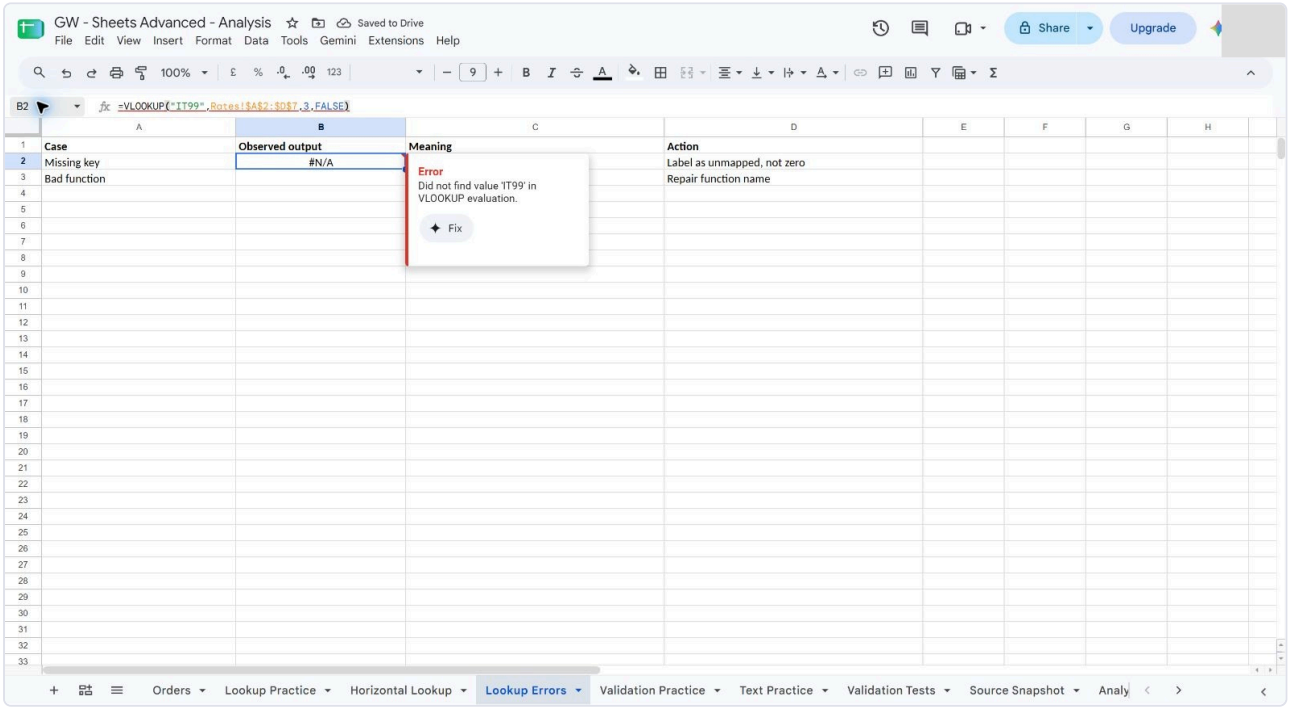


4. Restore F4 to **IT03** and confirm30. Preserve this restored key for the later three-method comparison.

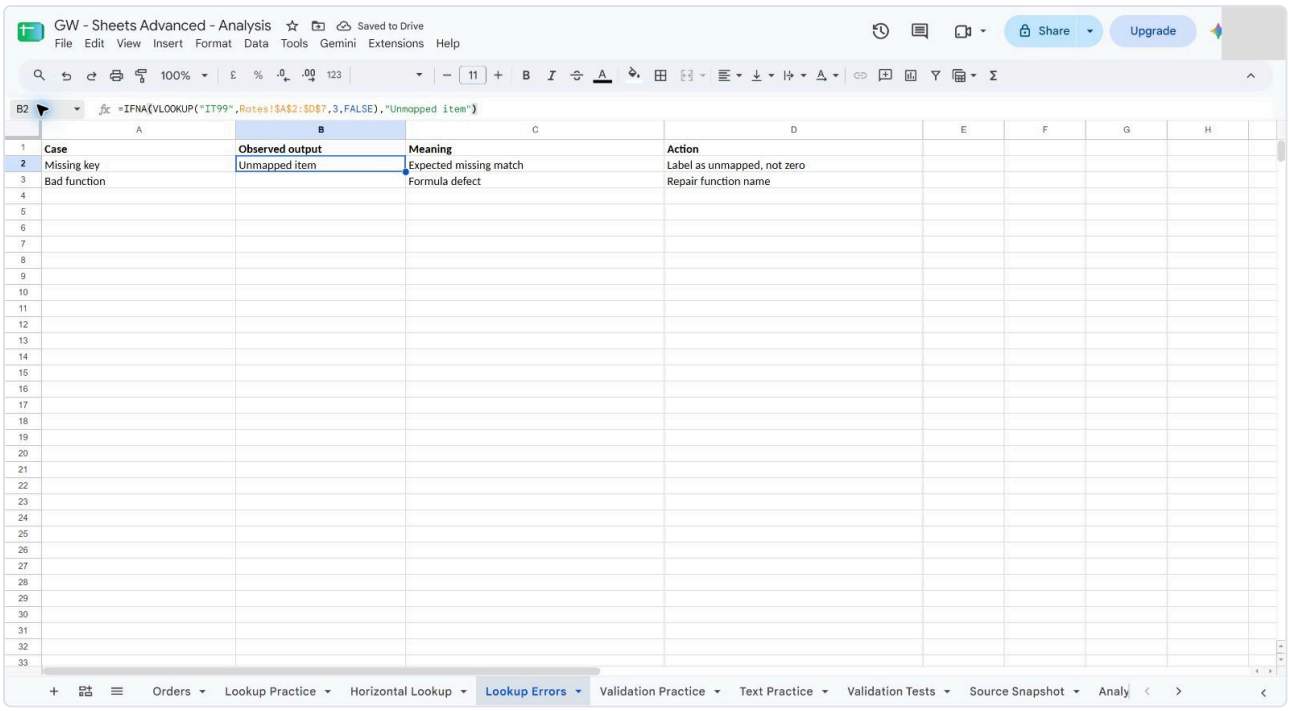


5.3 Handle an Expected Missing Key

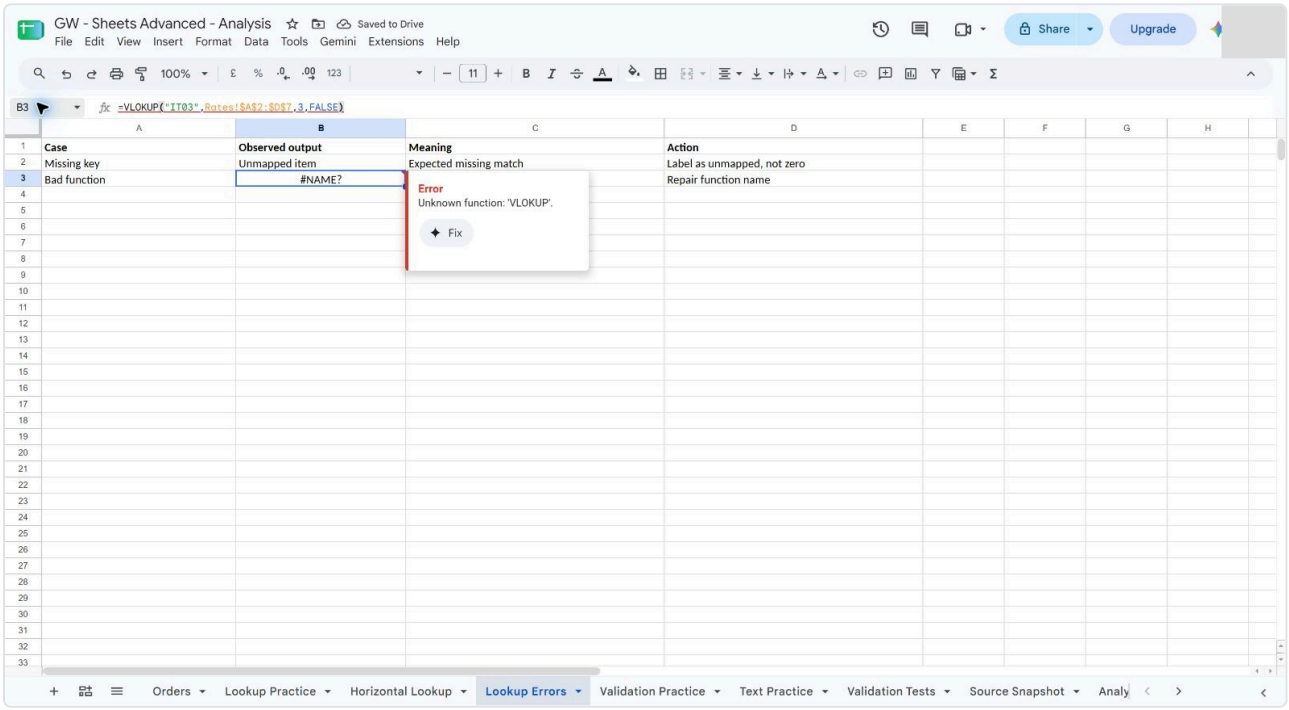
1. Create **Lookup Errors** with the case/meaning/action card. In B2 enter **=VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE)**. Observe the actual **#N/A** missing-match result; it is not an evidenced zero rate.



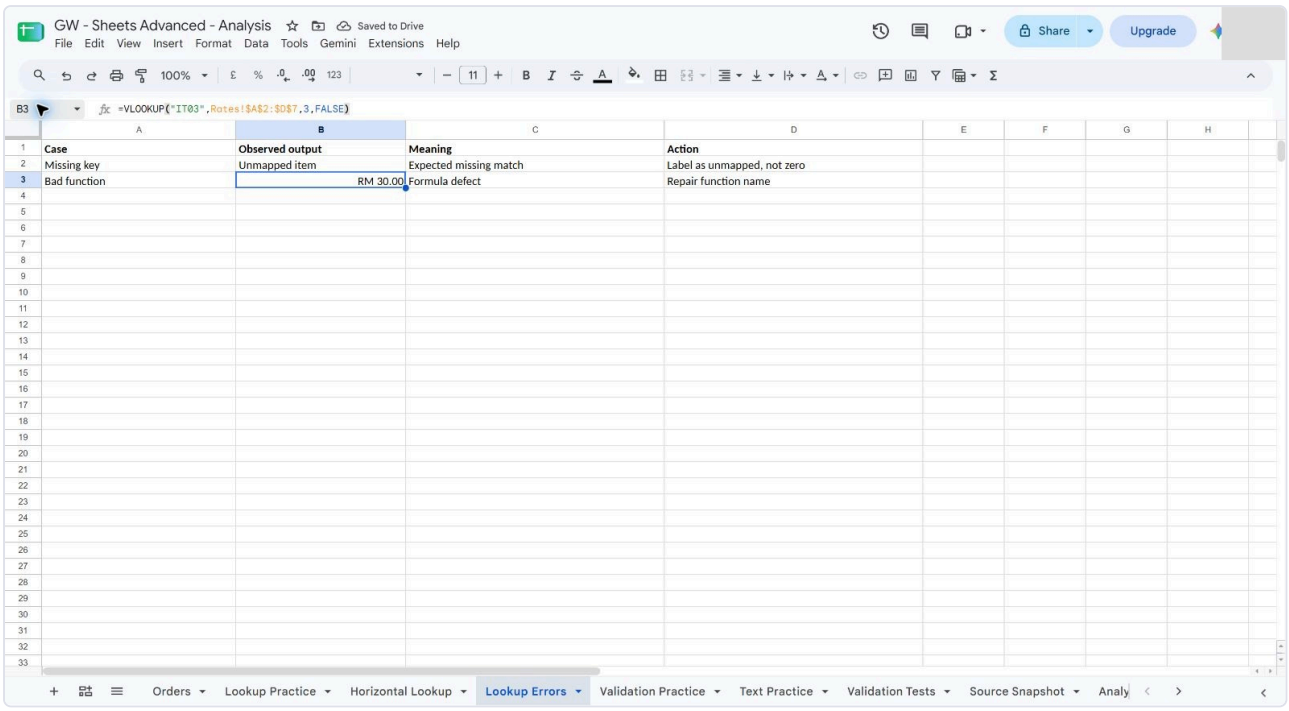
2. Change B2 to **=IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")**. This labels the intended missing-match case.



3. In B3 deliberately enter **=VLOKUP("IT03",Rates!$A$2:$D$7,3,FALSE)**. The native result is **#NAME?**, with the message **Unknown function: 'VLOKUP'.** Capture the actual defect before repair.

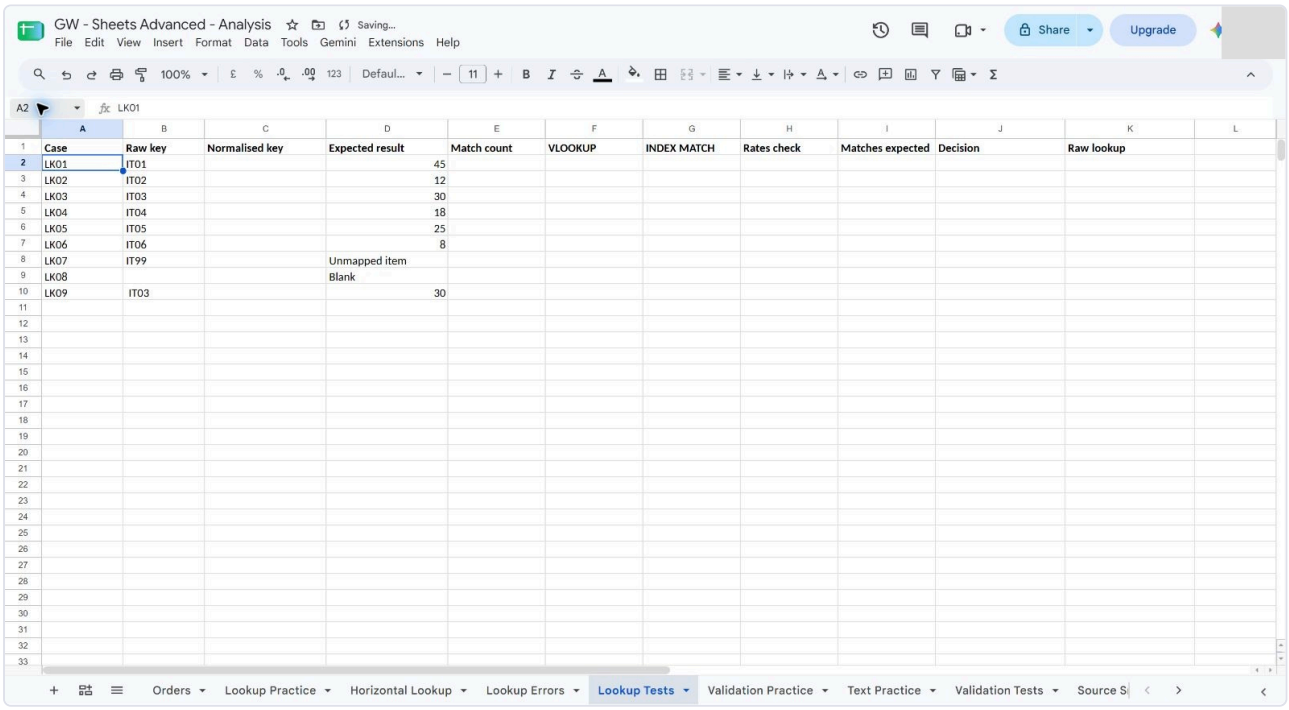


4. Correct VLOKUP to **VLOOKUP**. B3 now returns30. Repairing a formula defect is different from labelling a genuinely absent key.

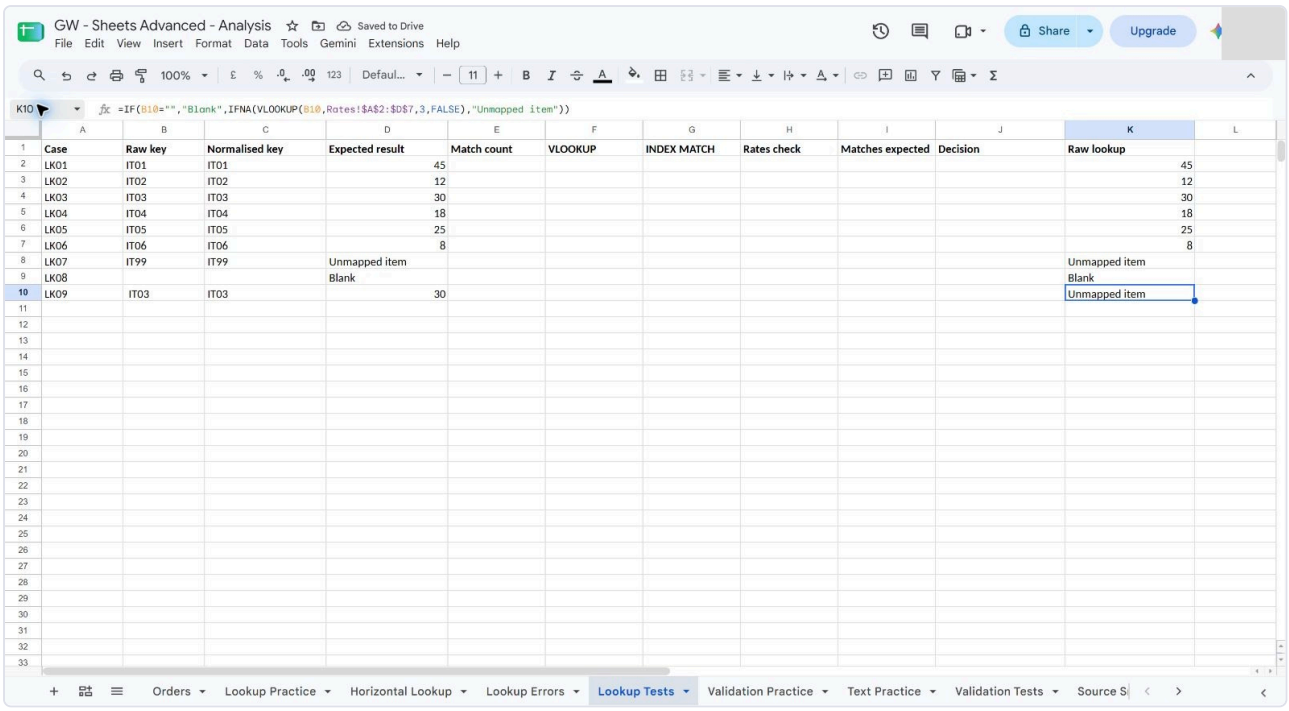


5.4 Challenge Exact Lookups With Missing and Duplicate Keys

1. Create **Lookup Tests** from the nine-row expectation card. It includes all six known keys, IT99, a truly blank key and IT03 with one leading/trailing ordinary space. Preserve raw input B beside normalised C. Enter the padded B10 separately if block paste strips its spaces. Keep expected D values before populating actual formula columns.

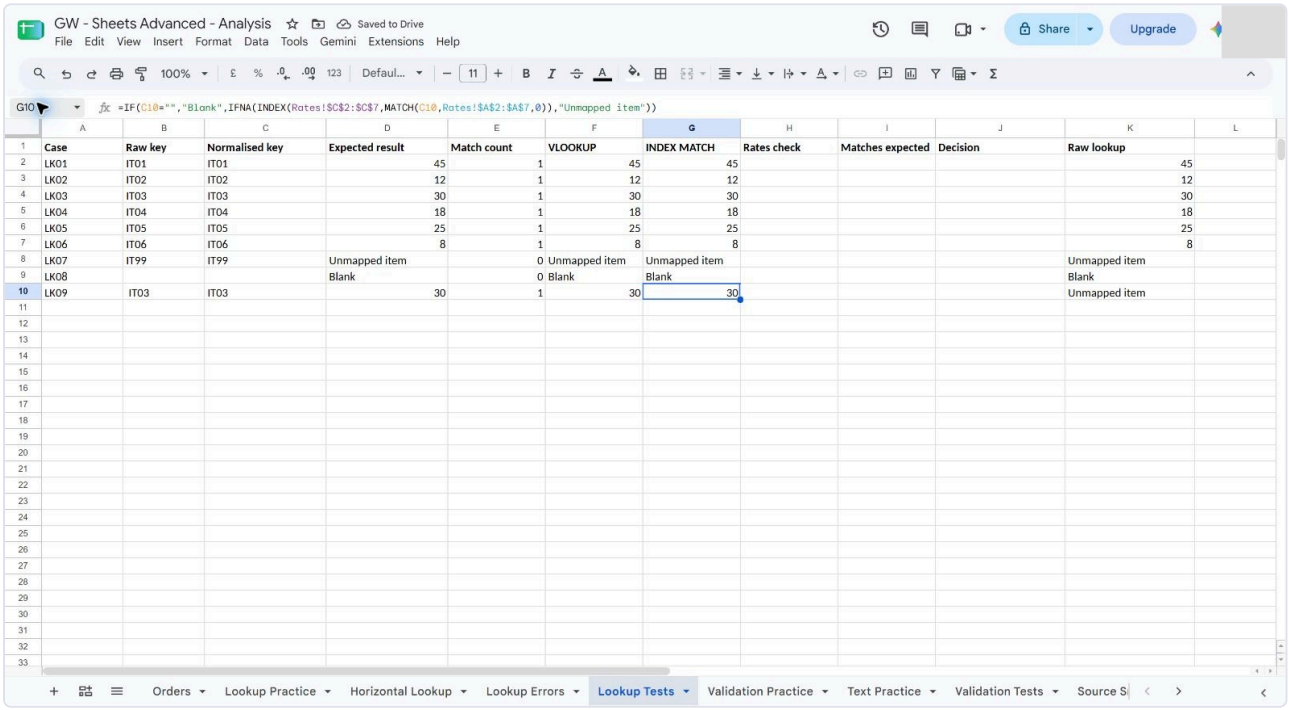


2. In C2 use **=TRIM(B2)** and fill down. In K2 use the raw-key exact lookup card and fill through row10. The raw padded key returns **Unmapped item**, while its normalised key is IT03. Keep that distinction visible instead of overwriting raw B10.



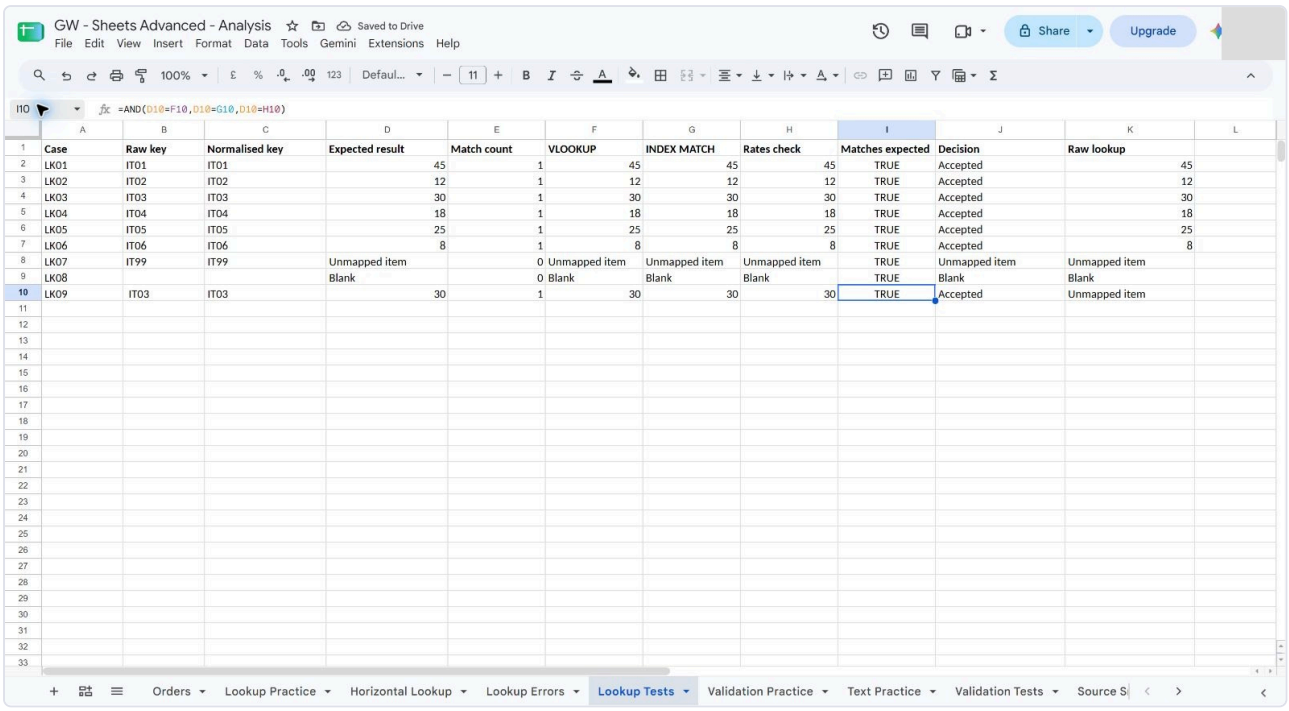
Case	Raw key	Normalised key	Expected result	Match count	VLOOKUP	INDEX MATCH	Rates check	Matches expected	Decision	Raw lookup
LK01	IT01	IT01	45							45
LK02	IT02	IT02	12							12
LK03	IT03	IT03	30							30
LK04	IT04	IT04	18							18
LK05	IT05	IT05	25							25
LK06	IT06	IT06	8							8
LK07	IT99	IT99	Unmapped item							Unmapped item
LK08			Blank							Blank
LK09	IT03	IT03	30							Unmapped item

3. Fill E2:G10 using the card: count matches only for nonblank keys, then use exact VLOOKUP and INDEX/MATCH on normalised C. Handle an explicit blank separately from a missing mapping. The six known keys and cleaned IT03 return their numeric rates; IT99 remains Unmapped item.



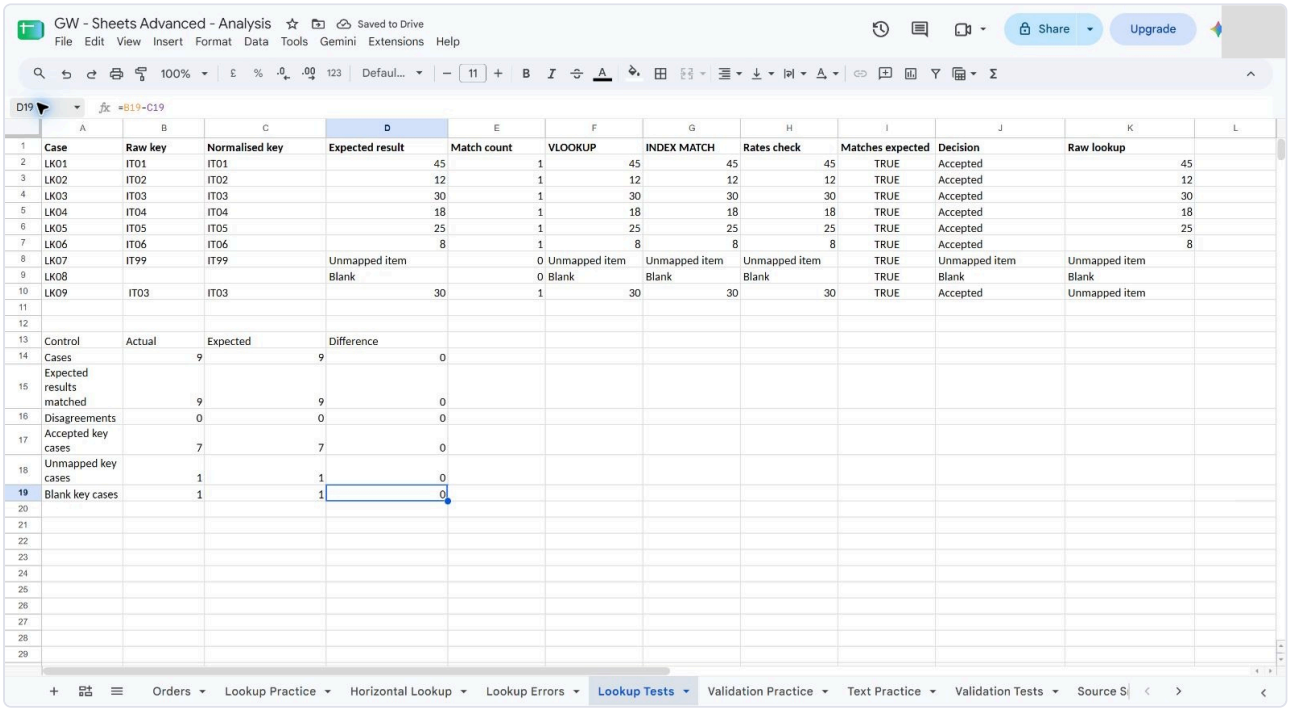
Case	Raw key	Normalised key	Expected result	Match count	VLOOKUP	INDEX MATCH	Rates check	Matches expected	Decision	Raw lookup
LK01	IT01	IT01	45	1	45	45				45
LK02	IT02	IT02	12	1	12	12				12
LK03	IT03	IT03	30	1	30	30				30
LK04	IT04	IT04	18	1	18	18				18
LK05	IT05	IT05	25	1	25	25				25
LK06	IT06	IT06	8	1	8	8				8
LK07	IT99	IT99	Unmapped item	0	Unmapped item	Unmapped item				Unmapped item
LK08			Blank	0	Blank	Blank				Blank
LK09	IT03	IT03	30	1	30	30				Unmapped item

4. H checks the bounded Rates values only after the match count is one. I compares the expected result with both lookup methods and H; J records Accepted / Unmapped item / Blank. Fill all three columns through row10. All nine expected-versus-actual checks are TRUE. This is guided author casework, not an independent learner attempt.



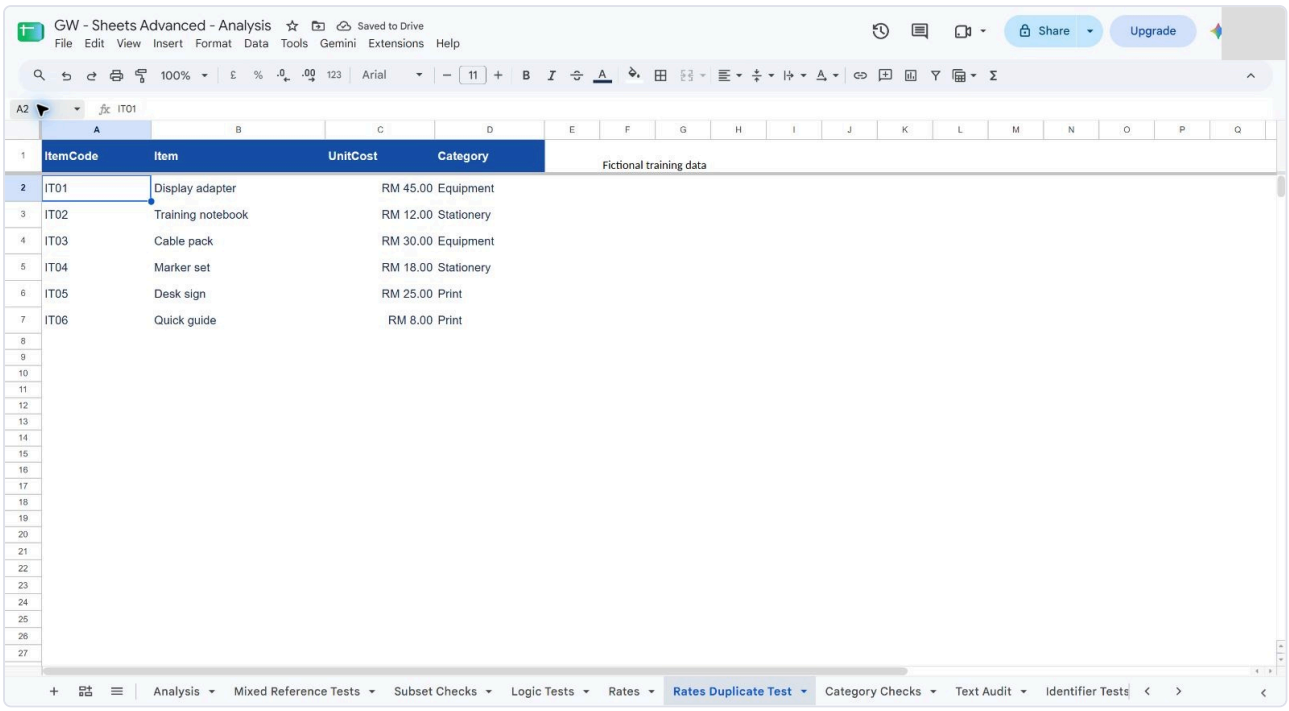
	A	B	C	D	E	F	G	H	I	J	K	L
1	Case	Raw key	Normalised key	Expected result	Match count	VLOOKUP	INDEX MATCH	Rates check	Matches expected	Decision	Raw lookup	
2	LK01	IT01	IT01	45	1	45	45	45	TRUE	Accepted	45	
3	LK02	IT02	IT02	12	1	12	12	12	TRUE	Accepted	12	
4	LK03	IT03	IT03	30	1	30	30	30	TRUE	Accepted	30	
5	LK04	IT04	IT04	18	1	18	18	18	TRUE	Accepted	18	
6	LK05	IT05	IT05	25	1	25	25	25	TRUE	Accepted	25	
7	LK06	IT06	IT06	8	1	8	8	8	TRUE	Accepted	8	
8	LK07	IT99	IT99	Unmapped item	0	Unmapped item	Unmapped item	Unmapped item	TRUE	Unmapped item	Unmapped item	
9	LK08			Blank	0	Blank	Blank	Blank	TRUE	Blank	Blank	
10	LK09	IT03	IT03	30	1	30	30	30	TRUE	Accepted	Unmapped item	

5. Add A13:D19 controls: **9 cases /9 matches /0 disagreements /7 accepted cases /1 unmapped /1 blank**. All six differences are zero. The seven accepted cases include the repeated, space-normalised IT03 input; they are not seven unique source keys.

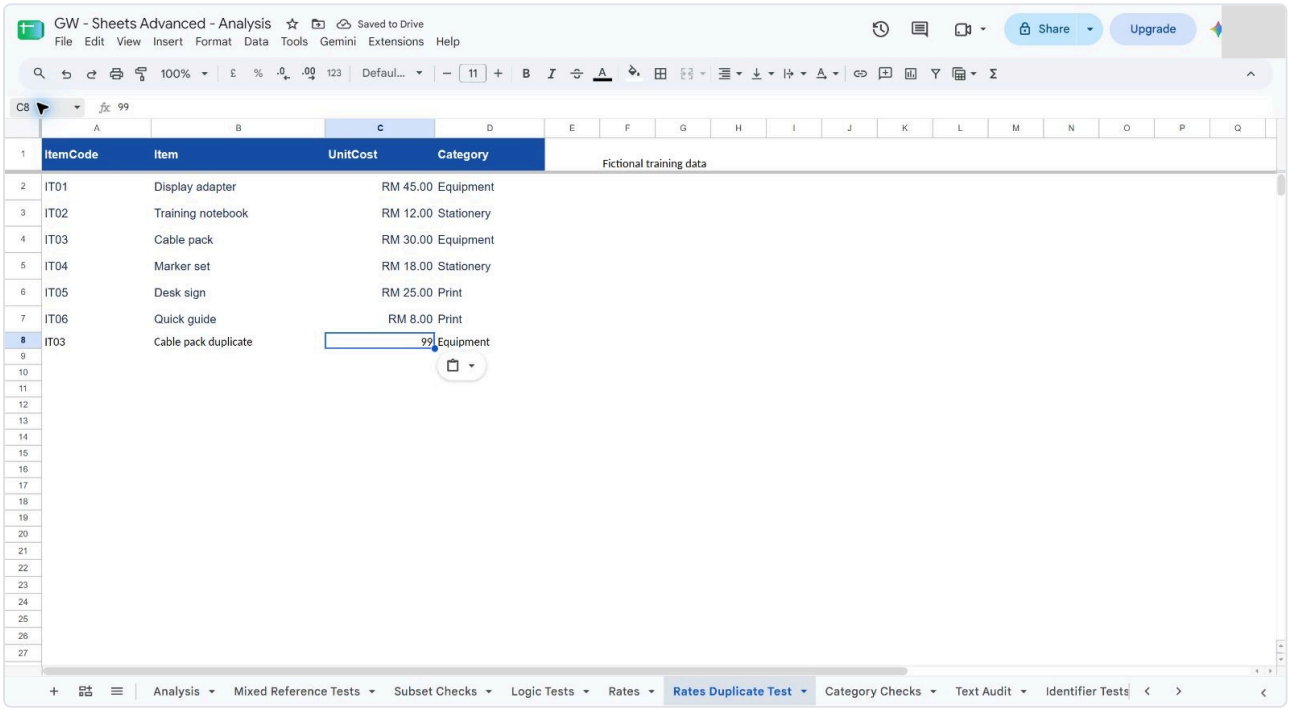


	A	B	C	D	E	F	G	H	I	J	K	L
1	Case	Raw key	Normalised key	Expected result	Match count	VLOOKUP	INDEX MATCH	Rates check	Matches expected	Decision	Raw lookup	
2	LK01	IT01	IT01	45	1	45	45	45	TRUE	Accepted	45	
3	LK02	IT02	IT02	12	1	12	12	12	TRUE	Accepted	12	
4	LK03	IT03	IT03	30	1	30	30	30	TRUE	Accepted	30	
5	LK04	IT04	IT04	18	1	18	18	18	TRUE	Accepted	18	
6	LK05	IT05	IT05	25	1	25	25	25	TRUE	Accepted	25	
7	LK06	IT06	IT06	8	1	8	8	8	TRUE	Accepted	8	
8	LK07	IT99	IT99	Unmapped item	0	Unmapped item	Unmapped item	Unmapped item	TRUE	Unmapped item	Unmapped item	
9	LK08			Blank	0	Blank	Blank	Blank	TRUE	Blank	Blank	
10	LK09	IT03	IT03	30	1	30	30	30	TRUE	Accepted	Unmapped item	
11												
12												
13	Control	Actual	Expected	Difference								
14	Cases	9	9	0								
15	Expected results matched	9	9	0								
16	Disagreements	0	0	0								
17	Accepted key cases	7	7	0								
18	Unmapped key cases	1	1	0								
19	Blank key cases	1	1	0								

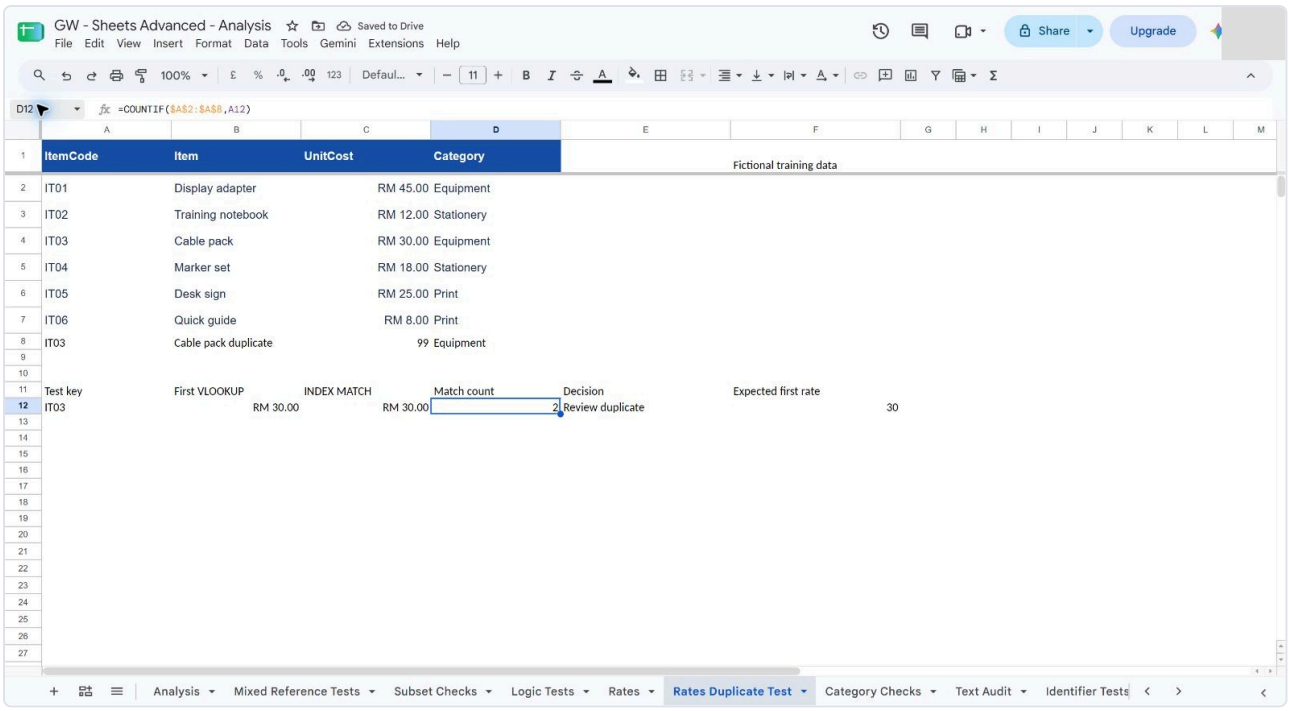
6. Duplicate baseline **Rates** as **Rates Duplicate Test**. Confirm the separate name and six original rows before introducing a conflict. Do not edit maintained Rates.



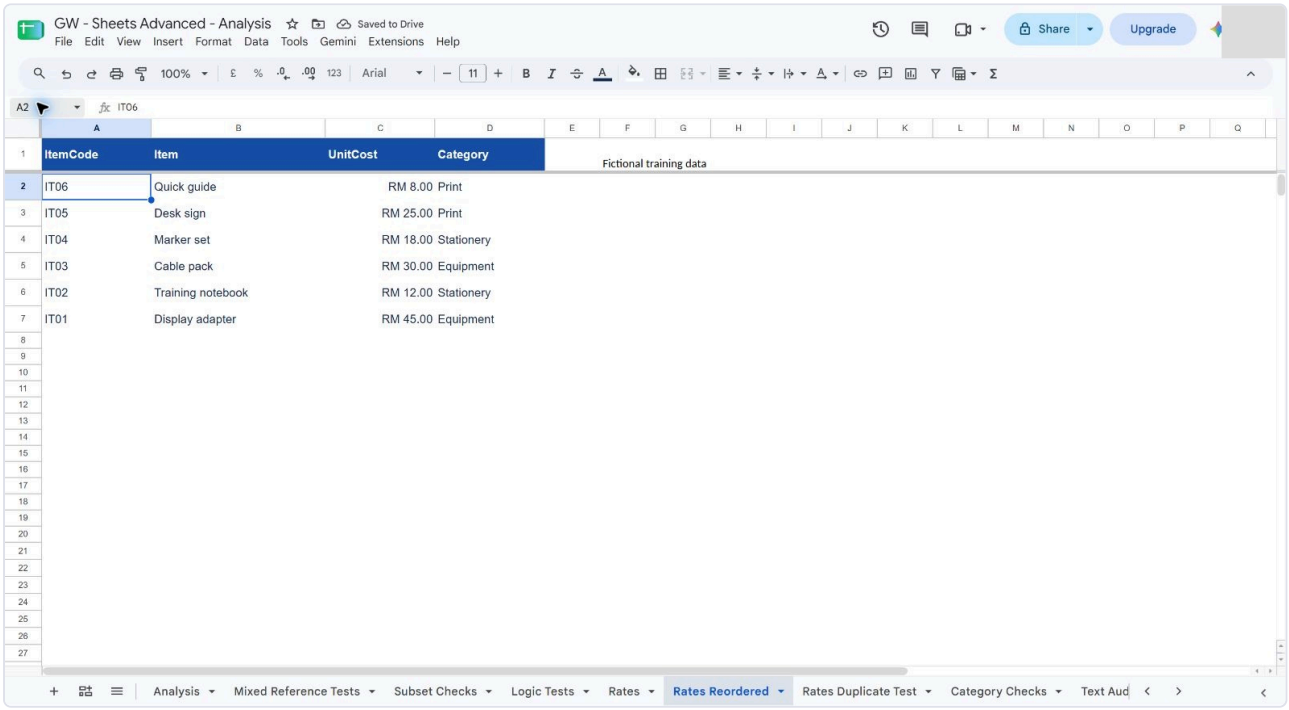
7. Append **IT03 /Cable pack duplicate /99 /Equipment** to A8:D8 on the copy only. The original IT03 row still has30.



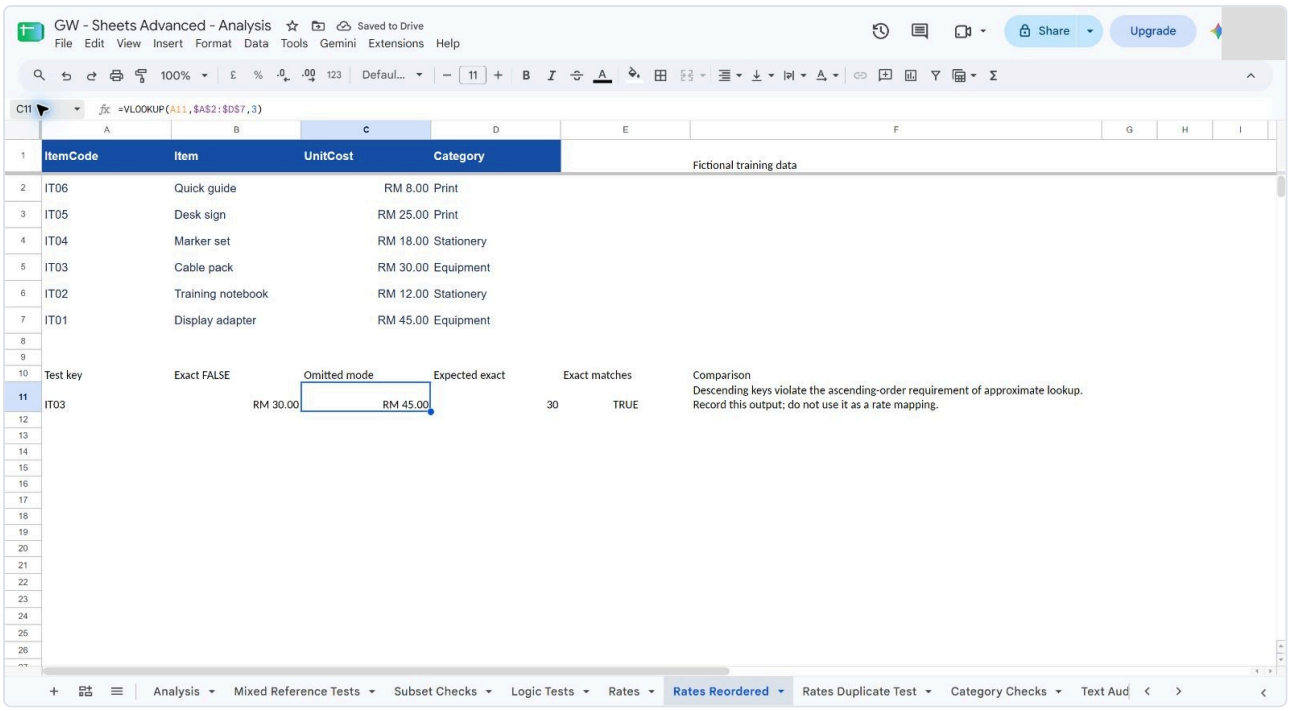
8. Add the duplicate diagnostic card at A11:F12. Exact VLOOKUP and INDEX/MATCH both return the first IT03 rate **30**, but COUNTIF over A2:A8 returns **2**. The decision is **Review duplicate**. Returning a value does not establish uniqueness; do not accept either30 or99 as an authoritative repair. Keep the labelled experimental copy.



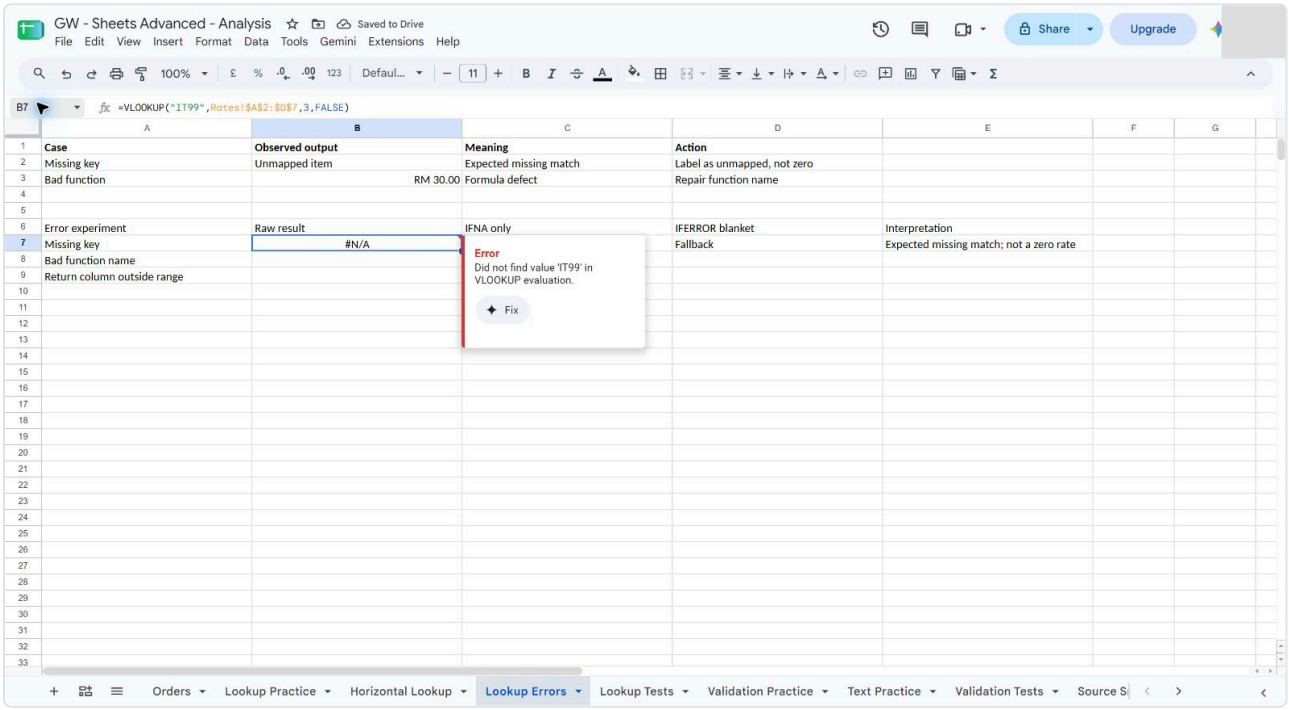
9. Make a second copy of baseline Rates named **Rates Reordered**. Select **A2:D7 only**, then **Data > Sort range > Sort range by column A (Z to A)**. Keep the header and outside-table data label separate. The keys become IT06 down to IT01, with the corresponding rows moving together.



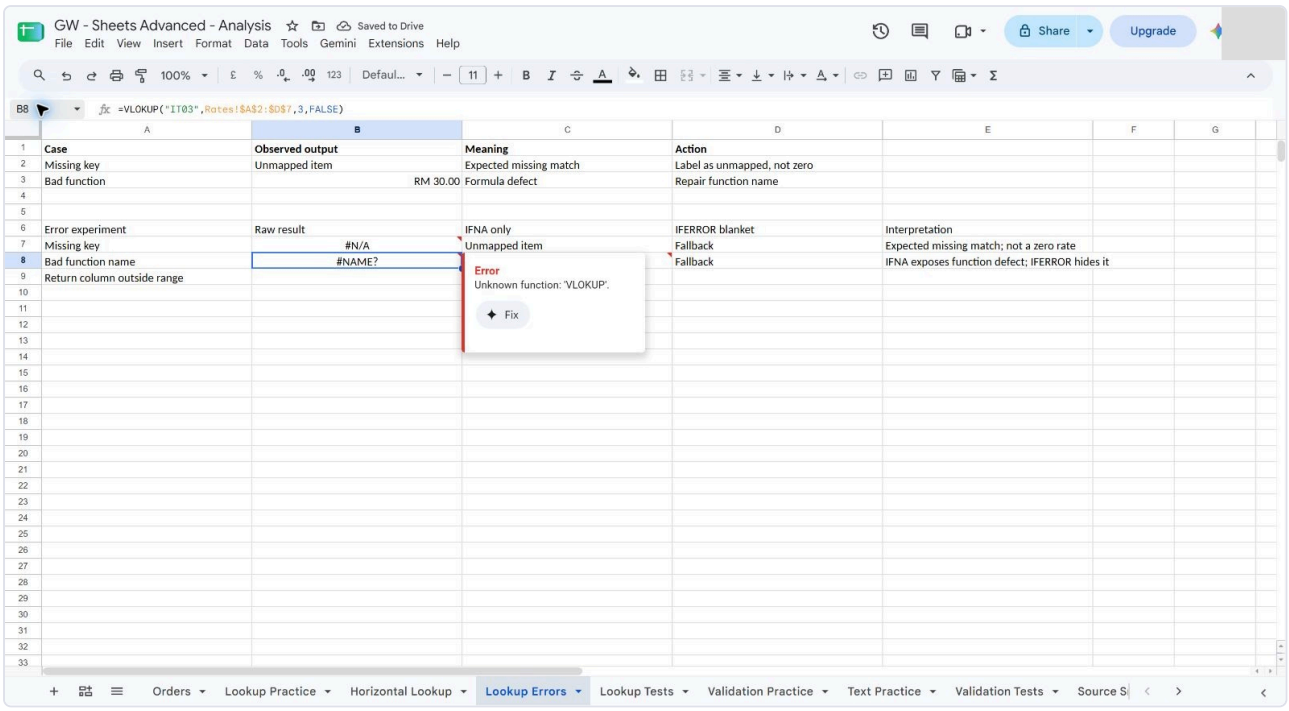
10. At A10:F11 compare IT03 using **=VLOOKUP(A11,$A$2:$D$7,3,FALSE)** with **=VLOOKUP(A11,$A$2:$D$7,3)**. In this actual descending-copy test, exact mode returns **30** while omitted mode returns the plausible but wrong **45**. The omitted mode assumes approximate matching on ascending keys; its precondition is violated here. Record this observation without generalising that every unsorted table will return45. Keep the experiment separate from the authoritative mapping.



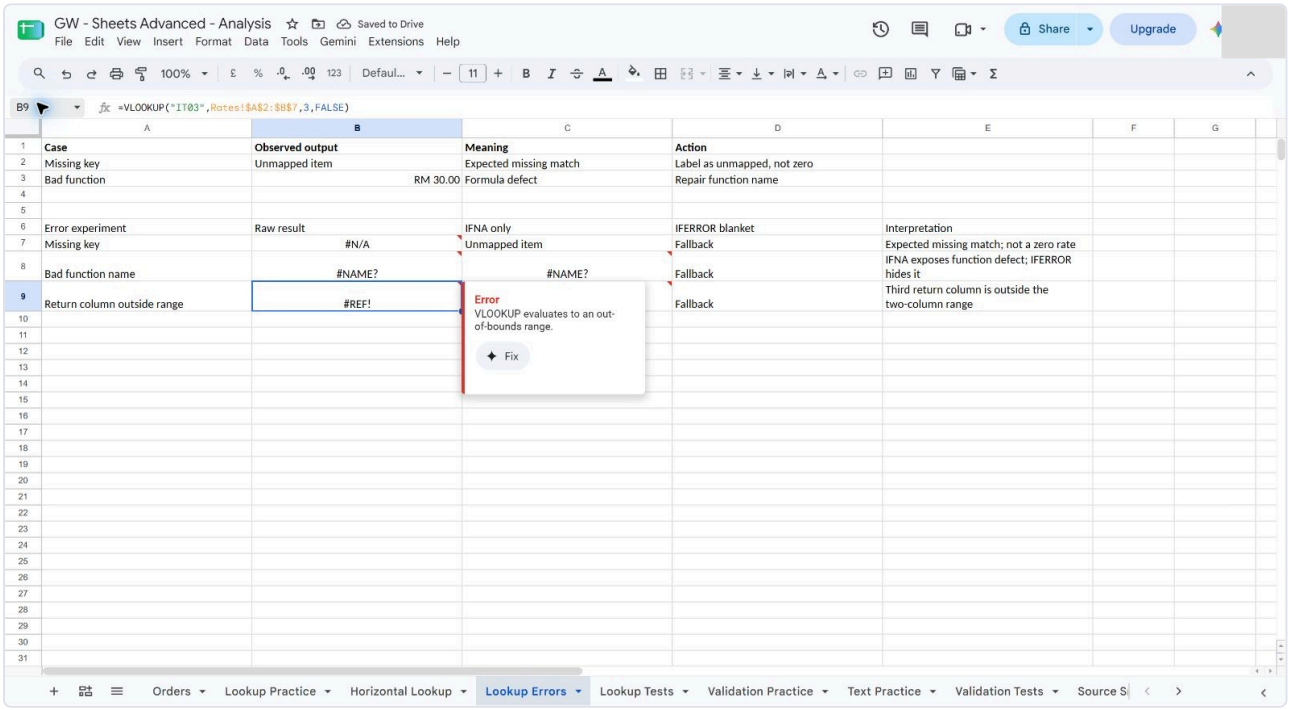
11. Return to Lookup Errors. Create the A6:E9 wrapper comparison. First run the raw missing-key formula in B7, **=IFNA(B7,"Unmapped item")** in C7 and **=IFERROR(B7,"Fallback")** in D7. NA becomes an explicit unmapped result under IFNA; a generic fallback is less specific.



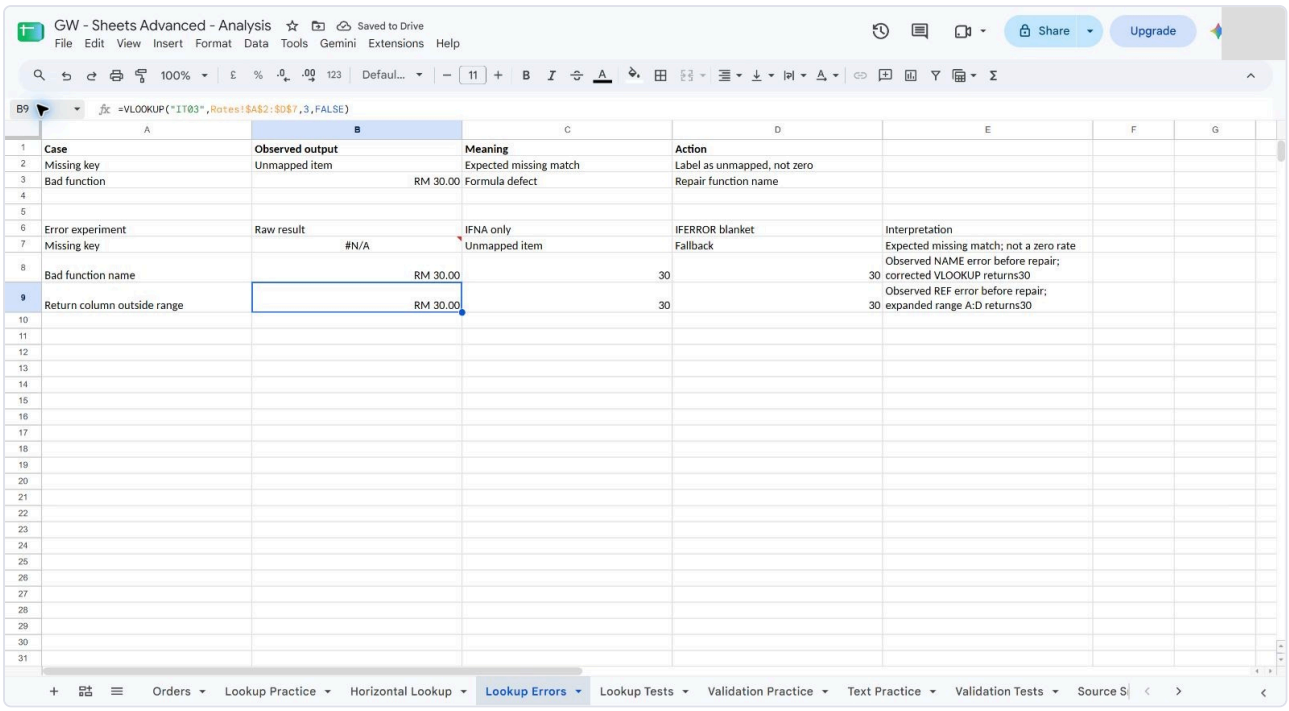
12. In B8 enter the misspelled VLOKUP case. Apply the same IFNA/IFERROR wrappers in C8/D8. The native **#NAME?** persists through IFNA but IFERROR hides it as Fallback. Retain actual error evidence before correcting it.



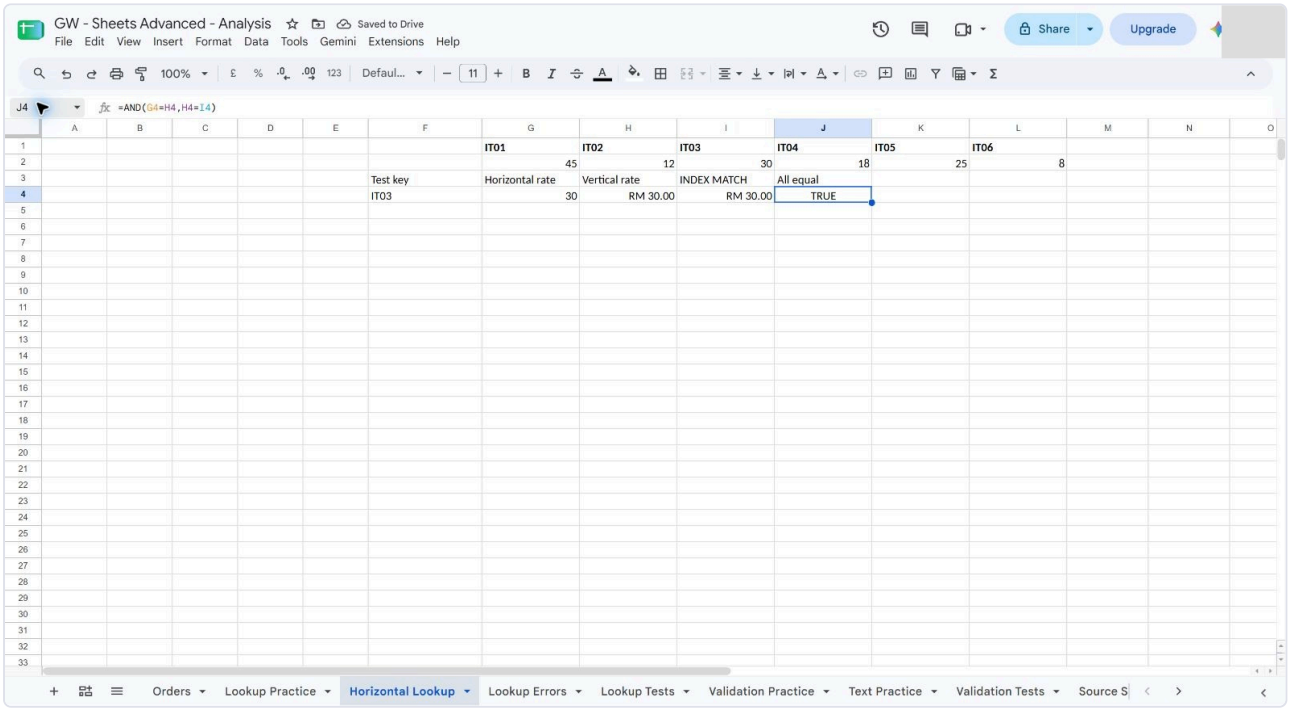
13. In B9 enter **=VLOOKUP("IT03",Rates!$A$2:$B$7,3,FALSE)**. Return column3 lies outside the two-column range. The native **#REF!** message is **VLOOKUP evaluates to an out of bounds range.** C9's IFNA preserves that defect; D9's IFERROR hides it. This is a range-shape defect, not a missing item.



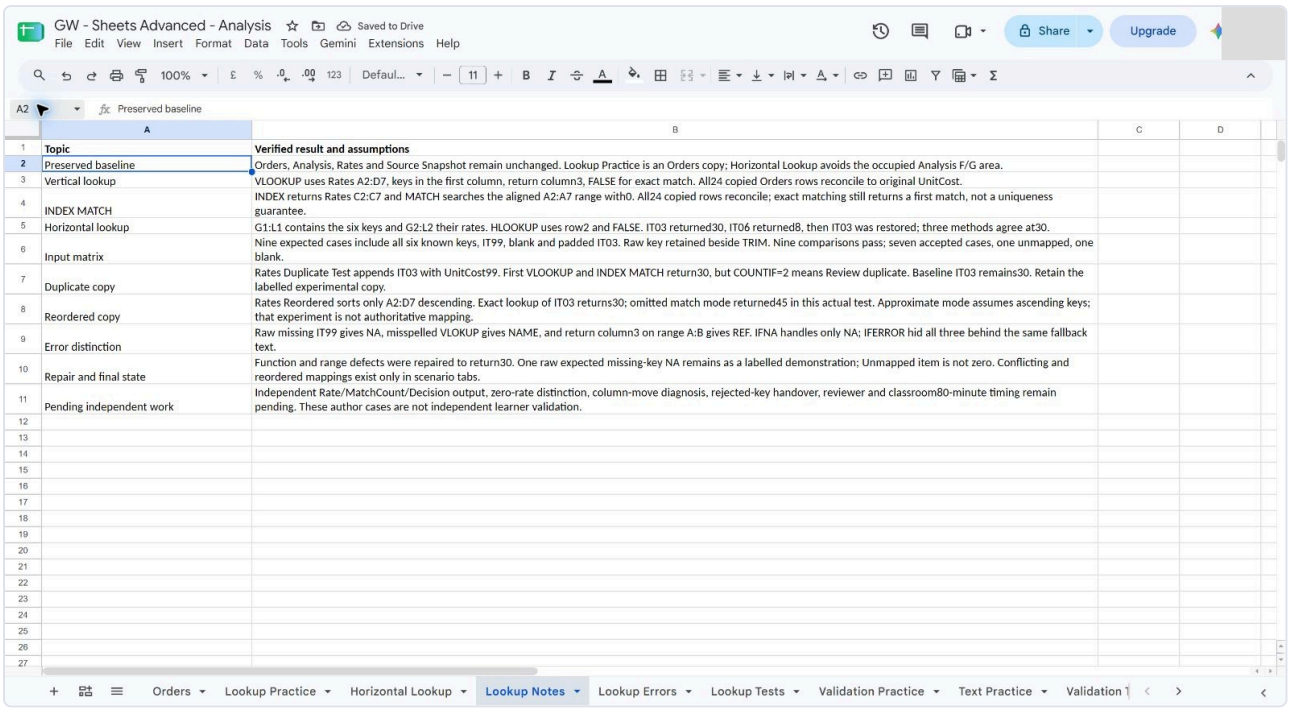
14. Repair B8's function spelling and expand B9's source range back to **Rates A2:D7**. B8:C9 and their broad wrappers now return30. Add the observed-error/repair notes. Retain B7's raw **#N/A** only as a clearly labelled expected missing-key demonstration; do not claim all error cells were removed.



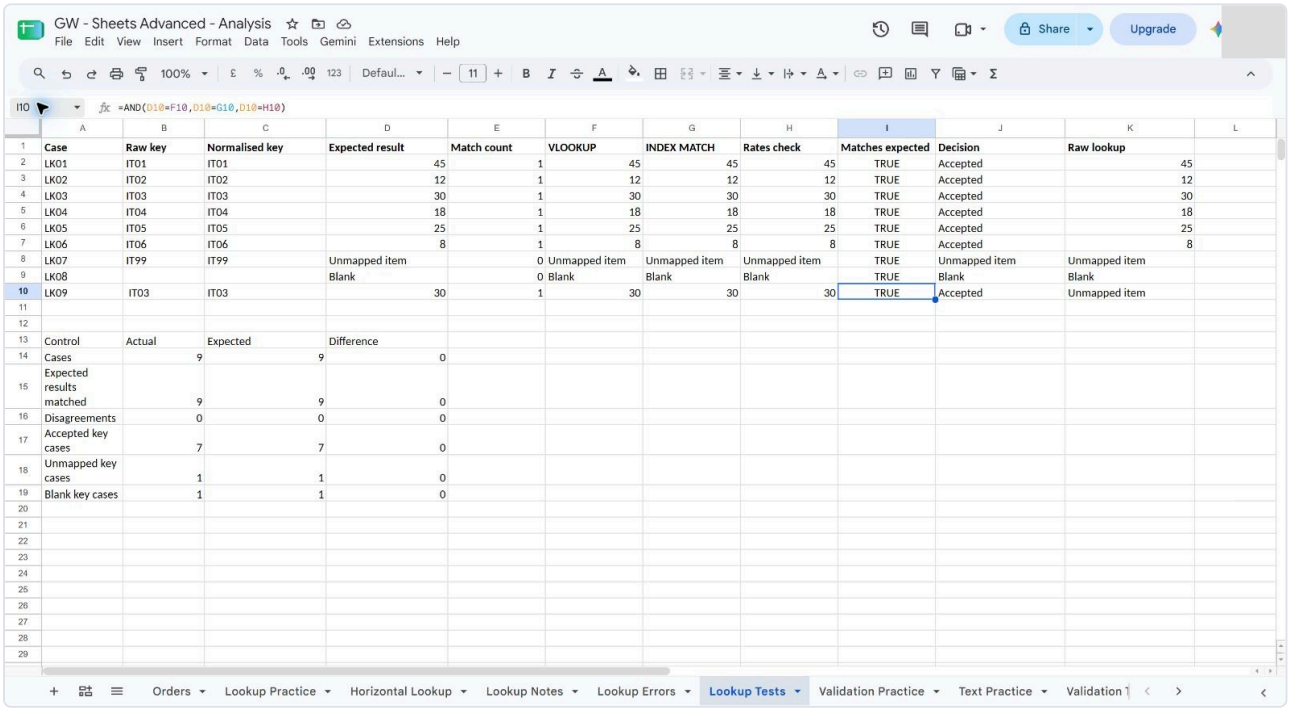
15. On Horizontal Lookup, add Vertical rate / INDEX MATCH / All equal in H3:J3. With F4 restored to IT03, use the card's exact vertical and INDEX/MATCH formulas in H4/I4 and **=AND(G4=H4,H4=I4)** in J4. All three results are30 and the comparison is TRUE.



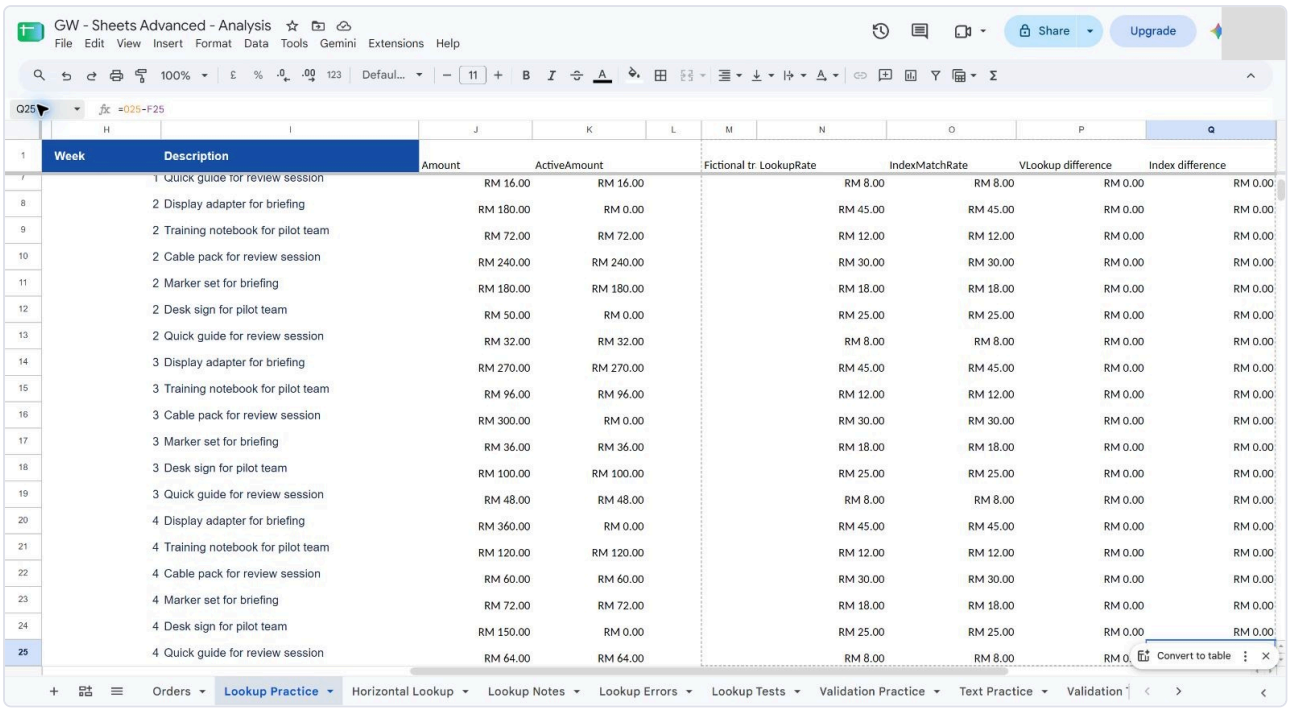
16. Create **Lookup Notes**. Record key/return range alignment, VLOOKUP first-column and column-index assumptions, HLOOKUP row orientation, explicit exact modes, raw-key handling, uniqueness diagnostic, experiment results and the final repair state. Exact matching also does not, by itself, prove a matched return value is populated; these supplied baseline rates were checked as numeric.

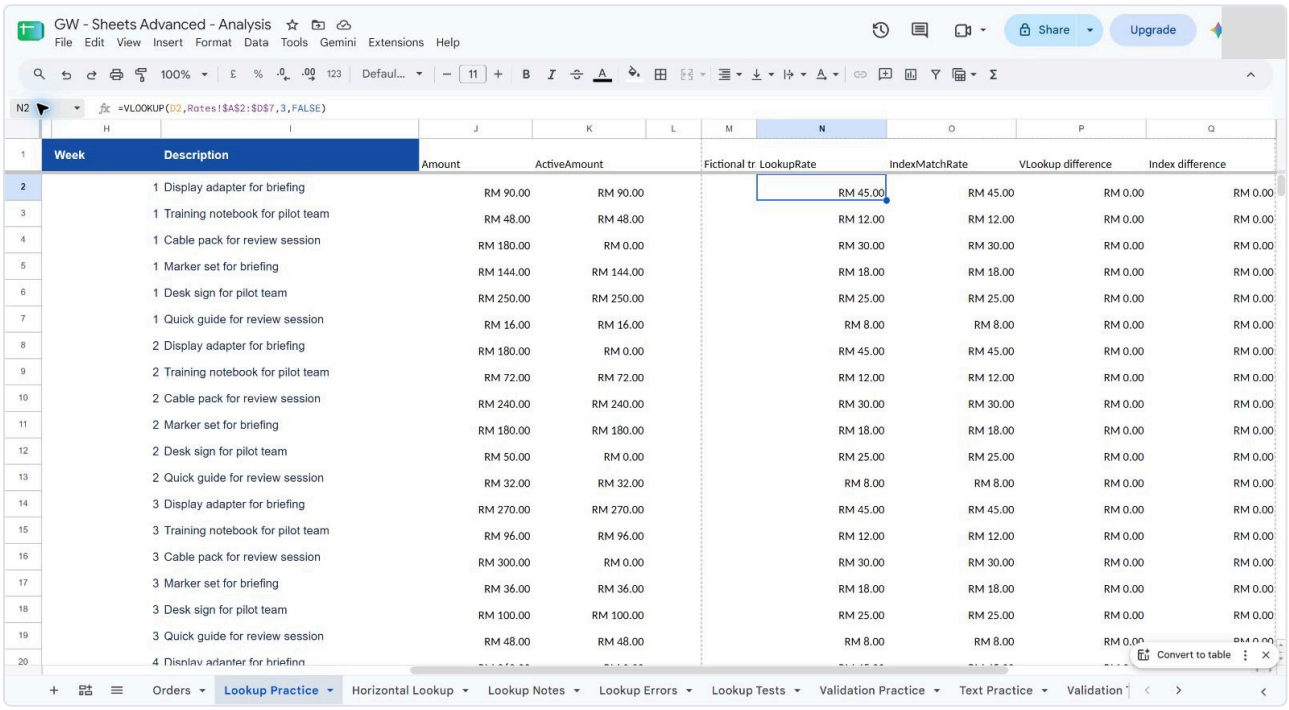


17. Inspect the final nine-key matrix and controls. Unmapped item and Blank remain distinct from a numeric rate. The independent zero-rate/output/column-move challenge is still pending.

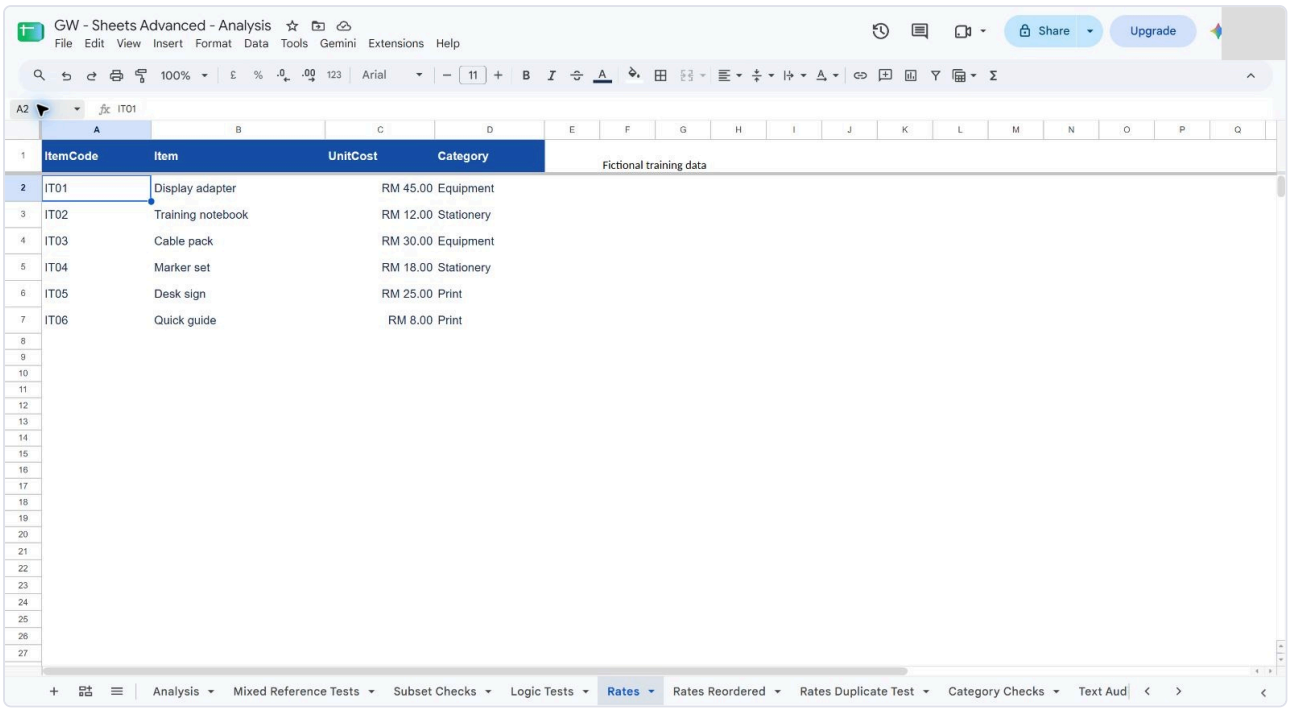


18. Recheck the final copied order's two zero rate differences, then the first order. All24 rows reconcile to the maintained source.

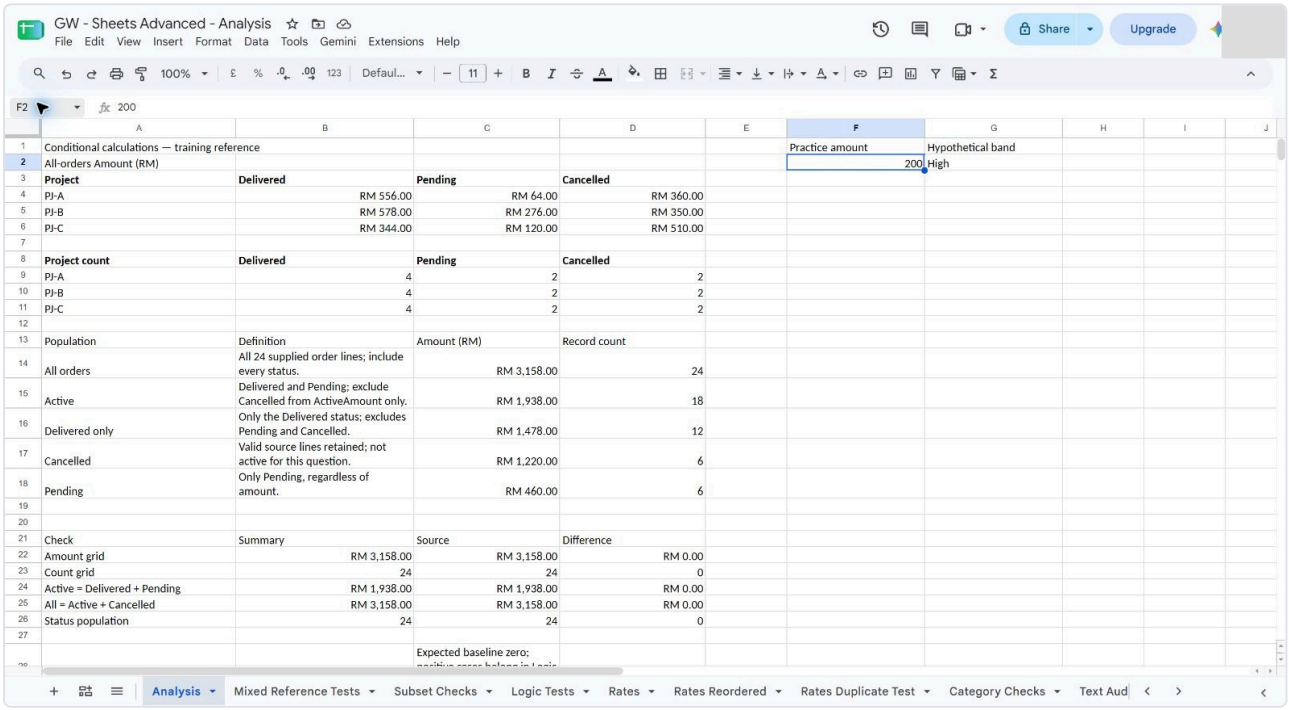




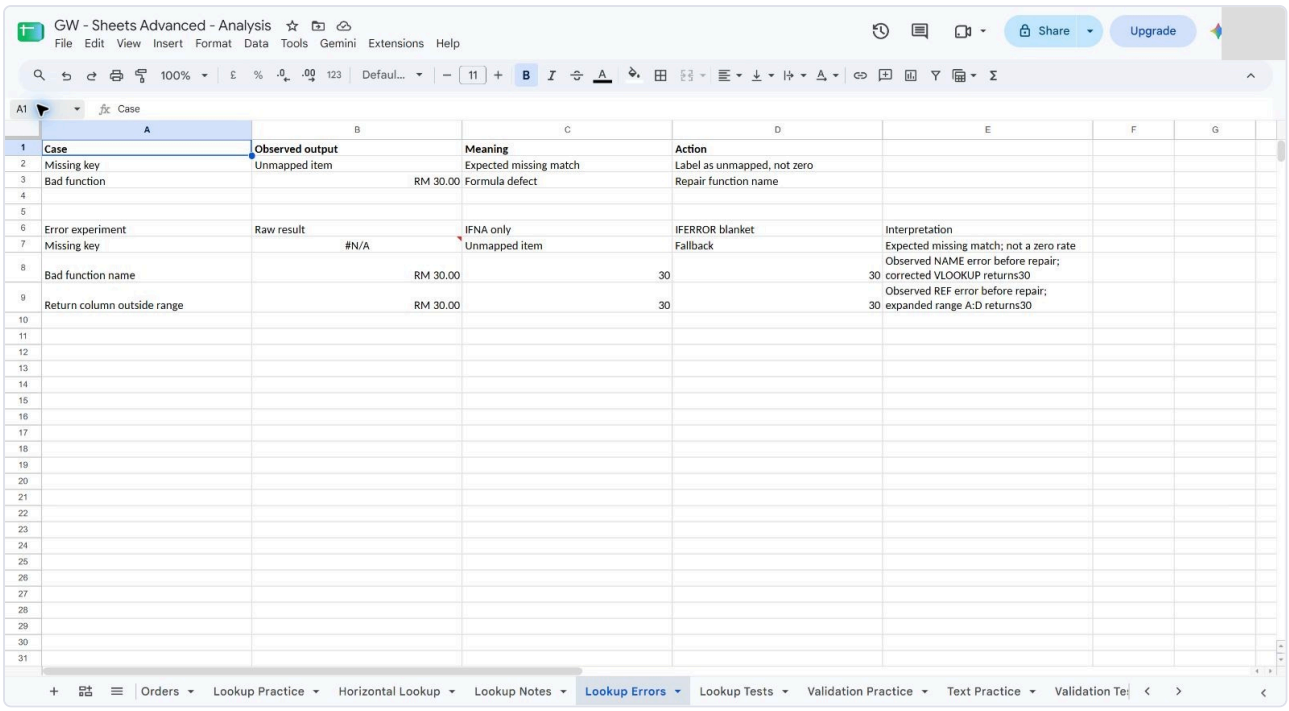
19. Inspect maintained Rates: exactly the original six unique items, with IT03 still30. No duplicate or reordering experiment changed this mapping.



20. Inspect maintained Analysis F2. Its Lab2 band test remains200/High and the earlier grids are intact; the horizontal exercise used a separate tab.

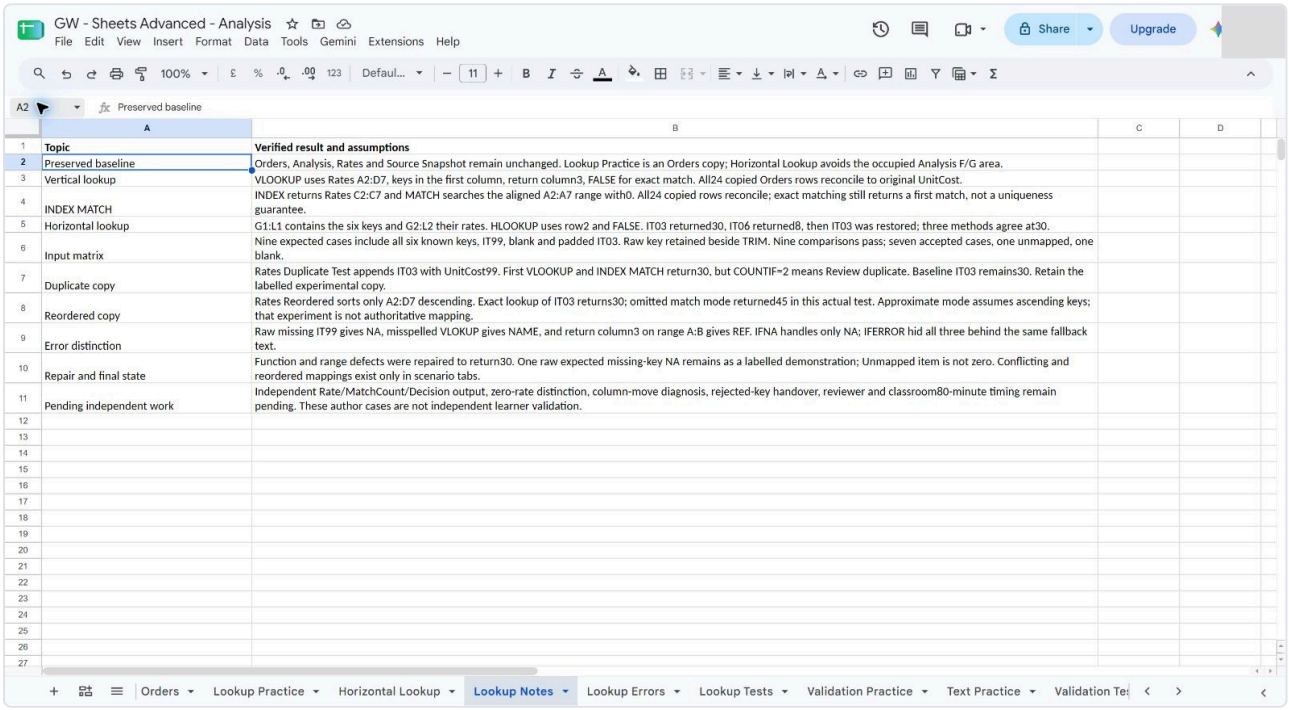


21. Leave Lookup Errors with repaired function/range cases and the one labelled raw missing case. The broad wrapper did not repair the defects; the underlying formulas were corrected.



	A	B	C	D	E
1	**Case**	**Observed output**	**Meaning**	**Action**	
2	Missing key	Unmapped item	Expected missing match	Label as unmapped, not zero	
3	Bad function	RM 30.00	Formula defect	Repair function name	
4					
5					
6	Error experiment	Raw result	IFNA only	IFERROR blanket	Interpretation
7	Missing key	#N/A	Unmapped item	Fallback	Expected missing match; not a zero rate
8	Bad function name	RM 30.00	30	30	Observed NAME error before repair; corrected VLOOKUP returns30
9	Return column outside range	RM 30.00	30	30	Observed REF error before repair; expanded range A:D returns30

22. Leave saved Lookup Notes ready for handover. Retain the labelled duplicate/reordered scenario tabs as teaching evidence, with their limitations stated.



	A	B	C	D
1	**Topic**	**Verified result and assumptions**		
2	Preserved baseline	Orders, Analysis, Rates and Source Snapshot remain unchanged. Lookup Practice is an Orders copy; Horizontal Lookup avoids the occupied Analysis F/G area.		
3	Vertical lookup	VLOOKUP uses Rates A2:D7, keys in the first column, return column3, FALSE for exact match. All24 copied Orders rows reconcile to original UnitCost.		
4	INDEX MATCH	INDEX returns Rates C2:C7 and MATCH searches the aligned A2:A7 range with0. All24 copied rows reconcile; exact matching still returns a first match, not a uniqueness guarantee.		
5	Horizontal lookup	G1:L1 contains the six keys and G2:L2 their rates. HLOOKUP uses row2 and FALSE. IT03 returned30, IT06 returned8, then IT03 was restored; three methods agree at30.		
6	Input matrix	Nine expected cases include all six known keys, IT99, blank and padded IT03. Raw key retained beside TRIM. Nine comparisons pass; seven accepted cases, one unmapped, one blank.		
7	Duplicate copy	Rates Duplicate Test appends IT03 with UnitCost99. First VLOOKUP and INDEX MATCH return30, but COUNTIF=2 means Review duplicate. Baseline IT03 remains30. Retain the labelled experimental copy.		
8	Reordered copy	Rates Reordered sorts only A2:D7 descending. Exact lookup of IT03 returns30; omitted match mode returned45 in this actual test. Approximate mode assumes ascending keys; that experiment is not authoritative mapping.		
9	Error distinction	Raw missing IT99 gives NA, misspelled VLOKUP gives NAME, and return column3 on range A:B gives REF. IFNA handles only NA; IFERROR hid all three behind the same fallback text.		
10	Repair and final state	Function and range defects were repaired to return30. One raw expected missing-key NA remains as a labelled demonstration; Unmapped item is not zero. Conflicting and reordered mappings exist only in scenario tabs.		
11	Pending independent work	Independent Rate/MatchCount/Decision output, zero-rate distinction, column-move diagnosis, rejected-key handover, reviewer and classroom80-minute timing remain pending. These author cases are not independent learner validation.		

> **Validation boundary:** guided5.1–5.3 and author casework5.4 were executed through learner UI and bounded read-only native-cell evidence. Independent output, zero-rate distinction, column-move diagnosis, rejected-key handover, reviewer and planned80-minute learner timing remain unverified. Baseline Orders/Analysis/Rates/Source Snapshot remain unchanged.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Design an independent lookup output with Rate, MatchCount and Decision for the mixed intake. A zero rate, unknown key and duplicate key must remain distinguishable.
2. Move a column only on a disposable mapping copy and inspect which lookup formula needs repair. Explain why the original column-index assumption failed.
3. Hand over a rejected-key list and source-mapping check, leaving baseline Rates unchanged.

Evidence to Hand Over

- Known/missing/blank/padded-key test matrix with exact-method reconciliation.
- Duplicate-key and column-move diagnosis on labelled copies.
- Independent lookup result that never silently turns unknown rates into zero.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A lookup returns a plausible but wrong rate	Inspect exact-match mode, key uniqueness, whitespace and the lookup column position.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Reliable lookups depend on clean keys, explicit match behaviour and deliberate error handling.

Check yourself: Does an exact-match lookup prove that a matched source value is populated?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
LookupRate
```

```
=VLOOKUP(D2,Rates!$A$2:$D$7,3,FALSE)
```

```
=INDEX(Rates!$C$2:$C$7,MATCH(D2,Rates!$A$2:$A$7,0))
```

```
=HLOOKUP(F4,$G$1:$L$2,2,FALSE)
```

```
=IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Lookup challenge: Known IT01 and IT06; unknown IT99; blank; space-padded IT03. Duplicate-test
mapping adds IT03 with UnitCost 99 only on a disposable copy. Baseline IT03 remains 30. Keep
the supplied six-key Rates source intact.
```

Captured Author-Run Reference

Fictional training material. Use separate practice/copy tabs, preserve raw keys and record expectations before actual results. This is not independent learner validation.

Lookup Practice N1:Q2 — seed formulas, fill row2 through25

```
LookupRate      IndexMatchRate  VLookup difference      Index difference
=VLOOKUP(D2,Rates!$A$2:$D$7,3,FALSE)    =INDEX(Rates!$C$2:$C$7,MATCH(D2,Rates!$A$2:$A$7,0))
=N2-F2  =O2-F2
```

Horizontal Lookup G1:L2 reference

```
IT01    IT02    IT03    IT04    IT05    IT06
45      12      30      18      25      8
```

Lookup Errors A1:D3 — repaired introductory state

```
Case    Observed output Meaning Action
Missing key     =IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")  Expected
missing match   Label as unmapped, not zero
Bad function    =VLOOKUP("IT03",Rates!$A$2:$D$7,3,FALSE)        Formula defect  Repair
function name
```

Lookup Tests A1:K10 — expectations, blank actual columns

```
Case    Raw key Normalised key  Expected result Match count     VLOOKUP INDEX MATCH     Rates
check   Matches expected        Decision        Raw lookup
LK01    IT01            45
LK02    IT02            12
LK03    IT03            30
LK04    IT04            18
LK05    IT05            25
LK06    IT06            8
LK07    IT99            Unmapped item
LK08                    Blank
LK09     IT03           30
```

Lookup Tests A13:D19 controls

```
Control Actual  Expected        Difference
Cases   =COUNTA(A2:A10) 9       =B14-C14
Expected results matched        =COUNTIF(I2:I10,TRUE)   9       =B15-C15
Disagreements   =COUNTIF(I2:I10,FALSE)  0       =B16-C16
Accepted key cases      =COUNTIF(J2:J10,"Accepted")     7       =B17-C17
Unmapped key cases      =COUNTIF(J2:J10,"Unmapped item")        1       =B18-C18
Blank key cases =COUNTIF(J2:J10,"Blank")        1       =B19-C19
```

Rates Duplicate Test A8:D8 — copy only

```
IT03    Cable pack duplicate    99      Equipment
```

Duplicate diagnostic A11:F12

```
Test key        First VLOOKUP   INDEX MATCH     Match count     Decision        Expected first
rate
IT03    =VLOOKUP(A12,$A$2:$D$8,3,FALSE) =INDEX($C$2:$C$8,MATCH(A12,$A$2:$A$8,0))
=COUNTIF($A$2:$A$8,A12) =IF(D12<>1,"Review duplicate","Accepted")       30
```

Descending-copy experiment A10:F11

```
Test key        Exact FALSE     Omitted mode    Expected exact  Exact matches   Comparison
IT03    =VLOOKUP(A11,$A$2:$D$7,3,FALSE) =VLOOKUP(A11,$A$2:$D$7,3)       30      =B11=D11
Descending keys violate the ascending-order requirement of approximate lookup. Record this
output; do not use it as a rate mapping.
```

Actual error comparison A6:E9 — before repair

```
Error experiment        Raw result      IFNA only       IFERROR blanket Interpretation
Missing key     =VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE)        =IFNA(B7,"Unmapped item")
=IFERROR(B7,"Fallback") Expected missing match; not a zero rate
Bad function name       =VLOKUP("IT03",Rates!$A$2:$D$7,3,FALSE) =IFNA(B8,"Unmapped item")
=IFERROR(B8,"Fallback") IFNA exposes function defect; IFERROR hides it
Return column outside range     =VLOOKUP("IT03",Rates!$A$2:$B$7,3,FALSE)
=IFNA(B9,"Unmapped item")       =IFERROR(B9,"Fallback") Third return column is outside the
two-column range
```

Lookup Notes A1:B11

```
Topic   Verified result and assumptions
Preserved baseline      Orders, Analysis, Rates and Source Snapshot remain unchanged. Lookup
Practice is an Orders copy; Horizontal Lookup avoids the occupied Analysis F/G area.
Vertical lookup VLOOKUP uses Rates A2:D7, keys in the first column, return column3, FALSE for
exact match. All24 copied Orders rows reconcile to original UnitCost.
INDEX MATCH     INDEX returns Rates C2:C7 and MATCH searches the aligned A2:A7 range with0.
All24 copied rows reconcile; exact matching still returns a first match, not a uniqueness
guarantee.
Horizontal lookup       G1:L1 contains the six keys and G2:L2 their rates. HLOOKUP uses row2
and FALSE. IT03 returned30, IT06 returned8, then IT03 was restored; three methods agree at30.
Input matrix    Nine expected cases include all six known keys, IT99, blank and padded IT03.
Raw key retained beside TRIM. Nine comparisons pass; seven accepted cases, one unmapped, one
blank.
Duplicate copy  Rates Duplicate Test appends IT03 with UnitCost99. First VLOOKUP and INDEX
MATCH return30, but COUNTIF=2 means Review duplicate. Baseline IT03 remains30. Retain the
labelled experimental copy.
Reordered copy  Rates Reordered sorts only A2:D7 descending. Exact lookup of IT03 returns30;
omitted match mode returned45 in this actual test. Approximate mode assumes ascending keys;
that experiment is not authoritative mapping.
Error distinction       Raw missing IT99 gives NA, misspelled VLOKUP gives NAME, and return
column3 on range A:B gives REF. IFNA handles only NA; IFERROR hid all three behind the same
fallback text.
Repair and final state  Function and range defects were repaired to return30. One raw expected
missing-key NA remains as a labelled demonstration; Unmapped item is not zero. Conflicting and
reordered mappings exist only in scenario tabs.
Pending independent work        Independent Rate/MatchCount/Decision output, zero-rate
distinction, column-move diagnosis, rejected-key handover, reviewer and classroom80-minute
timing remain pending. These author cases are not independent learner validation.
```

Lookup Tests seed formulas — fill each column from row2 through10

C2:C10

```
=TRIM(B2)
```

E2:E10

```
=IF(C2="",0,COUNTIF(Rates!$A$2:$A$7,C2))
```

F2:F10

```
=IF(C2="","Blank",IFNA(VLOOKUP(C2,Rates!$A$2:$D$7,3,FALSE),"Unmapped item"))
```

G2:G10

```
=IF(C2="","Blank",IFNA(INDEX(Rates!$C$2:$C$7,MATCH(C2,Rates!$A$2:$A$7,0)),"Unmapped item"))
```

H2:H10

```
=IF(C2="","Blank",IF(E2=0,"Unmapped
item",IF(E2>1,"Duplicate",SUMIF(Rates!$A$2:$A$7,C2,Rates!$C$2:$C$7))))
```

I2:I10

```
=AND(D2=F2,D2=G2,D2=H2)
```

J2:J10

```
=IF(C2="","Blank",IF(E2=0,"Unmapped item",IF(E2>1,"Duplicate","Accepted")))
```

K2:K10

```
=IF(B2="","Blank",IFNA(VLOOKUP(B2,Rates!$A$2:$D$7,3,FALSE),"Unmapped item"))
```

Horizontal comparison formulas

Horizontal Lookup G4

```
=HLOOKUP(F4,$G$1:$L$2,2,FALSE)
```

Horizontal Lookup H4

```
=VLOOKUP(F4,Rates!$A$2:$D$7,3,FALSE)
```

Horizontal Lookup I4

```
=INDEX(Rates!$C$2:$C$7,MATCH(F4,Rates!$A$2:$A$7,0))
```

Horizontal Lookup J4

```
=AND(G4=H4,H4=I4)
```

Enter padded Lookup Tests B10 individually as one ordinary space, IT03, then one ordinary space. Keep B9 truly empty. Error comparison formulas intentionally create errors; capture and inspect them before repairing B8/B9 to exact VLOOKUP on Rates A2:D7. Independent zero-rate/column-move/output and reviewer checks remain pending.

06

CHAPTER 6

Complete the Formula Clinic

In this chapter you will debug supplied defects, explain the correction and reconcile a checked sample.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 65 minutes: 15 explanation/demonstration, 22 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 50 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Three repaired defects with independent baseline reconciliations.

> **Status:** author reference with 31 staged application captures through 6.4. Three defects are observed and repaired in a separate workbook copy. Zero, missing, numeric-text, duplicate-key and final-row tests are verified; original source remains unchanged. Independent diagnosis, partner/reviewer and learner timing remain pending.

What You Will Learn

- Diagnose formula defects.
- Test zero and missing values.
- Reconcile source totals independently.

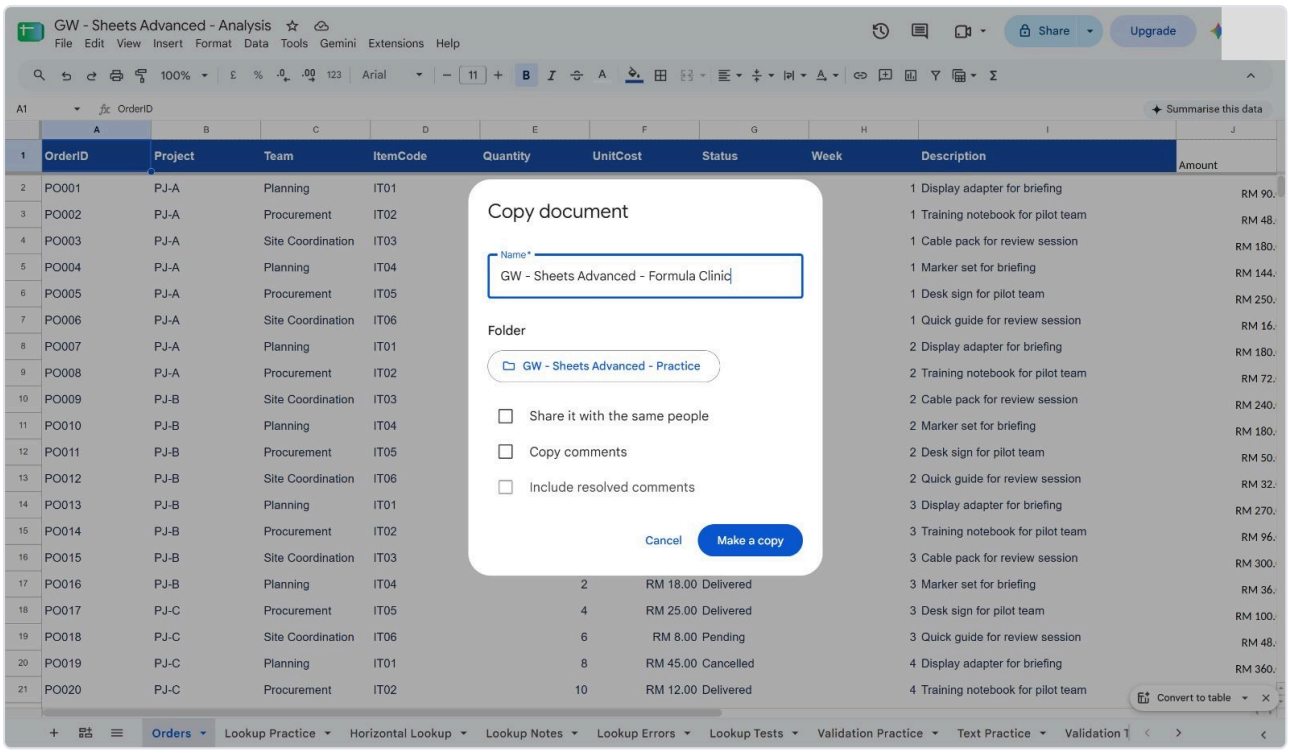
Before You Begin

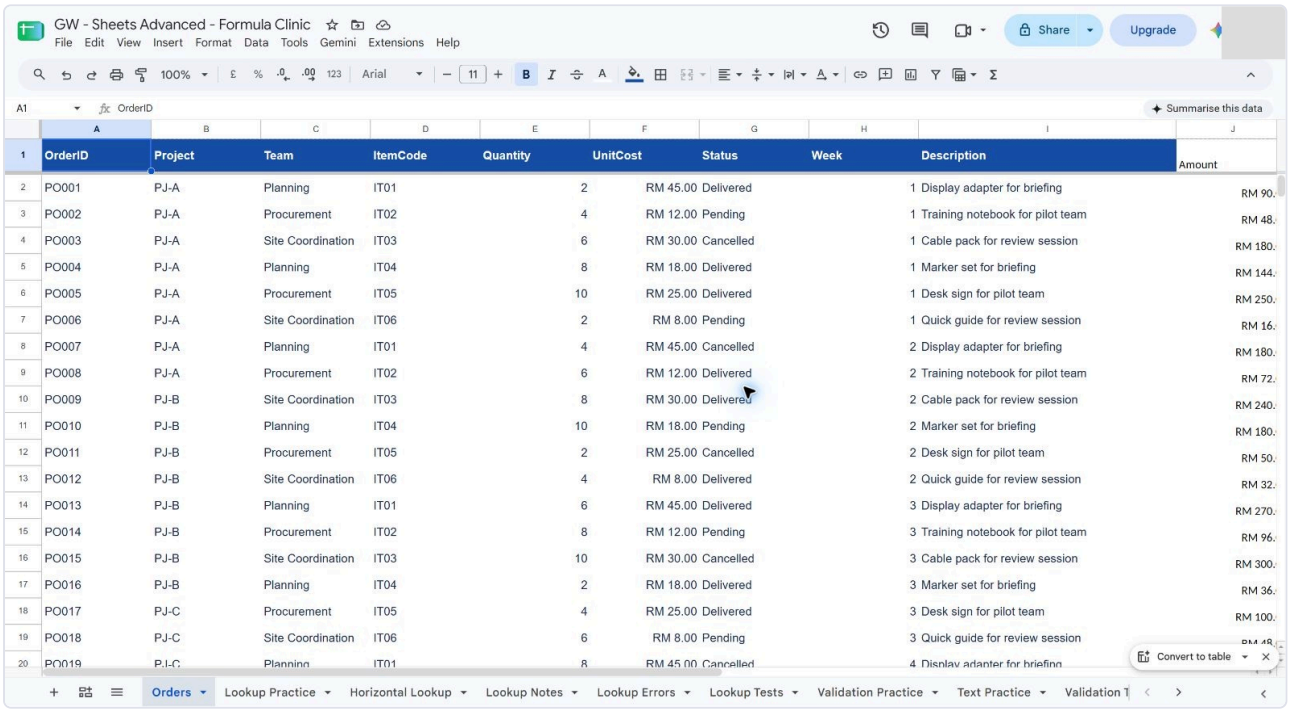
Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use a challenge copy of your advanced workbook. Keep the baseline original untouched.

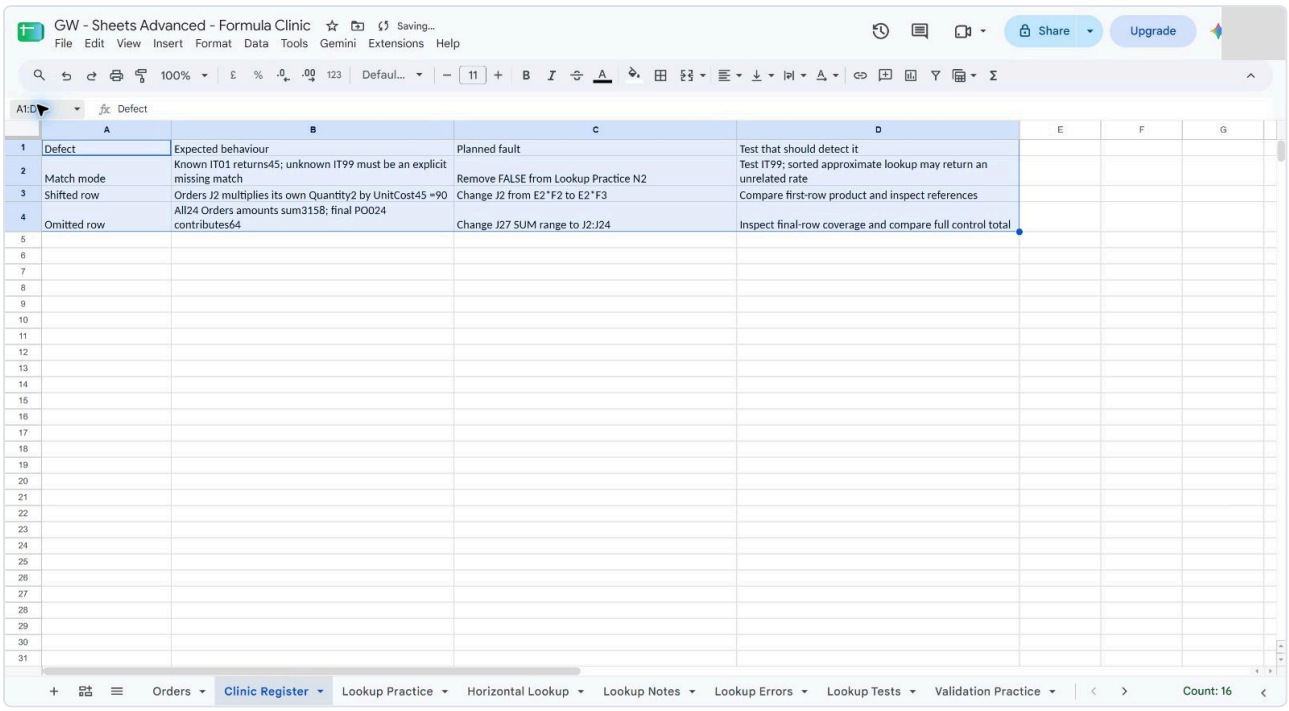
6.1 Create Three Test Defects

1. File > Make a copy of the existing Advanced workbook. Name it GW - Sheets Advanced - Formula Clinic in the practice folder. Keep sharing/comments unchecked. Verify the new workbook title and different file URL before editing.



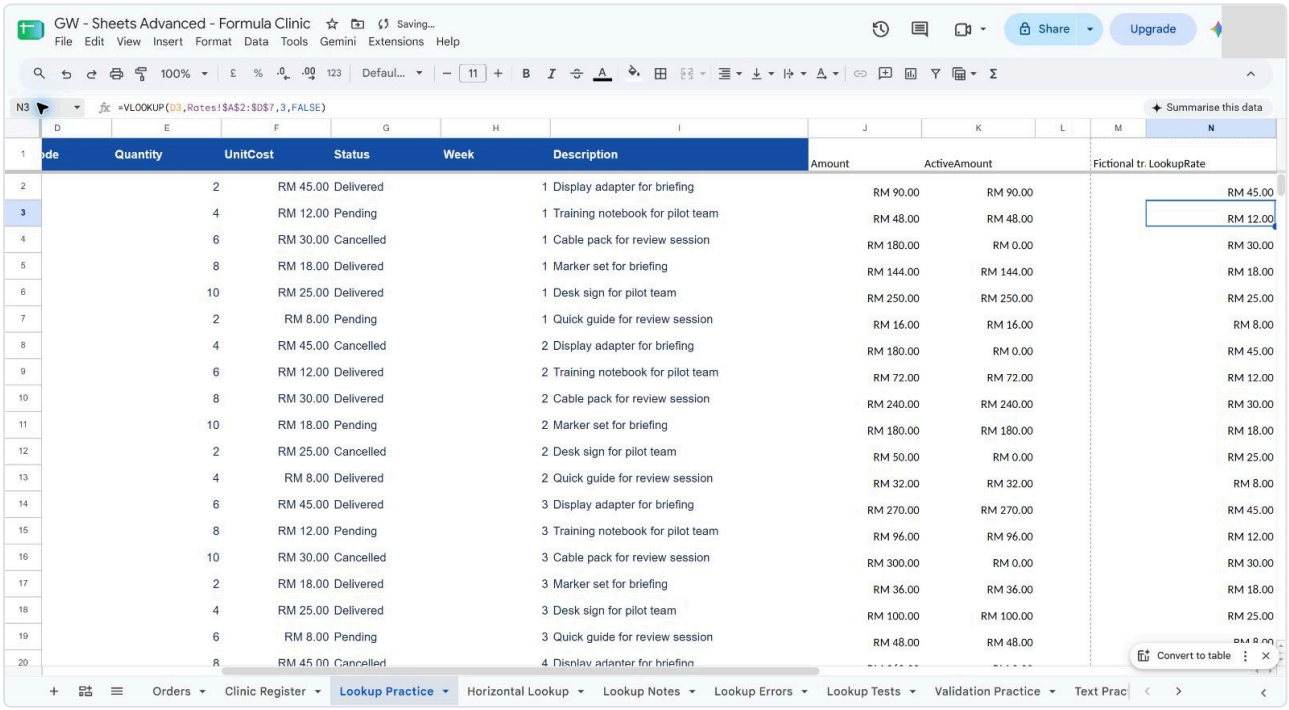


2. Create Clinic Register and record intended behaviour, fault and detecting test before mutation. Known IT01=45 is not sufficient to detect omitted match mode; missing IT99 must remain explicit. First line is 2×45=90; final order contributes 64 to 3158.

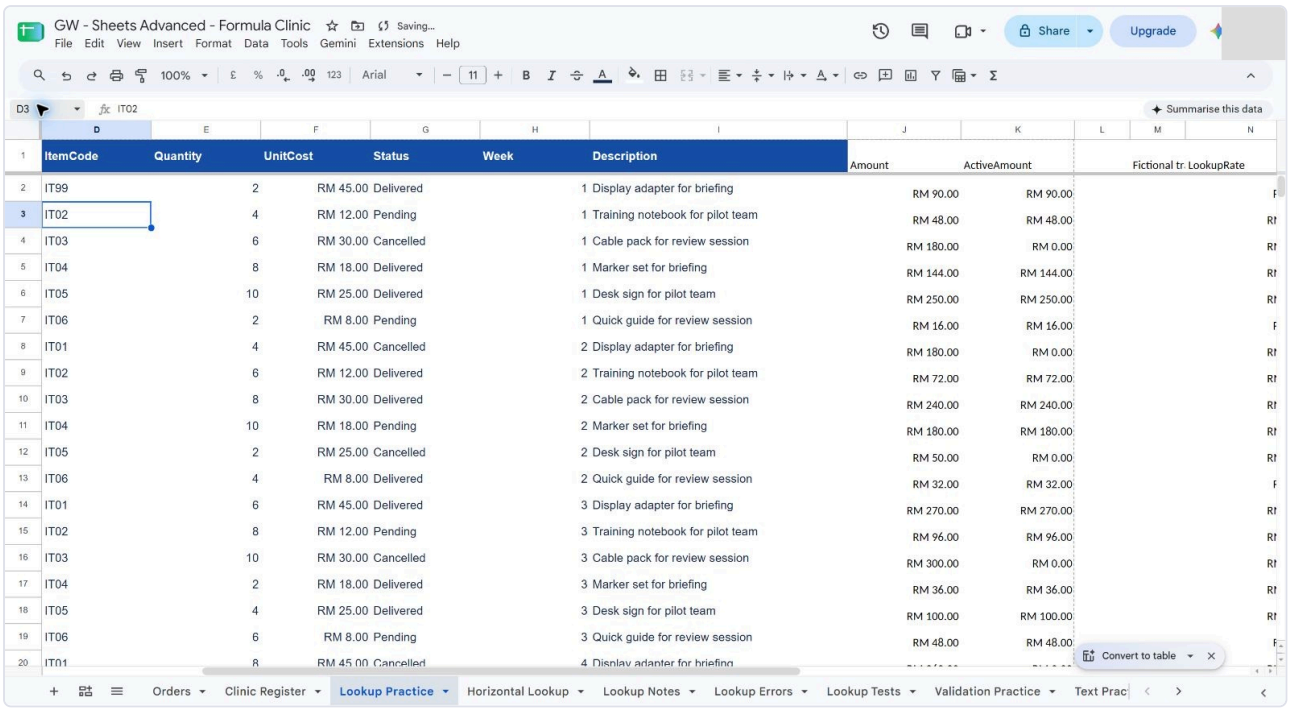


	A	B	C	D
1	Defect	Expected behaviour	Planned fault	Test that should detect it
2	Match mode	Known IT01 returns45; unknown IT99 must be an explicit missing match	Remove FALSE from Lookup Practice N2	Test IT99; sorted approximate lookup may return an unrelated rate
3	Shifted row	Orders J2 multiplies its own Quantity2 by UnitCost45 =90	Change J2 from E2*F2 to E2*F3	Compare first-row product and inspect references
4	Omitted row	All24 Orders amounts sum3158; final PO024 contributes64	Change J27 SUM range to J2:J24	Inspect final-row coverage and compare full control total

3. On the copy Lookup Practice N2 remove FALSE. With known IT01 it still returns 45. Temporarily test D2=IT99: the omitted mode returns8 from the sorted mapping, an incorrect rate for an absent key. Preserve that observation, then restore D2=IT01.

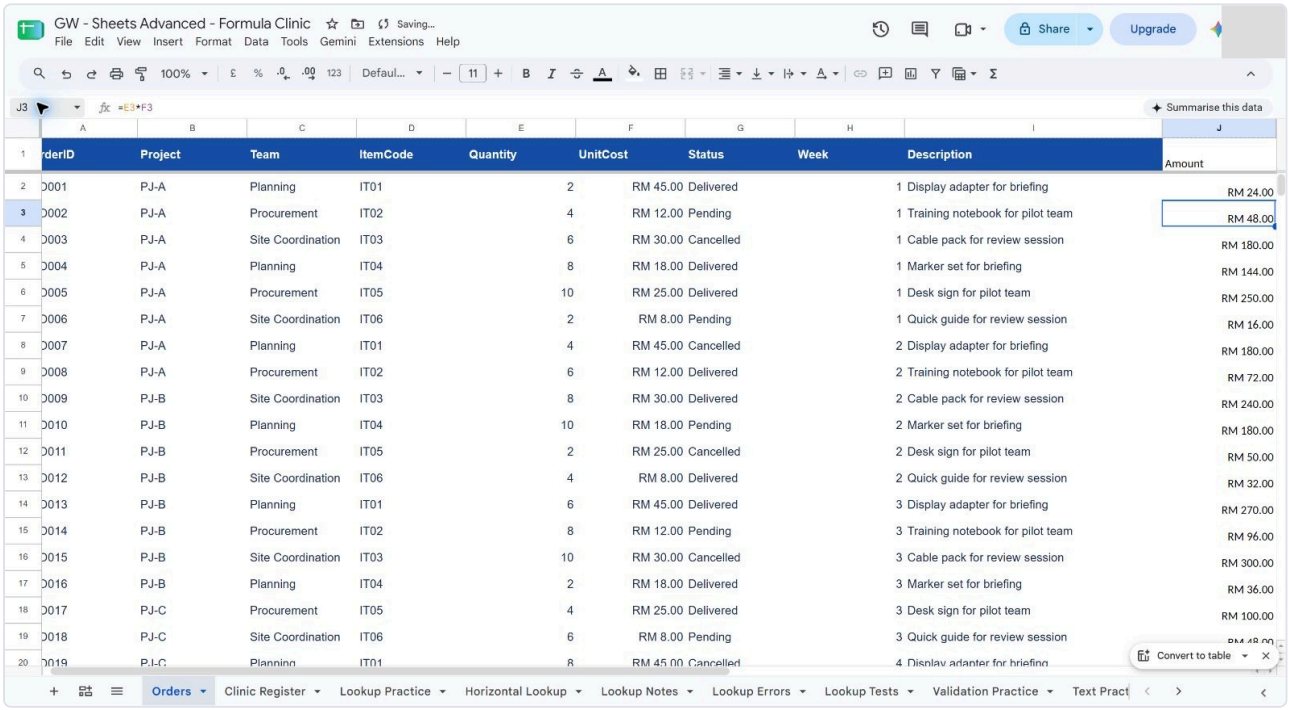


	ode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	Fictional tr. LookupRate
2		2	RM 45.00	Delivered	1	Display adapter for briefing	RM 90.00	RM 90.00	RM 45.00
3		4	RM 12.00	Pending	1	Training notebook for pilot team	RM 48.00	RM 48.00	RM 12.00
4		6	RM 30.00	Cancelled	1	Cable pack for review session	RM 180.00	RM 0.00	RM 30.00
5		8	RM 18.00	Delivered	1	Marker set for briefing	RM 144.00	RM 144.00	RM 18.00
6		10	RM 25.00	Delivered	1	Desk sign for pilot team	RM 250.00	RM 250.00	RM 25.00
7		2	RM 8.00	Pending	1	Quick guide for review session	RM 16.00	RM 16.00	RM 8.00
8		4	RM 45.00	Cancelled	2	Display adapter for briefing	RM 180.00	RM 0.00	RM 45.00
9		6	RM 12.00	Delivered	2	Training notebook for pilot team	RM 72.00	RM 72.00	RM 12.00
10		8	RM 30.00	Delivered	2	Cable pack for review session	RM 240.00	RM 240.00	RM 30.00
11		10	RM 18.00	Pending	2	Marker set for briefing	RM 180.00	RM 180.00	RM 18.00
12		2	RM 25.00	Cancelled	2	Desk sign for pilot team	RM 50.00	RM 0.00	RM 25.00
13		4	RM 8.00	Delivered	2	Quick guide for review session	RM 32.00	RM 32.00	RM 8.00
14		6	RM 45.00	Delivered	3	Display adapter for briefing	RM 270.00	RM 270.00	RM 45.00
15		8	RM 12.00	Pending	3	Training notebook for pilot team	RM 96.00	RM 96.00	RM 12.00
16		10	RM 30.00	Cancelled	3	Cable pack for review session	RM 300.00	RM 0.00	RM 30.00
17		2	RM 18.00	Delivered	3	Marker set for briefing	RM 36.00	RM 36.00	RM 18.00
18		4	RM 25.00	Delivered	3	Desk sign for pilot team	RM 100.00	RM 100.00	RM 25.00
19		6	RM 8.00	Pending	3	Quick guide for review session	RM 48.00	RM 48.00	RM 8.00
20		8	RM 45.00	Cancelled	4	Display adapter for briefing			



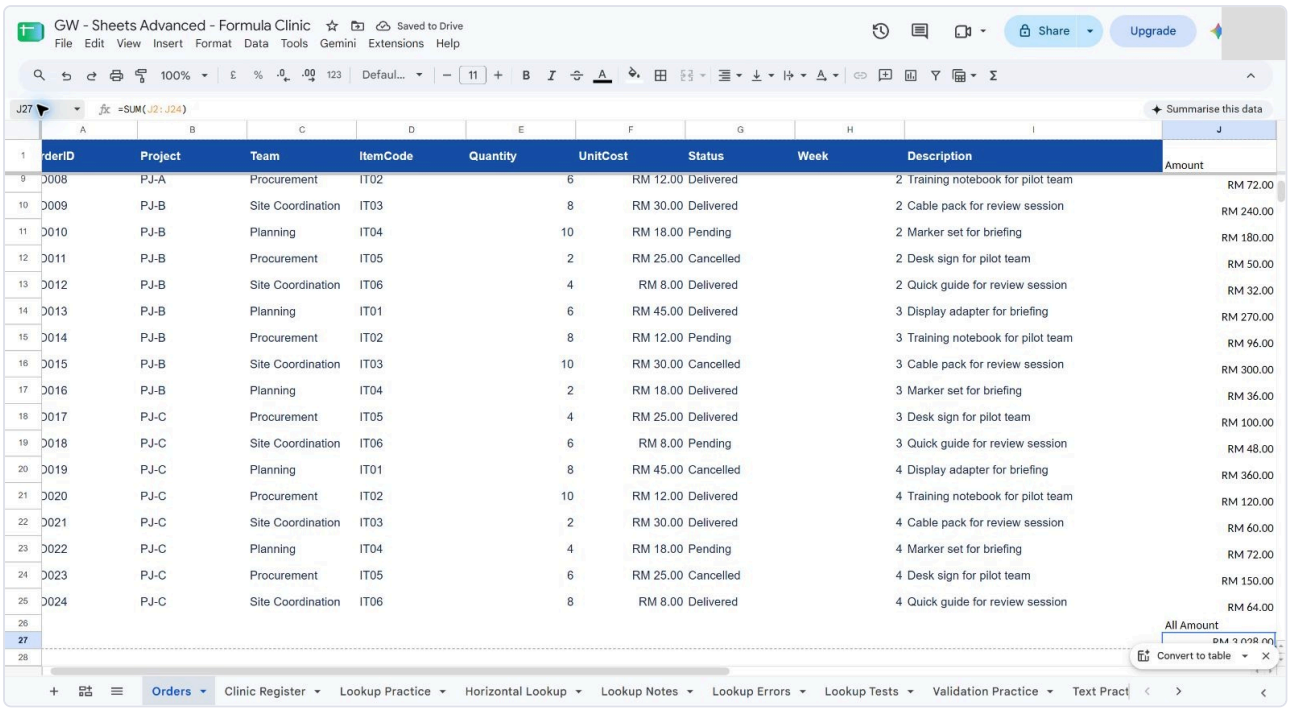
	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount	Fictional tr LookupRate
2	IT99	2	RM 45.00	Delivered	1	Display adapter for briefing	RM 90.00	RM 90.00	
3	IT02	4	RM 12.00	Pending	1	Training notebook for pilot team	RM 48.00	RM 48.00	
4	IT03	6	RM 30.00	Cancelled	1	Cable pack for review session	RM 180.00	RM 0.00	
5	IT04	8	RM 18.00	Delivered	1	Marker set for briefing	RM 144.00	RM 144.00	
6	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team	RM 250.00	RM 250.00	
7	IT06	2	RM 8.00	Pending	1	Quick guide for review session	RM 16.00	RM 16.00	
8	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing	RM 180.00	RM 0.00	
9	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team	RM 72.00	RM 72.00	
10	IT03	8	RM 30.00	Delivered	2	Cable pack for review session	RM 240.00	RM 240.00	
11	IT04	10	RM 18.00	Pending	2	Marker set for briefing	RM 180.00	RM 180.00	
12	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team	RM 50.00	RM 0.00	
13	IT06	4	RM 8.00	Delivered	2	Quick guide for review session	RM 32.00	RM 32.00	
14	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing	RM 270.00	RM 270.00	
15	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team	RM 96.00	RM 96.00	
16	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session	RM 300.00	RM 0.00	
17	IT04	2	RM 18.00	Delivered	3	Marker set for briefing	RM 36.00	RM 36.00	
18	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team	RM 100.00	RM 100.00	
19	IT06	6	RM 8.00	Pending	3	Quick guide for review session	RM 48.00	RM 48.00	
20	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing			

4. On copy Orders J2 use =E2*F3. The numeric 24 is plausible but wrong because the next row's cost is12.



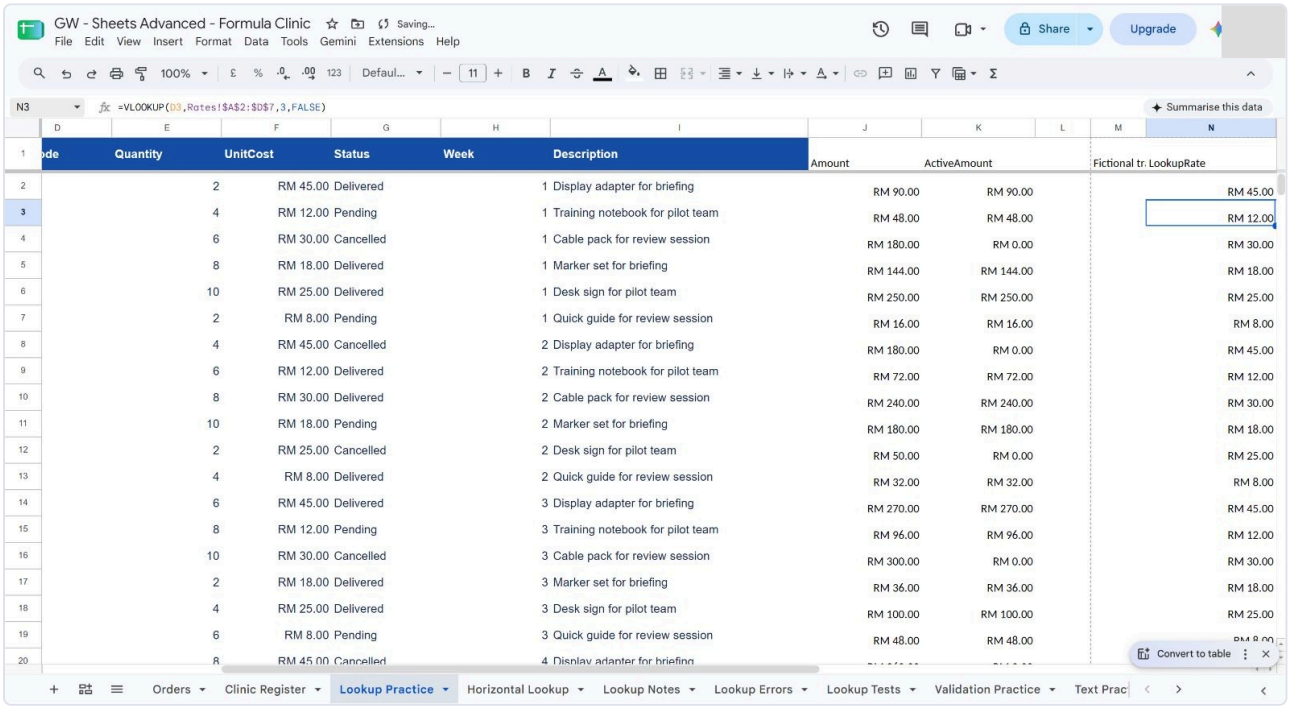
	rderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount
2	0001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter for briefing	RM 24.00
3	0002	PJ-A	Procurement	IT02	4	RM 12.00	Pending	1	Training notebook for pilot team	RM 48.00
4	0003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled	1	Cable pack for review session	RM 180.00
5	0004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for briefing	RM 144.00
6	0005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team	RM 250.00
7	0006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session	RM 16.00
8	0007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing	RM 180.00
9	0008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team	RM 72.00
10	0009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session	RM 240.00
11	0010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing	RM 180.00
12	0011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team	RM 50.00
13	0012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session	RM 32.00
14	0013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing	RM 270.00
15	0014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team	RM 96.00
16	0015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session	RM 300.00
17	0016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing	RM 36.00
18	0017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team	RM 100.00
19	0018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session	
20	0019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing	

5. Shorten copy Orders J27 to =SUM(J2:J24). With both amount and range faults present, observe 3028. Do not attribute this combined discrepancy solely to the last row.

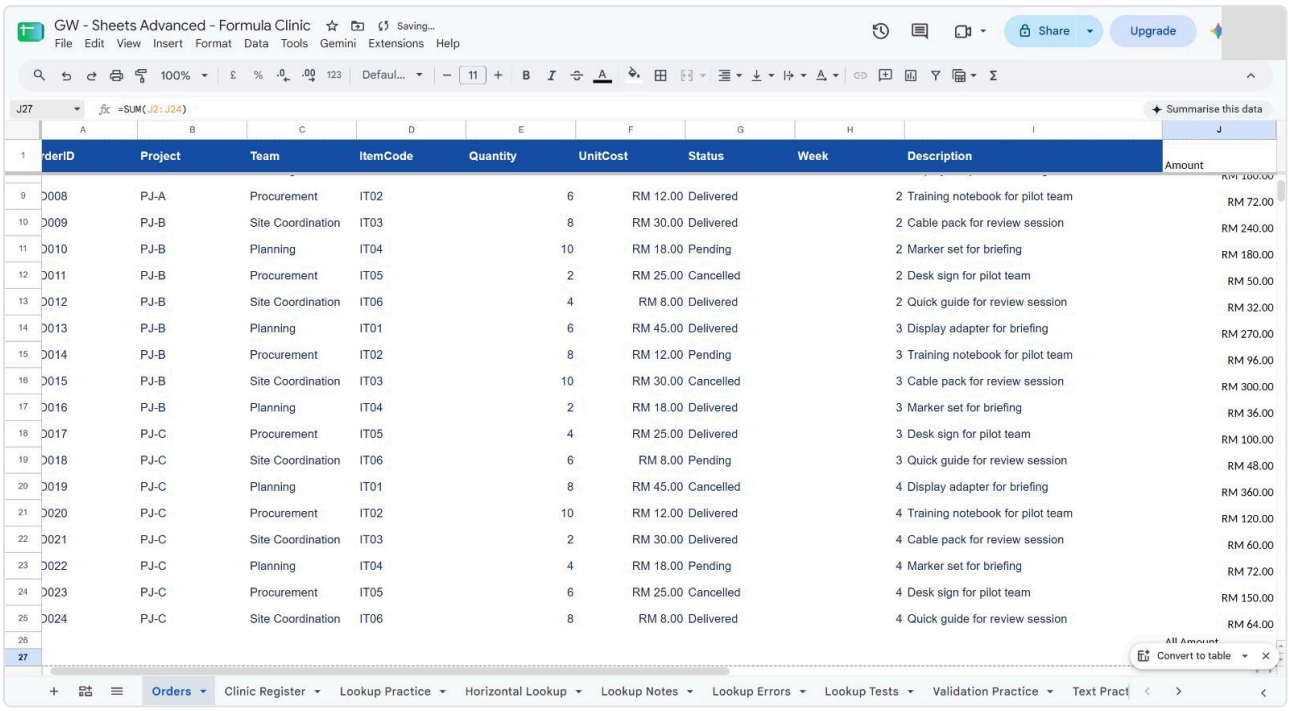


6.2 Repair and Test

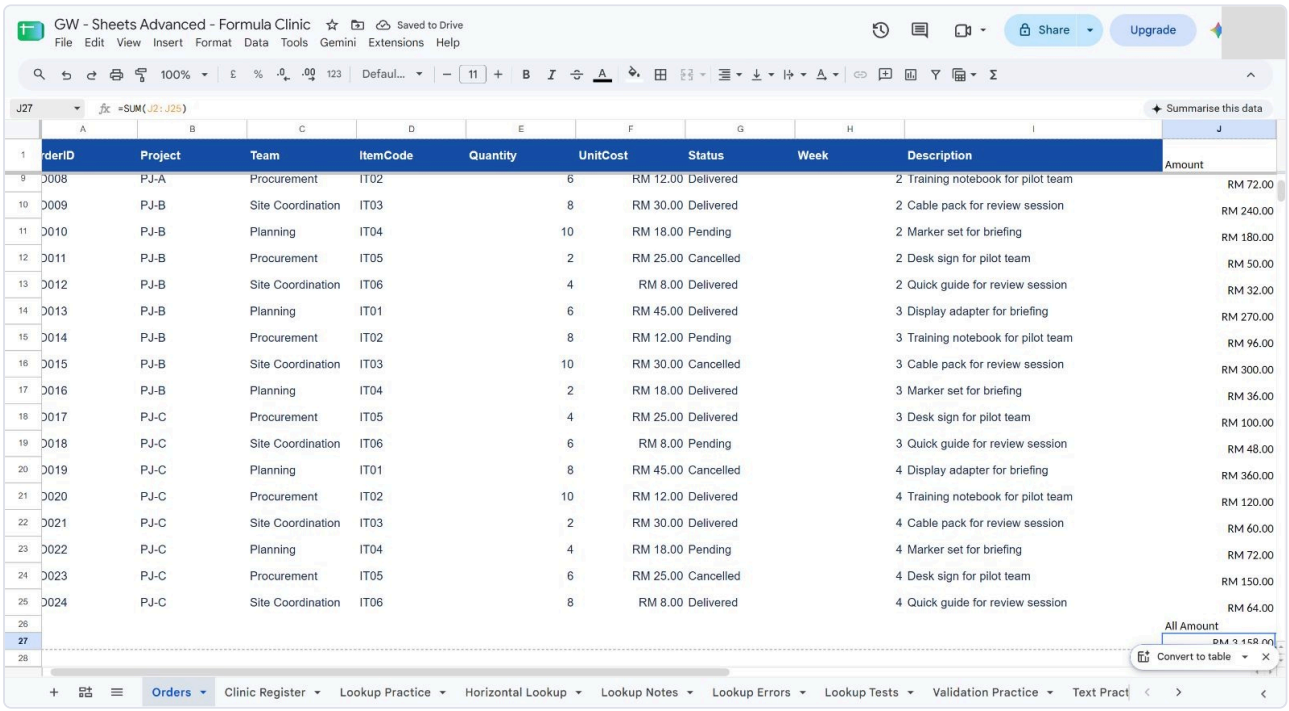
1. Restore exact FALSE in Lookup Practice N2; IT01 returns 45.



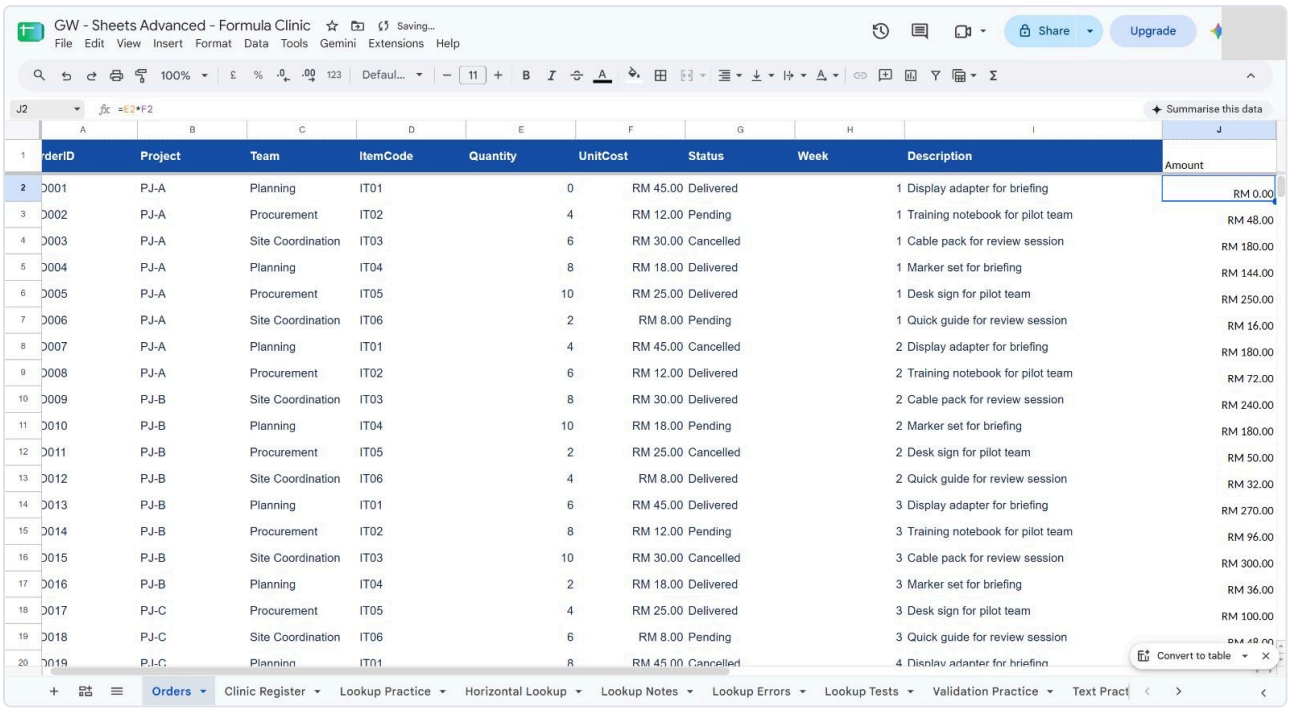
2. Restore Orders J2=E2*F2=90. The shortened total is now 3094, isolating the missing 64.

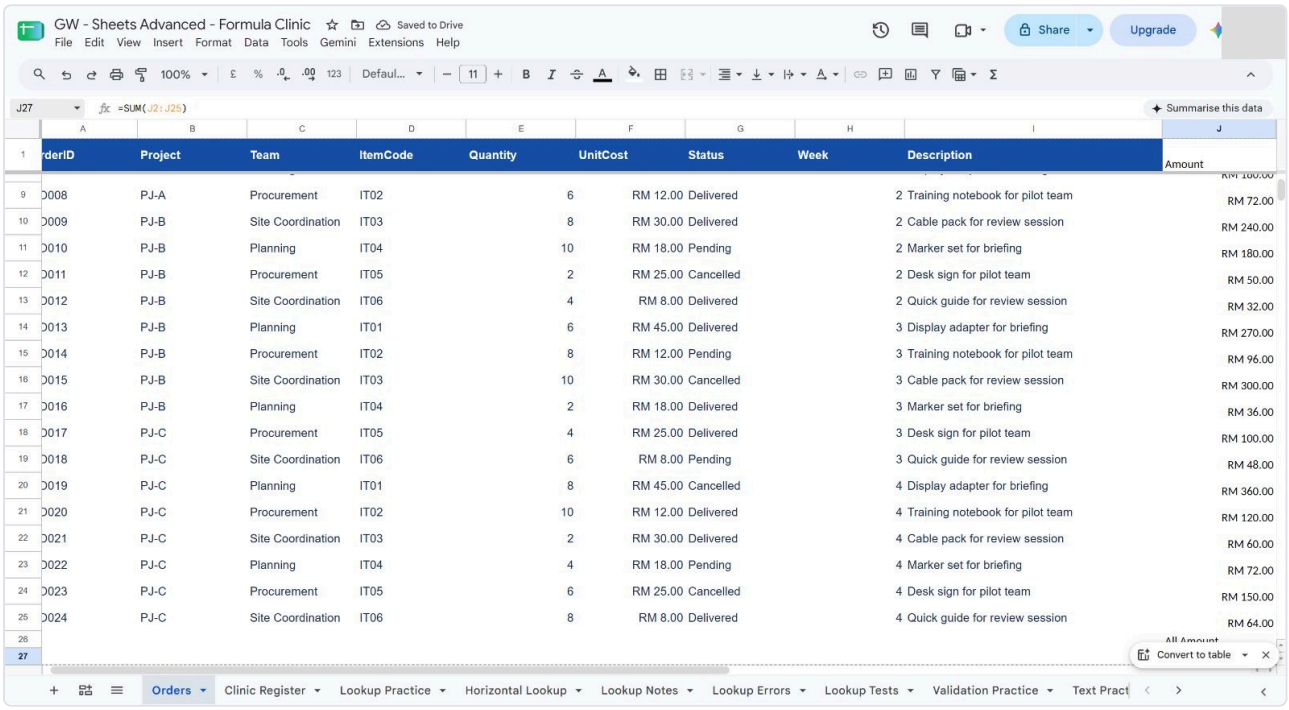


3. Restore J27=SUM(J2:J25)=3158.

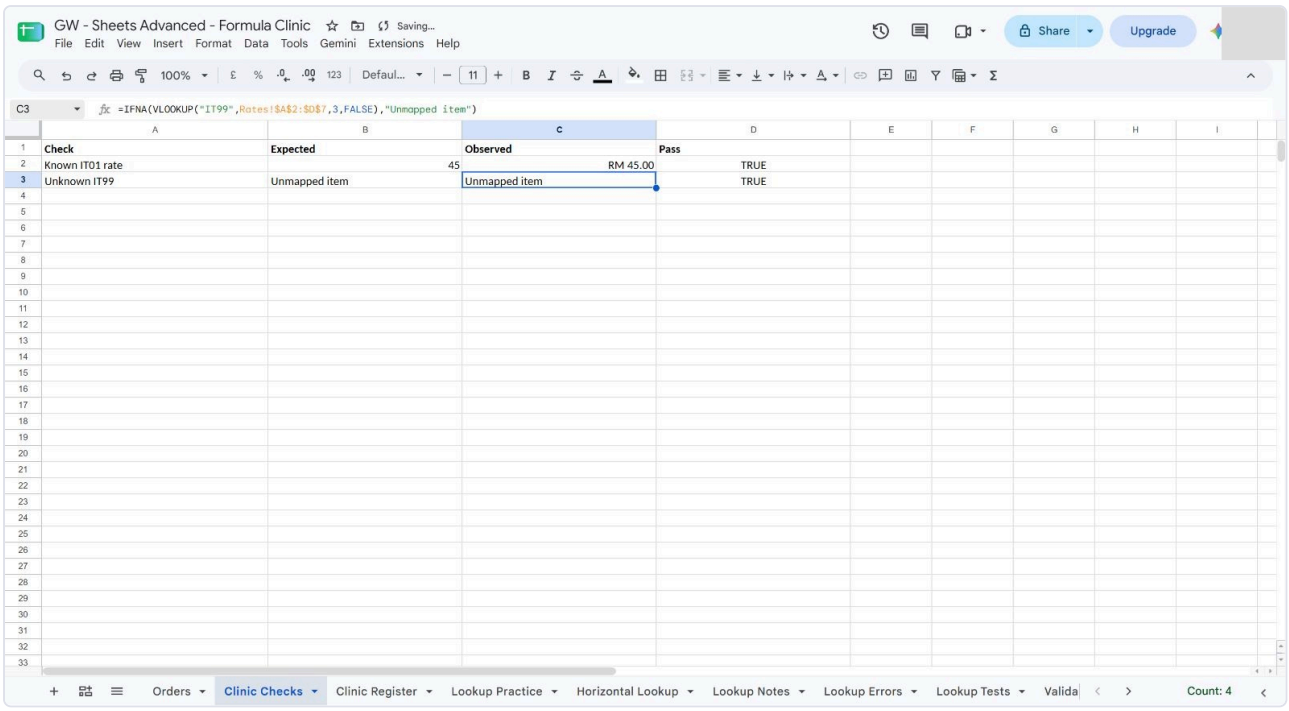


4. Temporarily set Orders E2=0. J2/K2 become a legitimate numeric 0, all-orders total 3068. Save the observed test before restoring E2=2, J2=90 and total 3158.



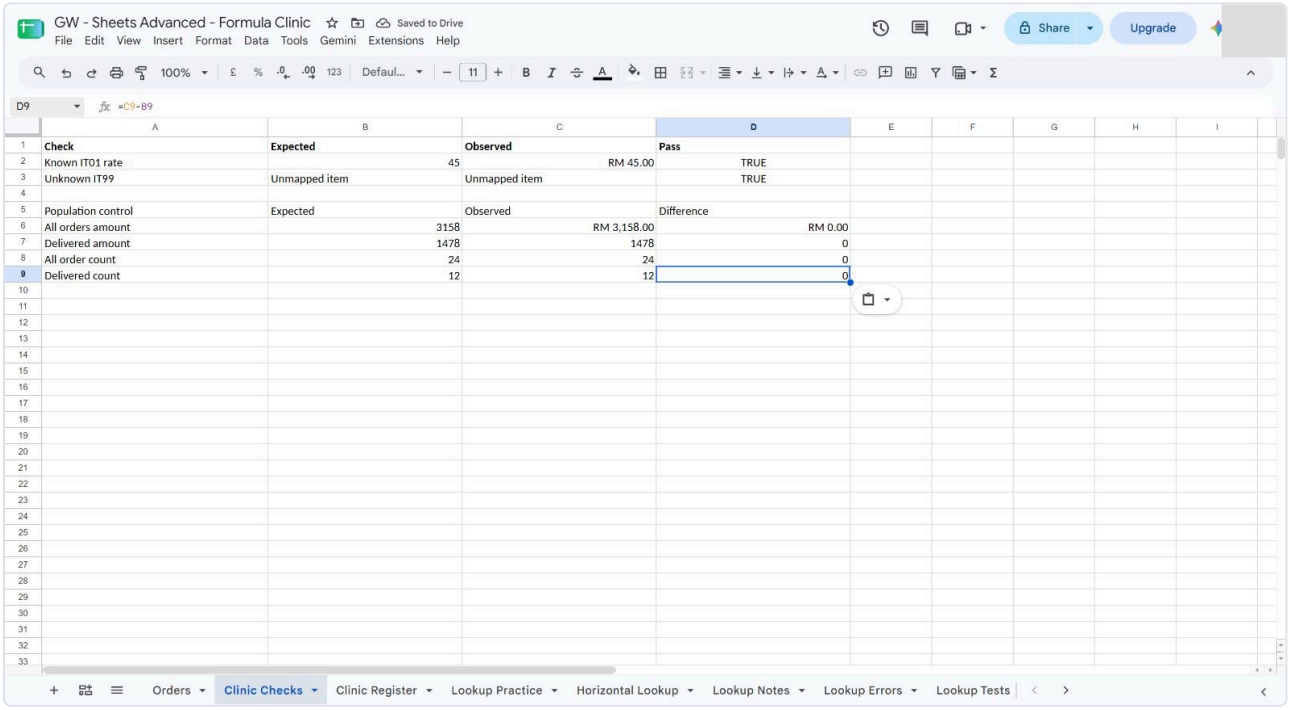


5. In Clinic Checks compare known IT01 VLOOKUP result 45 and unknown IT99 IFNA result Unmapped item against prior expectations. Both checks pass. Unmapped item is not a zero price.

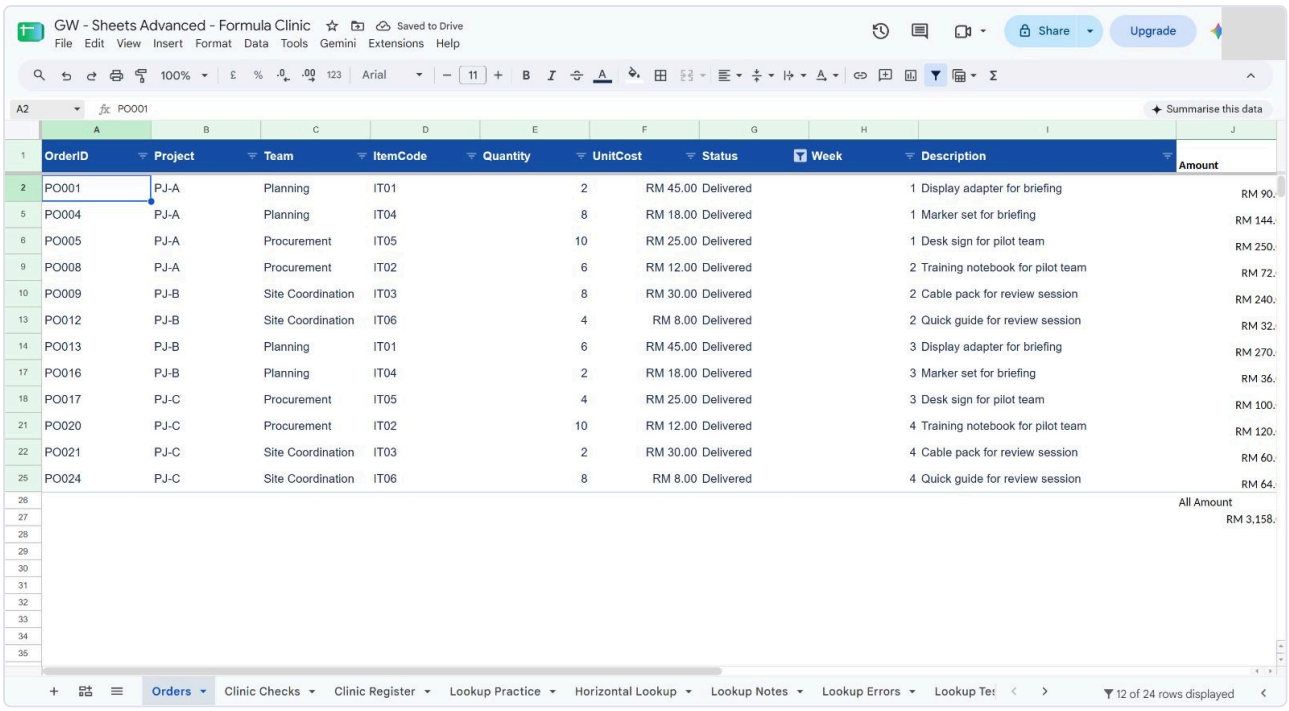


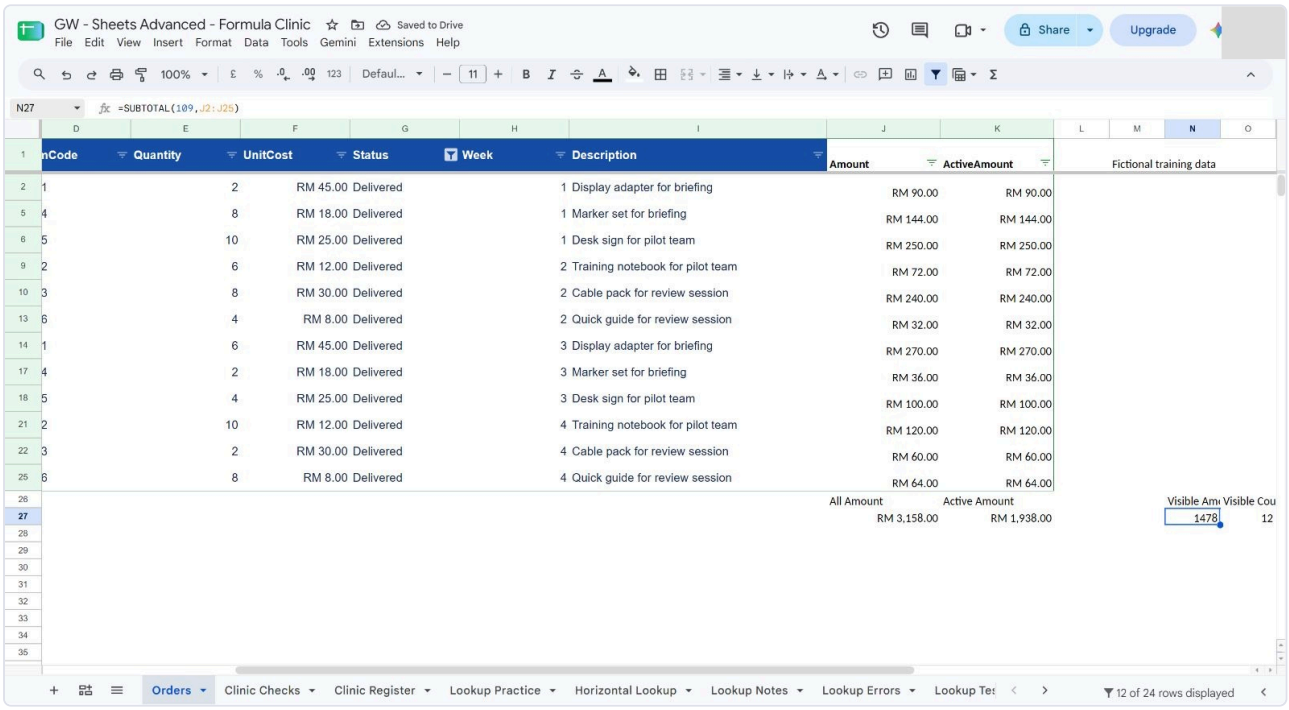
6.3 Reconcile and Explain

1. Clinic Checks independently calculates SUM(Orders!J2:J25)=3158, SUMIF status Delivered=1478, COUNTA OrderID=24 and COUNTIF Delivered=12. Compare each to its stated population control.

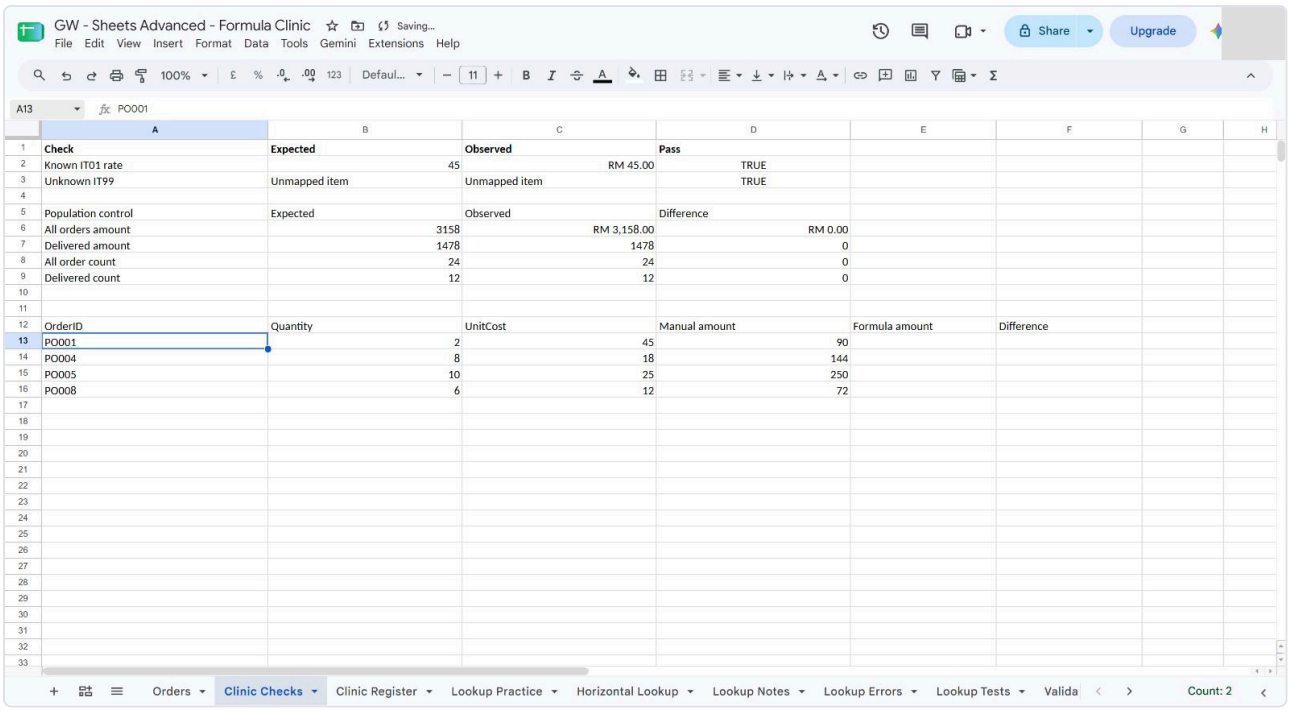


2. On copy Orders create a bounded A1:K25 filter and select Delivered only from Status. Use outside-table N27=SUBTOTAL(109,J2:J25) and O27=SUBTOTAL(103,A2:A25). The actual filtered result is 1478 and 12 visible records. Keep helpers outside the filtered source; source SUM is not a filtered subtotal. Remove filter after the observation, leaving 3158/24 in the helper cells.



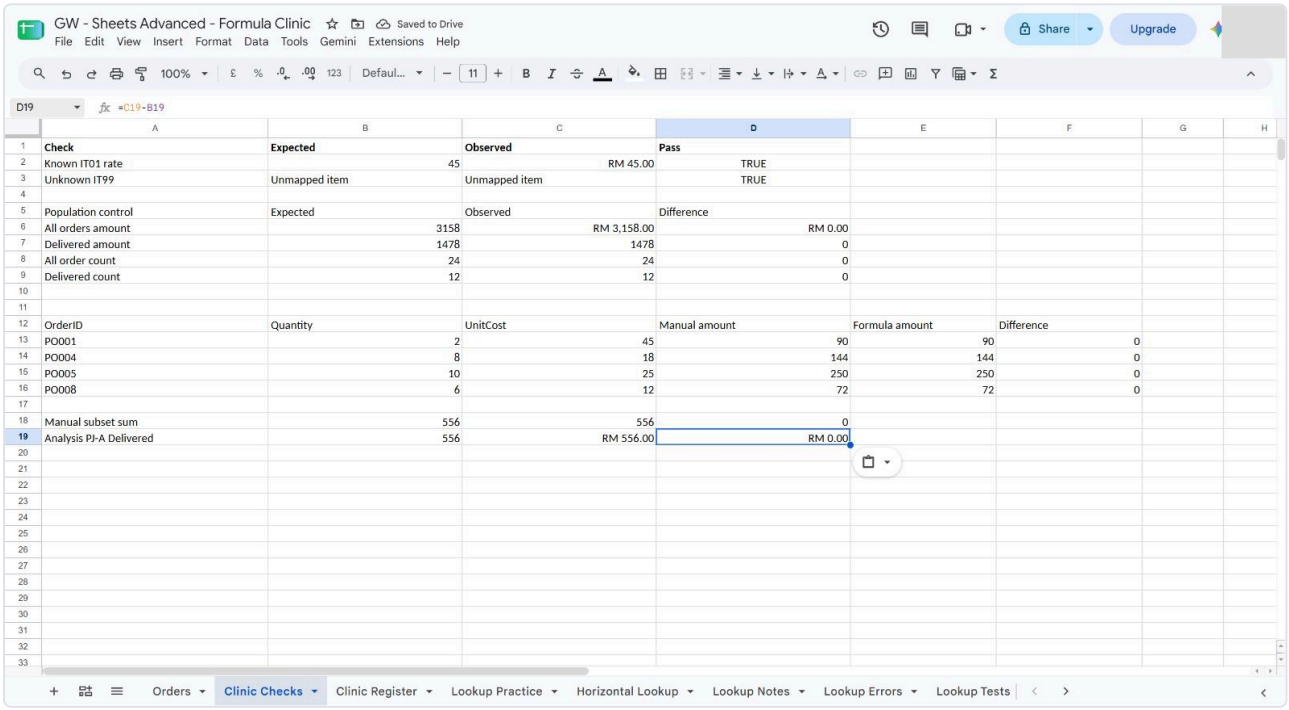


3. Inspect PJ-A Delivered IDs PO001/PO004/PO005/PO008 and record manual 2×45=90,8×18=144,10×25=250,6×12=72 before formula comparison.



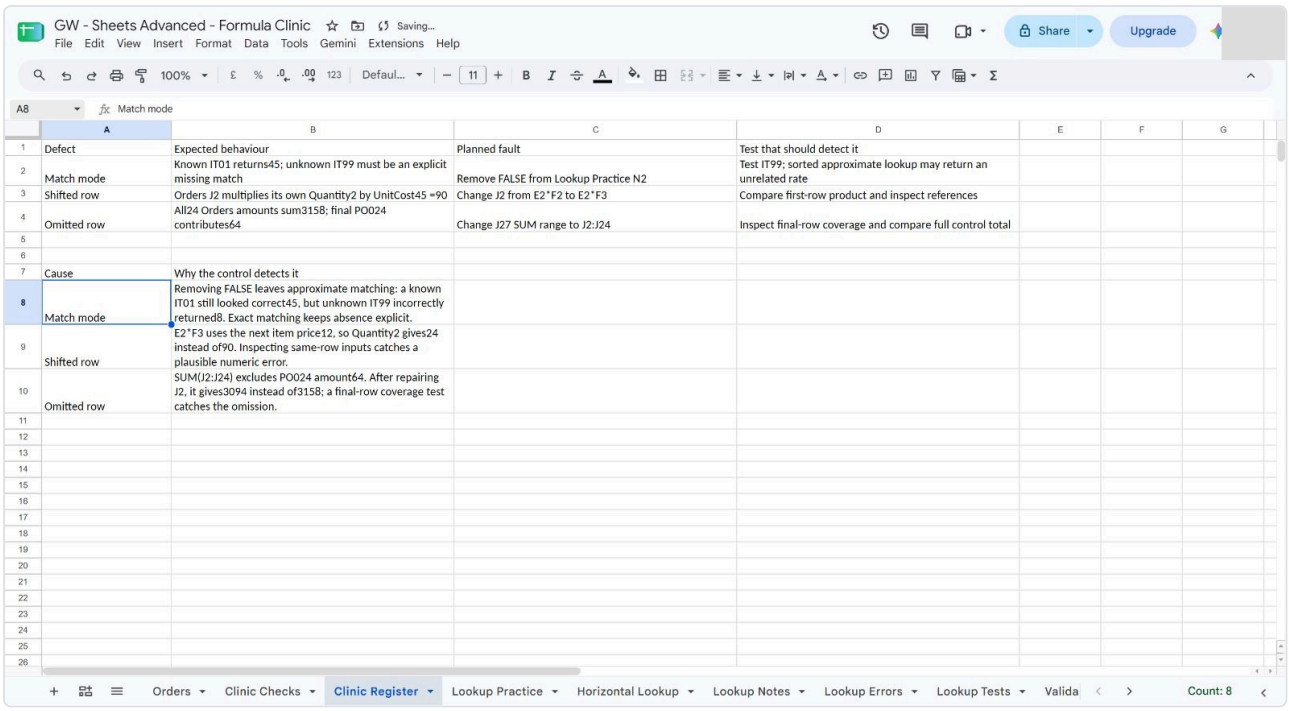
	A	B	C	D	E	F
1	**Check**	**Expected**	**Observed**	**Pass**		
2	Known IT01 rate	45	RM 45.00	TRUE		
3	Unknown IT99	Unmapped item	Unmapped item	TRUE		
4						
5	Population control	Expected	Observed	Difference		
6	All orders amount	3158	RM 3,158.00	RM 0.00		
7	Delivered amount	1478	1478	0		
8	All order count	24	24	0		
9	Delivered count	12	12	0		
10						
11						
12	OrderID	Quantity	UnitCost	Manual amount	Formula amount	Difference
13	PO001	2	45	90		
14	PO004	8	18	144		
15	PO005	10	25	250		
16	PO008	6	12	72		

4. Calculate each product and its difference against the manually recorded amount. All four differences 0; manual subtotal 556 matches Analysis B4 PJ-A Delivered 556. These are author-supported manual checks, not independent learner validation.



	A	B	C	D	E	F
1	**Check**	**Expected**	**Observed**	**Pass**		
2	Known IT01 rate	45	RM 45.00	TRUE		
3	Unknown IT99	Unmapped item	Unmapped item	TRUE		
4						
5	Population control	Expected	Observed	Difference		
6	All orders amount	3158	RM 3,158.00	RM 0.00		
7	Delivered amount	1478	1478	0		
8	All order count	24	24	0		
9	Delivered count	12	12	0		
10						
11						
12	OrderID	Quantity	UnitCost	Manual amount	Formula amount	Difference
13	PO001	2	45	90	90	0
14	PO004	8	18	144	144	0
15	PO005	10	25	250	250	0
16	PO008	6	12	72	72	0
17						
18	Manual subset sum	556	556	0		
19	Analysis PJ-A Delivered	556	RM 556.00	RM 0.00		

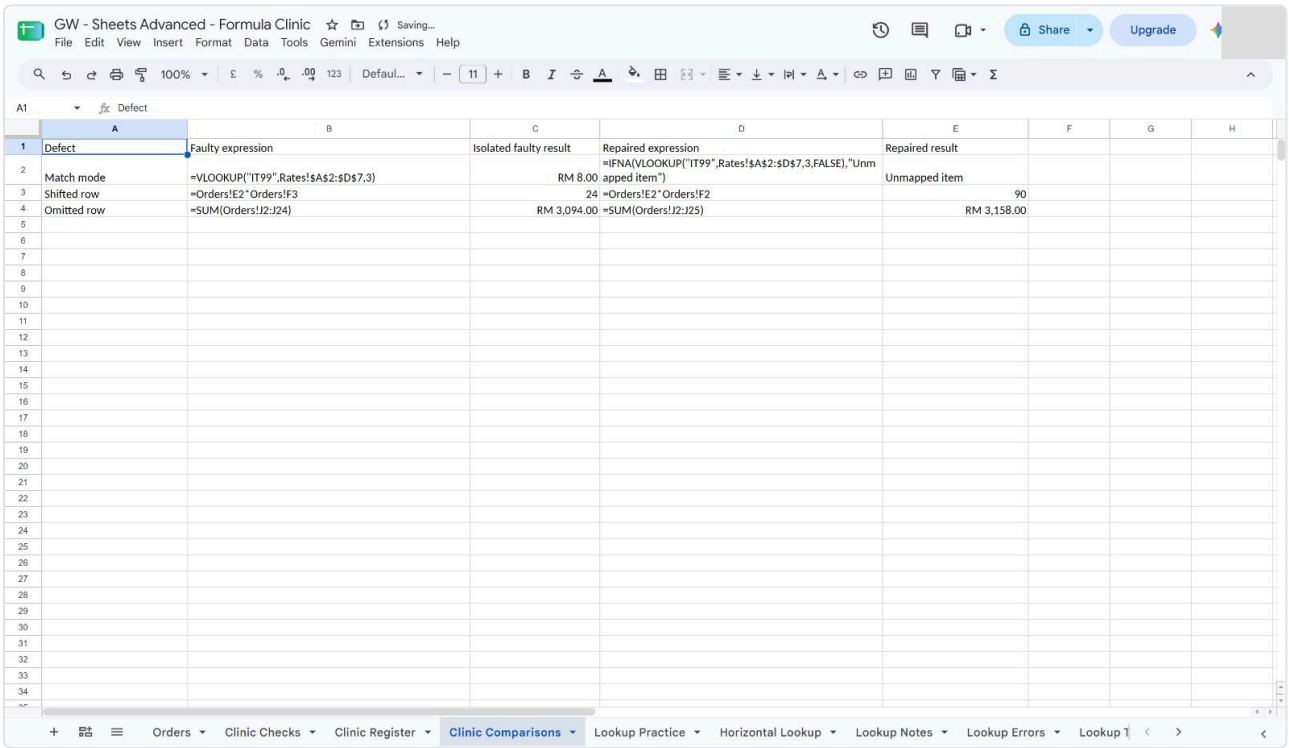
5. Explain the match-mode, row-reference and range-coverage causes and their detecting tests in Clinic Register.



	A	B	C	D
1	Defect	Expected behaviour	Planned fault	Test that should detect it
2	Match mode	Known IT01 returns45; unknown IT99 must be an explicit missing match	Remove FALSE from Lookup Practice N2	Test IT99; sorted approximate lookup may return an unrelated rate
3	Shifted row	Orders J2 multiplies its own Quantity2 by UnitCost45 =90	Change J2 from E2*F2 to E2*F3	Compare first-row product and inspect references
4	Omitted row	All24 Orders amounts sum3158; final PO024 contributes64	Change J27 SUM range to J2:J24	Inspect final-row coverage and compare full control total
5				
6				
7	Cause	Why the control detects it		
8	Match mode	Removing FALSE leaves approximate matching: a known IT01 still looked correct45, but unknown IT99 incorrectly returned8. Exact matching keeps absence explicit.		
9	Shifted row	E2*F3 uses the next item price12, so Quantity2 gives24 instead of90. Inspecting same-row inputs catches a plausible numeric error.		
10	Omitted row	SUM(J2:J24) excludes PO024 amount64. After repairing J2, it gives3094 instead of3158; a final-row coverage test catches the omission.		

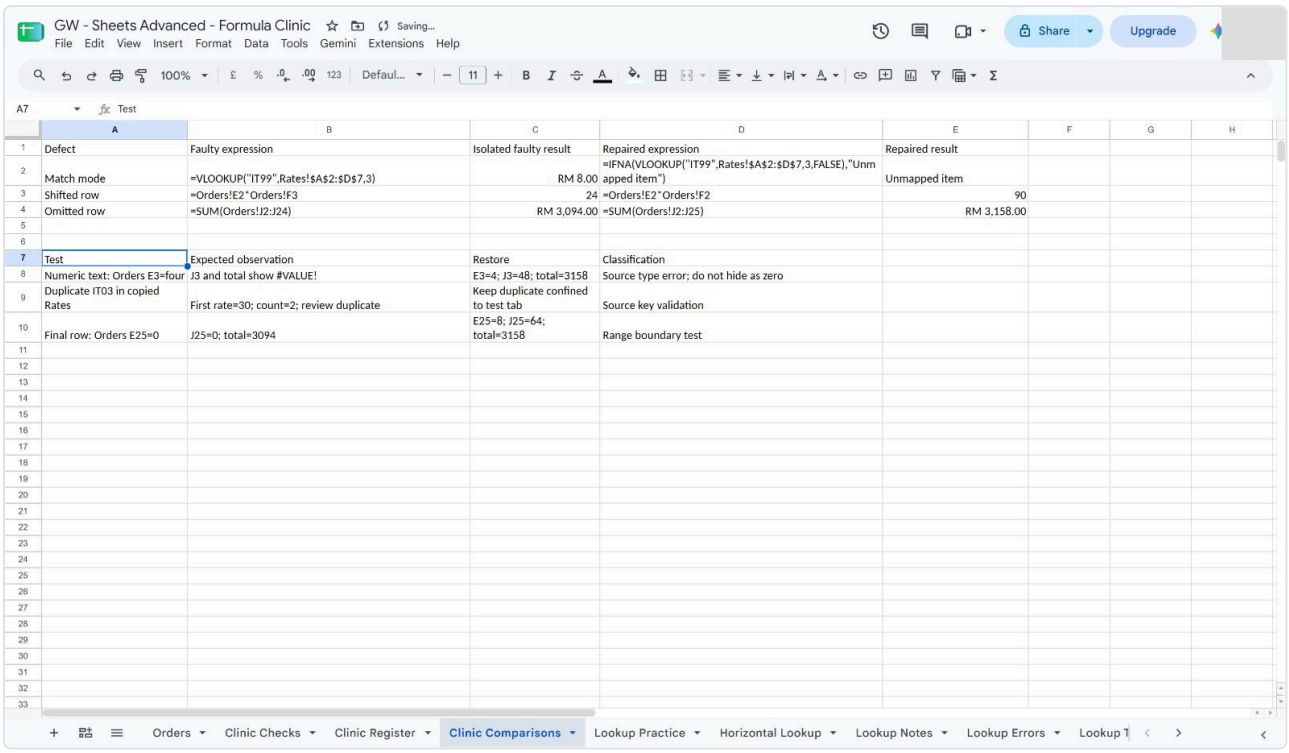
6.4 Run the Formula Fault Clinic

1. Create **Clinic Comparisons A1:E4** from the side-by-side card. Put live faulty expressions in C and repaired expressions in E; use **FORMULATEXT(C2)** and **FORMULATEXT(E2)** in B/D and fill through row 4. Wrap A1:E4. Observe **8 versus Unmapped item**, **24 versus 90**, and **3094 versus 3158**. These are isolated re-evaluations against the restored baseline; the earlier combined-fault total was 3028.

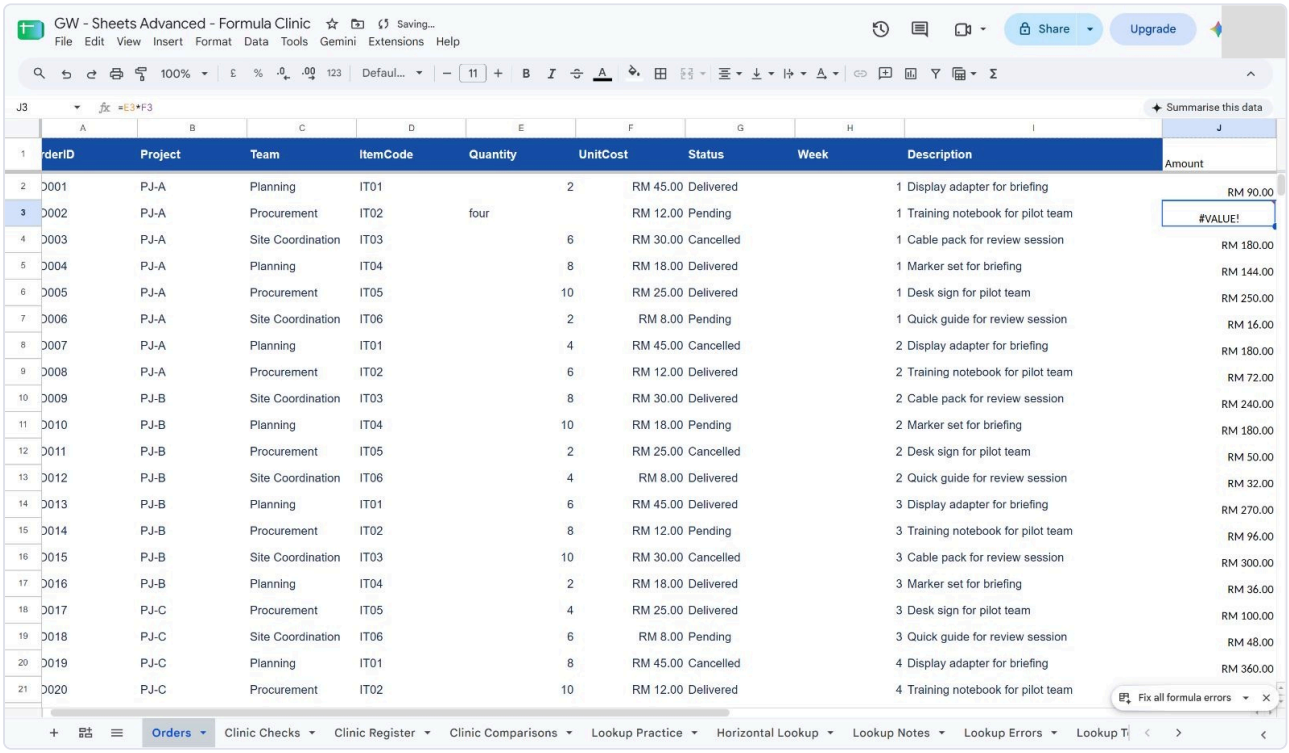


	A	B	C	D	E
1	Defect	Faulty expression	Isolated faulty result	Repaired expression	Repaired result
2	Match mode	=VLOOKUP("IT99",Rates!$A$2:$D$7,3)	RM 8.00	=IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")	Unmapped item
3	Shifted row	=Orders!E2*Orders!F3	24	=Orders!E2*Orders!F2	90
4	Omitted row	=SUM(Orders!J2:J24)	RM 3,094.00	=SUM(Orders!J2:J25)	RM 3,158.00

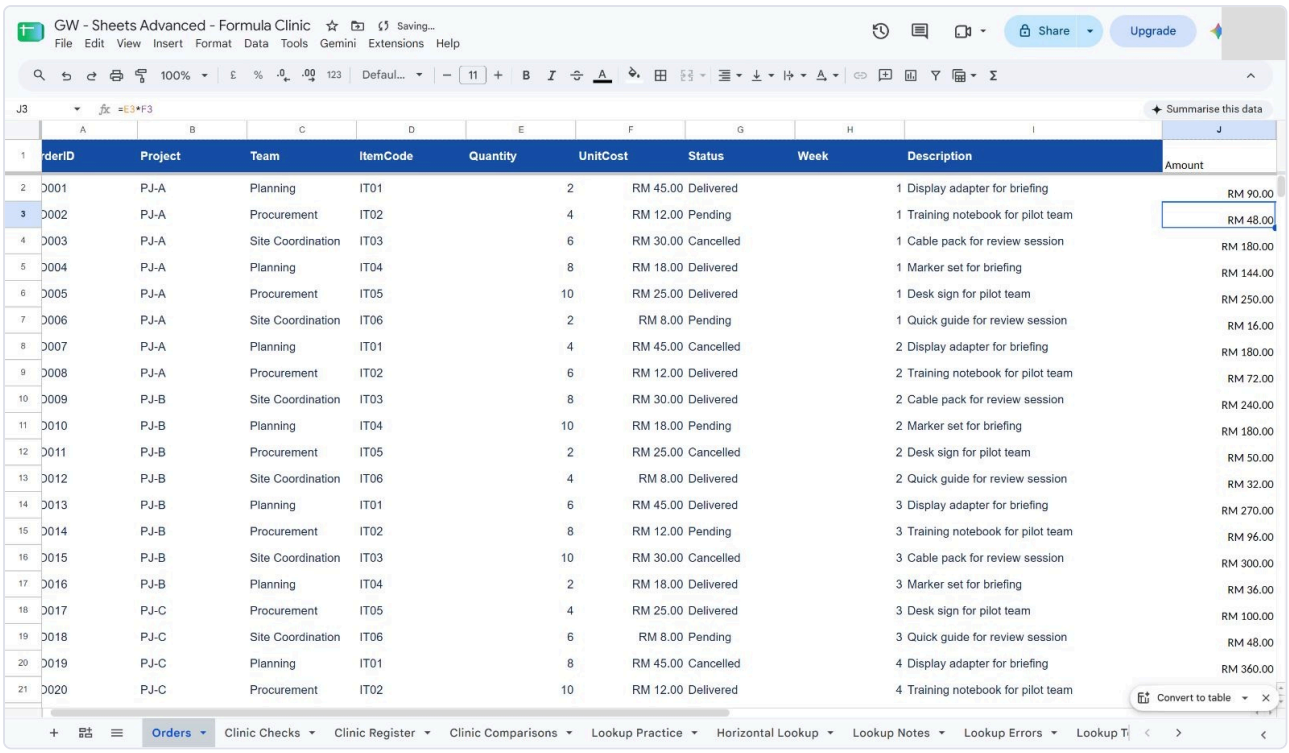
2. Before changing inputs, record the numeric-text, duplicate-key and final-row expectations in A7:D10. A source type error must remain visible; do not replace it with zero.



3. On clinic-copy **Orders**, temporarily enter literal **four** in E3. J3's **=E3*F3** produces **#VALUE!**, which propagates to J27. Inspect the error before repairing its source.

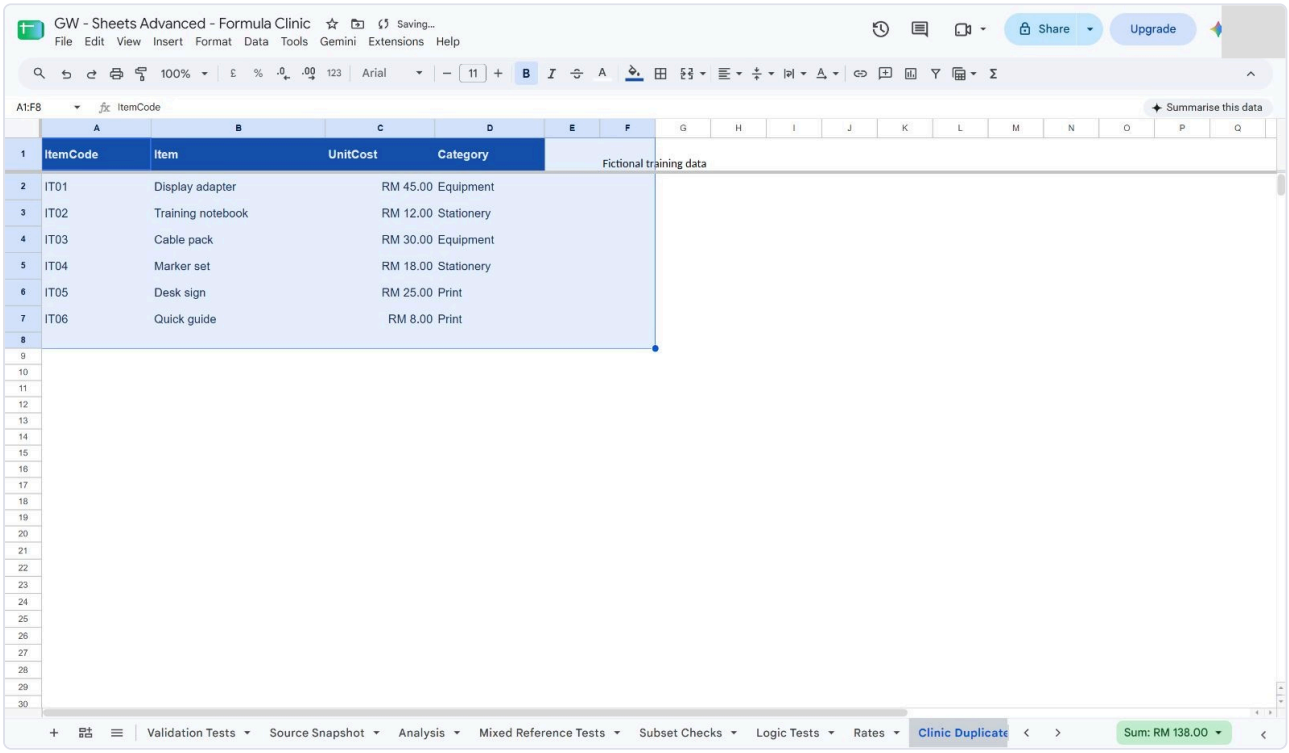


4. Restore **E3=4**. J3 returns 48 and the all-orders total returns 3158. This is source cleaning, not a reason to wrap the multiplication in IFERROR(...,0).



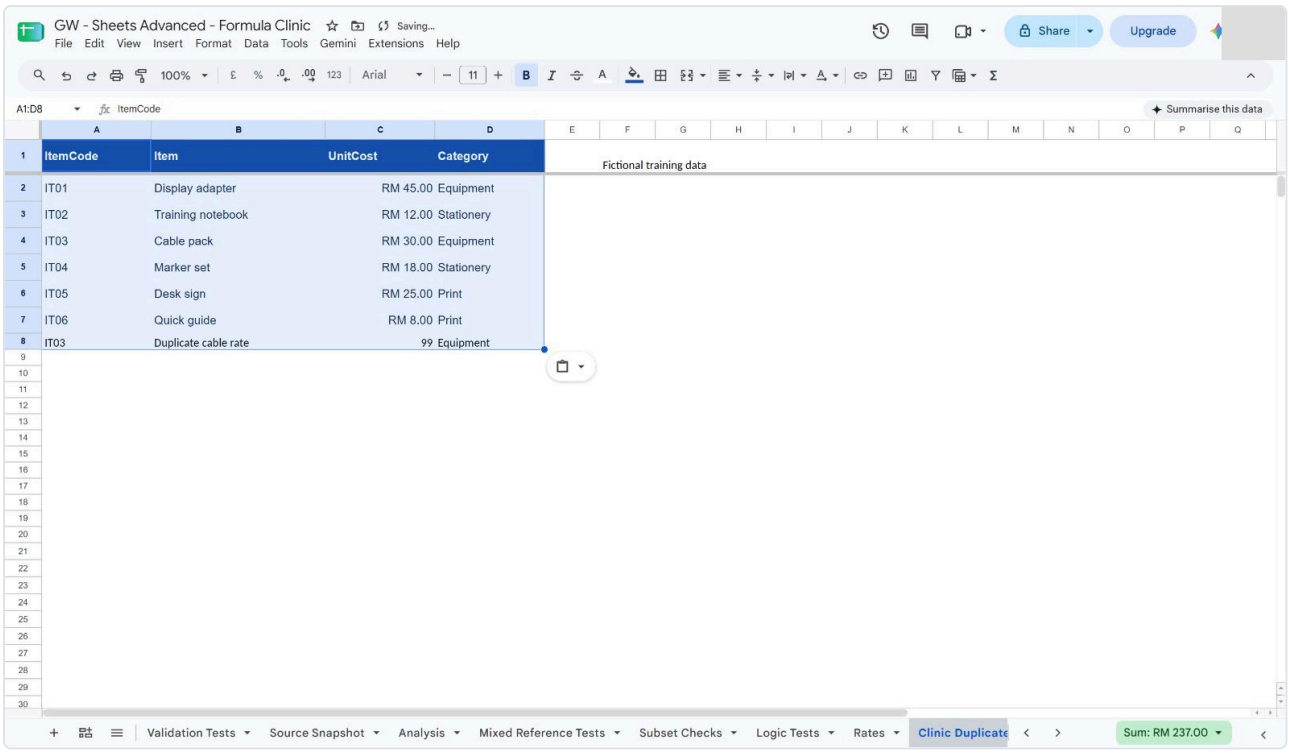
rderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount
0001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter for briefing	RM 90.00
0002	PJ-A	Procurement	IT02	4	RM 12.00	Pending	1	Training notebook for pilot team	RM 48.00
0003	PJ-A	Site Coordination	IT03	6	RM 30.00	Cancelled	1	Cable pack for review session	RM 180.00
0004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for briefing	RM 144.00
0005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilot team	RM 250.00
0006	PJ-A	Site Coordination	IT06	2	RM 8.00	Pending	1	Quick guide for review session	RM 16.00
0007	PJ-A	Planning	IT01	4	RM 45.00	Cancelled	2	Display adapter for briefing	RM 180.00
0008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training notebook for pilot team	RM 72.00
0009	PJ-B	Site Coordination	IT03	8	RM 30.00	Delivered	2	Cable pack for review session	RM 240.00
0010	PJ-B	Planning	IT04	10	RM 18.00	Pending	2	Marker set for briefing	RM 180.00
0011	PJ-B	Procurement	IT05	2	RM 25.00	Cancelled	2	Desk sign for pilot team	RM 50.00
0012	PJ-B	Site Coordination	IT06	4	RM 8.00	Delivered	2	Quick guide for review session	RM 32.00
0013	PJ-B	Planning	IT01	6	RM 45.00	Delivered	3	Display adapter for briefing	RM 270.00
0014	PJ-B	Procurement	IT02	8	RM 12.00	Pending	3	Training notebook for pilot team	RM 96.00
0015	PJ-B	Site Coordination	IT03	10	RM 30.00	Cancelled	3	Cable pack for review session	RM 300.00
0016	PJ-B	Planning	IT04	2	RM 18.00	Delivered	3	Marker set for briefing	RM 36.00
0017	PJ-C	Procurement	IT05	4	RM 25.00	Delivered	3	Desk sign for pilot team	RM 100.00
0018	PJ-C	Site Coordination	IT06	6	RM 8.00	Pending	3	Quick guide for review session	RM 48.00
0019	PJ-C	Planning	IT01	8	RM 45.00	Cancelled	4	Display adapter for briefing	RM 360.00
0020	PJ-C	Procurement	IT02	10	RM 12.00	Delivered	4	Training notebook for pilot team	

5. Duplicate baseline **Rates** as **Clinic Duplicate Rates**. Verify the new tab has the six original keys before altering it. Keep maintained Rates unchanged.

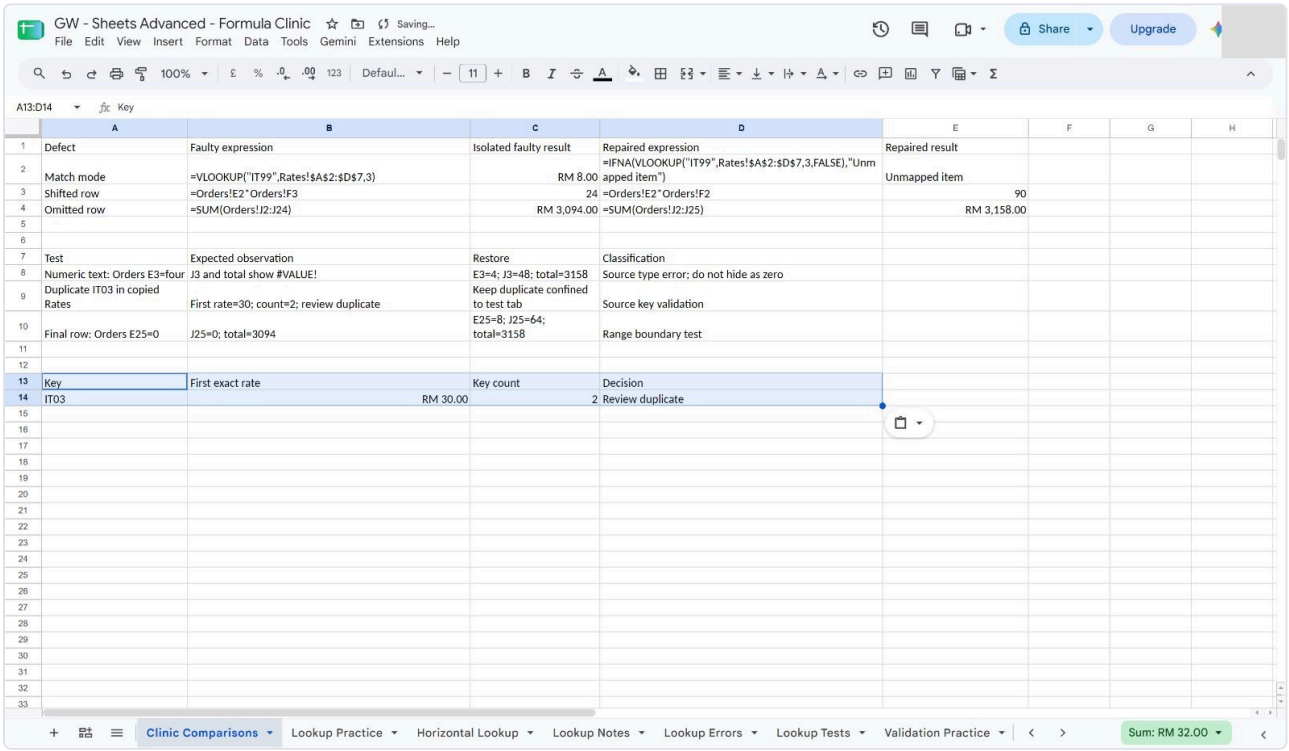


ItemCode	Item	UnitCost	Category
IT01	Display adapter	RM 45.00	Equipment
IT02	Training notebook	RM 12.00	Stationery
IT03	Cable pack	RM 30.00	Equipment
IT04	Marker set	RM 18.00	Stationery
IT05	Desk sign	RM 25.00	Print
IT06	Quick guide	RM 8.00	Print

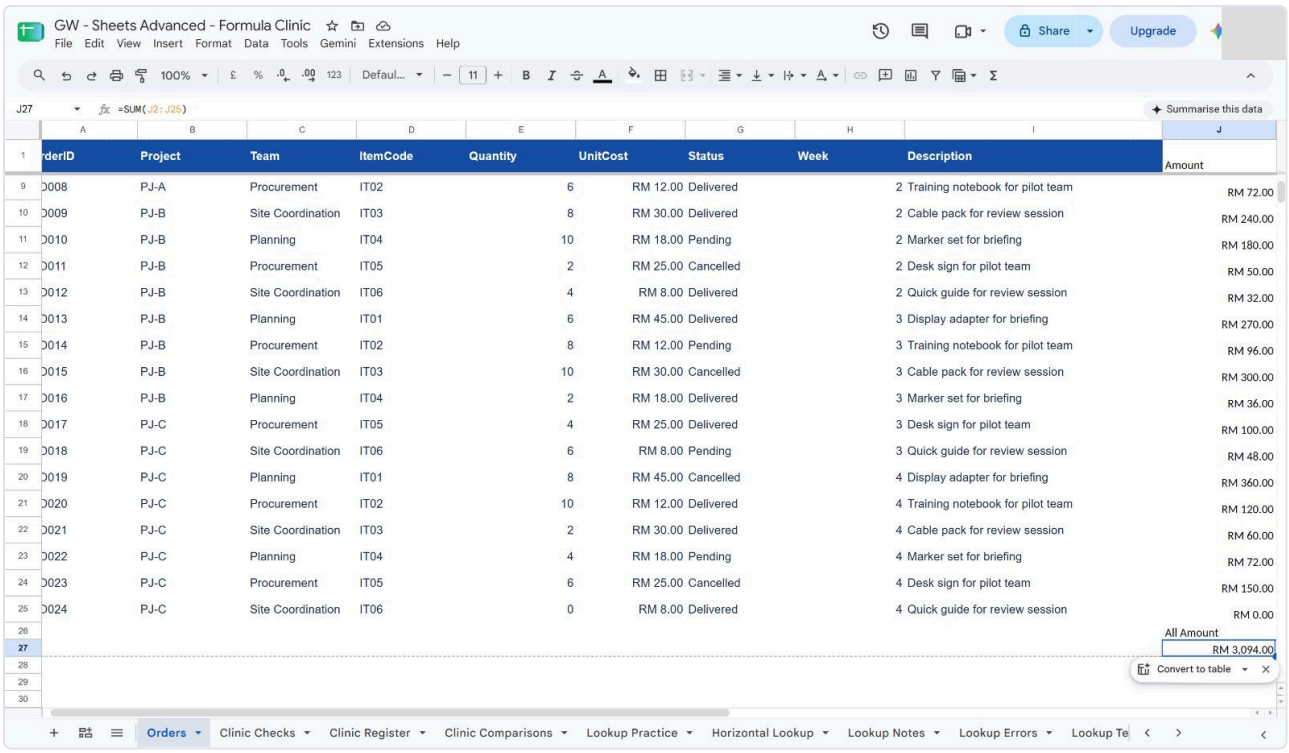
6. Append **IT03 / Duplicate cable rate /99 /Equipment** at A8:D8 on that test tab only. Original IT03 remains30.



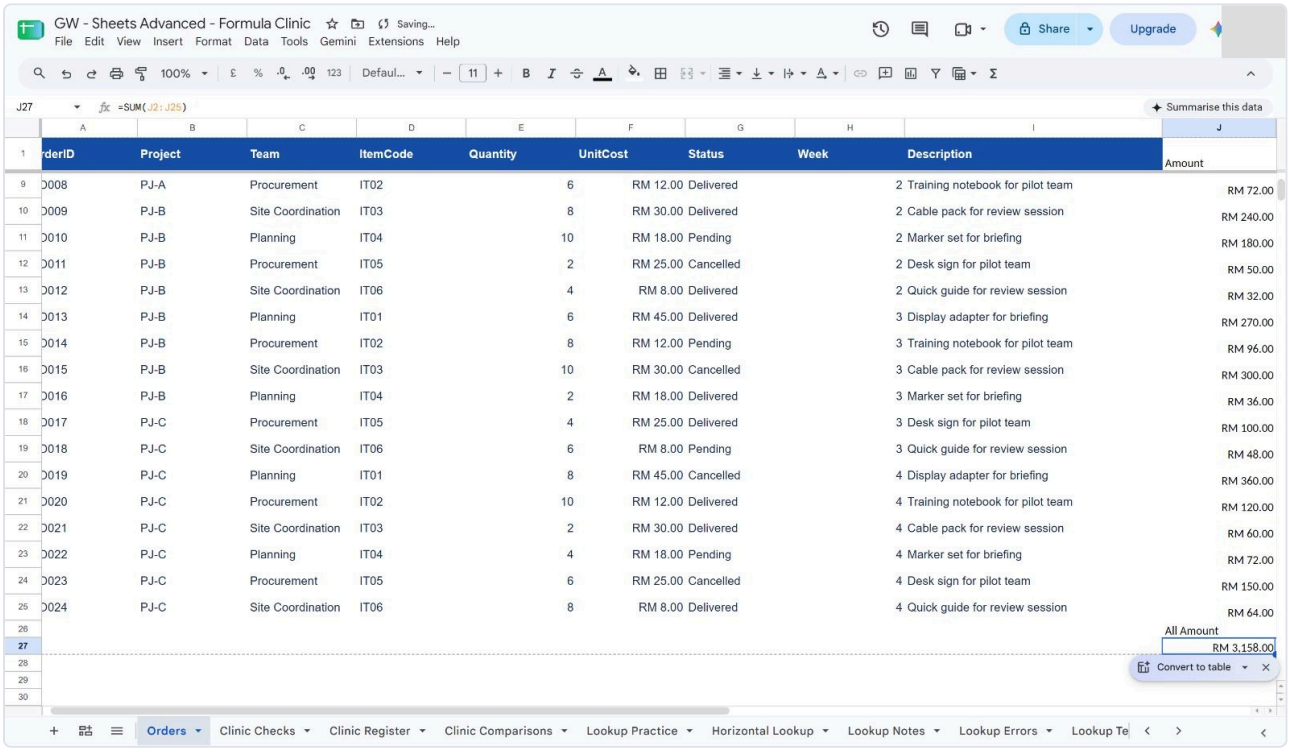
7. In **Clinic Comparisons A13:D14**, run the duplicate diagnostic card over the seven data rows. Exact VLOOKUP returns the first rate 30, COUNTIF returns 2, and the decision is **Review duplicate**. Neither 30 nor 99 becomes an approved repair. Retain the clearly named scenario tab.



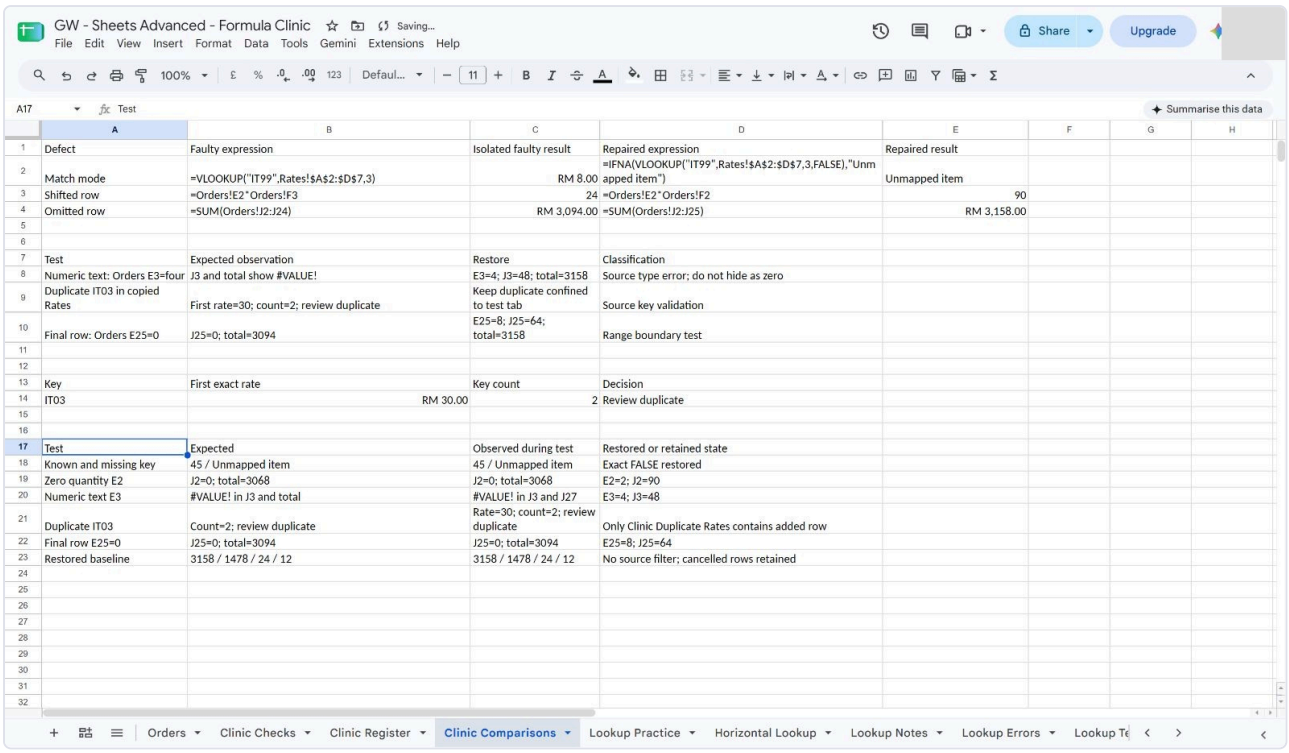
8. Temporarily change **Orders E25 to 0**. J25 becomes 0 and the repaired full-range total J27 changes to 3094. The shortened range would not respond to this last-row change. Inspect J27 in the formula bar; scroll down if it is below the viewport.



9. Restore **E25=8**. J25 returns 64 and J27 returns 3158. Keep the full **=SUM(J2:J25)** formula.

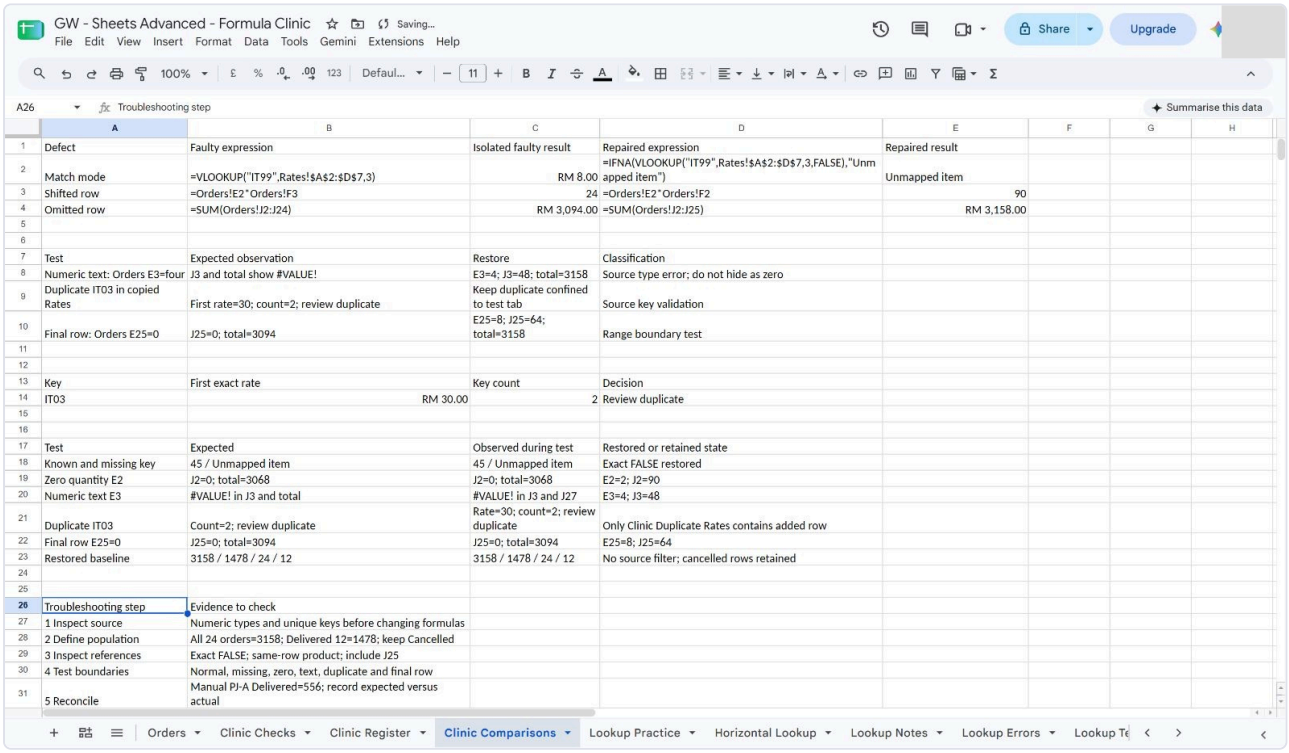


10. Record actual observations in **A17:D23** after each experiment. Distinguish captured historical results from formulas that recalculate with the current baseline. The normal/missing lookup, normal/zero multiplication and normal/final-row total tests all have recorded outcomes.



	A	B	C	D	E
1	Defect	Faulty expression	Isolated faulty result	Repaired expression	Repaired result
2	Match mode	=VLOOKUP("IT99",Rates!$A$2:$D$7,3)	RM 8.00	=IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")	Unmapped item
3	Shifted row	=Orders!E2*Orders!F3	24	=Orders!E2*Orders!F2	90
4	Omitted row	=SUM(Orders!J2:J24)	RM 3,094.00	=SUM(Orders!J2:J25)	RM 3,158.00
5					
6					
7	Test	Expected observation	Restore	Classification	
8	Numeric text: Orders E3=four	J3 and total show #VALUE!	E3=4; J3=48; total=3158	Source type error; do not hide as zero	
9	Duplicate IT03 in copied Rates	First rate=30; count=2; review duplicate	Keep duplicate confined to test tab	Source key validation	
10	Final row: Orders E25=0	J25=0; total=3094	E25=8; J25=64; total=3158	Range boundary test	
11					
12					
13	Key	First exact rate	Key count	Decision	
14	IT03	RM 30.00	2	Review duplicate	
15					
16					
17	Test	Expected	Observed during test	Restored or retained state	
18	Known and missing key	45 / Unmapped item	45 / Unmapped item	Exact FALSE restored	
19	Zero quantity E2	J2=0; total=3068	J2=0; total=3068	E2=2; J2=90	
20	Numeric text E3	#VALUE! in J3 and total	#VALUE! in J3 and J27	E3=4; J3=48	
21	Duplicate IT03	Count=2; review duplicate	Rate=30; count=2; review duplicate	Only Clinic Duplicate Rates contains added row	
22	Final row E25=0	J25=0; total=3094	J25=0; total=3094	E25=8; J25=64	
23	Restored baseline	3158 / 1478 / 24 / 12	3158 / 1478 / 24 / 12	No source filter; cancelled rows retained	
24					
25					
26					
27					
28					
29					
30					
31					
32					

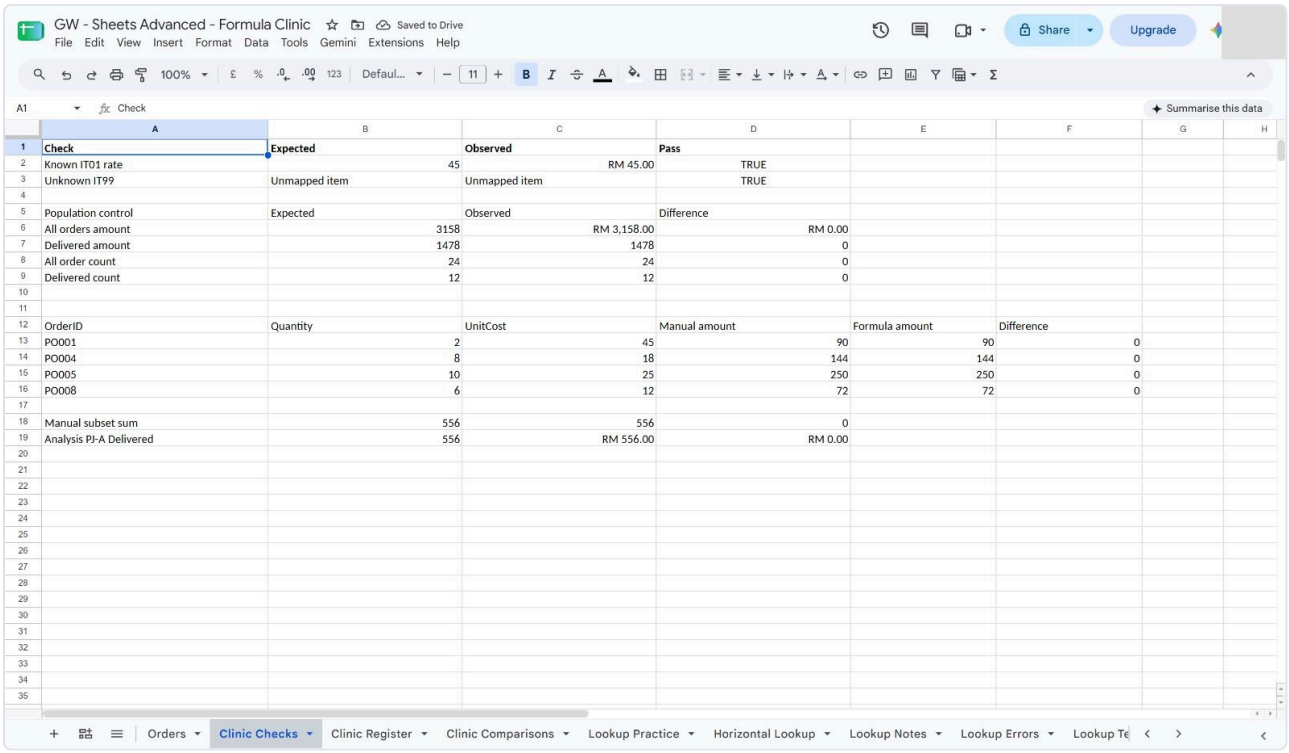
11. Write the troubleshooting path in A26:B31: inspect source → define population → inspect references → test boundary → reconcile. Explain which control detects each defect and why a plausible numeric result is insufficient.



	A	B	C	D	E
1	Defect	Faulty expression	Isolated faulty result	Repaired expression	Repaired result
2	Match mode	=VLOOKUP("IT99",Rates!$A$2:$D$7,3)	RM 8.00	=IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")	Unmapped item
3	Shifted row	=Orders!E2*Orders!F3	24	=Orders!E2*Orders!F2	90
4	Omitted row	=SUM(Orders!J2:J24)	RM 3,094.00	=SUM(Orders!J2:J25)	RM 3,158.00
5					
6					
7	Test	Expected observation	Restore	Classification	
8	Numeric text: Orders E3=four	J3 and total show #VALUE!	E3=4; J3=48; total=3158	Source type error; do not hide as zero	
9	Duplicate IT03 in copied Rates	First rate=30; count=2; review duplicate	Keep duplicate confined to test tab	Source key validation	
10	Final row: Orders E25=0	J25=0; total=3094	E25=8; J25=64; total=3158	Range boundary test	
11					
12					
13	Key	First exact rate	Key count	Decision	
14	IT03	RM 30.00	2	Review duplicate	
15					
16					
17	Test	Expected	Observed during test	Restored or retained state	
18	Known and missing key	45 / Unmapped item	45 / Unmapped item	Exact FALSE restored	
19	Zero quantity E2	J2=0; total=3068	J2=0; total=3068	E2=2; J2=90	
20	Numeric text E3	#VALUE! in J3 and total	#VALUE! in J3 and J27	E3=4; J3=48	
21	Duplicate IT03	Count=2; review duplicate	Rate=30; count=2; review duplicate	Only Clinic Duplicate Rates contains added row	
22	Final row E25=0	J25=0; total=3094	J25=0; total=3094	E25=8; J25=64	
23	Restored baseline	3158 / 1478 / 24 / 12	3158 / 1478 / 24 / 12	No source filter; cancelled rows retained	
24					
25					
26	Troubleshooting step	Evidence to check			
27	1 Inspect source	Numeric types and unique keys before changing formulas			
28	2 Define population	All 24 orders=3158; Delivered 12=1478; keep Cancelled			
29	3 Inspect references	Exact FALSE; same-row product; include J25			
30	4 Test boundaries	Normal, missing, zero, text, duplicate and final row			
31	5 Reconcile	Manual PJ-A Delivered=556; record expected versus actual			

12. Return to **Clinic Checks**. Confirm all-orders 3158, Delivered 1478, total count 24, Delivered count 12, and manually inspected PJ-A Delivered 556. All differences are 0; known/missing checks are TRUE.

Remove any active source filter and retain Cancelled records. Orders N27/O27 SUBTOTAL helpers now show 3158/24.



	A	B	C	D	E	F
1	**Check**	**Expected**	**Observed**	**Pass**		
2	Known IT01 rate	45	RM 45.00	TRUE		
3	Unknown IT99	Unmapped item	Unmapped item	TRUE		
4						
5	Population control	Expected	Observed	Difference		
6	All orders amount	3158	RM 3,158.00	RM 0.00		
7	Delivered amount	1478	1478	0		
8	All order count	24	24	0		
9	Delivered count	12	12	0		
10						
11						
12	OrderID	Quantity	UnitCost	Manual amount	Formula amount	Difference
13	PO001	2	45	90	90	0
14	PO004	8	18	144	144	0
15	PO005	10	25	250	250	0
16	PO008	6	12	72	72	0
17						
18	Manual subset sum	556	556	0		
19	Analysis PJ-A Delivered	556	RM 556.00	RM 0.00		

> **Validation boundary:** guided 6.1–6.3 and author casework 6.4 were executed through learner UI with bounded native-cell evidence. Original maintained ranges are unchanged; temporary clinic inputs are restored. The copied workbook also retains earlier labelled teaching scenarios, including a raw missing-key error from Lab 5. This is not independent learner diagnosis or proof of the planned 65-minute classroom allocation. Complete the independent challenge separately.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Ask a partner to introduce a defect on a separate clinic copy without telling you its cell; if alone, use the card and hide your defect note before diagnosis.
2. Locate it through controls, isolate the cause, repair it and run a regression check on a previously correct population.
3. Present the evidence trail in two minutes, including why your chosen control would catch the defect on a future refresh.

Evidence to Hand Over

- Defect register with preserved faulty/repaired formulas and test evidence.
- Normal/boundary checks for every repair and restored baseline controls.
- Independent diagnosis, regression check and an explainable troubleshooting path.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The total is corrected but the defect remains	Require a formula/source repair and a new input test rather than a hard-coded total.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

A useful formula check tests meaning and boundaries rather than merely the absence of an error message.

Check yourself: Which of your checks would catch a missing final row?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Use the bounded formulas and test cards below on the separate clinic copy. Record expected behaviour before introducing a fault. Restore every temporary source change.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Extra fault choices, clinic copy only: change a numeric quantity to text “four”; insert a
duplicate lookup key; replace a blank/missing-rate outcome with 0; shorten one delivered
summary range by two rows. Diagnose the cause before restoring from Source Snapshot.
```

Captured Author-Run Reference

Create a whole-workbook copy named **GW - Sheets Advanced - Formula Clinic**. Keep sharing/comments unchecked. Use the new copy for every mutation below. These are guided reference cards, not a completed independent challenge.

Clinic Register A1:D4 — record before faults

```
Defect  Expected behaviour      Planned fault   Test that should detect it
Match mode      Known IT01 returns45; unknown IT99 must be an explicit missing match    Remove
FALSE from Lookup Practice N2   Test IT99; sorted approximate lookup may return an unrelated
rate
Shifted row     Orders J2 multiplies its own Quantity2 by UnitCost45 =90        Change J2 from
E2*F2 to E2*F3  Compare first-row product and inspect references
Omitted row     All24 Orders amounts sum3158; final PO024 contributes64 Change J27 SUM range
to J2:J24       Inspect final-row coverage and compare full control total
```

Clinic Checks A1:D3 — expected values before actuals

```
Check   Expected        Observed        Pass
Known IT01 rate 45
Unknown IT99    Unmapped item
```

Clinic Checks A5:D9 — population expectations

```
Population control      Expected        Observed        Difference
All orders amount       3158
Delivered amount        1478
All order count 24
Delivered count 12
```

Clinic Checks A12:F16 — manual amounts before formulas

```
OrderID Quantity        UnitCost        Manual amount   Formula amount  Difference
PO001   2       45      90
PO004   8       18      144
PO005   10      25      250
PO008   6       12      72
```

Clinic Checks A18:D19 — reconciliation formulas

```
Manual subset sum       556     =SUM(D13:D16)   =C18-B18
Analysis PJ-A Delivered =SUM(D13:D16)   =Analysis!B4    =C19-B19
```

Clinic Register A7:B10 — causes after repairs

```
Cause   Why the control detects it
Match mode      Removing FALSE leaves approximate matching: a known IT01 still looked
correct45, but unknown IT99 incorrectly returned8. Exact matching keeps absence explicit.
Shifted row     E2*F3 uses the next item price12, so Quantity2 gives24 instead of90.
Inspecting same-row inputs catches a plausible numeric error.
Omitted row     SUM(J2:J24) excludes PO024 amount64. After repairing J2, it gives3094 instead
of3158; a final-row coverage test catches the omission.
```

Clinic Comparisons A1:E4 — live isolated expressions

```
Defect  Faulty expression       Isolated faulty result  Repaired expression     Repaired
result
Match mode      =FORMULATEXT(C2)        =VLOOKUP("IT99",Rates!$A$2:$D$7,3)
=FORMULATEXT(E2)        =IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")
Shifted row     =FORMULATEXT(C3)        =Orders!E2*Orders!F3    =FORMULATEXT(E3)
=Orders!E2*Orders!F2
Omitted row     =FORMULATEXT(C4)        =SUM(Orders!J2:J24)     =FORMULATEXT(E4)
=SUM(Orders!J2:J25)
```

Clinic Comparisons A7:D10 — expectations before tests

```
Test    Expected observation    Restore Classification
Numeric text: Orders E3=four    J3 and total show #VALUE!       E3=4; J3=48; total=3158 Source
type error; do not hide as zero
Duplicate IT03 in copied Rates  First rate=30; count=2; review duplicate        Keep duplicate
confined to test tab    Source key validation
Final row: Orders E25=0 J25=0; total=3094       E25=8; J25=64; total=3158       Range boundary
test
```

Clinic Duplicate Rates A8:D8 — append only on the test tab

```
IT03    Duplicate cable rate    99      Equipment
```

Clinic Comparisons A13:D14 — duplicate diagnostic

```
Key     First exact rate        Key count       Decision
IT03    =VLOOKUP(A14,'Clinic Duplicate Rates'!$A$2:$D$8,3,FALSE)        =COUNTIF('Clinic
Duplicate Rates'!$A$2:$A$8,A14) =IF(C14>1,"Review duplicate","Unique")
```

Clinic Comparisons A17:D23 — test log, actual columns blank

```
Test    Expected        Observed during test    Restored or retained state
Known and missing key   45 / Unmapped item
Zero quantity E2        J2=0; total=3068
Numeric text E3 #VALUE! in J3 and total
Duplicate IT03  Count=2; review duplicate
Final row E25=0 J25=0; total=3094
Restored baseline       3158 / 1478 / 24 / 12
```

Clinic Comparisons A26:B31 — troubleshooting path

```
Troubleshooting step    Evidence to check
1 Inspect source        Numeric types and unique keys before changing formulas
2 Define population     All 24 orders=3158; Delivered 12=1478; keep Cancelled
3 Inspect references    Exact FALSE; same-row product; include J25
4 Test boundaries       Normal, missing, zero, text, duplicate and final row
5 Reconcile     Manual PJ-A Delivered=556; record expected versus actual
```

Actual fault and repair sequence

Target on clinic copy	Fault	Repair / restore
Lookup Practice N2	=VLOOKUP(D2,Rates!$A$2:$D$7,3)	=VLOOKUP(D2,Rates!$A$2:$D$7,3,FALSE)
Lookup Practice D2	IT99 to expose approximate result8	IT01; exact result45
Orders J2	=E2*F3 returns24	=E2*F2 returns90
Orders J27	=SUM(J2:J24); combinedfault3028, after rowrepair3094	=SUM(J2:J25) returns3158
Orders E2	0 gives valid zero line amount; total3068	2; total3158
Orders E3	four gives #VALUE! in J3 and total	4; J3=48; total3158
Orders E25	0 gives J25=0 and total3094	8; J25=64; total3158

Filtered population helpers outside the source range

Create the filter on **Orders A1:K25**, select Status **Delivered** only, and add the helpers outside that range:

```
Visible Amount  Visible Count
=SUBTOTAL(109,J2:J25)   =SUBTOTAL(103,A2:A25)
```

Paste in **N26:O27**. With the Delivered filter active, expect1478/12. Remove the filter after recording the result; the helpers then show3158/24. SUM over the source remains an all-record total. Keep Cancelled records.

In Clinic Checks **E13** use **=B13*C13** and F13 **=E13-D13**, then fill both through row16. The manual amounts must be recorded before these comparisons. Expected differences are0; subset sum556.

The numeric-text and duplicate-key cases classify source type/uniqueness problems separately from formula logic. Do not replace missing or invalid values with0. Independent hidden-fault diagnosis, partner presentation, reviewer work and learner duration remain pending.

07

CHAPTER 7

Summarise with Pivot Tables

In this chapter you will group project costs and status; inspect filters, calculated fields and source coverage.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 75 minutes: 15 explanation/demonstration, 32 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 60 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A native pivot with checked filters and a calculated-field demonstration or recorded limitation.

> **Status:** author reference with 34 staged application captures through 7.4. Native amount/count pivots, Delivered filter, Custom average field, actual drilldown and PO025 coverage tests are verified. Maintained baseline unchanged; new record retained only on a separate scenario copy. Independent challenge, reviewer and learner timing remain pending.

What You Will Learn

- Create a native pivot summary.
- Choose filters and aggregation deliberately.
- Check a calculated field against row calculations.

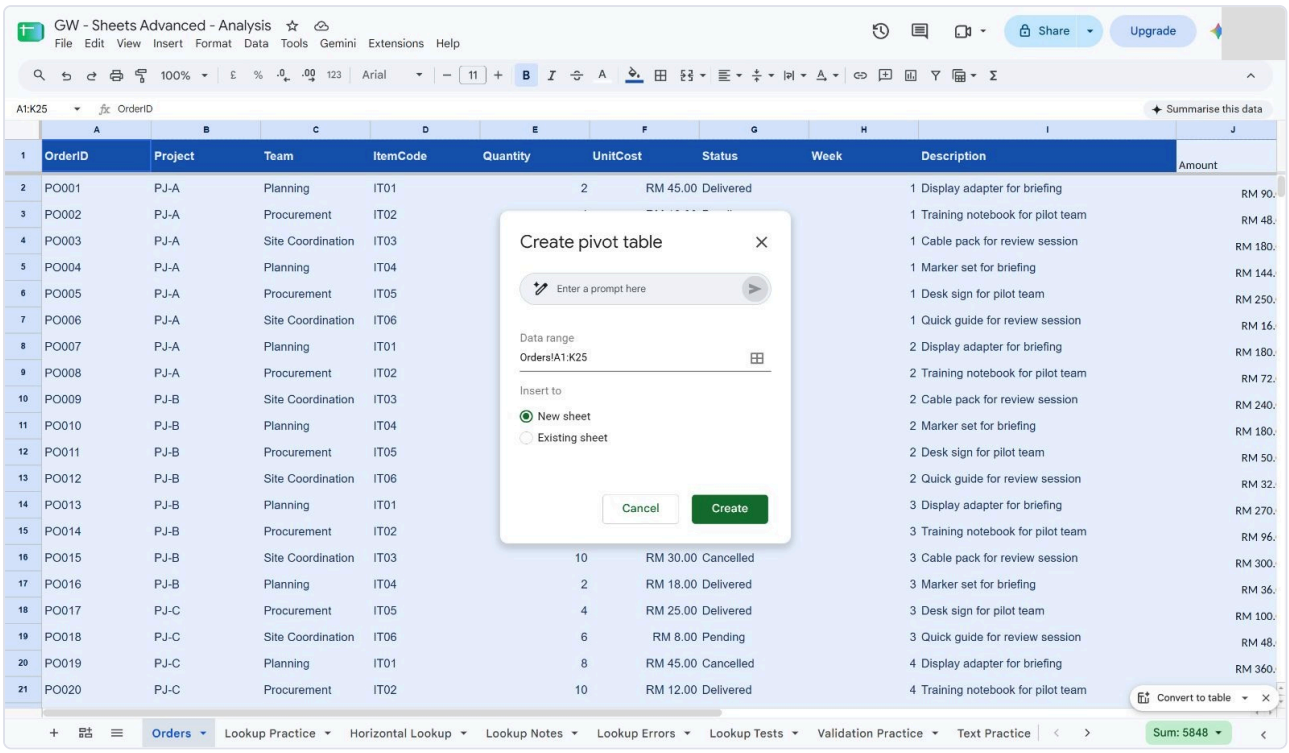
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

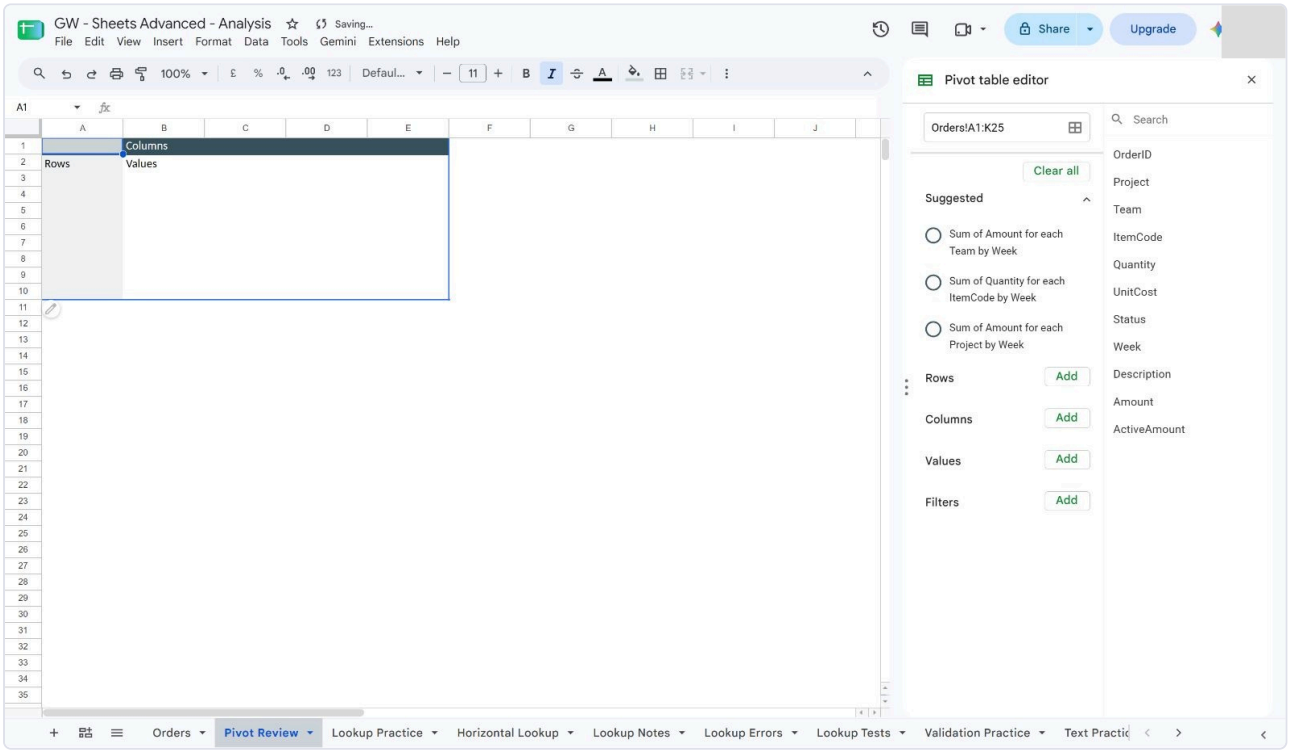
Use Orders A1:K25 after calculating Amount in J and ActiveAmount in K. A recovery route is to repeat Advanced Chapter 2 on the supplied workbook.

7.1 Build the Pivot

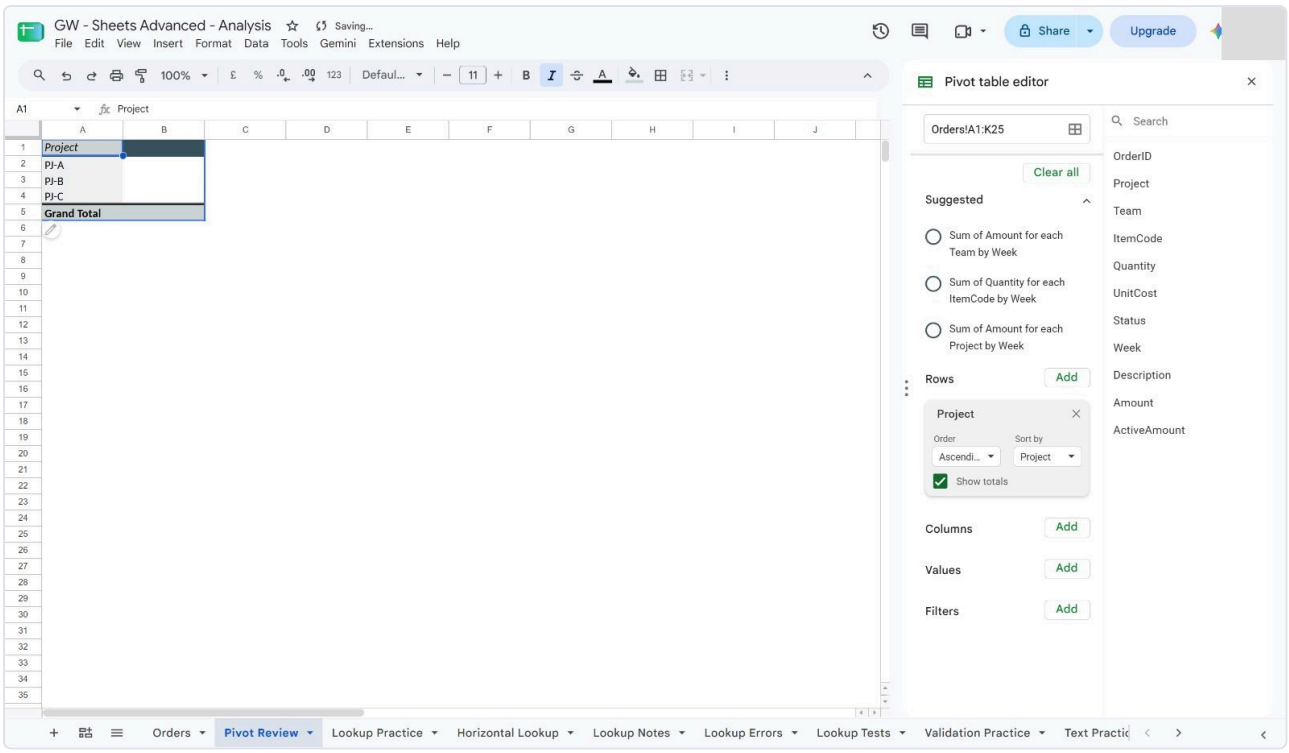
1. On the maintained Advanced workbook, select **Orders A1:K25**. Choose **Insert > Pivot table**. Verify the data range and select **New sheet** before Create. This includes 24 records plus their header and the existing Amount/ActiveAmount columns; it excludes the summary cells below them.



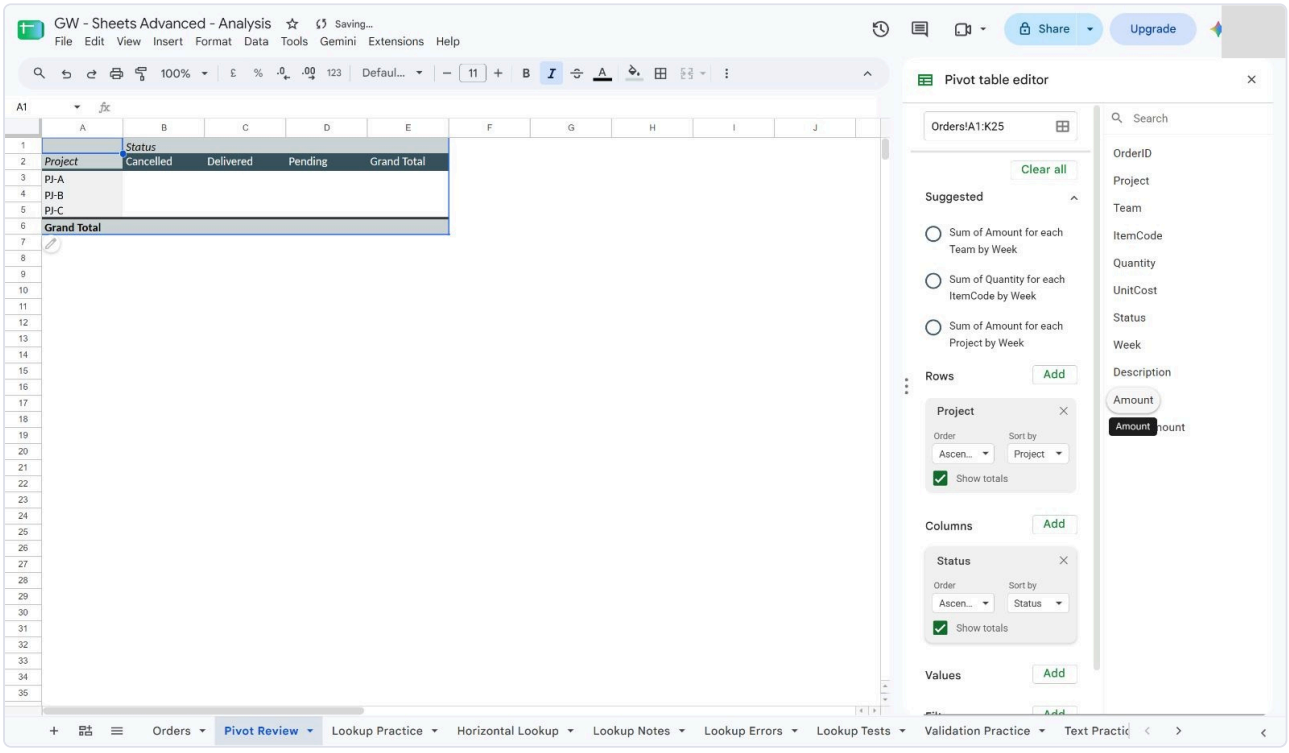
2. Rename the new **Pivot Table 1** tab **Pivot Review**. Keep the empty native pivot and editor visible before adding fields. Do not type a summary grid in place of a pivot.



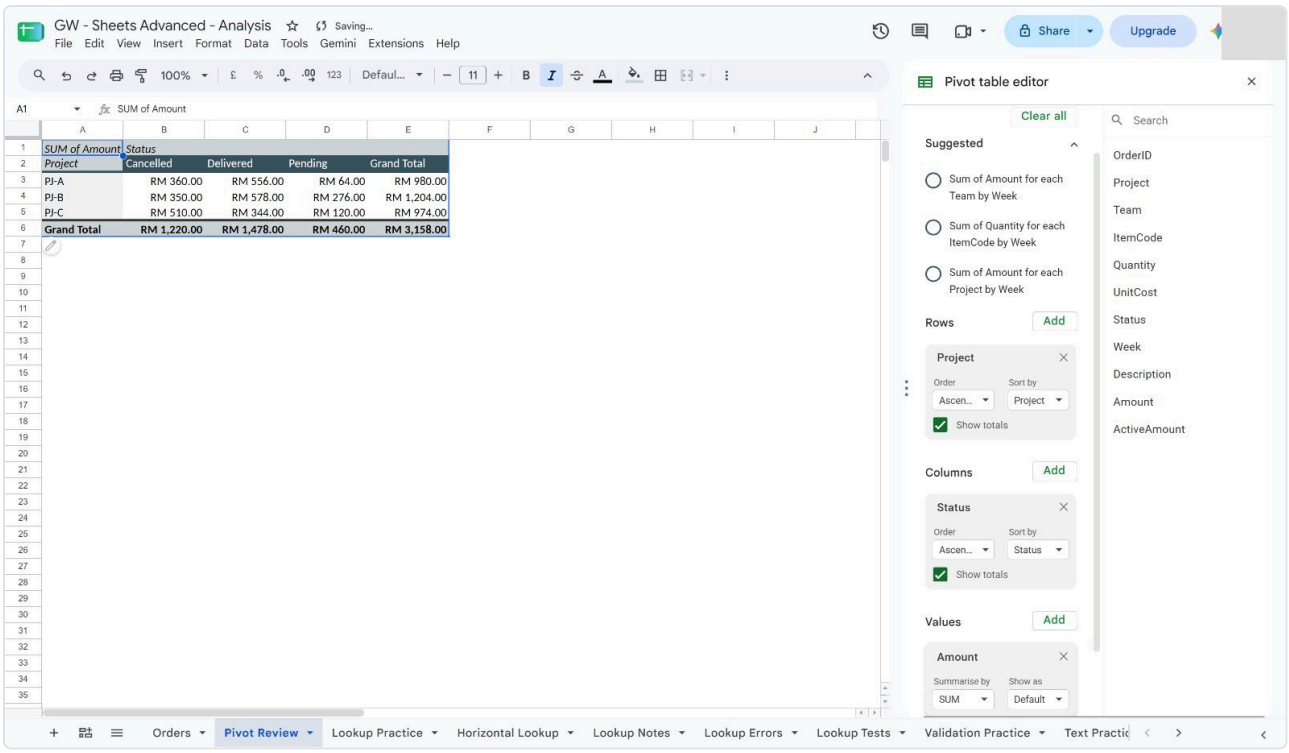
3. In the editor, use **Rows > Add > Project**.



4. Use **Columns > Add > Status**. Keep Show totals enabled for both groups.

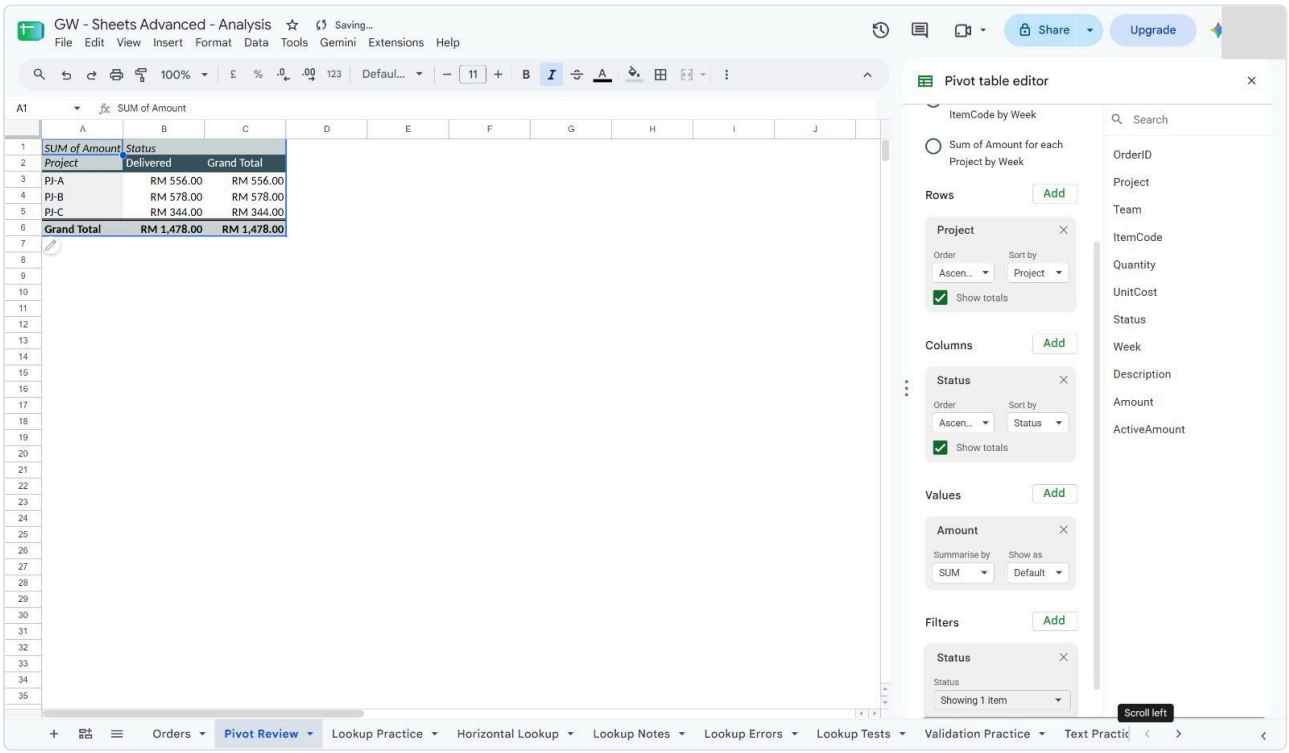


5. Use **Values > Add > Amount** and inspect **Summarise by SUM**. The native grand total is **3,158**: Cancelled1,220, Delivered1,478 and Pending460. PJ-A totals980, PJ-B1,204 and PJ-C974. Cancelled records remain part of the all-orders population.



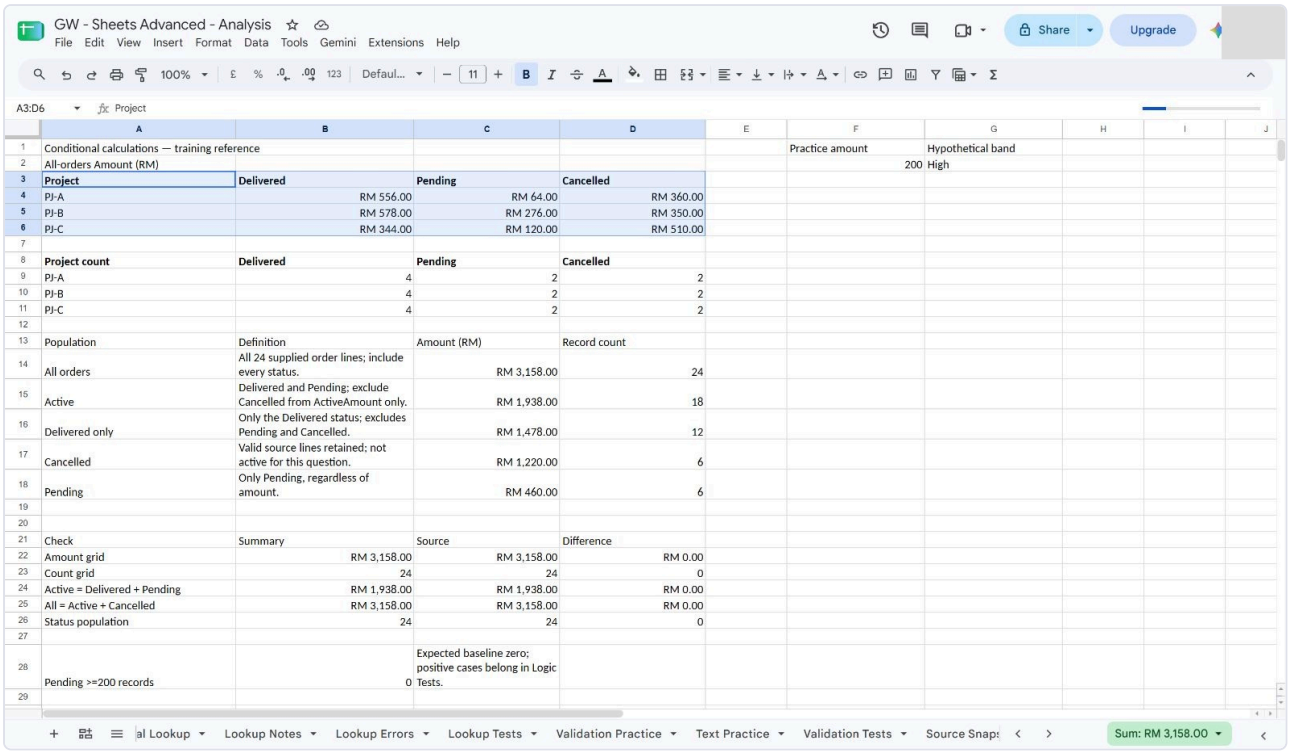
7.2 Filter and Compare

1. Use **Filters > Add > Status**. Open Showing all items, choose Clear, select **Delivered** only and confirm OK. The grand total becomes **1,478**; project amounts are **556 /578 /344**. This is a pivot filter, not deletion from Orders.

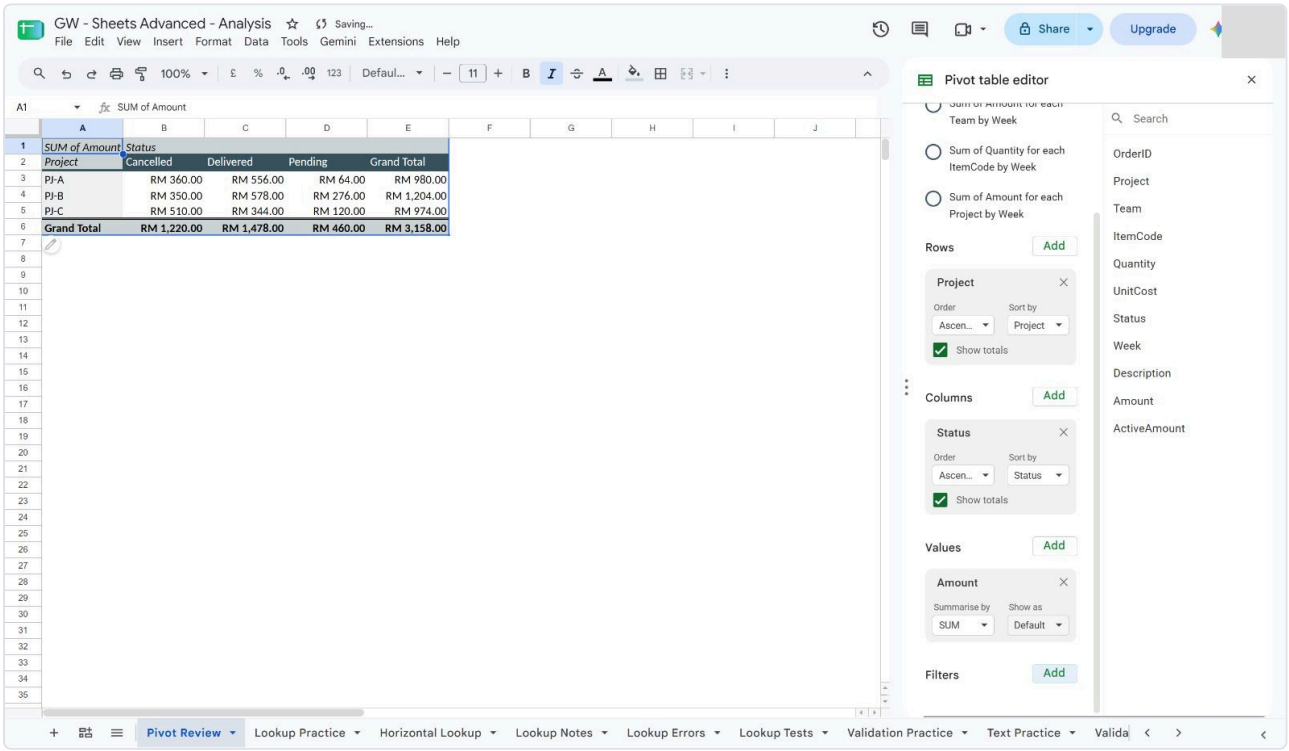


2. On **Analysis A3:D6**, inspect the earlier SUMIFS grid. Its Delivered column gives the same three project amounts. Note that this grid orders its status columns differently from the pivot; compare labels rather

than column positions.



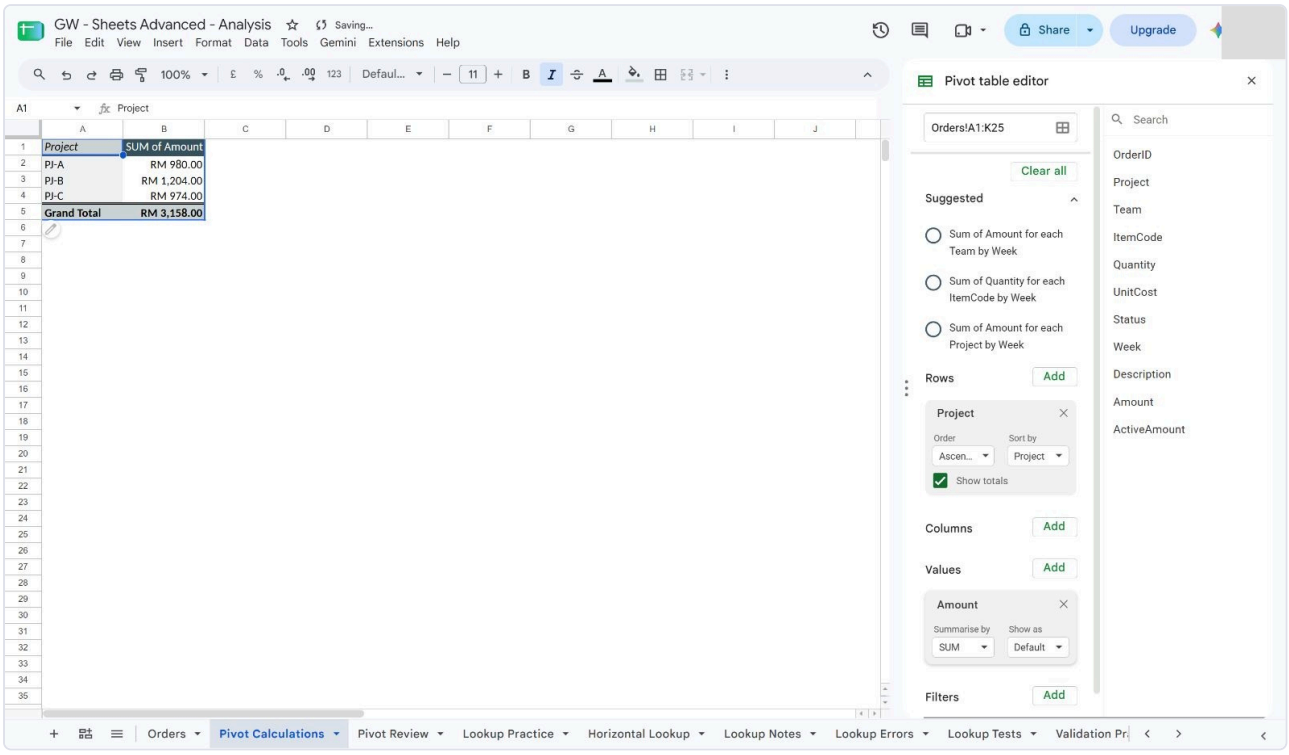
3. Return to Pivot Review and remove only the **Filter Status** card with its close/remove button. Do not remove the Status column grouping. All statuses and the **3,158** grand total return.



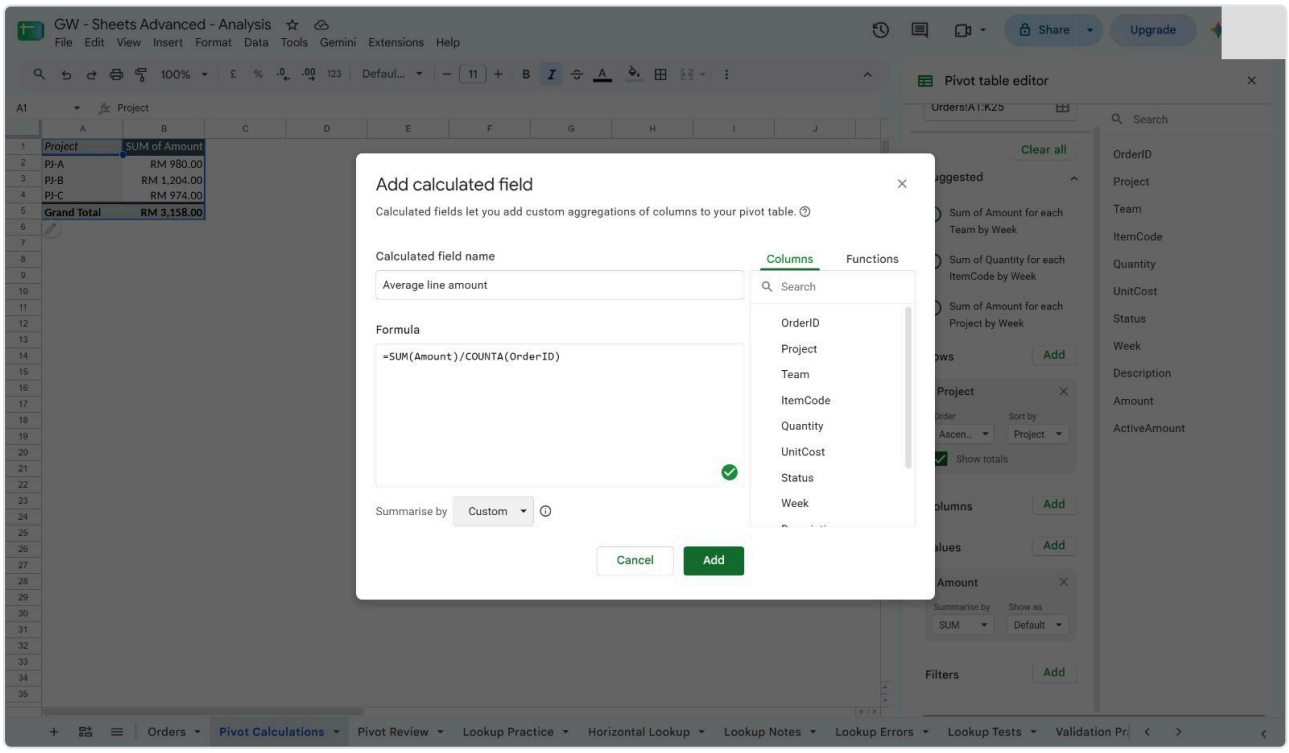
7.3 Inspect a Calculated Field

1. Select Orders A1:K25 and create another native pivot on a new sheet named **Pivot Calculations**. Add Project as rows and Amount as values, SUM. Keep this authoritative row-amount sum alongside the

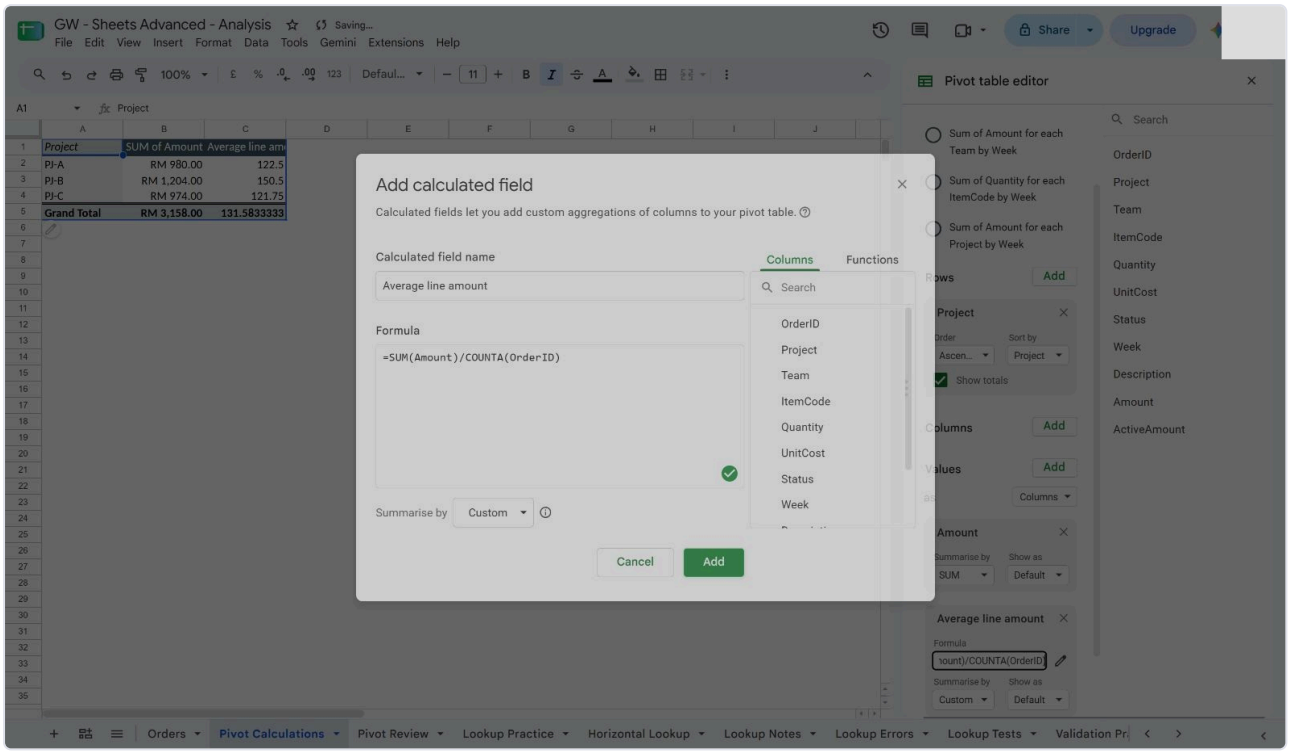
demonstration.



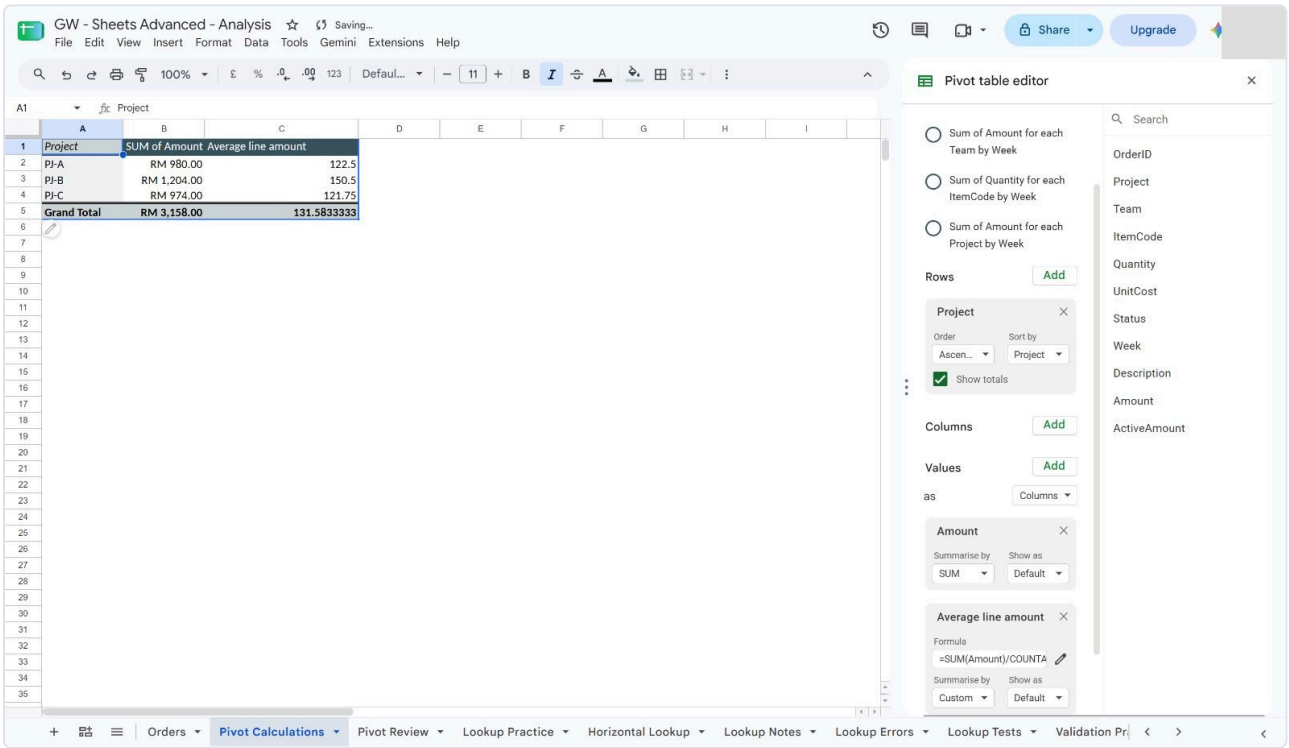
2. Choose **Values > Add > Calculated Field**. The current UI opens **Add calculated field**. Name it **Average line amount**, enter **=SUM(Amount)/COUNTA(OrderID)**, choose **Summarise by Custom**, and inspect Formula is valid before Add. Use the bounded source with populated unique OrderID values; COUNTA counts nonblank lines, not distinct identifiers.



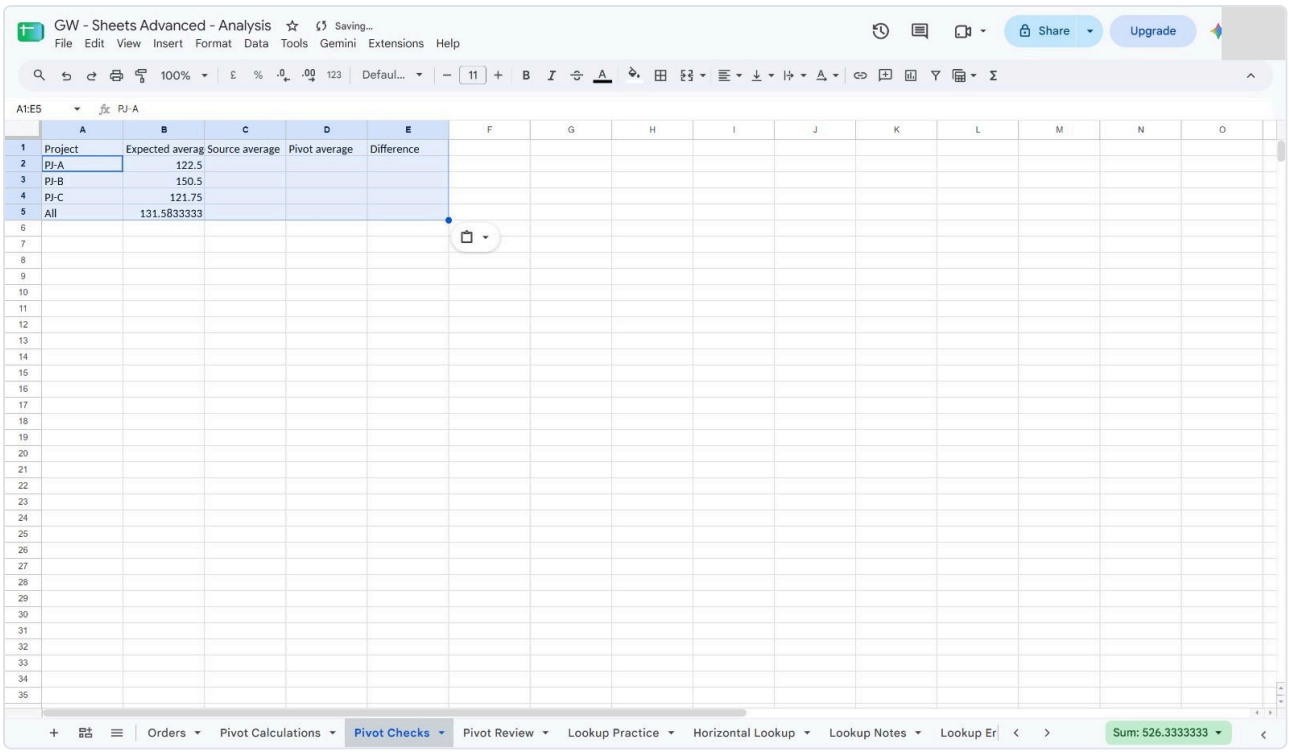
3. Inspect the native results: **122.5 /150.5 /121.75** for PJ-A/B/C and **131.5833333** overall. The overall result is total Amount divided by all24 lines; it is not a sum of the project averages.



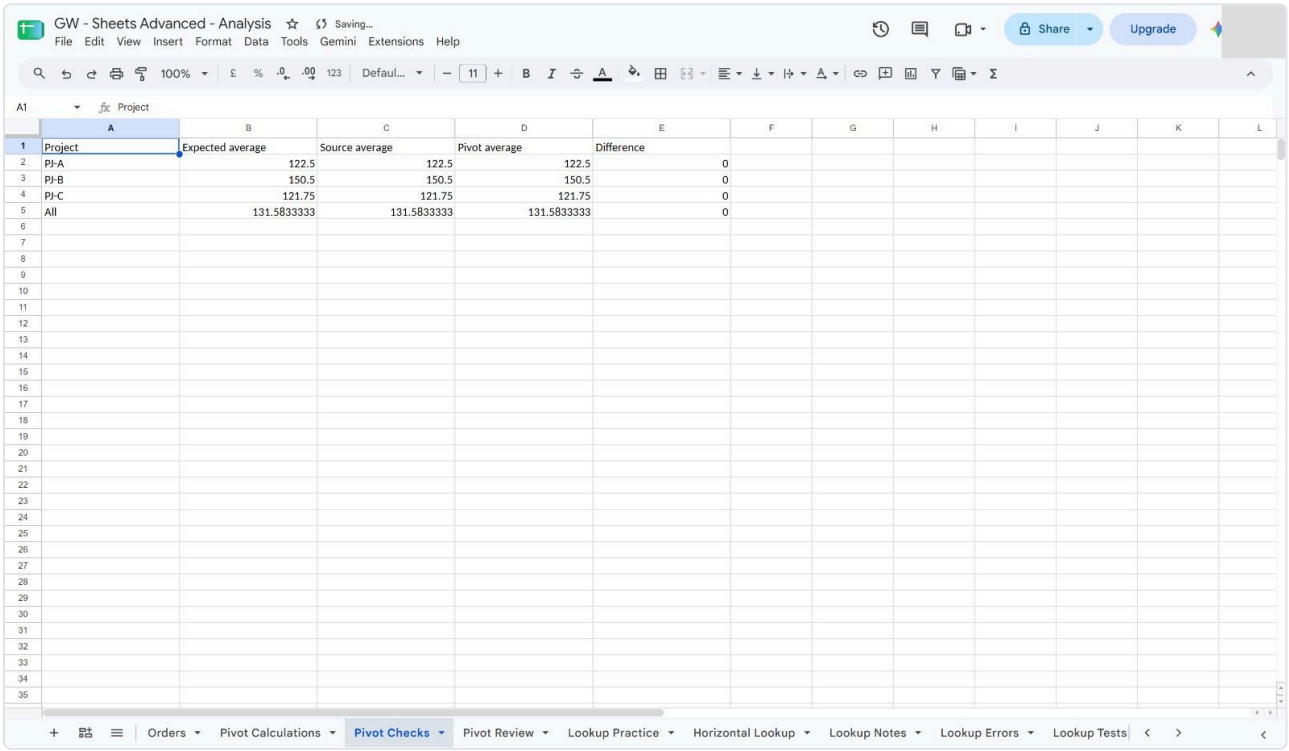
4. Widen column C to190 pixels so its heading is readable. Preserve the unrounded calculated result for reconciliation; display rounding alone is not a formula repair.



5. Create **Pivot Checks**. Record the expected project and overall averages in A1:B5 before adding actual calculations. Use the copy-paste card; C:E start blank.



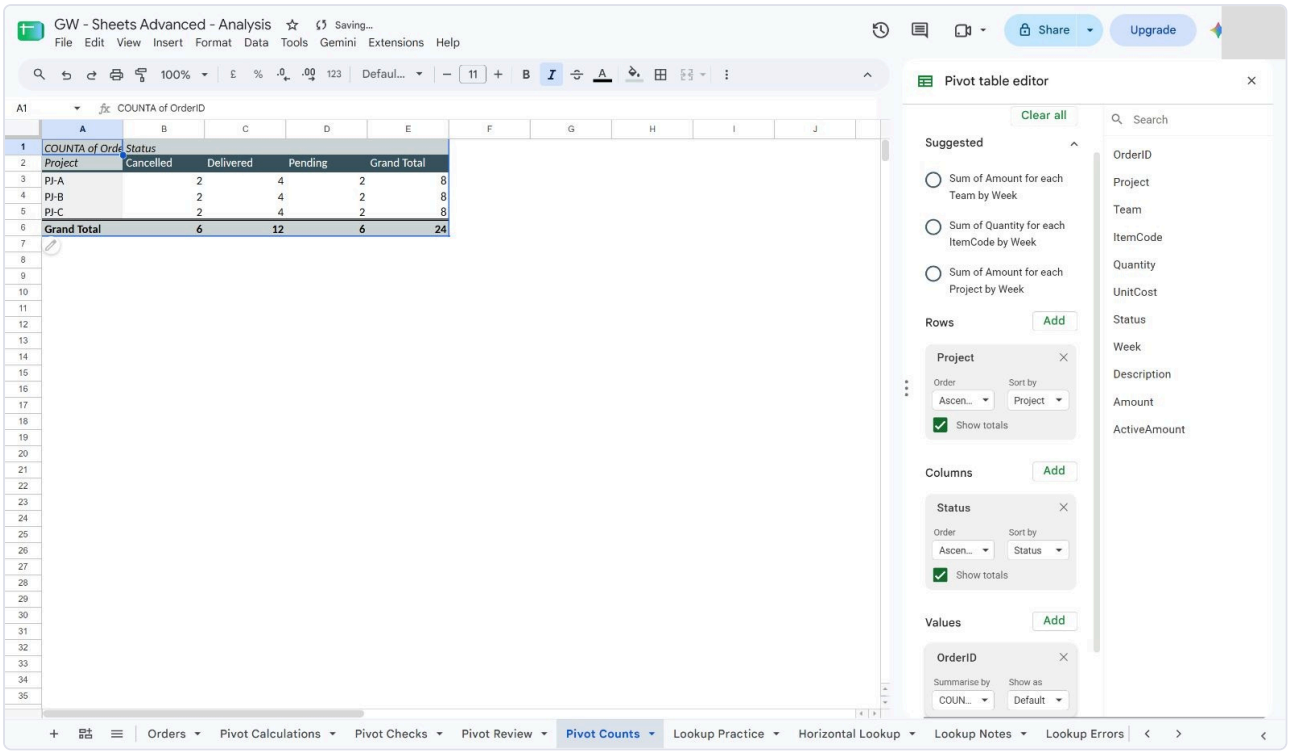
6. Add source SUMIF/COUNTIF averages in C, native Pivot Calculations references in D, and D minus C in E. All four differences are0. Widen A:E to170 pixels. This target UI supports the specified calculated field, so an unavailable-feature fallback was not needed in this author run.



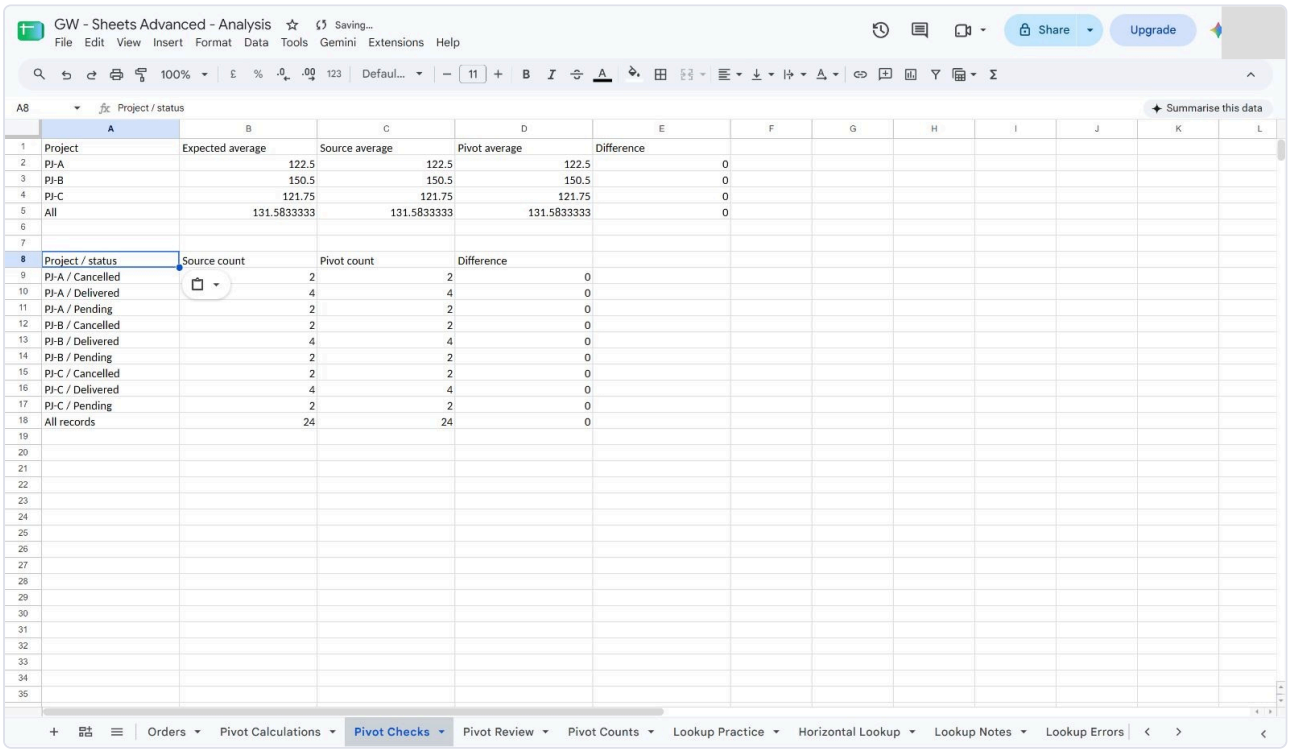
7.4 Reconcile Pivot Semantics and New-Row Coverage

1. Duplicate **Pivot Review** as **Pivot Counts**. Select A1 and open its Edit control if the editor is closed. Remove Amount from Values, then add **OrderID**, inspecting **COUNTA**. Keep Project rows, Status

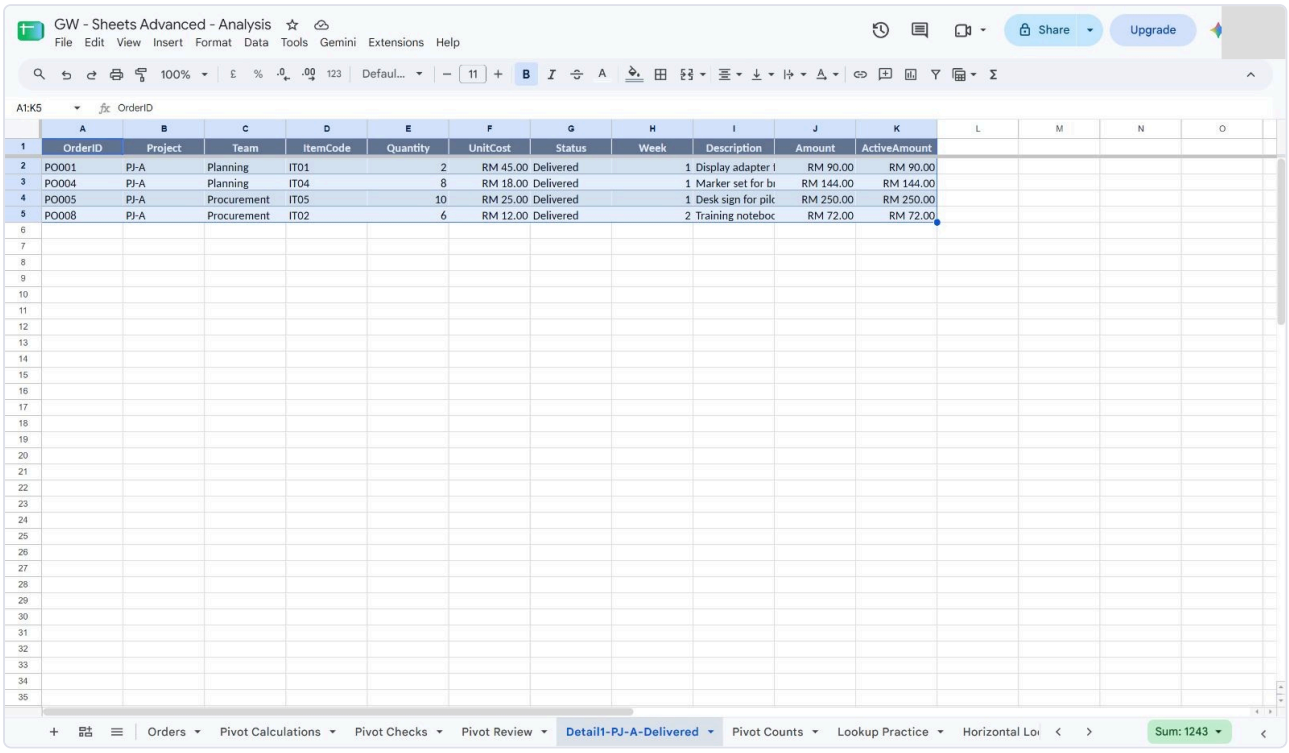
columns and source Orders A1:K25. The grand total is24: Cancelled6, Delivered12 and Pending6. Each project has8 lines.



2. In Pivot Checks A8:D18, compare all nine project/status categories with bounded source **COUNTIFS** and the corresponding count-pivot cells. Add a total COUNTA check. All ten differences are0. COUNT on text OrderID would answer a different question, so inspect the aggregation explicitly.

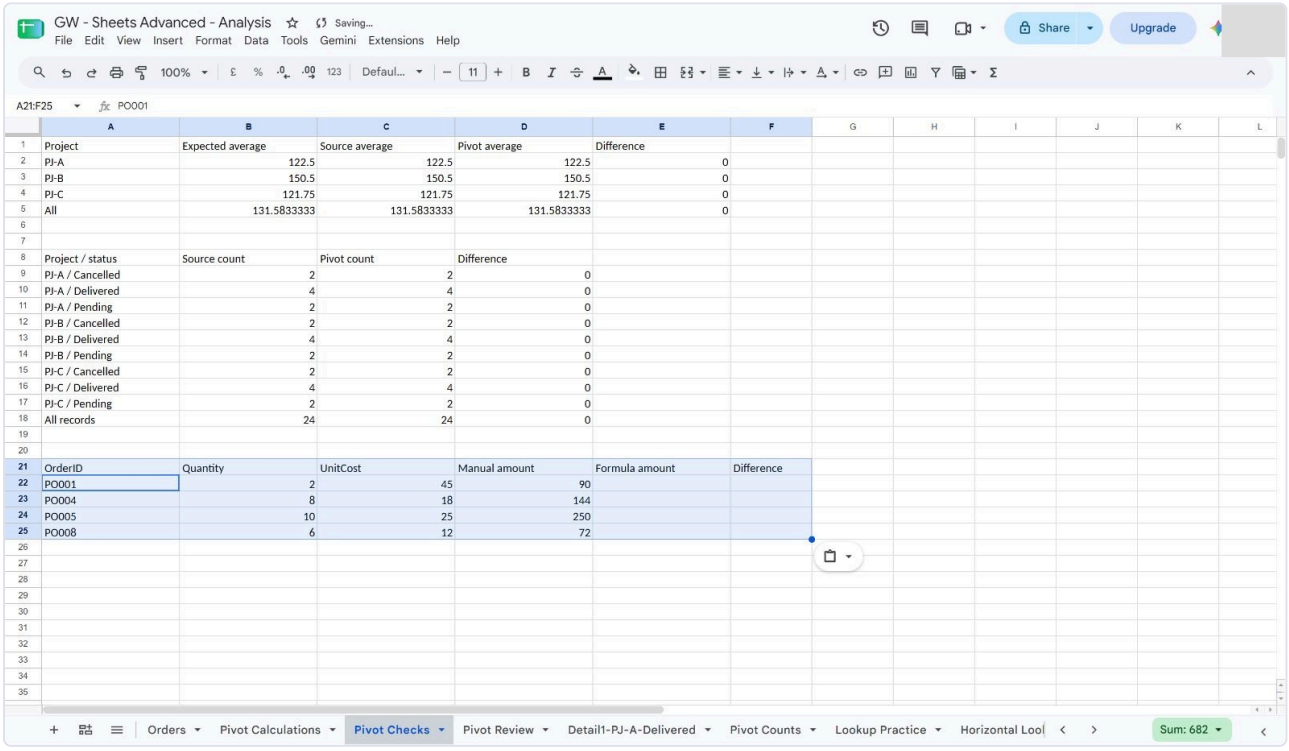


3. Return to Pivot Review and double-click the **PJ-A / Delivered** value556 in C3. The application creates **Detail1-PJ-A-Delivered** with actual contributing records **PO001 /PO004 /PO005 /PO008**. This is a snapshot of the drilldown at this point; do not treat it as a live refreshable source table.



OrderID	Project	Team	ItemCode	Quantity	UnitCost	Status	Week	Description	Amount	ActiveAmount
PO001	PJ-A	Planning	IT01	2	RM 45.00	Delivered	1	Display adapter	RM 90.00	RM 90.00
PO004	PJ-A	Planning	IT04	8	RM 18.00	Delivered	1	Marker set for bo	RM 144.00	RM 144.00
PO005	PJ-A	Procurement	IT05	10	RM 25.00	Delivered	1	Desk sign for pilo	RM 250.00	RM 250.00
PO008	PJ-A	Procurement	IT02	6	RM 12.00	Delivered	2	Training noteboo	RM 72.00	RM 72.00

4. In Pivot Checks A21:F25, record the four manually inspected quantities, costs and amounts **90 /144 /250 /72** before formula comparisons. These are guided author checks; they do not replace an independent learner attempt.

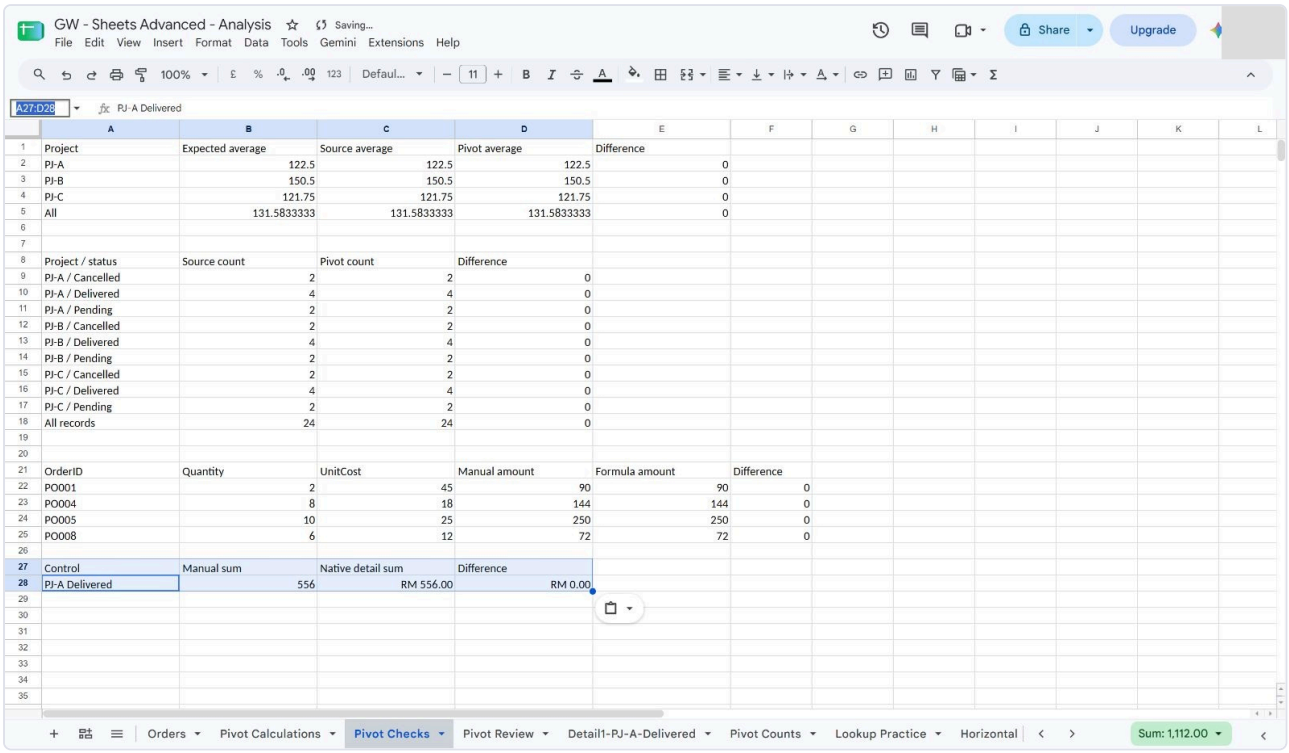


Project	Expected average	Source average	Pivot average	Difference
PJ-A	122.5	122.5	122.5	0
PJ-B	150.5	150.5	150.5	0
PJ-C	121.75	121.75	121.75	0
All	131.5833333	131.5833333	131.5833333	0

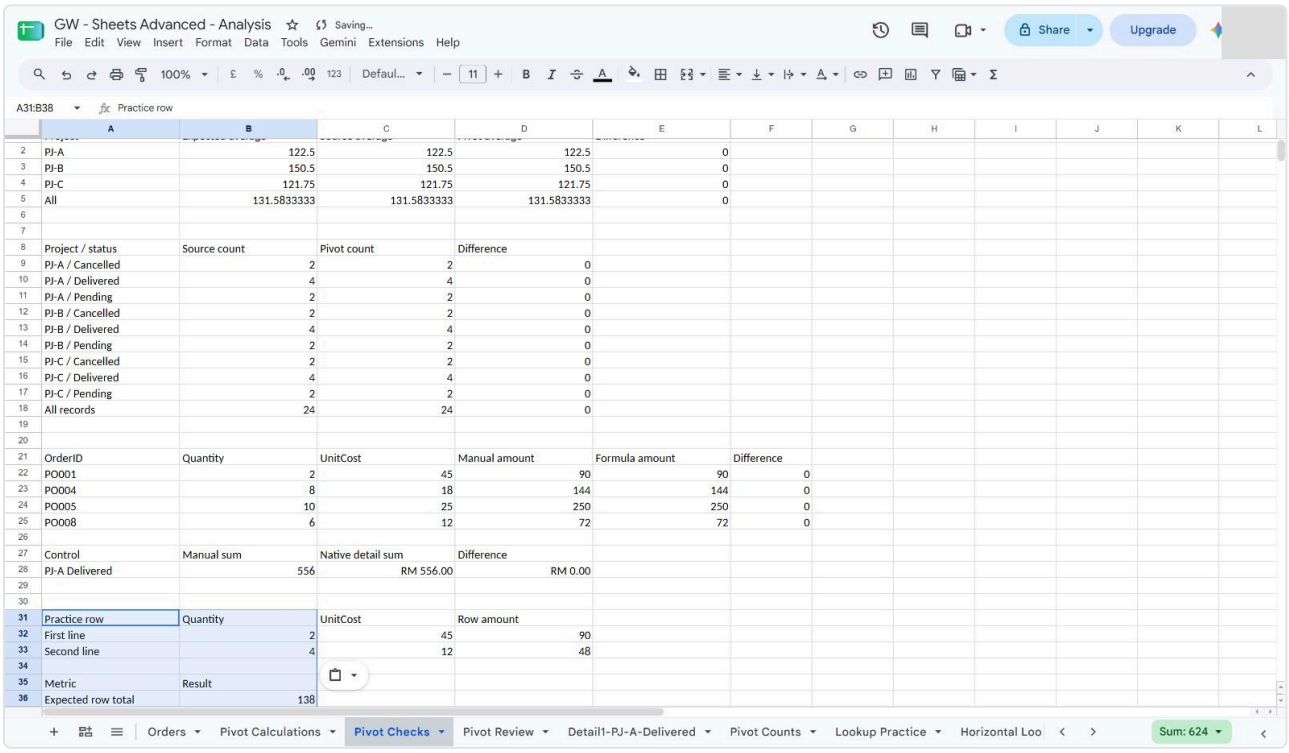
Project / status	Source count	Pivot count	Difference
PJ-A / Cancelled	2	2	0
PJ-A / Delivered	4	4	0
PJ-A / Pending	2	2	0
PJ-B / Cancelled	2	2	0
PJ-B / Delivered	4	4	0
PJ-B / Pending	2	2	0
PJ-C / Cancelled	2	2	0
PJ-C / Delivered	4	4	0
PJ-C / Pending	2	2	0
All records	24	24	0

OrderID	Quantity	UnitCost	Manual amount	Formula amount	Difference
PO001	2	45	90		
PO004	8	18	144		
PO005	10	25	250		
PO008	6	12	72		

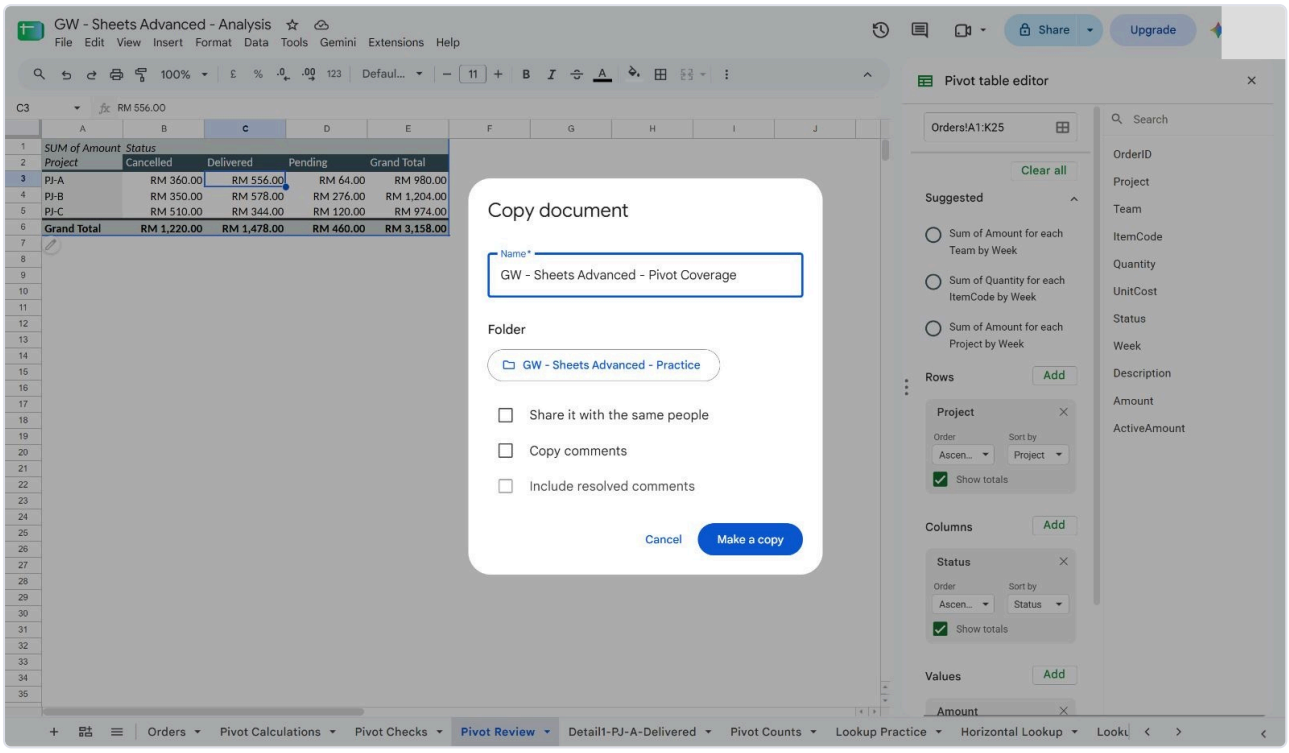
5. Calculate each row product and difference in E/F. At A27:D28 compare the manual sum556 with SUM of the actual detail sheet's J2:J5. All four row differences and the total difference are0.



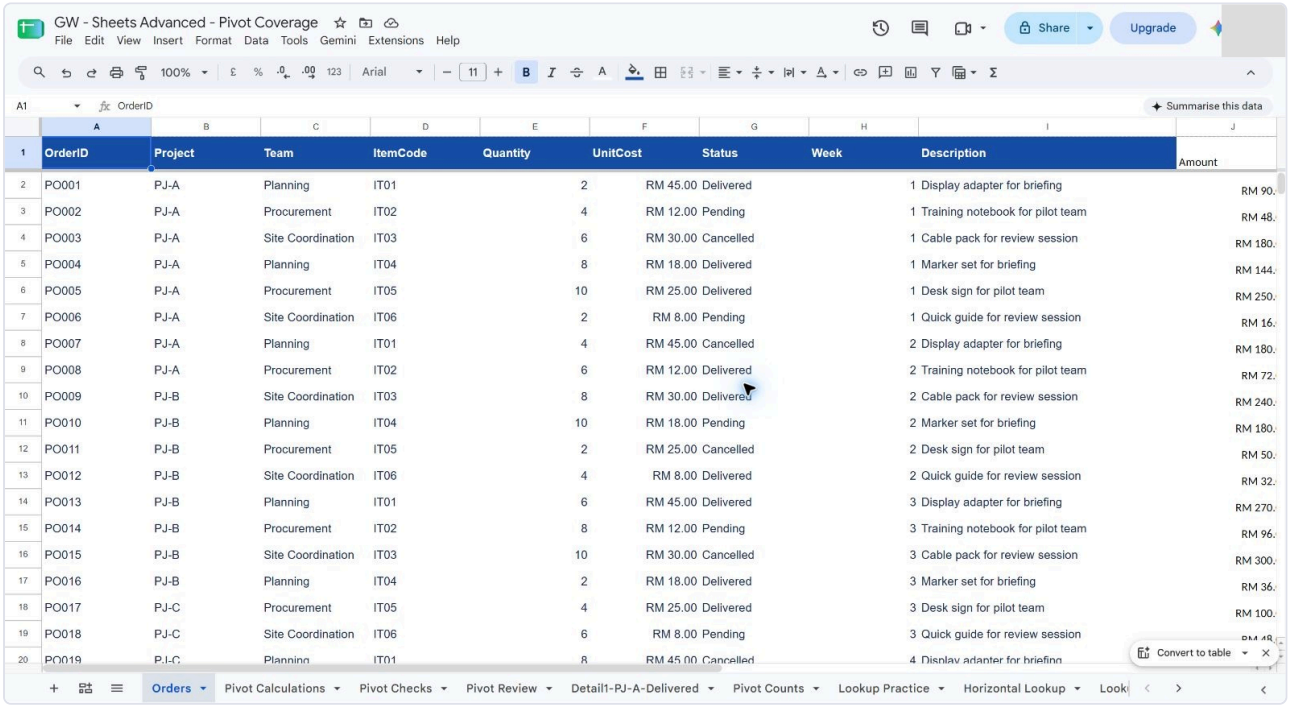
6. Build the separate two-row practice table at A31:D33: **2×45=90** and **4×12=48**. Record expected138, SUM the row products, and separately calculate **SUM(Quantity)*SUM(UnitCost)**. The first returns138; the second342 includes cross-products that do not belong to either order. Retain SUM of source row Amount as the authoritative total. Do not substitute this product-of-sums calculation for it.



7. Use **File > Make a copy** of the workbook. Name it **GW - Sheets Advanced - Pivot Coverage** in the practice folder. Keep sharing and comments unchecked. This copies the whole workbook, including native pivots; keep the maintained workbook unchanged during the following source experiment.

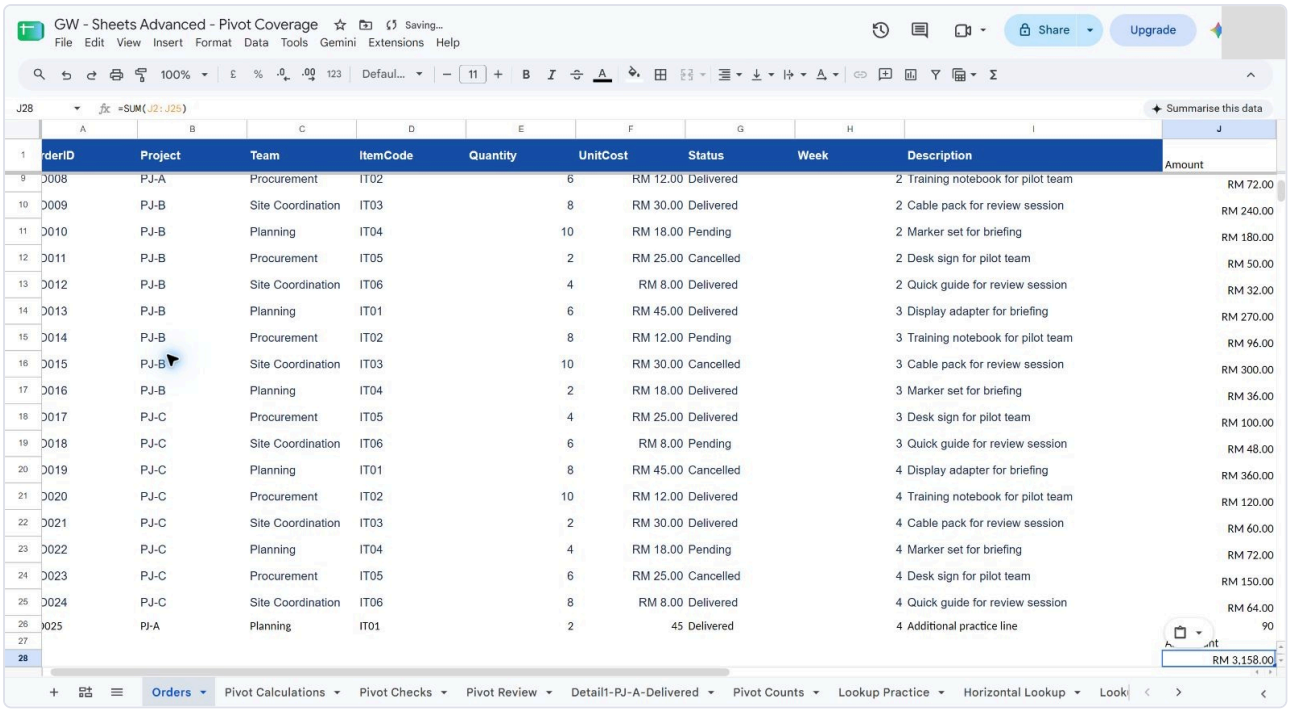


8. Verify the new title and separate file URL before editing. On its Orders, select row26 and **Insert1 row above**. This reserves a new data row before the existing summary labels; those labels move to row27 and their totals to row28.

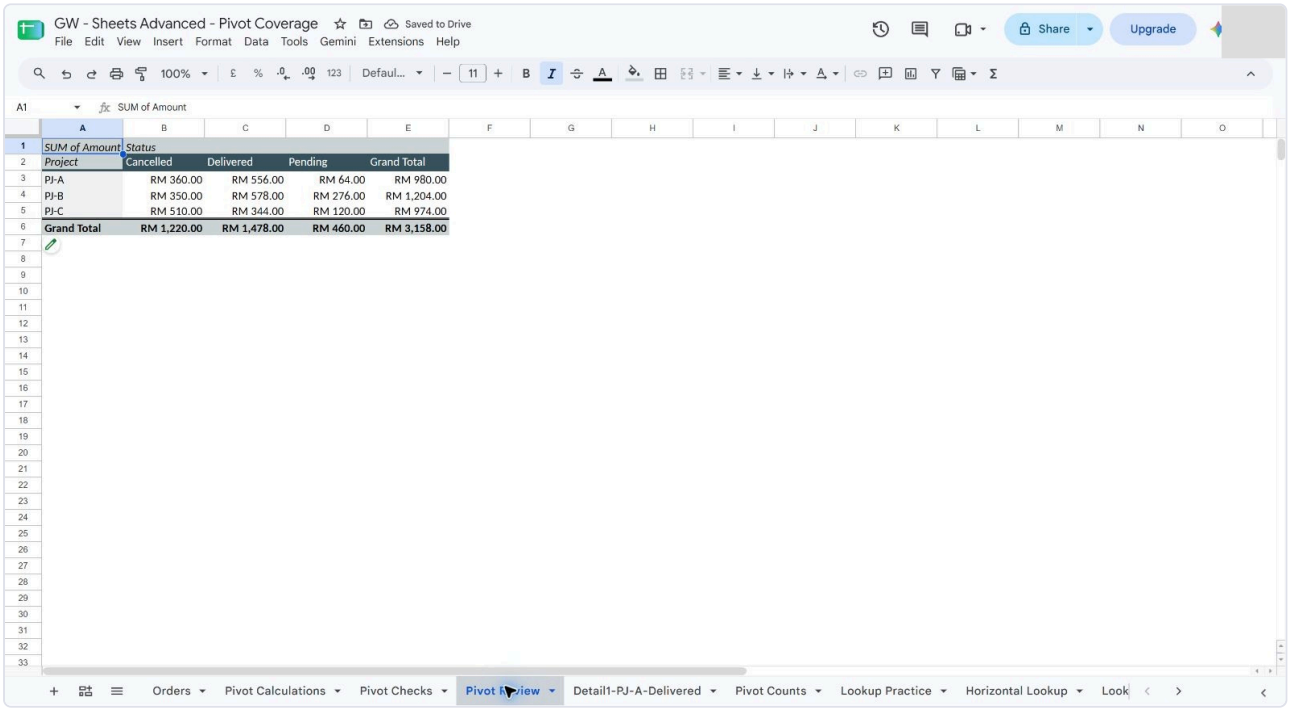


9. Enter the supplied **PO025** record in A26:I26, **=E26*F26** in J26 and **=IF(G26="Cancelled",0,J26)** in K26. Its amount is90. Expect all Amount3,248, Delivered1,568, count25 and Delivered count13 once

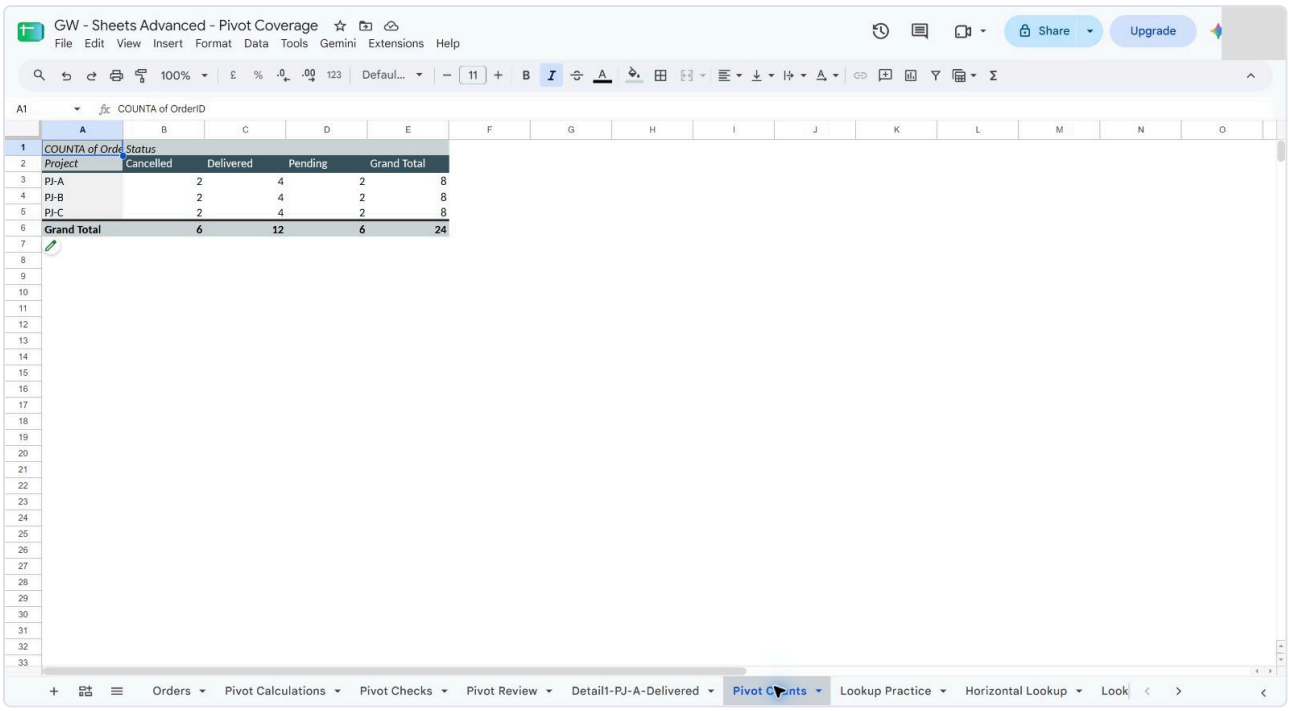
coverage is repaired. The copied summary J28 still uses J2:J25 at this point, so it remains3,158.



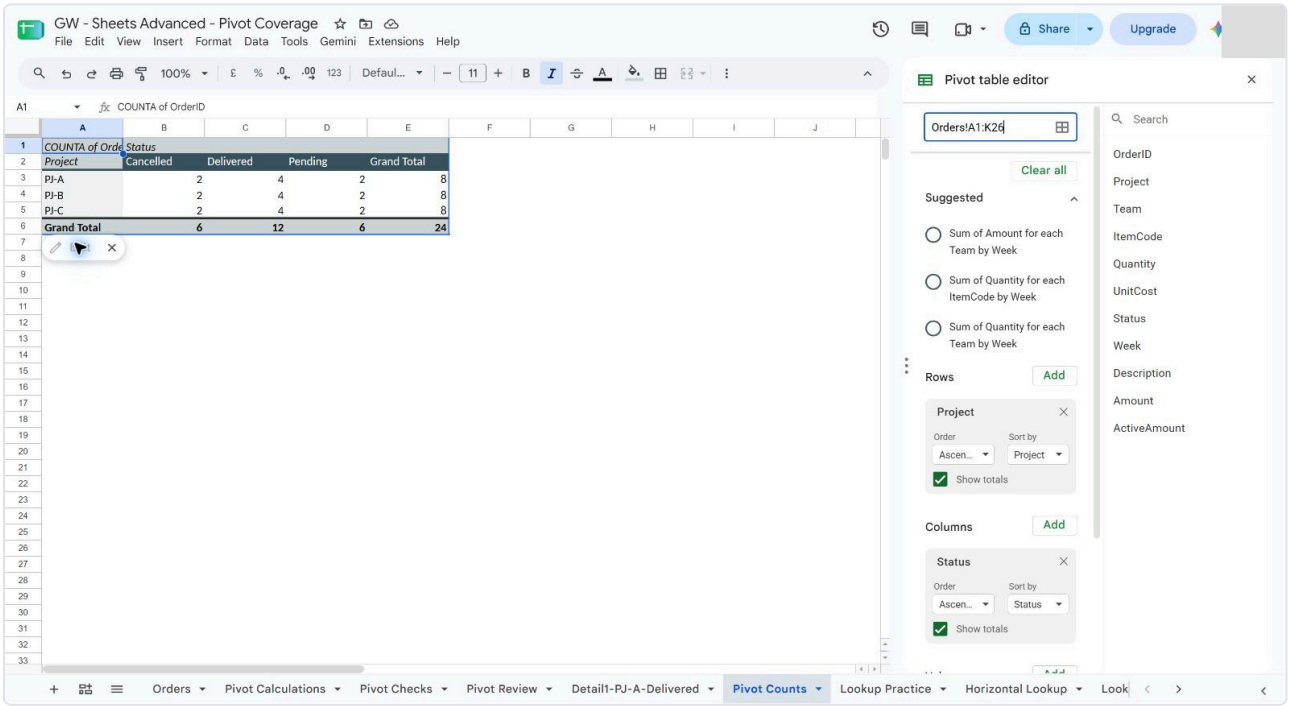
10. Inspect the copied **Pivot Review**. Its source is still Orders A1:K25, its total3,158 and PJ-A Delivered556: the new record has not entered the pivot.



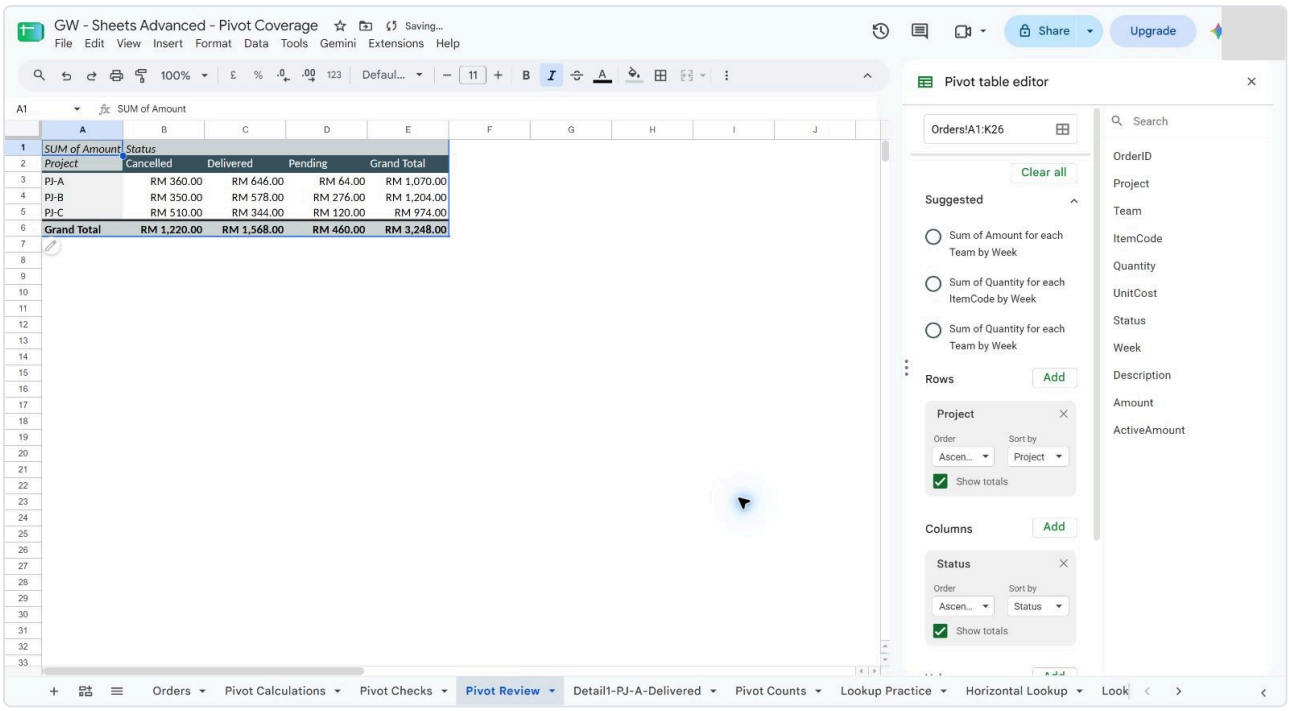
11. Inspect copied **Pivot Counts**. It still counts24 and Delivered12. Matching the old total is not proof that a new source record was included.



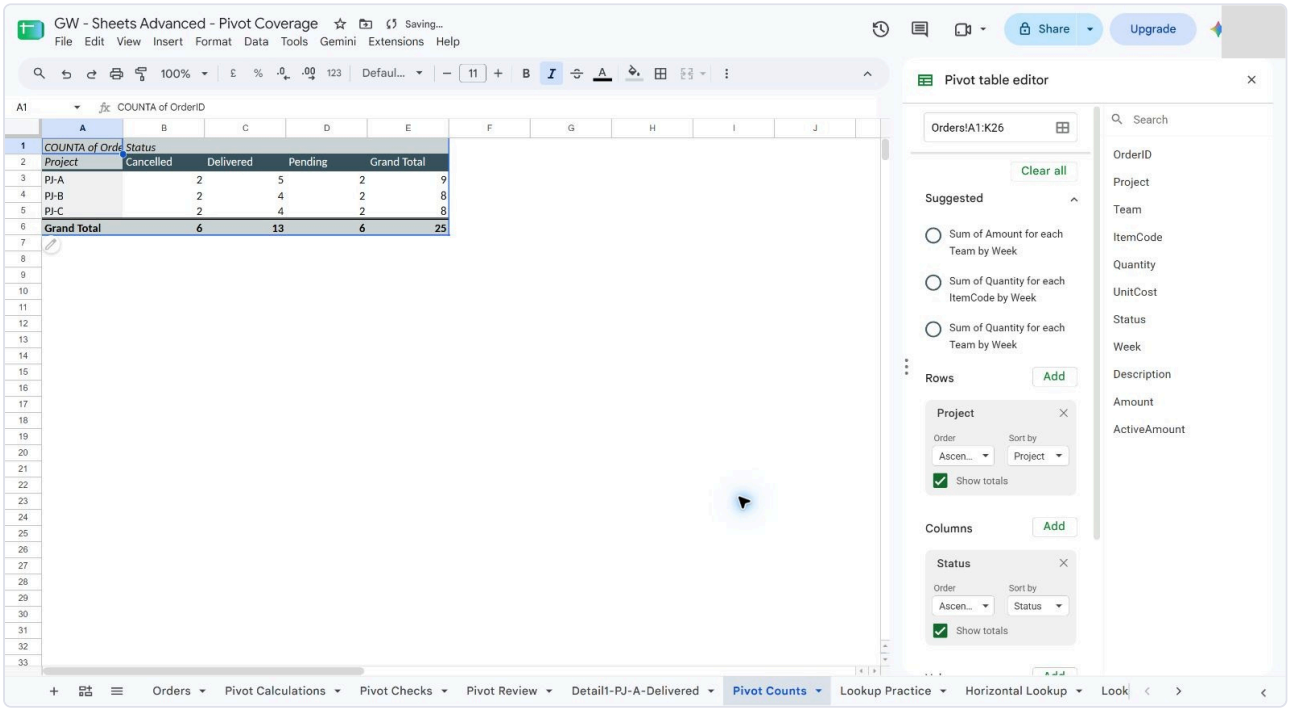
12. Use the editor's **Select data range** button, enter **Orders!A1:K26** in the Range dialog and confirm **OK** for each pivot. In the captured intermediate state, the count editor displayed typed A1:K26 text but the count was still24. Verify the applied native definition and recalculated results; typed field text alone is insufficient evidence.



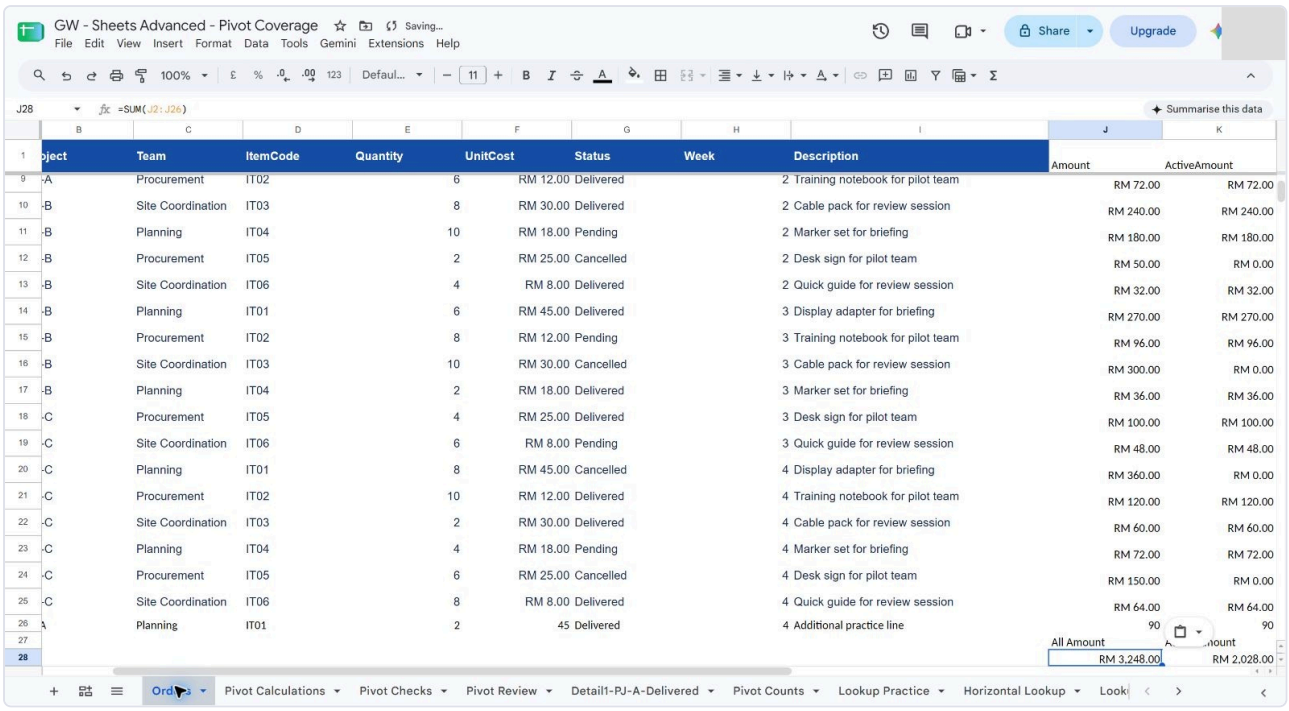
13. After confirming the Amount pivot range, inspect **3,248** overall, **1,568** Delivered, and **646** PJ-A Delivered. This is the original556 plus the new90.



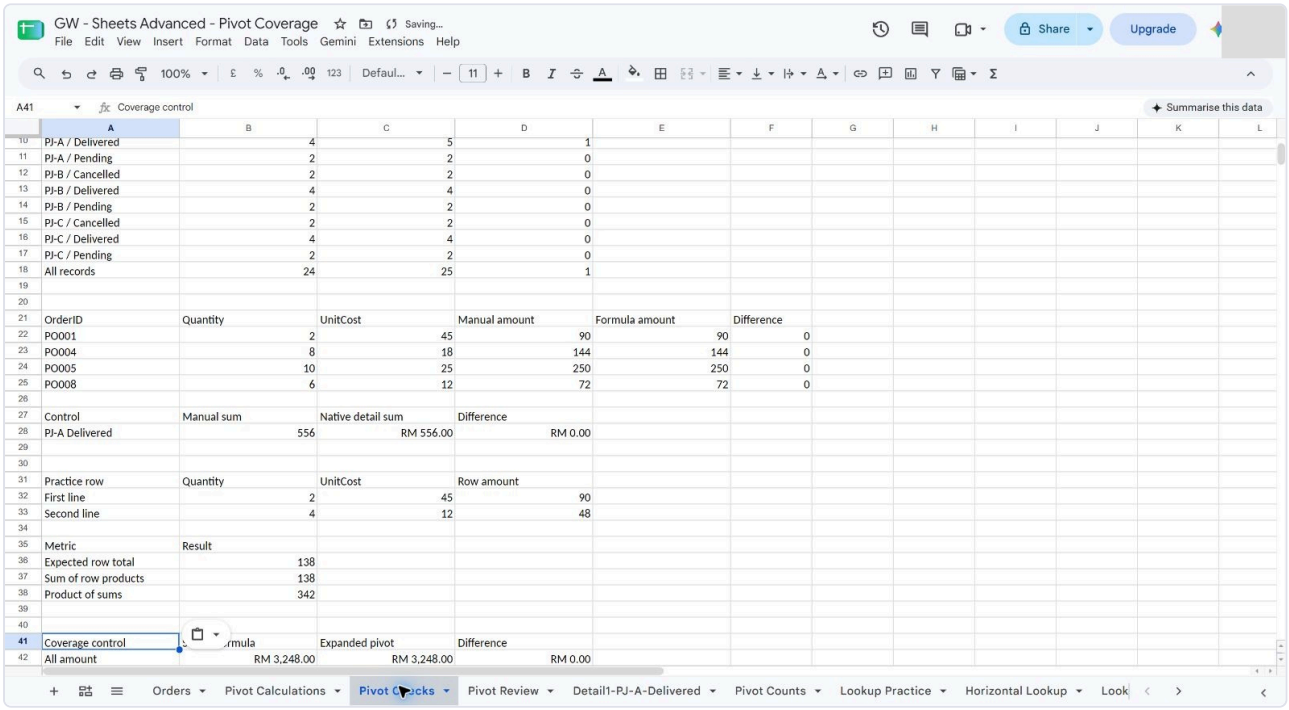
14. Confirm the count pivot's range through the same dialog. It now counts **25** overall, **13** Delivered and **5** PJ-A Delivered. Both pivots must cover row26.



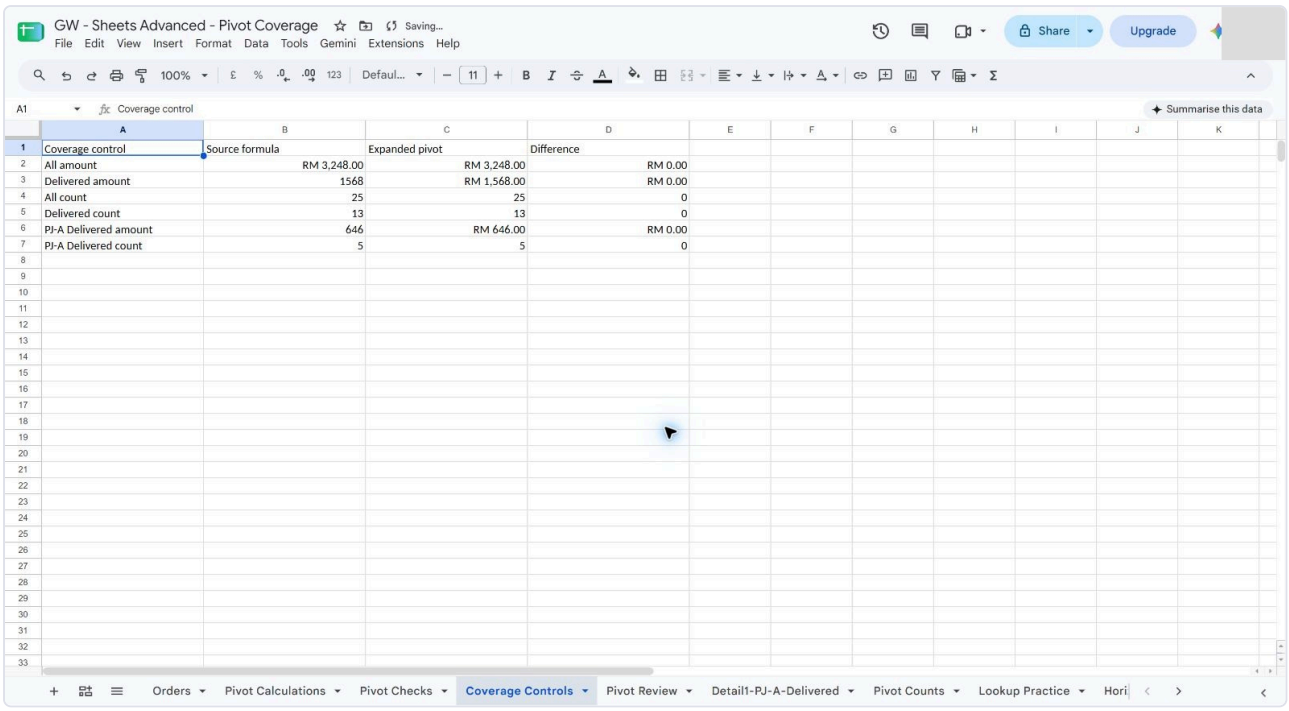
15. Repair only the scenario Orders summaries to **=SUM(J2:J26)** and **=SUM(K2:K26)** in J28/K28. They become **3,248 /2,028**. ActiveAmount excludes Cancelled; it is not the same population as Delivered.



16. In the scenario Pivot Checks A41:D47, compare expanded source formulas and both native pivots for all/Delivered amount and count plus the changed PJ-A contribution. All six differences are0.

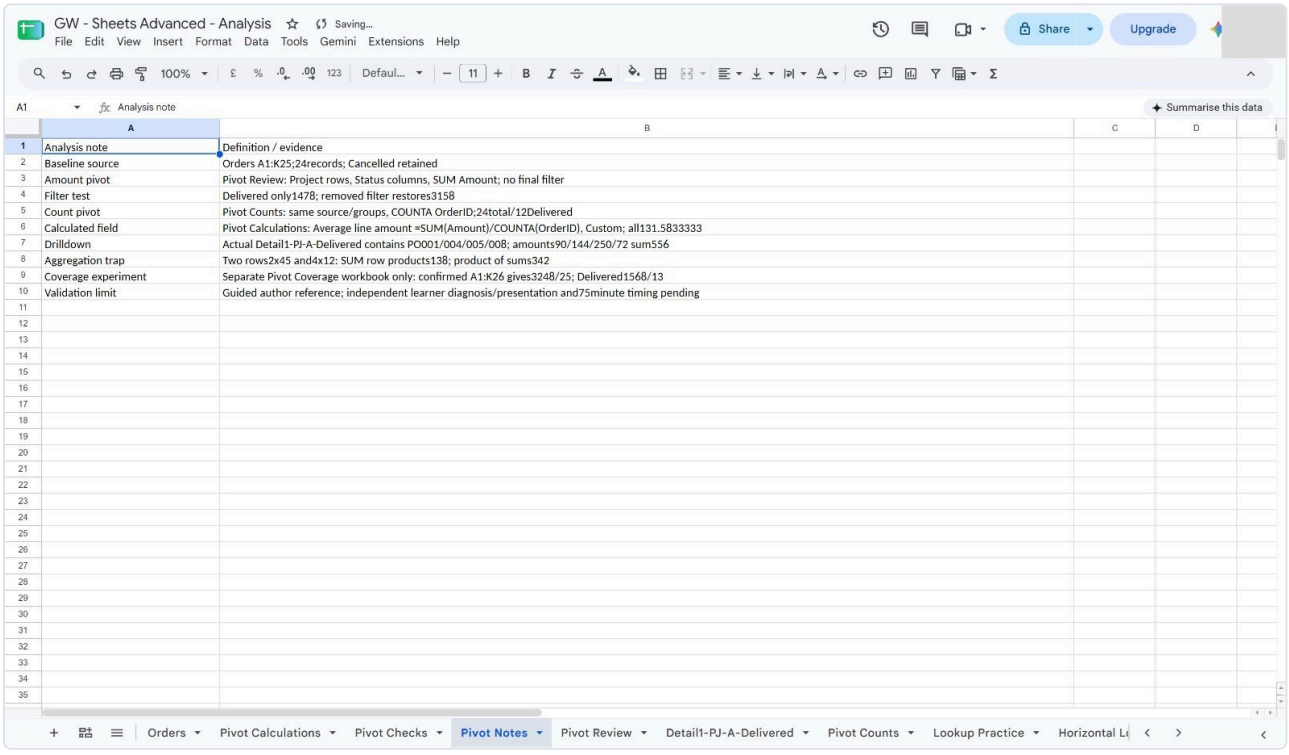


17. Create a clearly separate **Coverage Controls** tab containing the same six expanded controls at A1:D7. Widen A:D to200 pixels. Record scenario scope below it: earlier copied checks/calculated pivots still describe the24-row baseline and must not be presented as fully refreshed25-row analysis. The expanded controls are the authoritative checks for this experiment. Retain PO025 only on this named scenario copy.



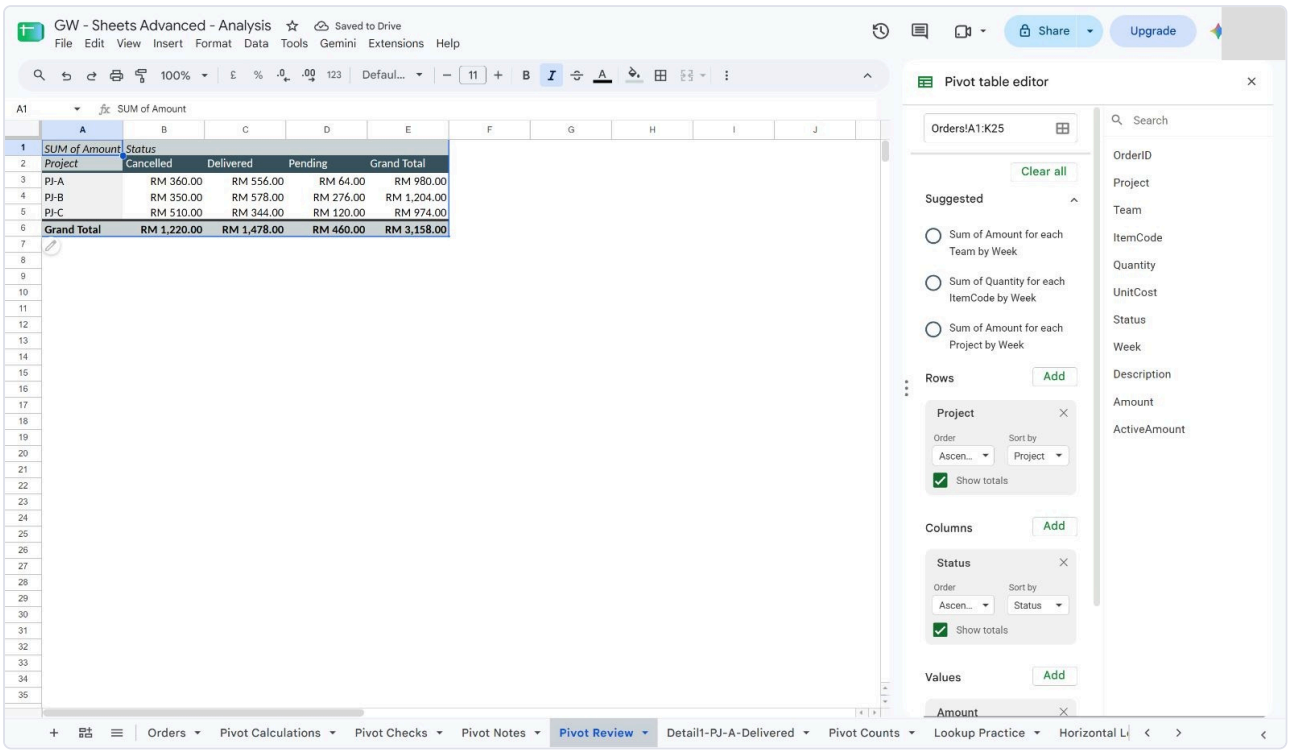
	A	B	C	D
1	Coverage control	Source formula	Expanded pivot	Difference
2	All amount	RM 3,248.00	RM 3,248.00	RM 0.00
3	Delivered amount	1568	RM 1,568.00	RM 0.00
4	All count	25	25	0
5	Delivered count	13	13	0
6	PJ-A Delivered amount	646	RM 646.00	RM 0.00
7	PJ-A Delivered count	5	5	0

18. In the maintained workbook create **Pivot Notes**. Record source range, groups, filter experiment/restoration, SUM versus COUNTA, Custom average definition, drilldown evidence, aggregation trap and the separate coverage result. State the validation boundary.

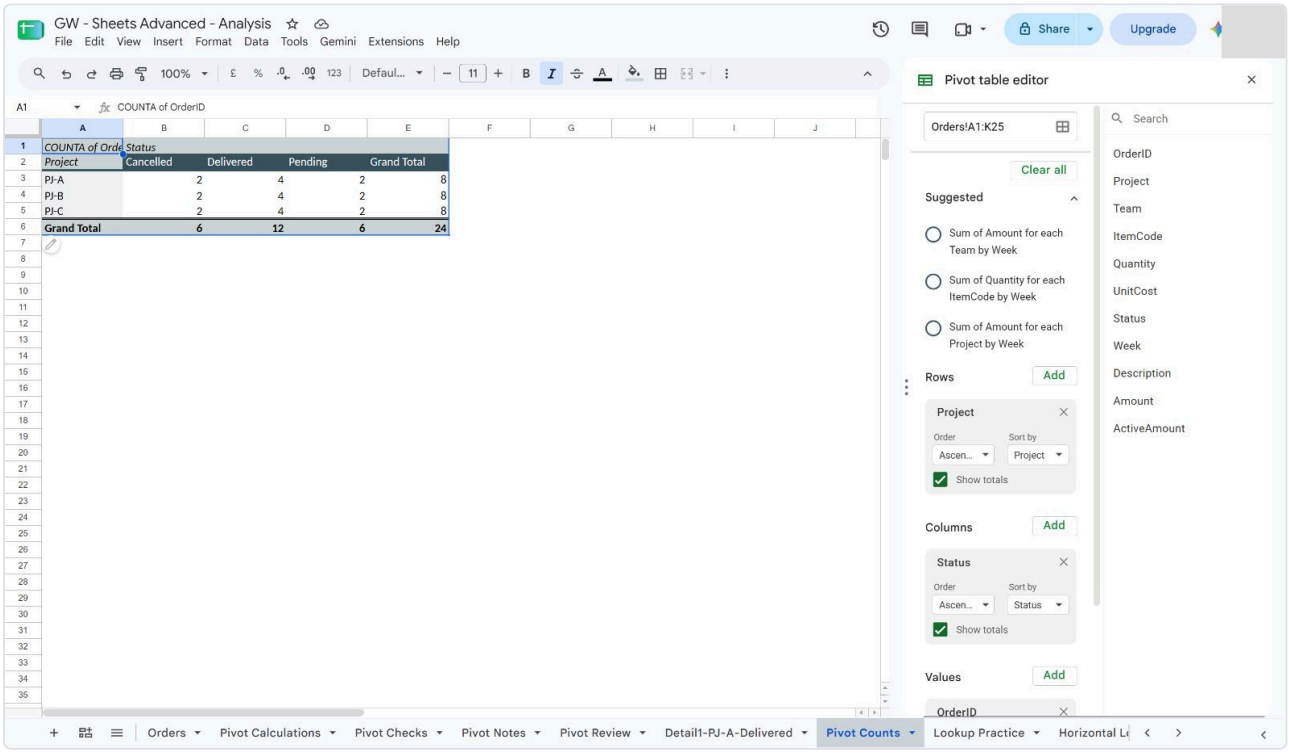


	A	B
1	Analysis note	Definition / evidence
2	Baseline source	Orders A1:K25;24records; Cancelled retained
3	Amount pivot	Pivot Review: Project rows, Status columns, SUM Amount; no final filter
4	Filter test	Delivered only1478; removed filter restores3158
5	Count pivot	Pivot Counts: same source/groups, COUNTA OrderID;24total/12Delivered
6	Calculated field	Pivot Calculations: Average line amount =SUM(Amount)/COUNTA(OrderID), Custom; all131.5833333
7	Drilldown	Actual Detail1-PJ-A-Delivered contains PO001/004/005/008; amounts90/144/250/72 sum556
8	Aggregation trap	Two rows2x45 and4x12: SUM row products138; product of sums342
9	Coverage experiment	Separate Pivot Coverage workbook only: confirmed A1:K26 gives3248/25; Delivered1568/13
10	Validation limit	Guided author reference; independent learner diagnosis/presentation and75minute timing pending

19. Inspect the maintained Pivot Review: still **3,158** overall and **1,478** Delivered. Original Orders, Rates, Source Snapshot and Analysis values/formats are unchanged.



20. Inspect maintained Pivot Counts: still **24 /12**. Leave the baseline pivots and named coverage experiment available for trainer review. The independent challenge below remains a fresh learner task.



> **Validation boundary:** guided7.1–7.3 and author casework7.4 were executed through learner UI and bounded native pivot/cell evidence. Custom average is supported in this target UI. PO025 is retained only in the separate coverage workbook; baseline source and pivots are preserved. Independent Delivered-output construction, diagnosis/presentation, human reviewer and the planned75-minute learner allocation remain unverified.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Build a fresh pivot independently answering Delivered amount by Project, with a count of contributing order lines.
2. On the challenge copy include PO025 and show the source row, count, total and changed project contribution. Leave the baseline pivot unchanged.
3. Explain one aggregation trap and use contributing rows to prove the correct result.

Evidence to Hand Over

- Amount and count pivots reconciled to formulas and contributing source rows.
- Safe calculated-field comparison or documented unavailable route.
- Separate new-record pivot with repaired coverage and demonstrated aggregation semantics.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A pivot total is wrong even though the row appears	Inspect aggregation, filters, source range and calculated-field meaning separately.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Native pivots summarise a defined source. Their filters and formulas need the same checks as ordinary calculations.

Check yourself: Why must the pivot's source range include every record and calculated column?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Pivot Review
```

```
=SUM(Amount)/COUNTA(OrderID)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
PO025, challenge copy only: PJ-A | Planning | IT01 | Quantity 2 | UnitCost 45 | Delivered |
week 4 | Additional practice line. Amount is quantity times cost. Extend both formulas and
pivot source range; do not alter the baseline workbook.
```

Captured Author-Run Reference

Use native **Insert > Pivot table**, not a typed summary. Baseline source **Orders A1:K25** contains24records. Keep source edits for the named whole-workbook **Pivot Coverage** copy only. These cards support guided casework, not an independent learner submission.

Calculated field — Average line amount, Summarise by Custom

```
=SUM(Amount)/COUNTA(OrderID)
```

Pivot Checks A1:E5 — expectations first, actual columns blank

```
Project Expected average        Source average  Pivot average   Difference
PJ-A    122.5
PJ-B    150.5
PJ-C    121.75
All     131.583333333333
```

Pivot Checks C2:E5 — source/native comparison formulas

```
=SUMIF(Orders!$B$2:$B$25,A2,Orders!$J$2:$J$25)/COUNTIF(Orders!$B$2:$B$25,A2)    ='Pivot
Calculations'!C2        =D2-C2
=SUMIF(Orders!$B$2:$B$25,A3,Orders!$J$2:$J$25)/COUNTIF(Orders!$B$2:$B$25,A3)    ='Pivot
Calculations'!C3        =D3-C3
=SUMIF(Orders!$B$2:$B$25,A4,Orders!$J$2:$J$25)/COUNTIF(Orders!$B$2:$B$25,A4)    ='Pivot
Calculations'!C4        =D4-C4
=SUM(Orders!$J$2:$J$25)/COUNTA(Orders!$A$2:$A$25)       ='Pivot Calculations'!C5        =D5-C5
```

Pivot Checks A8:D18 — category counts and total

```
Project / status        Source count    Pivot count     Difference
PJ-A / Cancelled        =COUNTIFS(Orders!$B$2:$B$25,"PJ-A",Orders!$G$2:$G$25,"Cancelled")
='Pivot Counts'!B3      =C9-B9
PJ-A / Delivered        =COUNTIFS(Orders!$B$2:$B$25,"PJ-A",Orders!$G$2:$G$25,"Delivered")
='Pivot Counts'!C3      =C10-B10
PJ-A / Pending  =COUNTIFS(Orders!$B$2:$B$25,"PJ-A",Orders!$G$2:$G$25,"Pending") ='Pivot
Counts'!D3      =C11-B11
PJ-B / Cancelled        =COUNTIFS(Orders!$B$2:$B$25,"PJ-B",Orders!$G$2:$G$25,"Cancelled")
='Pivot Counts'!B4      =C12-B12
PJ-B / Delivered        =COUNTIFS(Orders!$B$2:$B$25,"PJ-B",Orders!$G$2:$G$25,"Delivered")
='Pivot Counts'!C4      =C13-B13
PJ-B / Pending  =COUNTIFS(Orders!$B$2:$B$25,"PJ-B",Orders!$G$2:$G$25,"Pending") ='Pivot
Counts'!D4      =C14-B14
PJ-C / Cancelled        =COUNTIFS(Orders!$B$2:$B$25,"PJ-C",Orders!$G$2:$G$25,"Cancelled")
='Pivot Counts'!B5      =C15-B15
PJ-C / Delivered        =COUNTIFS(Orders!$B$2:$B$25,"PJ-C",Orders!$G$2:$G$25,"Delivered")
='Pivot Counts'!C5      =C16-B16
PJ-C / Pending  =COUNTIFS(Orders!$B$2:$B$25,"PJ-C",Orders!$G$2:$G$25,"Pending") ='Pivot
Counts'!D5      =C17-B17
All records     =COUNTA(Orders!A2:A25)  ='Pivot Counts'!E6      =C18-B18
```

Pivot Checks A21:F25 — manual amounts before comparison

```
OrderID Quantity        UnitCost        Manual amount   Formula amount  Difference
PO001   2       45      90
PO004   8       18      144
PO005   10      25      250
PO008   6       12      72
```

Pivot Checks E22:F25 — row comparison formulas

```
=B22*C22        =E22-D22
=B23*C23        =E23-D23
=B24*C24        =E24-D24
=B25*C25        =E25-D25
```

Pivot Checks A27:D28 — actual drilldown reconciliation

```
Control Manual sum      Native detail sum       Difference
PJ-A Delivered  =SUM(D22:D25)   =SUM('Detail1-PJ-A-Delivered'!J2:J5)    =C28-B28
```

Pivot Checks A31:D33 — two-row aggregation practice

```
Practice row    Quantity        UnitCost        Row amount
First line      2       45      =B32*C32
Second line     4       12      =B33*C33
```

Pivot Checks A35:B38 — aggregation comparison

```
Metric  Result
Expected row total      138
Sum of row products     =SUM(D32:D33)
Product of sums =SUM(B32:B33)*SUM(C32:C33)
```

Scenario Orders A26:K26 — after inserting row26

```
PO025   PJ-A    Planning        IT01    2       45      Delivered       4       Additional
practice line   =E26*F26        =IF(G26="Cancelled",0,J26)
```

Scenario Orders J28:K28 — repaired source totals

```
=SUM(J2:J26)    =SUM(K2:K26)
```

Coverage Controls A1:D7 — expanded source/pivot reconciliation

```
Coverage control        Source formula  Expanded pivot  Difference
All amount      =SUM(Orders!J2:J26)     ='Pivot Review'!E6      =C2-B2
Delivered amount        =SUMIF(Orders!G2:G26,"Delivered",Orders!J2:J26) ='Pivot Review'!C6
=C3-B3
All count       =COUNTA(Orders!A2:A26)  ='Pivot Counts'!E6      =C4-B4
Delivered count =COUNTIF(Orders!G2:G26,"Delivered")     ='Pivot Counts'!C6      =C5-B5
PJ-A Delivered amount   =SUMIFS(Orders!J2:J26,Orders!B2:B26,"PJ-A",Orders!G2:G26,"Delivered")
='Pivot Review'!C3      =C6-B6
PJ-A Delivered count    =COUNTIFS(Orders!B2:B26,"PJ-A",Orders!G2:G26,"Delivered")
='Pivot Counts'!C3      =C7-B7
```

Coverage Controls A10:B14 — scenario scope note

```
Scenario scope  Record
Source  Orders A1:K26 includes PO025, amount90
Pivots  Pivot Review SUM Amount; Pivot Counts COUNTA OrderID; both A1:K26
Earlier sheets  Copied baseline checks and calculated pivot still target24rows; historical
comparison only
Authoritative expanded checks   Coverage Controls A1:D7; six differences0; independent learner
challenge pending
```

Pivot Notes A1:B10 — baseline handover

```
Analysis note   Definition / evidence
Baseline source Orders A1:K25;24records; Cancelled retained
Amount pivot    Pivot Review: Project rows, Status columns, SUM Amount; no final filter
Filter test     Delivered only1478; removed filter restores3158
Count pivot     Pivot Counts: same source/groups, COUNTA OrderID;24total/12Delivered
Calculated field        Pivot Calculations: Average line amount =SUM(Amount)/COUNTA(OrderID),
Custom; all131.5833333
Drilldown       Actual Detail1-PJ-A-Delivered contains PO001/004/005/008; amounts90/144/250/72
sum556
Aggregation trap        Two rows2x45 and4x12: SUM row products138; product of sums342
Coverage experiment     Separate Pivot Coverage workbook only: confirmed A1:K26 gives3248/25;
Delivered1568/13
Validation limit        Guided author reference; independent learner diagnosis/presentation
and75minute timing pending
```

Native pivot definitions

Tab	Source	Rows	Columns	Values	Final filter
Pivot Review	Orders A1:K25	Project	Status	SUM of Amount	None; Delivered-only1478 test then removed
Pivot Counts	Orders A1:K25	Project	Status	COUNTA of OrderID	None
Pivot Calculations	Orders A1:K25	Project	None	SUM of Amount plus Average line amount, Custom	None
Scenario Pivot Review	Orders A1:K26	Project	Status	SUM of Amount	None
Scenario Pivot Counts	Orders A1:K26	Project	Status	COUNTA of OrderID	None

Use **Select data range > Range > OK** to confirm range edits. Inspect recalculation, not just input text. Expected baseline3158/1478/24/12; scenario3248/1568/25/13. PJ-A Delivered changes556/4 to646/5. Do not include summary rows27–28 in the scenario pivot source.

The two-row trap has row products90 and48, sum138; multiplying the summed quantities6 by summed costs57 returns342. These metrics have different meanings. The original ambiguous multiplication prompt has been replaced by the actually verified average field; retain SUM of row Amount for authoritative totals.

Earlier copied scenario checks/calculated pivots remain24-row references. Only the expanded source summaries and Coverage Controls A1:D7 establish25-row coverage. Independent Delivered pivot/output, learner diagnosis/presentation and timing remain pending.

08

CHAPTER 8

Query Structured Criteria

In this chapter you will use DSUM, DCOUNT and DAVERAGE with a visible criteria range and reconcile against a manual subset.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 60 minutes: 15 explanation/demonstration, 17 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A visible criteria area and reconciled database function results.

> **Status:** author reference with 36 staged application captures through 8.4. DSUM/DCOUNT/DAVERAGE, exact AND/OR ID sets, overlap, field types, weighted means and an actual header fault/repair are verified. A wrong mean prediction is explicitly investigated and corrected. Maintained source and earlier pivots unchanged. Independent question, reviewer and learner timing remain pending.

What You Will Learn

- Build a visible database criteria range.
- Use DSUM, DCOUNT and DAVERAGE.
- Reconcile criteria-based results.

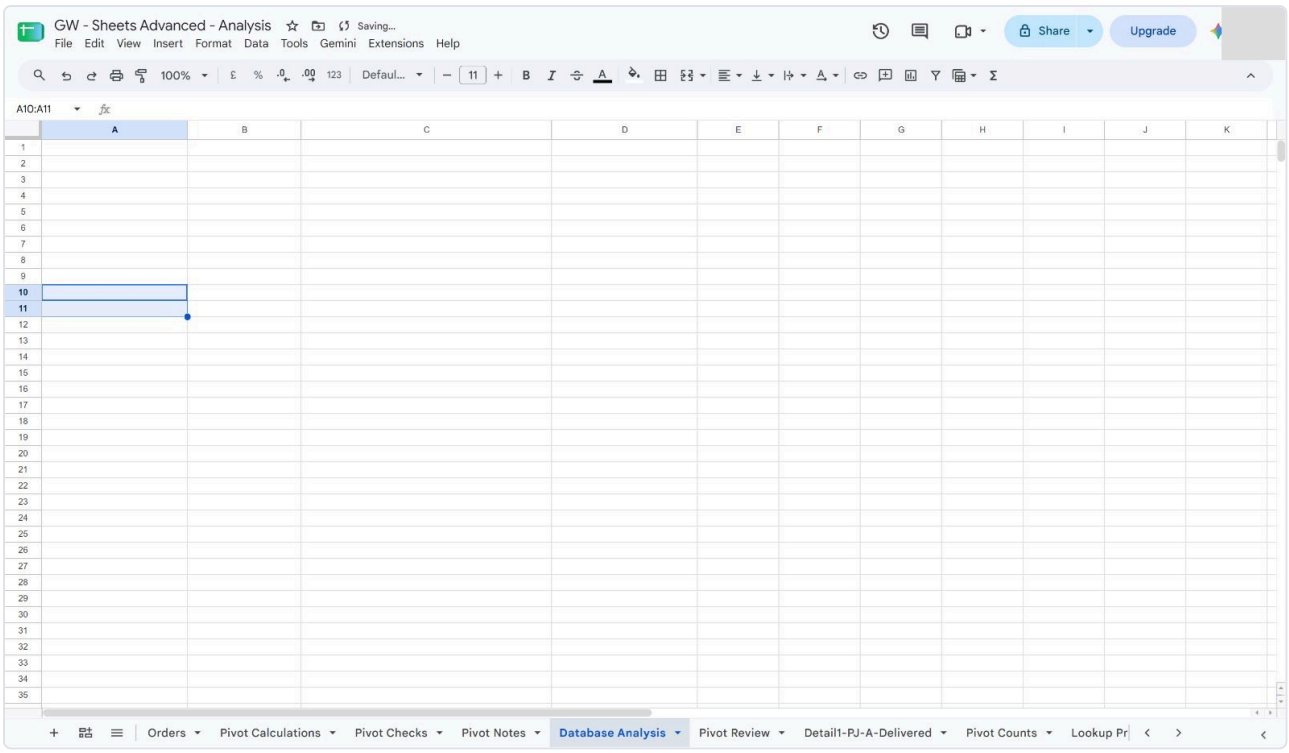
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

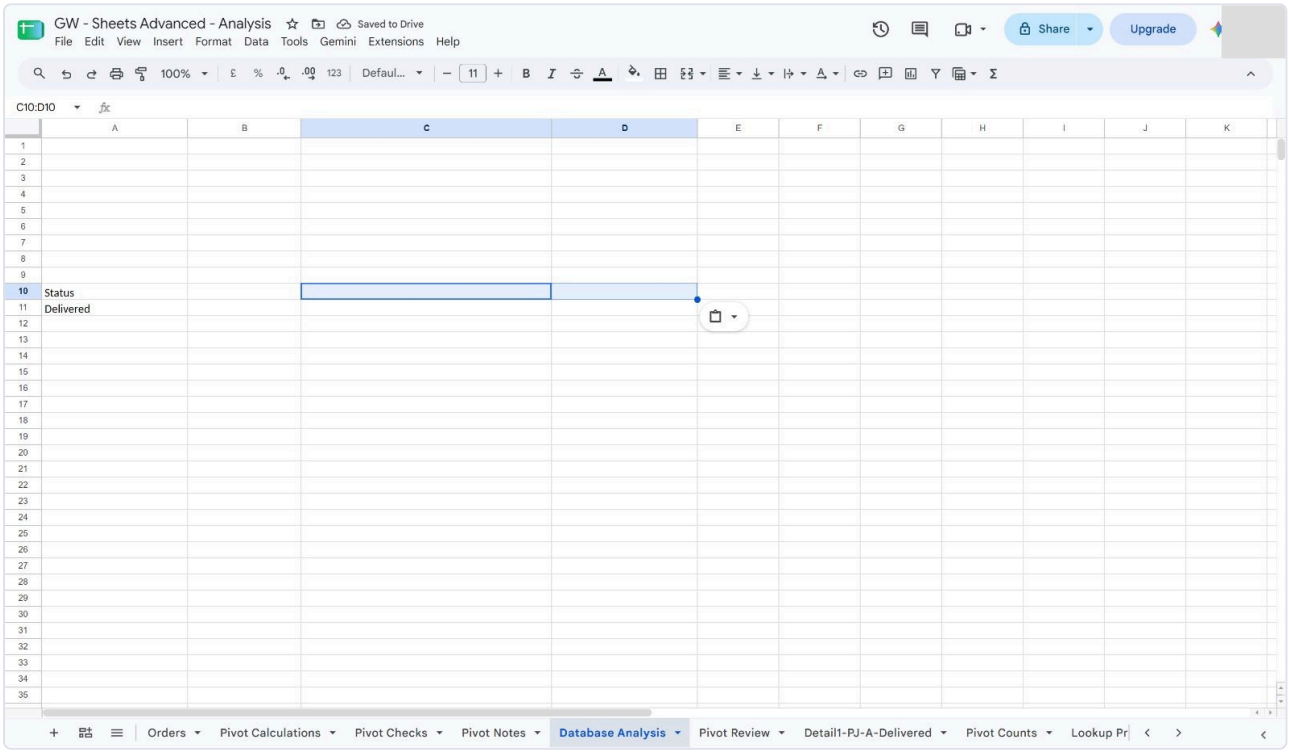
Use Orders A1:J25, including Amount. Keep criteria outside the source table.

8.1 Build the Criteria

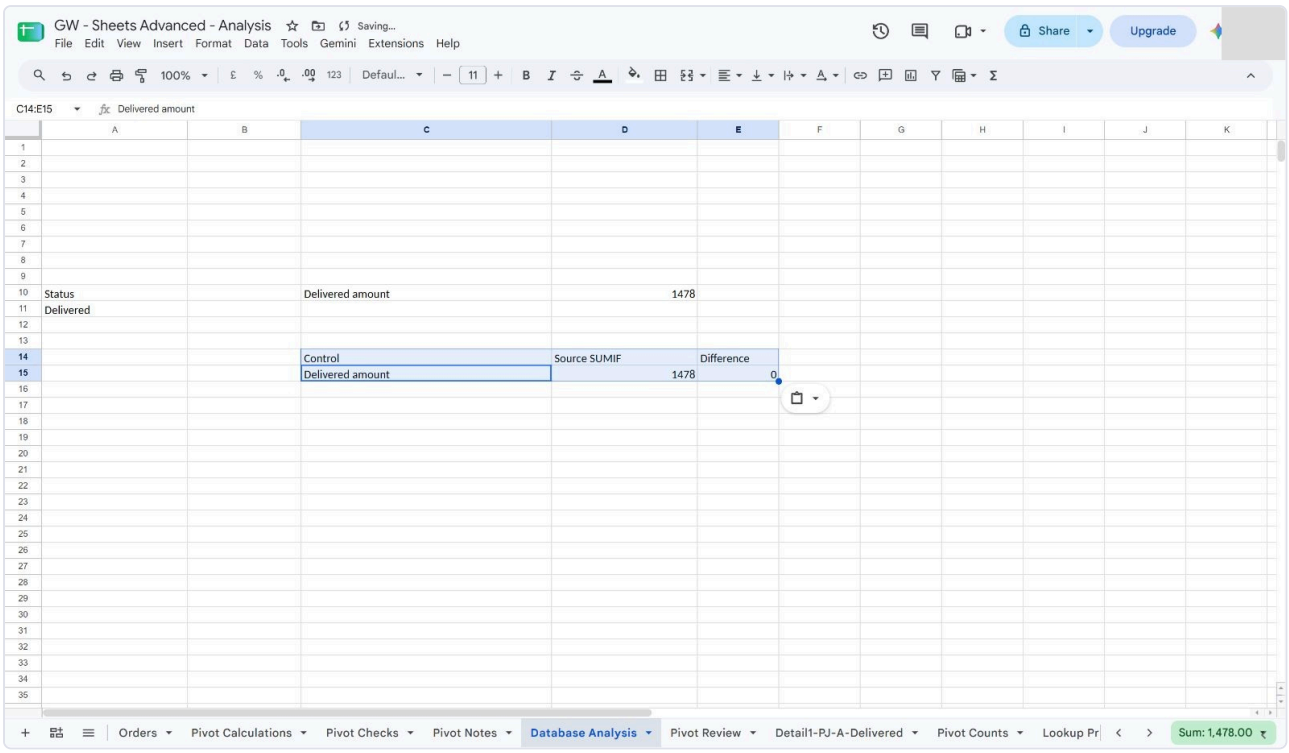
1. Create a separate tab named **Database Analysis**. The draft's Analysis A10:D12 locations already contain earlier count/population work, so use the same addresses on this new tab. Keep maintained Analysis unchanged. Widen A/B/C/D to180/140/310/180 pixels. Create the blank tab before entering criteria.



2. In **A10:A11** enter **Status / Delivered**. Match the Orders header exactly. This visible one-column criteria block is ready before entering the result formulas.

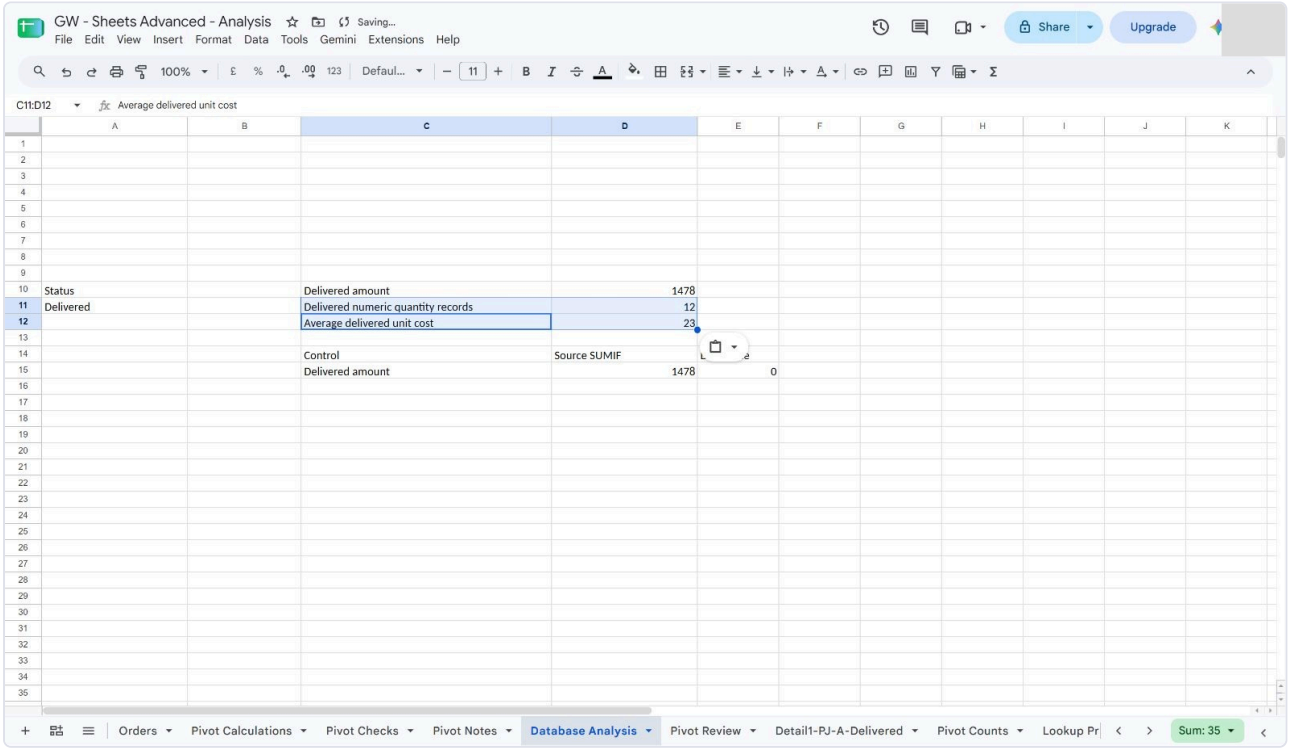


3. In C10 enter **Delivered amount** and D10 **=DSUM(Orders!A1:J25,"Amount",A10:A11)**. Add the C14:E15 comparison card. Both DSUM and source **SUMIF** return **1,478**; their difference is0. The source includes the header and24 data records, with criteria outside it.

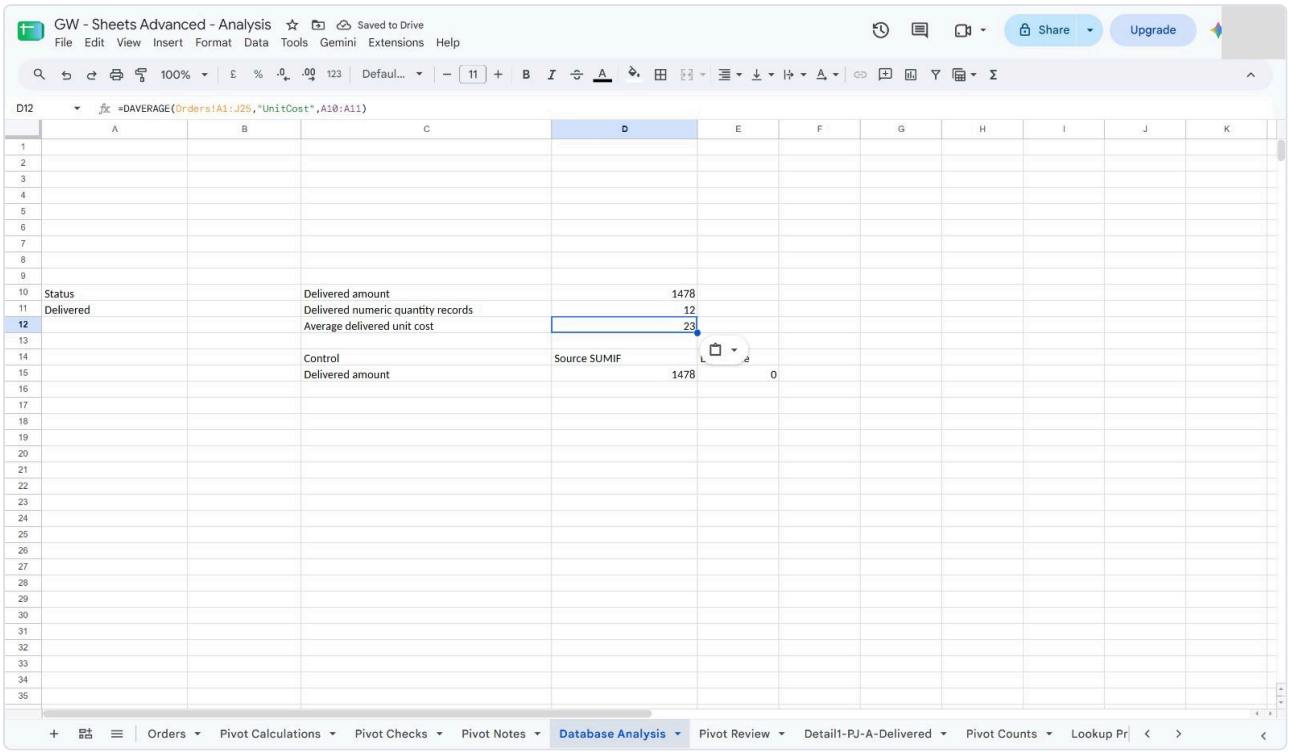


8.2 Count and Average

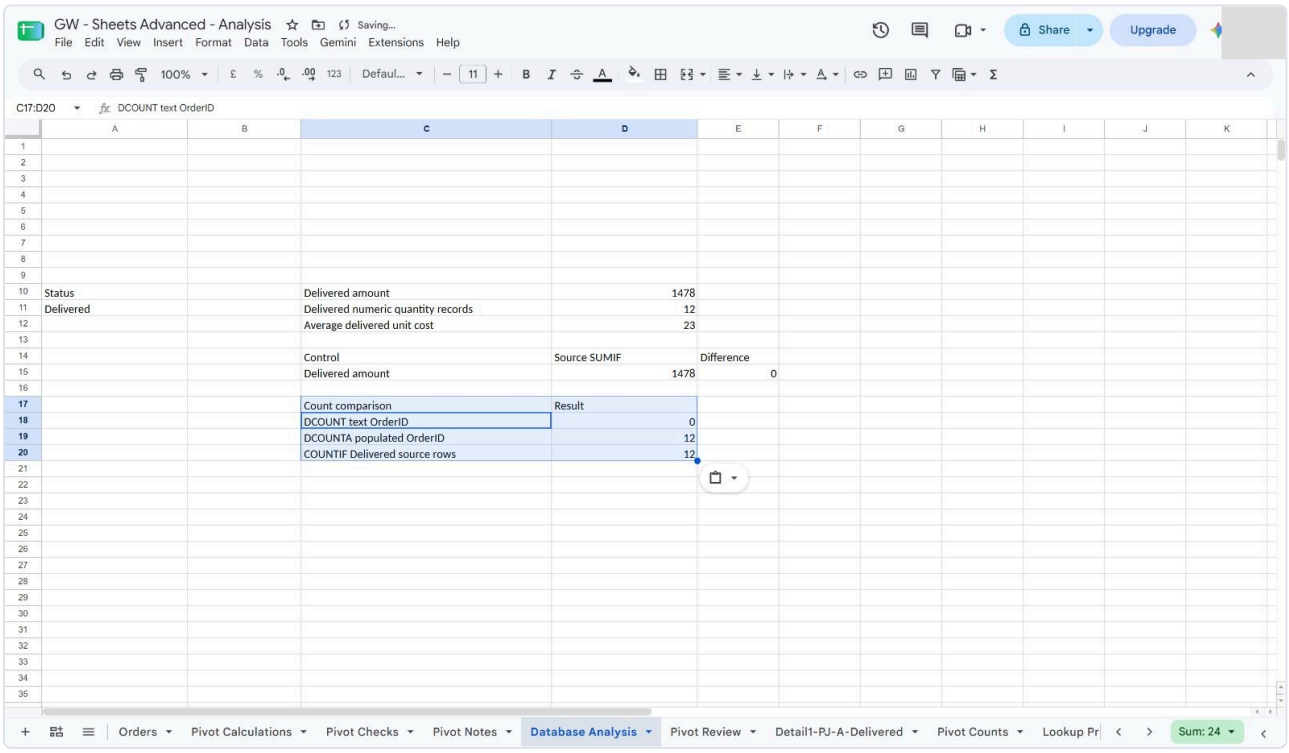
1. Enter both metric rows: C11 **Delivered numeric quantity records**, D11 **=DCOUNT(Orders!A1:J25,"Quantity",A10:A11)**; C12 **Average delivered unit cost**, D12 **=DAVERAGE(Orders!A1:J25,"UnitCost",A10:A11)**. Inspect D11: **12 numeric Quantity entries**, not a count of text identifiers.



2. Select D12 and inspect its DAVERAGE formula. It returns **23**, the unweighted mean of12 numeric UnitCost values. It is not an average line Amount or a quantity-weighted price.

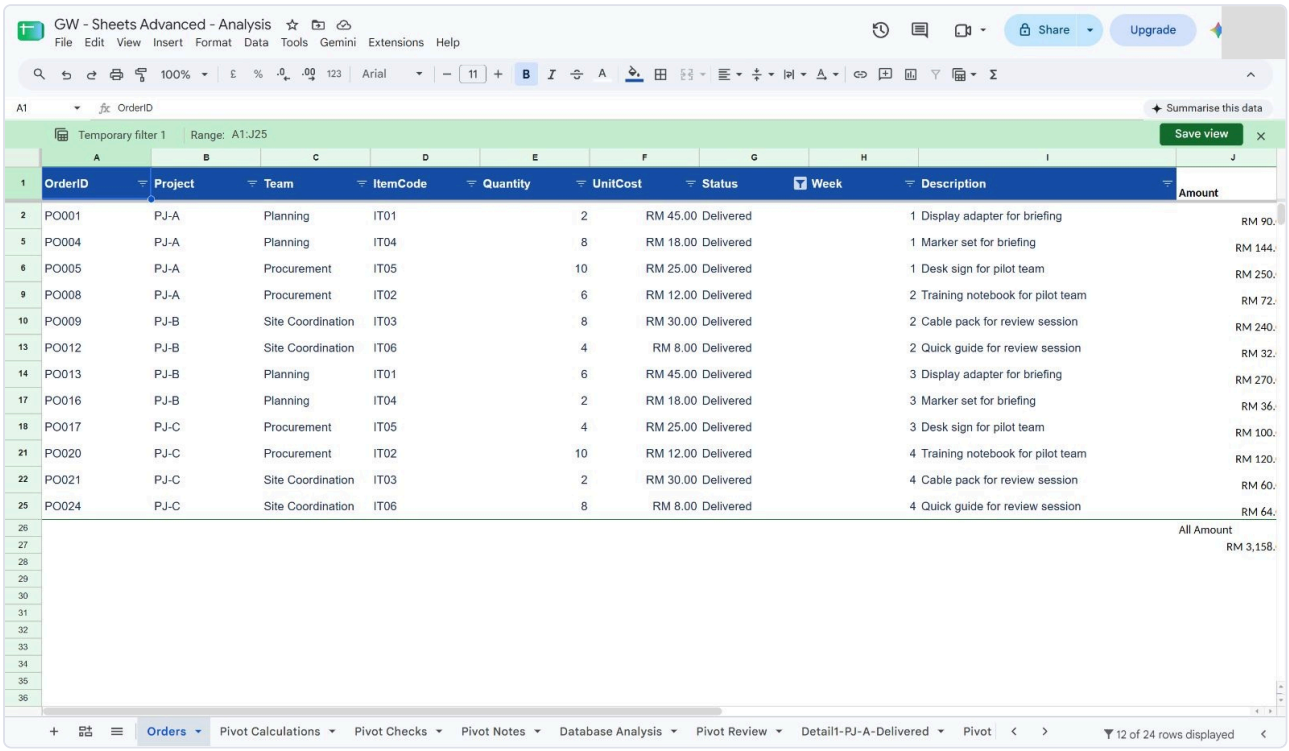


3. Add C17:D20. **DCOUNT of OrderID returns0**, **DCOUNTA of OrderID returns12**, and source COUNTIF Delivered returns12. This baseline has populated text IDs and numeric quantities; those different fields happen to identify the same12 records only when counted with the appropriate function.

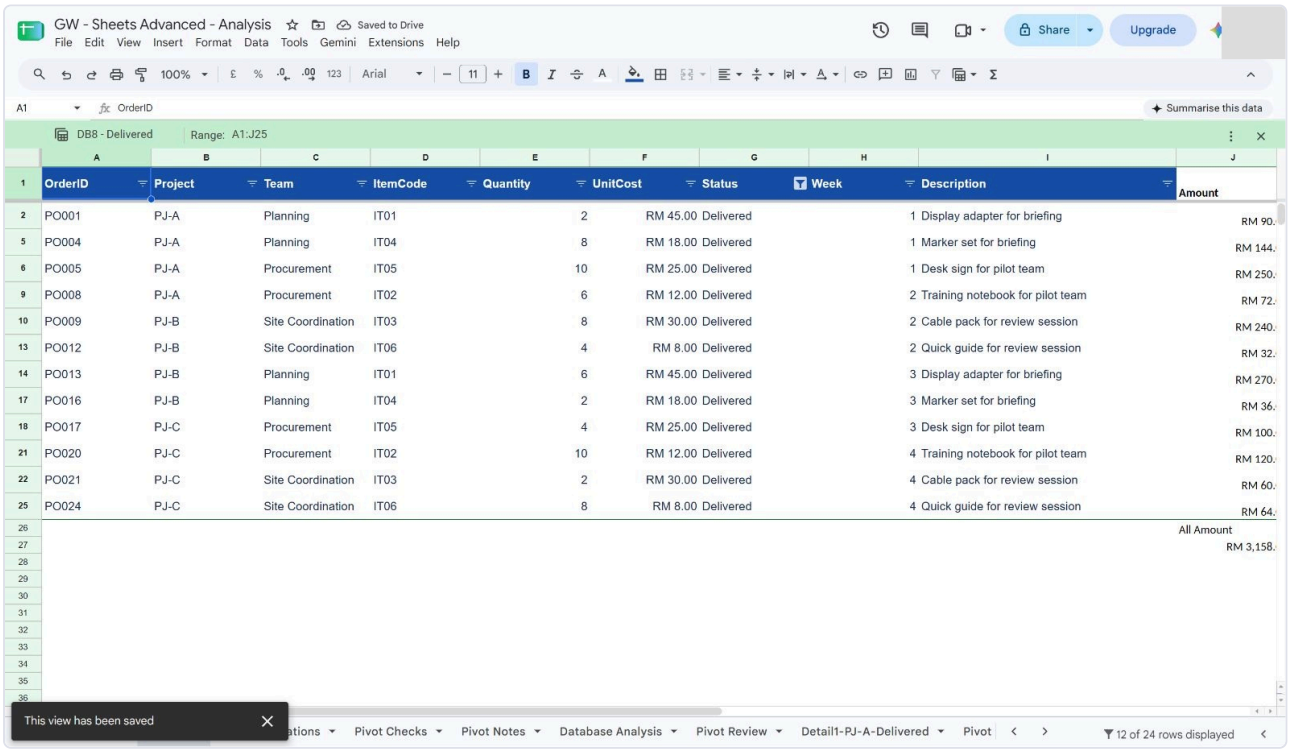


8.3 Check Criteria Semantics

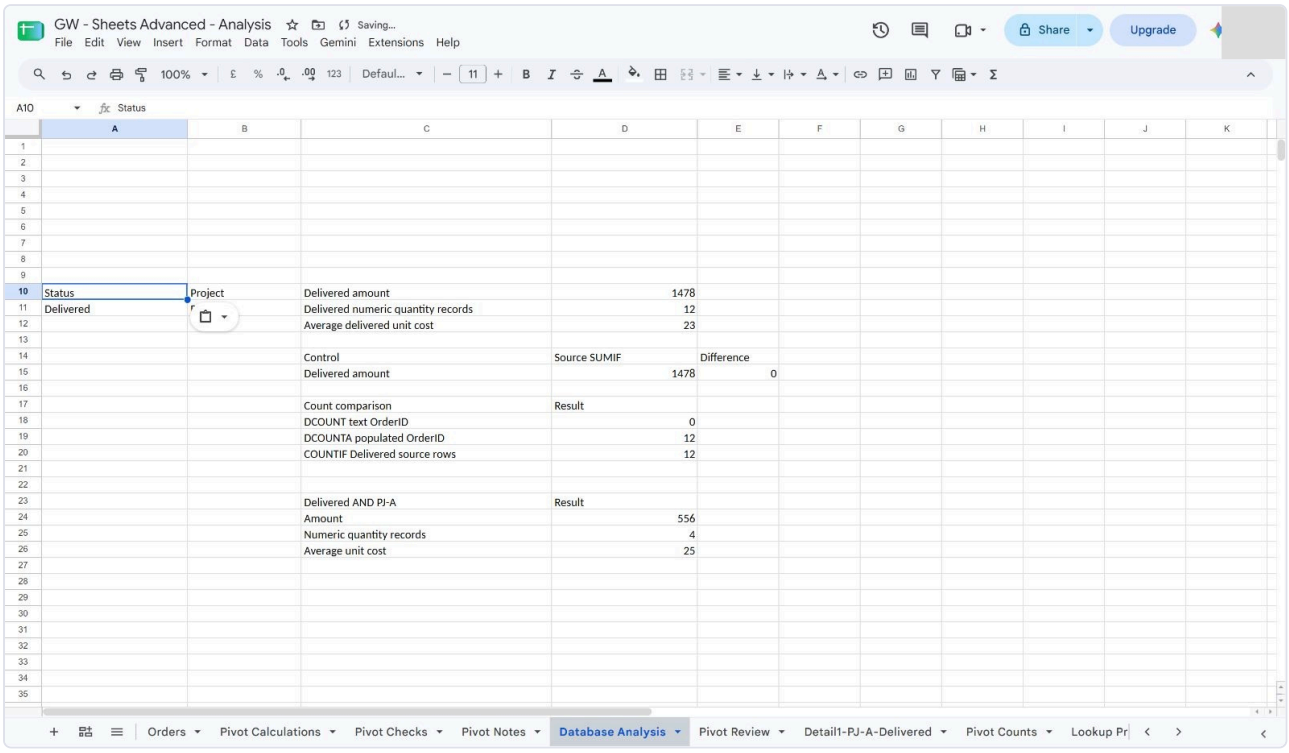
1. On Orders select **A1:J25**, then **Data > Create filter view**. In the Status dropdown choose Clear, select Delivered only and OK. Inspect the12 visible source records; the ordinary SUM below the table still totals all records. Do not delete the other statuses.



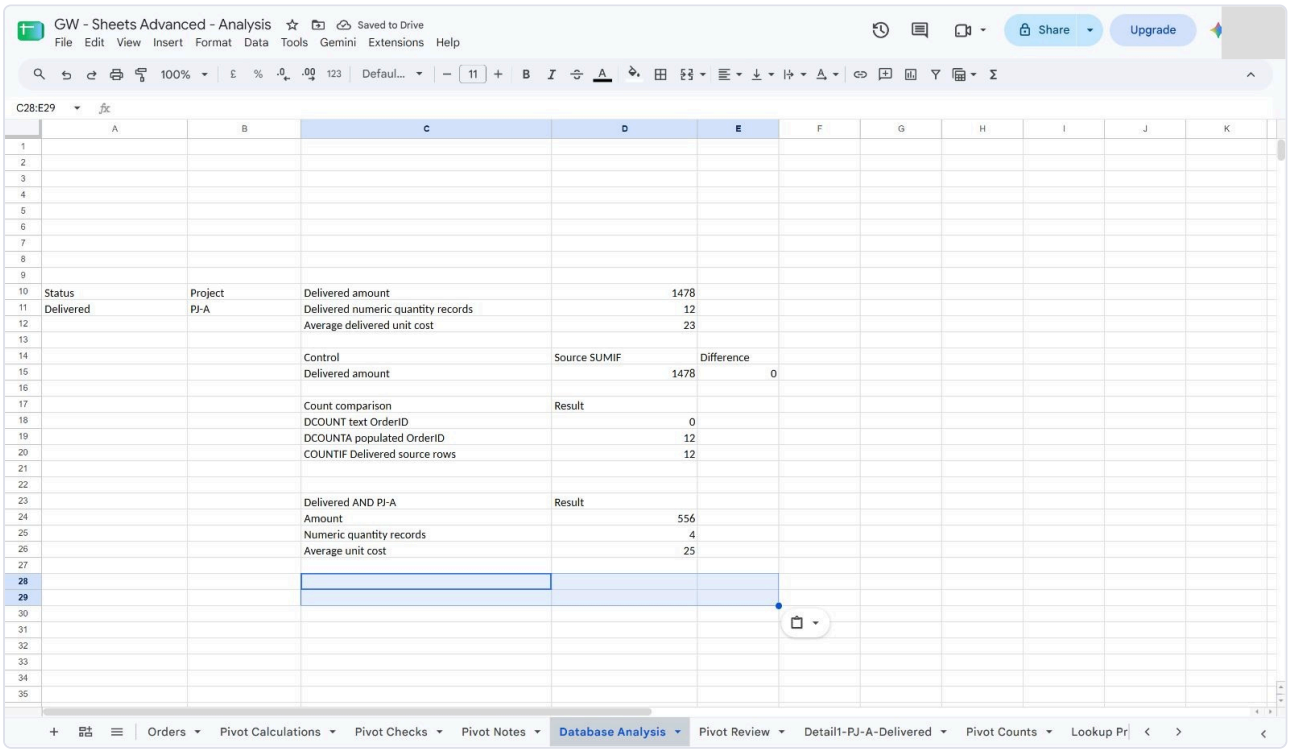
2. Choose **Save view**, name it **DB8 - Delivered**, and Save. Inspect the named view and then Exit the view toolbar so later work starts from the unfiltered source.



3. Return to Database Analysis. Add **Project / PJ-A** in **B10:B11**, on the same header/data rows as Status/Delivered. Keep the earlier one-column formulas unchanged. Add the new C23:D26 block using **A10:B11**: DSUM Amount **556**, DCOUNT Quantity **4**, DAVERAGE UnitCost **25**. Two conditions on the same criteria row mean both must hold.



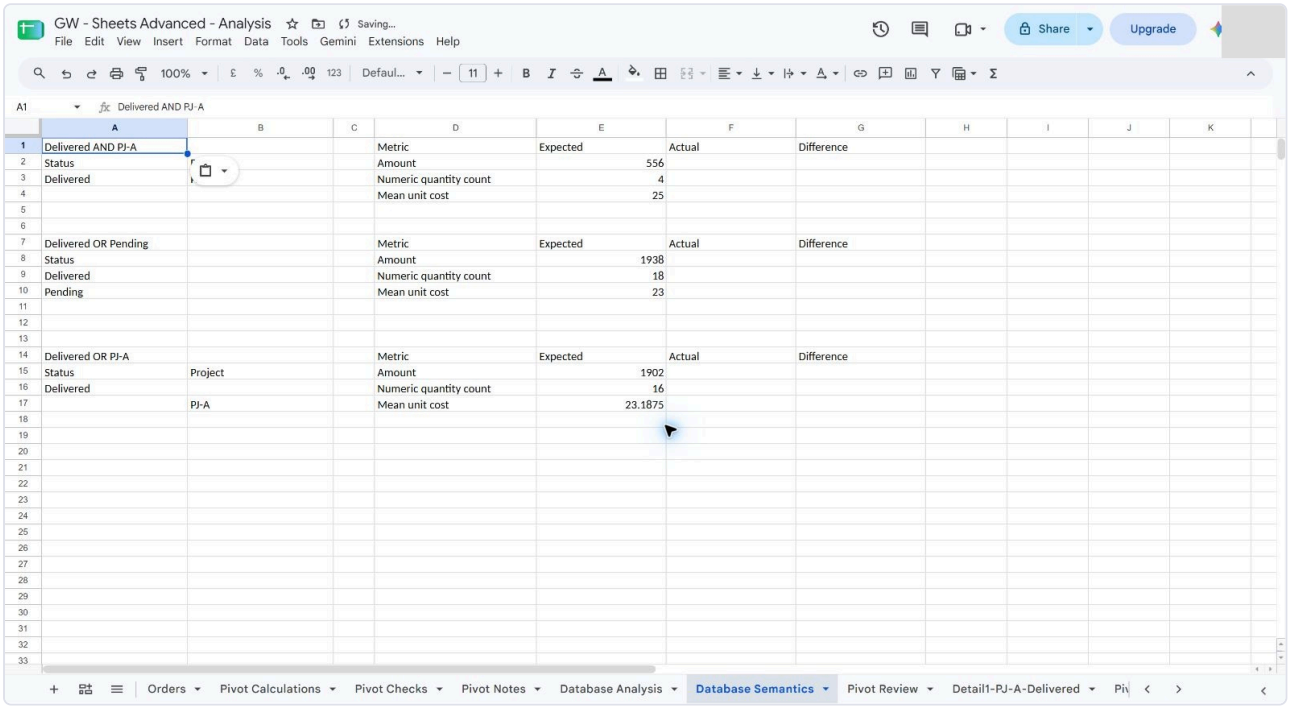
4. Add C28:E29. Source **SUMIFS** for Delivered and PJ-A returns556 and the difference is0. Label this expanded criterion explicitly; the original A10:A11 formulas still answer Delivered across all projects.



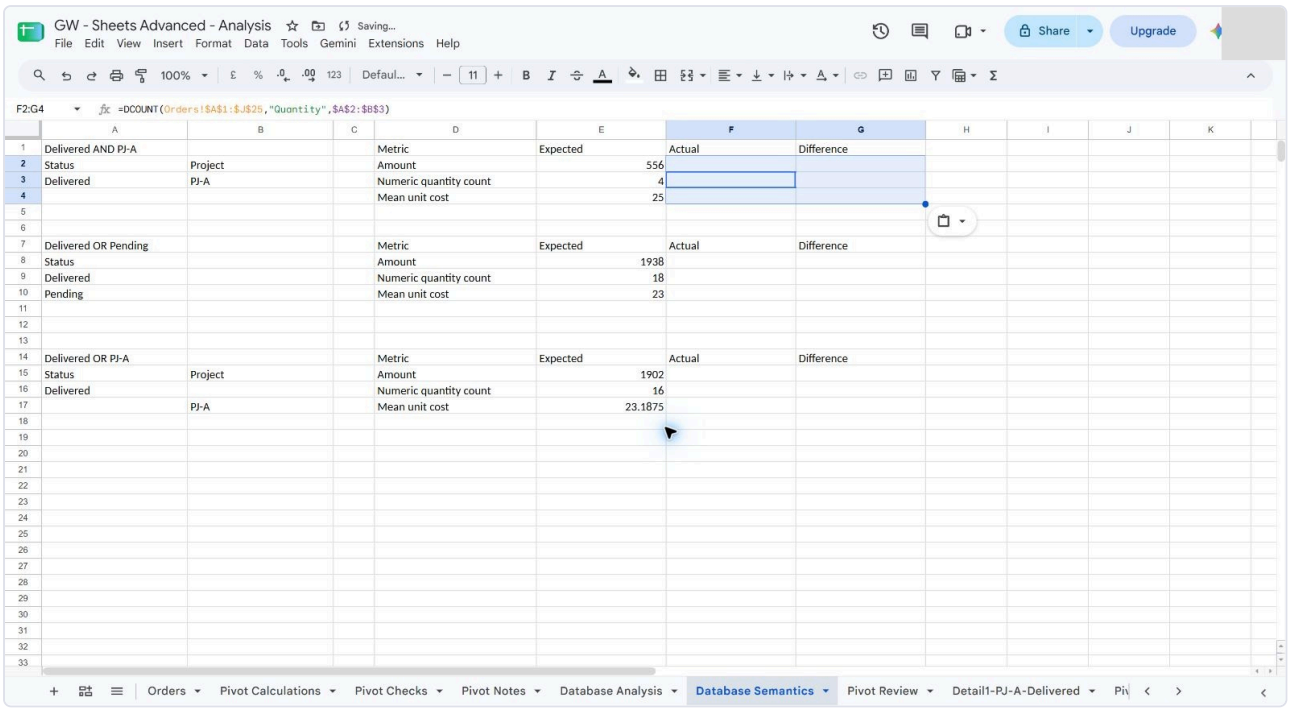
8.4 Prove AND, OR and Overlapping Criteria

1. Create **Database Semantics**, with separate labelled blocks: A2:B3 for **Delivered AND PJ-A**; A8:A10 for **Delivered OR Pending**; A15:B17 for **Delivered OR PJ-A**. The OR cases use separate data rows. In the overlapping block, put Delivered alone on row16 and PJ-A alone in B17; leave the other cell on each row blank. Those blanks mean the corresponding field is unrestricted on that OR row, not a requirement for a blank source value.

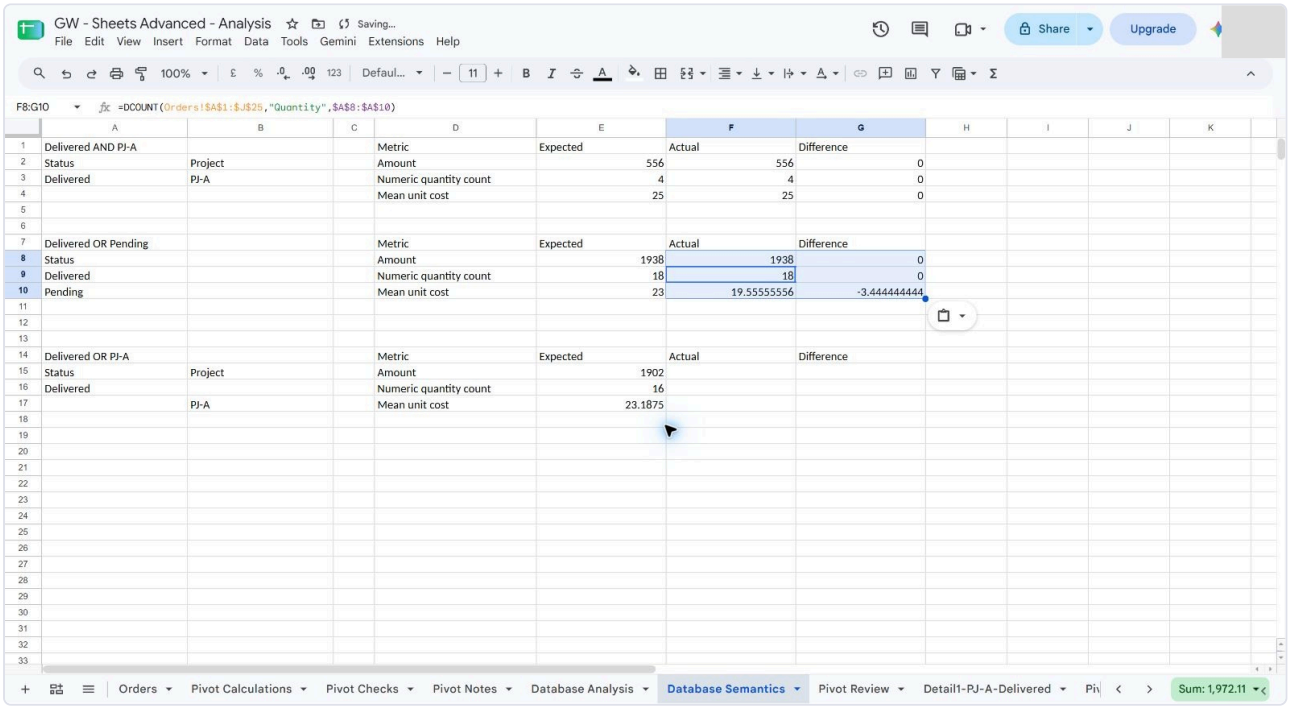
2. Record predictions before actual formulas. The reference's initial prediction card below contains a mistaken disjoint-OR mean of23. Treat it as a prediction to investigate, not an answer key. Final source-derived expectations are **556 /4 /25**, **1938 /18 /352÷18**, and **1902 /16 /23.1875**. The later audit-repair stages show why that one prediction changed.



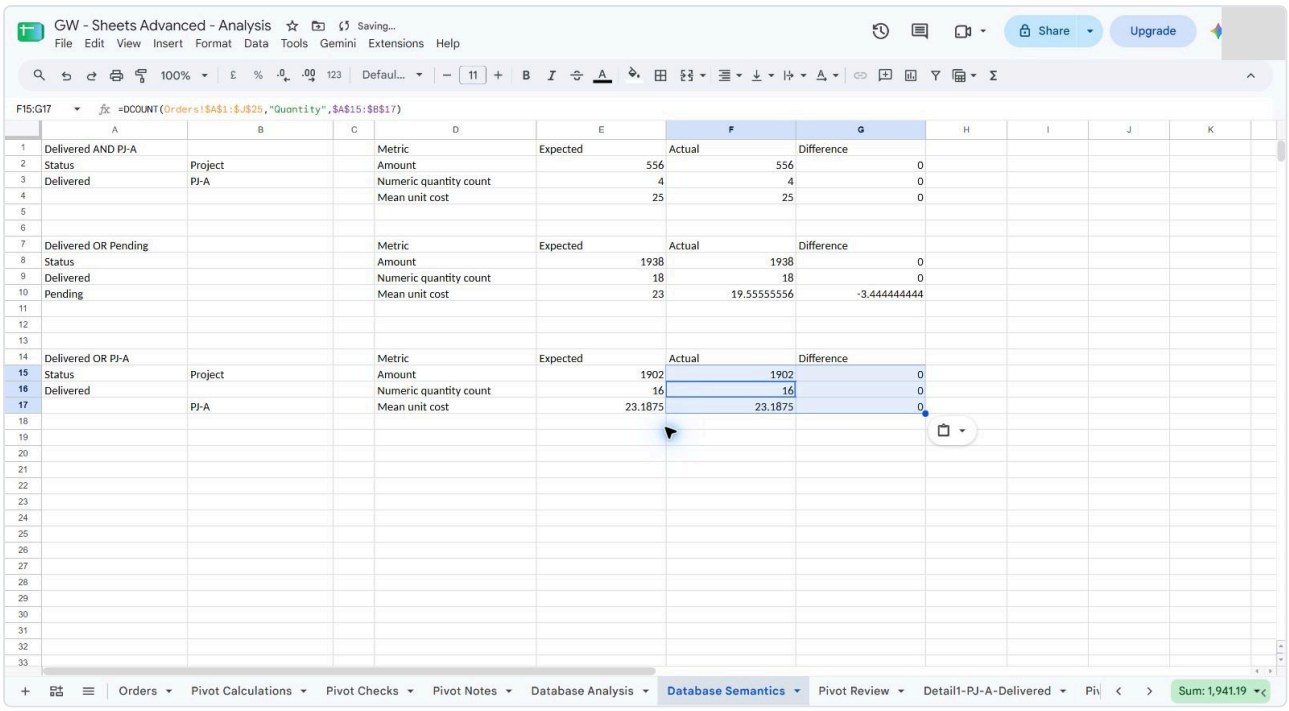
3. For the AND block, calculate DSUM Amount, DCOUNT Quantity and DAVERAGE UnitCost against **A2:B3**. The three results are **556 /4 /25**, with zero differences.



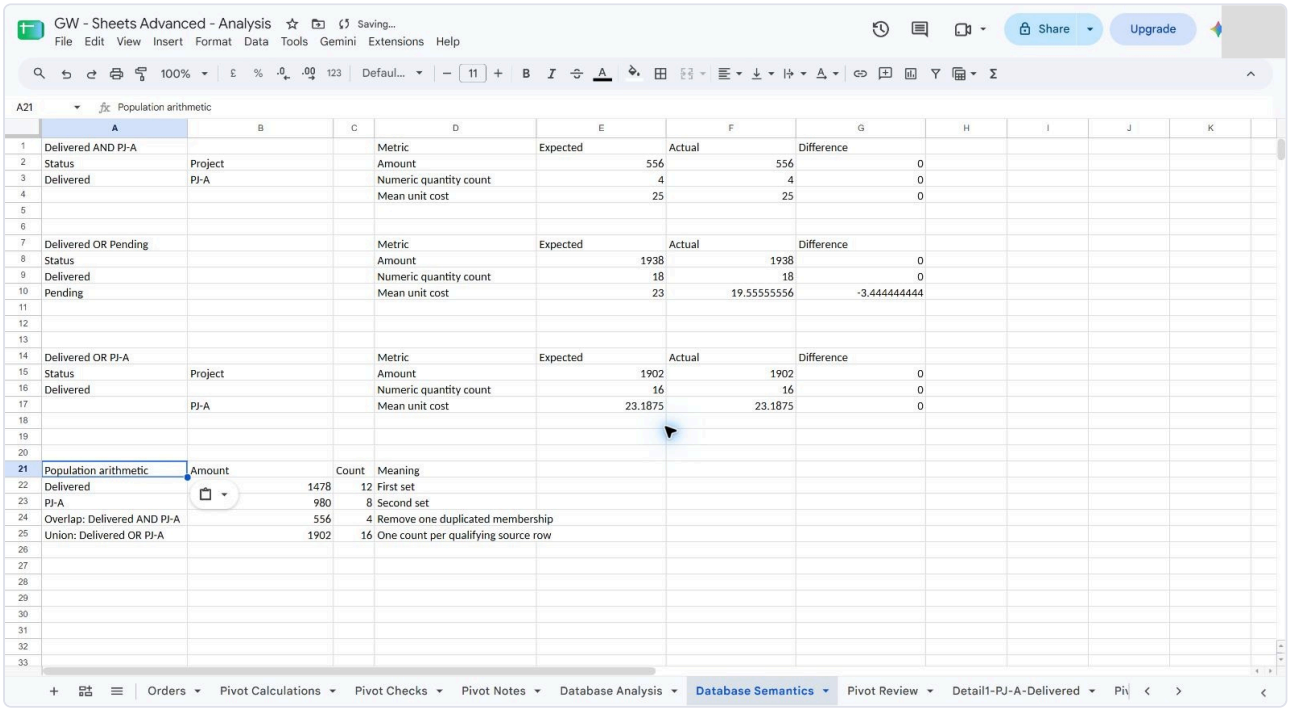
4. For disjoint OR, use **A8:A10**. Amount is **1,938**, numeric count **18**, and the observed mean is **19.55555556**. In the captured first pass, its mean difference is nonzero because the prediction23 was wrong. Keep that mismatch visible until source costs explain it; do not force the actual formula to the prediction.



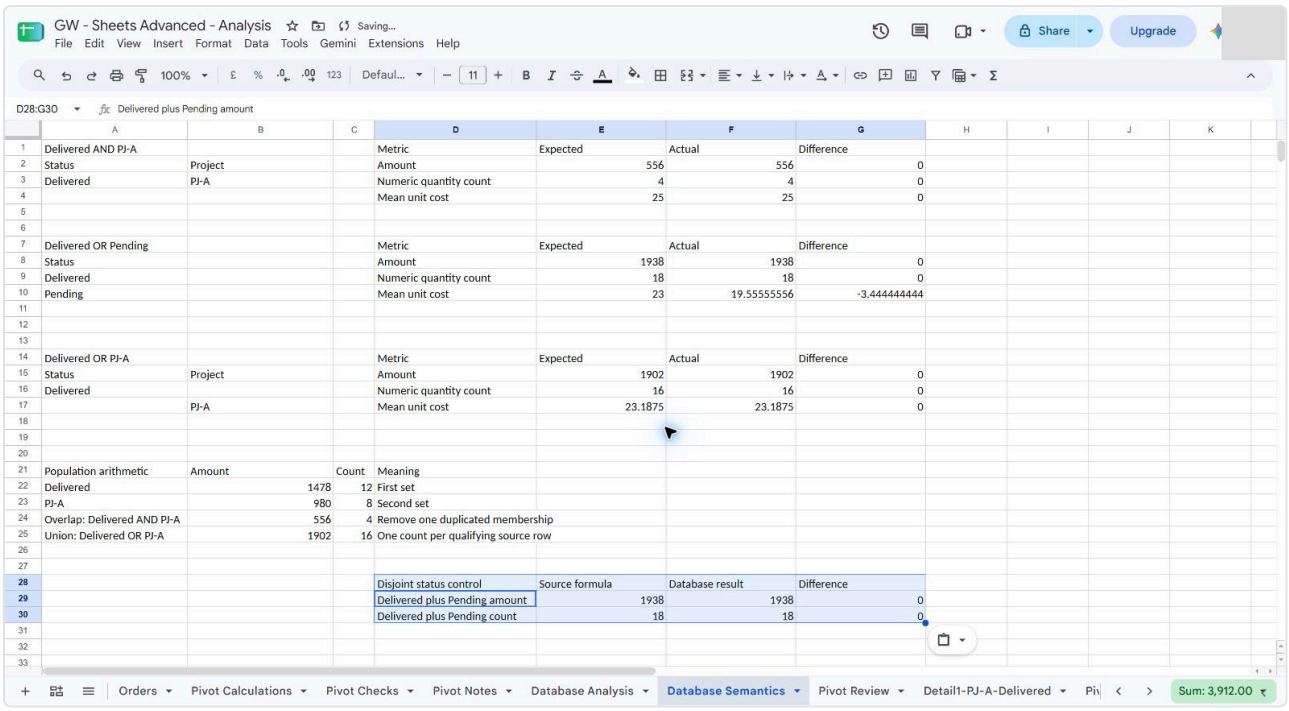
5. For overlapping OR, use **A15:B17**. Results are **1,902 /16 /23.1875**. A record satisfying both Delivered and PJ-A qualifies once.



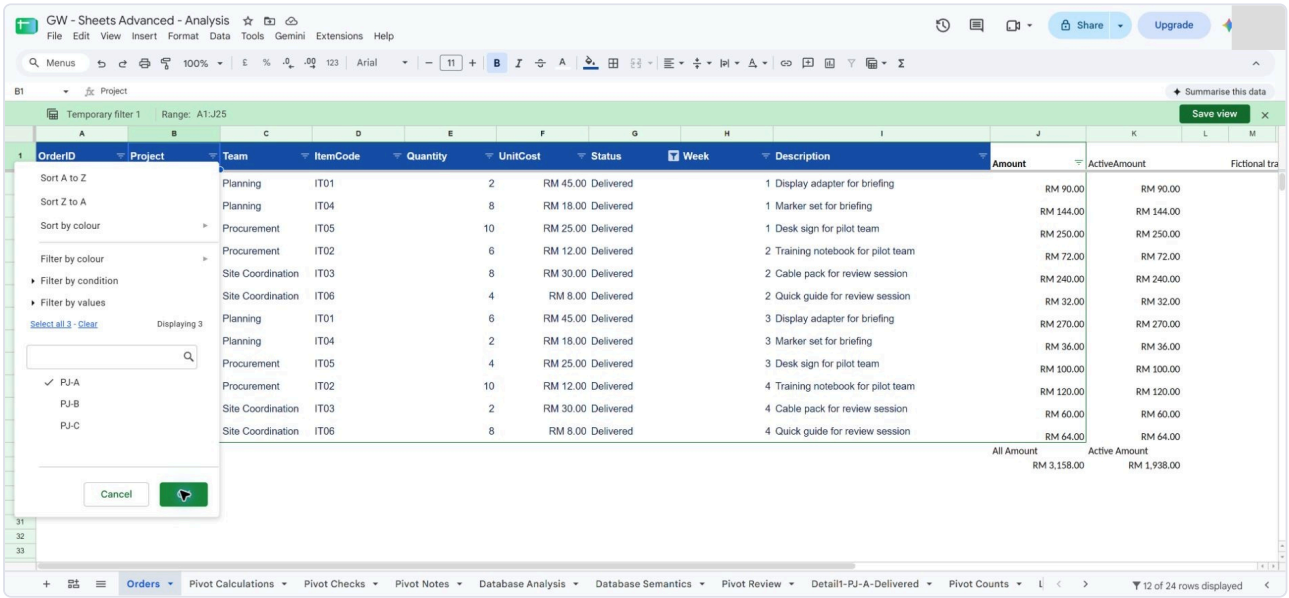
6. Add the inclusion–exclusion card at A21:D25. Delivered1,478 plus PJ-A980 minus their shared556 gives **1,902**. Counts12+8−4 give **16**. Adding the two sets without removing the shared membership would overstate the population.



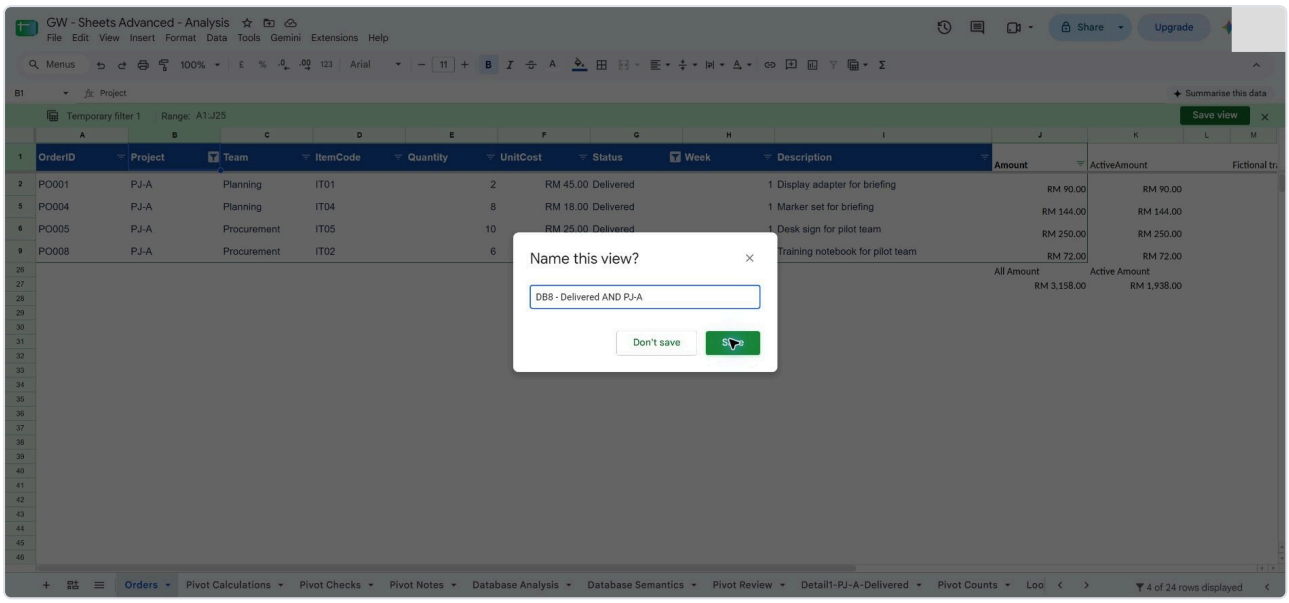
7. Reconcile the disjoint statuses in D28:G30: Delivered1,478 + Pending460 = **1,938**, and12+6 = **18**. These status sets do not overlap. This checks amount and count; it does not yet explain the mean prediction.



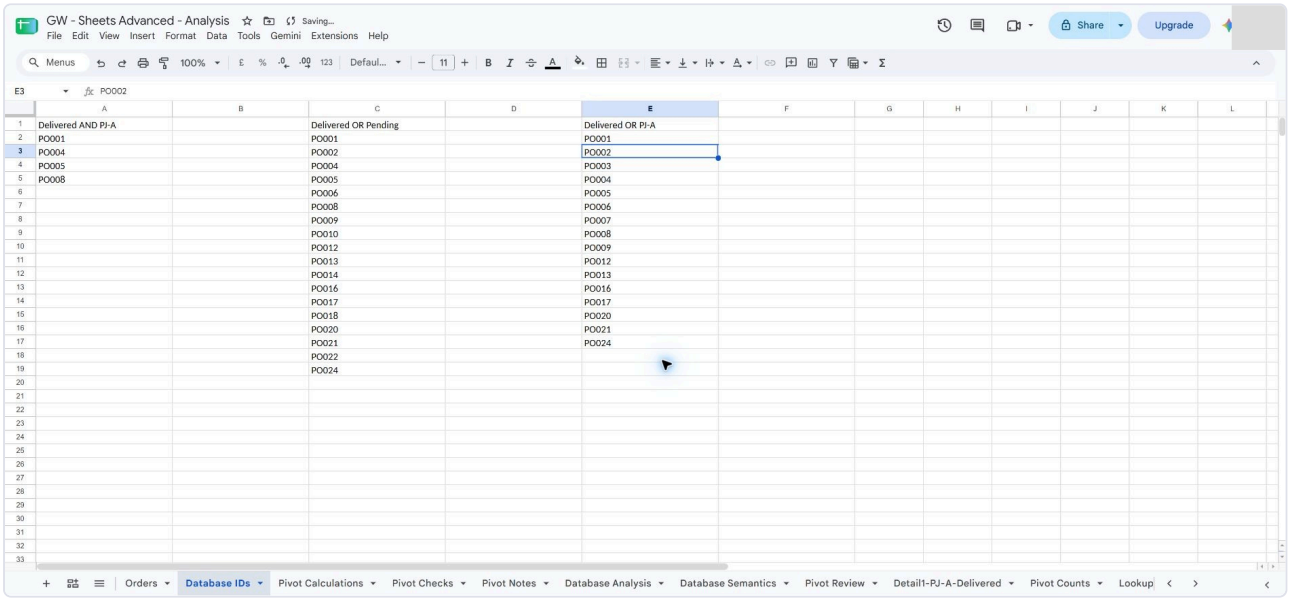
8. Create a second actual Orders **A1:J25** filter view. Select Delivered in Status and PJ-A in Project. Inspect the four visible IDs **PO001 /PO004 /PO005 /PO008** and their amounts90/144/250/72, which sum to556.



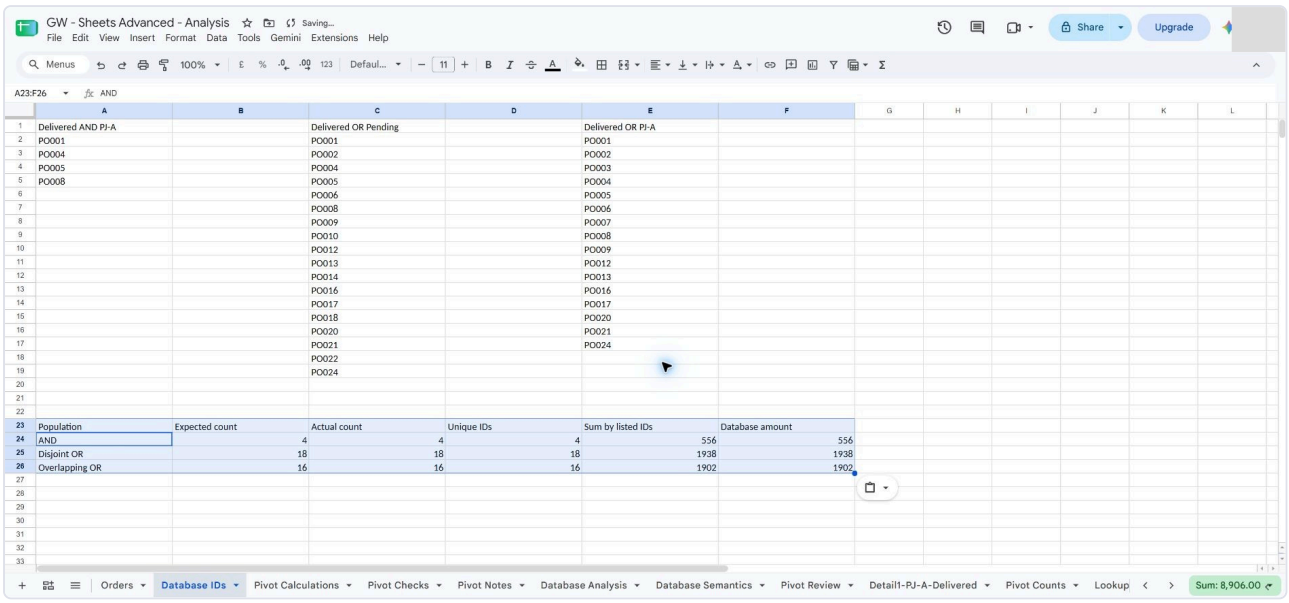
9. Save it as **DB8 - Delivered AND PJ-A** and Exit. Retain both named views for trainer inspection while leaving the source unfiltered.

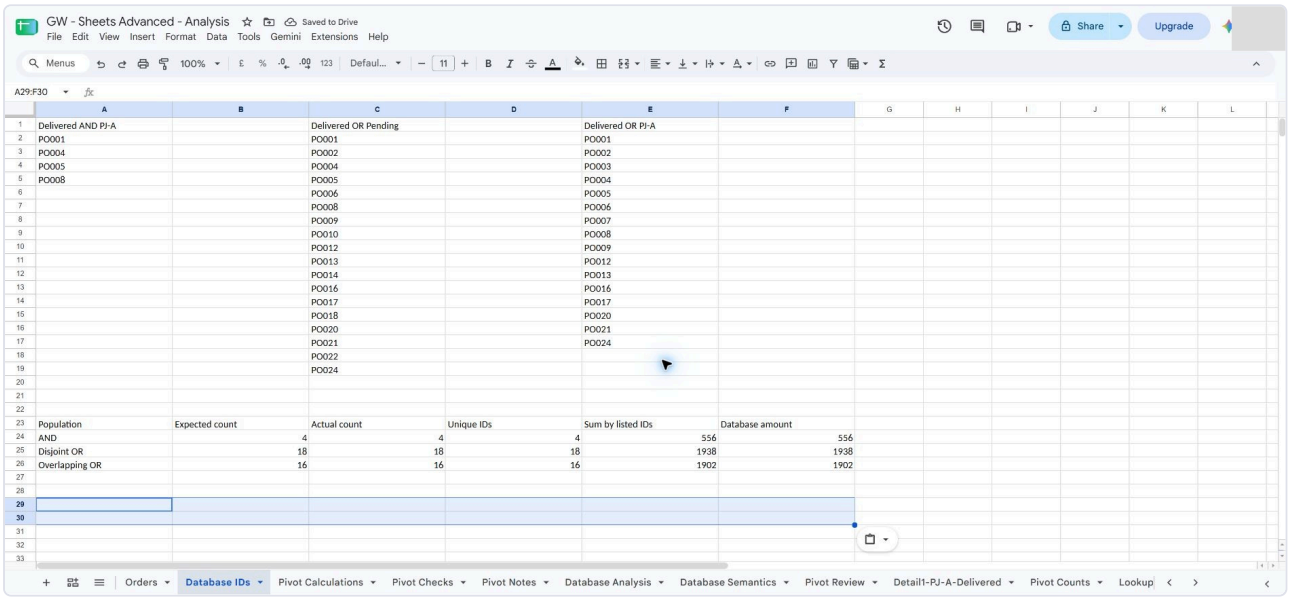


10. Create **Database IDs**. Use the FILTER cards in A2/C2/E2 to list the AND, disjoint-OR and overlapping-OR IDs from the source in their original order. Each OR condition is converted to a boolean membership test rather than concatenating two lists with duplicates.

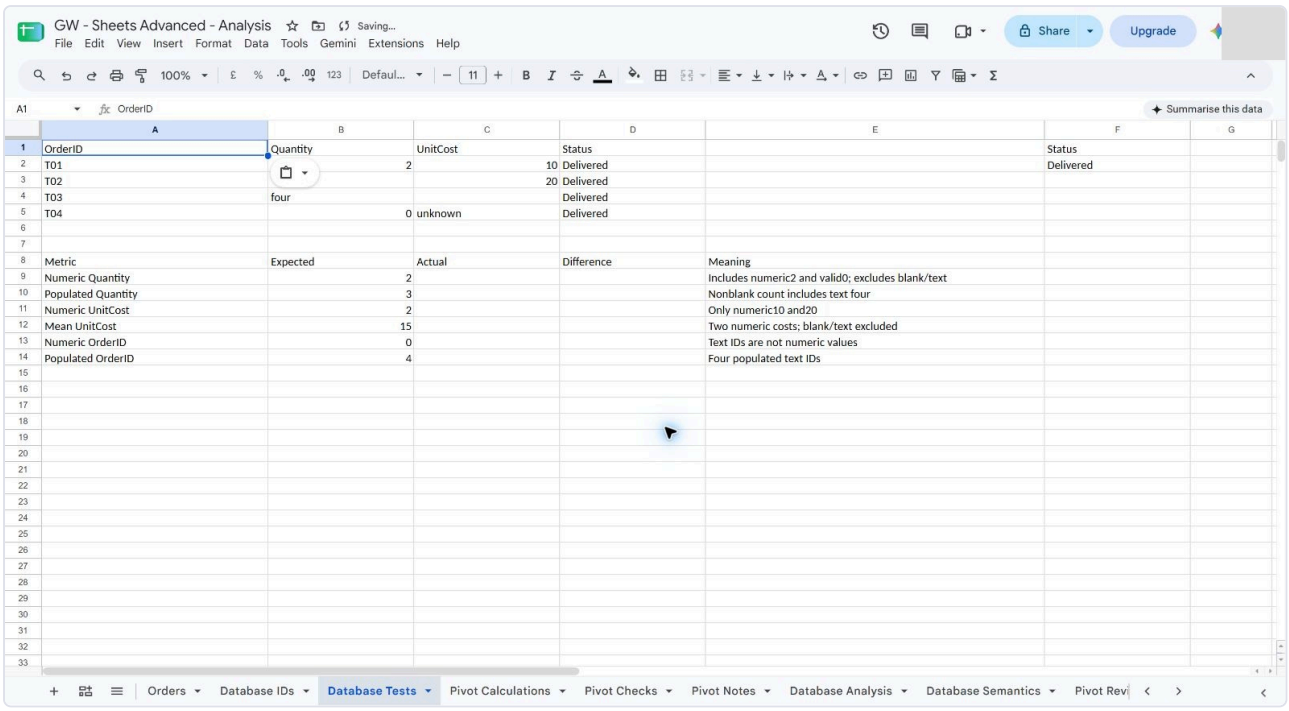


11. Add A23:F26 controls. COUNTA and COUNTUNIQUE both return **4 /18 /16** for the three ID sets. Sum Amount by the listed IDs and compare with the database results: **556 /1938 /1902**. The baseline has unique source IDs, so this key-based reconciliation is valid for these supplied data.

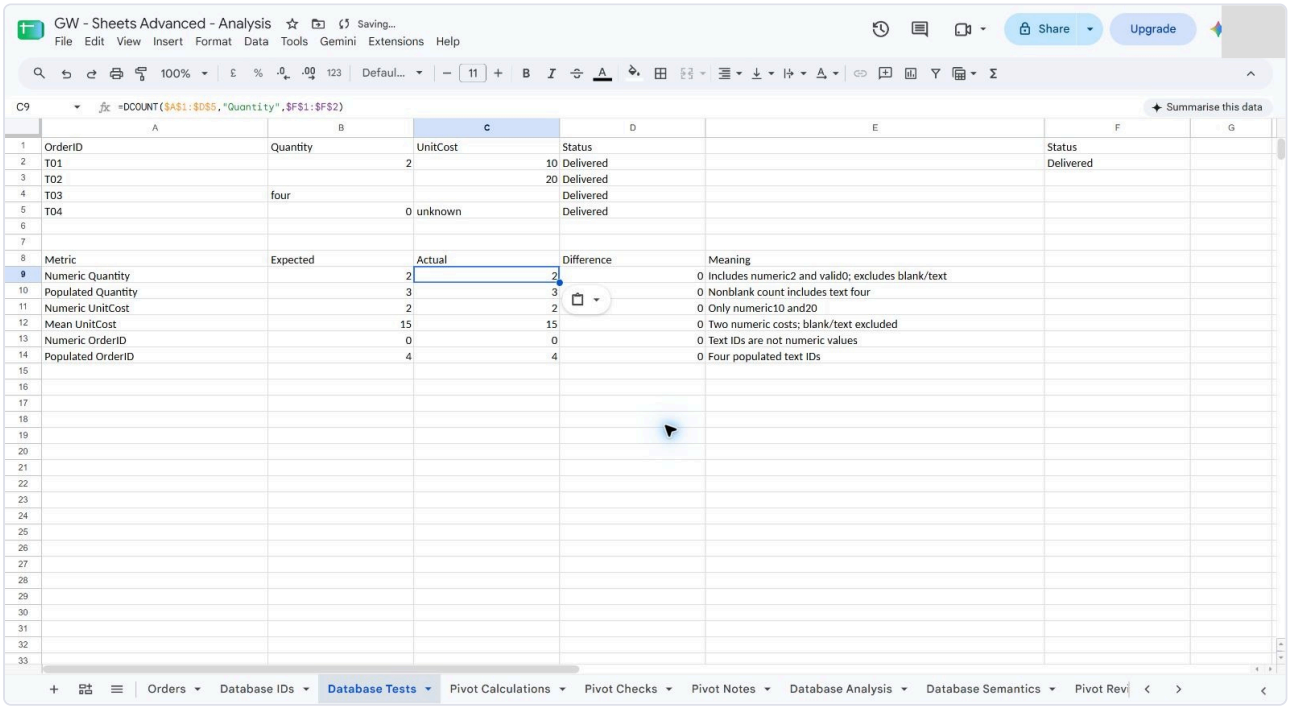


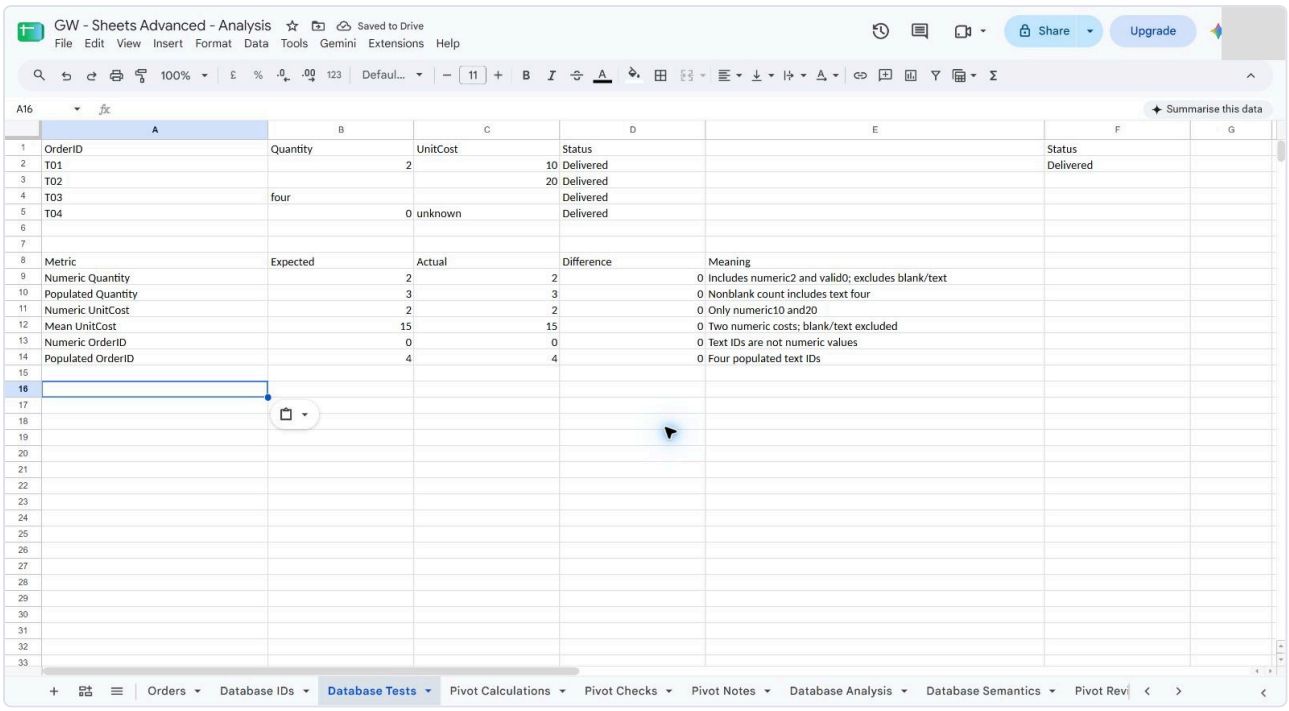


12. Create **Database Tests** with the explicit four-row type table at A1:D5 and Status/Delivered criteria in F1:F2. Keep its intentionally invalid text/blank cases separate from maintained Orders. Record expected field counts and means before filling actuals: numeric Quantity2, populated Quantity3, numeric UnitCost2, mean UnitCost15, numeric OrderID0 and populated OrderID4.



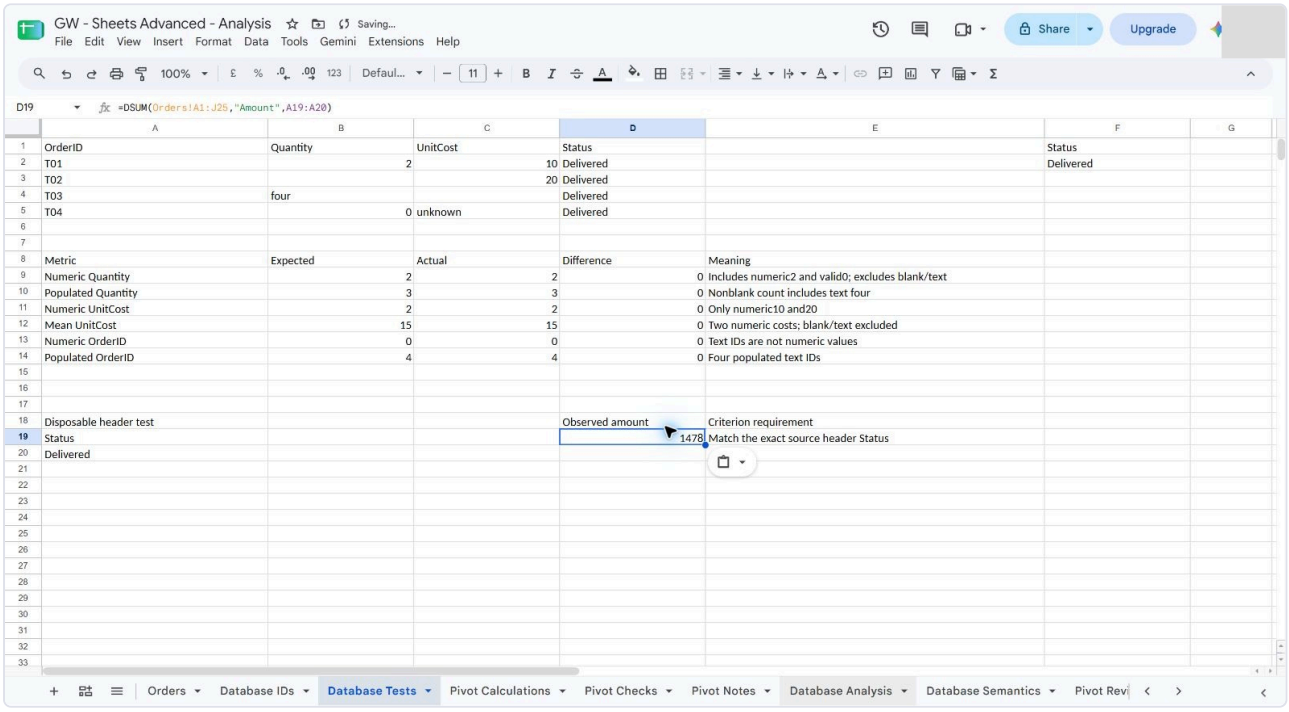
13. Enter the six database formulas in C9:C14 and C minus B differences in D. Numeric zero is included by DCOUNT; a blank and literal four are excluded from its numeric Quantity count. DCOUNTA includes the nonblank text four. DAVERAGE uses only the two numeric costs10 and20, giving15; blank/text costs do not enter that denominator. Invalid source types still require review even when a numeric result appears.





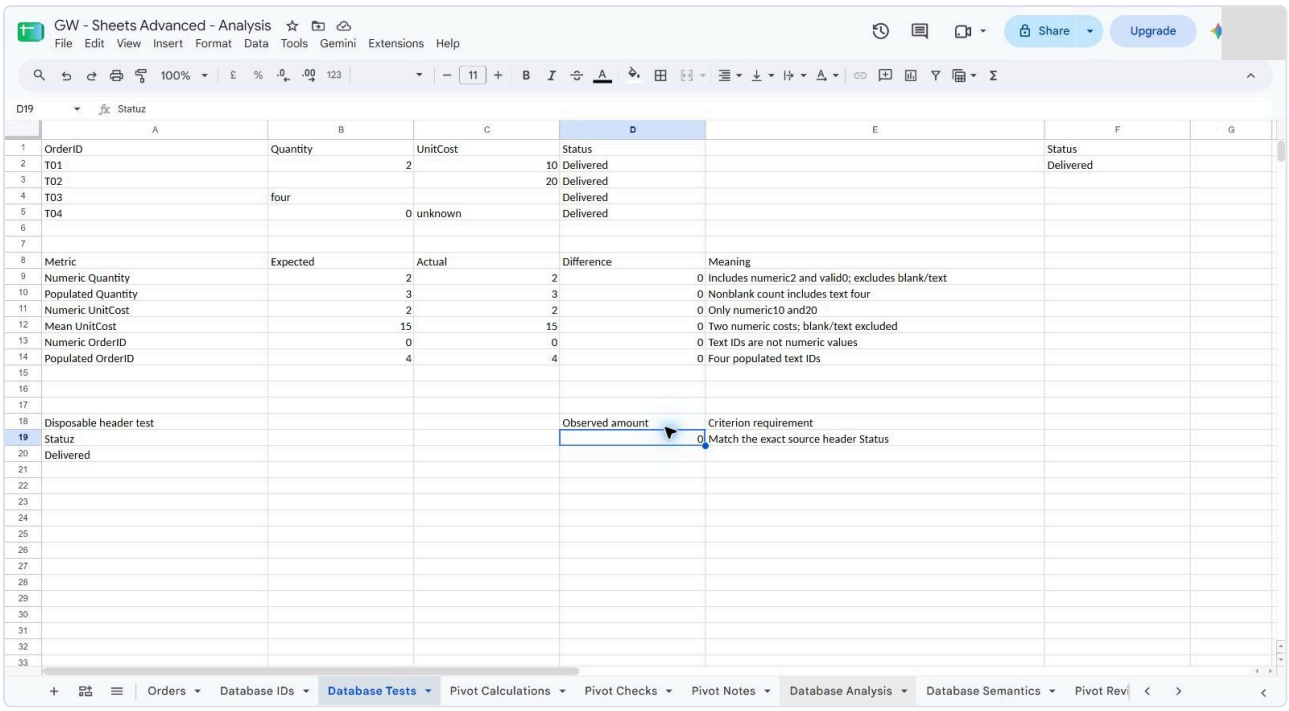
	A	B	C	D	E	F
1	OrderID	Quantity	UnitCost	Status		Status
2	T01	2	10	Delivered		Delivered
3	T02		20	Delivered		
4	T03	four		Delivered		
5	T04	0	unknown	Delivered		
6						
7						
8	Metric	Expected	Actual	Difference	Meaning	
9	Numeric Quantity	2	2	0	Includes numeric2 and valid0; excludes blank/text	
10	Populated Quantity	3	3	0	Nonblank count includes text four	
11	Numeric UnitCost	2	2	0	Only numeric10 and20	
12	Mean UnitCost	15	15	0	Two numeric costs; blank/text excluded	
13	Numeric OrderID	0	0	0	Text IDs are not numeric values	
14	Populated OrderID	4	4	0	Four populated text IDs	
15						
16						
17						
18						
19						
20						
21						
22						
23						
24						
25						
26						
27						
28						
29						
30						
31						
32						
33						

14. Build a separate disposable header test at A18:A20: label, exact **Status**, then Delivered. D19 uses **=DSUM(Orders!A1:J25,"Amount",A19:A20)** and returns1,478.

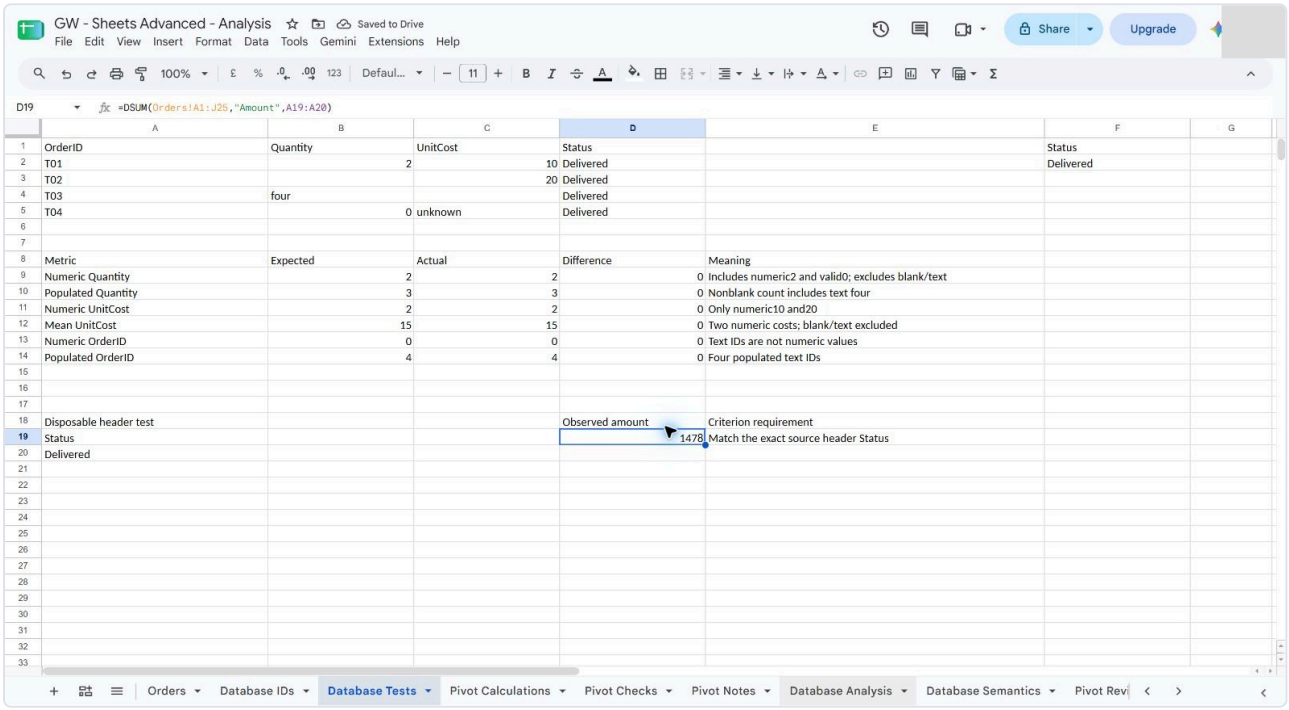


	A	B	C	D	E	F
1	OrderID	Quantity	UnitCost	Status		Status
2	T01	2	10	Delivered		Delivered
3	T02		20	Delivered		
4	T03	four		Delivered		
5	T04	0	unknown	Delivered		
6						
7						
8	Metric	Expected	Actual	Difference	Meaning	
9	Numeric Quantity	2	2	0	Includes numeric2 and valid0; excludes blank/text	
10	Populated Quantity	3	3	0	Nonblank count includes text four	
11	Numeric UnitCost	2	2	0	Only numeric10 and20	
12	Mean UnitCost	15	15	0	Two numeric costs; blank/text excluded	
13	Numeric OrderID	0	0	0	Text IDs are not numeric values	
14	Populated OrderID	4	4	0	Four populated text IDs	
15						
16						
17						
18	Disposable header test			Observed amount	Criterion requirement	
19	Status			1478	Match the exact source header Status	
20	Delivered					
21						
22						
23						
24						
25						
26						
27						
28						
29						
30						
31						
32						
33						

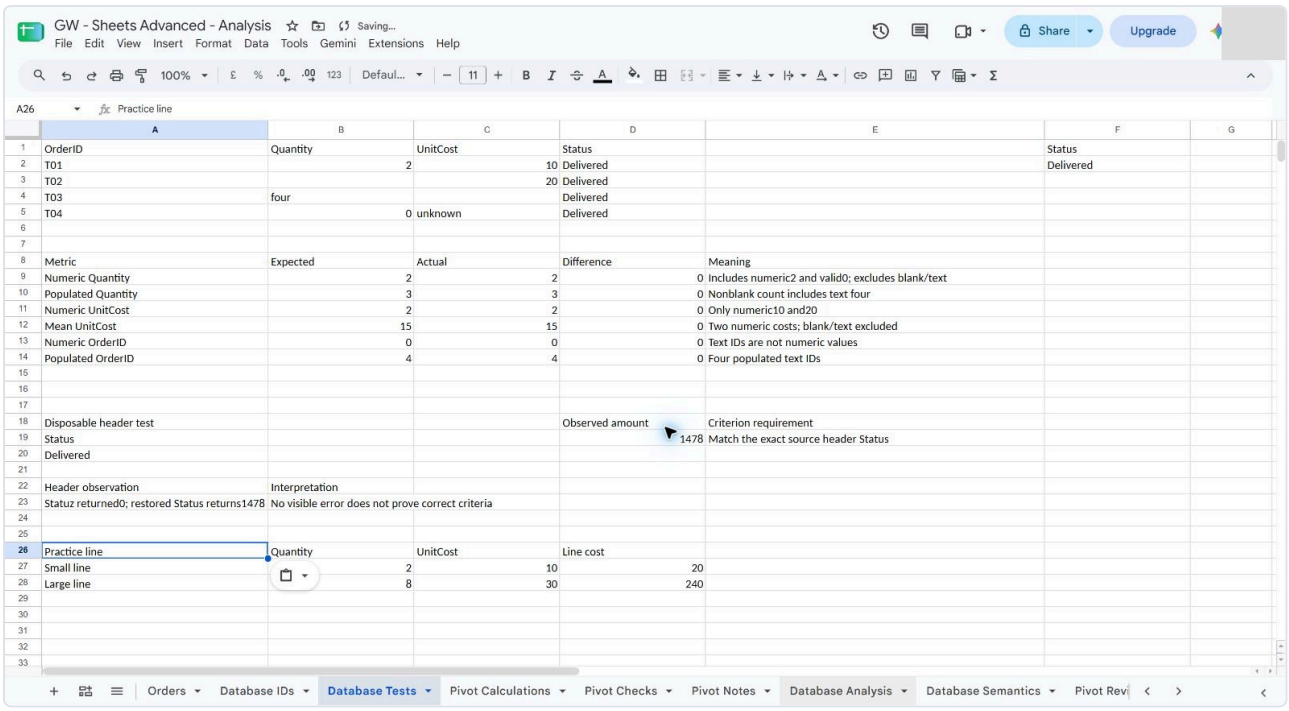
15. Change only test header A19 to **Statuz**. In this actual native experiment, D19 returns **0**, with no formula error. Save the observation before repair. Do not generalise this one result to every malformed header; a plausible or error-free output is insufficient evidence of correct criteria.



16. Restore A19 to exact **Status**. The same DSUM formula returns **1,478** again. Record the observed zero and repair rather than hiding it with an error wrapper.

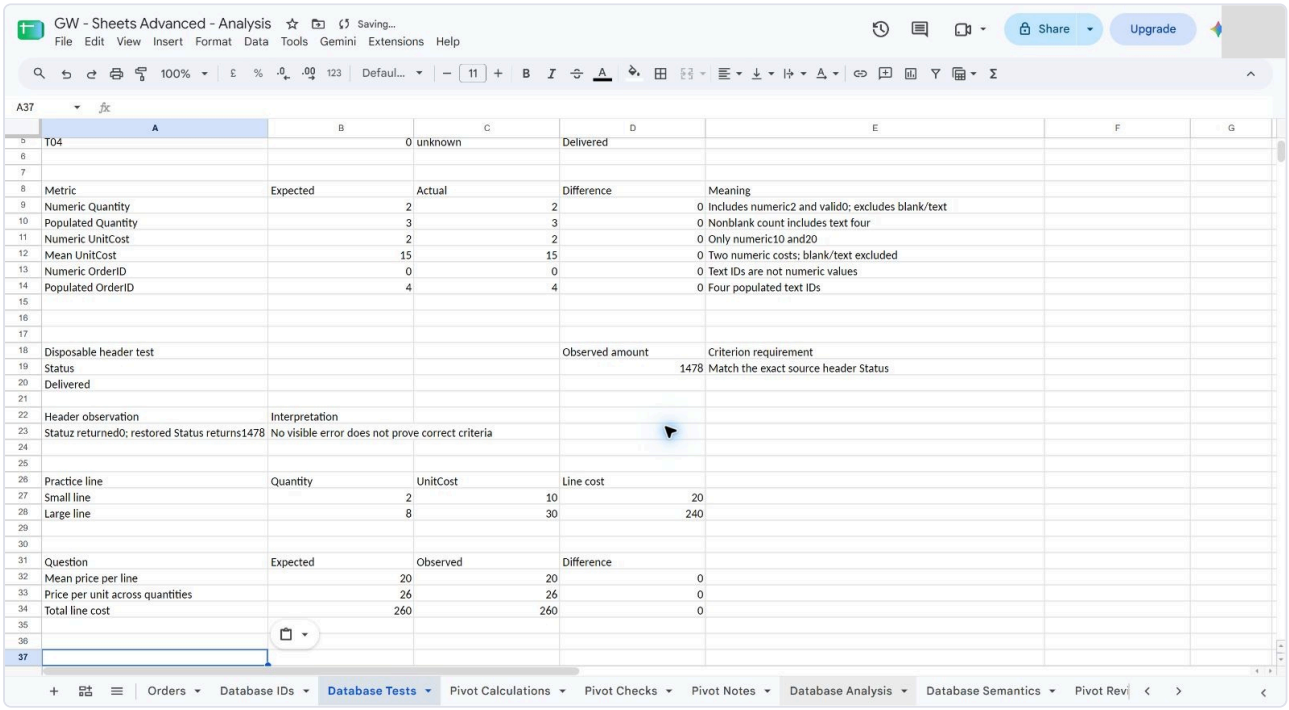


17. Add the explicit two-row weighted-price table at A26:D28: quantity2 at cost10 and quantity8 at cost30. Row costs are20 and240. The rows are labelled practice data, not replacements for source Orders.



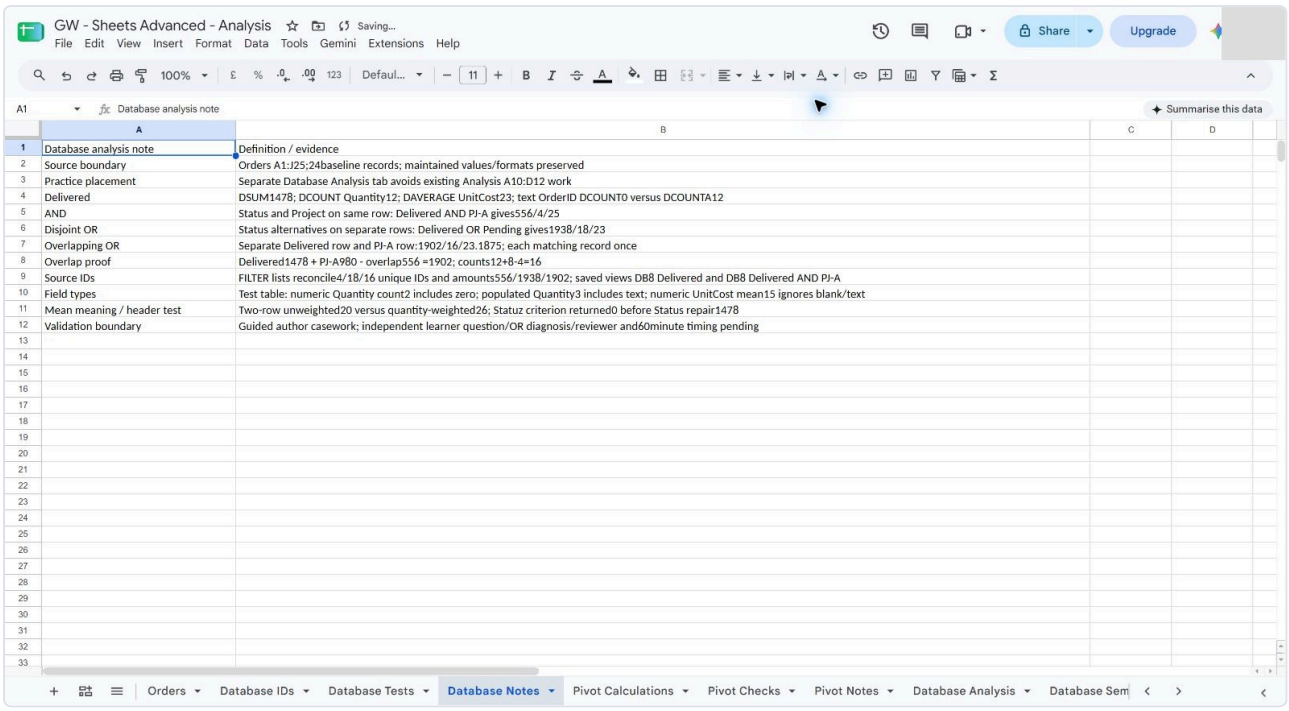
	A	B	C	D	E	F	G
1	OrderID	Quantity	UnitCost	Status		Status	
2	T01	2	10	Delivered		Delivered	
3	T02		20	Delivered			
4	T03	four		Delivered			
5	T04	0	unknown	Delivered			
6							
7							
8	Metric	Expected	Actual	Difference	Meaning		
9	Numeric Quantity	2	2	0	Includes numeric2 and valid0; excludes blank/text		
10	Populated Quantity	3	3	0	Nonblank count includes text four		
11	Numeric UnitCost	2	2	0	Only numeric10 and20		
12	Mean UnitCost	15	15	0	Two numeric costs; blank/text excluded		
13	Numeric OrderID	0	0	0	Text IDs are not numeric values		
14	Populated OrderID	4	4	0	Four populated text IDs		
15							
16							
17							
18	Disposable header test			Observed amount	Criterion requirement		
19	Status			1478	Match the exact source header Status		
20	Delivered						
21							
22	Header observation	Interpretation					
23	Statuz returned0; restored Status returns1478	No visible error does not prove correct criteria					
24							
25							
26	Practice line	Quantity	UnitCost	Line cost			
27	Small line	2	10	20			
28	Large line	8	30	240			
29							
30							
31							
32							
33							

18. Compare **AVERAGE(C27:C28)=20** with **SUM(D27:D28)/SUM(B27:B28)=26**. The first asks the mean price per line; the second asks cost per unit across10 units. Total line cost is260. Both results are correct for their stated questions, and their expected-versus-actual differences are0.



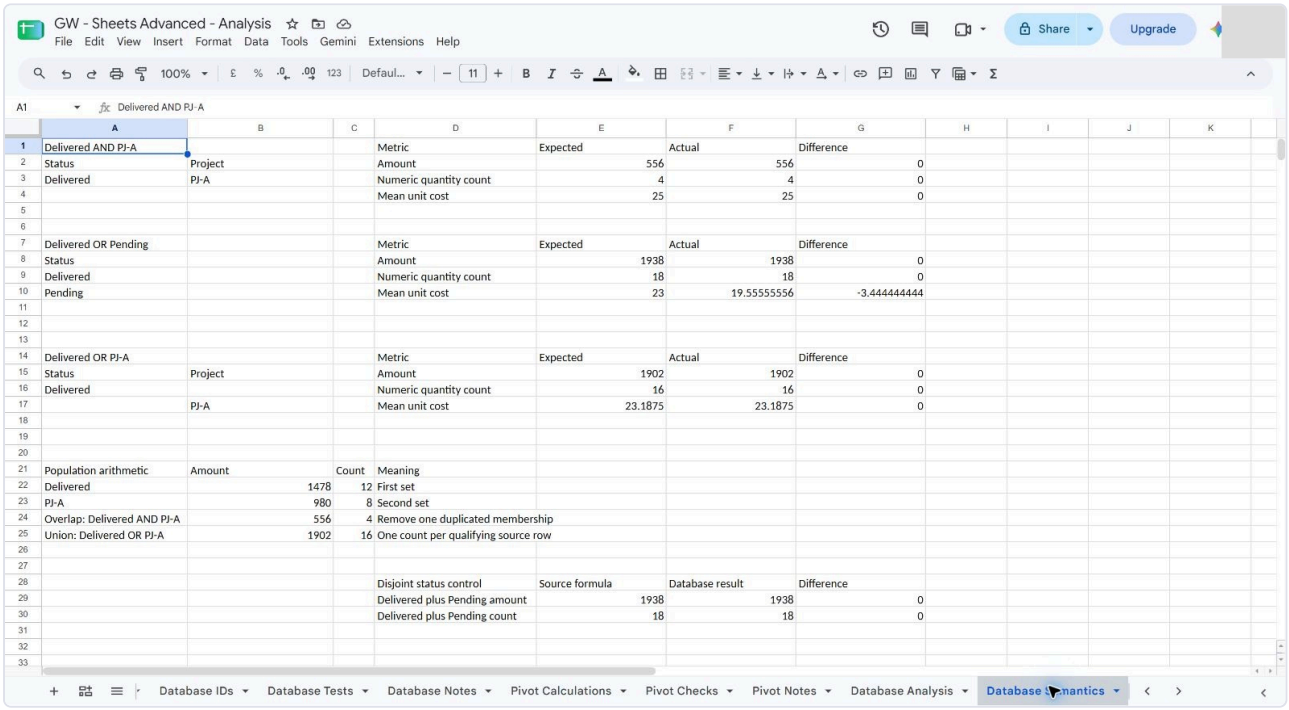
	A	B	C	D	E	F	G
5	T04	0	unknown	Delivered			
6							
7							
8	Metric	Expected	Actual	Difference	Meaning		
9	Numeric Quantity	2	2	0	Includes numeric2 and valid0; excludes blank/text		
10	Populated Quantity	3	3	0	Nonblank count includes text four		
11	Numeric UnitCost	2	2	0	Only numeric10 and20		
12	Mean UnitCost	15	15	0	Two numeric costs; blank/text excluded		
13	Numeric OrderID	0	0	0	Text IDs are not numeric values		
14	Populated OrderID	4	4	0	Four populated text IDs		
15							
16							
17							
18	Disposable header test			Observed amount	Criterion requirement		
19	Status			1478	Match the exact source header Status		
20	Delivered						
21							
22	Header observation	Interpretation					
23	Statuz returned0; restored Status returns1478	No visible error does not prove correct criteria					
24							
25							
26	Practice line	Quantity	UnitCost	Line cost			
27	Small line	2	10	20			
28	Large line	8	30	240			
29							
30							
31	Question	Expected	Observed	Difference			
32	Mean price per line	20	20	0			
33	Price per unit across quantities	26	26	0			
34	Total line cost	260	260	0			
35							
36							
37							

19. Create **Database Notes** with source scope, criteria definitions, ID reconciliation, type tests and validation limits. The first handover retained the incorrect disjoint mean prediction, so the final audit below is required before accepting it.



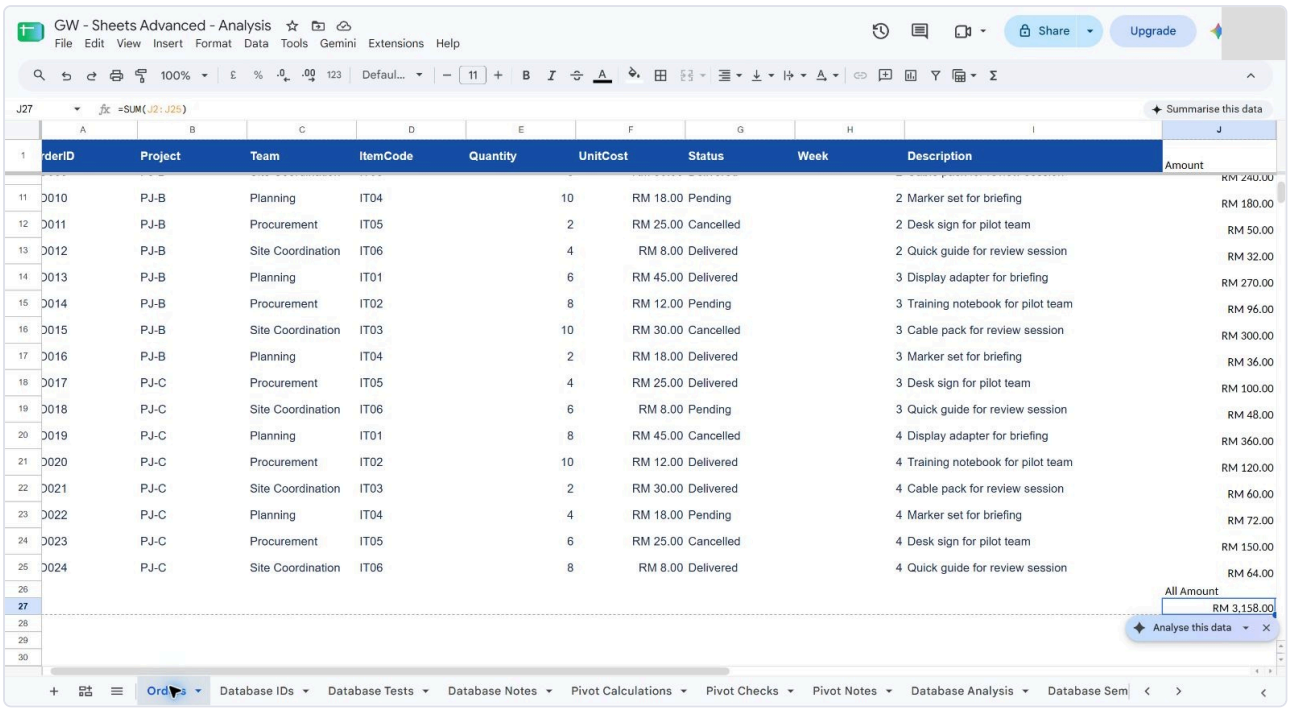
	A	B
1	Database analysis note	Definition / evidence
2	Source boundary	Orders A1:J25;24baseline records; maintained values/formats preserved
3	Practice placement	Separate Database Analysis tab avoids existing Analysis A10:D12 work
4	Delivered	DSUM1478; DCOUNT Quantity12; DAVERAGE UnitCost23; text OrderID DCOUNT0 versus DCOUNTA12
5	AND	Status and Project on same row: Delivered AND PJ-A gives556/4/25
6	Disjoint OR	Status alternatives on separate rows: Delivered OR Pending gives1938/18/23
7	Overlapping OR	Separate Delivered row and PJ-A row:1902/16/23.1875; each matching record once
8	Overlap proof	Delivered1478 + PJ-A980 - overlap556 =1902; counts12+8-4=16
9	Source IDs	FILTER lists reconcile4/18/16 unique IDs and amounts556/1938/1902; saved views DB8 Delivered and DB8 Delivered AND PJ-A
10	Field types	Test table: numeric Quantity count2 includes zero; populated Quantity3 includes text; numeric UnitCost mean15 ignores blank/text
11	Mean meaning / header test	Two-row unweighted20 versus quantity-weighted26; Statuz criterion returned0 before Status repair1478
12	Validation boundary	Guided author casework; independent learner question/OR diagnosis/reviewer and60minute timing pending

20. Review the three criteria blocks and their difference cells. The initial disjoint-OR mean mismatch remains visible in this first-pass checkpoint. The next steps investigate it rather than accepting the handover as complete.

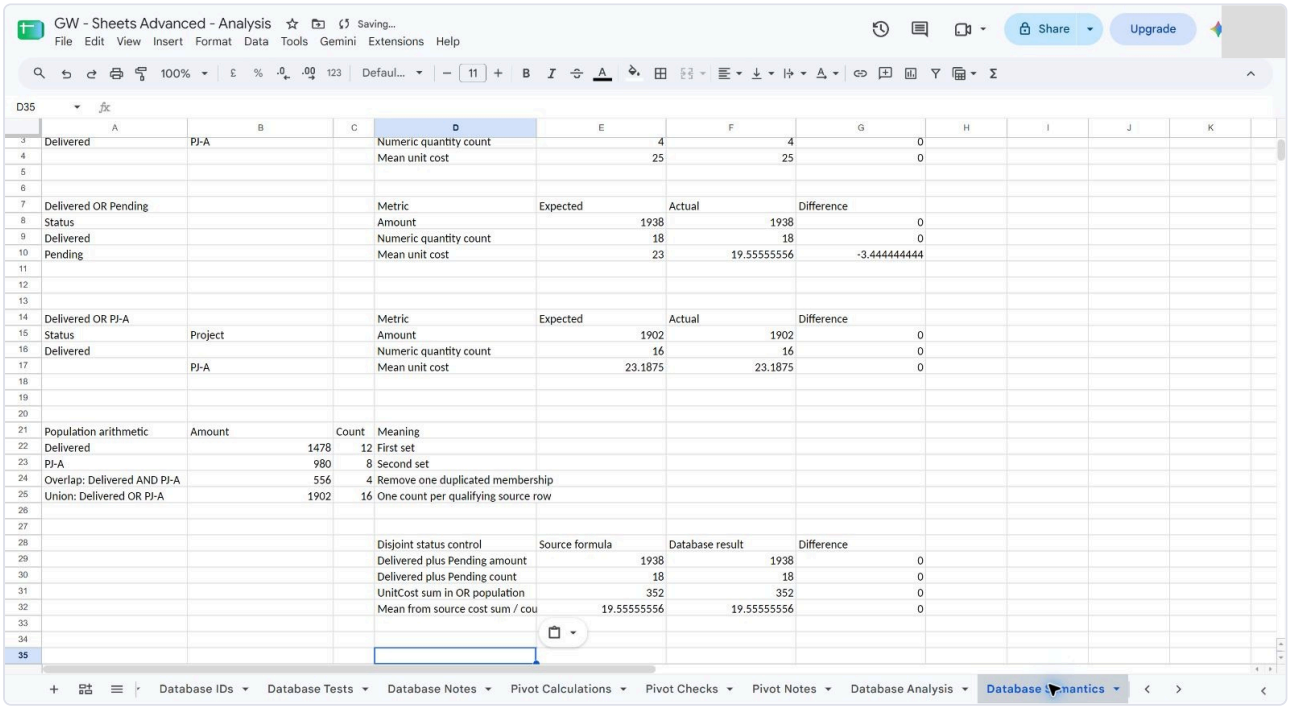


	A	B	C	D	E	F	G
1	Delivered AND PJ-A			Metric	Expected	Actual	Difference
2	Status	Project		Amount	556	556	0
3	Delivered	PJ-A		Numeric quantity count	4	4	0
4				Mean unit cost	25	25	0
5							
6							
7	Delivered OR Pending			Metric	Expected	Actual	Difference
8	Status			Amount	1938	1938	0
9	Delivered			Numeric quantity count	18	18	0
10	Pending			Mean unit cost	23	19.55555556	-3.444444444
11							
12							
13							
14	Delivered OR PJ-A			Metric	Expected	Actual	Difference
15	Status	Project		Amount	1902	1902	0
16	Delivered			Numeric quantity count	16	16	0
17		PJ-A		Mean unit cost	23.1875	23.1875	0
18							
19							
20							
21	Population arithmetic	Amount	Count	Meaning			
22	Delivered	1478	12	First set			
23	PJ-A	980	8	Second set			
24	Overlap: Delivered AND PJ-A	556	4	Remove one duplicated membership			
25	Union: Delivered OR PJ-A	1902	16	One count per qualifying source row			
26							
27							
28				Disjoint status control	Source formula	Database result	Difference
29				Delivered plus Pending amount	1938	1938	0
30				Delivered plus Pending count	18	18	0
31							
32							
33							

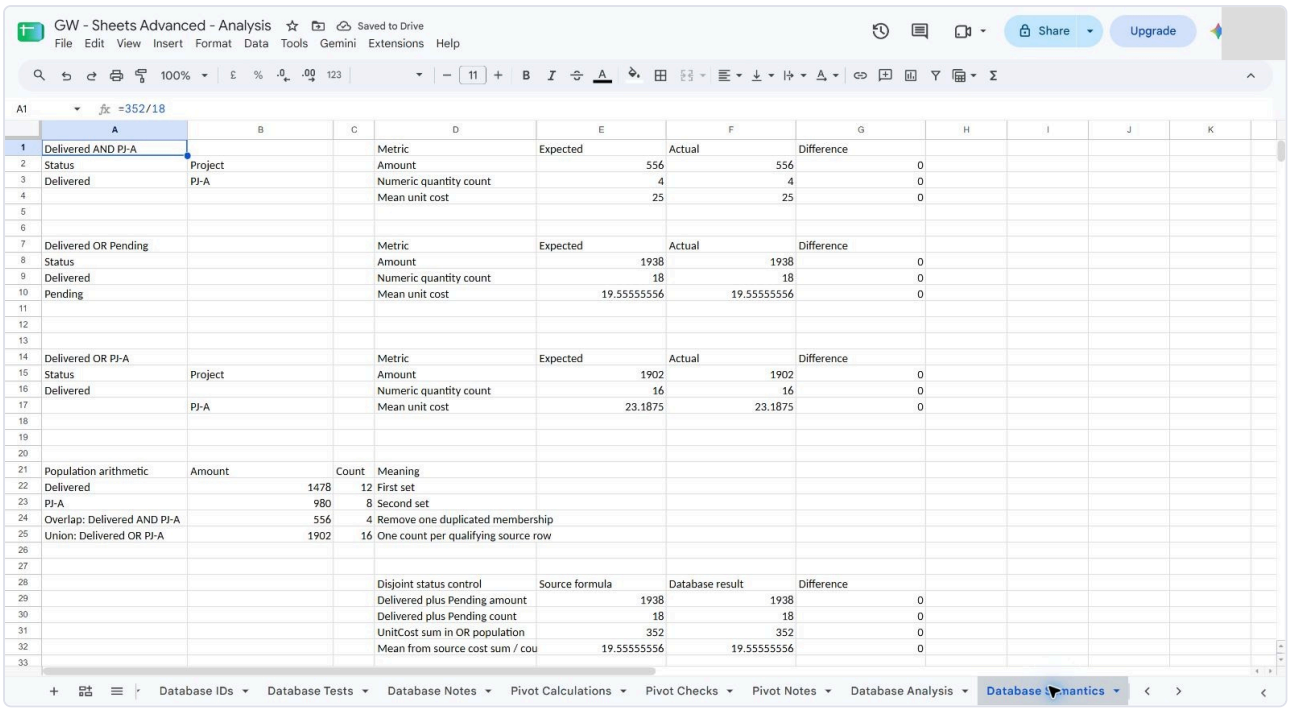
21. Inspect maintained Orders unfiltered: the all-record Amount remains **3,158**. Earlier Analysis, Rates, Source Snapshot and native pivots remain intact. The discrepancy is in the prediction for a defined subset, not a changed source baseline.



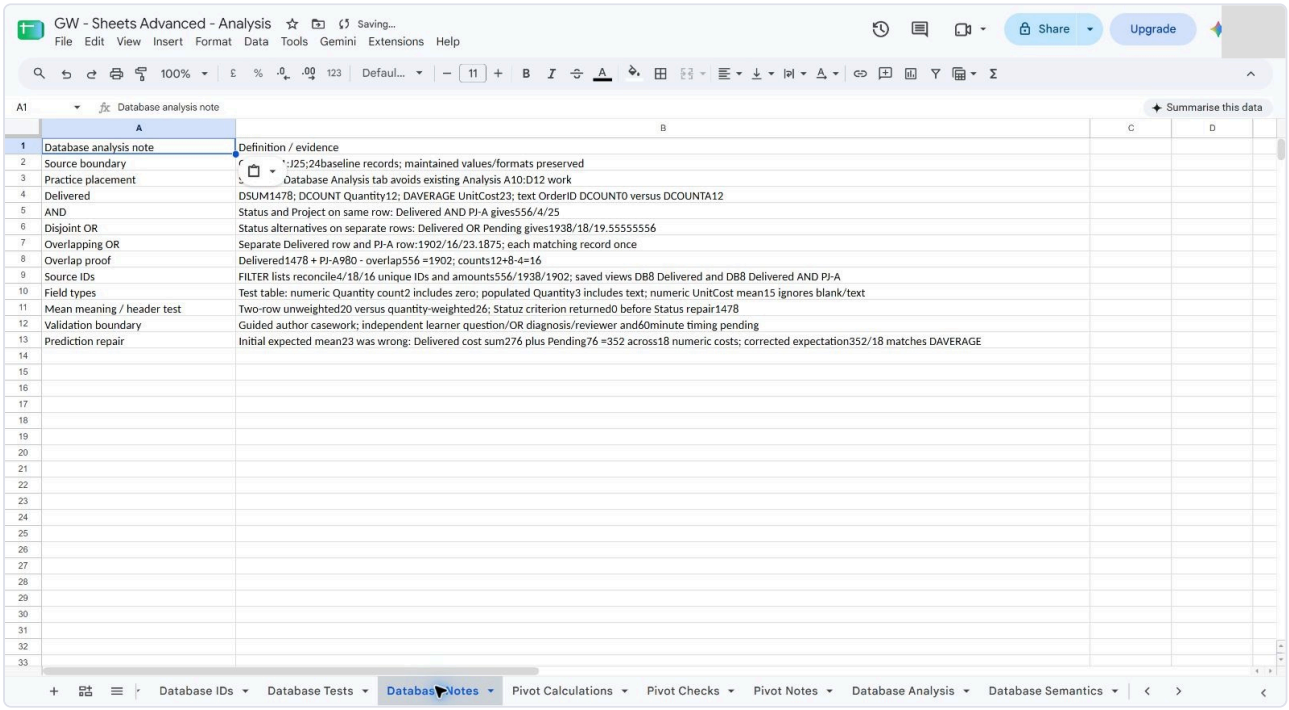
22. In Database Semantics D31:G32, reconcile the OR population's UnitCost sum. Source SUMIF totals and DSUM both give **352**: Delivered contributes276 and Pending76. Divide by18 numeric costs to obtain **19.55555556**, matching DAVERAGE. The statuses do not have the same UnitCost distribution, so copying Delivered's mean23 was an invalid expectation.



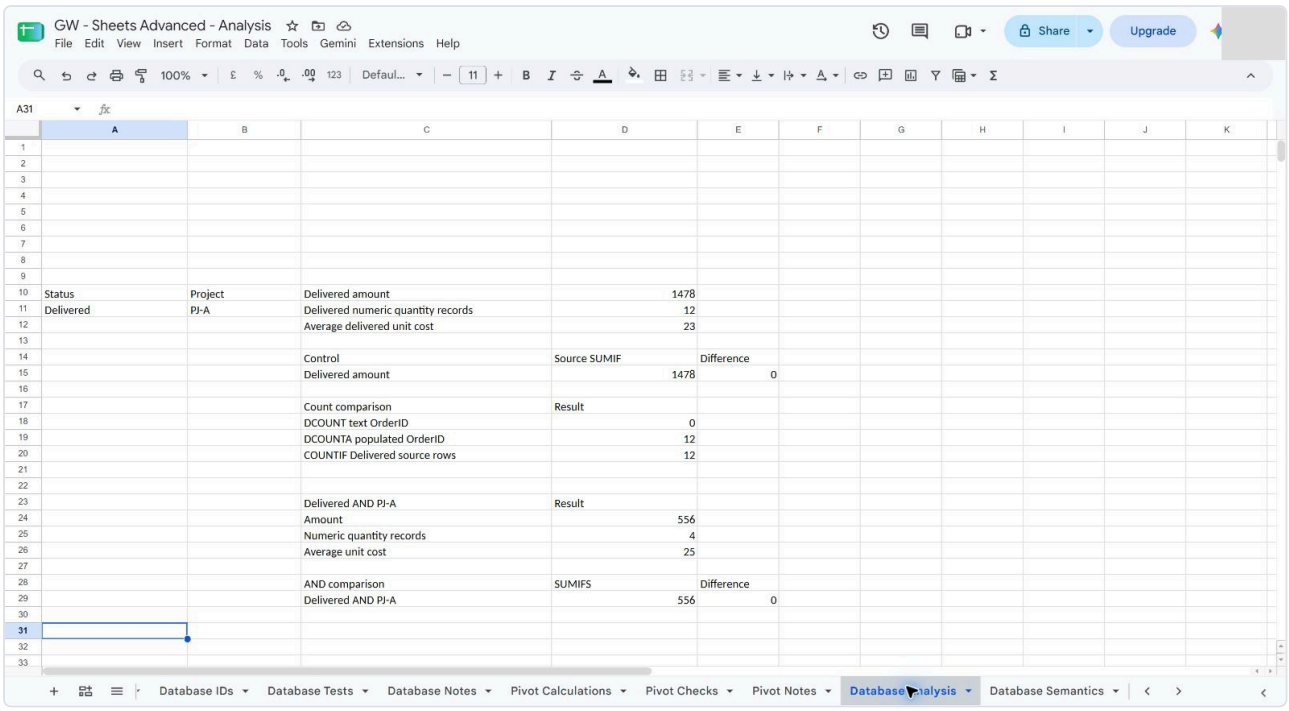
23. Correct **E10 to =352/18**. Keep F10 as the original DAVERAGE formula. All nine metric differences now reconcile to0, and the separate source-cost mean check also has zero difference. The preserved first-pass evidence distinguishes a wrong prediction from a formula repair.



24. Correct the disjoint-OR entry in Database Notes and append the prediction-repair explanation. Leave the exact Status test header restored and the source unfiltered, with the two named filter views retained. The independent challenge below remains unperformed.



25. Finish by inspecting **Database Analysis** with all Delivered and AND formulas visible. DSUM/SUMIF1478 and AND DSUM/SUMIFS556 have zero differences; the numeric/text count distinction remains explicit. This final view confirms the completed results after the earlier entry-stage captures.



> **Validation boundary:** guided8.1–8.3 and author casework8.4 were executed through learner UI with bounded native-cell evidence. A mistaken author prediction was caught and corrected using source UnitCost sums; it was not a source-data defect. The Statuz0/Status1478 observation is specific to the captured test. Independent learner question/OR diagnosis, human reviewer and the planned60-minute allocation remain unverified.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Write a new analysis question requiring two conditions and design its database criteria without copying a finished block.
2. Create a second version that uses OR instead of AND, predict the population difference and verify it through contributing IDs.
3. Explain the numeric field counted by DCOUNT, the average’s denominator and one overlap risk.

Evidence to Hand Over

- Three labelled criteria blocks with source-ID reconciliation.
- AND/OR and overlap proof, plus weighted/unweighted meaning comparison.
- Independent question and corrected criteria-header test.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check

An OR total double-counts overlapping rows	Compare the contributing ID sets and count each qualifying source record once.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Database functions are useful when their header matching, criteria logic and field meaning are explicit.

Check yourself: Why is an average unit cost different from a quantity-weighted average?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Delivered amount
```

```
=DSUM(Orders!A1:J25,"Amount",A10:A11)
```

```
Delivered numeric quantity records
```

```
=DCOUNT(Orders!A1:J25,"Quantity",A10:A11)
```

```
Average delivered unit cost
```

```
=DAVERAGE(Orders!A1:J25,"UnitCost",A10:A11)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Criteria cases: Status=Delivered AND Project=PJ-A; Status=Delivered OR Status=Pending;
Status=Delivered OR Project=PJ-A. The last case overlaps. DCOUNT uses numeric Quantity;
OrderID is text and must not be treated as a numeric record count.
```

Captured Author-Run Reference

Use a separate **Database Analysis** tab because the earlier Analysis cells are occupied. Maintained source is **Orders A1:J25**. Predict results before actual formulas. These cards show the final corrected source-derived mean352/18; the historical first-pass prediction23 in screenshots is investigated in the guide. Independent learner work remains a separate task.

Database Analysis A10:B11 — criteria

```
Status  Project
Delivered       PJ-A
```

Database Analysis C10:D12 — original Delivered formulas

```
Delivered amount        =DSUM(Orders!A1:J25,"Amount",A10:A11)
Delivered numeric quantity records      =DCOUNT(Orders!A1:J25,"Quantity",A10:A11)
Average delivered unit cost     =DAVERAGE(Orders!A1:J25,"UnitCost",A10:A11)
```

Database Analysis C14:E15 — amount control

```
Control Source SUMIF    Difference
Delivered amount        =SUMIF(Orders!G2:G25,"Delivered",Orders!J2:J25) =D10-D15
```

Database Analysis C17:D20 — field-aware counts

```
Count comparison        Result
DCOUNT text OrderID     =DCOUNT(Orders!A1:J25,"OrderID",A10:A11)
DCOUNTA populated OrderID       =DCOUNTA(Orders!A1:J25,"OrderID",A10:A11)
COUNTIF Delivered source rows   =COUNTIF(Orders!G2:G25,"Delivered")
```

Database Analysis C23:D26 — AND metrics

```
Delivered AND PJ-A      Result
Amount  =DSUM(Orders!A1:J25,"Amount",A10:B11)
Numeric quantity records        =DCOUNT(Orders!A1:J25,"Quantity",A10:B11)
Average unit cost       =DAVERAGE(Orders!A1:J25,"UnitCost",A10:B11)
```

Database Analysis C28:E29 — SUMIFS comparison

```
AND comparison  SUMIFS  Difference
Delivered AND PJ-A      =SUMIFS(Orders!J2:J25,Orders!G2:G25,"Delivered",Orders!B2:B25,"PJ-A")
=D24-D29
```

Database Semantics A1:B3 — AND

```
Delivered AND PJ-A
Status  Project
Delivered       PJ-A
```

Database Semantics A7:A10 — disjoint OR

```
Delivered OR Pending
Status
Delivered
Pending
```

Database Semantics A14:B17 — overlapping OR

```
Delivered OR PJ-A
Status  Project
Delivered
        PJ-A
```

Database Semantics D1:G4 — AND expectations, actuals blank

```
Metric  Expected        Actual  Difference
Amount  556
Numeric quantity count  4
Mean unit cost  25
```

Database Semantics D7:G10 — corrected OR expectations, actuals blank

```
Metric  Expected        Actual  Difference
Amount  1938
Numeric quantity count  18
Mean unit cost  =352/18
```

Database Semantics D14:G17 — overlap expectations, actuals blank

```
Metric  Expected        Actual  Difference
Amount  1902
Numeric quantity count  16
Mean unit cost  23.1875
```

Database Semantics A21:D25 — inclusion–exclusion

```
Population arithmetic   Amount  Count   Meaning
Delivered       =SUMIF(Orders!G2:G25,"Delivered",Orders!J2:J25)
=COUNTIF(Orders!G2:G25,"Delivered")     First set
PJ-A    =SUMIF(Orders!B2:B25,"PJ-A",Orders!J2:J25)      =COUNTIF(Orders!B2:B25,"PJ-A")  Second
set
Overlap: Delivered AND PJ-A     =F2     =F3     Remove one duplicated membership
Union: Delivered OR PJ-A        =B22+B23-B24    =C22+C23-C24    One count per qualifying
source row
```

Database Semantics D28:G32 — independent source controls

```
Disjoint status control Source formula  Database result Difference
Delivered plus Pending amount
=SUMIF(Orders!G2:G25,"Delivered",Orders!J2:J25)+SUMIF(Orders!G2:G25,"Pending",Orders!J2:J25)
=F8     =F29-E29
Delivered plus Pending count
=COUNTIF(Orders!G2:G25,"Delivered")+COUNTIF(Orders!G2:G25,"Pending")    =F9     =F30-E30
UnitCost sum in OR population
=SUMIF(Orders!G2:G25,"Delivered",Orders!F2:F25)+SUMIF(Orders!G2:G25,"Pending",Orders!F2:F25)
=DSUM(Orders!A1:J25,"UnitCost",A8:A10)  =F31-E31
Mean from source cost sum / count       =E31/E30        =F10    =F32-E32
```

Database IDs A23:F26 — ID counts and amounts

```
Population      Expected count  Actual count    Unique IDs      Sum by listed IDs
Database amount
AND     4       =COUNTA(A2:A5)  =COUNTUNIQUE(A2:A5)
=SUM(ARRAYFORMULA(SUMIF(Orders!A2:A25,A2:A5,Orders!J2:J25)))    ='Database Semantics'!F2
Disjoint OR     18      =COUNTA(C2:C19) =COUNTUNIQUE(C2:C19)
=SUM(ARRAYFORMULA(SUMIF(Orders!A2:A25,C2:C19,Orders!J2:J25)))   ='Database Semantics'!F8
Overlapping OR  16      =COUNTA(E2:E17) =COUNTUNIQUE(E2:E17)
=SUM(ARRAYFORMULA(SUMIF(Orders!A2:A25,E2:E17,Orders!J2:J25)))   ='Database Semantics'!F15
```

Database Tests A1:D5 — isolated type cases

```
OrderID Quantity        UnitCost        Status
T01     2       10      Delivered
T02             20      Delivered
T03     four            Delivered
T04     0       unknown Delivered
```

Database Tests A8:E14 — expectations before actuals

```
Metric  Expected        Actual  Difference      Meaning
Numeric Quantity        2                       Includes numeric2 and valid0; excludes
blank/text
Populated Quantity      3                       Nonblank count includes text four
Numeric UnitCost        2                       Only numeric10 and20
Mean UnitCost   15                      Two numeric costs; blank/text excluded
Numeric OrderID 0                       Text IDs are not numeric values
Populated OrderID       4                       Four populated text IDs
```

Database Tests C9:D14 — six actual formulas

```
=DCOUNT($A$1:$D$5,"Quantity",$F$1:$F$2) =C9-B9
=DCOUNTA($A$1:$D$5,"Quantity",$F$1:$F$2)        =C10-B10
=DCOUNT($A$1:$D$5,"UnitCost",$F$1:$F$2) =C11-B11
=DAVERAGE($A$1:$D$5,"UnitCost",$F$1:$F$2)       =C12-B12
=DCOUNT($A$1:$D$5,"OrderID",$F$1:$F$2)  =C13-B13
=DCOUNTA($A$1:$D$5,"OrderID",$F$1:$F$2) =C14-B14
```

Database Tests A18:A20 — correct header test

```
Disposable header test
Status
Delivered
```

Database Tests D18:E19 — header test formula

```
Observed amount Criterion requirement
=DSUM(Orders!A1:J25,"Amount",A19:A20)   Match the exact source header Status
```

Database Tests A22:B23 — actual fault observation

```
Header observation      Interpretation
Statuz returned0; restored Status returns1478   No visible error does not prove correct
criteria
```

Database Tests A26:D28 — weighted-price inputs

```
Practice line   Quantity        UnitCost        Line cost
Small line      2       10      =B27*C27
Large line      8       30      =B28*C28
```

Database Tests A31:D34 — mean interpretation checks

```
Question        Expected        Observed        Difference
Mean price per line     20      =AVERAGE(C27:C28)       =C32-B32
Price per unit across quantities        26      =SUM(D27:D28)/SUM(B27:B28)      =C33-B33
Total line cost 260     =SUM(D27:D28)   =C34-B34
```

Database Notes A1:B13 — corrected handover

```
Database analysis note  Definition / evidence
Source boundary Orders A1:J25;24baseline records; maintained values/formats preserved
Practice placement      Separate Database Analysis tab avoids existing Analysis A10:D12 work
Delivered       DSUM1478; DCOUNT Quantity12; DAVERAGE UnitCost23; text OrderID DCOUNT0 versus
DCOUNTA12
AND     Status and Project on same row: Delivered AND PJ-A gives556/4/25
Disjoint OR     Status alternatives on separate rows: Delivered OR Pending
gives1938/18/19.55555556
Overlapping OR  Separate Delivered row and PJ-A row:1902/16/23.1875; each matching record once
Overlap proof   Delivered1478 + PJ-A980 - overlap556 =1902; counts12+8-4=16
Source IDs      FILTER lists reconcile4/18/16 unique IDs and amounts556/1938/1902; saved views
DB8 Delivered and DB8 Delivered AND PJ-A
Field types     Test table: numeric Quantity count2 includes zero; populated Quantity3
includes text; numeric UnitCost mean15 ignores blank/text
Mean meaning / header test      Two-row unweighted20 versus quantity-weighted26; Statuz
criterion returned0 before Status repair1478
Validation boundary     Guided author casework; independent learner question/OR
diagnosis/reviewer and60minute timing pending
Prediction repair       Initial expected mean23 was wrong: Delivered cost sum276 plus
Pending76 =352 across18 numeric costs; corrected expectation352/18 matches DAVERAGE
```

AND F2:G4

```
=DSUM(Orders!$A$1:$J$25,"Amount",$A$2:$B$3)     =F2-E2
=DCOUNT(Orders!$A$1:$J$25,"Quantity",$A$2:$B$3) =F3-E3
=DAVERAGE(Orders!$A$1:$J$25,"UnitCost",$A$2:$B$3)       =F4-E4
```

Disjoint OR F8:G10

```
=DSUM(Orders!$A$1:$J$25,"Amount",$A$8:$A$10)    =F8-E8
=DCOUNT(Orders!$A$1:$J$25,"Quantity",$A$8:$A$10)        =F9-E9
=DAVERAGE(Orders!$A$1:$J$25,"UnitCost",$A$8:$A$10)      =F10-E10
```

Overlapping OR F15:G17

```
=DSUM(Orders!$A$1:$J$25,"Amount",$A$15:$B$17)   =F15-E15
=DCOUNT(Orders!$A$1:$J$25,"Quantity",$A$15:$B$17)       =F16-E16
=DAVERAGE(Orders!$A$1:$J$25,"UnitCost",$A$15:$B$17)     =F17-E17
```

Database IDs AND — A2

```
=FILTER(Orders!A2:A25,Orders!G2:G25="Delivered",Orders!B2:B25="PJ-A")
```

Database IDs Disjoint OR — C2

```
=FILTER(Orders!A2:A25,((Orders!G2:G25="Delivered")+(Orders!G2:G25="Pending"))>0)
```

Database IDs Overlapping OR — E2

```
=FILTER(Orders!A2:A25,((Orders!G2:G25="Delivered")+(Orders!B2:B25="PJ-A"))>0)
```

Filter views and test boundaries

Select **Orders A1:J25 > Data > Create filter view**. Save **DB8 - Delivered** after selecting Delivered in Status. Create a separate **DB8 - Delivered AND PJ-A** with both Status Delivered and Project PJ-A; its IDs are PO001/PO004/PO005/PO008. Exit the view after inspection. Do not delete excluded source rows.

Database Tests criteria are **F1:F2 = Status / Delivered**. Preserve genuine blanks B3/C4, text four/unknown and numeric zero B5 in that isolated table. DCOUNT counts numbers; DCOUNTA counts populated fields; DAVERAGE's denominator includes only numeric values of the selected field.

For the disposable header test, capture D19 with exact Status, then temporarily change A19 to Statuz. The observed result was0; restore Status and verify1478. Do not treat an error-free numeric result as proof of correct criteria.

The disjoint-OR mean is352/18 =19.55555556, not23. Final expected E10 is **=352/18**, while actual F10 remains DAVERAGE. Delivered UnitCost sum276 plus Pending76 explains the mean. Keep prediction errors distinct from formula/source defects.

Final populations: Delivered1478/12/23; Delivered AND PJ-A556/4/25; Delivered OR Pending1938/18/19.55555556; Delivered OR PJ-A1902/16/23.1875. Amount/numeric Quantity count/mean UnitCost are different metrics. Independent question, learner diagnosis and timing remain pending.

09

CHAPTER 9

Build a Dashboard

In this chapter you will choose charts, sparklines and filter views; distinguish source-linked charts from generated analysis snapshots.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 70 minutes: 15 explanation/demonstration, 27 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 55 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A compact source-linked dashboard with a tested update path.

> **Status:** author reference with31 staged application captures through9.4. Native column chart, weekly sparkline, distinct KPI populations and recalculating Checks are verified. Existing-row +45 and scenario-copy +90 tests were restored. Independent PO025 coverage, reviewer and learner timing remain pending.

What You Will Learn

- Choose source-linked visuals.
- Use a sparkline for a defined time sequence.
- Explain how a dashboard refreshes.

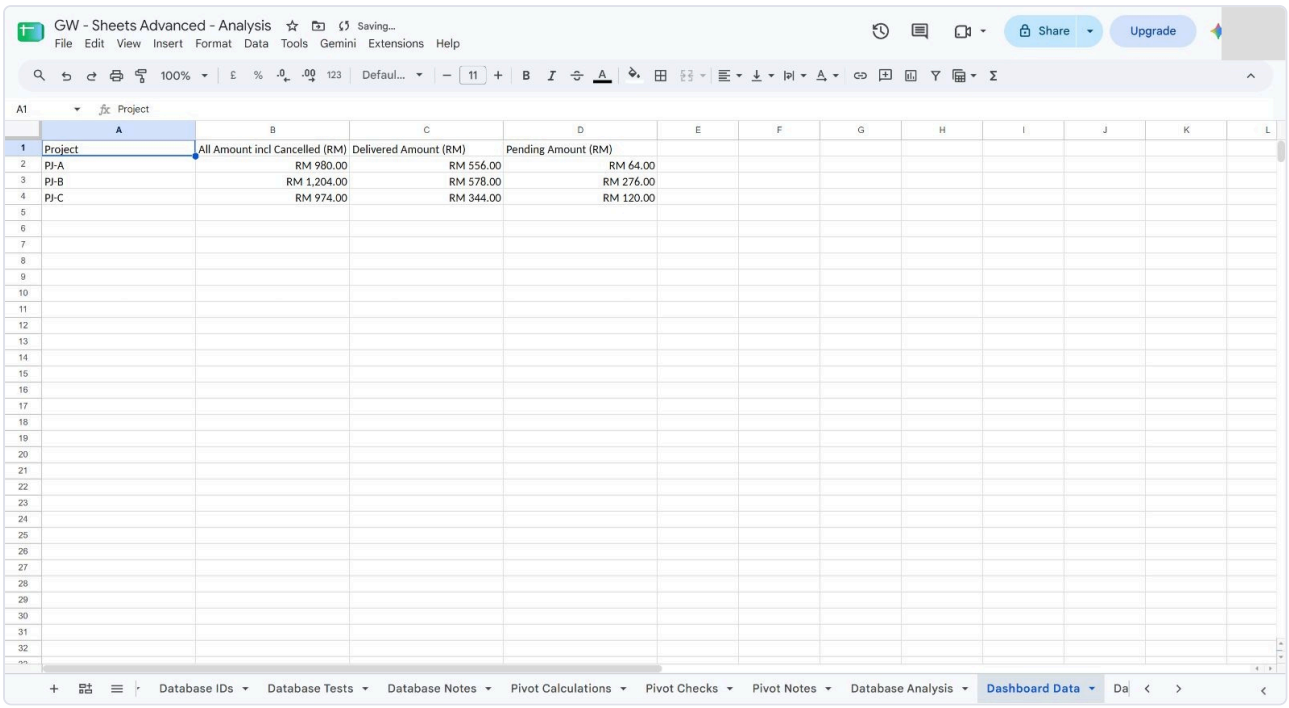
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

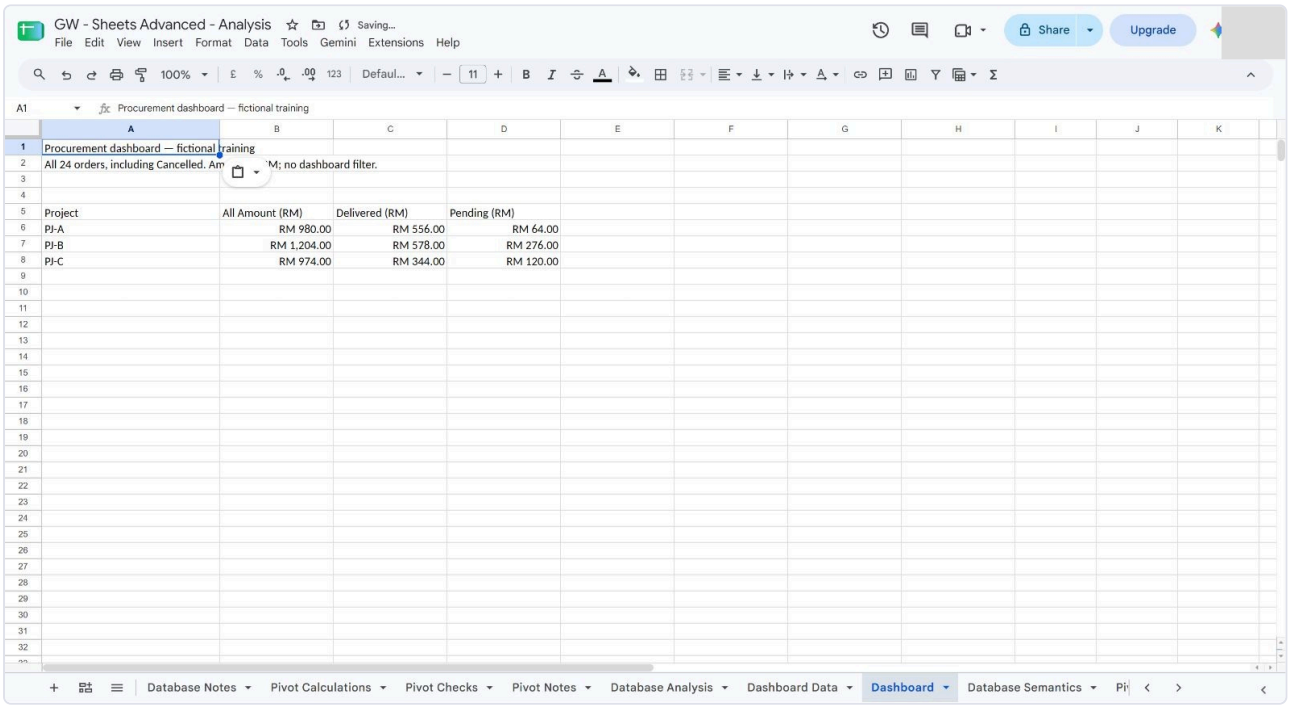
Use the checked advanced workbook with all-orders and delivered controls reconciled.

9.1 Build a Compact View

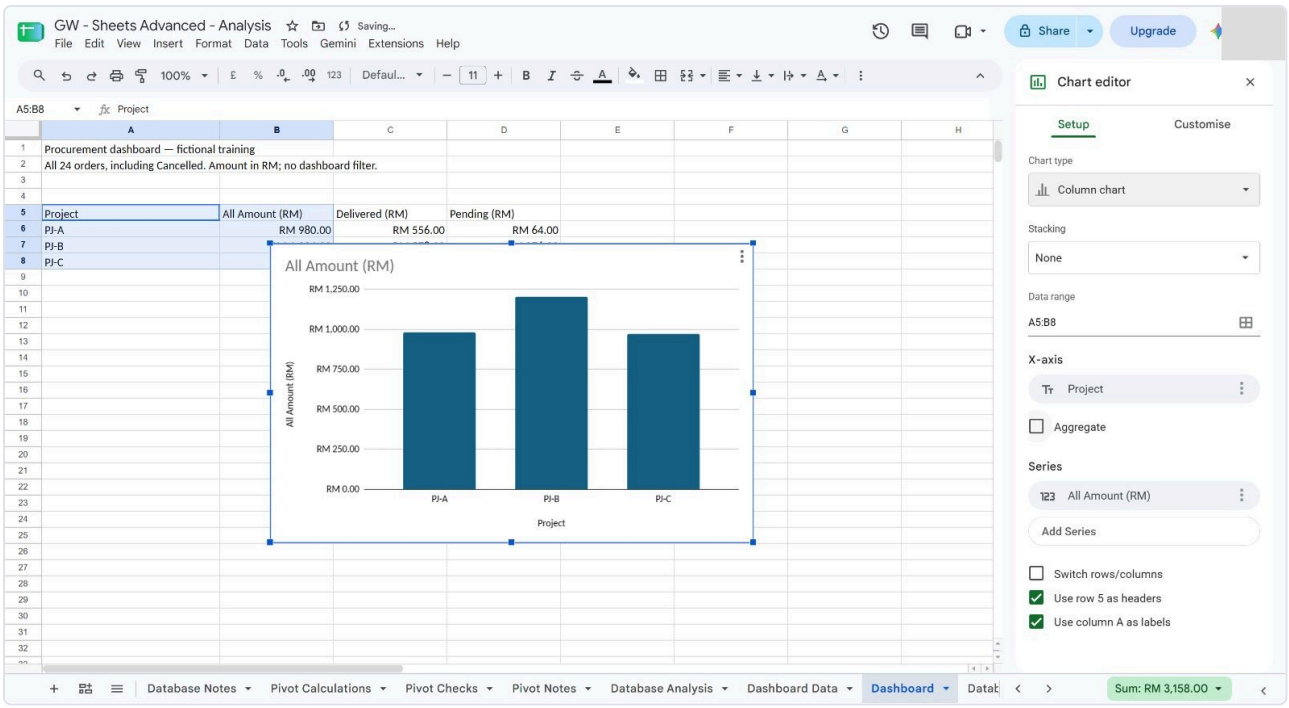
1. Create **Dashboard Data** separately from the maintained Analysis tab. Link A2:A4 to the three project labels PJ-A/PJ-B/PJ-C. In B2:B4 sum each checked Analysis project row across Delivered/Pending/Cancelled; C links Delivered and D links Pending. All project amounts are **980 /1204 /974**. Keep **Cancelled included** explicit.



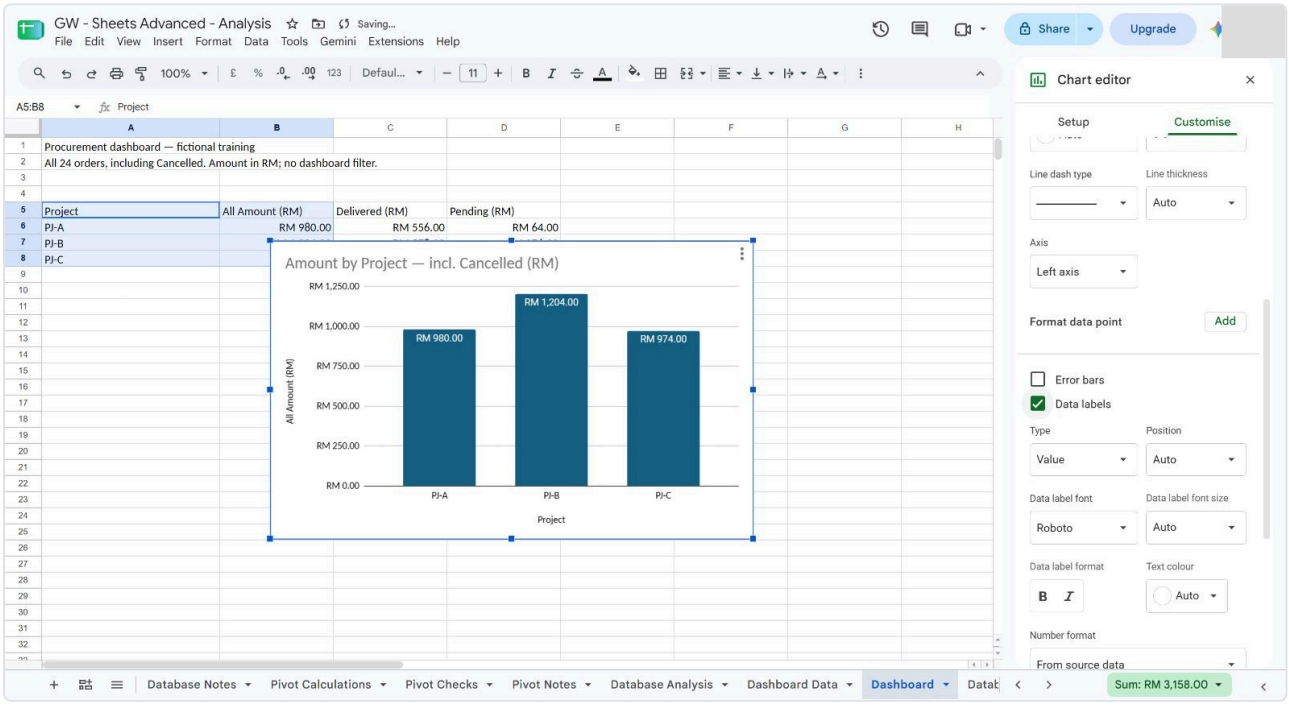
2. Create **Dashboard**. Set column A to220 pixels and B:H to140. Add the title and scope label. Link **A5:D8** to Dashboard Data A1:D4; this is a formula-linked view rather than copied static numbers.



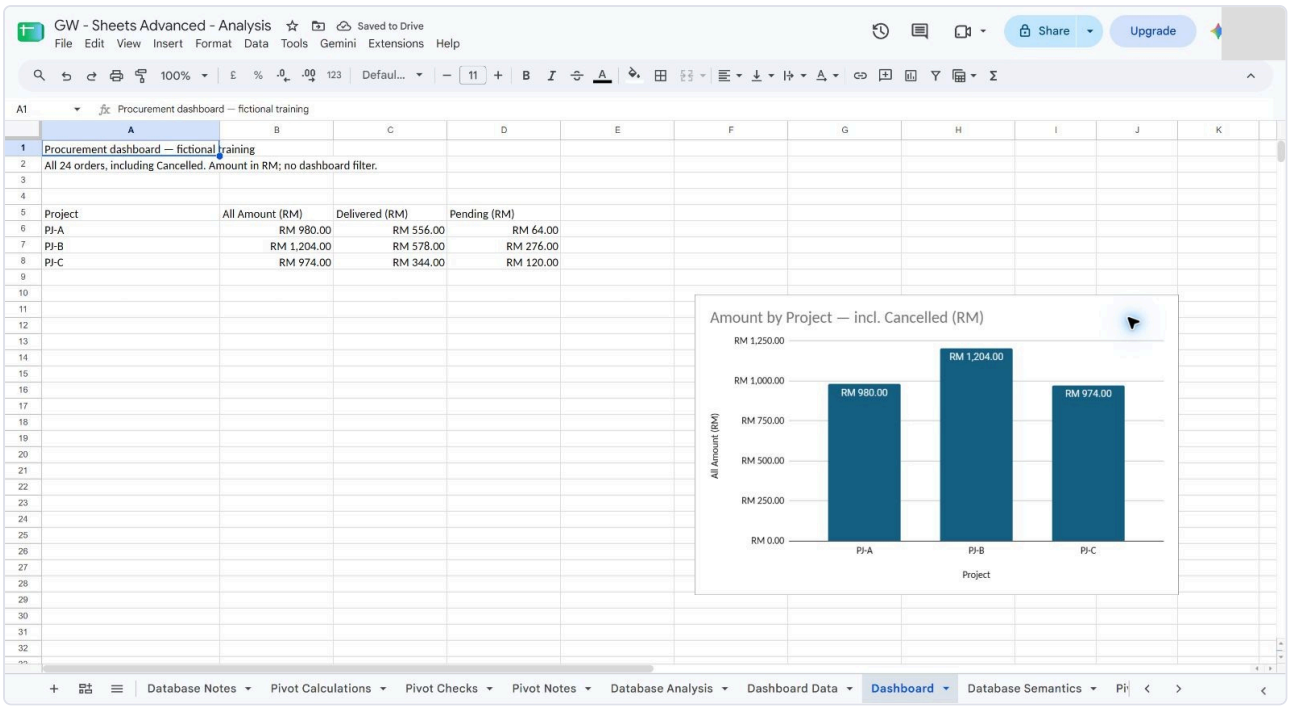
3. Select Dashboard **A5:B8 > Insert chart**. In the Chart editor Setup, choose **Column chart**, retaining header row5 and project labels in column A. The automatic initial suggestion may be Pie; change the type explicitly.



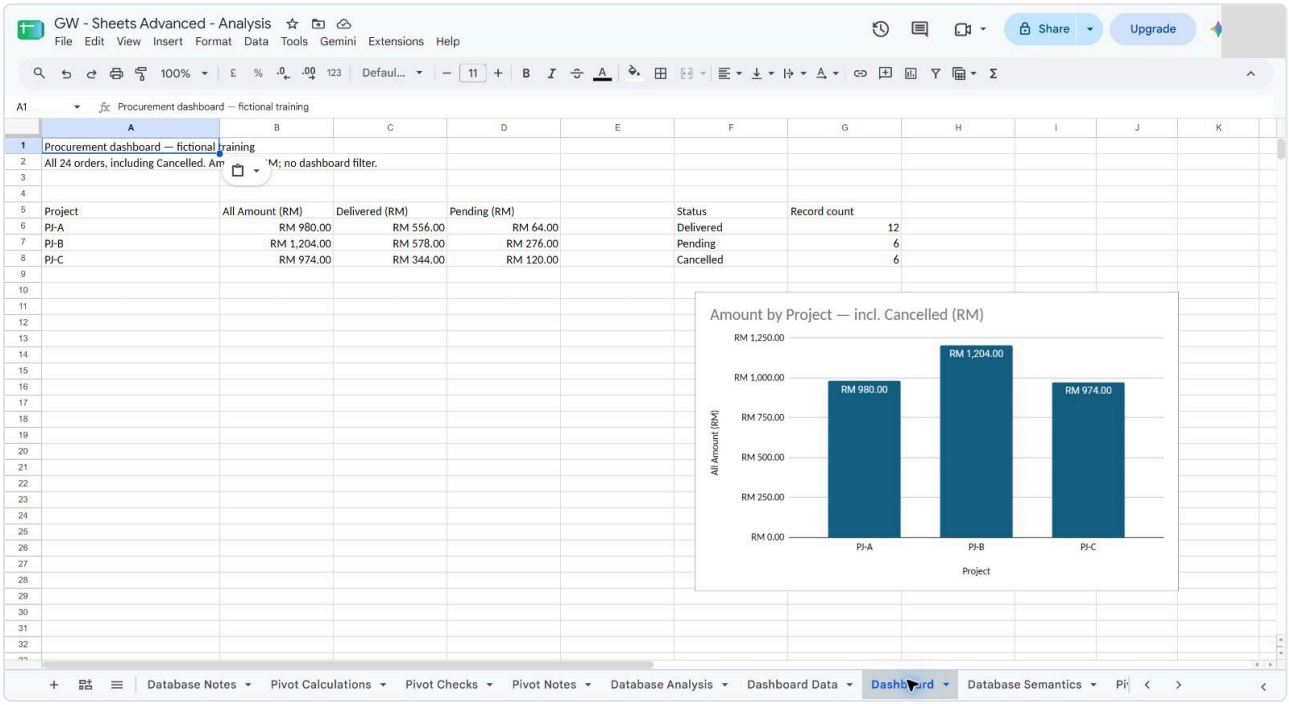
4. Under **Customise > Chart and axis titles**, set **Amount by Project - incl. Cancelled (RM)**. Under **Series**, enable **Data labels**. The captured title uses an em dash separator; its meaning and population are the same.



5. Select the chart border and drag its blank upper margin to the right of the working tables. Verify all three amount labels are visible and that the chart does not cover source cells or later controls.

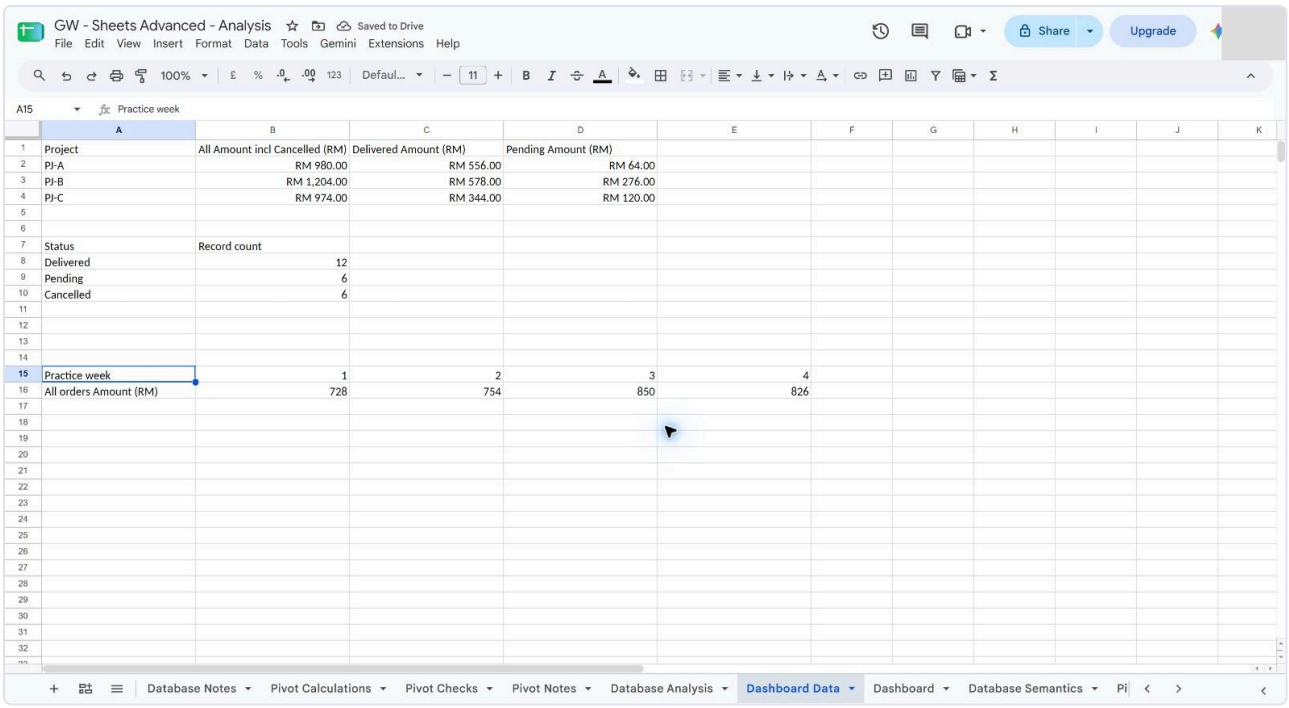


6. In Dashboard Data A7:B10 create a status record-count table using **COUNTIF(Orders!$G$2:$G$25,A8)** and corresponding status labels. Link it to Dashboard F5:G8. **Delivered12 /Pending6 /Cancelled6** reconcile to24 records.



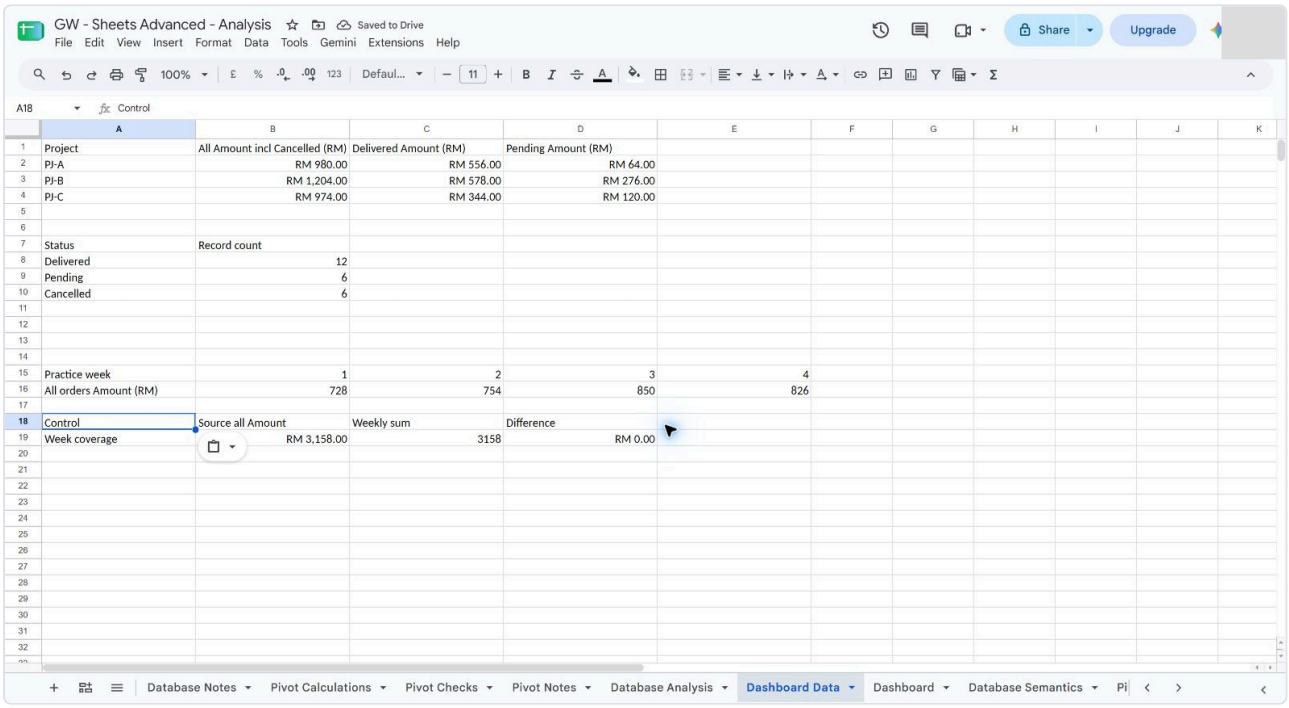
9.2 Show the Weekly Sequence

1. Use **Dashboard Data A15:E16**, because the draft's Analysis B16:E16 cells already contain earlier work. Put numeric week labels1/2/3/4 in B15:E15. B16 is **=SUMIFS(Orders!$J$2:$J$25,Orders!$H$2:$H$25,B$15)**; fill across. The four practice-week totals are **728 /754 /850 /826**.



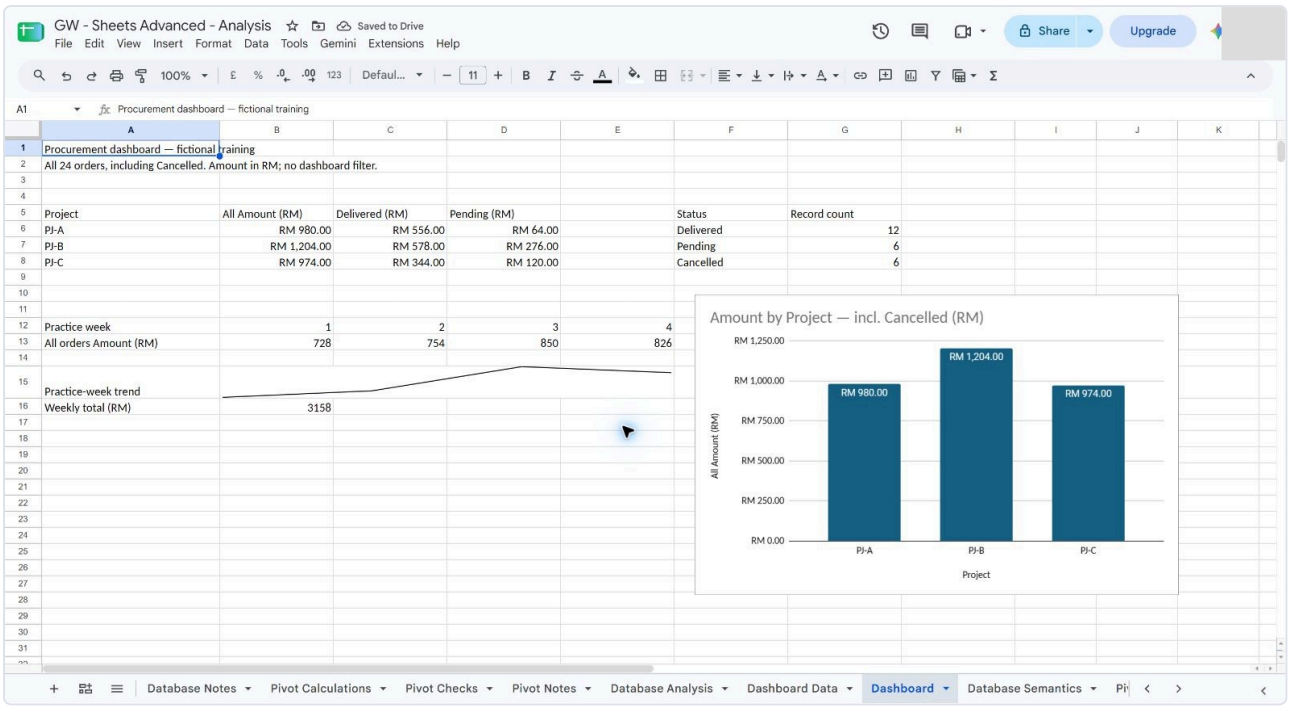
	A	B	C	D	E
1	Project	All Amount incl Cancelled (RM)	Delivered Amount (RM)	Pending Amount (RM)	
2	PJ-A	RM 980.00	RM 556.00	RM 64.00	
3	PJ-B	RM 1,204.00	RM 578.00	RM 276.00	
4	PJ-C	RM 974.00	RM 344.00	RM 120.00	
5					
6					
7	Status	Record count			
8	Delivered	12			
9	Pending	6			
10	Cancelled	6			
11					
12					
13					
14					
15	Practice week	1	2	3	4
16	All orders Amount (RM)	728	754	850	826

2. Add the A18:D19 control: source SUM of Orders J2:J25 and SUM of B16:E16 both return **3158**; their difference is **0**. These weeks are numeric practice periods, not an invented calendar.



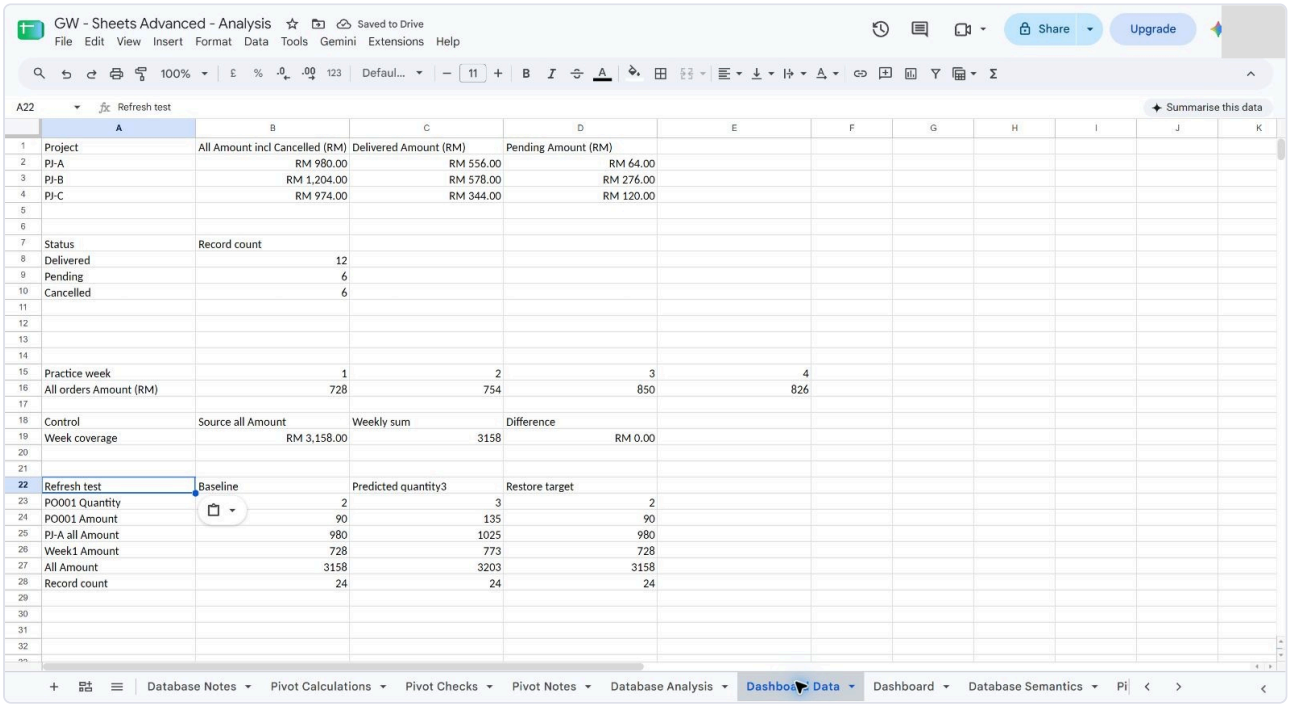
	A	B	C	D	E
1	Project	All Amount incl Cancelled (RM)	Delivered Amount (RM)	Pending Amount (RM)	
2	PJ-A	RM 980.00	RM 556.00	RM 64.00	
3	PJ-B	RM 1,204.00	RM 578.00	RM 276.00	
4	PJ-C	RM 974.00	RM 344.00	RM 120.00	
5					
6					
7	Status	Record count			
8	Delivered	12			
9	Pending	6			
10	Cancelled	6			
11					
12					
13					
14					
15	Practice week	1	2	3	4
16	All orders Amount (RM)	728	754	850	826
17					
18	Control	Source all Amount	Weekly sum	Difference	
19	Week coverage	RM 3,158.00	3158	RM 0.00	

3. Link Dashboard B12:E13 to the week labels and amounts. Merge B15:E15 and enter **=SPARKLINE('Dashboard Data'!B16:E16)** in B15. Set row15 height40 and label the trend in A15. B16 **=SUM(B13:E13)** returns3158.

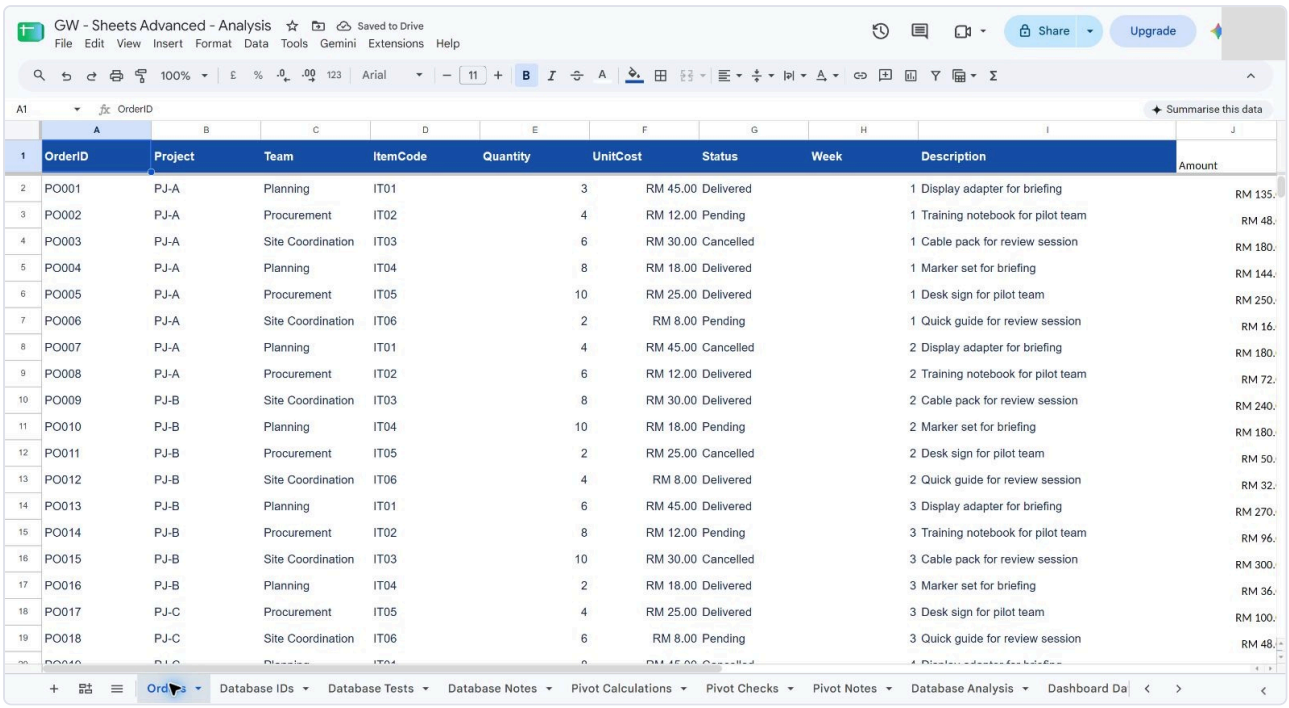


9.3 Test a Refresh

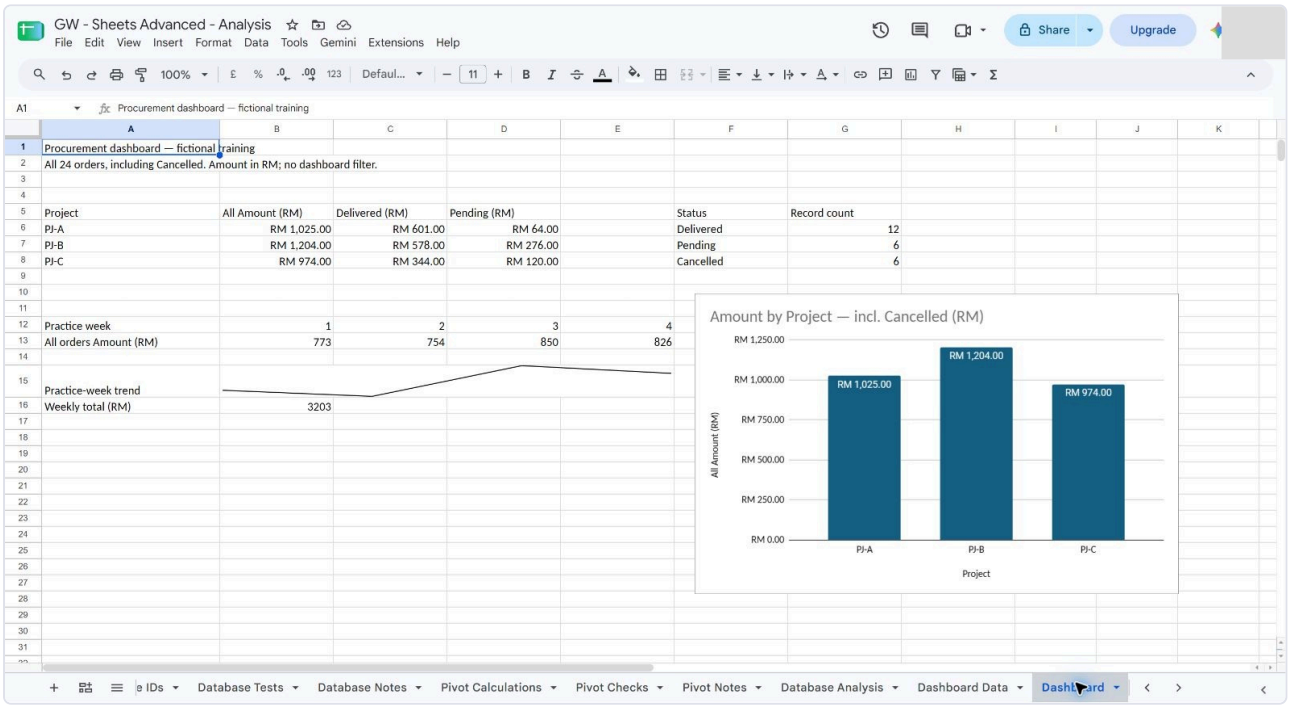
1. Before changing source, record the predicted **+45** delta for PO001 Quantity **2 to3** at UnitCost45. Expected line135, PJ-A all amount1025, Week 1 amount 773, total3203 and unchanged record count24.



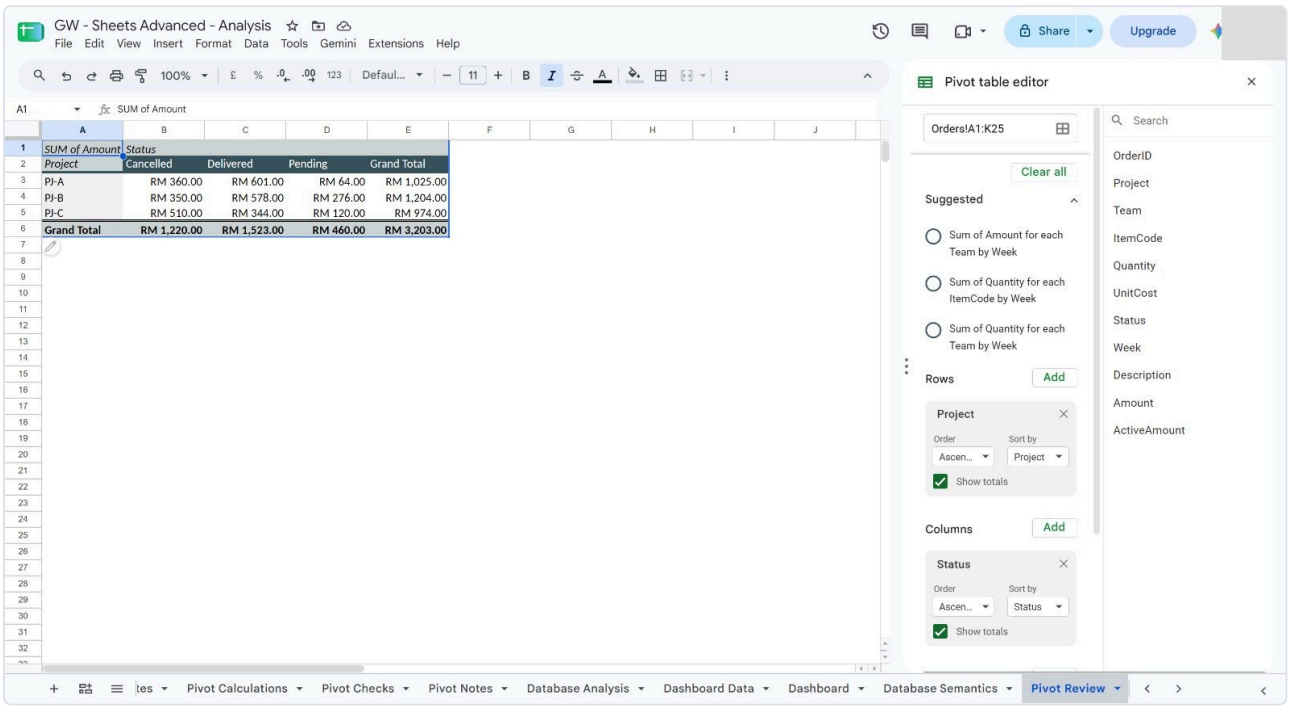
2. On maintained Orders, change **E2 to3**. Inspect the PO001 source line and its calculated **Amount135**. This is a temporary change to an existing record, not a new-row coverage test.



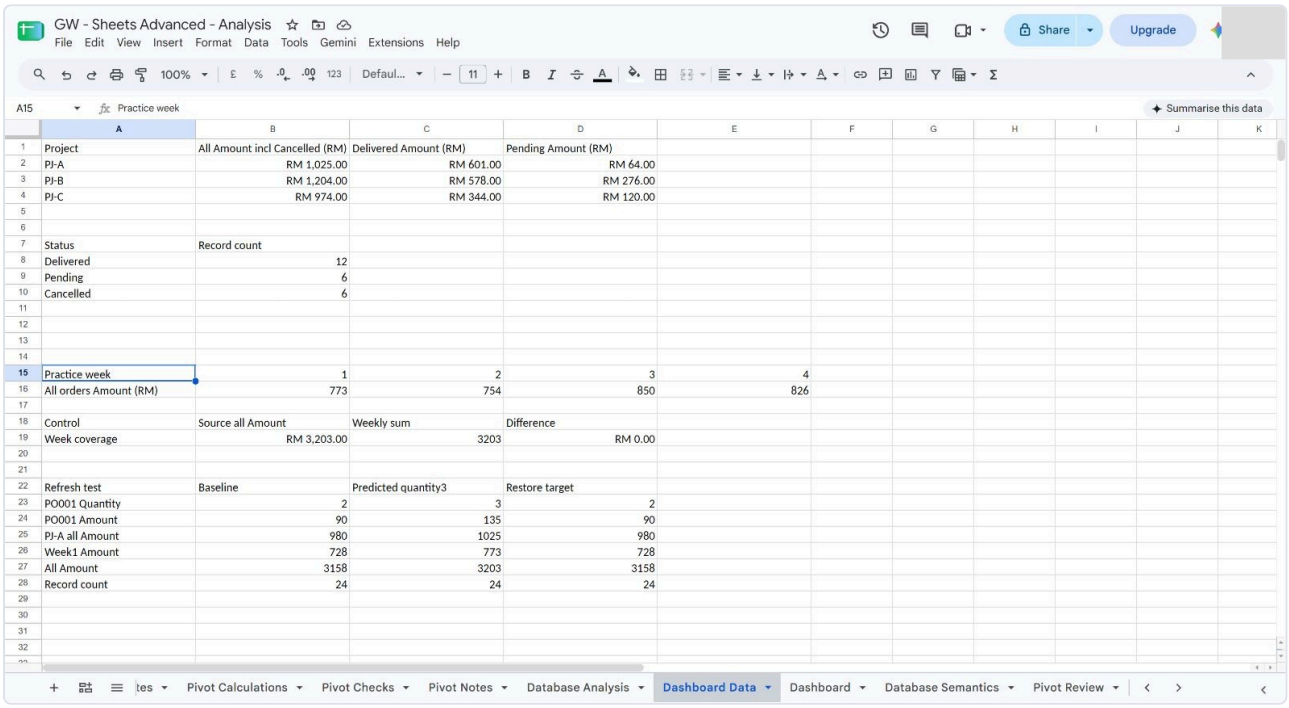
3. Inspect Dashboard. The project table and chart update PJ-A to **1025**; Week 1 becomes **773**, the sparkline shape changes and the weekly total becomes **3203**.



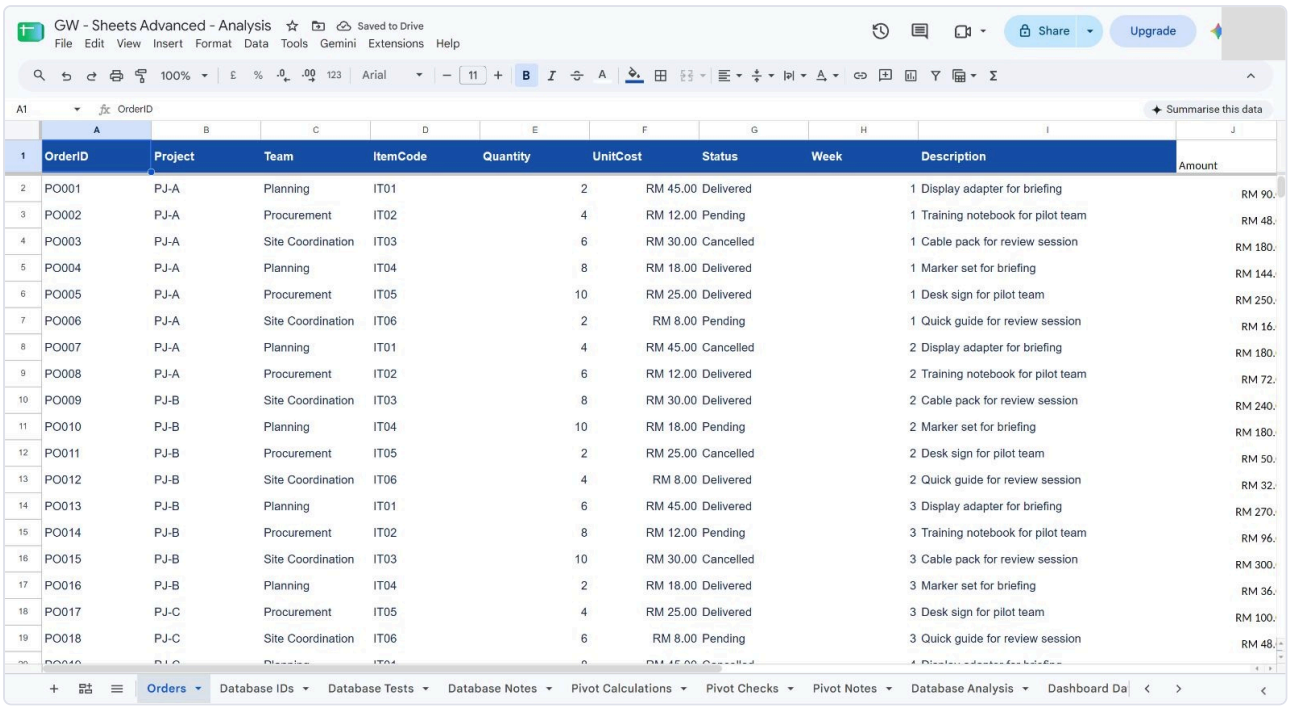
4. Inspect the existing native **Pivot Review**: its all-record total also becomes3203 without a manual refresh in this captured test. Its source bounds remain the same24-record table.



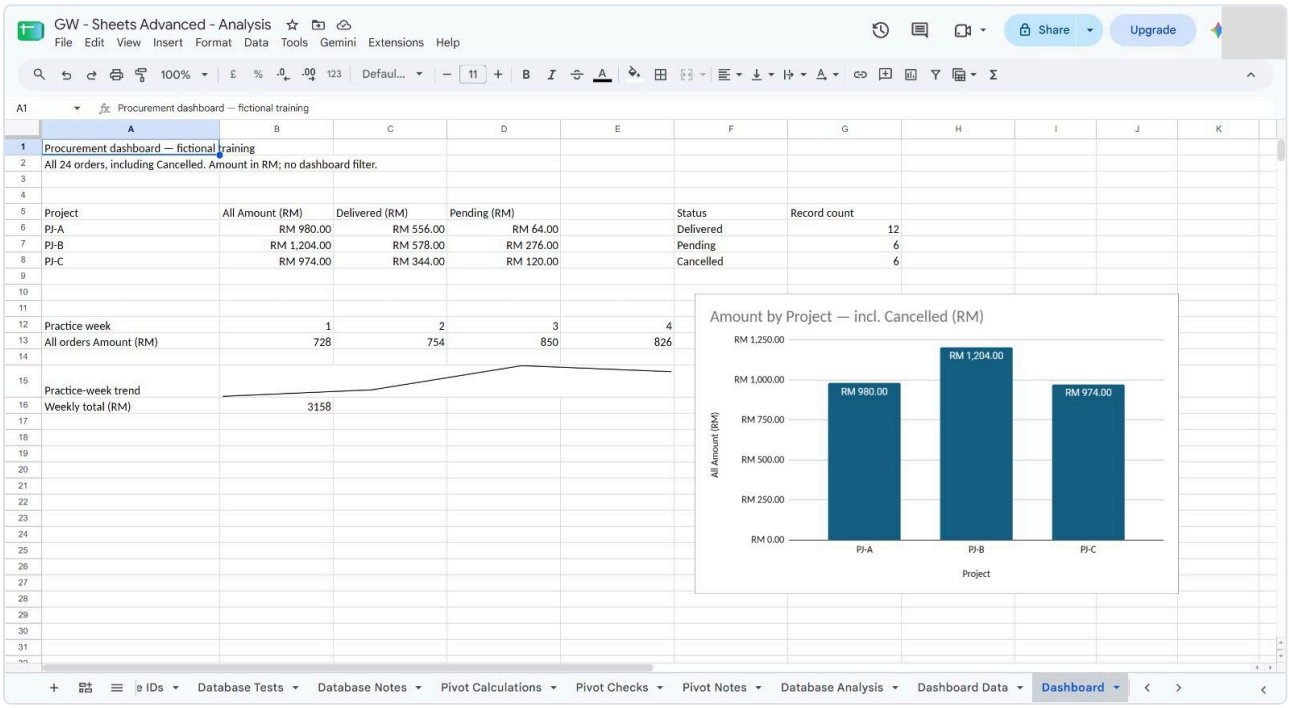
5. Inspect Dashboard Data's weekly totals and source comparison. Both totals are3203 and the difference remains0. Preserve the transient observation before restoration.



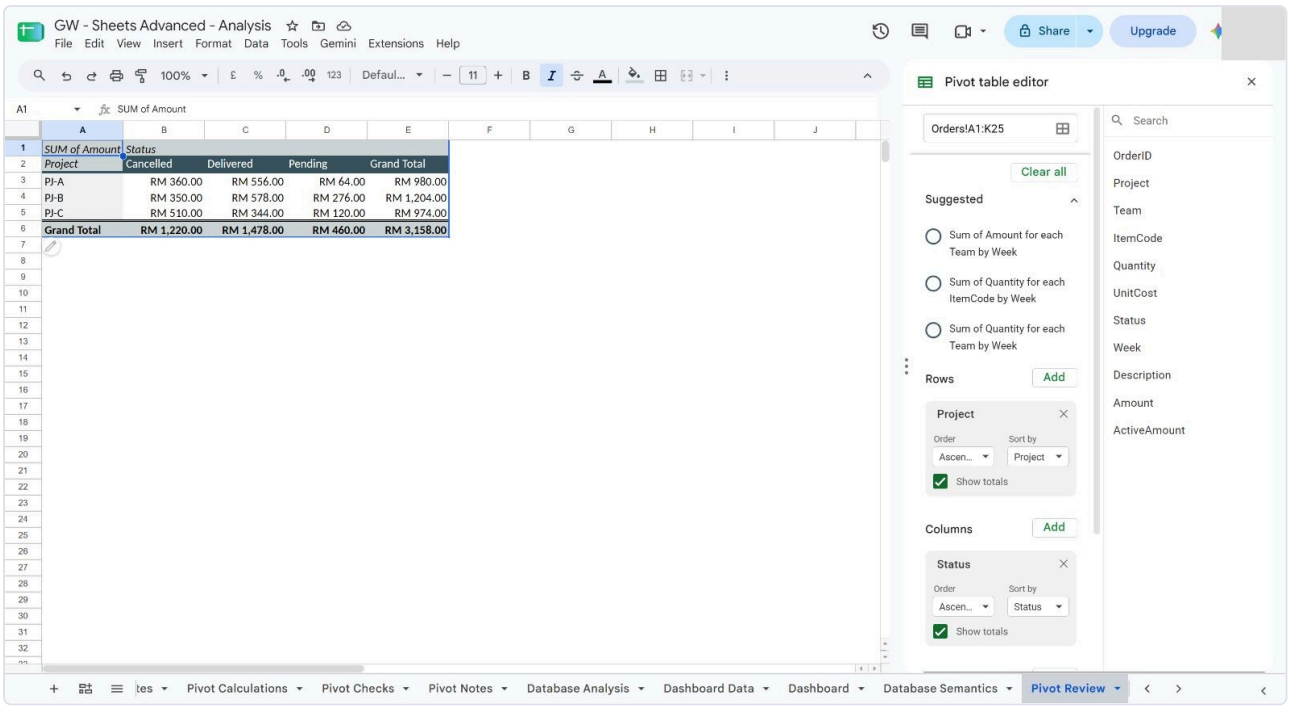
6. Restore **Orders E2 to2**, returning the line Amount to90.



7. Verify Dashboard returns to PJ-A980, Week 1 = 728 and total3158; the chart and sparkline return to baseline.

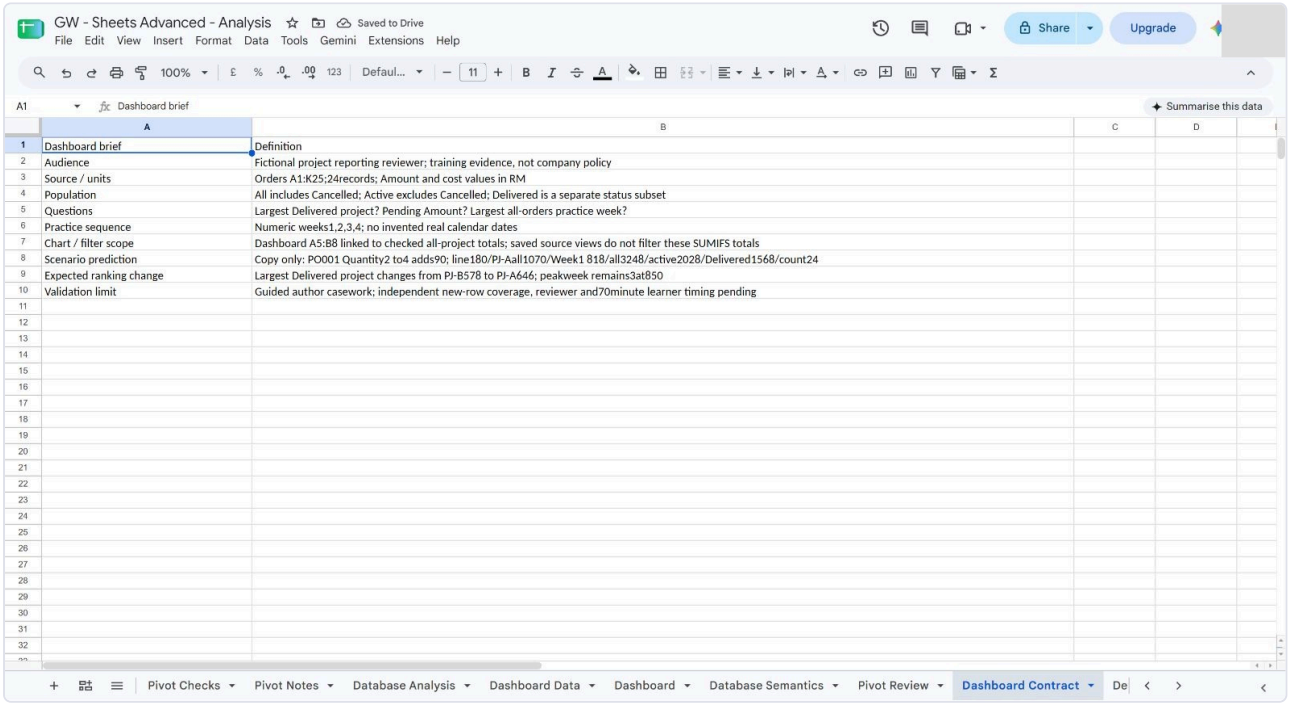


8. Confirm Pivot Review also returns to **3158**. Existing-row value refresh was automatic in this run; this does not establish automatic range extension for appended records.



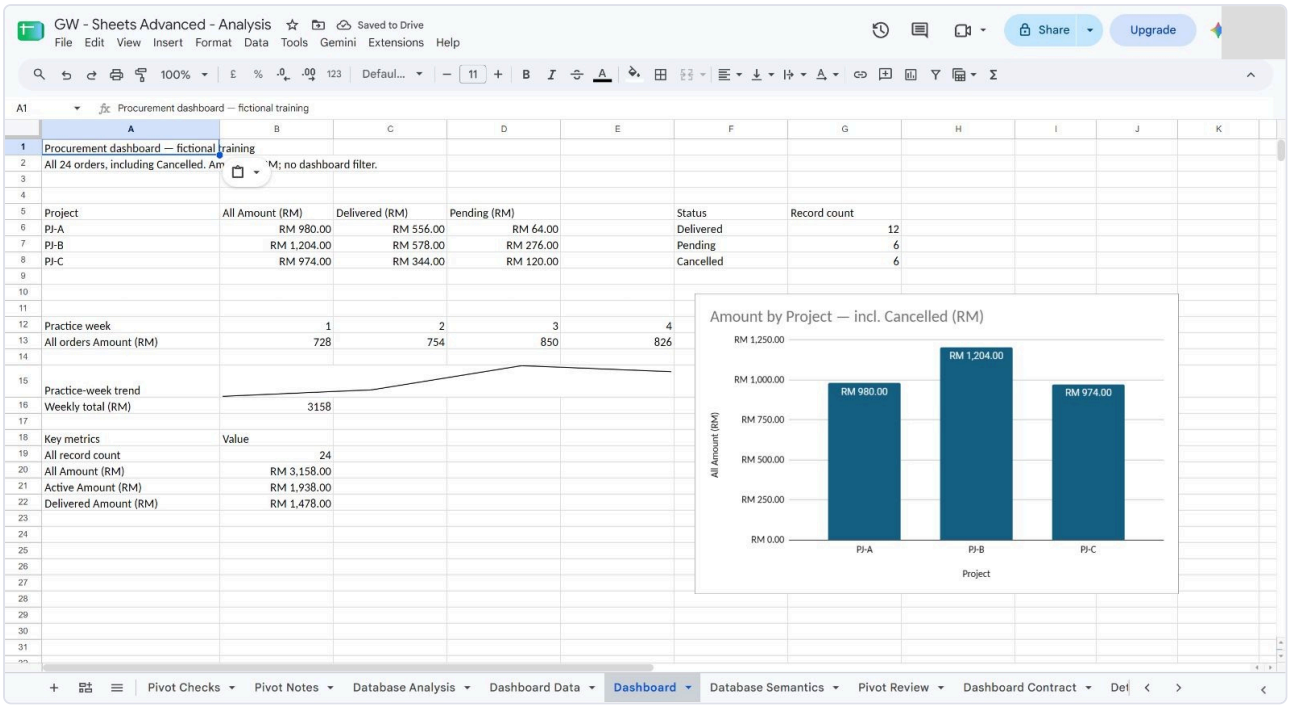
9.4 Design a Dashboard With a Visible Refresh Contract

1. Create **Dashboard Contract** and record the audience, source **Orders A1:K25**, units RM,24-record population and practice weeks. Keep **all orders including Cancelled**, **ActiveAmount excluding Cancelled**, and **Delivered only** distinct. Predict the separate copy test before editing: Quantity2 to4 adds90, giving line180, PJ-A = 1,070, Week 1 = 818, total3248, Active2028 and Delivered1568. Count remains24.

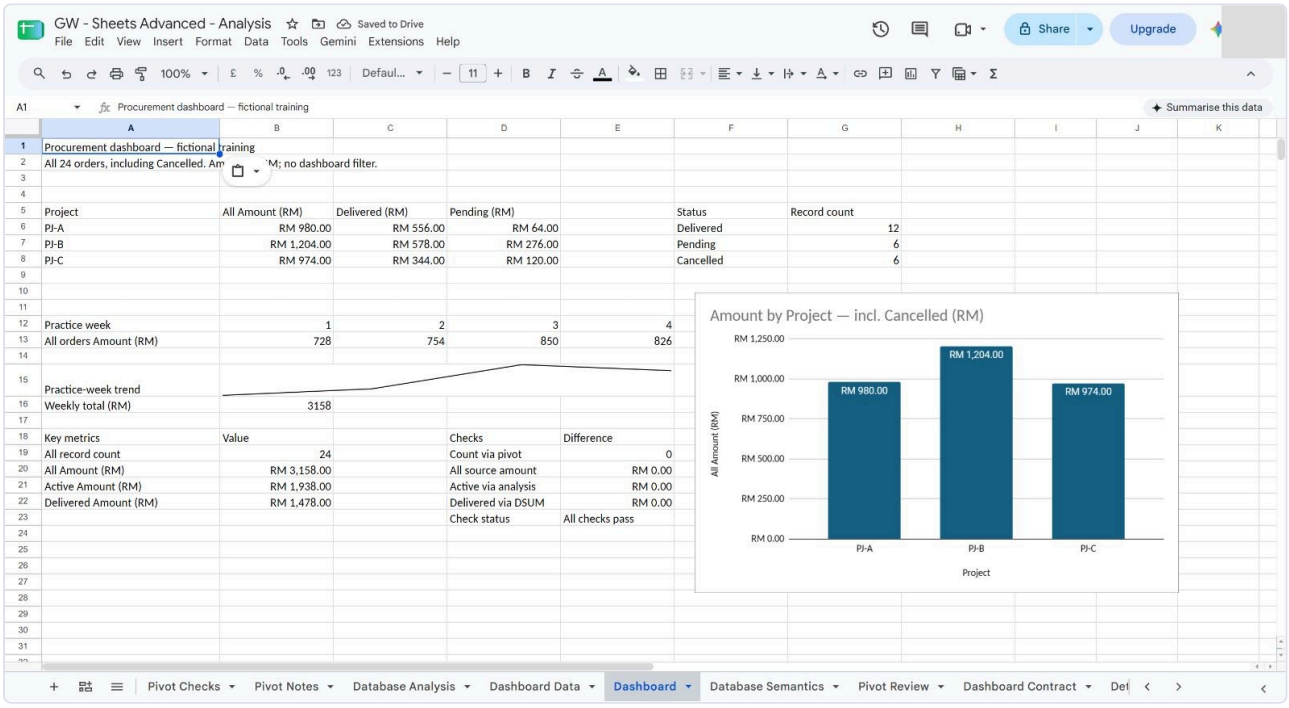


2. Add Dashboard A18:B22 KPI formulas: **COUNTA(Orders!A2:A25)=24**, **SUM('Dashboard Data'!B2:B4)=3158**, **SUM(Orders!K2:K25)=1938**, and **SUM('Dashboard Data'!C2:C4)=1478**. Label

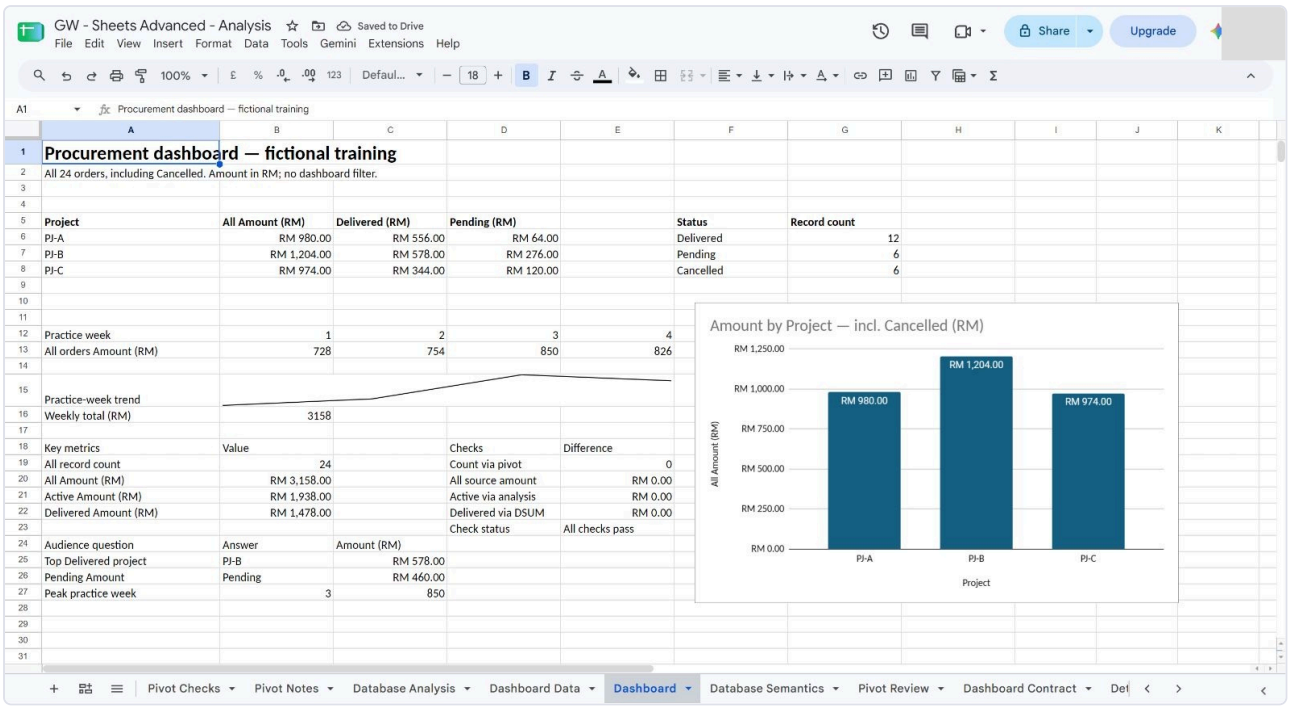
each population.



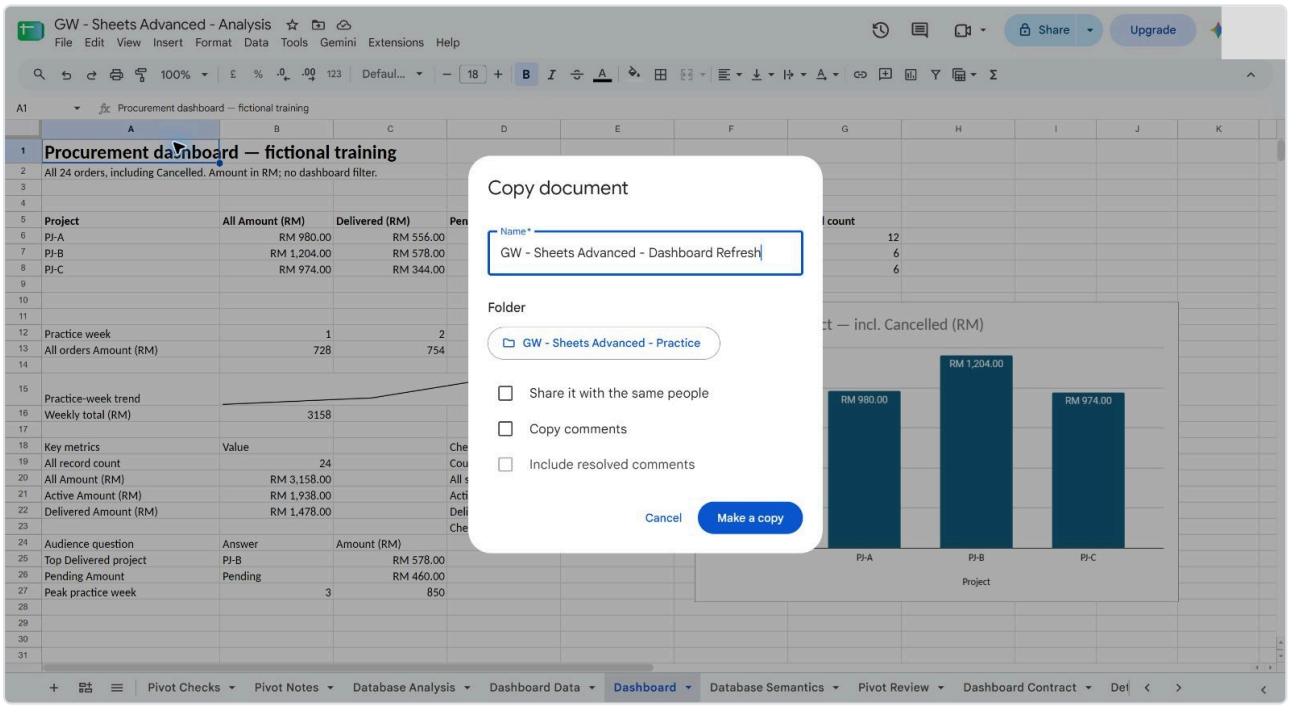
3. Add D18:E23 **Checks**. Compare count with Pivot Counts E6, all amount with source SUM J2:J25, active with Analysis C15, and Delivered with Database Analysis D10. All four differences are0. E23 uses **=IF(COUNTIF(E19:E22,0)=4,"All checks pass","Review mismatch")**; it is a recalculating result, not typed assurance.



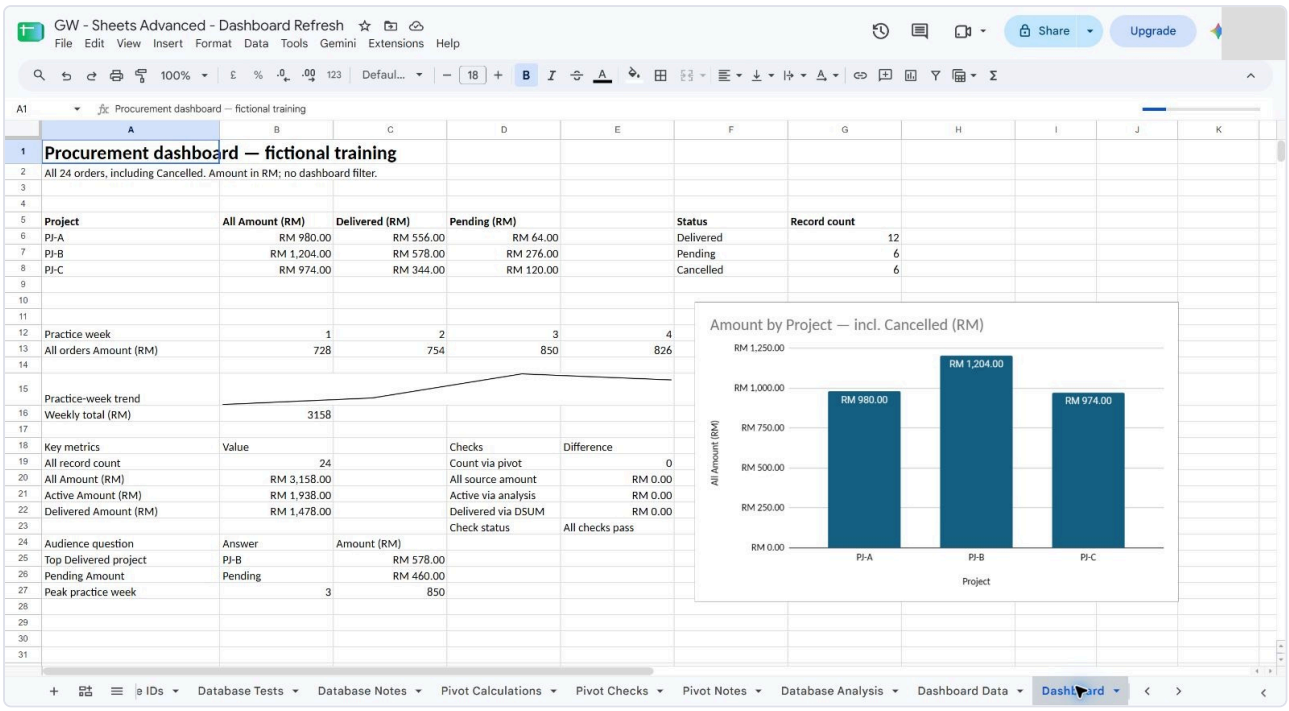
4. Answer the audience questions using formulas in A24:C27. Largest Delivered project is **PJ-B = 578**; total Pending Amount is **460**; largest practice week is **3 at850**. Keep the chart's all-orders comparison distinct from the Delivered ranking.



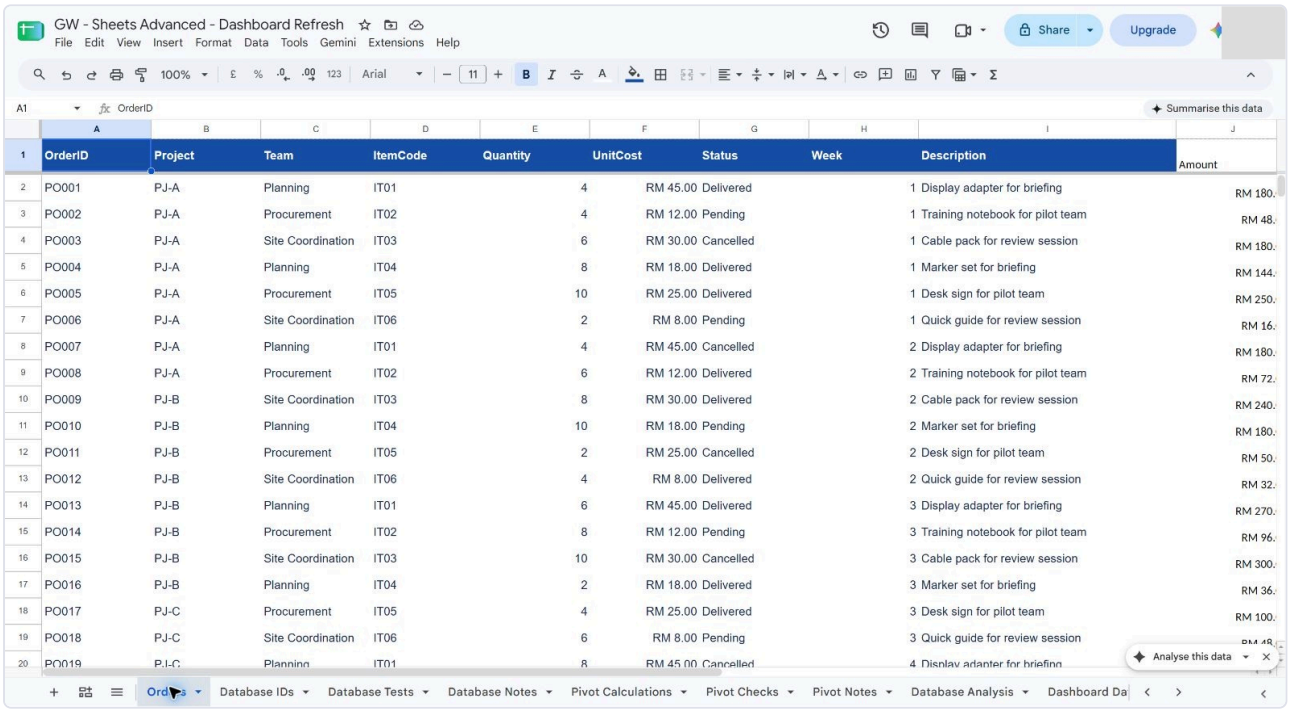
5. Choose **File > Make a copy**, name it **GW - Sheets Advanced - Dashboard Refresh**, and retain the default practice folder. Leave sharing and comment-copy boxes unchecked.



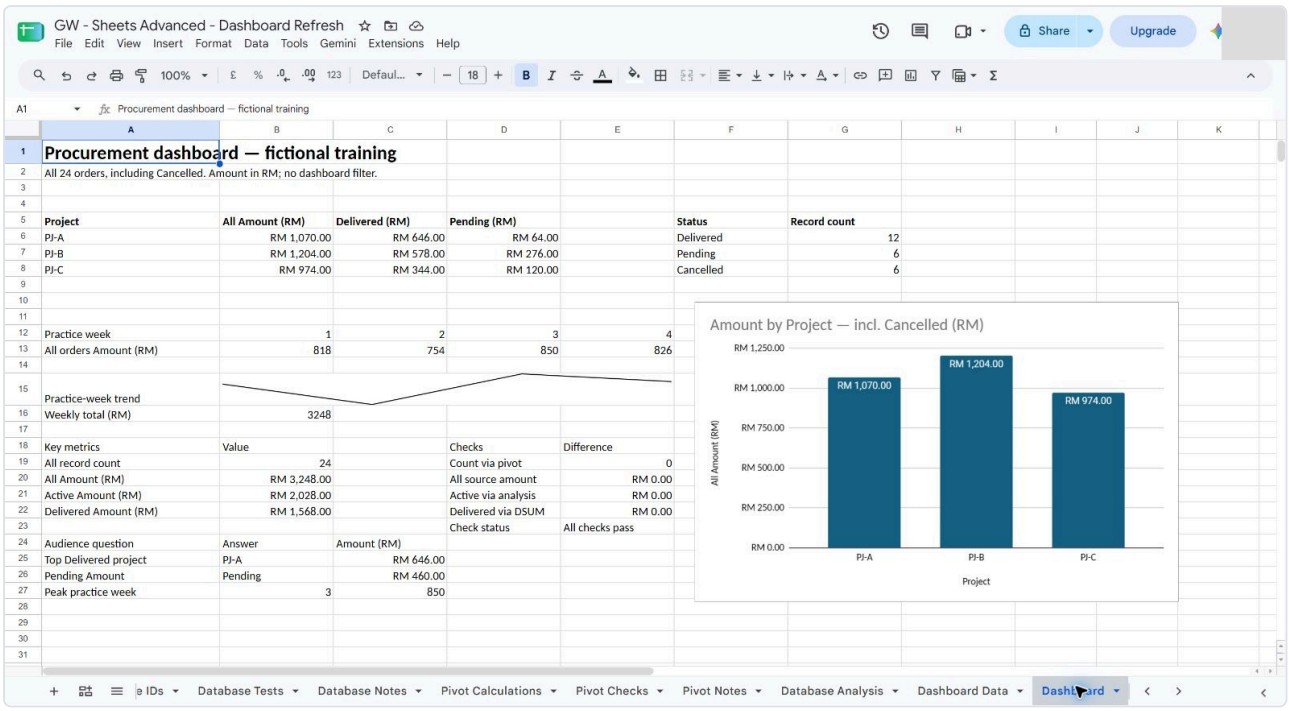
6. Open the new workbook and inspect Dashboard before editing. It starts at the same24/3158/1938/1478 baseline and has its own source, formulas and chart.



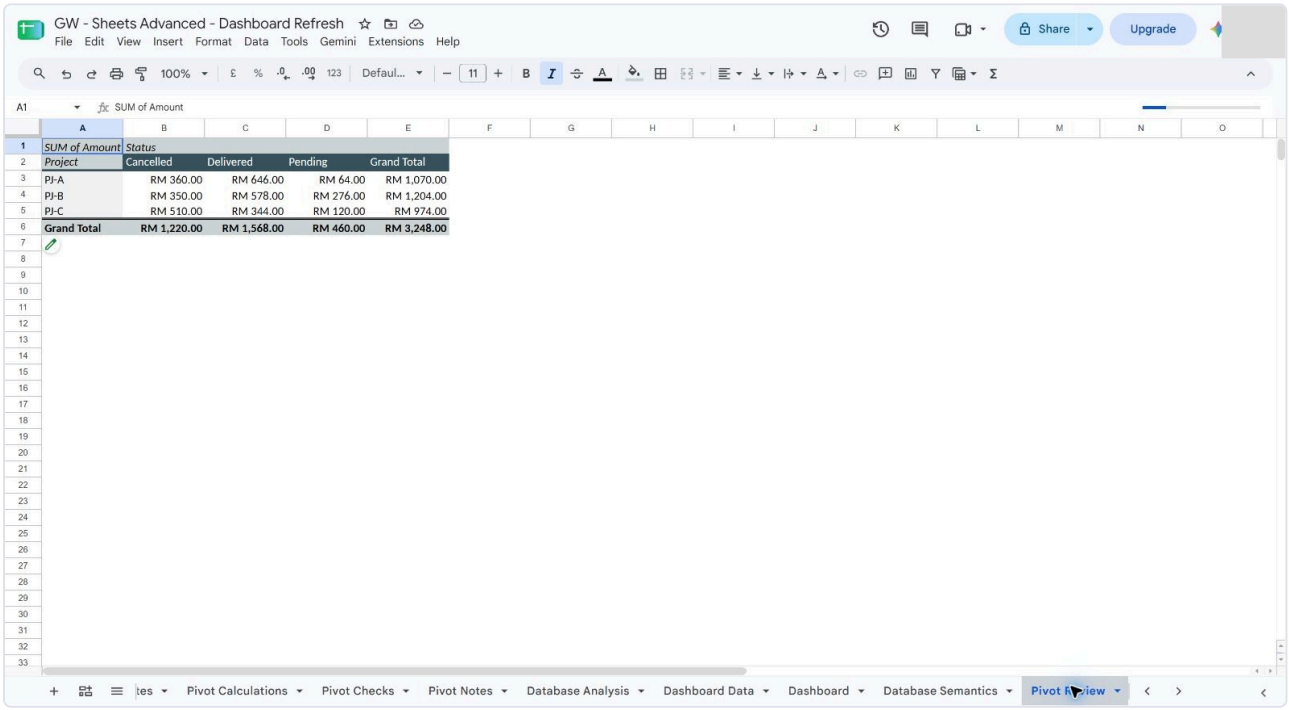
7. In this **copy only**, change Orders **E2 to4**. PO001 Amount becomes **180**, a90 increase.



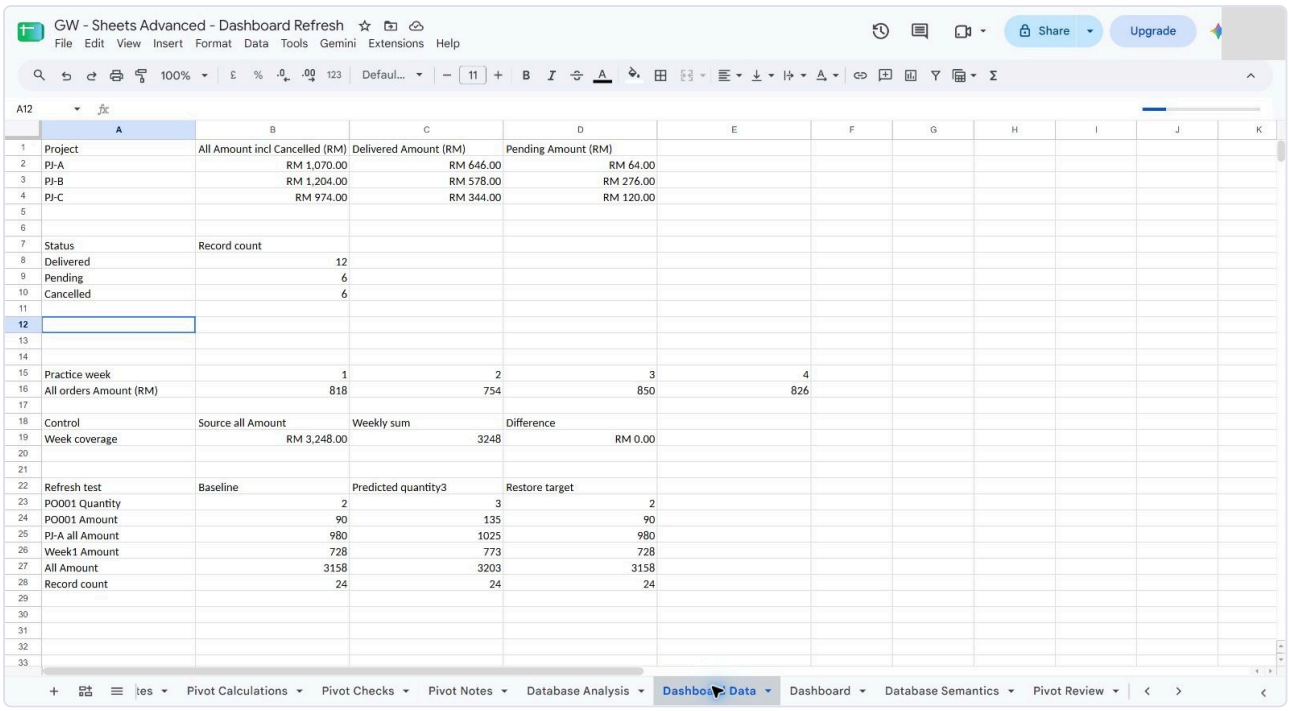
8. Inspect the copied dashboard: PJ-A all amount **1070**, Week1 **818**, all amount **3248**, active **2028**, Delivered **1568**, count24 and all Checks differences0. The chart label and sparkline change. Largest Delivered project changes to **PJ-A = 646**; Pending remains460 and week 3 remains largest at850.



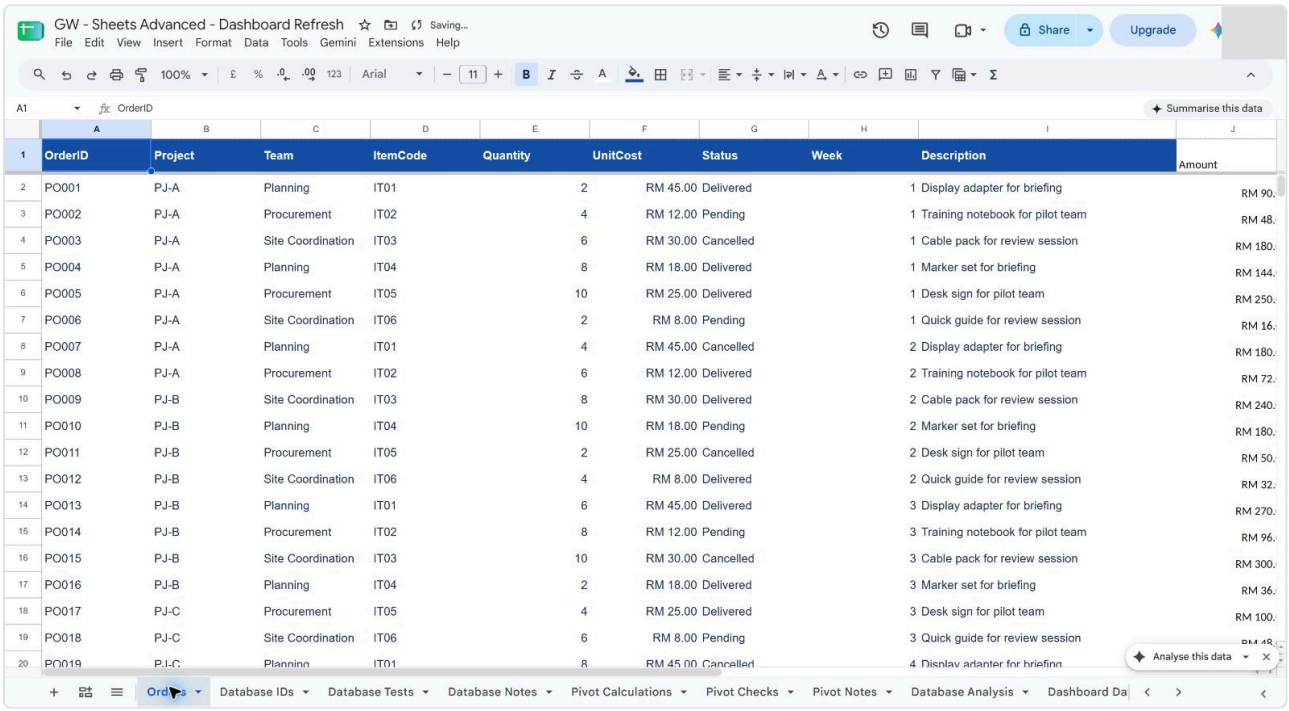
9. Check the copied native Pivot Review total **3248**. Pivot Counts remains24. The copy retains its native source and pivot relationships.



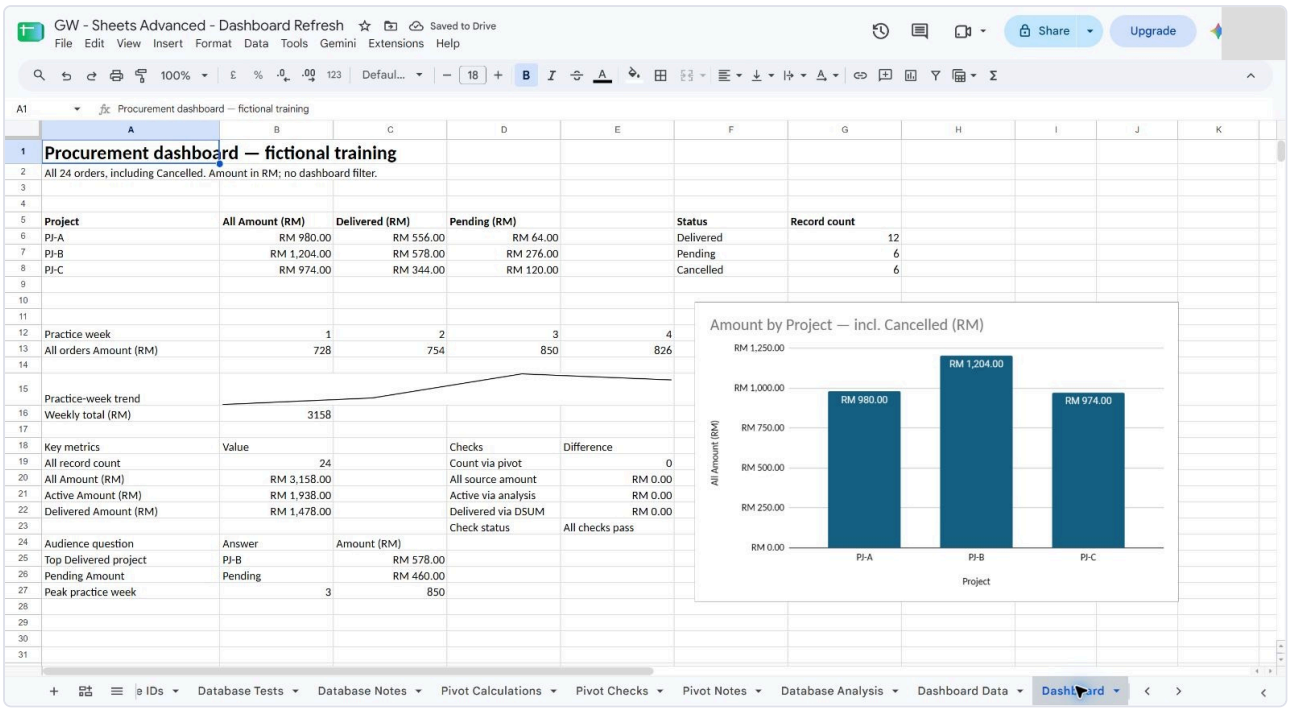
10. Check copied Dashboard Data: weekly amounts **818 /754 /850 /826**, weekly sum and source amount3248, difference0.



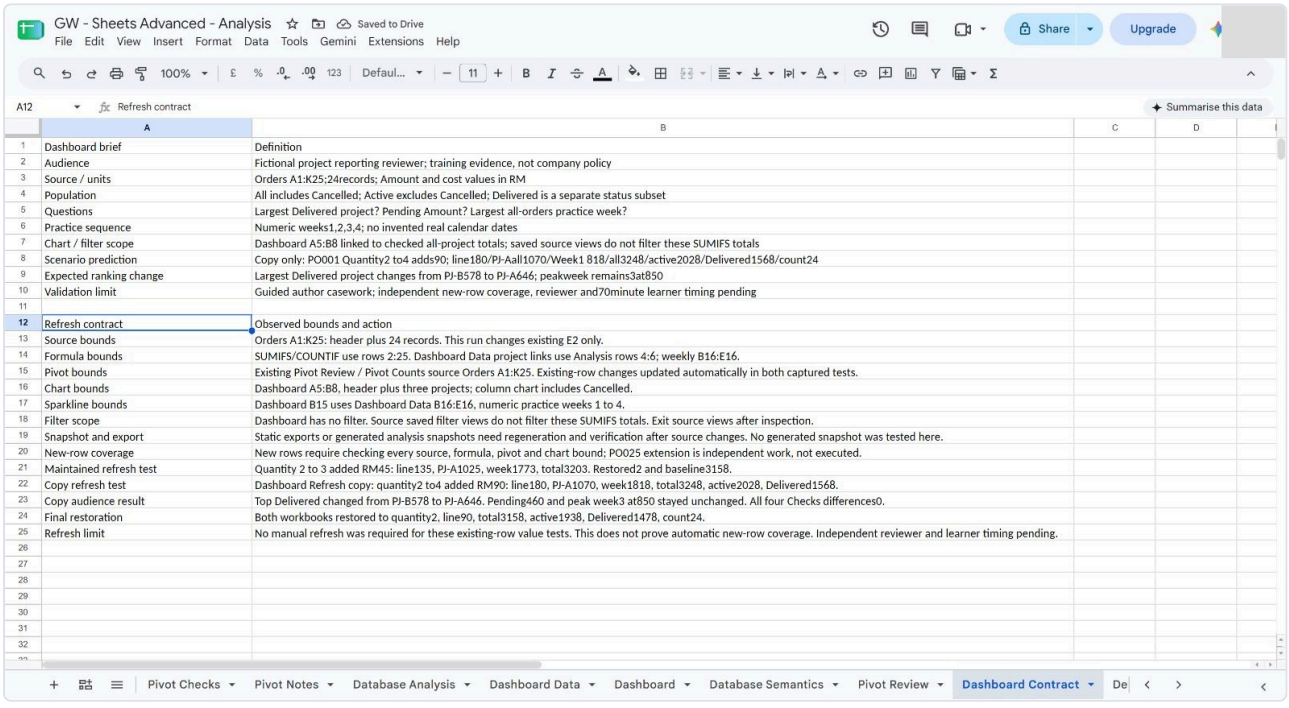
11. Restore the copy's **Orders E2 to2**, returning line Amount90.



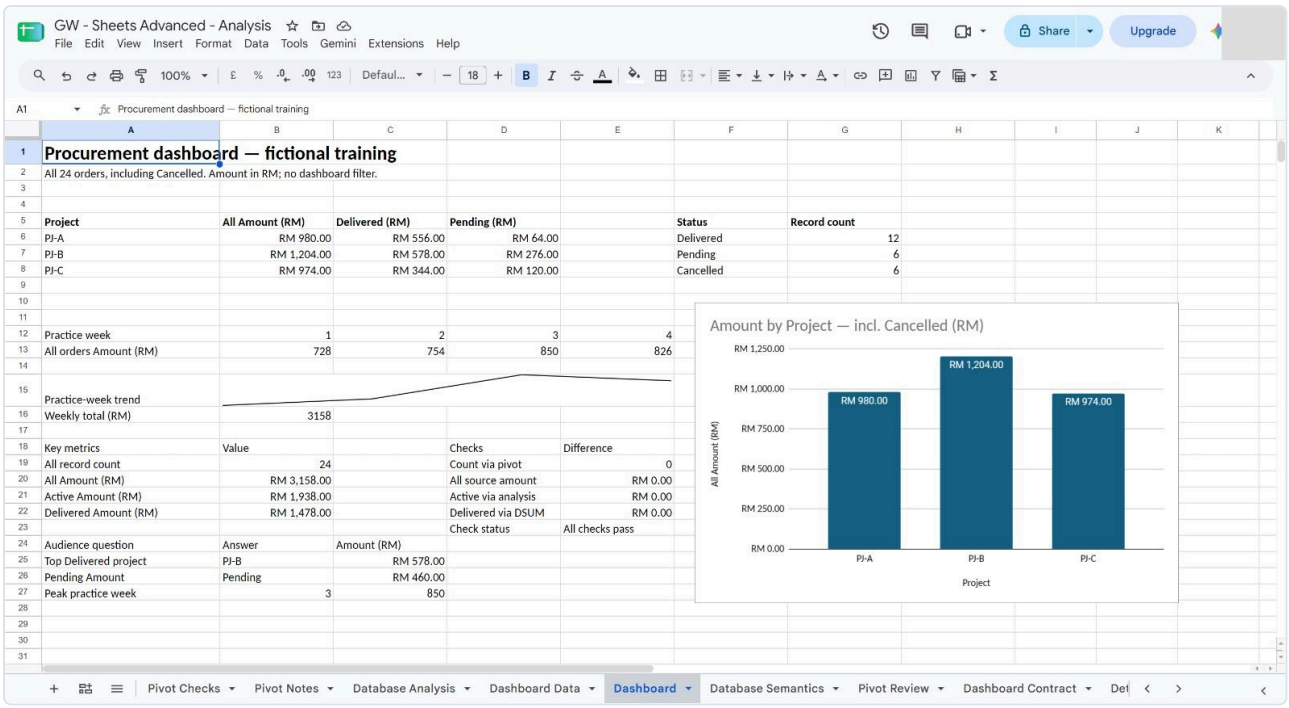
12. Confirm the copied Dashboard returns to **24 /3158 /1938 /1478**, PJ-B = 578, Pending = 460, peak week 3 at850 and four zero differences. Retain the restored copy as evidence of the controlled test.



13. Complete the visible refresh contract in both workbooks. Record source/formula/pivot bounds, **Dashboard A5:B8** chart source, **Dashboard Data B16:E16** sparkline source and no dashboard filter. Named source filter views do not filter these SUMIFS totals. Static exports or generated snapshots need regeneration and verification; no generated snapshot was tested. New records require checking every dependent range. The independent PO025 challenge below remains unperformed.



14. Finish on the maintained Dashboard and verify the reconciled baseline. Native bounded readback confirms source values/formats and earlier pivot definitions match their pre-test state. The two captured value-change tests needed no manual refresh; future row coverage and human comprehension still require separate checks.



> **Validation boundary:** guided9.1-9.3 and author casework9.4 were executed through learner UI, with read-only native evidence of both temporary changes and restoration. This is not an independent learner attempt. PO025 range extension, human reviewer and the planned70-minute classroom allocation remain unverified.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Build an independent dashboard view answering the three audience questions on the card, with at least one visible reconciliation check.
2. Append PO025 only on a challenge copy and repair source-range coverage across KPIs, pivots, charts and the weekly sequence.
3. Ask a reviewer to identify the population and a stale component using the labels/checks, then revise any misleading element.

Evidence to Hand Over

- Defined dashboard with distinct populations, units and recalculating controls.
- Source-change/restoration trace across line, summary and visuals.
- Independent new-row coverage test and a usable refresh contract.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check

A dashboard looks current but omits the new line	Trace range coverage through every KPI and visual rather than checking only one total.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

A dashboard is useful when its totals, populations and update path are understandable.

Check yourself: Which dashboard objects depend on which source calculations?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=SPARKLINE(Analysis!B16:E16)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Audience questions: Which project has the largest Delivered amount? How much Amount remains
Pending? Which practice week has the largest all-orders Amount? PO025 extension: PJ-A,
Planning, IT01, Quantity 2, UnitCost 45, Delivered, week 4.
```

Captured Author-Run Reference

Use separate **Dashboard Data**, **Dashboard**, and **Dashboard Contract** tabs. Earlier Analysis B16:E16 is occupied. Formulas below describe the final baseline; build them in guide order and record predictions before changing source. All input data is fictional.

Dashboard Data A1:D4 - project view

```
Project All Amount incl Cancelled (RM)  Delivered Amount (RM)   Pending Amount (RM)
PJ-A    =SUM(Analysis!B4:D4)    =Analysis!B4    =Analysis!C4
PJ-B    =SUM(Analysis!B5:D5)    =Analysis!B5    =Analysis!C5
PJ-C    =SUM(Analysis!B6:D6)    =Analysis!B6    =Analysis!C6
```

Dashboard Data A7:B10 - status counts

```
Status  Record count
Delivered       =COUNTIF(Orders!$G$2:$G$25,A8)
Pending =COUNTIF(Orders!$G$2:$G$25,A9)
Cancelled       =COUNTIF(Orders!$G$2:$G$25,A10)
```

Dashboard Data A15:E16 - numeric weeks

```
Practice week   1       2       3       4
All orders Amount (RM)  =SUMIFS(Orders!$J$2:$J$25,Orders!$H$2:$H$25,B$15)
=SUMIFS(Orders!$J$2:$J$25,Orders!$H$2:$H$25,C$15)
=SUMIFS(Orders!$J$2:$J$25,Orders!$H$2:$H$25,D$15)
=SUMIFS(Orders!$J$2:$J$25,Orders!$H$2:$H$25,E$15)
```

Dashboard Data A18:D19 - weekly control

```
Control Source all Amount       Weekly sum      Difference
Week coverage   =SUM(Orders!J2:J25)     =SUM(B16:E16)   =C19-B19
```

Dashboard Data A22:D28 - first test predictions

```
Refresh test    Baseline        Predicted quantity3     Restore target
PO001 Quantity  2       3       2
PO001 Amount    90      135     90
PJ-A all Amount 980     1025    980
Week1 Amount    728     773     728
All Amount      3158    3203    3158
Record count    24      24      24
```

Dashboard A5:D8 - project links

```
Project All Amount (RM) Delivered (RM)  Pending (RM)
='Dashboard Data'!A2    ='Dashboard Data'!B2    ='Dashboard Data'!C2    ='Dashboard Data'!D2
='Dashboard Data'!A3    ='Dashboard Data'!B3    ='Dashboard Data'!C3    ='Dashboard Data'!D3
='Dashboard Data'!A4    ='Dashboard Data'!B4    ='Dashboard Data'!C4    ='Dashboard Data'!D4
```

Dashboard F5:G8 - status links

```
Status  Record count
='Dashboard Data'!A8    ='Dashboard Data'!B8
='Dashboard Data'!A9    ='Dashboard Data'!B9
='Dashboard Data'!A10   ='Dashboard Data'!B10
```

Dashboard A12:E16 - weekly links and sparkline

```
Practice week   1       2       3       4
All orders Amount (RM)  ='Dashboard Data'!B16   ='Dashboard Data'!C16   ='Dashboard Data'!D16
='Dashboard Data'!E16

Practice-week trend     =SPARKLINE('Dashboard Data'!B16:E16)
Weekly total (RM)       =SUM(B13:E13)
```

Dashboard A18:B22 - KPIs

```
Key metrics     Value
All record count        =COUNTA(Orders!A2:A25)
All Amount (RM) =SUM('Dashboard Data'!B2:B4)
Active Amount (RM)      =SUM(Orders!K2:K25)
Delivered Amount (RM)   =SUM('Dashboard Data'!C2:C4)
```

Dashboard D18:E23 - live Checks

```
Checks  Difference
Count via pivot =B19-'Pivot Counts'!E6
All source amount       =B20-SUM(Orders!J2:J25)
Active via analysis     =B21-Analysis!C15
Delivered via DSUM      =B22-'Database Analysis'!D10
Check status    =IF(COUNTIF(E19:E22,0)=4,"All checks pass","Review mismatch")
```

Dashboard A24:C27 - audience answers

```
Audience question       Answer  Amount (RM)
Top Delivered project   =INDEX(A6:A8,MATCH(MAX(C6:C8),C6:C8,0)) =MAX(C6:C8)
Pending Amount  Pending =SUM(D6:D8)
Peak practice week      =INDEX(B12:E12,1,MATCH(MAX(B13:E13),B13:E13,0)) =MAX(B13:E13)
```

Dashboard Contract A1:B10 - brief

```
Dashboard brief Definition
Audience        Fictional project reporting reviewer; training evidence, not company policy
Source / units  Orders A1:K25;24records; Amount and cost values in RM
Population      All includes Cancelled; Active excludes Cancelled; Delivered is a separate
status subset
Questions       Largest Delivered project? Pending Amount? Largest all-orders practice week?
Practice sequence       Numeric weeks1,2,3,4; no invented real calendar dates
Chart / filter scope    Dashboard A5:B8 linked to checked all-project totals; saved source
views do not filter these SUMIFS totals
Scenario prediction     Copy only: PO001 Quantity2 to4 adds90; line180/PJ-Aall1070/Week1
818/all3248/active2028/Delivered1568/count24
Expected ranking change Largest Delivered project changes from PJ-B578 to PJ-A646; peakweek
remains3at850
Validation limit        Guided author casework; independent new-row coverage, reviewer
and70minute learner timing pending
```

Dashboard Contract A12:B25 - refresh bounds

```
Refresh contract        Observed bounds and action
Source bounds   Orders A1:K25: header plus 24 records. This run changes existing E2 only.
Formula bounds  SUMIFS/COUNTIF use rows 2:25. Dashboard Data project links use Analysis rows
4:6; weekly B16:E16.
Pivot bounds    Existing Pivot Review / Pivot Counts source Orders A1:K25. Existing-row
changes updated automatically in both captured tests.
Chart bounds    Dashboard A5:B8, header plus three projects; column chart includes Cancelled.
Sparkline bounds        Dashboard B15 uses Dashboard Data B16:E16, numeric practice weeks 1 to
4.
Filter scope    Dashboard has no filter. Source saved filter views do not filter these SUMIFS
totals. Exit source views after inspection.
Snapshot and export     Static exports or generated analysis snapshots need regeneration and
verification after source changes. No generated snapshot was tested here.
New-row coverage        New rows require checking every source, formula, pivot and chart
bound; PO025 extension is independent work, not executed.
Maintained refresh test Quantity 2 to 3 added RM45: line135, PJ-A1025, week1773, total3203.
Restored2 and baseline3158.
Copy refresh test       Dashboard Refresh copy: quantity2 to4 added RM90: line180, PJ-A1070,
week1818, total3248, active2028, Delivered1568.
Copy audience result    Top Delivered changed from PJ-B578 to PJ-A646. Pending460 and peak
week3 at850 stayed unchanged. All four Checks differences0.
Final restoration       Both workbooks restored to quantity2, line90, total3158, active1938,
Delivered1478, count24.
Refresh limit   No manual refresh was required for these existing-row value tests. This does
not prove automatic new-row coverage. Independent reviewer and learner timing pending.
```

Controlled refresh tests

Maintained workbook E2:2 to3, observe +45 then restore2. Separate whole-workbook copy E2:2 to4, observe +90 then restore2. Never perform the copy test on maintained source. Copy-test expected KPIs:24 /3248 /2028 /1568; chart PJ-A1070; Week1818; Delivered leader PJ-A646. Final baseline:24 /3158 /1938 /1478; weeks728/754/850/826; Delivered leader PJ-B578, Pending460, peak week3 at850. Independent PO025 extension is not covered by existing-row refresh.

10

CHAPTER 10

Use Gemini as an Analysis Partner

In this chapter you will generate and inspect formula proposals and chart summaries; test exceptions before accepting.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 55 minutes: 10 explanation/demonstration, 25 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A checked AI formula proposal and evidence-based findings.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Ask AI for a bounded formula proposal.
- Verify a generated finding.
- Distinguish a source-linked result from a generated snapshot.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use the native checked workbook and usable Gemini in Sheets access. If unavailable, critique the supplied examples manually and record the AI objective as incomplete.

10.1 Request a Formula Proposal

1. Open Ask Gemini. Paste the Part 2 prompt describing the actual source columns and range.
2. Read the proposed formula before applying it. Check its row bounds, exact status spelling and use of Amount.
3. Test the formula in a spare labelled cell and compare with the known Delivered total of 1,478.

What should you see?

The proposal is accepted only after it matches the actual schema and control total.

10.2 Check an Interpretation

1. Ask for two findings using the checked project/status summary, without web research.
2. For each finding, identify the cells that support it. Reject unsupported causal claims, forecasts or benefit percentages.
3. Write a final finding in your own words with its population and units.

What should you see?

Each retained finding is traceable to values rather than AI confidence.

10.3 Inspect Generated Visuals

1. If asking Gemini to generate a chart, inspect where its data was written and what range the chart references.
2. Temporarily change a source quantity and observe whether that generated analysis follows the original source.
3. Restore the quantity. Label a static/generated analysis snapshot appropriately or use your already checked source-linked chart.

What should you see?

The final dashboard does not imply automatic refresh where it has not been demonstrated.

10.4 Evaluate AI Proposals Against Known Test Cases

1. Prepare a test matrix before prompting: Delivered total, Pending >=200, unknown ItemCode and a new record outside the original bounds.
2. Ask Gemini for a formula and explanation using the actual headers and range. If unavailable, use the deliberately flawed proposals on the card for the same audit.
3. Inspect every proposed formula's population, field, match mode, bounds and copied-reference behaviour before using it.
4. Run the proposal in labelled spare cells against known controls and a manually verified subset. Record pass/fail and the cause of each failure.
5. Request a correction for one identified defect, or repair it manually. Keep the original proposal alongside the checked replacement so improvement is demonstrable.
6. For any generated chart inspect its data source and test a source-cell change. Record whether it follows the maintained data, separate generated cells or a static snapshot.
7. Write a final two-finding commentary grounded in checked values, rejecting causal claims, forecasts and unsupported ROI.

What should you see?

Prewritten test matrix and observed proposal results. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Design an independent prompt that includes the population, fields, intended output and tests, then run or manually review its result.
2. Challenge it with IT99 and PO025 in separate test contexts. Keep missing mapping distinct from zero and inspect new-row coverage.
3. Produce a short audit comparing original proposal, correction, evidence and remaining limitation.

Evidence to Hand Over

- Prewritten test matrix and observed proposal results.
- Original/corrected formula trace with independent verification.
- Generated-chart source check or stated unavailable route, and supported final findings.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The AI explanation sounds right while the formula differs	Audit the actual formula and contributing rows, not its prose description.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

AI can propose analysis; source tests establish whether it is suitable.

Check yourself: What evidence would establish that a generated chart follows the maintained source?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 2: Propose a Delivered Total

Session: new chat | **Grounding:** explicit Orders schema

```
Goal: Propose a formula for delivered order amount.
Context: Fictional Google Sheets analysis.
Source: Orders rows 2 to 25; Status is G and Amount is J. Header row is 1.
Expectations: Reply with a formula and explanation only. Do not edit cells. Include only
Status equal to Delivered. Use bounded ranges. I will test the formula against the source.
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Flawed formula proposals to audit if AI is unavailable: `=SUM(Orders!J2:J24)` described as all
24 rows; `=IFERROR(VLOOKUP("IT99",Rates!A2:D7,3,FALSE),0)` described as a confirmed zero rate;
`=SUMIFS(Orders!J2:J25,Orders!G2:G25,"Delivered")` described as active amount excluding only
Cancelled.
```

11

CHAPTER 11

Complete the Analysis Capstone

In this chapter you will answer a new project question, substantiate the finding and deliver a checked dashboard.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 80 minutes: 15 explanation/demonstration, 37 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 65 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A source-traceable capstone analysis and checked presentation.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Answer a new analysis question.
- Reconcile a result independently.
- Communicate assumptions and limitations.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Create a challenge copy. Use the same 24-order source and six-item Rates table; do not import real purchasing data.

11.1 Define and Calculate

1. Answer: which project has the largest Delivered amount, and which Pending records have Amount at least 200?
2. Write the population and criteria before choosing formulas or a pivot.
3. Calculate the result and retain a visible trace to qualifying order IDs.

What should you see?

The ranking and exception list are based on explicit delivered/pending definitions.

11.2 Challenge the Result

1. Reconcile the delivered total to 1,478 and all-orders total to 3,158.
2. If no baseline pending row qualifies, record zero matches explicitly. Inspect a non-qualifying baseline row and use a separate synthetic row to test the positive boundary without altering the baseline.
3. Test a duplicate or missing lookup key on a disposable copy and explain its effect without changing the baseline workbook.

What should you see?

Your result survives an independent check and identifies meaningful data assumptions.

11.3 Present the Analysis

1. Show one clear chart or table with project names and currency units.
2. State the source population, result and one limitation. Do not equate order amount with profit or budget approval.
3. Explain which sources and calculations must update for a future record.

What should you see?

A reviewer can trace the finding to source records and understand its limits.

11.4 Complete a Decision Analysis With Independent Evidence

1. Write a one-page analysis specification for the card's three questions, naming population, metric, threshold and exclusions.
2. Build answers with formulas and a pivot or filtered-source reconciliation. Retain the contributing OrderIDs for the exception list.
3. Compare project Delivered amounts and identify the largest using checked numbers. Do not infer profitability or project performance from order amount alone.
4. For Pending Amount >=200, first report the actual baseline match count, including zero if none qualify. Inspect a non-qualifying baseline row and use a separate synthetic positive/boundary test. Keep synthetic tests outside the baseline analysis.
5. Run a separate PO025 scenario and describe its effect on ranking, delivered totals and week 4. Do not overwrite the baseline result.
6. Prepare an executive table/chart and an audit appendix showing source range, assumptions, controls and unresolved data issues.
7. Ask a reviewer to challenge one answer and prove it from the source rows and a second calculation method.

What should you see?

Three-question specification and independently reconciled results. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently compare all-orders and active populations and explain how excluding Cancelled changes the question and results.
2. Stress-test an unknown rate key and a duplicate lookup key in a scenario copy. Demonstrate the warning/disposition rather than forcing a numeric answer.
3. Deliver a three-minute recommendation limited to what the data supports, plus one next question requiring additional evidence.

Evidence to Hand Over

- Three-question specification and independently reconciled results.
- Separate added-row and key-quality scenarios with audit evidence.
- Executive output, reviewer challenge response and a bounded recommendation.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A recommendation claims savings or profitability	Return to dataset meaning and request missing cost/outcome evidence before making that claim.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

The capstone combines reliable data, explicit calculations and a defensible explanation.

Check yourself: Would the same result follow if Cancelled records were included?

Zero-Match Check

The supplied baseline has no Pending record with Amount at least 200. Record that result explicitly. Use a separately labelled synthetic case to prove the positive branch; do not edit baseline records just to obtain a highlight.

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Capstone questions: Largest Delivered amount by project; Pending records with Amount ⩾200 and
their IDs; highest all-orders Amount by practice week. Scenario PO025: PJ-A, Planning, IT01,
Quantity 2, UnitCost 45, Delivered, week 4. No actual budget or profit data is supplied.
```

12

CHAPTER 12

Review and Handover

In this chapter you will explain assumptions, refresh steps and known data limitations.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 50 minutes: 10 explanation/demonstration, 20 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 40 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A restored baseline and an update-tested workbook handover.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Explain the workbook update path.
- Identify future maintenance work.
- Leave a checked final state.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use your baseline workbook and separate capstone copy.

12.1 Review the Baseline

1. Clear temporary test entries and restore all source values changed during exercises.
2. Confirm the 24-record count and control totals.
3. Inspect filter states and remove or name any view that could mislead the next reader.

What should you see?

The baseline is consistent and temporary tests no longer affect it.

12.2 Write the Handover

1. Record where the source lives, what one row represents and which columns users may edit.

2. Describe formula, validation, pivot and chart ranges that must include a new row.
3. Document unmapped keys and other unresolved data issues rather than converting them to zero.

What should you see?

A future editor can update the analysis without silently omitting records.

12.3 Demonstrate One Update

1. On the capstone copy, add one new fictional order line and walk through the documented update steps.
2. Check the resulting calculations and visuals, then clearly label that copy as an extension test.
3. Keep the original baseline for future class recovery.

What should you see?

The handover has been tested on an actual extension in a separate copy.

12.4 Make the Analysis Operable by Another Analyst

1. Create a **Runbook** with source location, row grain, accepted inputs, editable fields, formula columns, exception handling and release controls.
2. Create a range inventory for validation, conditional formatting, SUMIFS, lookups, database functions, pivots, chart sources and weekly summaries used in your workbook.
3. On a separately named extension copy add PO025 and follow only the runbook to extend the analysis. Record every instruction that needed interpretation.
4. Reconcile the 25-record scenario by count, all-orders total, Delivered total and project/week contribution, then compare with the preserved 24-record baseline.
5. Test the unknown-key route without overwriting the supplied Rates mapping. Record the issue and leave a visible unresolved disposition.
6. Give a partner the runbook and ask them to locate a source, run a check and explain a failed key. If alone, use a fresh read-through with previous working tabs closed.
7. Repair one handover gap and issue a manifest identifying the maintained baseline, scenario test and current fixed summary.

What should you see?

Complete range inventory and a runbook tested on PO025. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Use the different PO026 line on the card in another disposable copy and update the analysis without returning to the lesson steps.

2. Verify how Pending amount and exception flags change, including any new-row coverage failure. Record and repair the cause.
3. Hand over the runbook, controls, open issues and manifest. Another analyst must be able to reproduce both a successful update and an unresolved-key outcome.

Evidence to Hand Over

- Complete range inventory and a runbook tested on PO025.
- Independent PO026 update with reconciled controls and exception coverage.
- Recorded handover repair, open-issue route and clear scenario manifest.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The next analyst must guess a range extension	Record exact current bounds, the new-row action and the expected checks in the runbook.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

You can maintain and hand over a checked analysis rather than a collection of unexplained formulas.

Check yourself: Which assumption would you check first if a future total unexpectedly drops?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Independent PO026, separate baseline-derived copy: PJ-B | Procurement | IT03 | Quantity 10 | UnitCost 30 | Pending | week 4 | Handover exception test. Do not combine it with PO025 unless you explicitly define and reconcile that additional scenario.
```

Google Sheets Advanced

A practical 2-day course with step-by-step draft labs, copy-ready prompts, fictional inputs and expected-result checks. Use the supplied practice pack. Full lab execution, screenshots and classroom timing validation remain required before participant release.

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